INDUSTRY STARTER PACKS FOR ENTERPRISE ARCHITECTS

Pharmaceuticals

What the industry is, how it runs, and how to model it — for architects who have never worked in it

A working guide for an enterprise architect walking into a pharmaceutical company — and far more likely a generics manufacturer or a contract developer than an originator. The eight archetypes and why they are different businesses. The decade between spend and revenue. The molecule that has four identities before it reaches a shelf. The layer no other industry has: a regime where the system itself is regulated, validation is a maintained state and "just patch it" does not exist. Then the part that matters most: how to turn all of it into a model a regulatory affairs manager would recognise as true.

South African lens. Sourced throughout. October 2026.

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CHAPTER 1

The industry in one chapter

A pharmaceutical company is a research-and-development business wearing a manufacturer's clothes. Almost everything strange about its systems estate follows from that one sentence — and so does most of what an architect gets wrong in the first month.

An R&D business wearing a manufacturer's clothes

A pharmaceutical originator spends money for ten to fifteen years discovering and proving a molecule, then makes and sells it for a period that ends on a largely known date. The factory is the visible part. It is regulated, capital-intensive and impressive, and it is not where the economics live.

Here is the arithmetic that gives the game away. A small-molecule tablet whose active ingredient costs cents per dose can carry a price measured in rands or dollars per dose. The factory is not expensive. The price has to recover a decade of programmes that failed, plus the one that worked.

So: **a pharmaceutical company is a research-and-development business wearing a manufacturer's clothes.** Four things follow, and between them they generate most of pharmaceutical architecture.

One: a decade or more between spend and revenue. Pharmaceutical accounting expenses research as it is incurred and capitalises very little of it. The revenue that spend produces arrives, if at all, ten to fifteen years later, from a different product, under a different management team. There is no accounting period in which the research line and the revenue line describe the same cohort. Every productivity metric that divides this year's approvals by this year's research spend compares two unrelated populations, and you will be asked to build one anyway. Build it with the cohort year visible on the face of it.

Management, therefore, does not steer on profit. It steers on portfolio value and on stage gates. The architectural consequence surprises people: **the systems of record for the most important decisions in the company are not financial systems.** They are the portfolio management system, the clinical trial management system and the regulatory information management system.

Two: most candidates fail, and the failures are the cost base. The cost of an approved medicine is, by construction, the cost of that medicine plus the cost of everything that failed on the way. The data consequence is immediate. **The cost object that matters is the programme, not the product**, because a programme that fails produces cost and no product record at all. Systems built around a product master cannot hold the cost of the candidates that never became products. Ask a client what it spent on a terminated programme in total, and watch what happens: the spend was booked to cost centres, studies and purchase orders, and the programme identifier was optional on half of them.

Three: cost of goods is often trivial against the cost of discovery — and the exception is probably where you work. For small molecules, cost of goods sold is a modest share of net revenue. For biologics it is materially higher: cell-culture manufacture, cold chain, single-use consumables, and a quality-control cost per batch that dwarfs anything in tablet making. And for a generics manufacturer or a contract developer — the two archetypes a South African company is most likely to be — **cost of goods is the entire business.** There is no discovery cost to dwarf it. Chapter 2 is about nothing else.

The sourced anchor here is worth carrying into a meeting. Aspen Pharmacare, which is not a research originator but a commercial and manufacturing business, reported a gross profit margin of 58.5% on Commercial Pharmaceuticals for the six months to 31 December 2025. In the same period its Manufacturing segment turned R4,499 million of revenue into R208 million of normalised earnings before interest, tax, depreciation and amortisation — revenue down 23%, normalised EBITDA down 84% [1]. A 23% revenue decline producing an 84% earnings decline is the arithmetic of a very high fixed-cost base in a sterile facility. **Sterile manufacturing is utilisation economics, not margin economics.** An architect designing capacity planning or product costing for a sterile site should have that sentence in front of them on the first day.

Four: a patent cliff ends a product's economics on a known date. Exclusivity is time-bounded. When it ends, generic or biosimilar entry collapses price and volume for the originator. The date is known years in advance, which means the loss is forecastable, which means it is already in the plan — which means the entire commercial architecture of an originator is organised around replacing revenue that is scheduled to disappear.

Pharmaceutical Executive reported on 2 June 2026 that roughly 200 medicines will lose patent or exclusivity protection between 2026 and 2030, with an estimated USD 400 billion of revenue at stake, describing it as the largest cliff the industry has faced [2]. Moody's has separately put USD 300 billion on the cliff, in the narrower context of hospital drug spending [3]. **Name both and do not reconcile them.** They measure different baskets over different windows, and an architect who averages them has invented a number.

> Exclusivity is not one thing. It is a stack: the composition-of-matter patent on the molecule; secondary patents on formulation, polymorph, salt form, process, combination, regimen and device; regulatory data and market exclusivity, which is a period in which a regulator will not accept or approve a generic relying on the originator's data; orphan designation for rare disease; and patent term extension compensating for regulatory review time. The stack differs per country and it is litigated, so the date moves.

The loss-of-exclusivity date is therefore planning master data of the highest consequence, it differs per country per product, and it is contested. Its system of record is usually the intellectual property function's own database, and frequently a spreadsheet. **Finding where exclusivity dates actually live is one of the more revealing first questions you can ask an originator.**

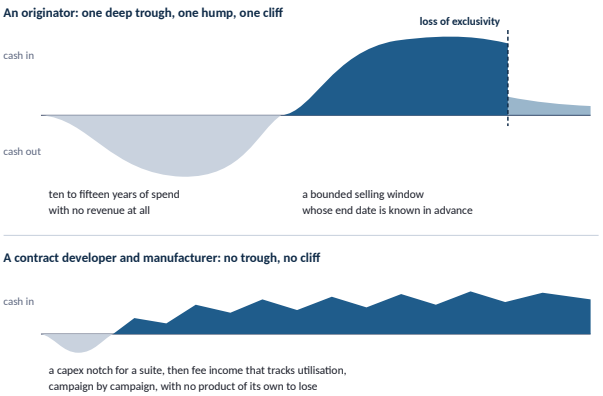


Figure 1.1 Money in, money out. An originator's cash shape is a deep trough, a bounded hump and a cliff on a known date. A contract developer has neither trough nor cliff: it earns fees as capacity is used, and owns no registration to lose. Shapes, not quantities — the vertical scales are illustrative.

The portfolio is the unit of economics, not the product

A discovery programme is best understood as a sequence of purchased options. You spend to reach the next decision point, and at that point you either pay again to continue or you abandon. The value of the portfolio is the value of those options, not the expected value of the pipeline treated as a set of certainties.

That is why pharmaceutical management does not discuss programmes in terms of revenue or margin. It discusses them in terms of **risk-adjusted net present value**, usually written rNPV and sometimes eNPV. If you cannot follow an rNPV conversation you cannot sit in a portfolio review, so here is the construction, step by step.

1. **Forecast the launched product's cash flows** by year — addressable patients, diagnosed and treated share, market share, net price per patient-year, compliance and persistence, then cost of goods, selling costs and distribution. Run it from launch to loss of exclusivity, then apply an erosion curve past that date.
2. **Lay out the remaining development costs** by year and by phase: trial costs, manufacturing scale-up, regulatory fees, launch preparation.
3. **Discount both** at a cost of capital.
4. **Risk-adjust** by multiplying the cash flows beyond each gate by the cumulative probability of technical and regulatory success from the current phase. That multiplication is the step that makes it rNPV rather than NPV.
5. **Sum, and compare against the alternatives — including stopping now.**

Three words you will hear and must not misread. **Peak sales** is the maximum annual revenue the forecast reaches, used as shorthand for a programme's size: "a two-billion-dollar asset". It is a forecast, not a fact, and its probability weighting is usually dropped when the number is quoted in a corridor. **Probability of technical and regulatory success**, also PTRS, PoS or likelihood of approval, is the risk adjustment in step four. **Cost of goods as a percentage of net sales** is a technical-operations target, often set as a gate criterion before Phase III, because a product that cannot be made at an acceptable cost at the price the market will bear should be stopped before the expensive trial starts.

Now the part that matters to you. rNPV is a *model*, and it is assembled from inputs that live in five different places: epidemiology and market forecasts in commercial; trial cost estimates in clinical operations and procurement; probability of success in portfolio or statistics; cost-of-goods estimates in technical operations; and exclusivity dates in intellectual property. **The rNPV is the most dependency-rich artefact in the company and it is almost always maintained in a spreadsheet.** When a client asks for "a portfolio management capability", what they usually want is for that spreadsheet to become governed — versioned inputs, traceable assumptions, and an audit trail on who changed the probability of success in the week before the gate meeting. That is a data-governance project, not a tool purchase.

The second architectural consequence of the portfolio view is smaller to state and larger to build. **The pipeline data model must be able to represent a programme that stops.** "Terminated" is a first-class lifecycle state, not an exception. The programme must retain its full cost and its full evidence and never produce a product, and that evidence must stay retrievable for decades, because it may bear on a later safety question about a related molecule, or on a regulator's question about what the company already knew.

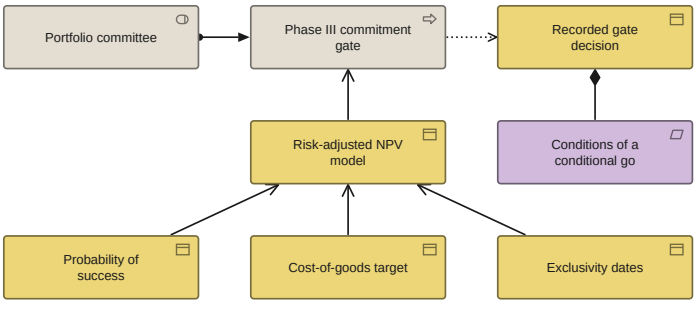


Figure 1.2 The stage gate is an authorisation to spend, not a status review. The risk-adjusted net present value is the artefact the decision turns on, and it is assembled from five functions' data. Conditional go is the commonest outcome and the hardest to track, because its conditions become obligations someone must evidence later.

DO NOT INVENT AN ATTRITION NUMBER

The standard industry picture is a funnel: ten to twenty thousand screened compounds yield a handful of clinical candidates, of which one reaches approval. The research behind this book could not retrieve a citable primary dataset for phase-level success rates, and **phase-level attrition and probability of approval are therefore unverified here.** The named sources to go to are BIO/Informa/QLS *Clinical Development Success Rates*, the Wong, Siah and Lo analysis from MIT, and Norstella/Citeline's annual series — and when you quote one, quote the study, the window and the dataset, because the figures differ by several percentage points between studies and by a factor of two or more between therapeutic areas. What is safe to assert without a figure: attrition is severe; it is worst in oncology and in central-nervous-system disease; it concentrates in Phase II, where efficacy is first tested against a control; and Phase III failures are the expensive ones, because by then the spend is already sunk.

"USD 2.6 billion to develop a drug" is one estimate, not the fact

This is the most-cited and least-settled number in the industry, and a chapter that prints one figure as *the* cost of developing a medicine is wrong whichever figure it picks. The credible published estimates differ by a factor approaching fifty.

Estimate	Figure	Method and sample
DiMasi et al. 2016	USD 1,395m out-of-pocket; **USD 2,558m capitalised at 10.5%**; USD 2,870m with post-approval cost, all in 2013 dollars	Self-reported data from 10 multinational firms on 106 new drugs, 19 of them biologics, first tested in humans 1995–2007
Prasad and Mailankody 2017	**Median USD 648.0m out-of-pocket** (range USD 157.3m to USD 1,950.8m); USD 757.4m with a 7% opportunity cost	10 cancer drugs from 10 companies with no other US-marketed product, approved 2006–2015, reconstructed from filings with the Securities and Exchange Commission
DiMasi et al. 2003	USD 403m out-of-pocket; USD 802m capitalised at 11%, in 2000 dollars	68 new drugs from 10 firms entering clinical development 1983–1994
Public Citizen 2001	USD 57–71m after tax; USD 163m pre-tax average	Public filings, industry association research spending across the 1990s, 857 new molecular entities. An advocacy estimate; label it as one
DNDi 2019	EUR 4–32m for a new combination or formulation; **EUR 60–190m for a new chemical entity**	A non-profit developer's own experience since 2003 across eight medicines; not capitalised, excludes post-registration studies

All five rows come from one compiled source, which also carries the critiques [4]. The "USD 2.6 billion" in general circulation is the DiMasi 2016 capitalised figure, and the single most important thing to know about it is this: **roughly half of it is not cash that anyone spent. It is the imputed opportunity cost of capital over the development period.** Light and Warburton's 1996-era critique of the earlier study, re-adjusting the inputs, produced a lowest estimate of USD 43.4 million — eighteen times lower [4].

Four methodological choices explain almost the whole spread. Whether the cost of failures is included, and at what attribution rate. Whether capital is charged for the development period, and at what rate. Whether the sample is self-reported industry data or mandatory public filings. And whether the medicine is a novel chemical entity from a large portfolio or a reformulation from a non-profit.

> Name the range, name at least DiMasi 2016 and Prasad and Mailankody 2017, and say what drives the difference. On the figures in the table above the honest span runs from well under USD 100 million to about USD 2.9 billion, and the published floor is lower than the frequently quoted "USD 60 million to USD 2.9 billion" because of the Public Citizen and Light and Warburton numbers.

Why should an architect care about an argument between health economists? Because the company's own internal number — the one that actually decides gates — is **none of the published ones.** It is a company-specific model built from its own historical programme costs, its own attrition rates and its own cost of capital, maintained by the portfolio or research-finance function, and it is an input to the rNPV that decides which programmes live. Finding that model, understanding it, and not breaking it is a real obligation on any research-systems engagement.

A product is four things, and they do not have the same cardinality

This is the single most useful picture in the volume, and the error it prevents is the most expensive one in pharmaceutical data modelling.

A **molecule** is the active pharmaceutical ingredient: dolutegravir. A **formulation** is the molecule plus excipients at a strength in a dosage form: dolutegravir 50 mg film-coated tablet. A **registration**, which South Africans usually call a registration and most of the world calls a marketing authorisation, is one formulation approved by one regulator for one set of indications, manufactured at a specified set of named sites, with specified specifications and a specified patient information leaflet. A **SKU** or pack is that registration in a pack configuration — 30s blister, 90s bottle, three-month patient-ready pack — with its own barcode, its own single exit price and its own stock-ledger identity.

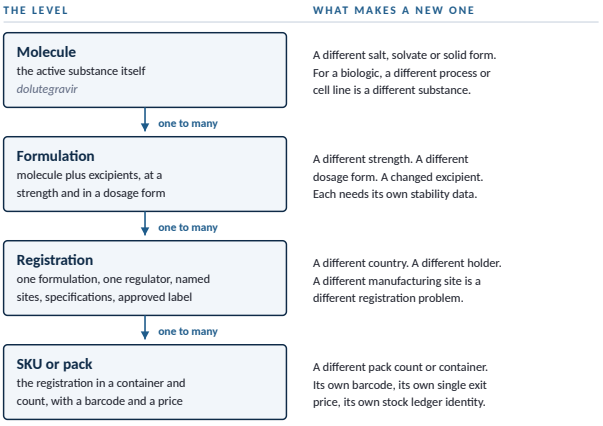


Figure 1.3 The four identities of a product. Molecule, formulation, registration and SKU are four objects with four cardinalities: one to many down each step, and many-to-many once you look across the whole set. A single product record as the unit of everything will be rebuilt. Chapter 9 models all four.

The ratio between those four is many-to-many in every direction. One molecule can carry twenty registrations across strengths and dosage forms. One registration can carry six packs. One pack can be sold under two brand names, as a clone or duplicate registration. **Any data model that assumes one product record is the unit of everything will be rebuilt.**

There is a test for this, and it takes ten seconds. When a business user says "we have 180 products", ask which of the four they mean. The answer is usually something like "180 registrations, about 450 SKUs, about 60 molecules" — three different numbers, all true, routinely reported as one. Chapter 9 builds the conceptual data model that holds all four properly.

The value chain in one page



Figure 1.4 The value chain, with the decision that closes each stage. The two squares are external authorisation gateways: permissions, not stages whose duration you control. The spend column shows relative magnitude only — no absolute duration or cost figure in this chapter is sourceable, and none is given.

Target identification and validation. A biological target — a protein, receptor, enzyme or pathway — is proposed as causally implicated in a disease, and evidence is assembled that modulating it would help. The output is a target validation package and a target product profile: a statement of what the eventual medicine would have to do to be worth having. This stage runs on an electronic laboratory notebook, literature subscriptions and compute, and its output is mostly unstructured. **The traceability problem starts here**, because the dossier submitted fifteen years later may need to point back to the original rationale, and almost no company can.

Discovery and screening. Compounds are tested against the target, by high-throughput screening of large libraries, or structure-based design, or biologics discovery. Hits become leads through iterative chemical modification; the output is a development candidate. The closing decision is **candidate selection**, and it is a real gate with a real committee. It is where a molecule acquires a programme code, a budget and a place in the portfolio review. Architecturally, note this: **the compound registration system is the first authoritative identity for the thing that will eventually be a product**, and it is almost never linked to the material master that will carry the same substance twelve years later. That gap is a standing finding.

Preclinical development. The candidate is characterised well enough to justify giving it to a human: toxicology in at least two species under Good Laboratory Practice, safety pharmacology, genotoxicity, and absorption, distribution, metabolism and excretion work. In parallel, chemistry, manufacturing and controls work begins — a scalable synthetic route, a formulation, specifications, and manufacture of the clinical trial material itself under Good Manufacturing Practice.

The IND or CTA gateway. An application is filed for permission to dose humans. In the United States this is an Investigational New Drug application, effective by default if the regulator does not object within 30 days. In the European Union it is a Clinical Trial Application through a single portal with coordinated assessment. In South Africa it is a clinical trial application to SAHPRA, which must be accompanied by approval from a registered health research ethics committee. SAHPRA's target is 80 working days; in 2024/25 it received 166 applications, had 130 due, and met the target on 94% against a target of 80% [5]. That 166 is the best available indicator of South African clinical trial volume, and it beats any market-research estimate.

This gateway is also the first point at which the regulatory information management system and the clinical trial management system must agree with each other — and the first point at which a company discovers that they do not. The authorisation number, the protocol version it covers and the sites it covers are three facts that live in two systems and diverge.

Phase I. First administration to humans, usually healthy volunteers, except in oncology where it is patients because the drugs are toxic by design. Tens of subjects. The questions are safety, tolerability, pharmacokinetics and dose.

Phase II. First test of efficacy in patients with the disease, against a control. Phase IIa asks whether it does anything; Phase IIb asks how much and how often. Hundreds of patients. This stage produces the evidence that the mechanism works in humans, the dose and endpoint and population for Phase III, and — frequently — the decision to stop.

The gate at the end of Phase II is **the most consequential decision in the industry.** Phase III is where the great majority of development cost sits, it is effectively irreversible once started, and it fixes the label the product will eventually carry. This is where rNPV does its real work and where the cost-of-goods target is enforced.

Phase III. Confirmatory trials: large, randomised, usually multi-centre and multi-country, powered to demonstrate the claims that will appear on the label. Typically two pivotal trials. In parallel, the commercial manufacturing process is developed and validated, stability studies run on registration batches, launch supply is built, and the submission is prepared. What closes it is **database lock, unblinding and the primary analysis read-out** — one moment that decides whether a decade of spend produced an asset or a write-off.

Submission and review. A dossier in the Common Technical Document structure — five modules: regional and administrative, summaries, quality, nonclinical, clinical — goes to each regulator electronically. SAHPRA's evaluation targets are expressed in **working days, not months**: 360 for new chemical entities and 250 for generic medicines [5]. Three hundred and sixty working days is roughly 17 to 18 calendar months of regulator time, before any sponsor response time is added. Quote the working days, as SAHPRA does, and then convert for your audience; do not convert silently.

> **THE NUMBER TO CARRY INTO A SOUTH AFRICAN ENGAGEMENT**
>
> In 2024/25 SAHPRA **received 2,691 generic applications and finalised 568** [5]. Inflow exceeded finalisation by roughly five to one on that stream. Whatever the backlog clearance programme has achieved, the queue regenerates unless capacity or process changes. **Registration lead time in South Africa is a capacity-constrained queue, not a backlog that was cleared**, and a client's launch plan that assumes a timeline is assuming a queue position.

Launch, lifecycle management, loss of exclusivity. Launch sets the price, arranges reimbursement, releases stock and deploys the field force, and pharmacovigilance switches from trial-based to spontaneous-report-based. Lifecycle management is everything afterwards that extends or defends the product's economics — new indications, new formulations, fixed-dose combinations, paediatric programmes, new geographies, cost reduction — and on the regulatory side it is a continuous stream of variations, periodic safety reports, renewals and commitments. **Variations are the volume business of regulatory affairs**, and they are where the architecture either works or does not. Then exclusivity ends, generics or biosimilars enter, price and volume fall together, and the originator decides whether to defend, harvest or divest.

> **ON DURATIONS AND COSTS**
>
> The research behind this book gives stage durations and cost orders of magnitude as explicitly unsourced, illustrative ranges, and this chapter does not reproduce them. **The shape and the sequence hold; the absolute numbers are not sourceable here.** What is safe to say is relative: discovery is cheap against Phase III, and Phase III is the largest single line in a development programme by a wide margin. If you need a number for a business case, get it from the client's own historical programme costs, and date it.

South African pharma's actual shape

Start with an uncomfortable admission, because it is more useful than a confident wrong answer. **There is no current, authoritative, publicly sourced figure for the size of the South African pharmaceutical market.** Who Owns Whom's industry report editions give total sector sales revenue of approximately R60.4 billion for 2021 and approximately R56.4 billion for 2019; the 2025 edition gives no market value at all [6]. Use the R60.4 billion figure only with its date attached, and note that it predates two single-exit-price cycles, a substantial currency move and nine contract manufacturer closures. Market-research vendors publish South African forecasts; those are marketing figures from models you cannot inspect and they should not appear in a client deliverable.

Next, the structural fact that most commentary gets backwards.

	Private sector	Public sector
Revenue share	**More than 80%** of manufacturing and distribution revenue (Who Owns Whom 2023 edition, 23 August 2023) [6]	Under 20%
Volume share	The minority	**The overwhelming majority**, driven by the antiretroviral and tuberculosis programmes
Price mechanism	A gazetted single exit price per pack	A tender price won competitively
Measured as	SEP multiplied by volume	Tender price multiplied by volume

Revenue share and volume share point in opposite directions, and the two chains are measured on different bases. Adding them produces a number that means nothing. This is the commonest single error in South African pharmaceutical market commentary, and it matters to an architect because the two chains are also structurally separate in the systems — the private price lives in one place and the tender contract in another, usually a different system entirely.

The defining volume fact is the antiretroviral programme. As at the Thembisa version 5.0 model published 12 May 2026, **6.18 million people were on antiretrovirals in 2025**, up from 6.14 million in 2024 — a gain of about 40,000 [7]. Roughly 7.9 million people were estimated to be living with HIV in mid-2025, about 12.3% of the population [7].

But read the next sentence slowly, because it is the one that changes what you build. **The programme is no longer growing fast.** The last year-on-year growth below 100,000 was twenty years ago. In 2024/25 more than 700,000 people restarted treatment and 180,000 started for the first time, and almost as many stopped [7]. The "Close the Gap" campaign aimed to add 1.1 million people by the end of 2025 and fell about 800,000 short [7].

> The programme processes on the order of 900,000 treatment starts and restarts a year against a net gain of 40,000. **The operational story is churn, not growth.** A demand model built on a growth curve will be wrong; a demand model built on re-entry, interruption and re-initiation has a chance. And three molecules in one fixed-dose combination therefore carry a material share of national tablet volume, which makes a supply interruption in any one of them a national event.

Two more facts about local capacity, both of which must carry their dates. **About 70% of pharmaceutical products consumed in South Africa are locally produced** — but that figure comes from the Who Owns Whom edition of 11 December 2020 and **must be flagged as dated**, given the plant and contract-manufacturer closures since [6]. And "locally produced" means **locally formulated and packed, not locally synthesised.** On the industry bodies' evidence to Parliament in August 2026, there is no remaining South African capacity to produce the active ingredient for paracetamol or to formulate penicillin, nine contract manufacturing organisations have closed recently, the country's only oral contraceptive facility has shut, and more than 2,500 jobs were lost in the preceding 18 months [8]. Those are the industry bodies' claims as reported and were not independently verified in the source.

> NEVER APPLY THE CONTINENTAL FIGURE TO SOUTH AFRICA
>
> African countries import 70% to 100% of finished pharmaceutical products and about 99% of vaccines. **South Africa does not.** It is far less import-dependent for finished product than the continental average, and an architect who quotes the continental number for a South African client has misdescribed the market by a wide margin. The import dependence that *is* near-total is in active ingredients — and **a sourced percentage for South African active-ingredient import dependence was not available to this book.** Say "near-total, and the percentage is not something I could source", not a number.

One consequence runs through everything. Active ingredients, excipients, primary packaging, filters, single-use consumables and capital equipment are priced in dollars or euros. Single exit price revenue is in rand and is capped by gazette, adjusted once a year — twice in 2026. **Cost reprices continuously; selling price reprices annually, by permission.** That one asymmetry is the best single-sentence explanation of why South African pharmaceutical manufacturing capacity is contracting, and you should be able to state it without notes.

Single exit price, in one page

The single exit price governs everything commercial in the South African private market, and three facts about it are routinely stated wrongly.

In the Act's own terms the SEP is "the only price at which manufacturers shall sell medicines and scheduled substances to any person other than the state" [9]. Three things follow at once. **One price regardless of buyer volume**: a large private hospital group and a single independent pharmacy pay the same, and private buyers cannot use purchasing power to negotiate a lower price [9]. The SEP is also the price the final seller charges the patient or medical scheme [9]. And **the SEP applies to the private sector and not to the State, which buys by tender entirely outside the regime** [9]. That last point is the most commonly misstated fact in South African pharmaceutical commerce.

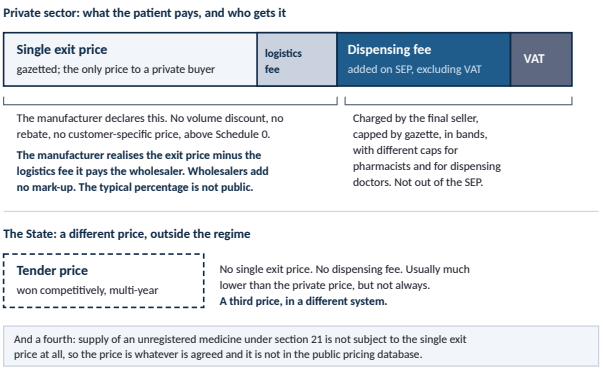


Figure 1.5 The single exit price stack, and the three other prices. The logistics fee comes *out of* the exit price and the dispensing fee is added *on top of* it. The State pays a tender price outside the regime, and section 21 supply has no regulated price at all.

Each year the Pricing Committee determines the **maximum** percentage increase manufacturers may apply to existing prices. It is a maximum, not a mandate; a manufacturer may take less, or none. **There were two adjustments in 2026, and any statement citing one of them is wrong**: the ordinary annual adjustment of **1.47% in January 2026** [9], and an **extraordinary second adjustment of 2.88% effective 1 October 2026** [10][11]. The extraordinary one covers registered medicines and scheduled substances on the national database as at 30 September, including related pack sizes, and it explicitly does not apply to medicines bought by the State [10][11]. The next ordinary cycle is 2027 [11]. One outlet rendered the January figure as 1.4%; use 1.47% [8].

Two more pieces of the stack, each with a trap in it.

The logistics fee. Wholesalers add no mark-up to the SEP. Instead the manufacturer pays the wholesaler a logistics fee, which is a portion of the exit price. The amount "varies considerably", it appears as a column in the national pricing database, and where a seller owns its own wholesaler it can earn that income undisclosed to buyers [9]. **The typical percentage is not something this book could source**, and it matters, because it means **SEP multiplied by volume is gross revenue, not net revenue, for a South African manufacturer.** It is the one genuine gross-to-net item in the local private market.

The dispensing fee is charged in addition to the SEP, not out of it. The final seller may add it, capped by gazette, calculated on the SEP exclusive of value-added tax, with different caps for pharmacists and for licensed dispensing practitioners, structured as a flat amount plus a percentage across four price bands with the percentage falling as the price rises [9][12]. If you are implementing the calculation, be careful which gazette your bands came from: a February 2025 notice published *proposed* bands for comment at different values, and the later-published bands differ. Check the document before quoting a band.

Three closing notes on the regime, because architects design against them. First, **a SEP adjustment is a scheduled, high-blast-radius, date-effective master-data event** across every pack in the country at once — treat it the way you would treat a tax-rate change, not as a pricing decision. Second, **South Africa does not operate external reference pricing.** The 1996 National Drug Policy proposed monitoring prices against other countries and no SEP has ever been declared unacceptable; the new-medicine declaration form asks for international prices, but no external reference pricing system is in place [9]. Third, **bonus, rebate and incentive schemes are prohibited above Schedule 0 — but the regulations defining what is prohibited were published for comment in 2014 and 2017 and never finalised**, so the extent of co-marketing fees, shelf fees and off-invoice rebates is unknown [9]. The prohibition is real; the definition is absent. That is a compliance-architecture problem of the first order and chapter 6 returns to it.

Approval is not launch

A registration is permission to sell. It is not the ability to sell.

In South Africa a price must be declared and gazetted into the SEP database before private-sector sale, and for the public sector a tender must be won. The initial launch price of a new medicine is not itself regulated — the manufacturer sets it, and only subsequent increases are controlled — but it must be declared [9]. And it is not always declared. **As at February 2026 no single exit price had been declared for cabotegravir or lenacapavir** [9]: approved HIV medicines with no private-sector price, and therefore not available in the private market at all.

Carry that example. "Approved" has at least five meanings and they are months or years apart: submitted, accepted for review, recommended by the committee, registered, and launched. A pipeline report that says "12 approvals this year" has to declare which gate it counted. SAHPRA's own performance reporting counts **applications finalised within a working-day target, not registrations granted**, and its 2024/25 report to Parliament gives finalisation counts and percentages but explicitly no registration numbers [5].

One structural warning: National Health Insurance

You will be asked about this in your first week, and the honest answer is uncomfortable.

National Health Insurance is judicially suspended, not being implemented. By an order recorded on 24 February 2026, the Minister of Health and the President are prohibited from proclaiming or implementing *any* provision of the NHI Act until the pending public-participation challenges are resolved [13]. The Constitutional Court heard the procedural public-participation challenges on 5 to 7 May 2026 and **reserved judgment** [14] — and as at the compilation date of this book that judgment had not been traced, so the position is unresolved. Separately, the Court **struck down the certificate-of-need provisions** of the Act, reported by 21 May 2026, without striking down the whole Act [15]. The substantive constitutional challenges remain alive.

TREAT NHI AS A SCENARIO, NEVER A ROADMAP ITEM

How medicines would be priced and purchased under NHI is not settled. The Fund is expected to be an "active purchaser", but the mechanism is unclear, and informed commentary arguing that tenders would not work under full NHI is commentary, not policy [9]. There is no timetable that can be relied on. **Any client proposal that sequences investment on an assumed NHI commencement date is unsound, and you should say so in writing.** The defensible preparatory work is capability that pays regardless of NHI's fate: patient and provider master data quality, coding standards, claims and dispensing data integrity, formulary management, and the ability to supply at different price points to different channels.

While you are being careful about legal state, be careful about tenders too. On 22 October 2026 the High Court set aside the one-month and three-month pack awards of a R15.5 billion antiretroviral tender as unlawful and unconstitutional, ordering the Department to redo the split within 120 days with the set-aside taking effect 90 days after the new awards [16]. And on 17 September 2026 the Constitutional Court struck down the entire Public Procurement Act 28 of 2024 for inadequate public participation, without suspending the declaration of invalidity [17]. **A tender award is not master data. It is a contract with a validity, a challenge window and a litigation status**, and the production plan, ingredient purchase commitments and cash forecast built on it inherit that state. Which procurement framework now governs health-product tenders in practice is genuinely unresolved.

What the rest of this book does

You now have the shape. The rest is detail in a deliberate order.

Part I finishes the industry. Chapter 2 is the chapter that earns the book: eight archetypes, because a South African company is far more likely to be a generics manufacturer or a contract developer than an originator, and the systems estate differs completely. Chapter 3 gives you the forty-odd words whose everyday meaning is wrong. Chapter 4 works the business model, and chapter 5 names the jobs, which is what lets you run a workshop.

Part II is how a company actually runs — where the money comes from, the portfolio cadence and the stage gate, and the process model.

Part III is the technical core. The conceptual data model, the information model, the application model conceptual and real, then chapter 13, which is the chapter that makes this volume different from every other in the series: a regime in which the system itself is regulated, validation is a maintained state, and "just patch it" does not exist. Chapter 14 reports an awkward finding honestly — pharma has mandatory, dated, machine-enforced data standards at the regulatory interface, and no enterprise reference architecture at all.

Part IV is constraints: the provider categories, and what the inspector actually asks for.

Part V is modelling it. ArchiMate for pharmaceuticals, then Veldtkruin Group end to end, then your first ninety days.

Veldtkruin Group, the worked example used throughout this volume, is **invented**. It describes no real organisation and every figure in it is illustrative. It is deliberately a composite of two archetypes — a generics manufacturer and a contract developer with a small clinical-stage portfolio — because the archetype distinction is the thing an architect most needs to see. Chapter 2 introduces it properly.

THE CHAPTER IN TEN LINES

- A pharmaceutical company is a research-and-development business wearing a manufacturer's clothes; the factory is visible and is not where the economics live.
- There is no accounting period in which the research line and the revenue line describe the same cohort, so management steers on portfolio value and stage gates rather than on profit.
- The cost object that matters is the programme, not the product, because a programme that fails produces cost and no product record at all.
- USD 2.6 billion is one estimate from one study; credible published figures run from well under USD 100 million to about USD 2.9 billion, and roughly half the headline is imputed cost of capital rather than cash.
- Phase-level attrition and probability of approval are not sourced in this book; name the study or give no number.
- A product is four things with different cardinality - molecule, formulation, registration and SKU - and a single product record will be rebuilt.
- In South Africa more than 80% of pharmaceutical revenue is private-sector while the overwhelming majority of volume is public-sector; the two figures point in opposite directions and must never be added.
- The antiretroviral programme reached 6.18 million people in 2025 and is no longer growing fast; the operational story is churn, not growth.
- The single exit price applies to sales to anyone other than the State, there were two adjustments in 2026, wholesalers take no mark-up, and the dispensing fee is added on top rather than taken out.
- Approval is not launch: as at February 2026 no single exit price had been declared for cabotegravir or lenacapavir.
- National Health Insurance is judicially suspended with a reserved Constitutional Court judgment outstanding; treat it as a scenario, never a roadmap item.

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CHAPTER 2

The archetypes

Eight businesses share one name. The systems estate of a generics manufacturer has almost nothing in common with a contract developer's, and neither looks like the originator that every reference architecture describes. Getting this wrong is the most expensive mistake available to you in the first month.

Why this is the chapter that earns the book

"Pharmaceutical company" names at least eight businesses. They have different revenue mechanics, different regulatory burdens, different cost structures, different systems of record and different dominant architectural problems. An architect who assumes the originator model and walks into a generics plant will design the wrong estate, and will be three months into a programme before anyone notices.

The reason this matters more in South Africa than anywhere else is arithmetic. Almost all pharmaceutical reference material — the textbooks, the vendor decks, the conference talks, the reference architectures — describes the originator. Almost no South African pharmaceutical company is one. **A South African pharmaceutical company is far more likely to be a generics manufacturer, a contract developer and manufacturer, or a commercial and licensing operation than an originator.** About 70% of pharmaceutical products consumed in South Africa are locally produced, on a figure dated 11 December 2020 [1], and essentially none of that local production is originator discovery output.

So the first question on any pharmaceutical engagement is not "what systems do you have". It is **which of these eight is this client, and which of them does it also partly do.** Most real companies are two or three at once, and where a company spans archetypes, the estate contains the architectural problems of each, and the conflicts between them are where the defects are.

For each archetype below: where the money comes from, what the regulatory burden is, what the cost structure looks like, what the system of record is, and the dominant architectural problem.

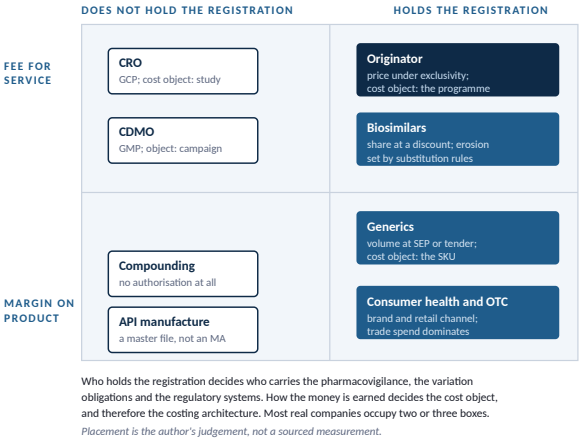


Figure 2.1 The eight archetypes on the two axes that matter. Who holds the registration decides who carries the pharmacovigilance, the variations and the regulatory systems. How the money is earned decides the cost object. Most real companies sit in two or three boxes at once.

Originator, or research-based pharmaceutical company

Where the money comes from. Patent-protected products sold at a price that recovers research spend across the whole portfolio, in a window bounded by loss of exclusivity. Revenue is concentrated: a handful of products typically carry most of it. Supplemented by out-licensing income — upfront payments, development and regulatory milestones, sales milestones and royalties.

Regulatory burden. The heaviest and the broadest in the industry. Full dossiers to multiple regulators. Clinical trial authorisations and ethics approvals in every participating country. Good Laboratory Practice for preclinical, Good Clinical Practice for trials, Good Manufacturing Practice for manufacture, Good Pharmacovigilance Practice for safety, Good Distribution Practice for the supply chain. After approval: periodic safety update reports, risk management plans, post-authorisation studies, and a continuous stream of variations. Advertising and medical information are separately regulated.

Cost structure. Research is the dominant discretionary spend. Selling, general and administrative cost — especially the field force and medical affairs — is comparable. Cost of goods is modest for small molecules and material for biologics. Capital sits in research facilities and in specialised manufacture.

System of record. There is no single one, and that is the point. Six systems of record carry hard boundaries between them: the compound and programme master; the study record, meaning the clinical trial management system and the electronic trial master file; the safety database; the regulatory information management system; the quality system; and the product, batch and supply record spanning the enterprise resource planning system, the manufacturing execution system and the laboratory information management system. The integrity question in originator architecture is whether a fact asserted in one of those six is consistent with the same fact in the others.

Dominant architectural problem. **End-to-end traceability from a research observation to a regulatory claim to a label to a safety signal.** That chain crosses six systems of record, four regulatory regimes and two decades. The secondary problem is the boundary between trial data and commercial data, which must not be joined.

In South Africa. Multinational originators operate here predominantly as **commercial affiliates**: a registration holder, a regulatory affairs team, a locally accountable pharmacovigilance person, medical affairs and medical science liaisons, market access and sales — with manufacturing either absent or limited to secondary packaging. If you are engaged by a multinational's South African arm, you are engaged by a sales-and-regulatory organisation whose systems are mostly group systems it does not control. The local architecture question is almost always integration, local regulatory compliance and data residency, not core system design. Saying so in the first week saves everybody a quarter.

Generics manufacturer

Where the money comes from. Volume at low unit margin on off-patent molecules, won either at the single exit price in the private market, where a doctor's or a pharmacist's substitution decision picks the product, or at a tender price in the public market, where a procurement committee picks it. **Both are price-led.** First to market after loss of exclusivity earns a window of higher margin, and after that margin erodes with each additional entrant. The generic business is a business of being early and being cheap, so the systems that matter are regulatory submission throughput, which is how you are early, and cost control, which is how you are cheap.

Regulatory burden. Real, but narrower than an originator's. The dossier is an **abbreviated** one: quality and manufacturing data in full, plus a **bioequivalence** study showing the generic delivers the same active substance to the bloodstream at the same rate and extent as the reference product. No efficacy trials. But: full Good Manufacturing Practice, full pharmacovigilance obligations on the registration holder, full variation management, and — this is the constraint people miss — **the registration is site-specific**, so manufacturing flexibility is bounded by what is registered.

> **A GENERIC IS NOT THE SAME PRODUCT AS THE BRAND**
>
> It is bioequivalent, not identical. It is a different registration, manufactured at different named sites under different specifications, with its own stability data, its own artwork, its own pharmacovigilance obligation and its own annual product quality review. Treating the generic and the reference as one product in a data model collapses four separate regulatory objects.

Cost structure. **Cost of goods is the business.** The active ingredient is usually the largest single input and is usually imported. Conversion cost, packaging, quality control and regulatory maintenance follow. Research exists, but it is formulation development, analytical method development and bioequivalence studies — a few percent of revenue, not a quarter. Fixed cost in plant is high, so utilisation determines profitability.

System of record. The enterprise resource planning system for material, batch and cost; the manufacturing execution system where there is one; the laboratory system for release; the quality system for deviations and changes; and the regulatory information management system for the registration portfolio. In practice, and this is worth checking on day one, **the authoritative registration register is frequently a spreadsheet.** Ask to see it.

Dominant architectural problem. **Cost to serve per SKU across a very wide, very shallow portfolio, against a regulated price ceiling.** A mid-size generics company may hold hundreds of registrations and over a thousand SKUs, most selling small volumes, each with its own stability programme, its own annual product quality review, its own artwork and its own variation history. The problem is **portfolio administration at scale**: artwork and labelling management, change control throughput, master data for product, registration, SKU and market, and the ability to answer "is this SKU still worth having". The secondary problem is planning against tender awards that are lumpy, three years long and contested.

Biosimilars developer and manufacturer

Where the money comes from. Market share taken from an originator biologic after its exclusivity ends, at a discount to the originator's price but at a far higher absolute price than a small-molecule generic. Fewer competitors per molecule than in small-molecule generics, because the barrier to entry is much higher.

Regulatory burden. **Far heavier than a generic and much closer to an originator.**

> **BIOSIMILARS ARE NOT GENERIC BIOLOGICS**
>
> A biologic cannot be exactly copied, so the regulatory standard is **similarity, not identity**. It is demonstrated through an extensive analytical **comparability** exercise plus comparative non-clinical and clinical studies — typically a pharmacokinetic and pharmacodynamic study and at least one comparative clinical study in a sensitive indication. The reason runs deeper than regulation: for a biologic, the identity rule is effectively same process, same source, same sequence, which means **the process is part of the substance's identity.** A change of cell line is not a manufacturing change; it is a change of substance.

That has a commercial consequence an architect has to model. **Biosimilar erosion is slower and less complete than small-molecule generic erosion**, and the reasons are structural: fewer entrants; clinician caution about switching a stable patient; device and patient-support switching costs; a genuine cost floor, because biosimilars cost a great deal to make; and effective originator defence through price concessions, device upgrades and next-generation formulations. Above all, **substitution rules drive erosion speed, and they are regulatory and jurisdiction-specific.** A pharmacist can usually substitute a small-molecule generic at the counter; substituting a biosimilar typically needs a prescriber decision or an explicit interchangeability determination, and the rules differ by country. A quantified erosion comparison was not something this book could source, so **state the mechanism, not a percentage.**

The architectural consequence: an originator's post-exclusivity forecast must carry **two distinct erosion models**, small molecule and biologic, with different curve shapes and different per-market substitution parameters. A single erosion assumption applied across a mixed portfolio will be wrong in both directions.

Cost structure. Development cost per biosimilar is orders of magnitude above a generic, and manufacturing is cell-culture-based, capital-intensive, cold-chain and quality-cost-heavy. Cost of goods as a share of revenue is far higher than for a small molecule.

System of record. The analytical dataset is the asset. The laboratory information management system, the scientific data management system holding raw instrument files, and the statistical environment that supports the comparability argument.

Dominant architectural problem. **Analytical comparability data management.** A biosimilar programme generates an enormous, heterogeneous analytical dataset — structure, charge variants, glycosylation, potency, impurities — across many lots of both the biosimilar and the reference product, and **the regulatory argument is that dataset.** Managing it with integrity, lineage and statistical traceability is the programme. The secondary problem is process characterisation and comparability across scale-up.

In South Africa. Local development is limited; biosimilars reach the country predominantly as imported registered products. SAHPRA treats biosimilars as eligible for its clinical abridged reliance review where a reference regulatory authority has already approved them [2].

Contract development and manufacturing organisation

Where the money comes from. **Fee for service, not margin on product.** This is the single most important sentence in the chapter for anyone arriving from manufacturing or retail. The contract developer does not own the registration, does not own the product and usually does not own the inventory. It is paid for development work — process development, analytical method development, formulation, scale-up, technology transfer, stability studies — on a time-and-materials or milestone basis, and for manufacture on a per-batch, per-unit or capacity-reservation basis. Contracts commonly carry minimum volume commitments, take-or-pay reservations, and pass-through of materials at cost plus a handling percentage.

Regulatory burden. Full Good Manufacturing Practice and a manufacturing licence, inspected by the regulator — but **no marketing authorisation of its own.** The client holds the registration; the contract developer is a named manufacturing site *on the client's registration*. That is the defining asymmetry: **its regulatory exposure is distributed across every client's registration, so a GMP finding at the site is a problem for every client simultaneously.** The quality agreement with each client allocates responsibility for release, deviation handling, change-control approval rights and regulatory notification, which is why those agreements are its most important commercial documents.

Cost structure. Capacity cost, utilisation-driven — but with a quality overhead a non-GMP contract manufacturer does not carry. Gross margin per campaign depends on changeover and cleaning time between products as much as on run rate, because cleaning validation and cross-contamination control are mandatory and slow. What actually drives margin: utilisation of the asset; non-billable changeover; first-time-right batch rate, since a failed batch is usually at the contract developer's cost; and scope control on development work, where the client's experiments expand without a change order.

System of record. One quality system, one laboratory system, one manufacturing execution system — and many clients' products inside each of them. Plus a project accounting system, which is not optional here and which an architect arriving from product manufacturing will under-scope.

Dominant architectural problem. **Multi-tenancy under Good Manufacturing Practice.** One physical site, one quality system, many clients' products, each with a different registration, different specifications, different client-specific documentation requirements, different change-control approval rights, and an absolute requirement that one client's data and material never touch another's. Segregation is simultaneously physical, procedural and informational.

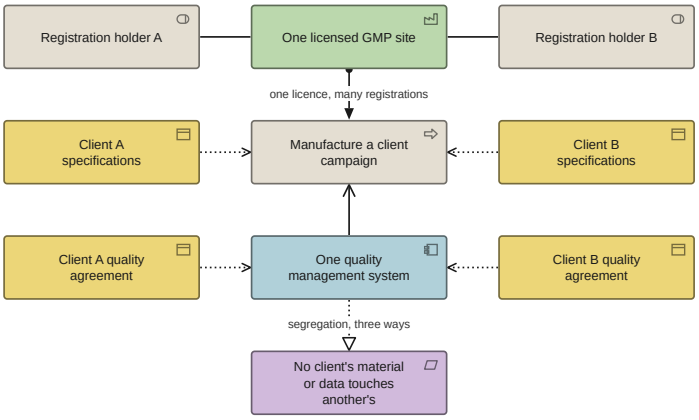


Figure 2.2 Multi-tenancy under GMP. One licensed site and one quality system carry many clients' registrations, specifications and approval rights. The segregation requirement is not a preference and it is not only about data — it is physical, procedural and informational at once.

The secondary problem is costing, and it is the one that breaks imported designs. **The cost object is the campaign or the client programme, not the batch and not the product**, because the product belongs to the client. The right analogue is job costing in engineer-to-order manufacturing, not standard costing in repetitive manufacturing.

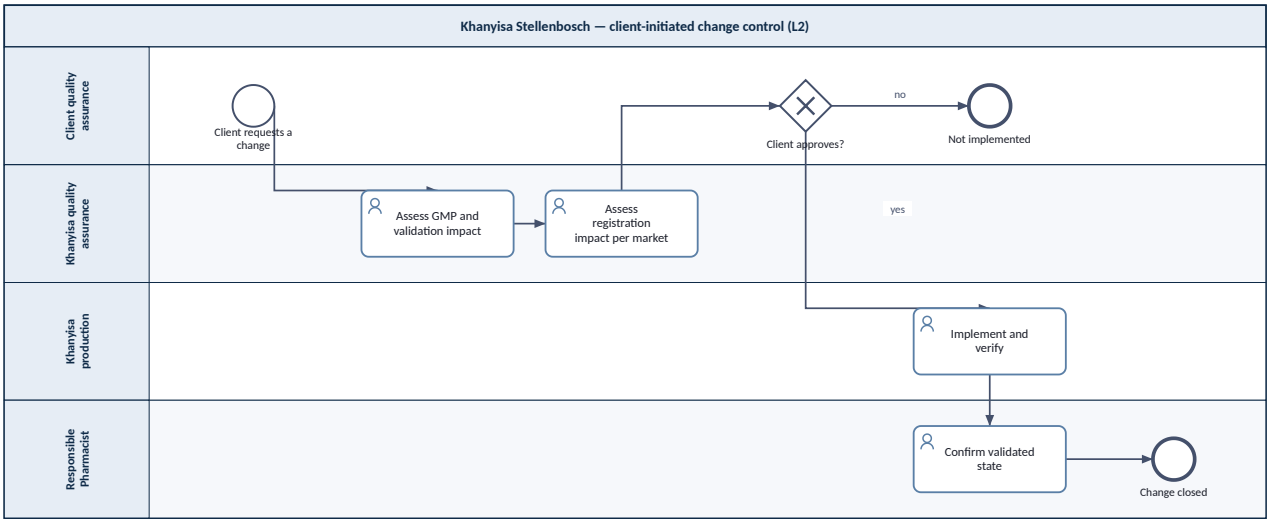


Figure 2.3 Why a quality agreement has to be enforced in workflow, not in a procedure document. At a contract developer the client, not the site, holds the change-control approval right — so an approval gate belonging to an external party sits in the middle of the site's own quality process, once per client.

In South Africa. Relevance is high and the position is deteriorating. On the industry bodies' evidence to Parliament in August 2026, **nine contract manufacturing organisations have closed recently**, disrupting supply of folic acid, plasmoquin and various eye drops, and there is no remaining domestic capacity to formulate penicillin or to produce the active ingredient for paracetamol [3]. Those are the industry bodies' claims as reported and were not independently verified in the source.

Contract research organisation

Where the money comes from. **Fee for service.** Paid to run clinical trials, or parts of them, on a sponsor's behalf: protocol and medical writing, site identification and feasibility, monitoring, data management, biostatistics, pharmacovigilance case processing, regulatory submission support, central laboratory and bioanalysis. Contracts are unit-priced — per subject, per visit, per monitoring visit, per case processed — with pass-through costs such as site payments, investigator grants, couriers and laboratory kits handled separately and sometimes at a different margin.

> **A CRO IS NOT A KIND OF CDMO**
>
> This is the commonest error an outsider makes, and the two share almost nothing architecturally. **Good Clinical Practice against Good Manufacturing Practice. Labour against capacity. The study against the batch. Sponsor-system multi-tenancy against physical segregation.** A contract research organisation holds no registrations and does not manufacture, except where it also runs a bioanalytical or small-scale clinical-supply operation. An architect who proposes one services platform covering both has proposed to merge two different inspection regimes.



Figure 2.4 The comparison an outsider gets wrong. A contract research organisation and a contract developer are both fee-for-service and share nothing else: different inspection regime, different cost driver, different cost object, different segregation problem, different deliverable.

Regulatory burden. Good Clinical Practice, a completely different inspection regime from everything else in this chapter. A contract research organisation is inspected on whether trials were conducted to protocol, whether subject consent and safety were properly handled, whether data were accurately transcribed, and whether its own quality system functions.

Cost structure. **Labour.** It is a professional-services business, so utilisation, bill rate, realisation and pass-through leakage are the metrics. The architecture that matters is resource management, timesheets, project accounting and change-order control, as much as it is clinical systems. The twin killers of margin are non-billable rework — a monitoring visit redone, a dataset re-cleaned — and unpriced change orders when a protocol is amended.

System of record. Partly the sponsor's systems rather than its own, which is an ownership and retention problem before it is an integration problem. Its own estate is the clinical trial management system, the electronic data capture system, the electronic trial master file, and the project accounting system.

Dominant architectural problem. **Sponsor-system multi-tenancy and data hand-back.** It works in many sponsors' systems as well as its own, must keep each sponsor's data separate and blinded as required, and must hand over a complete, validated, audit-ready dataset and trial master file at study end. That obligation is contractual, regulatory and brutally detailed, and getting it wrong is the most common source of disputes in the sector. The secondary problem is change orders: trials change constantly, and an organisation that cannot price and bill a protocol amendment promptly bleeds margin.

In South Africa. An established clinical trial location with mature investigator networks in HIV, tuberculosis, oncology, diabetes, cardiovascular disease and respiratory medicine. SAHPRA received **166 human clinical trial applications in 2024/25** and met its 80-working-day target on 94% of them against a target of 80% [4]. Use that 166 rather than a market-research estimate; it is the best available indicator of South African trial volume.

Consumer health and over-the-counter

Where the money comes from. Brand-led, retail-channel, consumer-marketed. Price is set commercially within regulatory limits, promotion goes to the public rather than to prescribers, and the buyer is a shopper rather than a prescriber or a procurement committee. Revenue depends on brand equity, shelf presence, trade terms and category management. **This archetype's commercial model is closer to fast-moving consumer goods than to pharmaceuticals**, and it is the one place in this volume where retail vocabulary legitimately applies.

Regulatory burden. Lower and more variable by country. Schedule 0 products in South Africa may be sold outside pharmacies; higher schedules are pharmacy-only or pharmacist-supervised. Notably, **the prohibition on bonus, rebate and incentive schemes under the South African Act has its enforcement paused for Schedule 0** [5] — meaning trade-promotion practices that are illegal for prescription medicines are tolerated for Schedule 0. Complementary and traditional medicines carry their own regime. Advertising to the public is permitted and separately regulated.

Cost structure. Marketing and trade spend dominate. Cost of goods is moderate. Research is minimal and is largely formulation and claims-substantiation work.

System of record. The enterprise resource planning system and a trade promotion management capability, alongside the same GMP estate as any other manufacturer.

Dominant architectural problem. **Trade promotion management sitting beside a regulated-price estate.** A company that sells prescription medicines at a gazetted single exit price *and* Schedule 0 consumer products with trade discounts has two incompatible commercial models inside one order-to-cash process. **The controls that prevent a rebate reaching a prescription line are the architecture.** The secondary problem is demand sensing for consumer seasonality against GMP manufacturing lead times that cannot flex.

Active pharmaceutical ingredient manufacture

Where the money comes from. Business-to-business supply of the active substance to formulators, at a price per kilogram, under long-term supply agreements or spot purchase. Margins are commodity-like for established molecules and much better for complex or newly off-patent ingredients. Scale is everything.

Regulatory burden. Good Manufacturing Practice for active ingredients, which is a distinct part of the GMP regime from finished-dose manufacture, plus the obligation to support customers' registrations. The ingredient maker supplies either an open dossier or a **drug master file**, also called an active substance master file, whose restricted part the regulator reads and the customer does not. **A change at the ingredient maker therefore propagates into every customer's registration as a variation**, and that is the single most under-modelled dependency in pharmaceutical supply architecture.

Cost structure. Chemical process economics: raw materials, solvents, energy, effluent treatment, capital-intensive multipurpose plant. Environmental compliance cost is material in a way it is not for tablet compression.

System of record. The enterprise resource planning system and the laboratory estate, plus — and this is the one that is usually missing — a customer-and-registration relationship dataset.

Dominant architectural problem. **Change propagation and customer qualification.** An ingredient maker must know, for every change it contemplates, which customers' registrations in which countries name it as the source, and what variation each would need to file. Most ingredient makers hold that relationship dataset poorly or not at all.

In South Africa. Critically thin, and this is one of the sector's defining facts. There is **no domestic capacity remaining to produce the active ingredient for paracetamol** and none to formulate penicillin [3]. In October 2026 the Council for Scientific and Industrial Research launched a facility intended to reduce reliance on imported ingredients [6] — which is an admission of the dependency. **Treat any claim that South Africa has meaningful active-ingredient capacity as false unless it is specifically sourced.**

Compounding

Where the money comes from. Preparation of a medicine for an individual patient, or in anticipation of patient need, by a pharmacist — paid as a professional service plus materials, not as a registered product sale. Economically small; legally and architecturally interesting out of all proportion to its size.

Regulatory position, and it is widely misunderstood. Section 14(1) of the Medicines and Related Substances Act 101 of 1965 requires registration before sale. **Section 14(4) exempts a medicine "compounded in the course of carrying on his or her professional activities by a pharmacist"** [7]. The exemption is bounded: a compounded medicine may not contain components whose sale is prohibited or whose registration was refused, and may not be advertised to the public or to health professionals [7]. The 2017 General Regulations limited compounding to 30 consecutive days' use; a High Court ruling in December 2021 ordered redrafting; amended regulations finalised in 2022 deleted the provision barring compounding to circumvent section 14 and recognised **anticipatory compounding** based on past volumes [7].

> **"COMPOUNDING PHARMACY" IS NOT A LICENSED CATEGORY IN SOUTH AFRICA**
>
> The Pharmacy Act regulations recognise community, institutional, wholesale, manufacturing and consultant pharmacies. **Specialist compounding operations are licensed as community pharmacies**, and community and institutional pharmacies may perform compounding, manipulation or preparation of any medicine or scheduled substance [7]. **SAHPRA licenses manufacturers and wholesalers under section 22C(1)(b) and does not license community pharmacies** [7]. Whether a compounding pharmacy may import active ingredients is contested [7]. Draft guidelines on good compounding practice were published for comment in **June 2023 and were never finalised**, and as at June 2026 were no longer accessible on SAHPRA's website [7]. If you are told a client is "a licensed compounding pharmacy", the licence is a community pharmacy licence and it is held under a different Act by a different body.

The live controversy, which tells you where the fault line is. SAHPRA and the South African Pharmacy Council announced action against a compounding pharmacy for critical regulatory non-compliance in compounding unregistered medicines, concerning unregistered GLP-1 products — the weight-loss and diabetes class [8][7]. **At what volume does compounding become unlicensed manufacturing?** In South Africa that is not settled in regulation.

Cost structure. Labour and materials.

System of record. Pharmacy practice records, and in a scaled operation a de facto batch record system in a business that is not licensed as a manufacturer.

Dominant architectural problem. **Record-keeping with no registration to hang it on.** Every other archetype anchors its data to a marketing authorisation. Compounding has none, so traceability of ingredient lots, formula versions, patient attribution and stability justification has to be built from pharmacy practice records.

How to tell which archetype you are in, in the first week

Six questions, in this order. You can ask all of them in one morning and they will tell you more than a month of document review.

1. **"Who holds the marketing authorisation for the products made on this site?"** If the answer is "we do", you are in an originator, a generics manufacturer, a biosimilars business or a consumer business. If the answer is "our clients do", you are in a contract developer, and everything about the estate follows from that. If the answer is "it depends on the product", you are in a company that in-licenses, and you should ask for the list before you ask anything else.
2. **"What is the smallest thing you work out a margin on?"** The answer names the cost object: programme, SKU, campaign or study. If nobody can answer, that is itself the finding.
3. **"Who approves a change to a manufacturing process here?"** If an external party has an approval right, you are in a multi-tenant quality system and the quality agreements are load-bearing documents, not legal boilerplate.
4. **"How many registrations do you hold, and how many are you not currently selling?"** The second number is the dormant-registration problem, and every generics business has one, because reinstating a registration is slower than retaining it. Each dormant registration still consumes retention fees, stability programme slots and review effort.
5. **"What share of revenue is state tender and what share is private?"** Two commercial models, two pricing mechanisms, two order profiles, two packaging designs, two delivery obligations, two payment behaviours. If the split is material, expect them to be modelled as one channel and to be wrong.
6. **"Do you run clinical trials, or supply them?"** Running them puts Good Clinical Practice in scope with its own quality system and its own inspectorate. Supplying them puts Good Distribution Practice and clinical labelling in scope. They are not the same answer and companies sometimes give both without noticing.

The matrix

Two tables, because one of eight rows and six columns does not survive a page boundary.

Archetype	Revenue basis	Heaviest regulatory load	Cost dominated by
Originator	Price under exclusivity	Everything, everywhere	Research plus selling cost
Generics	Volume at SEP or tender price	GMP, bioequivalence, pharmacovigilance	Cost of goods, mostly ingredient
Biosimilars	Share at a discount to the originator	Comparability plus biologics GMP	Development plus biologics cost of goods
CDMO	Fee for service	GMP, on clients' registrations	Capacity and its utilisation
CRO	Fee for service	GCP	Labour
Consumer and OTC	Brand, retail channel	Advertising rules plus scheduling	Marketing and trade spend
API manufacture	Price per kilogram	API GMP plus master files	Chemicals, capital, effluent
Compounding	Professional fee	Pharmacy Act, not SAHPRA licensing	Labour and materials

Archetype	System of record	Cost object	Dominant architectural problem
Originator	Six, with hard boundaries	The programme	Research to label to safety traceability
Generics	ERP, plus a registration register that is often a spreadsheet	The SKU	Portfolio administration at scale against a price ceiling
Biosimilars	The analytical dataset	The programme	Comparability data integrity and lineage
CDMO	One quality system holding many clients' products	The campaign	Multi-tenancy under GMP, and campaign costing
CRO	Partly the sponsor's systems	The study	Sponsor multi-tenancy and data hand-back
Consumer and OTC	ERP plus trade promotion management	The SKU and the promotion	Trade terms beside a regulated-price estate
API manufacture	ERP plus a customer-registration dataset	The kilogram and the campaign	Change propagation into customers' registrations
Compounding	Pharmacy practice records	The preparation	Traceability with no authorisation to anchor it



Figure 2.5 The cost object decides the costing architecture. Four archetypes, four units of cost, four different things that have to exist in the estate. Standard cost per finished item fits exactly one of these columns.

Three worked failure modes

Each of these is a real shape: a design that was correct in the archetype it came from, dropped into an archetype where it was not.

One: a United States revenue-management architecture in a South African generics manufacturer

What was imported. In the United States the gap between the price a manufacturer invoices and the price it realises is enormous. Drug Channels Institute's estimate, published 7 October 2026 for calendar 2025, puts total gross-to-net reductions for brand-name medicines at **USD 416 billion**, with an unweighted average discount off list of **51.7%** among the manufacturers that disclose it [9]. A manufacturer reporting USD 100 of gross sales may realise under USD 50. The machinery that produces that gap — contract management, eligibility determination, claim validation, chargeback processing, rebate accrual and true-up against claim files arriving months later — is one of the largest system estates in a United States pharmaceutical company.

What broke. In the South African private market there is **no gross-to-net gap by design.** The single exit price is the only price at which a manufacturer or importer may sell to anyone other than the State, it may not be discounted by volume, and bonus, rebate and incentive schemes are prohibited by the Act for everything above Schedule 0 [5]. **The transactions the imported architecture exists to process are largely illegal here.** The programme delivered contract hierarchies with no contracts to hold, eligibility rules with no eligibility to determine, and an accrual engine accruing against a rebate population of zero. Meanwhile the one genuine deduction — the logistics fee the manufacturer pays the wholesaler out of the exit price [5] — was handled in a spreadsheet, because it is not a rebate and did not fit the imported model.

The lesson, precisely stated. **Importing a United States pharmaceutical revenue-management reference architecture into a South African generics manufacturer builds an estate for transactions that cannot occur.** Build the capability anyway if the company exports or consolidates into a multinational group, because the group will demand it — but do not build it for the South African private book.

> And note what is genuinely unknown. The regulations defining which acts the rebate prohibition actually prohibits were published for comment in 2014 and 2017 and **were never finalised**, so the extent of co-marketing fees, shelf fees and off-invoice rebates in South Africa is unknown [5]. The prohibition is real; the definition is absent. If you are asked to design controls against it, say in writing that the control set is being designed against an undefined prohibition.

Two: an originator product master and standard costing in a contract developer

What was imported. A costing and master-data design built for a company that owns its products: a product master as the spine, a standard cost per finished item, variance analysis against that standard, and margin reported by product and by customer.

What broke. Three things, in order of how long they took to surface.

First, **the products are not the company's.** The contract developer holds no marketing authorisations; it is a named site on each client's registration. A product master owned by the contract developer, carrying specifications and release criteria, immediately competed with 38 clients' versions of the same facts, and the quality agreements gave the clients the approval rights.

Second, **the cost object was wrong.** The economic unit is the campaign — a run of one client's product, bounded by changeover and cleaning at each end. In a low-volume, high-variety site, changeover and cleaning consume a large share of available suite time and are not billable. A standard cost per finished item hides exactly the cost the business is managed on, so the costing system reported healthy unit costs while the site lost money on the clients whose products were hardest to change over.

Third, **there was no project accounting.** Development work, milestones, capacity reservations recognised whether used or not, and pass-through materials at cost plus a handling percentage are revenue by performance obligation, not revenue on despatch. Unapproved change orders — the real margin leak — had nowhere to live.

The lesson. **A contract developer earns fee for service, not margin on product, and its cost object is the campaign.** Design it as engineer-to-order job costing with a regulated quality overlay, not as repetitive manufacturing.

Three: a manufacturing quality and batch estate in a contract research organisation

What was imported. The GMP estate of a manufacturing business, presented as "one quality platform for our services division": a quality management system built around deviations against batches, batch-centric document control, and a release workflow ending in a signature by the Responsible Pharmacist.

What broke. Everything downstream of the word "batch". A contract research organisation's work has no batches. Its unit is the **study**, and its regime is Good Clinical Practice, which is inspected on an entirely different basis: whether the trial was conducted to protocol, whether consent and subject safety were handled properly, whether data were accurately transcribed. The deviation model had nothing to attach to, because a protocol deviation attaches to a subject at a visit, not to a quantity of material. Document control around the batch record could not hold the trial master file, whose **completeness and filing timeliness are inspection findings in their own right** — a concept with no GMP equivalent. And the release signature had no object: nothing is released. What has to happen instead is a complete, validated, audit-ready **hand-back** of the dataset and the trial master file at study end.

The cost model failed in parallel. Capacity utilisation metrics were meaningless for a labour business whose levers are bill rate and realisation, and the monitoring visit that had to be redone — the single largest non-billable cost in the business — was invisible.

The lesson. **A CRO is not a kind of CDMO.** When a client has both arms, expect two quality systems, and interrogate hard any proposal to merge them. It is occasionally right. It is never simple.

The archetypes against South African reality

Now place the eight against the market you are actually working in.

Who is here. Aspen Pharmacare is the largest, and is simultaneously a commercial originator-licensee, a generics manufacturer *and* a contract manufacturer. Adcock Ingram is a generics manufacturer *and* a consumer-health business. Cipla's South African operations are a generics manufacturer. Biovac is a vaccine manufacturer. Multinational originators are here as commercial affiliates. Retail pharmacy groups are major customers, not manufacturers. **Most of the named companies occupy two or three archetypes, and the conflicts between them are the architecture.**

In-licensing is the dominant local pattern. A South African company licenses a product from an originator for the South African or sub-Saharan territory, pays a transfer price or a royalty, and carries the local pharmacovigilance and regulatory obligations. **Whether the local entity or the licensor holds the South African registration is the single most consequential fact about such a deal for architecture**, because the registration holder carries the obligations and therefore needs the systems. Expect confusion internally about which products sit which way round. Expect a spreadsheet.

"Locally produced" does not mean what it sounds like. About 70% of products consumed in South Africa are locally produced [1] — a figure dated 11 December 2020 that **must be flagged as dated** given the closures since. And it means **locally formulated and packed, not locally synthesised.** Active-ingredient import dependence is near-total; a sourced percentage for it was not available to this book, so do not state one.

AFRICA IMPORTS 70% TO 100% OF ITS FINISHED MEDICINES. SOUTH AFRICA DOES NOT.

The 70%-to-100% finished-product figure and the roughly 99% vaccine figure are **continental**. South Africa is far less import-dependent for finished product than the continental average. Applying the continental number to a South African client misdescribes the market by a wide margin, and it is a mistake that appears in a great deal of otherwise competent material.

Local content has not protected local manufacturers. This is the most quotable pair of trends in the local-content debate, and it has to be attributed carefully. On the industry bodies' evidence to Parliament in August 2026, **the 2025 antiretroviral tender awarded 28% of its value to local manufacturers, down from 72% in 2008**, and in the pills and capsules tender **the local share by value fell from 56% in 2014 to 18% in 2026**, with procurement prioritising price above all [3]. **Those are the industry bodies' claims as reported and were not independently verified in the source.** Attribute them that way every time you use them.

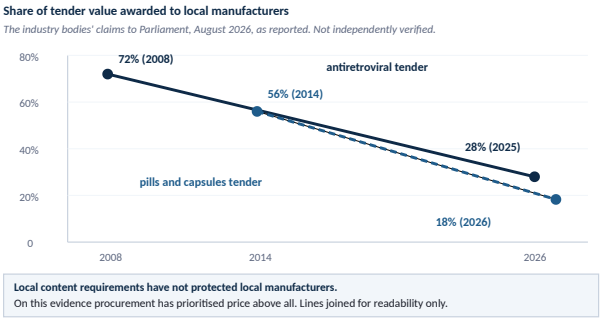


Figure 2.6 Local content has not protected local manufacturers. Two trend pairs from the industry bodies' evidence to Parliament in August 2026, reported in the press and **not independently verified**. Attribute them as claims every time.

Add the rest of the operating environment, with its dates. Nine contract manufacturers closed, more than 2,500 jobs lost over 18 months, and the country's only oral contraceptive facility shut, all on the same parliamentary evidence [3]. SAHPRA's single missed performance target in 2024/25 was GMP and wholesale licensing — 90 received, 60 due, 35 finalised, 30 within the 125-working-day timeline against a 60% target [4] — which is exactly the pathway a new site or a site transfer depends on. And the regulator itself can cause shortages: SAHPRA interventions including facility closures have disrupted supply [10]. **Model "our own site or our supplier's site is suspended" as a live scenario, not an unthinkable one.**

There is one piece of good news for an architect in all of this, and it is commercial. After the October 2026 antiretroviral tender judgment, an inadequately evidenced competitor's award can be set aside — the court found, among other things, that two bidders **did not hold a valid licence for their manufacturing site at bid closure**, so they should not have passed the first stage [11]. **A bid-management capability with governed evidence** — valid site licences covering every product and activity bid at bid-closure date, ownership verification, local-content declarations, past-performance evidence, price build-ups that survive a negotiation round — is a legitimate, valuable and now defensible architecture deliverable in this sector. Most clients assemble that evidence by hand for every tender.

Veldtkruin Group, and why it is two archetypes

Volume 1's worked example is **Veldtkruin Group**, and it is deliberately a composite, because the archetype distinction is the thing you most need to see from the inside.

Veldtkruin Pharmaceuticals is a generics manufacturer. Revenue about R2.1 billion, 184 registrations, 498 SKUs, 1,810 batches a year. Two GMP sites: an oral solid dose site at Isando and a separate sterile fill-and-finish site at Brackenfell. Public-sector tender exposure on three molecules, private-sector single-exit-price products alongside. It imports all its active ingredients. Its problems are **portfolio-scale problems against a regulated price**: SKU proliferation, dormant registrations maintained because reinstatement is slower than retention, single-sourced ingredients, manual annual product quality review assembly, a registration register that exists authoritatively only as a spreadsheet, and a manufacturing execution system two major versions behind because every upgrade is a revalidation.

Khanyisa Bio-Development is a contract developer and manufacturer with a small clinical-stage portfolio of its own, on a campus at Stellenbosch. Revenue about R265 million, work on 38 clients' registrations, 310 batches a year. Fee-for-service economics, campaign costing, and client data it holds but does not own. Its problems are **multi-tenancy and project-economics problems**: 38 quality agreements with different change-control approval rights that have to be enforced in workflow rather than in a procedure document, client-specific review and batch-documentation formats, unapproved change orders, capacity scheduling where changeover dominates, and a clinical-supply arm running under Good Clinical Practice and Good Distribution Practice alongside a manufacturing arm running under Good Manufacturing Practice, with one quality system trying to cover both.

Put them in one group, as the book does, and you get the third problem: **the group's reporting layer has to consolidate a product business and a services business whose cost objects are different words.** The chief financial officer wants one margin view. There is no honest one.

Nothing about Veldtkruin Group is real. It describes no organisation, and every figure in it is illustrative and internally consistent rather than sourced. It exists so that the rest of this book can be concrete. Chapter 18 walks the whole group end to end.

THE CHAPTER IN TEN LINES

- The first question on any pharmaceutical engagement is which of the eight archetypes the client is, and which of them it also partly does.
- A contract developer earns fee for service, not margin on product, and its cost object is the campaign rather than the batch or the product.
- A contract research organisation is not a kind of contract developer: GCP against GMP, labour against capacity, study against batch.
- Biosimilars are not generic biologics; the standard is similarity, a comparability package is required, and erosion speed is driven by substitution rules that are regulatory and jurisdiction-specific.
- United States gross-to-net has no South African analogue, and the transactions that produce it are largely illegal here above Schedule 0.
- Compounding pharmacy is not a licensed category in South Africa; such operations are licensed as community pharmacies and SAHPRA does not license community pharmacies.
- Locally produced means locally formulated and packed, not locally synthesised, and the 70% figure is dated December 2020.
- The continental figure that Africa imports 70% to 100% of finished medicines does not apply to South Africa.
- On the industry bodies' claims as reported, antiretroviral tender local share by value fell from 72% in 2008 to 28% in 2025 and pills and capsules from 56% in 2014 to 18% in 2026.
- Multinational originators are present in South Africa as commercial and regulatory affiliates whose systems are group systems they do not control.
- Veldtkruin Group is invented and is deliberately a composite of a generics manufacturer and a contract developer, because that is where most of the South African estate sits.

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CHAPTER 3

The vocabulary

Not a glossary — chapter 21 is that. This is the forty or so words whose everyday meaning is wrong, or that mean different things in pharma's four dialects. The expensive misunderstandings are never about what an architect does not know. They are about words an architect is sure they already understand.

What this chapter is, and what it is not

This is not the glossary. Chapter 21 is the glossary, at more than 170 terms.

This chapter is the forty or so words whose everyday meaning is wrong, or that mean different things in pharma's four dialects — **discovery, clinical, regulatory, and quality and manufacturing** — with commercial as a fifth that an architect in South Africa cannot avoid. The dialects grew up in different buildings, under different inspectors, and use the same words for different objects. The most expensive misunderstandings here are not about things an architect does not know. They are about words an architect is confident they already understand.

So the chapter runs by dialect, not alphabetically, and for each word it says what the object is and what breaks when you model it as another dialect means it.

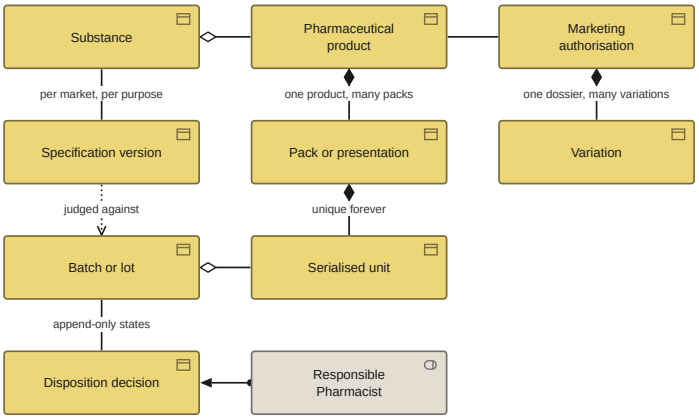


Figure 3.1 The core business objects, and the words that name them. Every box is a separate object with its own grain. The two that outsiders merge are the pharmaceutical product, which is country-independent, and the marketing authorisation, which is not. Chapter 9 adds cardinality and lifecycle.

The words that cross every dialect

Product, molecule, formulation, registration, SKU

Carried forward from chapter 1, because it is the one that costs most. **A molecule** is the active substance. **A formulation** is the molecule plus excipients at a strength in a dosage form. **A registration**, called a marketing authorisation nearly everywhere outside South Africa, is one formulation approved by one regulator, manufactured at named sites, with specified specifications and an approved label. **A SKU** or pack is that registration in a container and count, with its own barcode, its own single exit price and its own stock-ledger identity. Each step down is one to many. And the word "product" names a different object in each of six places.

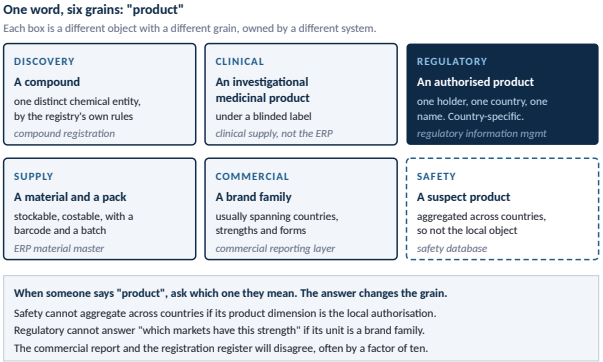


Figure 3.2 The word "product" names six different objects. Each has its own grain and its own owning system. Forcing the word to be qualified everywhere it appears is the cheapest defect prevention available in this industry.

The working rule: **force the word to be qualified wherever it appears** — not "product" but "authorised product", "pack", "material", "brand family", "suspect product".

Validation, qualification, verification

These three are not interchangeable, and getting them wrong marks you as an outsider in the first meeting.

Word	What it applies to	What it means
Qualification	Infrastructure, equipment, facilities, utilities, instruments	Documented evidence that the thing was installed and operates as specified
Validation	Applications, processes, analytical methods, cleaning	Documented evidence that the thing consistently does what it is required to do for its regulated purpose
Verification	An individual activity or outcome	A check that something was done, or is as expected, against a stated criterion

Infrastructure is qualified; applications are validated. Say "we will validate the servers" and the quality assurance manager will quietly reclassify you.

> **VALIDATION IS A MAINTAINED STATE, NOT A PHASE**
>
> The single most important sentence in the whole vocabulary. A computerised system is not "validated" on a date in the past; it is *held* in a validated state for its operational life, through change control, periodic review and controlled decommissioning. The consequences reach everything an architect normally treats as free: no unattended dependency bump, no administrator-made configuration change without a change record, no auto-updating tenant on the vendor's own cadence. **"We will just upgrade it" is not an available sentence here.** A system upgrade is a regulatory project with an IT component, and chapter 13 is the whole argument. One corollary surprises people: **more documentation is not more assurance.**

The discovery dialect

Compound, substance, API, drug substance, drug product

Five words for things that overlap and are not the same.

A **compound** is a chemical or biological entity under investigation, identified first by its structure. A **substance**, regulatorily, is that entity as a regulated object independent of any company, carrying an international non-proprietary name and a standards-based identifier. An **active pharmaceutical ingredient**, or API, is the substance in its role as the active part of a medicine. **Drug substance** and **drug product** are the dossier's words: the active ingredient as manufactured and specified, and the finished dosage form.

Now the distinction the chapter exists for. **A research identifier and a regulatory substance identifier are different objects solving different problems, and neither can do the other's job.** The research compound number is company-internal, structure-derived and uniqueness-enforced; its job is to stop two chemists registering the same molecule twice, and no regulator will accept a key only you can resolve. The regulatory substance identifier is global, authority-curated and definition-derived; its job is to let two regulators agree that two dossiers concern the same substance. It does not exist yet for a novel entity, and it is coarser than structure, so it will not distinguish two tautomers that the registry must.

So **the map between them is a first-class entity**, it is many-to-many, and — the part nobody builds — **it is dated.** External identifiers get reassigned, merged and deprecated by their issuing authorities, so a safety database that resolved a substance code in 2019 and re-resolves it in 2026 can get a different answer with no change on its own side.

Two more traps, both of which look like rounding errors and are not. **Form matters**: a parent structure has children — salt, solvate, hydrate, polymorph — and biological activity belongs to the parent while solubility, stability, assay and dose belong to the form. Collapse the form into the parent and dose calculations are wrong by the salt factor, because 100 mg of a hydrochloride is not 100 mg of the free base. And **batch is not substance**: a notebook saying "compound ABC-123456 gave 42% inhibition" records a test on *one sample of one batch*. The shape that survives is parent, then form, then batch, then sample.

For a biologic the identity rule is in effect same process, same source, same sequence — the architectural root of why biosimilars are not generics, and the reason a change of cell line is **a change of substance**.

The clinical dialect

Study, trial, protocol, amendment, arm, visit, schedule of assessments

Word	The object	The trap
Study or **trial**	The investigational undertaking itself	Not the protocol, not the database, not the project. One row is one study, keyed on its protocol identifier
Protocol	The controlled document describing it	It has versions. One study, many protocol versions
Amendment	A new approved protocol version	Not a document revision. See below
Site	One principal investigator at one location under one set of approvals	**A site is not a hospital.** One hospital can hold two sites in one study
Arm	A *planned* treatment sequence	Actual treatment is a different fact. Collapse them and the analysis population is undefinable
Visit	A planned occasion, and separately an actual one	The planned visit belongs to the protocol version; the actual one to the subject
Schedule of assessments	Which procedure happens at which visit for which arm under which protocol version	A four-way dated relationship, and the thing that makes "missing data" computable rather than an opinion

A PROTOCOL AMENDMENT IS A DATA EVENT, NOT A DOCUMENT REVISION

An amendment can change the arms, the visit schedule, the eligibility criteria, the endpoints or the assessments — so it **invalidates downstream structures**: the expected-data inventory, the out-of-window detection, the derived analysis datasets, the randomisation blocks, the supply forecast. Treat it as a document version and the study cannot re-derive what was expected under the earlier version, which is exactly what an inspector asks about.

One more thing about a site: the grain that gets lost is the **delegation log** — site, role, person, from-date, to-date. That is precisely what a Good Clinical Practice inspector asks for.

Subject, patient, participant

Three words for overlapping but distinct things. A **patient** is a person receiving care. A **participant** is a person who has consented to take part in research, and it is the word most ethics committees and modern guidelines prefer because it carries agency. A **subject** is the *record* of that individual within one study — and that record is deliberately **not the person**. It is a pseudonymous shell carrying identifiers, arm, dates and observations, with the link back to the identified person held separately.

Within a study there are at least three identifiers and they are routinely confused.

Identifier	Minted when	Grain	The trap
Screening number	At consent, before eligibility is known	One per screening episode	A person re-screened gets a second one, so it is not a subject key
Randomisation number	At randomisation, only for eligible subjects	One per randomisation	Key the subject on it and you lose the screen-failure population entirely
Subject identifier	By sponsor convention	One per enrolled subject	The submission-level version must be unique across the whole submission, not the study

And the identifier the sponsor should *not* hold: the site's own link from subject number to the patient's name, hospital number and contact details. That is **the re-identification key**, and where it lives is the most consequential privacy design decision in a clinical architecture. The pattern that works: the site holds it in the investigator site file; the sponsor holds pseudonymised data with direct identifiers excluded by design; monitors see identified data at the site and do not take it away.

WHY THE SUBJECT IS THE MOST SENSITIVE RECORD THE COMPANY HOLDS

Stack the attributes. Identity, health condition, genetic data in many studies, biological samples, sometimes reproductive history, sometimes HIV status. Held for ten years or more, shared with a contract research organisation, a central laboratory, a biostatistics vendor, a safety-database vendor and at least one foreign regulator, across borders, under a consent given once, in a system the sponsor will replace twice before the retention clock runs out.

The subject record is the one entity in the enterprise where the retention obligation, the sensitivity, the cross-border exposure, the number of onward recipients and the duration of custody all peak at the same time. Two further facts implementations get wrong: **consent is a versioned state, not an event**, and **withdrawal of consent does not delete collected data** in a trial — what stops is further collection. A deletion request executed naively against a locked clinical database is both impossible and a breach.

Phase I, II, III, IV, and what each actually tests

Phase	Population	The question it answers
I	Tens of healthy volunteers, except in oncology where it is patients	Is it safe and tolerable, what does the body do to it, and what dose?
II	Hundreds of patients with the disease, against a control	Does it do anything, at what dose, how often? IIa is proof of concept, IIb dose-finding
III	Hundreds to thousands of patients, usually multi-country	Does it deliver the effect claimed on the label, and what is the safety profile at scale?
IV	Patients in real use after approval	Post-authorisation safety, effectiveness, new populations, and commitments accepted at approval

Phase IV is used loosely: post-authorisation safety studies, efficacy studies, and company-initiated work that is no commitment at all. When someone says "it's a Phase IV", ask whether a regulator asked for it, because the answer decides whether a missed milestone breaches an approval condition.

Adverse event, serious adverse event, ADR, ICSR, signal, expectedness, causality

Word	What it means
Adverse event	Any untoward medical occurrence in a patient or participant given a medicinal product, **whether or not it is causally related** [1]
Adverse drug reaction	An adverse event where a causal relationship is at least a reasonable possibility [1]
Serious	At any dose: death, life-threatening, requiring or prolonging hospitalisation, persistent or significant disability, congenital anomaly, or otherwise medically important [1]
Individual case safety report	The exchangeable record of one patient's adverse experience **at one point in time**
Expectedness	Whether the reaction is consistent with a **named reference document at a named version** — the local label or the investigator's brochure
Causality	An assessment per suspect product, made **separately by reporter and company**; both are retained
Signal	Information suggesting a new or changed causal association, warranting evaluation

Three things here are architecture, not vocabulary.

"Serious" is not "severe". A severe headache may be non-serious; a mild event that caused an overnight admission is serious. The criterion that was met is the audit evidence, so the model stores the criterion, not just the flag.

Expectedness is a function of a document, computed from the coded term, the reference document, that document's version and the country. So the reference document set is versioned reference data inside the safety system; the same event can be expected in one market and unexpected in another; and when a label changes, **historical expectedness assessments do not change but future ones do.** Store the assessment with the document version it was made against, or every historical reporting decision becomes indefensible.

A case is not one row. It is a case, with versions, each version holding events, each event holding coded terms. **The safety database is a bi-temporal store** — what we believed, as at when we believed it — because the only question that matters at a pharmacovigilance inspection is *what did you know, and when did you know it?*

MedDRA and WHODrug

MedDRA is the hierarchical terminology used to code adverse events, medical history, indications and conditions for regulatory purposes. Its hierarchy runs from the lowest level term that the verbatim text maps to, up through the preferred term that is the analysis unit, then high level term, high level group term, and system organ class. Curated groupings called standardised queries exist for finding cases of a defined medical concept. **WHODrug** is the equivalent for medicines, concomitant and suspect, with its own hierarchy and cadence.

Both are **licensed subscription dictionaries, and that constrains the architecture**: typically not redistributable, so a vendor processing your data needs its own licence and a public-facing application cannot embed the terminology.

> **A TERM CODED UNDER ONE MEDDRA VERSION DOES NOT MEAN THE SAME THING UNDER ANOTHER**
>
> MedDRA releases **twice a year, in March and September**; version 29.0 was released in March 2026 [2] and 29.1 followed later in 2026 [3]. Each release adds terms, and also demotes, merges and changes the primary system organ class of existing ones. So the identity of a coded value includes the dictionary version.
>
> **Re-coding historical safety data is a project, not a refresh.** The design that works is easy to state and almost never done from the start: every coded value carries its dictionary, version, coding date, coder and method; every aggregate report declares the version it was run under; and the dictionary itself is a **stored, versioned asset in the estate**, not a library call, because a report produced in 2023 must be reproducible in 2029 and that needs the 2023 release to still be loadable.

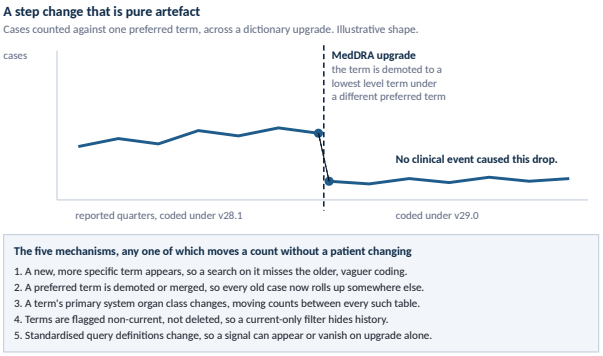


Figure 3.3 Why dictionary versioning is the most under-modelled thing in pharmaceutical data. The step change here is caused by a terminology release, not by a patient. Without the version stored per coded value, no aggregate safety report can be reproduced or defended.

Expedited versus periodic reporting

Expedited reporting means a single case or event sent within a short clock. **Periodic** reporting means an aggregate report covering a window. They are different obligations with different triggers, and most safety systems model only one.

South African timelines are their own, and an architect who imports European or American ones will build the wrong workflow. The verified positions, from the current post-marketing reporting guideline [1]:

Obligation	Clock
Serious reactions and events following immunisation **occurring in South Africa**, related or not, expected or not	**15 calendar days** of receipt
Medication errors, whether or not a reaction occurred	**15 calendar days**
Significant safety issues from listed regulators, and regulatory action abroad	**5 calendar days**
A change in the nature, severity or frequency of reactions	**5 calendar days**
Non-serious reactions	**Not routinely reported**; in periodic line listings, or on request within 30 days
Foreign individual case reports	**Not routinely forwarded**
Periodic safety reports	**Prepared** per the reference-date list or annually; **submitted only on request**, within 30 days
Surveillance after discontinuation	Until **six months after the last marketed batch expires**

Note three things. **The days are calendar days.** The five-day clock runs on foreign *regulatory action*, not foreign cases. And **preparation and submission of a periodic report have different triggers.**

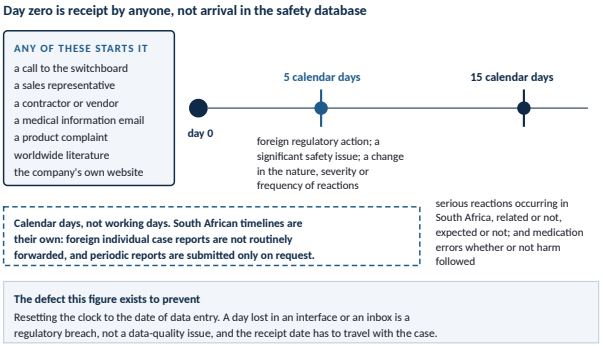


Figure 3.4 The South African pharmacovigilance clocks, and where they start. Day zero is receipt by any employee or contractual partner — including a sales representative — not arrival in the safety database. The days are calendar days.

And day zero. **The clock starts on the date the company first became aware** — awareness by *any* employee or contractual partner, including sales representatives and contractors, by mailbox rather than by a validated gateway [1]. So the receipt date must be captured at first contact and transported forward. **Resetting it to the data-entry date is the most common and most serious defect in pharmacovigilance data.**

Database lock and unlock

Database lock, also freeze or hard lock, is the point at which a study's clinical database is declared complete and closed to change, so that the analysis runs on a fixed dataset.

> **AN UNLOCKED DATABASE IS AN ARCHITECTURAL EVENT, NOT AN ADMINISTRATIVE ONE**
>
> At the instant of lock the database stops being a transaction system and becomes **evidence.** Write access is removed, the blind is broken, and the locked state becomes an *object* in its own right: study, lock identifier, lock date and time, scope, authoriser, record counts, a manifest, and the extract produced from it.
>
> Databases do get unlocked, for good reasons. The rules that make it survivable: unlock is a controlled change with a requester, a reason, a scope and an approver; **the pre-unlock snapshot is never overwritten**, so a study with three unlocks has four locked states and four lineages and every downstream artefact declares which one it used; the analysis is re-run or explicitly assessed as unaffected, and that assessment is a record.
>
> Treat lock as a status field and unlock as flipping it back, and you have destroyed the ability to say which data produced which result. Every other data-integrity finding is recoverable. That one is not.

Trial master file, and eTMF

The **trial master file** is the body of documentation that demonstrates the trial was conducted as described and in compliance with Good Clinical Practice. The **eTMF** is it, held electronically.

The point to absorb: **it is not a document archive. It is the inspection artefact**, and **its completeness and the timeliness of filing are inspection findings in their own right**, independent of how well the trial was run. A perfectly conducted trial whose file was assembled six months late has a finding. South Africa's verified retention floor is **ten years** after final closure, with research ethics committee records at three years; the frequently quoted 25 years is European.

The CDISC words, in one line each

These point forward to chapter 14, where the standards question gets its proper treatment.

Word	What it governs
CDASH	Collection: what the case report form asks, and what the field is called at capture
SDTM	Tabulation: the submitted representation of collected data, in two-letter domains grouped into observation classes
ADaM	Analysis: analysis-ready datasets derived from the tabulation data, with every value traceable back to it
SEND	The nonclinical equivalent of tabulation, for animal toxicology studies
ODM	An operational model for exchanging and archiving study data and metadata
Define-XML	The machine-readable metadata describing every dataset, variable, codelist and derivation in a submission

One structural fact decides a platform choice: **the submitted form of clinical data is tall and thin, not wide** — one row per subject per test per timepoint.

The regulatory dialect

Dossier, submission, sequence, eCTD, variation, renewal

Word	The object	Mutability
Submission	One transmission to one authority for one purpose	An event, on a date
Sequence	One numbered transmission, with its own backbone	**Immutable once sent**
Dossier or application	One application to one authority for one product	**A maintained, cumulative state** — the current view
eCTD	The structure: five modules plus an XML backbone declaring every file, its location and its lifecycle operation	Versioned; the backbone travels with the sequence
Variation	Any change to an approved registration, classified by risk	Sets the lead time, and the lead time sets feasibility
Renewal	The periodic re-approval of the registration itself	A diarised obligation with a hard consequence

> **A SUBMISSION IS AN ASSEMBLED ARTEFACT; A DOSSIER IS A MAINTAINED STATE**
>
> The submission is what you sent on a date. The dossier is what the regulator currently holds, and no single transmission contains it. **The current view of a dossier is a computed projection over an append-only sequence history** — the lifecycle operations in each backbone, new, replace, append and delete, fold the sequences into a current state. "Delete" removes a document from the current view; the sequence that carried it stays intact forever.
>
> Say that out loud and the design falls out: the dossier is an event-sourced aggregate, the sequences are the events, and the projection must be recomputable from the log. A system that models dossier state but not submissions cannot answer "prove you submitted this"; one that models submissions but not state cannot answer "what is currently approved".

The South African specifics are precise. **South Africa is on eCTD version 3.2.2**, through the SAHPRA portal, which is **the only valid method** for submitting sequences; version 4.0 is not adopted [4]. PDF files must be version 1.4 to 1.7, and a structural error **rejects the whole sequence** [4]. A new application starts at **0001**, but **a baseline built from non-eCTD content starts at 0000**, must be its own sequence, and may carry only previously submitted content [4]. And on a transfer of application the acquiring applicant **must not restart at 0001** [4].

> **VARIATIONS: THE ABSENCE OF A RESPONSE IS AN EVENT**
>
> South African variation classes do not all require prior approval. A Type IA is notified within twelve months. **A Type IAIN is notified on 47 working days and is deemed implementable if SAHPRA gives no opinion in that window.** A Type IB runs 87 working days, after which the change may be implemented though SAHPRA may still require reversal. Only a Type II requires prior approval, at 220 working days [5]. **"Deemed implementable on silence" inverts how an approval workflow is normally built**: the system must detect a non-event on a clock, not wait for a message.

And registration does expire. A South African registration is valid **five years**, renewal must be submitted **six months before expiry**, and failure to submit a compliant renewal may result in expiration — after which a new registration application is required. The programme is phased from 2023 to 2035 [6].

Reliance versus recognition

This is the single most commonly fudged thing in South African regulatory conversation, so it gets its own entry.

Term	What it means
Abridged review	A streamlined review based primarily on full assessment reports from a reference regulatory authority [7]
Verified review	A streamlined review based primarily on *verifying*, instead of evaluating, the information submitted [7]
Recognition	A streamlined process based on **directly recognising the outcome** of another authority's review [7]

SAHPRA operates reliance — abridged and verified review — and remains responsible and accountable for the decisions it takes [7]. **Recognition is a separate pathway: SAHPRA is still negotiating recognition agreements and the framework is still to be published** [7]. So **SAHPRA does not recognise United States or European approvals**, and anyone who tells a client otherwise has misdescribed what the client can rely on.

Two consequences for a launch plan. Under reliance, all supporting documents and amended dossier sections must still be submitted locally [7]. And **if no full assessment reports arrive from the reference authority within three months of the request, the application is reviewed as a full review** [7] — a third-party regulator's responsiveness sitting on a company's critical path.

Section 21

Section 21 of the Medicines and Related Substances Act permits SAHPRA to authorise the sale of an unregistered medicine for a specified purpose, typically a named patient under a named prescriber.

> **SECTION 21 IS NOT A RARE EXCEPTION**
>
> In 2024/25 SAHPRA received **20,245 applications** for the sale of unregistered Category A medicines, finalised 20,225, and met its **three-working-day** target on 96% of them against a target of 90% [8] — roughly 80 applications a working day, and **the busiest single workflow in the authority.** Three facts follow: it runs on a three-day clock; **section 21 supply is not subject to the single exit price**, so the price is whatever is agreed and it is not in the public pricing database [9]; and it typically sits outside the registered-product master data entirely. Every South African client will have a section 21 process and it will almost always be the least well-systematised part of the estate.

Scheduling, Schedules 0 to 8

A substance's **schedule** is a dated classification that determines who may sell or supply it, on what authority, with what record, and whether it may be advertised to the public [10].

Schedule	Supply	Advertising to the public
0	Any open shop, including general dealers, and pharmacies	Permitted
1	Pharmacists and supervised interns or assistants; prescribers	Permitted
2	Pharmacists, with counselling; no prescription needed	Not to the public
3, 4	Authorised prescribers, dispensed on prescription	Not to the public
5, 6	Prescription only, with rising abuse-potential controls and registers	Not to the public
7, 8	Not for ordinary medical use; Schedule 8 only by the Director-General	Prohibited

The schedule carries a **record** obligation, which is why it belongs here. Sales of Schedule 1 to 6 products are recorded with prescribed particulars; specified Schedule 5 and Schedule 6 require a register **balanced at 31 March, 30 June, 30 September and 31 December, with balancing completed within 14 days of each date** [10]; and the last seller retains a Schedule 2 to 6 prescription for at least five years [10]. Schedule is **per substance, per strength, per indication and dated** — not a label attribute — and rescheduling is a variation with an effective date.

The quality and manufacturing dialect

Batch, lot, campaign

Batch and **lot** are the same thing: a defined quantity of material produced in one process or series of processes such that it is expected to be homogeneous. It is the unit of traceability, quality and liability, and everything physical is batched — the active ingredient, each excipient, the printed carton, the bulk tablet, the finished pack. A batch repacked is a new batch, linked to its parent. A bulk batch filled into several pack configurations yields several finished batches. And **each batch must be considered separately for sampling, testing and release** [11].

A **campaign** is a run of one product, bounded by changeover and cleaning at each end. It is a scheduling and costing concept, not a quality one, and a batch is not a smaller version of it.

> **A CONTRACT DEVELOPER'S COST OBJECT IS THE CAMPAIGN**
>
> Not the batch, and not the product — because the product belongs to the client. In a low-volume, high-variety site, changeover and cleaning consume a large share of available suite time and are not billable, so a standard cost per finished item hides exactly the cost the business is managed on. Chapter 2 works the failure mode.

The GxP family

Six regimes, six inspection traditions, six vocabularies for overlapping ideas.

Regime	What it governs	Its defining feature
GLP	Non-clinical safety studies	A study director as a single accountable person; decades-long raw-data archiving
GCP	Clinical trials in humans	Informed consent, ethics oversight, source data verification, the trial master file
GMP	Manufacture and quality control	The validated process, the batch record, release by an authorised person
GDP	Storage and transport (good wholesaling practice, locally)	Temperature control, supply-chain integrity, recall capability
GVP	Safety surveillance during a trial and after launch	A responsible person, risk management plans, statutory clocks in days
GDocP	Record-keeping, across all of the above	Contemporaneous, attributable, legible records

The first question about any system is not "is it validated". It is "which GxP, and for which record". The same software product carries different obligations in a toxicology laboratory under GLP and a quality-control laboratory under GMP.

ALCOA and ALCOA+

The data-integrity attributes. **A**ttributable, **l**egible, **c**ontemporaneous, **o**riginal, **a**ccurate — plus **c**omplete, **c**onsistent, **e**nduring and **a**vailable.

> **TWO CORRECTIONS ABOUT ALCOA**
>
> First, **the PIC/S Guide to GMP, Part I, does not use the term "ALCOA"** at all [11]. Its nearest statement is a principle on the accuracy, integrity, availability and legibility of documents, plus requirements for contemporaneous records and signed, dated alterations [11]. A slide citing "PIC/S Chapter 4 ALCOA" is citing something not there in those words. ALCOA is a real expectation, expressed in data-integrity guidance rather than in the GMP guide's chapters.
>
> Second, **the verified South African ALCOA statement is ALCOA-CCEA, in the South African Good Clinical Practice guidelines for source documents** [12]. **That is a GCP requirement, not a GMP one.** Quote ALCOA at a South African manufacturing site and you are quoting a clinical guideline at a manufacturing inspector.

Deviation, OOS, OOT, CAPA, change control, complaint

Word	The object	What closes it
Deviation	One departure from an approved procedure, specification or expectation	An investigation, a root cause, a product-impact decision, a disposition
Out of specification	A result outside the acceptance criterion	A two-phase investigation: laboratory cause first, then manufacturing
Out of trend	A result **within** specification but outside the expected behaviour of the process or stability profile	A statistical assessment against a versioned trend model
CAPA	One corrective and preventive action plan, with tasks	Implementation **plus an effectiveness check**
Change control	One planned change, assessed and approved **before** implementation	Implementation, verification, closure
Complaint	One report from outside the company about a specific unit or batch	An investigation, a reply, a decision on reportability and recall

Three things to carry out of that table.

Out of trend is not a weaker out of specification. It is a statistical concept, so it needs a trend model whose control limits, comparison population and method are themselves versioned. An alert raised against one parameter set is not comparable to one raised against another.

CAPA closure is not the same as fixed. A CAPA is complete when the action is shown to have worked, after a defined interval, against a defined measure. The effectiveness check is the step everyone drops, and the mechanism is mundane: it falls due 90 or 180 days later, is owned by someone else, and has no transaction to attach to. Three design moves fix it: make it a separate dated obligation rather than a field on the CAPA; make closure impossible without one, so "implemented" and "closed" are distinct states; and report on overdue effectiveness checks rather than on closure rate, because closure rate without effectiveness lets a quality system look healthy while repeating the same deviations. Note that **SAHPRA's own guideline on responding to an inspection report does not mention an effectiveness check at all** [13]. If the regulator does not ask for it, the company has to want it for itself.

A complaint can be two things at once. If it is a suspected adverse reaction it is also a pharmacovigilance case and both clocks run. The intake process must fork, and a complaint system that cannot fork to safety is a common cause of missed 15-day reports.

Specification, test, method, result, certificate of analysis, disposition, release

Object	The grain
Specification	One version of the tests and acceptance criteria, for one material or product, one purpose, one market
Test or attribute	One measurable property: assay, dissolution, water content, related substances, sterility
Method	One version of the analytical procedure, with its own validation and transfer records
Result	One sample, one test, one method version, **one determination**
Certificate of analysis	One attested summary for one batch, for one recipient — immutable once issued
Disposition	The recorded decision about what a batch may be used for
Release	A specific disposition: certified as saleable to a named market

A result is per determination, not per test. Two injections, three tablets, six dissolution vessels are determinations, and the reportable value is a derivation over them with its own rounding rules. A system storing only the reportable value cannot investigate an out-of-specification result, because the investigation is about those determinations.

Disposition is more granular than most models allow, and the states are append-only: in production; quarantine; under investigation; approved for further processing; certified for sale; released with restriction; rejected; on hold; recalled; destroyed. And **market-specific release is the normal case, not an exception** — one batch, several markets, several decisions, possibly different outcomes, because specifications and approved shelf lives differ per authorisation.

Retest date versus expiry date

Different objects, different consequences, and routinely conflated.

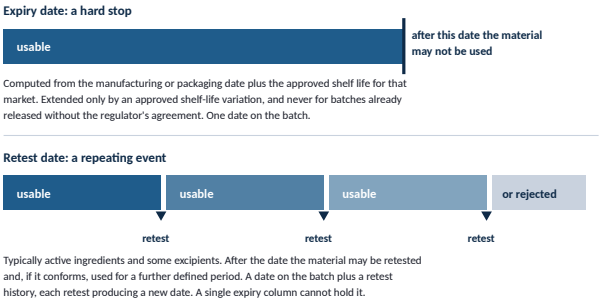


Figure 3.5 Retest date and expiry date are different objects. An expiry date is a hard stop on one date. A retest date is a repeating event with its own results and its own history, and it extends usability each time the material conforms.

Yield

In general manufacturing, improving yield is good. Here it is not.

> **HIGHER YIELD IS NOT GOOD UNDER GMP**
>
> A batch yield outside the **validated range** registered with the regulator is a deviation requiring investigation — **even when it is higher.** A plant cannot get better at making a registered product; a sustained change in process performance is a signal that the process has drifted from its validated state.
>
> That inverts ordinary manufacturing metric logic. **Continuous-improvement framing imported into a GMP site is wrong here**, and a yield dashboard with a green arrow for "up" will be asked about at the next inspection. Show yield against the validated range, with excursions in either direction flagged as events.

Qualified Person versus Responsible Pharmacist

South Africa has no European-style Qualified Person. Chapter authors get this wrong constantly and the difference is substantive, not a naming convention.

	Qualified Person (European Union)	Responsible Pharmacist (South Africa)
Who may hold it	A pharmacist, chemist or biologist with prescribed qualification	**A pharmacist registered with the South African Pharmacy Council** [14]
Accountable to	The medicines authority	**Both SAHPRA and the Pharmacy Council** [14][15]
Named on the licence	Yes	**Yes, and not more than one per licence** [15]
Per batch	Certifies each batch	**Must sign or certify, per batch, that the appropriate tests were done** [15]
Release mechanics	A legally defined register	Sign-off using a **named release checklist** [16]

Two further duties make the role an access-control design problem with legal consequences. The Responsible Pharmacist must supervise the pharmacy continuously and ensure every batch released for sale, supply or export meets the registration conditions and GMP, taking account of storage and transport conditions, sampling and quality-control results [16]. And the role carries a duty to ensure that **manufacturing and testing processes remain in a validated state** [16]. That is the sentence to remember: **validation status is not an IT matter in South Africa. It is a named pharmacist's personal duty.**

Serialisation, aggregation, commissioning, decommissioning, SGTIN, SSCC

Serialisation turns a batch — a set — into a population of individually addressable members, by putting a unique identifier on each saleable item. South Africa's traceability guideline sets the levels and the GS1 application identifiers: a global trade item number, expiry date, batch and serial at the primary, secondary and tertiary trade-item levels, and a serial shipping container code on the logistics unit [17].

Word	What it is
SGTIN	The serialised trade item: a global trade item number plus a serial, identifying one physical pack
SSCC	The serial shipping container code, identifying one logistics unit such as a pallet
Aggregation	The recorded parent-child relationship between a container and its contents
Disaggregation	Opening a container to pick individual packs — the event that breaks the tree
Commissioning	Bringing a serial into existence as valid and in use: allocated, printed, verified, batch-associated
Decommissioning	Retiring it, with a reason: dispensed, destroyed, sampled, exported, stolen, recalled

Three architectural points. **The trade item number and serial combination must be unique forever and may never be reused** [17], which makes the serial generator a system of record. **Aggregation is a time-varying tree, so the event log is the model and the current tree is a projection** — current-parent-as-a-field cannot represent a unit that was in one case, then loose, then in another. And **commissioning is irreversible as a record**: a serial is never un-issued, only decommissioned with a reason.

> The South African regime is **phased, batch-level first**, with imported products staged ahead of domestic ones and no transactional repository specified [17][18]. Chapter 12 covers the sequencing.

The commercial dialect

SEP, logistics fee, dispensing fee, tender

Word	What it is
Single exit price	The only price at which a manufacturer or importer may sell to anyone other than the State; gazetted per pack, identical regardless of buyer volume [9]
Logistics fee	A portion of the exit price the manufacturer **pays to the wholesaler**. Wholesalers add no mark-up [9]
Dispensing fee	A fee the final seller charges **in addition to** the exit price, capped by gazette, calculated on the price excluding value-added tax, in bands, with different caps for pharmacists and dispensing doctors [9]
Tender	How the State buys, at a price outside the single-exit-price regime entirely [9]

The two that get stated backwards: **the logistics fee comes out of the exit price, and the dispensing fee is added on top of it.** Say both aloud before a pricing workshop.

Where the same word means two things

Word	In one dialect	In another
Product	Discovery: a compound under investigation	Regulatory: a country-specific authorised product
Batch	Discovery: one physical preparation, for testing	Manufacturing: a quantity with a record, a genealogy, a disposition and an expiry
Validation	IT: a system held fit for its regulated purpose	Laboratory: a method shown to measure what it claims
Release	Manufacturing: certification that a batch may be sold	IT: a deployment, itself a controlled event here

When one of these comes up, the move is the same each time: **say which dialect you are in, and ask the other person which they are in.** It takes a sentence and it saves a rebuild.

THE CHAPTER IN TEN LINES

- Infrastructure is qualified; applications are validated; individual activities are verified. The words are not interchangeable.
- Validation is a maintained state rather than a phase, so there is no such thing as just upgrading a regulated system.
- A research compound identifier and a regulatory substance identifier are different objects, and the dated map between them is a first-class entity.
- A contract developer's cost object is the campaign, not the batch and not the product.
- A subject is a pseudonymous record, not a person; consent is a versioned state; and withdrawal stops further collection rather than deleting what was collected.
- A term coded under one MedDRA version does not necessarily mean the same thing under another, and MedDRA releases twice a year.
- South African safety clocks are 15 and 5 calendar days, and day zero is receipt by any employee or contractual partner, including sales representatives.
- A submission is an assembled artefact on a date; a dossier is a maintained state computed over an append-only sequence history.
- South Africa is on eCTD version 3.2.2, portal only, and a baseline sequence starts at 0000 rather than 0001.
- SAHPRA operates reliance and does not recognise United States or European approvals; recognition is a separate, unpublished pathway.
- Section 21 is not a rare exception: 20,245 applications in 2024/25 on a three-working-day clock, and that supply is not subject to the single exit price.
- CAPA closure is not the same as fixed; the effectiveness check is the step everyone drops.
- Higher yield is not good under GMP: a yield outside the validated range is a deviation even when it is higher.
- South Africa has no European-style Qualified Person; the Responsible Pharmacist is a Pharmacy Council registrant, dual-accountable, one per licence, and carries a duty to keep processes in a validated state.
- PIC/S PE 009 Part I never uses the term ALCOA, and the verified South African ALCOA statement is ALCOA-CCEA in the GCP guidelines, not a GMP requirement.
- The logistics fee comes out of the single exit price and the dispensing fee is charged in addition to it.

SOURCES FOR THIS CHAPTER

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CHAPTER 4

How the business model works

Not the arithmetic — chapter 6 does that. This chapter is the business model: who the customer actually is, what is really being sold, and what each way of configuring the firm demands of the architecture. In South Africa that means one gazetted price for the private market, a tender for the State, and the discovery that those are two different businesses inside one company.

Four customers at once

A blister of dolutegravir on a shelf in Mthatha got there through a transaction in which the person who chose it, the person who paid for it and the person who swallowed it were three different people. None of them was the manufacturer's customer in the sense a retailer or a bank would recognise.

In most industries there is one customer. In pharmaceuticals there are four at once, and an architect who collapses them into a single party has lost the model before the first workshop.

- **The prescriber chooses.** A doctor, a specialist, or in South African primary care a nurse working to an initiation protocol. Above Schedule 2 nothing moves without a prescription [24]. The prescriber decides, bears none of the cost, and is the only party a manufacturer may lawfully promote prescription medicines to.
- **The dispenser supplies, and may substitute.** A community pharmacist handed a prescription written for a brand may hand over a bioequivalent generic instead. That decision, taken at a counter in seconds by someone the manufacturer has never met, is where much of South African generics revenue is won and lost. A dispensing doctor is prescriber and dispenser in one person, with its own fee band [4].
- **The payer pays.** A medical scheme, a private hospital group out of its own tariff, a provincial department out of a tender contract, or the patient out of pocket. The payer rarely sees the patient and never sees the molecule.
- **The patient consumes.** Or does not: adherence is an assumption in every forecast and is measured by almost nobody. In the public sector the patient is counted in a programme register rather than invoiced.

Say the consequence plainly. **The party who decides is not the party who pays, and neither is the party who uses.** Every commercial system here has to carry all four: the influence sits with the first, the money with the third, the outcome with the fourth. A customer master holding only the entity on the invoice — the wholesaler, or the provincial depot — holds the least interesting of them.

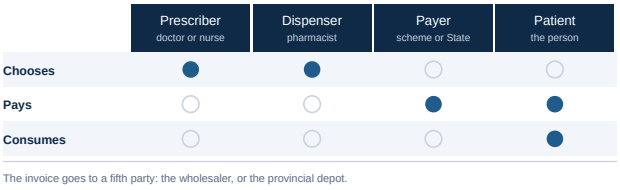


Figure 4.1 Four customers, three roles. The one who chooses, the one who pays and the one who consumes are usually three different parties — and the invoice is addressed to a fourth.

There is a fifth party that is not a customer but behaves like one, and architects leave it out: **the regulator**. It does not buy, but it decides whether the product may be sold at all, at what specification, from which site, with what wording on the leaflet. Its decision is a precondition of every transaction above and its clock does not run on the commercial calendar. Chapter 16 treats it as a stakeholder with a data demand.

> Four customers, three roles. The chooser, the payer and the consumer are usually three different parties, and the invoice is addressed to a fourth. A model with one customer entity cannot represent a prescription.

Two businesses inside one company

Now the South African split, because it is the structural fact of the market and it has nothing to do with segmentation.

A South African pharmaceutical manufacturer selling the same tablet into the private market and to the State is running two businesses. Not two channels. Two businesses, with different pricing law, pack configurations, delivery obligations, payment behaviour and kinds of risk.

In the private market the price is the **Single Exit Price** — one gazetted price per pack, the same to every private buyer, adjusted once a year by permission. In the public market it is whatever was bid and accepted in a **tender**, outside the Single Exit Price regime entirely [1][3]. The two are measured on different bases and do not add, so a single "South African medicines market" figure is almost always one or the other.

The share figures point in opposite directions, which is the trap. More than 80% of South African pharmaceutical manufacturing and distribution *revenue* came from the private sector, on Who Owns Whom's 2023 edition [12]. The overwhelming majority of *volume* is public sector, driven by roughly 6.18 million people on antiretrovirals in 2025 [17]. **A revenue view and a volume view of this market describe different companies.** Both are true; reports that mix them are not.

	Private market	State tender
Price set by	gazette, once a year	competitive bid
Same price to all buyers	yes, by law	no, per tender
Customer-specific pricing	prohibited	is the mechanism
Pack	standard retail packs	patient-ready, 3-month
Delivered to	wholesaler, pharmacy	provincial depot
Share of revenue	more than 80% (2023)	the remainder
Share of volume	the minority	the large majority
Main risk	price capped below cost	award set aside

Two order-to-cash flows, two planning regimes, one item master.

Figure 4.2 Private and State are two businesses, not two channels. Revenue share and volume share point in opposite directions, and both statements are true.

For an architect the reading is direct. Order to cash runs twice, on two pricing regimes, with a channel flag on the sale deciding which rule applies. Demand planning runs twice — on a forecast in one case and a contract in the other. Master data runs twice, because a tender pack is a different pack with its own artwork and often its own registration. And the controls differ in kind: preventing a discount from reaching a private prescription line is a compliance control, while preventing a tender price from reaching a private invoice is a correctness control.

Single Exit Price is a business model, not a pricing detail

Most architects meet the Single Exit Price as a line in a requirements document and treat it as a pricing rule. It is not. It is the shape of the whole commercial business, and it deletes most of what a commercial pricing system in another market exists to do.

What it actually is, and who it binds

The Single Exit Price is the only price at which a manufacturer or importer may sell a medicine or scheduled substance to any person other than the State [3]. Read that twice, because two things in it are routinely stated wrongly.

First, it binds importers as well as manufacturers. A South African company that in-licenses a finished product and imports it is inside the regime for that product exactly as a local manufacturer is — which matters, because in-licensing is the dominant local pattern and a company that thinks of itself as a distributor may be the party declaring the price.

Second, the State is outside it. The State buys by tender, at tendered prices, entirely outside the regime [1][3]. This is the most commonly misstated fact in South African pharmaceutical commerce, and the error runs both ways: that tender prices are capped by the Single Exit Price, and that the Single Exit Price is a maximum private buyers negotiate down from. Neither is so. It is not a maximum and not a recommendation. It is the price.

Three consequences follow immediately.

1. **One price, regardless of who is buying or how much.** A national pharmacy group and a single independent pharmacy in Kokstad pay the same price for the same pack, and a private hospital group with enormous volume cannot negotiate a cent off it [3]. Bulk purchasing power, the organising assumption of almost every other commercial architecture, does not exist here.
2. **It is also what the final seller charges.** The pharmacy does not mark it up; the dispensing fee is charged *in addition*.
3. **It is public.** Every competitor's price for every pack sits in the national pricing database, so price intelligence is a download rather than a research project — and that database is what a company's own records get reconciled against.

The dispensing fee sits on top, not inside

The final seller may charge a dispensing fee **in addition to** the Single Exit Price, capped by gazette, calculated on the price **excluding VAT**, with different bands for pharmacists and for licensed dispensing practitioners [3][4]. The pharmacist structure is a flat rand amount plus a percentage, across four price bands, with the percentage falling as the price rises; the dispensing doctor structure is simpler and has two bands [4].

> **Do not quote the February 2025 bands as current.** Government Gazette No. 52151 of 21 February 2025 published *proposed* dispensing fees for comment, at values that differ from the bands later published [5]. The two sets look almost identical — the proposed lowest band threshold was R152.16 against R159.00 in the published bands — which is why the error survives review. Establish which gazette each band came from, in writing, from the client's regulatory or legal function.
>
> A second trap: the published pharmacist bands leave small gaps at the band edges and the source does not address them [4]. A calculation has to do something there. Decide it deliberately and record the basis, rather than letting a developer pick a comparison operator by habit.

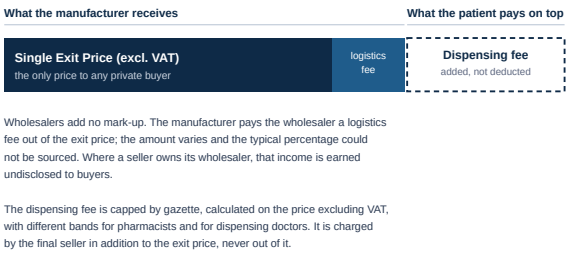


Figure 4.3 The exit price and the dispensing fee. The logistics fee comes *out of* the exit price; the dispensing fee is charged *on top of* it. Getting these the wrong way round is the commonest modelling error in South African pharmaceutical pricing.

The wholesaler takes no mark-up, and this is the one gross-to-net item

In the United States the gap between a manufacturer's list price and what it realises is roughly half on brand-name products: gross-to-net reductions of USD 416 billion in calendar 2025 on Drug Channels Institute's estimate published 7 October 2026, with disclosed discounts off list between –46% and –56% [15]. That machinery has no South African analogue, and the transactions that drive it are prohibited here above Schedule 0 [3]. An architect who imports a United States pharmaceutical revenue-management reference architecture into a South African generics manufacturer builds an estate for transactions that cannot lawfully occur.

But the South African private market is not quite gross-equals-net, and the reason is the **logistics fee**.

Wholesalers add no mark-up to the Single Exit Price. Instead, the manufacturer pays the wholesaler a logistics fee, which is a portion of the exit price [3]. Three facts about it matter to a model:

- The amount **varies considerably** between arrangements [3]. It is a negotiated number inside a regulated price.
- It appears as a column in the national pricing database spreadsheet [3], so it is not secret in the aggregate.
- **Where a seller owns its own wholesaler, it can earn that income undisclosed to buyers** [3]. The same rand is a cost in one legal entity and revenue in another inside one group.

> The typical logistics-fee percentage is something I could not source, and the research behind this book could not either. It is the figure every author wants and the one nobody publishes. Treat any percentage you are given as a client-specific input with a date on it, not as an industry constant.

The architectural reading: **price times volume is gross revenue, not net revenue, for a South African manufacturer.** The logistics fee is the only genuine gross-to-net deduction in the local private book. Build that one. Do not build the other twenty.

What the regime does not do

Three negatives, each of which an author working from general knowledge will get wrong.

- **South Africa does not operate external reference pricing.** The 1996 National Drug Policy proposed monitoring South African prices against other countries and declaring some unacceptable. The declaration form for a new medicine asks for international prices. But no system of external reference pricing is in place, and **no Single Exit Price has ever been declared unacceptable** [3]. If a client's requirements document asks for comparator-country price management, ask who the requirement is for.
- **The launch price of a new medicine is not regulated.** A manufacturer sets the initial price itself; only subsequent increases are controlled [3]. This is counter-intuitive to everyone who meets the regime as a price cap.
- **Section 21 supply is outside the regime.** Named-patient supply of an unregistered medicine under section 21 of the Act is not subject to the Single Exit Price [3], so the price is whatever is agreed and it does not appear in the public pricing database. Section 21 is not a rare exception: SAHPRA received 20,245 such applications in 2024/25 on a three-working-day clock, which is roughly 80 a working day and its busiest single workflow [8]. Every South African pharmaceutical client has a section 21 process, it sits outside the registered-product master data, and it is almost always the least systematised part of the estate.

The adjustment is an event, and 2026 had two of them

Each year a statutory pricing committee determines the **maximum percentage increase** a manufacturer may apply to existing prices [3]. It is a maximum, not a mandate: a company may take less or none, and across a portfolio of hundreds of packs that is a real commercial decision made once a year under time pressure.

2026 was not a normal year and any statement that cites one adjustment is wrong. There were two:

Adjustment	Percentage	Effective	Note
2025 annual	5.25%	2025	[3]
2026 annual	1.47%	January 2026	One outlet rendered it as 1.4%; use 1.47% [3][11]
2026 extraordinary second	2.88%	1 October 2026	[1][2]

The 2.88% adjustment was approved as an extraordinary second adjustment for 2026, covering registered medicines and scheduled substances on the national database as at 30 September, including related pack sizes. The Department of Health cited higher input costs, supply-chain pressure and

geopolitical disruption, and said explicitly that it applies to sales to private buyers and **not** to medicines bought by the State [1][2]. The next ordinary cycle is 2027 [2].

The January figure of 1.47% landed against consumer price inflation of 3.6% and sectoral wage increases of about 6%; industry had asked for an additional 1.73 percentage points in April 2026 and had no answer by August, and the extraordinary adjustment followed talks that month [9][10].

> A price adjustment in this market is a scheduled, date-effective, high-blast-radius master-data event across every pack a company sells, flowing on one day to every customer, every wholesaler, every scheme formulary and every point-of-sale system in the country. It is the sector's equivalent of a tax-rate change. Design for it explicitly, with an effective-dated price record at pack grain and an audit trail, because an adjustment applied to the wrong pack set or on the wrong date is a pricing non-compliance affecting every invoice issued afterwards.

What this deletes from the architecture

Collect the consequences, because this is what you are actually designing against.

- **No customer-specific pricing and no rebate processing for the private book.** No price lists by customer, no volume breaks, no negotiated discounts, no contract pricing for the regulated class; bonus, rebate and incentive schemes are prohibited above Schedule 0 [3]. The machinery every enterprise resource planning system ships with has to be switched off for these items.
- **Margin is managed from the cost side only.** Price is fixed by gazette; input costs move daily and are mostly dollar- or euro-denominated. The only levers are procurement, yield, capacity utilisation and mix. This is why cost accounting matters disproportionately here, and why a company with a weak standard-costing discipline has no steering wheel.
- **Portfolio pruning is a standing decision.** A price ceiling plus a long tail of low-volume packs, each carrying its own stability programme, annual quality review, artwork and variation history, makes "which registrations should we stop" a recurring question. Answering it needs fully loaded cost to serve per pack including regulatory and quality overhead — which is exactly the data most companies do not have.

Veldtkruin Pharmaceuticals, the invented generics manufacturer this book uses as its worked example, carries 184 registrations and 498 active packs against revenue of about R2.1bn, with 11 registrations held and not marketed because reinstating one is slower than retaining it. Nothing about Veldtkruin is real. The shape is a wide, shallow portfolio against a fixed price, where administrative cost per pack decides the business.

Public tender is a different business, and the award is not revenue

Public purchasers — the national and provincial departments of health, the military, correctional services — buy through published tenders. Only SAHPRA-registered medicines are eligible, tenders usually run three years, and tenders, awards and the Master Health Products List are published. High-volume lines such as first-line antiretrovirals may be split among several suppliers so that one supplier's failure does not stop a national programme [3].

The economic shape is the inverse of the private market: one buyer, price as the decision variable, volume large and committed, a multi-year contract, and the lowest margin available anywhere in the sector.

What an award does to a manufacturing plan

This is the part that matters architecturally, because an award changes the status of a number.

- **It converts a forecast into an obligation.** Before the award, volume is a planning assumption. After it, it is a commitment with penalties for non-supply, and the plant commits validated capacity, orders active ingredient on lead times measured in months, buys packaging components and plans labour around it.
- **It is lumpy and binary.** Winning or losing a line item moves a plant's utilisation by a step, not by a percentage. In the 2025 antiretroviral tender — R15.5bn in total, of which R12.6bn covered fixed-dose combinations in one-month and three-month packs — one established South African manufacturer won no one-month or three-month pack award at all, taking only three smaller awards worth R215 million [6]. For a plant sized around that pool, that is not a weak quarter. It is an existential event.
- **It drives pack design.** Tender packs are different products: patient-ready packs, three-month supplies, specific labelling, delivery to depots rather than pharmacies. They need their own pack variations and artwork, and they sit in the same item master as the private packs with none of the same rules.
- **It moves the one-month versus three-month dispensing decision into the supply chain.** Shifting patients between a one-month and a three-month supply changes pack demand by a factor of three without changing one patient number. A clinical policy decision is a capacity decision.
- **Payment is slow and the term is short.** One province owed suppliers more than R1 billion as at September 2026, and late public-sector payment is cited as a cause of supply failure [9]. Industry has asked for terms of up to seven years instead of three and reports the request going unanswered [10]; nobody validates a line and builds capacity for a three-year award they may lose.

And then the award can be taken away

> **A tender award is not secure revenue, and in South Africa as at October 2026 that is not a theoretical caution.**
>
> On **22 October 2026** the Gauteng Division of the High Court, Pretoria set aside the one-month and three-month pack awards of the national antiretroviral tender as unlawful and unconstitutional. The Department must redo the split and award **within 120 days**; the set-aside takes effect **90 days after the new awards are finalised**; and current suppliers must keep delivering until then [6].
>
> Separately, on **17 September 2026** the Constitutional Court struck down **the entire Public Procurement Act 28 of 2024** for inadequate public participation, and **did not suspend the declaration of invalidity** [7]. Existing procurement legislation continues to apply and Parliament may re-enact.
>
> **Which framework now governs health-product tenders in practice is something I could not establish, and it is live.** Do not write it down as settled in a client deliverable without a current National Treasury instruction in hand.

The findings read as a checklist of what a South African pharmaceutical tender actually tests, and each one is a data product a bidder has to be able to produce on demand [6]: two bidders **did not hold a valid licence for their manufacturing site at bid closure** and should not have passed the first stage, so the site licence is a gate rather than a scored criterion; past performance was not properly weighed; claims about historically disadvantaged ownership were **accepted without investigation**, which raised scores; and possible collusion between two bidders should have been investigated before substantial volumes were awarded — the court found a reasonable suspicion of collusive bidding without deciding whether collusion occurred.

The supplier-risk context matters as much: both bidders in the collusion question were subsidiaries of one group, both entered business rescue in December 2025, and one was out of stock of both items it was due to supply days afterwards [6]. A Competition Commission investigation into alleged collusion by three companies in state medicine tenders was reported in February 2026; it is early-stage and no conduct has been attributed [23].

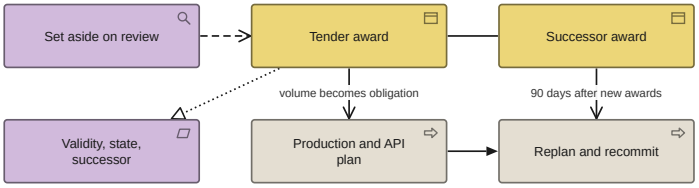


Figure 4.4 A tender award is a contract with a state, not master data. The production plan and the ingredient commitment inherit the award's legal status, and a successor award arrives with its own dates.

Model a tender award as a contract with a validity and a litigation state, not as master data. It needs an award date, contracted volume and price per line, a term, a challenge window, a current legal status, a successor award reference, and a flag for whether delivery continues during a set-aside. The

production plan, the ingredient purchase commitments and the cash forecast built on it inherit that state. Treat the award as a stable reference row and the plant will be holding ingredient against an award that no longer exists.

Fee for service: the contract developer and the contract researcher

Veldtkruin Group contains a contract development and manufacturing organisation, Khanyisa Bio-Development, because that is the archetype a South African architect is second most likely to meet. Its economics are not a variant of the product company's; they are a different business.

A contract developer earns a fee for service, not a margin on product. It does not own the registration, the product or usually the inventory. It is paid for development work on a time-and-materials or milestone basis, and for manufacture per batch, per unit or by capacity reservation, commonly with minimum volume commitments, take-or-pay reservations and materials passed through at cost plus a handling percentage.

Four consequences an architect has to carry.

The cost object is the campaign, not the product, because the product belongs to the client. Any costing design assuming a standard cost per finished item does not fit; the model is job and project costing. Revenue arrives against performance obligations — development milestones at a point in time, batches on release, capacity reservations spread over the term whether used or not, services as delivered. Whether a pass-through is recognised gross or net is a real accounting judgement with system consequences.

Capacity utilisation is the whole economics, and changeover eats it. For a product company, changeover time is a cost driver that reduces margin on goods sold. For a contract developer, suite time *is* the revenue line, so every hour of changeover and cleaning validation between campaigns is validated capacity consumed and not billable. At Khanyisa — invented, like everything about it — changeover and cleaning consume roughly 38% of available suite time, across about 95 products for 38 clients at an average of three batches per product per year. That one number is the most important economic fact about the business, and it is why a contract developer's planning problem is a scheduling problem and a product company's is a volume problem.

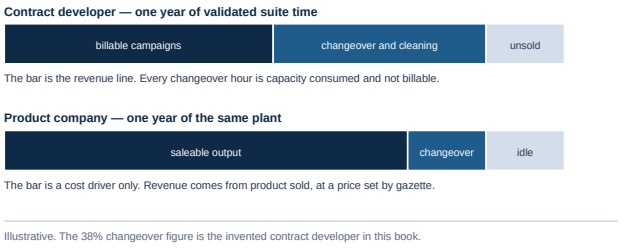


Figure 4.5 Why a contract developer's planning problem is not a product company's. For one, booked suite time is the revenue; for the other it is a cost driver behind a gazetted price.

Client data is held but not owned. One site, one quality system, one laboratory system, many clients' products, each with a different registration, different specifications, different documentation requirements and different change-control approval rights. Segregation is physical, procedural and informational at once. And one good-manufacturing-practice finding at the site is simultaneously a problem for every client whose registration names it — 38 registration holders, in Khanyisa's case, which is why its quality agreements are its most important commercial documents.

A contract researcher is not a kind of contract developer. They share almost nothing architecturally. A contract research organisation runs trials, or parts of them, for a sponsor: good clinical practice rather than good manufacturing practice, labour rather than capacity, the study rather than the batch as the cost object. Its hardest problems are working inside many sponsors' systems while keeping each sponsor's data separate and blinded as required, and handing back a complete, audit-ready dataset and trial master file at study end. Its margin killers are non-billable rework and unpriced change orders when a protocol is amended.

> For both archetypes a professional-services and project-accounting estate — resource management, timesheets, work breakdown, revenue recognition by performance obligation, change-order workflow — is not optional, and it is the part an architect arriving from product manufacturing reliably under-scopes.

Licensing, milestones and royalties

Pharmaceutical assets are traded constantly, and the deal structures create income and obligations that an architect has to be able to represent.

An out-licensing deal typically carries an upfront payment at signature; **development milestones** on defined technical events such as first patient dosed in a phase or a submission accepted; **regulatory milestones** on approval in named markets; **commercial milestones** on first achievement of annual net sales thresholds; and **royalties** as a percentage of net sales, tiered by band, running to a contractual expiry. Reversion rights, sublicence income shares and minimum diligence obligations sit alongside.

Two things follow that are genuinely hard.

The finance system depends on the clinical and regulatory systems for a revenue trigger. A development milestone is an objectively determinable event with a date, and the system that knows it happened is the trial management system or the regulatory information system, not the ledger. Recognising the revenue means knowing the fact, which means an interface nobody owns.

There is no "net sales" figure that satisfies all the contracts. Royalties are calculated on net sales *as each contract defines it* — which deductions are permitted, whether distributor sales count at the distributor's price or the transfer price, how combination products are apportioned, how currency is converted and at what rate. A company with fifty licence agreements has fifty definitions over overlapping product and country sets. The calculation is an interpretive layer over sales data, usually maintained by two people in a spreadsheet, audited by licensors, and a recurring source of restatements.

When a client says "we need a royalty system," the requirement is a contract-term repository with a calculation engine and an audit trail, and the hard part is modelling the terms, not the arithmetic. **A licensing deal creates obligations that live in no system anybody owns** — and the most consequential fact about an in-licensing deal, for architecture, is whether the local entity or the licensor holds the South African registration, because the registration holder carries the regulatory and safety obligations and therefore needs the systems.

Lifecycle management, and the one piece of planning that is deterministic

Everything after launch that extends or defends a product's economics is lifecycle management: new indications, each with its own trials and its own label variation; new strengths and formulations; fixed-dose combinations; paediatric programmes; new geographies; cost-reduction changes to the process. On the regulatory side it is a continuous stream of variations, periodic safety reports, renewals and commitments. **This is the volume business of regulatory affairs**, and it is where the architecture either works or does not.

Then exclusivity ends. Entrants file and are approved; price falls; volume migrates. For a small molecule the decline is multiplicative, not additive — a 60% volume loss with a 70% price reduction on what is left leaves roughly 12% of prior revenue. For a biologic the curve is shallower, for regulatory rather than commercial reasons: fewer entrants, substitution that usually needs a prescriber decision rather than a counter decision, perceived clinical risk in switching a stable patient, device and patient-support switching costs, and a cost floor, because a biosimilar genuinely costs a great deal to make. **Substitution mechanics drive erosion speed and they are jurisdiction-specific**, so one erosion assumption across a mixed portfolio is wrong in both directions.

Exclusivity is not one thing either. It is a stack: the composition-of-matter patent, secondary patents on formulation and polymorph and process, regulatory data and market exclusivity, orphan designation, patent term extension. Per country, and litigated. The date can move.

But — and this is the point for a planner — **the cliff arrives on a date known years in advance.** Roughly 200 medicines are expected to lose patent or exclusivity protection between 2026 and 2030, with an estimated USD 400 billion of revenue at stake, described as the largest such event the industry has faced [16]; a separate estimate puts USD 300 billion on it in a hospital-spending context [16]. The two measure different things over different windows and should not be reconciled.

> Loss of exclusivity is the one piece of strategic planning in this industry that is genuinely deterministic. The loss is forecastable, so it is in the plan, so the entire commercial architecture of an originator is organised around replacing revenue that is scheduled to disappear. And exclusivity dates very often live in the intellectual property function's own database, or a spreadsheet, rather than in any enterprise system. Finding where they actually live is one of the more revealing first questions an architect can ask.

Registered, priced, reimbursed: three different things

Approval is not launch, and an architect who models registration as the gate to revenue will build a pipeline report that lies.

A product that is **registered** may lawfully be sold. A product that is **priced** has a declared Single Exit Price in the national database and can reach a private pharmacy. A product that is **reimbursed** is on a scheme formulary at a benefit level that means the patient actually receives it. Three states, months or years apart, and a company can be in any one without the others.

The current South African illustration is uncomfortable and precise: **as at February 2026 no Single Exit Price had been declared for cabotegravir or lenacapavir** [3] — approved HIV medicines with no private-sector price, and therefore no private-sector availability. Registration happened. Pricing did not.

Market access manages the second and third states: the price declaration and its annual adjustment, scheme formulary listings and reimbursement positions, tender pricing strategy, and the health-economic argument behind it. In a South African company it is frequently one or two people, and interviewing them is the fastest route into the commercial architecture (chapter 5).

Pipeline reporting therefore has to declare which gate it counted. "Twelve approvals this year" is not a number until you know whether it means submitted, accepted for review, registered, priced or launched — and SAHPRA's own reporting counts applications *finalised within a working-day target*, not registrations granted [8].

The South African structural constraints, dated

Everything above is the business model. What follows bends it locally. Each item carries a date, because every one has moved in the last two years.

Import dependency and currency, and what "locally produced" means

About 70% of pharmaceutical products consumed in South Africa are locally produced, on a Who Owns Whom figure **dated 11 December 2020** [12]. It is almost certainly stale given the closures of 2025 and 2026, and it must be quoted with its date.

> African countries import 70–100% of finished pharmaceutical products and about 99% of vaccines. **South Africa does not.** Never apply the continental figure to South Africa; the two differ by an order of magnitude in the thing they describe.

But **"locally produced" means locally formulated and packed, not locally synthesised.** On the industry bodies' evidence to Parliament in August 2026, no domestic capacity remains to produce the active ingredient for paracetamol and none to formulate penicillin; nine contract manufacturing organisations had closed recently, disrupting supply of folic acid, plasmoquin and various eye drops; the only oral contraceptive facility had shut; and more than 2,500 jobs had been lost in 18 months [10][11]. **Those are the industry bodies' parliamentary claims as reported, not independently verified.** Attribute them that way.

Active-ingredient import dependence as a percentage is something I could not source. The qualitative position is firm; the number is not, and a number you cannot source should not appear in a client deliverable.

The currency exposure that follows is the best one-sentence explanation of why local capacity is contracting. Active ingredients, excipients, primary packaging, filters, single-use consumables and capital equipment are priced in dollars or euros. Revenue is in rand, capped by gazette, adjusted once or — in 2026 — twice a year. **Costs reprice continuously; the selling price reprices annually by permission.** A 10% rand depreciation lands on the cost base immediately and on the price at best at the next adjustment, which in January 2026 was 1.47% [9]. Any margin model without a dated exchange-rate assumption is incomplete, and supply-risk monitoring has to run at active-ingredient-source level rather than supplier level, because several formulators may depend on one plant in India or China without knowing it.

Local content has not protected local manufacturers

On the industry bodies' evidence to Parliament, the share of the antiretroviral tender by value awarded to local manufacturers fell from **72% in 2008 to 28% in 2025**, and in the pills and capsules tender from **56% in 2014 to 18% in 2026**, with procurement prioritising price above all [10]. **These are the industry bodies' claims as reported and were not independently verified.** They are also the most quotable figures in the local-content debate, which is a reason to attribute them carefully rather than to drop them.

On preferential procurement, be precise about the instruments. In the October 2026 antiretroviral judgment the operative mechanisms were **points for historically disadvantaged ownership** and **local manufacturing preference** — not a broad-based empowerment scorecard [6] — and qualifying for local manufacturing preference materially changed the price a bidder was held to [6]. **Check the tender document; do not assume which instrument applies**, particularly now that the statutory framework is in flux [7].

In October 2026 the Presidency ordered a government–business task force for the sector with four priorities: the regulatory framework for private-sector medicine prices, preferential local procurement, SAHPRA regulatory matters, and better use of data for policy [25]. Those four are, almost exactly, the constraints in this section.

The regulator can cause shortages

Architects design supply-risk monitoring as though regulation only protects supply. **SAHPRA interventions, including facility closures, have disrupted supply** [9]. A regulator doing its job correctly can take product off the market.

The September 2026 picture: the national department described public-sector availability as "above 80%" while saying it could not say how many patients had been turned away over six months, and while carrying outstanding orders of more than 1.6 million for simvastatin 10 mg, about 1.58 million for folic acid 5 mg and 1.93 million for mebendazole 500 mg [9]. Both statements are true. Availability is typically a weighted item-level fill rate at depot level; the patient experiences one item at one facility on one day. **Any availability metric has to declare its level, its item set and its weighting.**

Model "our own site or our supplier's site is suspended" as a live scenario, not an unthinkable one. It belongs on the risk register with a named mitigation — and the data the mitigation needs, qualified alternate sources per active ingredient with their registration status, is data most companies do not hold.

Rebates are prohibited and the prohibition is undefined

Bonus, rebate and incentive schemes are prohibited under the Act for medicines other than Schedule 0, where enforcement is paused. But **the regulations defining which acts are acceptable and which prohibited were published for comment in 2014 and 2017 and were never finalised**, so the extent of co-marketing fees, shelf fees, off-invoice rebates and similar arrangements is unknown [3]. The prohibition exists; the definition of what it prohibits does not; the practices continue unmeasured. The defensible architectural position is to make the transactions visible and dated rather than to assert that they are compliant — record what was paid, to whom, for what, against which agreement, and let the client's legal function form the view.

National Health Insurance is suspended, not being implemented

> **National Health Insurance is judicially suspended.** By an order recorded **24 February 2026**, the Minister of Health and the President are prohibited from proclaiming or implementing any provision of the Act until the pending public-participation challenges are resolved [19]. That account comes from a litigant; the order itself is needed for exact wording.
>
> The Constitutional Court heard the procedural public-participation challenges on **5–7 May 2026** and **reserved judgment** [20]; the position as at October 2026 was not established. In a **separate** matter reported by 21 May 2026 the Court **struck down the certificate-of-need provisions only** — not the whole Act [21]; that source is an advocacy organisation's press release and gives neither the judgment date nor the Court's reasoning.
>
> **Treat it as a scenario, never a roadmap item.** Any proposal that sequences investment on an assumed commencement date is unsound as at October 2026, and the architect should say so in writing.

How medicines would be priced under such a fund is unsettled. The fund is expected to be an active purchaser; informed commentary argues tenders would not work at full implementation and that something closer to scheme reference pricing would be needed [3]. That is commentary, not policy. The defensible preparatory work is capability that pays regardless: master data quality for patients and providers, coding standards, claims and dispensing data integrity, formulary management, and the ability to supply at different price points to different channels.

The marketing code, and the thing it cannot do

Promotion is governed by the South African Code of Marketing Practice for Health Products, maintained by the Marketing Code Authority, at version 19 approved in June 2026 [22]. It is **voluntary self-regulation** with a hook into the Act, and where the code and a law conflict the law prevails [22]. It prohibits compensation, payment, reward, benefit or inducement that is not legally due, prohibits personal incentives to health professionals in an advertisement, prohibits payment for interviews, and requires honoraria and fees for service to be paid at **fair market value** [22]. Records of financial transactions must sit in a signed agreement kept at least five years [22].

Here is the part with a direct architectural consequence:

> **The Marketing Code Authority says it cannot set fair market value, because competition law prevents it** [22]. Each company must set and substantiate its own. That makes fair market value a dated, documented, per-service, per-seniority reference dataset owned by the company — and the substantiation, not the number, is the record. It is also a competition-sensitive document in its own right, which means it has to be derived from the company's own data.

Whether the code imposes a public transfer-of-value disclosure obligation of the European or American kind is something I could not verify. The transparency chapter exists in the code's structure but its text was not available to this research, and "mandatory public disclosure" is the wrong framing for a voluntary scheme in any case [22]. What is verified is the signed agreement per financial transaction, kept five years [22] — so build the register, at recipient, activity, date, category, amount, agreement reference and fair-market-value basis, and leave the publication question to the client's compliance function.

What the business model asks of the architecture

Six things, straight out of the sections above.

1. **A price record at pack grain, effective-dated, with a channel flag**, because the rule differs by channel and one price column is a defect.
2. **A control that blocks customer-specific pricing for the regulated class**, with exceptions logged, because it is compliance rather than commercial preference.
3. **A tender award modelled as a contract with a validity, a status and a successor**, with the plan and purchase commitments visibly dependent on it.
4. **One gross-to-net calculation, not twenty** — the logistics fee.
5. **A cost object that matches the archetype**: the product for a generics manufacturer, the campaign for a contract developer, the study for a contract researcher.
6. **A dated register of the things that move** — the price adjustment, the tender's legal state, the exclusivity date, the procurement framework, the site's licence status.

Chapter 6 does the arithmetic: how the money adds up, line by line, for each of these shapes. Chapter 5 does the people, because none of the above is knowable without them.

THE CHAPTER IN TEN LINES

- There are four customers at once: the prescriber chooses, the dispenser may substitute, the payer pays and the patient consumes. The chooser, the payer and the consumer are usually three different parties.
- Single Exit Price is the only price at which a manufacturer or importer may sell to any buyer other than the State. The State buys by tender, entirely outside the regime.
- There were two adjustments in 2026: 1.47% in January and an extraordinary 2.88% effective 1 October 2026. Any statement citing one of them is wrong.
- Wholesalers take no mark-up. The manufacturer pays a logistics fee out of the exit price, and that is the only genuine gross-to-net item in the South African private market.
- The dispensing fee is charged in addition to the exit price, on the price excluding VAT, with different bands for pharmacists and dispensing doctors. Do not quote the February 2025 proposed bands as current.
- South Africa does not operate external reference pricing, and no Single Exit Price has ever been declared unacceptable.
- Model a tender award as a contract with a validity and a litigation state, not as master data. Both the antiretroviral awards and the procurement statute itself were set aside in late 2026.
- A contract developer earns a fee for service and its cost object is the campaign, not the product. Booked suite time is its revenue line, and changeover consumes it.
- Approval is not launch. Registered, priced and reimbursed are three different states, months or years apart.
- National Health Insurance is judicially suspended, not being implemented. Treat it as a scenario, never a roadmap item.

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CHAPTER 5

Who does what

This is the chapter that makes you able to run a workshop. Forty-odd roles, what each one does all day, what they will tell you, what they will not say unless you ask, and which system they live in — plus the order to see them in, because each interview supplies the vocabulary for the next.

Read this before you book the workshop

An architect who cannot name the jobs cannot run a workshop. That is the whole reason this chapter exists.

Pharmaceutical companies are organised by regulatory regime, not by process. Discovery runs under good laboratory practice, clinical under good clinical practice, manufacture under good manufacturing practice, safety under good pharmacovigilance practice, distribution under good distribution and wholesaling practice. Five regimes, five inspection traditions, five vocabularies for the same underlying ideas — and the people in each one have been trained to answer a different kind of question. Put the wrong person in the room and you will get a confident, well-evidenced answer to something you did not ask.

So: forty-odd roles below, grouped by where they sit. For each one, four things. What the job actually is. What they will tell you. What they will not say unless you ask. And which system they live in.

Two warnings before you start.

> **The same title means different things in different companies, and some titles mean opposite things in different functions.** A "data manager" in clinical development designs case report forms and takes a trial database to lock; a data manager in an information technology function governs master data. They have almost nothing in common and they will both answer "yes" when you ask whether they own data quality. "Quality assurance" in a pharmaceutical company is not quality assurance in software: it is a regulated function owning deviations, change control, corrective actions, supplier qualification and inspection readiness, and it has a statutory veto. "Validation" is not test automation. "Monitoring" is not observability.
>
> Write the person's actual responsibilities into your stakeholder map, not their title.

the word	in pharma it means	elsewhere it means
data manager	designs case report forms, cleans trial data, takes a database to lock	governs master data definitions and lifecycle
quality assurance	a regulated function with a statutory veto: deviations, change control, release	a test discipline inside delivery
validation	proving a system fit for regulated use, and keeping it so	checking user input
monitoring	visiting investigator sites to verify conduct and data	watching a system in production
protocol	the controlled document that defines a clinical trial	a transport or message standard
batch	one defined manufacturing cycle, with genealogy and disposition	a scheduled bulk job

Figure 5.1 The same word, four dialects. Pharmaceutical vocabulary overlaps with software vocabulary at exactly the points where the meanings diverge most. Write responsibilities into your stakeholder map, not titles.

> Role names here are the ones the industry uses. Where a role is specific to South Africa, or has a South African variant that differs from the European or American one, it is marked. The invented company in this book, Veldtkruin Group, supplies the names and the scale; nothing about it is real.

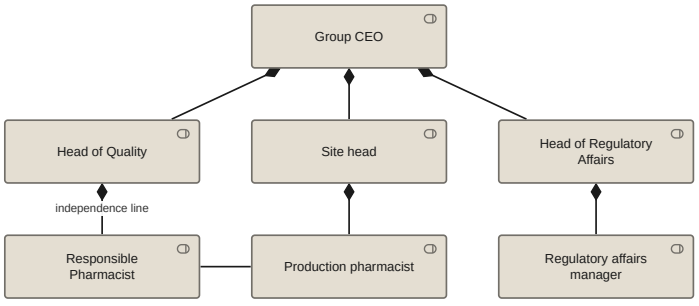


Figure 5.2 One plausible organisation shape, in ArchiMate Role elements. Quality independent of production is the structure the regime assumes. Whether the Responsible Pharmacist actually sits on that line is the question to ask out loud — this is not a template.

Discovery and preclinical

The people here produce the evidence that everything downstream rests on, decades before it matters, in systems nobody downstream has heard of.

Medicinal chemist. Designs and synthesises compounds to improve potency, selectivity and the properties that make a molecule behave like a medicine. *All day:* designing analogues, reading assay results from last week's batch, arguing about the next series. *Will tell you:* that structure-activity data is spread across an electronic laboratory notebook, a compound registry and personal spreadsheets, and that establishing whether a compound was made before is harder than making it again. *Will not say unprompted:* that the registry's naming conventions changed twice and the old records were never remapped. *Lives in:* the electronic laboratory notebook, compound registration, chemical drawing tools.

Biologist (assay and pharmacology). Builds and runs the assays that measure whether a compound does anything. *Will tell you:* that results from different runs are not comparable without the control data, and that the plate-reader instrument file is the real raw data, not the summary in the report. *Will not say unprompted:* how many assays have no formal specification at all. *Lives in:* a laboratory information management system, assay data management, instrument software.

Pharmacologist. Characterises what the drug does in a living system — dose-response, mechanism, the relationship between exposure and effect. *Will tell you:* that the dose chosen for the first human study is an inference from animal exposure and that the inference has to be auditable.

Toxicologist. Runs and interprets safety studies under good laboratory practice. *Will tell you:* that the study data lives with the contract laboratory that ran it, in that laboratory's system, and that the company holds the report and not always the underlying data. *Will not say unprompted:* that this becomes a serious problem twenty years later when a regulator asks a question the report does not answer. *Lives in:* study reports, the good-laboratory-practice archive.

Drug metabolism and pharmacokinetics scientist. Measures what the body does to the drug: absorption, distribution, metabolism, excretion. *Will tell you:* that the bioanalytical method has to be validated before any number from it means anything, and that method changes mid-programme are a traceability problem.

Computational chemist. Models binding, properties and liabilities before anything is synthesised. *Will tell you:* that the compute is the easy part and curating the training data is the job. *Architectural note:* this is the one place in a pharmaceutical company where a modern data-science stack exists, and it is usually outside the validated estate because it does not generate a regulatory record. Check that assumption rather than accepting it.

Laboratory head. Runs a laboratory: people, instruments, budget, throughput, and the qualification state of the equipment. *Will tell you:* which instruments are out of calibration and what that does to the queue. *Lives in:* calibration and maintenance records, the laboratory system.

Interview them for: data lineage from instrument to conclusion, the boundary between work done inside and work bought in, and the long-term archive obligation.

Clinical

This is where the systems estate gets expensive and where two roles are routinely conflated by people selling software.

Clinical study manager (clinical trial manager). Owns delivery of a trial: sites, timelines, budget, vendors, enrolment. *All day:* chasing site activation, approving invoices, re-forecasting enrolment, managing the contract research organisation. *Will tell you:* that enrolment is always behind, that the trial master file is never complete, and that site payments are painful. *Will not say unprompted:* which vendor is failing and why nobody has escalated it. *Lives in:* the clinical trial management system, the trial master file, the study budget.

Clinical research associate (the monitor). Visits investigator sites and verifies that the trial is being run to protocol: checks consent, verifies source data against what was entered, confirms drug accountability. *All day:* travelling, reading site files, writing monitoring visit reports, following up findings. *Will tell you:* exactly where the data does not match, site by site. *Will not say unprompted:* that remote monitoring changed the job and the procedures have not caught up.

> **The study manager and the monitor are not the same role and systems vendors conflate them.** The study manager runs the study from the sponsor side — one person, many sites, accountable for delivery. The monitor works the sites — many monitors, each with a handful of sites, accountable for the conduct and the data at those sites. Their system needs are different: the study manager needs plan, budget, milestone and vendor views; the monitor needs a visit plan, a report workflow, a findings-to-closure trail and source-data verification evidence. A product demonstration that shows one screen for both is showing you a screen neither will use.

Clinical data manager. Designs the case report form, builds the edit checks, cleans the data, manages queries to sites, and takes the database to lock. *Will tell you:* that "is the data clean" has a precise answer — open queries, outstanding pages, unreconciled serious adverse events — and that the safety database and the electronic data capture system never agree without manual reconciliation. *Will not say unprompted:* how many hours that reconciliation takes each week. *Lives in:* electronic data capture, the data management plan, query listings. **Interview this person early. They know where every data defect in the clinical estate is.**

Biostatistician. Designs the analysis, writes the statistical analysis plan, produces the tables, listings and figures and runs the primary analysis. *Will tell you:* that the analysis runs in a validated statistical computing environment with version-controlled programs, that the analysis datasets are a deliverable in their own right, and that the plan has to be final before unblinding or the analysis is not credible. *Lives in:* a validated statistical environment, standardised study datasets.

Medical monitor (study physician). Clinically responsible for subject safety in the trial; reviews adverse events and eligibility, answers investigator questions. *Will tell you:* which safety signals are being watched and why. *Will not say unprompted:* which investigator they do not trust.

Medical writer. Produces the protocol, the investigator's brochure, the clinical study report and the submission summaries from everyone else's outputs. *Will tell you:* where the document versions actually live and which figures in the report were pasted rather than generated. *Architectural note:* the medical writer is the integration point nobody models — a human assembling a regulated document from five systems' outputs, and the only person who knows which numbers came from where.

Clinical supply manager. Plans, labels, packages and distributes investigational product to sites, under good manufacturing practice, with randomisation and blinding to preserve. *Will tell you:* that the expiry date on the label is the constraint, that relabelling a shipment at a site is a regulated act, and that returns reconciliation at study end is nobody's favourite work. *Lives in:* randomisation and trial supply management, the manufacturing site's systems, the distribution records.

Site coordinator (study coordinator at the investigator site). Not an employee of the sponsor, and the person on whom the whole trial actually depends. Recruits and consents subjects, books visits, takes samples, enters data, answers queries. *Will tell you:* that they are running four studies on four different sponsors' systems with four different logins, and that the protocol's visit schedule does not fit a working clinic. *Will not say unprompted:* which data they entered from memory at the end of a long day. **Any design that adds site burden will fail here and you will not hear about it until a monitor finds it.**

Interview them for: the data flow from site to submission, the controls around unblinding, and the reconciliation between the trial database and the safety database.

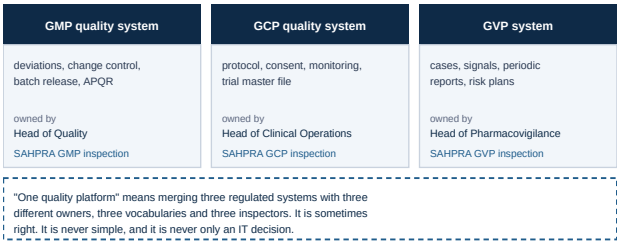


Figure 5.3 Three quality systems, three owners, three inspectors. An architect who proposes one quality platform is proposing to merge three regulated systems.

Regulatory affairs

The function that decides whether a business change is possible, and the one most often discovered too late.

Regulatory affairs manager. Prepares and maintains submissions, manages variations, answers regulator questions, and owns the registration portfolio and its commitments. *All day:* assembling and tracking submissions, assessing whether a proposed change needs a variation and of what class, answering the regulator's clock. *Will tell you:* the single most useful thing an architect can learn in a pharmaceutical company — which products are registered where, on what conditions, with what expiry, and which planned business change is blocked on a variation. *Will not say unprompted:* that the authoritative registration register is a spreadsheet. **Ask to see the spreadsheet.** *Lives in:* a regulatory information management system, a submission publishing tool, and that spreadsheet.

Be clear about what regulatory affairs actually owns, because it is not "the paperwork". It owns two things, and both are states rather than documents.

1. **The dossier as a maintained state.** The dossier is not a submission that happened; it is the live record of what the regulator approved — this formulation, these specifications, these named manufacturing sites, this leaflet wording, these conditions. The company's answer to "what are we allowed to do" *is* the dossier. The architectural question is whether there is one authoritative place where that answer lives per product per country, or whether it is reconstructed from folders each time someone asks.
2. **The commitment calendar.** Every registration carries obligations with dates: periodic safety reports on a fixed cycle, annual retention fees, five-yearly licence renewals applied for at least 180 days before expiry, stability time-points, post-approval commitments agreed at approval [2]. None of these cycles aligns to a financial year and **missing one can stop a product being sold.** It is the lowest-glamour, highest-consequence calendar in the company, and in many South African companies it is one person's spreadsheet.

Submission publisher. Assembles the electronic dossier to the regulator's technical specification, validates it and transmits it. *Will tell you:* that South Africa runs on an older electronic submission format, portal-only, with whole-sequence rejection on a single technical error — so a validation failure at five in the afternoon is a missed deadline. *Will not say unprompted:* that sequence numbering errors are the most common cause of rework. *Lives in:* the publishing tool and the regulator's portal.

Labelling specialist. Maintains the approved product information and the patient leaflet per market per version, and drives the artwork change that follows every label change. *Will tell you:* that a label change is a chain — approved text, artwork, printed component, pack in stock, pack on a shelf — and that the chain takes months. *Architectural note:* label version is a reference dataset that safety, promotion and manufacturing all depend on, and it is usually held in a document system with no identifiers.

Chemistry, manufacturing and controls regulatory specialist. The translator between the plant and the dossier. Assesses whether a proposed process, specification, supplier or site change is a notification or needs approval before implementation, and writes the variation. *Will tell you:* the lead time for each class of change, which is the number the supply chain actually needs. **This person is the gating authority on a surprisingly large number of apparently operational decisions** — changing an active-ingredient supplier, moving a product to another line, changing a packaging component, revising a specification.

> South African variation classes do not all work the way an approval workflow is normally built. One class is deemed implementable if the regulator gives no opinion within a set number of working days. **The absence of a response is an event the system has to detect**, which is the opposite of waiting for an approval to arrive. Chapter 16 has the classes and the clocks; the point here is that your regulatory affairs manager knows this and your workflow designer does not.

Safety and pharmacovigilance

A small function with statutory clocks measured in days and an intake surface that runs across the whole company.

Pharmacovigilance officer. Runs the safety system: intake, processing, assessment, reporting, periodic reports, signal detection, and the audit trail over all of it. *Will tell you:* the reporting clocks, where cases come from — spontaneous reports, literature, trials, partners, digital channels — and which inbound channel loses days. *Will not say unprompted:* that the sales force is a reporting channel and has never been audited as one. *Lives in:* the safety database and the system documentation.

Case processor (drug safety associate). Receives a report, books it in, codes it, requests follow-up, assesses seriousness and expectedness, and prepares the submission. *All day:* case after case against a clock. *Will tell you:* that day zero is the date of first awareness **by anyone in the company**, not the date the case reached the database — and that a day lost in an interface is a regulatory breach, not a data-quality issue.

Medical reviewer (safety physician). Assesses causality, clinical significance and whether the label already describes the event. *Will tell you:* that "expected" is a judgement against a specific version of a specific country's label, so the label version is part of the case record.

Signal detection scientist. Reviews aggregate safety data for patterns, maintains the signal register, and carries a signal through to a label change where one is warranted. *Will tell you:* that the register has to include **refuted** signals, because an inspector asks what you looked at and dismissed, not only what you escalated. *Will not say unprompted:* that coding dictionary versions make the historical series non-comparable.

Qualified person for pharmacovigilance, where one exists. In the European Union this is a named individual with a statutory role and a named back-up. *Will tell you:* where the accountability actually sits.

> Be careful here, because this is a place where confident writing outruns the evidence.
>
> **"SAHPRA requires a pharmacovigilance system master file" is wrong** — the term does not appear in the operative post-marketing reporting guideline, which requires a pharmacovigilance system with full documentation of its procedures and activities [3]. A master file is a sensible way to satisfy that; it is not, on the available evidence, the requirement. Two related statements to avoid: the operative guideline does not state a MedDRA mandate, and foreign individual case reports are **not routinely forwarded** to the regulator — what runs on a five-day clock is foreign *regulatory action* [3].
>
> **Whether South Africa requires a named, locally resident pharmacovigilance person for a registration holder is something I could not settle.** The research behind this book found one account of inspection practice expecting residency evidence for such a person, and a separate finding that the requirement could not be located in the retrieved guidelines. Treat it as an open question to put to the client's regulatory affairs manager on day one, not as a fact in your model.

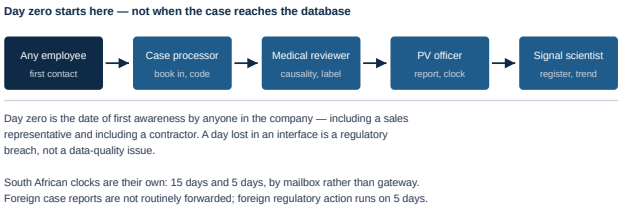


Figure 5.4 The safety case chain, and where the clock starts. First awareness by anyone in the company is day zero, which makes the medical information inbox and the sales force regulated intake channels.

Quality and manufacturing

The largest group, the one with statutory veto rights, and the one where an architect will meet the single most useful interview in the building.

Quality assurance officer or manager. Owns the quality system: deviations, change control, corrective and preventive actions, complaints, recalls, supplier qualification, audits, inspection readiness, document control, training records and validation governance. *All day:* reviewing and approving records, chasing overdue investigations, preparing for inspections. *Will tell you:* the deviation and change-control backlog, which is the best single proxy for whether a site is in control. *Will not say unprompted:* how many corrective actions were closed without an effectiveness check. *Lives in:* the electronic quality management system and the document management system.

Quality control analyst. Tests raw materials, in-process samples, finished product and stability samples against specification. *Will tell you:* that an out-of-specification result triggers a formal investigation, that instrument data has to be attributable and audit-trailed, and that the testing queue is the release bottleneck. *Lives in:* the laboratory information management system, chromatography data systems.

Responsible Pharmacist. South Africa's batch-release role, and the one chapter authors get wrong most often. It is **not** the European Qualified Person under another name.

- A **pharmacist registered with the South African Pharmacy Council**, accountable to the Council for compliance with the Pharmacy Act and with the legislation applicable to the pharmacy under their personal supervision [1].
- **Named on the SAHPRA manufacturing licence, with not more than one named per licence** [2]. One licence, one named individual.
- Must **sign or certify, for each batch, that the appropriate tests have been carried out** [2], and release of finished product for sale is signed off **using a release checklist** that is a named artefact in the guideline [5].
- Carries a duty to ensure that **manufacturing and testing processes remain in a validated state**, and that a self-inspection programme is active [5].
- **Dual-accountable** — to SAHPRA under the Medicines Act and to the Pharmacy Council under the Pharmacy Act [1][2][5]. The European Qualified Person answers to the medicines regulator alone, and may be a chemist or biologist rather than a pharmacist.

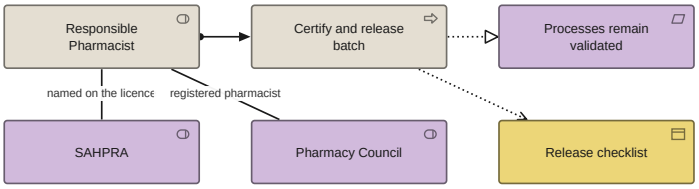


Figure 5.5 The Responsible Pharmacist's accountability lines. Named on the licence with not more than one per licence, registered with the Pharmacy Council, signing or certifying each batch against a release checklist, and carrying a duty that the processes stay in a validated state.

That last duty has a consequence most architects miss. **"Processes remain in a validated state" is a Responsible Pharmacist duty**, which makes computerised-system validation status a release-relevant fact [5]. A release view that does not surface validation status, open change records and periodic-review state is incomplete against the duty as written. And because the role requires Council registration and residency, the release signature cannot be delegated to an offshore shared-services centre — a design pattern that works elsewhere and does not work here.

Production pharmacist. A South African role: a pharmacist supervising manufacturing operations in a licensed plant, where other jurisdictions would use a production manager or engineer. *Will tell you:* the real batch cycle time, where the deviations come from, and which products are painful to make. *Will not say unprompted:* which line is being run outside its comfortable range to hit a tender delivery.

Validation engineer and validation lead. Qualifies equipment, facilities, utilities and processes, validates computerised systems, and keeps all of it in a validated state. *Will tell you:* which systems hold regulated records, what their validation state is, when each is next due for periodic review, and what a proposed change would cost in revalidation.

> **The validation lead is the person who knows which of your systems are actually in a validated state — and nobody else in the company knows.** Not the information technology director, not the quality head, not the system owners. The register lives with validation, and in most companies it is partly in a spreadsheet and partly in someone's head.
>
> In the manufacturing volume of this series the equivalent claim was made about the maintenance planner: the one person whose records tell you what the plant really is. In a pharmaceutical company it is the validation lead, and the reason is sharper. An inspector's first question is some version of *which of your systems hold regulated records?* If the answer takes three weeks to assemble, you have found the engagement. **In most pharmaceutical companies this is the single most useful first interview an architect can have**, and it should happen in the first week, before the application landscape workshop, not after it.
>
> At Veldtkruin the person is Anita Govender. She is invented. The role is not.

Qualification engineer. Qualifies new equipment and facilities — design, installation, operation, performance — and hands over a qualified asset with its documentation. *Will tell you:* the difference that matters: **infrastructure is qualified; applications are validated.** The words are not interchangeable and using them loosely in a steering pack will cost you credibility with this audience.

Calibration technician. Keeps instruments within their calibration state and records it. *Will tell you:* that a result produced on an out-of-calibration instrument is not a result, and that the calibration due date is therefore a production constraint.

Microbiologist. Runs environmental monitoring, water testing, sterility and bioburden work. *Will tell you:* that environmental monitoring generates a continuous data stream with alert and action limits, and that an excursion is an investigation.

Interview them for: the validated-system register, the deviation and change backlog, the release path, and what a proposed change would actually cost.

Supply and commercial

Supply planner. Balances demand against validated capacity, ingredient lead times, batch sizes and shelf life. *Will tell you:* that batch sizes are fixed by the registration, so there is no such thing as a half batch; that active-ingredient lead times run to months; that shelf life limits how far ahead you can build; and that a tender award changes everything. *Lives in:* enterprise planning, plus a spreadsheet.

Demand planner. Builds the forecast, separately for the private market and for tender commitments, because one is a forecast and the other is a contract. *Will not say unprompted:* that the tender number in the plan is the award as awarded, with no allowance for its legal state (chapter 4).

Market access and pricing manager. A South African role of disproportionate importance. Owns the Single Exit Price declaration and its annual adjustment, medical-scheme formulary listings and reimbursement positions, tender pricing strategy and the health-economic argument. *Will tell you:* how the company actually sets and administers its exit prices, what the logistics fee arrangement is with each wholesaler, and how a tender price is built up. **This is the most valuable single interview for understanding a South African pharmaceutical company's commercial architecture, and the role is often one or two people.**

Tender manager (bid manager). Assembles state tender submissions: pricing, site licences valid at bid closure, ownership and local-content declarations, past-performance evidence, samples. *Will tell you:* how long a bid takes to assemble and which piece of evidence is hardest to produce. *Will not say unprompted:* that the evidence is assembled by hand every time, and that a competitor's award was set aside in 2026 partly because its evidence was not checked.

Medical science liaison. A field-based scientific role engaging specialist clinicians on the science, strictly non-promotional and strictly separated from sales. *Will tell you:* what clinicians actually think of the product. *Architectural note:* **merging the medical science liaison's interaction records into the sales customer-relationship system creates a compliance finding.** The separation of scientific exchange from promotion is a boundary, not a preference.

Medical information scientist. Answers unsolicited enquiries from health professionals and patients from an approved response library. *Will tell you:* that every enquiry has to be screened for an adverse event or a product complaint and routed accordingly. **Architects routinely miss this: the medical information inbox is a pharmacovigilance intake channel and a regulated record.**

Key account manager and medical representative. Promote prescription products to prescribers within the advertising rules, or sell consumer products into retail. *Will tell you:* which products are substituted at the counter and why. *Will not say unprompted:* that they occasionally hear about side effects and are not sure what to do with that.

Data and systems

Four roles that pharmaceutical companies need and information technology organisations usually have not staffed.

Master data steward. Owns the definition and lifecycle of a master data domain — material, product, registration, site, customer, supplier. *Will tell you:* which of the four meanings of "product" each system is actually using (chapter 9), and where the mappings between them live.

GxP system owner. A real, named, accountable role, defined in the quality system: the person accountable for a specific regulated system being fit for use, validated, change-controlled, periodically reviewed and properly retired. *Will tell you:* whether the system's validation documentation matches what is in production. **Most information technology organisations do not staff this role.** They have application owners who are accountable for availability and cost, which is a different thing, and then nobody signs the periodic review. If your client cannot name the system owner for each regulated system, that is a finding on day one.

Computerised-system validation lead. The same person as the validation lead above, in many South African companies, which is itself the problem: one person holding both the plant qualification programme and the computerised-system estate. *Will tell you:* which systems are classed as regulated, their risk classification, and the periodic review schedule.

Business system owner. The business-side counterpart of the system owner: accountable for the processes the system supports, the access model and the data. *Will tell you:* who actually approves access, which is usually not who the access matrix says.

Above it all

The Responsible Pharmacist's reporting line. The one organisational question worth asking out loud. If the person who certifies batches reports to the person accountable for output, the independence that the regime assumes is organisational fiction. Draw the line and ask about it; the answer is informative whichever way it goes. The same question applies to the European Qualified Person, and to quality's reporting line generally.

The quality council (quality management review). A formal governance body, typically quarterly, reviewing the quality system itself: deviation, corrective-action, complaint and change metrics, inspection outcomes, and the state of control. It is minuted and an inspector may read the minutes. *Architectural significance:* it is the governance forum where an architecture change to a regulated system gets its quality mandate, and going around it is how programmes acquire findings.

The portfolio committee. Reviews the development and commercial portfolio, revises programme valuations, allocates resources and takes kill decisions. *Architectural significance:* it is the demand signal for every research and development system. If the committee's pack cannot be assembled without three weeks of manual collection, that is the brief.

The chief information officer. Accountable for the estate and usually not accountable for its validation state, which is the structural tension this chapter has been circling. *Will tell you:* what the backlog is and which vendor is the problem. *Will not say unprompted:* that the manufacturing execution system is two major versions behind because every upgrade is a revalidation project.

The enterprise architect. You. The value you add in this industry is not a target-state diagram. It is being the one person who can hold the regulatory calendar, the validated-system register and the commercial plan in the same view and say which of them will break first.

The interview order

The research behind this book carries a recommended order and it is worth following, because each interview supplies the vocabulary for the next.

#	Who	What it reveals	Why this early
1	Validation lead	Which systems hold regulated records, their validation state, periodic review dates	Nothing you plan is feasible until you know this
2	Regulatory affairs manager	The registration portfolio, conditions, expiries, and which business change is blocked on a variation	Defines what the company is allowed to do
3	Market access and pricing manager	How exit prices are set and administered, logistics fee arrangements, tender price build-up	The whole commercial architecture, from one or two people
4	Quality assurance manager	The quality system and its deviation and change backlog	The backlog is the best proxy for control
5	Responsible Pharmacist	What cannot be changed, and what release actually depends on	A veto you need to hear before you design
6	Supply planner	The physical constraint set: batch sizes, lead times, shelf life, capacity	Turns the plan into arithmetic
7	Clinical data manager, if there is clinical activity	Every data defect in the clinical estate, and the reconciliation burden	They already know; nobody has asked
8	GxP system owners, named per system	Whether the validation documentation matches production	If they cannot be named, that is the finding

Two notes on using it. **Do not start with the chief information officer**, however natural that feels: you will get the estate as the estate sees itself, and you will not know which of it is regulated. And **do not treat the first four as a sequence you can reorder** — the regulatory conversation is much more productive once you know which systems are validated, and the pricing conversation is much more productive once you know what is registered.

A role-to-system map

The short version of chapter 12, so that a stakeholder map and an application landscape can be drawn against each other from the first week. Product names are the invented ones used throughout this book.

Role	System class	Veldtkruin name	What they do in it
Regulatory affairs manager	Regulatory information management	Fynbos	Registrations, variations, commitments
Submission publisher	Submission publishing	Fynbos and the portal	Assemble, validate, transmit
Labelling specialist	Labelling and artwork	Silverleaf	Approved text and artwork versions
Pharmacovigilance officer	Safety database	Sentinel	Cases, signals, reporting obligations
Quality assurance manager	Quality management system	Rooibos	Deviations, change control, corrective actions
Quality control analyst	Laboratory information management	Aloe	Specifications, tests, results
Responsible Pharmacist	Quality system and batch record	Rooibos and Protea	Certify and release each batch
Production pharmacist	Manufacturing execution	Protea	Execute and record the batch
Validation lead	Validated-system register	Rooibos, plus a spreadsheet	Validation state, periodic review
Supply and demand planner	Enterprise planning	Kelpfield	Demand, supply, capacity, inventory
Market access and pricing manager	Pricing and tender	Kelpfield, plus a spreadsheet	Exit prices, logistics fees, bids
Clinical study manager and monitor	Trial management and data capture	Sable	Sites, visits, monitoring, trial data
Clinical data manager	Electronic data capture	Sable	Forms, edit checks, queries, lock
Medical science liaison	Field medical records	Separate from sales	Scientific interactions, kept apart

Notice how often the answer is "plus a spreadsheet". That is not laziness and it is not a finding on its own. It is where the real model lives, and it is the most honest artefact in the building.

The skills reality

This section is judgement, not finding. **South African pharmaceutical-specific skills data is something I could not source**, and anyone quoting a number for the supply of validation engineers or regulatory affairs professionals in this market is quoting an estimate. What follows is what the pattern looks like from the inside of engagements, offered as an opinion you should test against your own client.

Three capabilities are the binding constraint on any pharmaceutical programme, and none of them is software engineering.

Validation capacity. Every change to a regulated system consumes validation effort — impact assessment, protocol, execution, review, approval. A programme that plans twelve system changes in a year at a company with two validation engineers has planned a queue, not a programme. Validation capacity, not development capacity, sets the throughput of the whole estate, and the honest version of a roadmap in this industry is sequenced against it.

Quality assurance capacity. Every regulated record needs review and approval by a qualified reviewer. Deviation and change-control backlogs in pharmaceutical companies are usually capacity problems rather than process problems, and automating a workflow that is queueing at a person moves the queue without shortening it.

Regulatory affairs capacity. The variation workload is already at capacity in most mid-size companies. A change programme that generates variations — a site transfer, a supplier change, a specification revision, a packaging rationalisation — adds to a queue that already has a legal clock on it.

The second-order point is the one worth taking to a steering committee. **These three capacities are shared across every initiative in the company**, and no initiative's plan shows them. Each programme assumes it has the regulatory affairs manager's attention. Modelling those three as constrained resources, with a visible forward load, is a small piece of architecture work that reliably changes a portfolio decision — and it is the kind of thing an architect can do in the first month.

Two cautions. These people are scarce: a Responsible Pharmacist or a validation lead who leaves takes the register with them, and succession for a named statutory role is a continuity risk with a licence attached to it. And outsourcing does not relieve the constraint, it relocates it — a contract developer's validation lead is just as finite, and a company that has outsourced manufacture still owns the regulatory obligations.

The regulator is a stakeholder with a data demand

One last move before you close your stakeholder map. Add the bodies that will ask your client for evidence, because each of them is a stakeholder with a specific data demand and none of them is in the organisation chart:

- **SAHPRA**, which inspects for good manufacturing practice, good clinical practice, good pharmacovigilance practice and good wholesaling practice — four different inspections, four different record sets, and a single 30-calendar-day response deadline with no critical/major/minor classification scheme in its response guideline [6]. It is also a capacity-constrained organisation that publishes its own performance against working-day targets, and the one target it missed in 2024/25 was manufacturing and wholesaling licences: 35 finalised of 90 received, 30 of them within the 125-working-day timeline [4]. That is the pathway a new site or a site transfer depends on.
- **The South African Pharmacy Council**, which registers pharmacies, pharmacists and premises, and to which your Responsible Pharmacist is separately accountable [1].
- **The Information Regulator**, which asks for the lawful basis of each processing operation, the cross-border transfer register and the adequacy determination behind it [7].
- **The research ethics committee**, for anything clinical, which asks for dated approval before the data it covers was collected.
- **The Department of Employment and Labour**, which asks for exposure and surveillance records on clocks measured in decades.

Chapter 16 carries the full table — who inspects what, and exactly what each one asks to see, against the repository that holds the evidence. Read down that table's last column and the same handful of structures answer nearly every inspector. Build those well and inspection readiness follows; build one repository per inspector and none of them will be current.

Chapter 6 picks up the money. Nothing in the rest of this book is knowable at a client site without the people in this chapter, and the order you meet them in decides how much of it you get.

THE CHAPTER IN TEN LINES

- Pharmaceutical companies are organised by regulatory regime, not by process, and the people in each regime are trained to answer a different kind of question.
- The same title means different things in different functions. A data manager in clinical is not a data manager in IT, and quality assurance in pharma is not quality assurance in software.
- The study manager and the clinical research associate are different roles with different system needs. Systems vendors conflate them.
- Regulatory affairs owns two states rather than a pile of documents: the dossier as a maintained record of what is approved, and the commitment calendar whose dates can stop a product being sold.
- The Responsible Pharmacist is South Africa's batch-release role, not a renamed European Qualified Person: a registered pharmacist, named on the licence with not more than one per licence, dual-accountable to SAHPRA and the Pharmacy Council.
- That role carries a duty that processes remain in a validated state, which makes system validation status a release-relevant fact.
- The validation lead is the one person who knows which systems are actually in a validated state, and is the most useful first interview in most pharmaceutical companies.
- The GxP system owner is a real, named, accountable role that most IT organisations have not staffed. If your client cannot name one per regulated system, that is a finding on day one.
- Validation, quality assurance and regulatory affairs capacity are the binding constraint on any pharmaceutical programme, and they are shared across every initiative while appearing in no initiative plan.
- The regulator is a stakeholder with a data demand. SAHPRA, the Pharmacy Council, the Information Regulator, the ethics committee and the Department of Employment and Labour all ask for evidence, and chapter 16 has the table.

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CHAPTER 6

How pharmaceutical companies make money

The arithmetic chapter. The portfolio is the unit of value, not the product; the headline cost of developing a drug is one estimate among many that differ by a factor of fifty; and the revenue-management machinery that dominates American pharmaceutical architecture processes transactions that cannot legally occur in South Africa.

A pharmaceutical company's money arithmetic is not a manufacturer's with longer lead times. Three things about it are different in kind, and each of them lands somewhere in the architecture.

The first is that the thing being valued is not a product. It is a portfolio of options, most of which will expire worth nothing, and the company steers on the value of that portfolio rather than on this year's profit. The second is that the selling price, in South Africa, is a number in a government gazette rather than the outcome of a negotiation. The third is that there is a gate between cost incurred and revenue recognisable — a quality decision, signed by a named person — and nothing crosses it early.

Everything else in this chapter follows from those three.

The portfolio, not the product, is the unit

Start with a discovery programme and describe it accurately: it is a sequence of purchased options. Spend to reach the next decision point. At that point, either pay again to continue or abandon. The value of the programme is the value of that option chain, not the expected revenue of a launched product treated as a certainty.

This is why pharmaceutical management does not talk about revenue or margin in a portfolio meeting. It talks about **risk-adjusted net present value** — rNPV, sometimes eNPV. An architect who cannot follow an rNPV conversation cannot sit in a portfolio review, and will therefore never find out what the systems are actually for.

The construction has five steps.

1. **Forecast the launched product's cash flows by year.** Addressable patient population, diagnosed and treated share, market share, price per patient-year — net price, not list — then compliance and persistence, then cost of goods, selling and administrative cost and distribution. Run it from launch to loss of exclusivity, then apply an erosion curve past that date.
2. **Lay out the remaining development costs by year and by phase.** Trial costs, manufacturing scale-up, regulatory fees, launch preparation.
3. **Discount both** at a cost of capital. Rates in the low teens are commonly described for development-stage programmes and a marketed product may be discounted at the company's weighted average cost of capital, but I could not source a survey of actual practice, so treat any specific rate you are shown as that company's own choice rather than an industry norm.
4. **Risk-adjust.** Multiply the cash flows beyond each gate by the cumulative probability of technical and regulatory success from the current phase. This is the step that makes it rNPV rather than NPV.
5. **Sum, and compare against the alternatives** — including "stop now", which must be a real option on the page or the exercise is theatre.

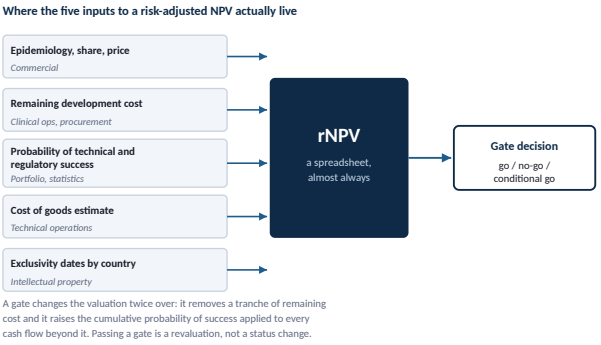


Figure 6.1 The risk-adjusted net present value is assembled from inputs owned by five different functions, and is **almost always maintained in a spreadsheet**. That makes it the most dependency-rich artefact in the company.

What a stage gate does to a valuation

A gate is not a status review. It is an authorisation to spend the next tranche, and passing one changes the number twice over. It removes a block of remaining cost from the model, and it raises the cumulative probability of success applied to every cash flow beyond it. The same forecast, unchanged, is worth materially more the day after a successful Phase II read-out than the day before — not because anybody learned anything about the market, but because the probability multiplier moved.

That has a blunt consequence for reporting. **A portfolio value that moves is not evidence of anything until you know whether the forecast changed, the probability changed, the discount rate changed or the cost profile changed.** Any portfolio dashboard that shows a single rNPV per programme and a period-on-period delta is useless and worse than useless, because somebody will act on it. The minimum honest design carries the four movement components separately. Ask for it at the first workshop; you will be told it is hard, and it is, and it is also the whole value of the exercise.

Attrition: the shape is knowable, the numbers are not

The industry picture is a funnel. Many thousands of screened compounds yield a handful of clinical candidates, of which perhaps one reaches approval. Attrition is severe. It is worst in oncology and central-nervous-system disease. It concentrates in Phase II, where efficacy is first tested against a control. And Phase III failures are the expensive ones, because by then the spend is already sunk.

> **DO NOT STATE A PHASE-LEVEL SUCCESS RATE**
>
> I could not source a citable primary dataset for phase-level attrition or cumulative probability of approval in the research behind this book. The commonly cited series are the BIO, Informa and QLS *Clinical Development Success Rates* reports, the Wong, Siah and Lo analysis from MIT, and Norstella and Citeline's annual series. They disagree with one another by several percentage points, and they differ by a factor of two or more between therapeutic areas. If a number matters to a client deliverable, re-source it to one named study and quote the study, its window and its dataset. **Teach the mechanism and the shape. Do not invent percentages.**

Why one asset and twenty assets are different businesses

A company with a single clinical-stage asset is a bet. Its reporting need is depth: every assumption in the one model, versioned, with an audit trail on who changed the probability of success in the week before the gate meeting. Its risk is binary and its investors know it.

A company with twenty assets is a statistical business. Its reporting need is comparability: the same probability model applied consistently across programmes, the same cost categories, the same treatment of indication expansion, so that the ranking means something. Its risk is that the twenty models were built by eleven people over nine years with different conventions, and the ranking is therefore an artefact of who built which spreadsheet.

The architectural difference is not scale. It is that the twenty-asset company needs a governed input layer and the one-asset company needs a versioned model. Those are different projects. When a client asks for "a portfolio management capability", the first question is which of the two they are, and the second is where the probability-of-success numbers come from and who is allowed to change them.

One more thing about the portfolio that the data model must carry: **a programme that stops still has a full cost history and no product record.** "Terminated" is a first-class lifecycle state, not an exception. The evidence from a terminated programme must stay retrievable for decades, because it may be relevant to a later safety question about a related molecule, or to a regulator asking what the company already knew. Systems built around a product master cannot hold the cost of the large majority of programmes that never became products, which is why so many companies cannot answer "what did we spend on that programme in total" — the spend went to cost centres, studies and purchase orders, and the programme identifier was optional on half of them.

What a drug costs to develop, honestly

This is the most-cited and least-settled number in the sector, and an architect who repeats the headline in a board paper will be corrected in the room.

> **"USD 2.6 BILLION TO DEVELOP A DRUG" IS ONE ESTIMATE, NOT THE FACT**
>
> The figure comes from DiMasi and colleagues at the Tufts Center for the Study of Drug Development, 2016: average out-of-pocket cost per approved drug **USD 1,395 million** in 2013 dollars, **USD 2,558 million** after capitalising at 10.5%, and **USD 2,870 million** including post-approval costs. The sample was self-reported data from ten multinational firms on 106 new drugs, nineteen of them biologics, first tested in humans between 1995 and 2007 [1].
>
> Set against it, Prasad and Mailankody, 2017, in *JAMA Internal Medicine*: median out-of-pocket cost **USD 648.0 million**, range **USD 157.3 million to USD 1,950.8 million**, or **USD 757.4 million** with a 7% opportunity cost applied. The sample was ten cancer drugs from ten companies that had no other United States-marketed product, approved 2006 to 2015, reconstructed from filings with the Securities and Exchange Commission rather than from company-supplied figures [1].
>
> Across credible published estimates the range runs from **well under USD 100 million to about USD 2.9 billion** — a factor of more than fifty. The frequently quoted floor of USD 60 million is itself too high: Public Citizen put the after-tax figure at USD 57 to 71 million, and Light and Warburton's adjusted lowest estimate is USD 43.4 million [1]. **Always name at least those two studies and say what drives the gap.** A chapter, or a slide, that cites one figure as "the" cost is wrong whichever figure it picks.

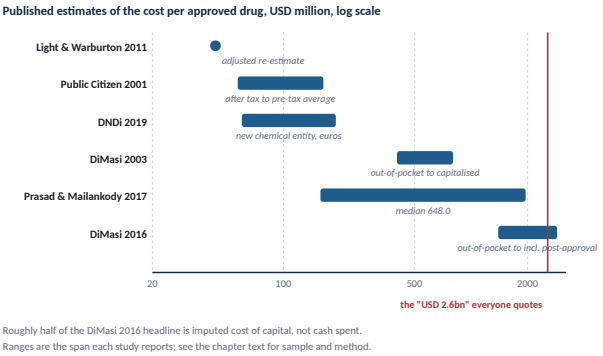


Figure 6.2 Published estimates of the cost per approved drug, on a log scale. The spread is nearly fiftyfold and is explained by four methodological choices, not by disagreement about arithmetic. **Name at least DiMasi 2016 and Prasad & Mailankody 2017** [1].

Four methodological choices explain almost the whole spread [1].

Whether failures are counted, and at what attribution rate. The cost of an approved drug is, by construction, the cost of that drug plus the cost of everything that failed on the way. Every credible estimate includes failures; they differ on how much failure to attribute to each success.

Whether capital is charged for the development period, and at what rate. This is where most of the disagreement sits. DiMasi 2016 capitalises at 10.5%; the earlier 2003 paper in the same series used 11% on sixty-eight drugs from ten firms entering clinical development 1983 to 1994, giving USD 403 million out-of-pocket and USD 802 million capitalised in 2000 dollars [1]. **Roughly half the headline figure is not cash spent. It is the imputed opportunity cost of capital over a decade.** Light and Warburton criticised the earlier study for inflated trial costs, exaggerated phase durations and a high capitalisation rate, and produced an adjusted lowest estimate of USD 43.4 million — eighteen times lower [1].

Whether the sample is self-reported industry data or public filings. DiMasi's data came from firms voluntarily; Prasad and Mailankody's came from mandatory disclosures. Each has a defensible claim and an obvious weakness, and the authors' critics made both.

What kind of drug. A novel chemical entity from a large portfolio is not the same object as a reformulation from a non-profit. DNDi, working from its own development experience since 2003 across eight drugs, gives EUR 4 to 32 million for a new combination or formulation of existing drugs and EUR 60 to 190 million for a new chemical entity, excluding post-registration studies, not capitalised, and excluding in-kind partner contributions [1]. Public Citizen in 2001, an advocacy estimate and worth labelling as one, gave USD 57 to 71 million after tax and an average of USD 163 million pre-tax [1].

Why an architect cares at all

Because **the company's internal number is none of the published ones.** The figure used in gate decisions is a company-specific model built from its own historical programme costs, its own attrition rates and its own cost of capital, maintained by portfolio or research finance, and fed into the rNPV that decides which programmes live. Finding that model, understanding it, and not breaking it is a real obligation on any research-systems engagement. It will be a spreadsheet. The useful deliverable is not to replace it but to govern its inputs: versioned assumptions, traceable provenance, and an audit trail on the probability figures.

Gross to net, and why it mostly does not apply here

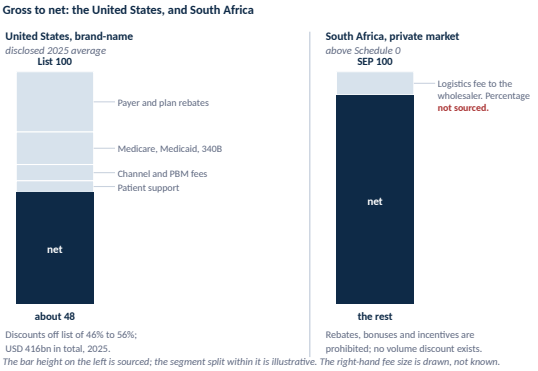


Figure 6.3 Gross to net in the United States against the South African private market. The United States bar height is sourced [2]; its internal split is illustrative. On the right, **the logistics fee is the only genuine gross-to-net item, and its size is not something I could source** [3].

In the United States, the price a manufacturer invoices and the price it realises differ by roughly half on brand-name products. The Drug Channels Institute's estimate published 7 October 2026 for calendar 2025: total gross-to-net reductions for brand-name drugs of **USD 416 billion**; among the manufacturers that disclose it, an unweighted average discount off list of **−51.7%**, with weighted averages between **−46% and −56%**; and one large manufacturer's disclosed discount widening from −49% in 2024 to −53% in 2025 [2].

What sits in that bucket is a list of separate contractual calculations: rebates, discounts and fees to commercial payers and plans; Medicare and Medicaid rebates; discounts under the 340B programme; payments to channel participants including pharmacy benefit manager administrative fees and wholesaler fees; and patient assistance and copayment support [2]. Each has its own claim stream arriving months later as data files that must be validated against contract terms. The revenue is booked net of an *estimate* and trued up as claims arrive. The architecture to support it — contract management, eligibility determination, claim validation, chargeback processing, accrual and true-up — is one of the largest system estates an American pharmaceutical company has.

Then the South African reality, which is total divergence

In the South African private market there is no gross-to-net gap by design. The Single Exit Price is the only price at which a manufacturer may sell to anyone other than the State. It does not vary by buyer volume. A large private hospital group and a single independent pharmacy pay the same number [3]. And bonus, rebate and incentive schemes are **prohibited** by the Act for medicines other than Schedule 0 [3].

So the machinery that produces American gross to net does not exist here, and most of the transactions that produce it are illegal above Schedule 0. That is the fact. But do not overstate it, for two reasons.

First, the prohibition is real and its definition is absent. Regulations defining acceptable and prohibited acts under that prohibition were published for comment in 2014 and again in 2017, and were never finalised. The published position is therefore that the extent of co-marketing fees, shelf fees, off-invoice rebates and similar arrangements **is unknown** [3]. A compliance architecture cannot be designed against a definition that does not exist, and an architect who tells a client the matter is settled is wrong in a way that will eventually matter.

Second, there is exactly one genuine South African gross-to-net item. Wholesalers take no mark-up on the Single Exit Price. Instead the manufacturer pays the wholesaler a **logistics fee out of the exit price**. The amount varies considerably; it appears as a column in the national pricing database; and where a seller owns its own wholesaler it earns that income undisclosed to buyers [3]. **The typical percentage is not something I could source, and an architect should not state one.** What follows structurally is still firm: for a South African manufacturer, SEP multiplied by volume is *gross* revenue, not net revenue, and the logistics-fee deduction has to be modelled per wholesaler with an effective date.

> **THE CONSEQUENCE IS BLUNT**
>
> Importing a United States pharmaceutical revenue-management reference architecture into a South African generics manufacturer builds an estate for transactions that cannot occur. Contract management, rebate accrual engines, chargeback processing, eligibility determination — none of it has a South African private-market counterpart above Schedule 0. **Build the gross-to-net capability anyway if the company exports or consolidates into a multinational group, because the group will demand it. Do not build it for the local private book.**

One further South African item that is often mistaken for a gross-to-net deduction and is not: the **dispensing fee**. It is charged by the final seller **in addition to** the Single Exit Price, not out of it, capped by gazette, calculated on the price excluding value-added tax, with different bands for pharmacists and for licensed dispensing practitioners [3]. It never touches the manufacturer's realised price. If you see it netted off manufacturer revenue in a model, the model is wrong.

Cost of goods, and where it actually sits

The single most useful sentence about pharmaceutical cost structure is that the same two words — "pharmaceutical company" — describe businesses whose cost structures are mirror images. The archetype question has to be answered before any costing conversation.

For a small-molecule originator, cost of goods is frequently trivial against the cost of discovery. A tablet whose active ingredient costs cents per dose can carry a price measured in rands or dollars per dose, not because the factory is expensive but because the price has to recover a decade of failed programmes and the one that worked.

For biologics it is materially higher. Cell-culture manufacture, cold chain, single-use consumables and very high quality-control cost per batch all land in cost of goods.

For a generics manufacturer, cost of goods is the business. There is no discovery cost to dwarf it. Active ingredient is usually the largest single input and is usually imported. Conversion cost, packaging, quality control and regulatory maintenance follow. Fixed cost in plant is high and utilisation determines profitability.

For a contract development and manufacturing organisation there is no cost of goods in the usual sense at all, because the customer owns the material. What the organisation sells is capacity and competence, and what it buys is people, suites and quality overhead.

> **PERCENTAGES AND P&L SHAPES**
>
> Cost of goods as a percentage of net revenue, and the illustrative profit-and-loss shapes that circulate for an originator and for a generics manufacturer, are not sourced in the research behind this book and are registered as unverified. **Give the structure. Do not print the numbers.** A reader who needs them should build them from audited segment reporting of three or more comparable companies in the same period, and say so.

What *is* sourced, and is worth more than any generic percentage, comes from Aspen Pharmacare's interim results for the six months to 31 December 2025 [4]. Commercial Pharmaceuticals revenue of R16,586 million, up 3%, at a gross profit margin of **58.5%**. And the Manufacturing segment: revenue **R4,499 million, down 23%**, with normalised earnings before interest, tax, depreciation and amortisation of **R208 million, down 84%** — a margin of roughly 4.6% [4].

Sit with that pair for a moment, because it is the most instructive sourced number in South African pharmaceutical economics. **A 23% revenue decline producing an 84% earnings decline is the arithmetic of a very high fixed-cost base in a sterile facility.** Sterile manufacturing is utilisation economics, not margin economics. An architect designing capacity planning or product costing for a sterile site must understand that the dominant variable is how full the facility is, and that a costing model which treats cost per unit as a property of the product rather than of the volume will mislead every decision made from it.

Line	Small-molecule originator	Generics manufacturer	CDMO
Dominant discretionary spend	research and development	formulation and bioequivalence	business development and qualification
Cost of goods	modest against price	the whole game	not applicable — client owns material
Largest input cost	people and trials	imported active ingredient	capacity and people
Price set by	the market, under exclusivity	gazette or tender	contract, per campaign
Margin lever	portfolio replacement	procurement, yield, mix, utilisation	utilisation and changeover time
What destroys margin	a failed Phase III	a lost tender	an unbilled change in scope

Campaign costing, and product costing

A contract development and manufacturing organisation earns **fee for service, not margin on product**. It does not own the registration, does not own the product and usually does not own the inventory. The most common conceptual error an architect imports from manufacturing is to assume otherwise.

Revenue is earned against performance obligations in a contract: development milestones, often recognised at a point in time on client acceptance; manufacturing batches, on release or over time as the batch progresses; capacity reservations, over time whether used or not; and pass-through materials, often at cost plus a handling percentage and sometimes recognised gross and sometimes net — a real accounting judgement with system consequences.

The cost object is the campaign or the client programme, not the product. The product belongs to the client. Any costing architecture that assumes a standard cost per finished item will not fit; the right analogue is job costing in engineer-to-order manufacturing, not standard costing in repetitive manufacturing.

What drives the margin is specific and almost entirely architectural in its implications: utilisation of the asset; changeover and cleaning time between campaigns, every hour of which consumes validated capacity and is non-billable; the first-time-right batch rate, because a failed batch is usually at the organisation's cost; and scope control on development work, where the client's experiments expand without a change order.

	Generics plant, product costing	CDMO, campaign costing
Cost object	the product, per unit	the campaign, per client programme
Standard exists for	material, labour, absorbed overhead per unit	nothing standard; each campaign is priced
Revenue trigger	sale of released stock	milestone, release, reservation period, pass-through
Inventory on balance sheet	yours	mostly the client's
Variance analysis	purchase price, usage, labour, absorption	billable against non-billable hours and suite time
Capacity question	did we absorb budgeted volume	was the suite reserved, used and billed
Change control cost	yours to absorb	often the client's, under the quality agreement
The margin leak	yield and absorption	unapproved change orders and cleaning time

That last row deserves one more sentence, because it is where the architecture earns its money. An unapproved change order is work done and not billed. At a small contract organisation the aged list of unapproved change orders is a standing board item, and it exists because the request arrives in an email to a project manager rather than as a workflow object with a price and an approval. The remedy is a change-order workflow with a commercial state, not a better spreadsheet.

A related correction worth making once: **a contract research organisation is not a kind of CDMO.** They share nothing architecturally. Good Clinical Practice against Good Manufacturing Practice, labour against capacity, the study against the batch, sponsor-system multi-tenancy against physical segregation. A research organisation is a professional-services business whose metrics are utilisation, bill rate, realisation and pass-through leakage, and whose killers are non-billable rework and unpriced change orders when a protocol is amended.

Money that surprises outsiders: milestones and royalties

Pharmaceutical assets are traded constantly, and the deal structures are an income and obligation stream an architect has to be able to represent. An out-licensing deal typically carries an upfront payment at signature; **development milestones** triggered by defined technical events — first patient dosed in Phase II, Phase III initiation, submission acceptance; **regulatory milestones** on approval in a named territory; **commercial milestones** on first achievement of annual net sales thresholds; **royalties** as a percentage of net sales, usually tiered by annual sales band; and then the quieter terms — sublicence income share, reversion rights on discontinuation, minimum royalty and diligence obligations.

Two of those create architecture work immediately.

A milestone means the finance system depends on the clinical and regulatory systems for a revenue trigger. Recognising a development milestone requires knowing, auditably, that a defined event occurred on a defined date. The event lives in the clinical trial management system or the regulatory information system. If the hand-off is a project manager's email, revenue recognition rests on an email, and the auditor will say so.

And royalties are calculated on net sales as defined in the contract, where each contract defines net sales differently: which deductions are permitted, whether distributor sales count at the distributor's price or the transfer price, how combination products are apportioned, how currency is converted and at which rate. A company with fifty licence agreements has fifty different definitions applied to overlapping product and country sets. **There is no net-sales figure in the enterprise system that satisfies all of them.** The royalty calculation is an interpretive layer over sales data, usually maintained by two people with a spreadsheet, routinely audited by licensors, and a recurring source of disputes and restatements. When a client asks for a royalty system, the real requirement is a contract-term repository with a calculation engine and an audit trail, and the hard part is modelling the contract terms rather than the arithmetic.

In-licensing is the dominant South African pattern: a local company licenses a product for South Africa or sub-Saharan Africa, pays a transfer price or a royalty, and may or may not hold the local registration. **Whether the local entity or the licensor holds the South African registration is the single most consequential fact about such a deal for architecture**, because the registration holder carries the regulatory and pharmacovigilance obligations and therefore needs the systems. Ask it in the first hour; the answer is frequently different from what the commercial team believes.

Two numbers that point in opposite directions

One last piece of arithmetic, because it reverses the intuition an architect brings from most industries. **More than 80% of South African pharmaceutical manufacturing and distribution revenue comes from the private sector**, per Who Owns Whom's 2023 edition [13] — while the overwhelming majority of *volume* is public-sector, driven by a treatment programme with roughly **6.18 million people on antiretrovirals in 2025**, up from 6.14 million in 2024 [14].

Revenue share and volume share therefore point in opposite directions, and this is the most common single error in South African pharmaceutical commentary. It matters architecturally because the two chains are measured on different bases — tender price times volume against Single Exit Price times volume — and adding them produces a number that means nothing. A "South African medicines market" figure is almost always one chain or the other. And on market size generally: **there is no current authoritative public figure.** Who Owns Whom gives approximately R60.4 billion for 2021 and R56.4 billion for 2019, and its 2025 edition gives no market value at all [13]. Use the dated figure with its date attached, say that it predates two pricing cycles and a currency move, and never put a market-research vendor's forecast in a client deliverable.

The patent cliff and the erosion curve

Exclusivity ends. That is the defining structural feature of originator economics and the reason the whole commercial architecture of an originator is organised around replacing revenue that is scheduled to disappear.

The cliff is deterministic in a way almost nothing else in this industry is. The date is known years in advance. The loss is therefore forecastable, which means it is in the plan, which means management is held to replacing it. Pharmaceutical Executive, writing on 2 June 2026, puts roughly **200 drugs** losing patent or exclusivity protection between 2026 and 2030, with an estimated **USD 400 billion** of revenue at stake — described as the largest patent cliff the industry has faced [5]. Moody's has separately put **USD 300 billion** on the cliff in the context of hospital drug spending [6]. **Name both figures and both sources and do not reconcile them**, because they measure different sets of products over different windows. An architect who averages them has invented a third number.

One caution before the curve: "the patent" is not one thing, and treating the composition-of-matter patent expiry as the loss-of-exclusivity date is a common and expensive error. Exclusivity is a stack — the composition-of-matter patent on the molecule, secondary patents on formulation, polymorph, salt form, process, combination, regimen and device, regulatory data and market exclusivity, orphan designation, and patent term extension or supplementary protection. The periods differ by jurisdiction, and the secondary patents are routinely litigated, so the date **moves**. The system of record for exclusivity dates is usually the intellectual property function's own database, and it is frequently a spreadsheet. Asking where exclusivity dates actually live is one of the more revealing first questions an architect can put to an originator.

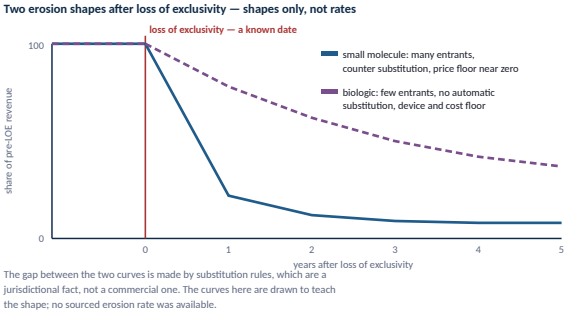


Figure 6.4 Two erosion shapes after loss of exclusivity. **The shapes are drawn to teach the difference; no sourced erosion rate was available.** The gap between them is made by substitution rules, which are jurisdictional facts rather than commercial ones.

Why the decline is multiplicative, not additive

At loss of exclusivity for a small molecule, multiple generics enter, often on day one. Price falls steeply. The originator loses most volume within the first year or two, stops promoting the product and either divests it or runs it as a cash harvest.

The arithmetic matters: **the cliff is a volume event and a price event at the same time, so the revenue decline is multiplicative.** A 60% volume loss combined with a 70% price reduction on what remains leaves roughly 12% of prior revenue. That calculation is mine and illustrative, but the form of it is not optional — a planning system that models volume erosion and price erosion as one combined percentage cannot be reconciled to either, and cannot answer the only question management asks, which is whether to defend on price or let the volume go.

Biosimilar erosion is slower, and the reason is regulatory

Biosimilars are not generic biologics. The standard is **similarity**, not identity, because a biologic cannot be exactly copied; the dossier requires an extensive analytical comparability exercise plus comparative non-clinical and clinical work. Development cost is orders of magnitude higher than a generic's.

Erosion after a biologic's exclusivity ends is slower and less complete than for a small molecule. I could not source a quantified comparison, and nor should you state one without sourcing it. The reasons, though, are structural and can be stated without a figure:

- **Fewer entrants.** Development cost and manufacturing complexity mean a handful of competitors per molecule, not twenty.
- **No automatic substitution in most jurisdictions.** A pharmacist can usually substitute a small-molecule generic at the counter. Substituting a biosimilar typically needs a prescriber decision or an explicit interchangeability determination, and the rules differ by jurisdiction. **Substitution mechanics are the single biggest driver of erosion speed, and they are regulatory, not commercial** — which means the manufacturer does not control them and cannot forecast them from its own behaviour.
- **Clinician caution.** Switching a stable patient on a biologic carries a perceived clinical risk that switching a tablet does not.
- **Device and patient-support lock-in.** Biologics are often injected with a device, supported by a patient programme, reimbursed through a specialty channel. Each is a switching cost.
- **More effective originator defence.** Price concessions, device upgrades, next-generation formulations and contracting strategies all slow the decline.
- **A cost floor.** A biosimilar cannot be priced near zero because it genuinely costs a great deal to make.

The architectural consequence is that an originator's post-exclusivity forecast needs two distinct erosion models with different curve shapes and per-market substitution parameters. A single erosion assumption across a mixed portfolio is wrong in both directions at once.

Tender economics, and why an award is not revenue

For a South African generics manufacturer the public sector is a different business, not a different customer. The buyer is effectively a single purchaser, price is the decision variable, volume is large and committed, and the contract usually runs three years. Only registered medicines are eligible, and high-volume lines such as first-line antiretrovirals may be split among several suppliers for supply security [3].

What a tender award does to a manufacturing plan is the architecturally important part.

It converts a forecast into an obligation. Before the award, volume is a projection; after it, it is a contract with penalties. The plant plans validated capacity, active-ingredient procurement with lead times measured in months, packaging components and labour around the award.

It is lumpy and binary. Winning or losing a line item moves utilisation by a step change, not a percentage. In the national antiretroviral tender — R15.5 billion in total, of which R12.6 billion covers fixed-dose combinations in one-month and three-month packages — one established local manufacturer won **no** one-month or three-month pack award at all, receiving only three smaller awards worth R215 million [7]. For a plant sized around that pool, that is not a bad quarter.

It drives SKU design. Tender packs are different products: patient-ready packs, three-month supplies, specific labelling, delivery to depots rather than pharmacies. They need their own pack variations and their own artwork.

Payment is slow. One provincial health department owed suppliers more than R1 billion as at September 2026, and late payment by the public sector is reported as affecting supplier availability [8].

And three years is too short for the capital it asks for. The industry's own position, put to Parliament, is a request for contracts of up to seven years instead of three, on the straightforward ground that a manufacturer will not validate a line and build capacity for a three-year award it may lose [9].

> **A TENDER AWARD IS NOT SECURE REVENUE, AND IT IS NOT MASTER DATA**
>
> On 22 October 2026 the Gauteng Division of the High Court, Pretoria **set aside** the one-month and three-month pack awards of that R15.5 billion antiretroviral tender as unlawful and unconstitutional, ordering the Department to redo the split and award within 120 days, with the set-aside taking effect 90 days after the new awards are finalised and current suppliers required to keep delivering until then [7]. The irregularities found read as a checklist of what a South African health tender actually tests: two bidders did not hold a valid licence for their manufacturing site at bid closure, so should not have passed the first stage; past performance was not properly weighed; historically disadvantaged individual claims were accepted without investigation, raising scores; and a reasonable suspicion of collusive bidding was not investigated before substantial volumes were awarded [7].
>
> Separately, on 17 September 2026, the Constitutional Court struck down the **entire Public Procurement Act 28 of 2024** for inadequate public participation, **without suspending the declaration of invalidity** [10]. Which framework now governs health-product tenders in practice is something I could not establish and must be treated as live and unresolved.
>
> **Model a tender award as a contract with a validity period, a challenge window, a litigation state and a possible successor award.** The production plan, the active-ingredient purchase commitments and the cash forecast all inherit that state. Model it as master data and the plant will be holding active ingredient against an award that no longer exists.

There is a second-order lesson in that judgment for anyone building a bid-management capability. The evidence a bidder must be able to produce, on demand and auditably, is a short list: a valid site licence covering every product and activity bid, at bid-closure date; ownership and historically-disadvantaged verification; local-content declarations in the form the tender specifies; past-performance evidence; and price build-ups that survive a negotiation round. **Each of those is a data product, and most companies assemble each one by hand for each tender.** After October 2026 that capability is also commercially defensible, because an inadequately evidenced competitor's award can be set aside.

Working capital and the long cycle

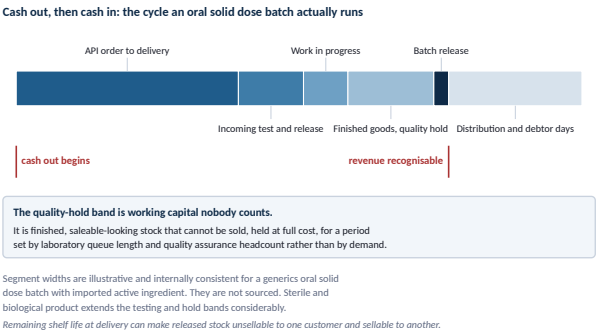


Figure 6.5 The cash cycle for one batch, with the quality hold shown at its real weight. **Segment widths are illustrative, not sourced.** Batch release is the gate between cost incurred and revenue recognisable.

The cash cycle in a pharmaceutical plant is longer than in general manufacturing, and the extra length is almost all regulatory rather than physical.

Inventory sits in more states than a general manufacturer's three. Active ingredient, received and awaiting incoming testing. Active ingredient, released. Intermediate. Bulk — compressed tablets or filled vials, made but not packed, and often held deliberately because bulk has a longer life than finished goods and packs can be decided later. Finished goods, packed. And then the state that matters most for the architecture: **finished goods on quality hold.**

That hold period is working capital nobody counts. It is finished, saleable-looking stock, carried at full cost, that cannot be sold. Its duration is set by the laboratory testing queue, the microbiological cycle time where sterility testing applies, the batch-record review queue and the headcount of quality assurance — not by demand, not by the market, and not by anything a planner controls. For a cold-chain or short-shelf-life product, those days are the business case for every integration improvement in the estate.

Batch release is the gate between cost incurred and revenue recognisable. Nothing crosses it early. And the quantity on hand is therefore never a single number: it is a number per batch per status per location, and the saleable subset depends on a quality decision and on remaining shelf life. A customer or a tender may contractually require a minimum remaining shelf life at delivery, which makes perfectly good stock unsellable to that customer and sellable to another. **"Stock on hand" without batch, status, expiry and the shelf-life rule is not an answer to any question worth asking.**

Two further working-capital facts that belong in any South African margin model.

Expiry-based provisioning is a real and material accounting judgement. Stock approaching its remaining-shelf-life threshold is provisioned, the thresholds are driven by the customer contracts rather than by the product, and the provision moves with the mix of customers rather than with the quality of the product.

And the currency exposure is asymmetric in time. Active ingredients, excipients, primary packaging, filters, single-use consumables and capital equipment are priced in dollars or euros. Single Exit Price revenue is in rand and moves only when the gazette says so. In 2026 it moved twice — the ordinary annual adjustment of **1.47%** in January and an **extraordinary second adjustment of 2.88% effective 1 October 2026**, which the Department of Health described as an extraordinary intervention and which explicitly does not apply to medicines bought by the State [11][12]. Any statement about 2026 pricing that mentions only one adjustment is wrong. Costs reprice continuously; the selling price reprices annually, by permission. **That one asymmetry is the best single-sentence explanation of why South African pharmaceutical manufacturing capacity is contracting, and an architect should be able to state it.** The practical requirement is that currency is a first-class dimension in the cost model, with the as-at date of the rate visible on every margin report.

One trap worth its own callout

> **HIGHER YIELD IS NOT GOOD**
>
> In general manufacturing, improving yield is good, and a dashboard that rewards yield improvement is a reasonable thing to build. Under Good Manufacturing Practice it is not. **A batch yield outside the validated range registered with the regulator is a deviation requiring investigation — even when it is higher.**
>
> A plant cannot get better at making a registered product. A sustained change in process performance is a signal that the process has drifted from its validated state, and the obligation is to investigate it, not to celebrate it. The validated range is part of what was registered, and changing it is a variation.
>
> So: **any dashboard that rewards yield improvement in a GMP plant is rewarding a deviation.** The correct target is yield *within range*, and the correct trend to watch is dispersion rather than level. This is among the most common errors an architect carries in from general manufacturing, and it is visible in the first sprint of any reporting project, so catch it there.

What the architect takes from this chapter

Six things, and each one is a design consequence rather than an observation.

Price is not an attribute of a product. It is an attribute of product, pack, channel and effective date. The private-sector price lives in the Single Exit Price record, the public-sector price lives in a tender contract, and section 21 supply of unregistered product is not subject to the Single Exit Price at all [3]. A single price column is a defect.

A Single Exit Price adjustment is a scheduled mass data-change event. It happens on a gazetted date, by a gazetted maximum, across the whole portfolio at once, and it flows to every customer, wholesaler, scheme formulary and point-of-sale system in the country simultaneously. Treat it the way you would treat a tax-rate change: a high-blast-radius, date-effective master-data event with a rehearsal.

The cost object follows the archetype. Product for a generics plant, campaign for a contract organisation, programme for anything in development. Get this wrong and the costing architecture cannot be repaired by configuration.

The portfolio data model must hold things that never became products. Terminated programmes with full cost and evidence, retained for decades.

Erosion and exclusivity are per-country, contested, and usually outside the systems of record. Find them before you promise a forecast.

And revenue recognition depends on the quality decision. Finance waits on a signature from a named person who is not in finance. That dependency, more than any other in this chapter, is the one that surprises an architect arriving from another industry — and it is the subject of chapter 8.

THE CHAPTER IN TEN LINES

- The unit of value is the portfolio of options, not the product, and most of those options expire worth nothing.
- Passing a stage gate revalues a programme twice over: it removes remaining cost and it raises the probability multiplier, so a portfolio dashboard must show the movement components separately.
- Phase-level attrition and probability of approval differ by several percentage points between published studies and by a factor of two between therapy areas, so teach the shape and never invent a percentage.
- USD 2,6bn is one estimate; credible published figures span roughly USD 60m to USD 2,9bn, about half the headline is imputed cost of capital, and the company's own internal number is none of them.
- United States gross to net reaches roughly half of list price, and the transactions that produce it are largely illegal in South Africa above Schedule 0.
- The only genuine South African gross-to-net item in the private market is the logistics fee the manufacturer pays out of the exit price, and its typical percentage could not be sourced.
- Cost of goods is trivial for a small-molecule originator, the whole game for a generics manufacturer, and absent for a contract manufacturer whose client owns the material.
- A contract organisation costs the campaign, not the batch or the product, and earns fee for service rather than margin on goods.
- The patent cliff is deterministic, the decline is multiplicative across price and volume, and biosimilar erosion is slower because substitution rules are jurisdictional rather than commercial.
- A tender award is a contract with a validity period and a litigation state, not master data, as the October 2026 setting aside of the antiretroviral pack awards demonstrates.
- The quality-hold period is working capital nobody counts, and batch release is the gate between cost incurred and revenue recognisable.
- A yield outside the validated range is a deviation even when it is higher, so any dashboard rewarding yield improvement in a GMP plant is rewarding a deviation.
- A milestone makes revenue recognition depend on the clinical and regulatory systems, and fifty licence agreements carry fifty different definitions of net sales.
- More than 80% of South African pharmaceutical revenue is private-sector while the overwhelming majority of volume is public-sector, so revenue share and volume share point in opposite directions.

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CHAPTER 7

The operating model and the portfolio cadence

What the company is made of, and the rhythm it runs on. A registered site is a licence with a scope and a named pharmacist rather than a building; and the eleven cadences that govern a pharmaceutical company answer to the regulatory calendar, not the financial one.

Chapter 6 was the arithmetic. This chapter is the machine that produces it: what the company is made of, and the rhythm it runs on.

Two framings before the detail. **A site is a regulated object, not a building.** And **the regulatory calendar runs independently of the financial calendar, and it wins.** Nearly every mistake an architect makes in the first month of a pharmaceutical engagement is a version of forgetting one of those two.

A registered manufacturing site is a regulated object

Walk into a plant and you see a building with machines in it. That is not what it is, for modelling purposes. A registered manufacturing site is a licence, a scope, an inspection history, a named individual and a set of registrations that point at it — and the building is the least interesting of those.

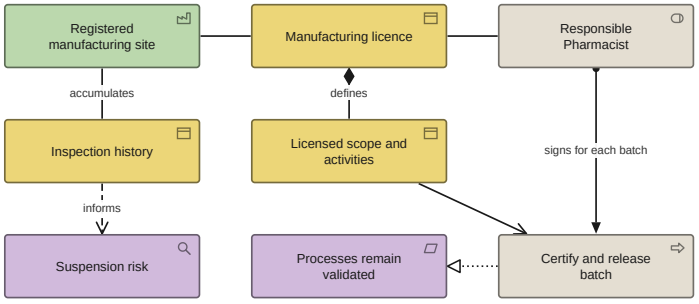


Figure 7.1 A registered manufacturing site, modelled as what it is. The licence, its scope, the inspection history and the named Responsible Pharmacist are the regulated objects; **the building is an attribute**. Keep ArchiMate figures to three grid columns so the labels stay legible.

In South Africa the pieces are specific and sourced. Manufacturing licences are issued under section 22C of the Medicines and Related Substances Act 101 of 1965, read with regulations 23 and 24 [1]. A **manufacturer licence** covers manufacture and packing. A separate licence covers **packaging only**. And **over-labelling, including of parallel imports, is a packaging activity and is therefore licensable** [1] — which catches out every company that assumed sticking a South African label on an imported finished pack was a warehouse operation.

A licence is **normally issued only following a GMP inspection**, inspectors carry out regular and repeated inspection, and serious non-compliance can lead to refusal, suspension or revocation [1]. It is renewed **every five years, applied for at least 180 days before expiry** [1]. Fees apply to new and amended applications, renewal, issue, inspections and **annual retention** [1]. The licence holder must inform the regulator's chief executive of changes to the particulars furnished in the application **within 30 days**, and must notify changes to premises, operations, listed personnel or products [1].

And one individual is named on it. **The Responsible Pharmacist must be named on the licence; not more than one may be named; and the Responsible Pharmacist must sign or certify, for each batch, that the appropriate tests have been carried out** [1].

THE RESPONSIBLE PHARMACIST IS NOT A RENAMED QUALIFIED PERSON

In the European Union a Qualified Person certifies each batch and may be a pharmacist, chemist or biologist. In South Africa the role is the **Responsible Pharmacist**, who **must be a pharmacist registered with the South African Pharmacy Council**, must reside in the Republic, and is **dual-accountable** — to the regulator under the Medicines Act and to the Pharmacy Council under the Pharmacy Act 53 of 1974 for the pharmacy under their personal supervision [2][1]. The GMP guideline states that any reference to the "Responsible Person" is equivalent to the Responsible Pharmacist, requires continuous supervision of the pharmacy, and requires that every batch released for sale, supply or export meets the marketing authorisation and GMP requirements, with release signed off **using a checklist** [3]. The duty extends to ensuring that manufacturing and testing processes **remain in a validated state** [3].

Architecturally: a single named individual's electronic signature on batch release is a legally significant act with two disciplinary authorities behind it. The identity, authority, delegation and continuity arrangements around that signature are an access-control design problem with legal consequences, not an administrative one. And **validation state is explicitly that person's duty, not the IT department's**, which changes who has to be in the room when you propose a system change.

GUIDELINE VERSIONS MOVE FASTER THAN THEIR STATED VALIDITY

The GMP guideline cited above, SAHPGL-INSP-02, was found in the research at **v9, published 17 November 2025, stated valid for five years** [3] — and then superseded by **v10 on 13 April 2026**, inside its own validity window, with the two retrieval passes disagreeing about which text is operative. **Every guideline citation in this book needs a version-and-date re-check before it reaches a client deliverable, and re-checking it is your standing job, not a one-off.** Nothing in this chapter is legal advice; every regulatory passage here frames the obligation as what the record has to show, never whether a particular company complies.

What each site is licensed to do

The licence has a **scope**, and the scope is what makes the estate shape itself. A company does not have "two factories". It has a set of licensed activities at named addresses, and moving a product between them is a regulatory project.

Site type	What it does	What its systems are built around
Research campus	discovery, formulation development, analytical methods	notebooks, compound registry, assay data
Active ingredient plant	chemical synthesis of the active substance	process control, effluent, master files
Oral solid dose site	tablets and capsules: granulate, compress, coat, pack	changeover and campaign scheduling
Sterile fill and finish	aseptic filling of injectables and biologicals	environmental monitoring, utilisation
Quality control laboratory	testing to registered specification, stability	sample and result management, chromatography
Distribution centre	storage and transport under a wholesale licence	batch, status, expiry and temperature

Why a sterile site is a different business from an oral solid dose site

Not a harder version of the same business. A different one.

An oral solid dose site is a **changeover business**. Its binding constraint is the time it takes to clean down and change over between campaigns, because cleaning validation and cross-contamination control are mandatory and slow, and that time is non-productive. Capacity planning is a scheduling problem, and the lever is sequencing.

A sterile site is a **utilisation business**. Its cost base is overwhelmingly fixed: classified areas, air handling, environmental monitoring, sterility testing with long cycle times, and the personnel qualification that aseptic work demands. The lever is how full it is.

Chapter 6 gave the sourced illustration: Aspen's Manufacturing segment revenue fell 23% in the half to 31 December 2025 while its normalised earnings fell 84% [4]. **That is sterile arithmetic.** It also helps explain why that company was reshaping sterile finished-dose facilities, with R695 million of restructuring cost booked in the same half [4].

The architectural difference falls out: cost per unit at a sterile site is a function of volume before it is a function of anything the product does, and a costing model that treats it the other way round will mislead every decision made from it. Sterility testing cycle time, not demand, sets the release lead time. And environmental monitoring data is location-and-time-series shaped, not sample shaped, which is why forcing it into a laboratory system's sample model makes trending nearly impossible [5].

A South African note worth having ready. A mid-size generics group frequently has **no sterile capacity at all** and imports injectables finished, over-labelling them under a packaging licence — which, again, is a licensable activity [1]. If the client tells you they "repack" imports, establish immediately whether that activity is on a licence, because the first-stage gate in a state tender is whether a valid site licence covered every product and activity bid at bid-closure date [6].

Contract manufacturing, in both directions, and the artefact that governs it

Most companies both give work out and take work in, and the two directions create different problems.

Giving work out. A product made for you at somebody else's site means their site is named on your registration. Their deviations are your deviations. Their inspection outcome is your supply risk. And their batch records live in their systems, so your product quality review cannot trend their in-process data unless somebody contracted for it [5].

Taking work in. A contract development and manufacturing organisation holds **no marketing authorisation of its own**. It is a named manufacturing site on its clients' registrations. That is the defining asymmetry: **its regulatory exposure is distributed across every client's registration, and one GMP finding is a problem for every client simultaneously.** A small site with work for thirty-eight clients' registrations is thirty-eight simultaneous problems on a bad inspection day, and the clients know it, which is why they audit it repeatedly.

The artefact that governs the relationship in both directions is the quality agreement. It allocates responsibility for release, deviation handling, change-control approval rights and regulatory notification. It is a commercial document with architectural force, because it specifies who may approve a change to a process on somebody else's premises.

> **THE QUALITY AGREEMENT IS A WORKFLOW SPECIFICATION, AND IT IS USUALLY ONLY A DOCUMENT**
>
> Thirty-eight quality agreements with different change-control approval rights must be enforced in workflow, not in a procedure manual. The commonest failure is a quality system configured with one change-control route, and a procedure document that says "obtain client approval where the quality agreement requires it" — which means the enforcement is a person remembering which client is which. Where client-specific approval rights are modelled as data and enforced by the workflow, this stops being a risk. It is one of the few genuinely cheap, genuinely high-value interventions available in a multi-tenant GMP site.

Also keep the archetype distinction clean. **A contract research organisation is not a kind of contract manufacturer.** Good Clinical Practice against Good Manufacturing Practice, labour against capacity, the study against the batch. A company doing both — clinical trial supply packaging alongside commercial manufacture — is running two regimes with one quality system, and that is a real architectural problem rather than a tidy efficiency.

Where the people are, and the one-deep problem

The organisation chart matters here more than it does in most industries, because several of the roles are statutory and several of them are one person deep.

The Responsible Pharmacist is one named individual per licence [1]. A locally accountable pharmacovigilance officer and a back-up must both be **resident in South Africa and trained** [16] — a statutory residency requirement that cannot be met by a remote hire. Market access and pricing, which owns the Single Exit Price declaration, its annual adjustment, scheme formulary positions and tender pricing strategy, is frequently one or two people in a company of a thousand. And the person who knows which systems are actually in a validated state is usually a single validation lead.

Each of those is a single point of failure with a regulatory consequence, and none of them appears on an application inventory. When you draw the operating model, draw the statutory roles as roles with named incumbents and a declared deputy, and then ask what happens on the day the incumbent resigns. For the Responsible Pharmacist the answer is that the site's release authority is in question until a new name is on the licence, which is a supply risk rather than an HR matter.

There is a related design rule that follows from scarcity rather than from regulation. The scarcest skills in a South African pharmaceutical estate are computerised-system validation specialists who understand both validation and software delivery, and the statutory roles above. **A design whose maintenance requires scarce skills will not be maintained.** Concretely: a bespoke validated application with no local support community is a worse choice here than a configured commercial product with a local partner, even where the bespoke option is technically better — and an estate that needs continuous effort to stay validated will drift, because the people who would do that work are the hardest people in the country to hire.

Centralised quality against site-autonomous quality

Every multi-site pharmaceutical company has made a choice here, often without articulating it, and the choice determines two things an architect cares about: what the master data looks like, and what the quality system is.

Centralised quality. One corporate quality system instance, one document hierarchy, one deviation and change-control process, corporate approval on significant events, and specifications held once. What it buys you is the enterprise view: the ability to see that the same supplier is causing deviations at three sites, and the ability to run a product quality review for a product made at two sites without reconciling two systems. What it costs is throughput — every site's change waits in one queue — and it requires harmonised master data that no acquisition ever arrives with.

Site-autonomous quality. Each site runs its own quality system and its own document set, with corporate setting policy rather than process. What it buys is local speed and local ownership, and it is often the honest answer for sites with genuinely different regimes — a sterile site and an active ingredient plant really do have different deviation profiles. What it costs is that there is no enterprise view, specifications exist in as many places as there are sites, and the same material code means different things in different plants [5].

The practical consequence is that **the decision is really about master data, and it is usually made as though it were about process.** Three objects decide it: the material and specification master, the product and registration master, and the document hierarchy. If those three are held once, a central quality system is viable. If they are not, a central quality system is a reconciliation engine wearing a governance badge.

There is a third pattern, and it is the commonest in practice: **declared hybrid**. One corporate system for change control, complaints, recalls, supplier quality and audits; site-level handling of deviations and local documents. It works, and it is legitimate when the split is declared and the interfaces are in somebody's validation scope — which, for the record, interfaces routinely are not [5].

The cadence, which is the spine of this chapter

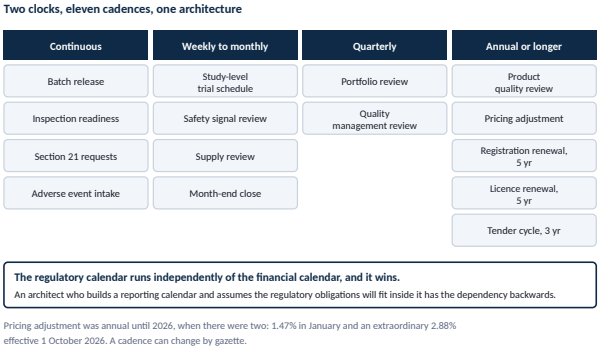


Figure 7.2 The eleven cadences, by frequency. Only three of them care when the financial year ends. **The regulatory calendar runs independently of the financial calendar, and it wins.**

Eleven rhythms run a pharmaceutical company, and only three of them care what month the financial year ends. Take them in order of how much trouble they cause an architect.

The portfolio review

A periodic review — usually quarterly, sometimes monthly for the largest assets — of the development and commercial portfolio by senior management: progress against plan, revisions to risk-adjusted net present value, resource allocation and prioritisation. Research, commercial, technical operations, regulatory and finance all attend.

It consumes the rNPV model and everything that feeds it. It produces priority rankings, resource reallocations and occasionally a kill decision outside a formal gate.

Its architectural significance is as a demand signal. If the portfolio review cannot be assembled without three weeks of manual data collection, that is the brief — and three weeks is a realistic number at a company of any size.

The stage gate

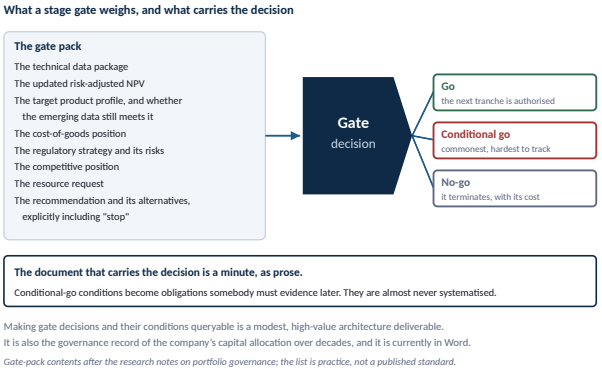


Figure 7.3 What a stage gate weighs, and the three outcomes. **Conditional go is the commonest and the hardest to track**, because its conditions become obligations nobody systematised.

A gate is a formal decision point at a defined transition — candidate selection, Phase II entry, Phase III commitment, submission, launch — with defined entry criteria, a defined decision body and a recorded decision. **A gate is not a status review. It is an authorisation to spend the next tranche.**

What it weighs is in the figure. What closes it is a decision: go, no-go, or **conditional go**. Conditional go is the commonest outcome and the hardest to track, because the conditions become obligations somebody must evidence later, and they frequently are not systematised anywhere.

Where the decision lives is a real question, and the answer is usually "in the minutes, as prose". Gate decisions and their conditions are the governance record of the company's capital allocation over decades. Making them queryable — decision, date, conditions, owner, evidence, closure — is a small deliverable with disproportionate value, and it is the kind of thing a client will not ask for because they have never seen it done.

The trial master schedule

The integrated plan for every clinical study in a programme or across the portfolio: study start, first patient in, last patient in, last patient out, database lock, study report, and the dependencies between them. A Phase III cannot start before the Phase II read-out. A submission cannot be assembled before the study reports exist.

It is reviewed monthly at programme level and weekly at study level during active enrolment. And it moves constantly, because it is a dependency network with external parties on the critical path — investigator sites, ethics committees, regulators, contract research organisations, central laboratories.

> **A CLINICAL PROGRAMME'S DATES MOVE FOR REASONS NO SYSTEM RECORDS**
>
> **Enrolment is the dominant uncertainty and it is almost always optimistic.** But the specific reason a date moved is usually a conversation: a principal investigator's competing study, a contract negotiation at one site, an ethics committee that meets monthly and has a full agenda, a shipment held at customs. None of that is in the clinical trial management system, which records only that the date changed.
>
> So a schedule-variance report in a clinical programme measures something real and explains nothing, and an architect who promises predictive analytics over it is promising to predict human negotiation. **What the architecture can honestly deliver is a reason code on every date change and a harmonised definition of each milestone.** The second of those is the unglamorous and genuinely valuable piece of work, because the trial master schedule is the integration point between the clinical trial management system, clinical supply planning and the regulatory submission plan — three systems with three different notions of what "study start" means [5].

The regulatory commitment calendar

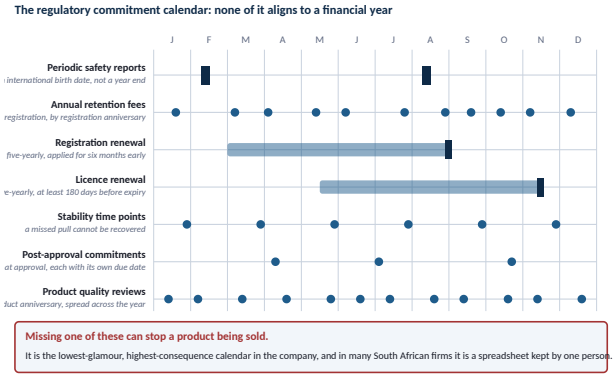


Figure 7.4 The regulatory commitment calendar over one year. Nothing on it aligns to a financial period, and its prerequisite is an authoritative registration register. Marker positions are illustrative; the five-yearly and 180-day rules are sourced [1].

The forward schedule of every obligation attached to every registration: periodic safety update reports on a cycle from a fixed international birth date, renewals, annual retentions, variation commitments, post-authorisation study milestones, labelling update deadlines, and licence renewals — five-yearly in South Africa, applied for at least 180 days before expiry [1].

This is the obligation set most companies cannot produce on demand, and missing one of them can suspend a product licence. It is a scheduling and alerting problem with a legal consequence, and its prerequisite is a complete, accurate registration inventory. **A commitment calendar that is not derived from an authoritative registration register is a liability dressed as a control** — it produces reassurance about the registrations somebody happened to type in.

The commonest real-world failure is quieter than a missed date: commitments living in the approval letter and nowhere else, owned by whoever read the letter, with a due date recorded and no evidence requirement, so the commitment is "closed" with nothing to show [5]. And an owner who has left, with the commitment reassigned to a vacant role.

The Annual Product Quality Review

An annual, per-product review of all quality data for that product across the year: batches made and their yields, in-process and finished-product results with trends, out-of-specification results and their investigations, deviations, changes, complaints, returns, recalls, stability results at every time point, equipment and utility qualification status, the adequacy of specifications, and the status of commitments. It concludes with an assessment of whether the process remains in a state of control, and a set of actions.

Quality assurance produces it, with input from quality control, production, engineering and regulatory. **It is an inspection document.** An inspector asks for the current reviews early and reads them for whether the company notices its own trends.

> **THE CORRECTION EVERY AUTHOR GETS WRONG HERE**
>
> **The South African GMP guideline sets no requirements for the Annual Product Quality Review.** It mentions the review only as a shared responsibility of the head of production and the head of quality control, with no detailed content and no frequency [3]. The same guideline **never uses the term "data integrity"** and sets no specific data-integrity requirements [3].
>
> **Do not infer from either silence that either is optional.** South African sites run product quality reviews, and PIC/S data-integrity expectations apply, because the regulator has adopted the PIC/S Guide to GMP and all its prospective adaptations [3]. **The annual frequency comes from the PIC/S guide, not from the South African guideline** — section 1.10 of PIC/S Part I [7]. Note also that South Africa **acceded to PIC/S in July 2007**, earlier than the United States Food and Drug Administration in January 2011, Japan in July 2014 and Brazil in January 2021 [8], so the framing that it adopts PIC/S because it is not a member has the history backwards.
>
> The practical rule for an architect: **do not cite the South African guideline for GMP detail.** It adopts and adapts the PIC/S guide, and the detail lives in the PIC/S guide. Cite the right document.

The architectural significance is that **this is the single most demanding reporting requirement in a pharmaceutical quality estate.** A year of data from the laboratory system, the manufacturing execution system, the enterprise system and the quality system, assembled per product, per site, with full traceability, read by an inspector. The characteristic failure is three months of manual extraction per product, with an analyst pulling data from eight systems into a spreadsheet, every product, every year [5]. For a generics company with a couple of hundred registered products that consumes a team, and the reviews are late — so their conclusions cannot inform anything.

Two further failures worth naming, because they are cheap to fix and nobody does. **Data pulled on inconsistent bases**: batches counted by manufacture date in one system and by release date in another, so the review's batch count does not match the ledger. And **actions not tracked**: the review identifies improvement opportunities, they are not raised as corrective actions, and next year's review repeats them [5].

This is the best single entry point for a data-architecture engagement in a GMP site, because the cost is concrete, the beneficiary is the inspector, and nobody has to be persuaded that it matters.

Medicine registration renewal

Registration does expire. Five years, renewed six months early, phased across 2023 to 2035 in South Africa. This surprises people who have modelled a registration as a permanent fact with a grant date and no end date, and it is a one-line data-model error with a product-availability consequence.

Model the registration with a validity period, a renewal window and a renewal state, and derive the renewal task from it. Do not let anybody tell you the renewal date is in a spreadsheet, because it will be, and that is the point.

Variation planning, and the correction that inverts a workflow

A variation is any change to an approved registration, classified by risk. Minor notifications may be implementable immediately or after a short notice period; major variations require approval before implementation. **The classification determines the lead time, and the lead time determines whether a supply-chain change is feasible at all.** The regulatory affairs manager is therefore the gating authority on a surprisingly large number of apparently operational decisions: changing an active-ingredient supplier, moving a product to another line, changing a packaging component, revising a specification, updating artwork.

> **TYPE IAIN VARIATIONS ARE DEEMED IMPLEMENTABLE ON SILENCE**
>
> A **Type IAIN** variation is **deemed implementable after 47 working days of SAHPRA silence.** That inverts how an approval workflow is normally built. The usual pattern waits for an approval event and blocks until it arrives. Here, the absence of a response **is** the outcome, which means the workflow needs a timer that computes a deemed-approval date from the submission date, an audit trail of the silence, and a defensible record of what was submitted and when — because the company's authority to have implemented the change rests on that record and on nothing else.
>
> Build it as a timer-driven state machine. If you build it as "await approval", the change sits in a queue forever and somebody implements it anyway, off-system.

The SAHPRA queue as a planning input

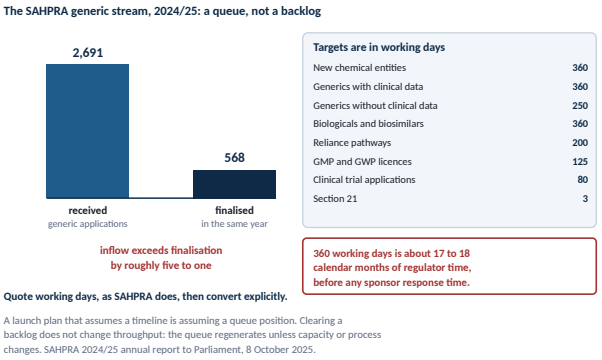


Figure 7.5 The generic registration stream in 2024/25: **2,691 applications received against 568 finalised** [6]. Registration lead time is a capacity-constrained queue, not a backlog that was cleared. Targets are in working days, as the regulator states them.

Review timelines are expressed in **working days to finalisation**, not calendar months to approval: **360 for new chemical entities, biologicals and biosimilars, 250 for generics without clinical data, and 200 for reliance pathways** [6][9]. Supporting streams have their own: 125 working days for GMP and wholesaling licences, 80 for clinical trial applications, 50 for lot release, 20 for permits, and 3 for section 21 [6].

360 working days is roughly 17 to 18 calendar months of regulator time, before any sponsor response time is added. Quote it in working days, as the regulator does, then convert explicitly. Silently converting to months is how a launch plan acquires six months of imaginary slack.

And then the fact that matters most:

> **CLEARING THE BACKLOG DID NOT FIX THROUGHPUT**
>
> In 2024/25 the regulator **received 2,691 generic applications and finalised 568** [6]. Inflow exceeded finalisation by roughly five to one on that stream.
>
> Whatever a backlog programme achieves, the queue regenerates unless capacity or process changes. **Registration lead time is a capacity-constrained queue, not a backlog that was cleared** — and a client's launch plan that assumes a timeline is assuming a queue position. The historic Backlog Clearance Programme addressed roughly 16,000 applications [9]; a separate project begun in September 2024 reported an 81% finalisation rate by October 2025 with closure targeted for end March 2026, and **whether that closure was achieved is not something I could confirm** [10]. Secondary sources conflate the two programmes; do not.

One structural observation to carry into any conversation about regulator throughput. The government grant share of the regulator's funding was projected to fall from 45% to 34% between 2022 and 2027, with the grant itself declining by about 11% over four years [9]. **A regulator becoming more fee-dependent is a regulator whose throughput and its customers' fees are coupled** — which is the context for a published industry survey in which 76% of 33 respondents would consider higher fees for faster reliance reviews and 67% favoured credits if timelines were missed [9].

Reliance is not recognition

This distinction is worth a paragraph of its own because getting it wrong changes what a client believes they can plan on.

SAHPRA does not recognise United States or European approvals. It operates **reliance**, in two forms: **abridged review**, a streamlined review based primarily on full assessment reports from reference regulatory authorities, and **verified review**, a streamlined review based on verifying rather than evaluating the information submitted [11]. **Recognition** — a streamlined process based on directly recognising another authority's outcome — is a **separate pathway, still being negotiated, with the framework still to be published** [11]. And under reliance the regulator **remains responsible and accountable for the decisions it takes** [11].

Three operational consequences an architect should carry.

All supporting documents and amended dossier sections must still be submitted locally, whichever pathway applies [11]. Reliance reduces assessment effort, not submission effort.

If no full assessment reports are received from the reference authority within three months of the request, the application is reviewed as a full review [11]. That is a dependency on a third-party regulator's responsiveness sitting inside your client's launch plan, and it should be on the risk register with a named owner.

And the benefit is real but it is measured in assessor hours, not calendar days. The regulator's own published research gives median assessor hours of 19.4 against 60.5 for new chemical entity chemistry data and 24.6 against 84.7 for generic data, with cost reductions of 77% and 81% respectively [9]. Those are the regulator's internal economics. **Do not translate them into a shorter calendar timeline for your client without evidence**, because the queue, not the assessment, is the constraint.

Section 21 as a continuous operational workflow

Section 21 of the Act permits the regulator to authorise sale of an unregistered medicine for a specified purpose, typically a named patient under a named prescriber. Architects routinely model it as an exception.

It is not. **In 2024/25 the regulator received 20,245 section 21 applications, finalised 20,225, and met its three-working-day target on 19,509 of them, or 96%** [6]. That is roughly **80 applications per working day**, and it is the **busiest single workflow in the authority**.

For a client, section 21 means a continuous, high-volume, three-day-clock process for product that sits outside the registered-product master data entirely, and whose supply is **not subject to the Single Exit Price** [12]. Every South African pharmaceutical client has a section 21 process and it is almost always the least systematised part of their estate — a mailbox, a spreadsheet and one person who knows how it works. Ask about it in week one.

The remaining cadences, briefly

Batch release is continuous, each batch certified individually. **Quality management review** is typically quarterly, with deviation, corrective-action, complaint and change metrics. **Safety signal review** is monthly, more often for a product under scrutiny. **Supply review** is monthly, constrained by validated batch sizes — you cannot make a half batch, because the batch size is fixed by the registration — and by active-ingredient lead times measured in months. **Month-end close** carries pharmaceutical specifics: inventory valued by batch with expiry-based provisioning, and milestone revenue triggered by clinical and regulatory events. **The pricing adjustment cycle** is annual, except that in 2026 it ran twice [13][14]. **The tender cycle** is roughly three-yearly per tender, plus a bid window and, as 2026 demonstrated, a possible litigation phase [15]. And **inspection readiness** is continuous, because an unannounced inspection is a real scenario and "inspection ready" is a steady state rather than a project.

The operating model, drawn for one company

Abstractions are easier to argue with than to use, so here is the same material as a shape. The group below is **invented. It describes no real organisation and every figure in it is illustrative and internally consistent rather than sourced.** It exists because an architect needs concrete texture at a plausible scale, and because the two halves of it are the two shapes that cover most of the South African pharmaceutical estate.

Veldtkruin Pharmaceuticals, a mid-size generics manufacturer. Revenue about R2.1 billion, split roughly 62% private market at the Single Exit Price, 34% state and institutional tender, 4% export into the region. 184 registrations with the regulator, of which 11 are held but not currently marketed — the dormant-registration problem that every generics client has, each one still consuming retention fees, a stability programme slot and a product quality review. 498 active SKUs, because one molecule at two strengths across five pack configurations is ten SKUs. 27 of those SKUs are tender-specific pack configurations with their own artwork and labelling.

Two manufacturing licences: an oral solid dose site at Isando, and a sterile fill-and-finish site at Brackenfell, with a separate quality control laboratory at Centurion and a wholesale licence for the distribution centre. One named Responsible Pharmacist per manufacturing licence, plus a wholesale responsible pharmacist. All 64 active ingredients imported from 31 suppliers, four of them single-sourced with no qualified alternate — the top item on the supply-risk register every month. About 1,810 batches a year group-wide, roughly seven a working day. Head office at Midrand.

Khanyisa Bio-Development, a small contract development and manufacturing organisation with a clinical-stage portfolio of its own, on one campus at Stellenbosch. Revenue about R265 million. It holds **no marketing authorisations**. It is a named manufacturing site on **38 clients' registrations**, so one GMP finding is simultaneously a problem for 38 registration holders, which is why its quality agreements are its most important commercial documents and why clients audit it roughly nine times a year. About 310 batches a year across 95 distinct products, so the average product runs three batches a year and the site is changeover-dominated: changeover and cleaning consume roughly 38% of available suite time, which is the single most important economic fact about the business.

Put the two cadences side by side and the architecture problems separate cleanly.

Veldtkruin's product quality review burden is **184 reviews a year**, scheduled by product anniversary at roughly fifteen a month, each taking a quality officer four to six working days of assembly from the laboratory system, the enterprise system, the quality system and three spreadsheets. **That is on the order of a thousand quality-assurance person-days a year, or about five full-time people doing data assembly.** The architecture business case is substantially that number, and it is a number a client can verify in an afternoon.

Khanyisa's burden is different in kind. It produces reviews **for its clients, to each client's template, under each quality agreement** — 38 formats across 95 products, with different data cuts, different trend-charting conventions, different deadlines and different approval routes, several of them contractually due within a fixed number of days of the product anniversary. **That is multi-tenant regulated reporting**, and it is the hardest architecture problem the business has: one quality system, one laboratory system, one set of batch records, and 38 non-negotiable output specifications.

Veldtkruin's problems are portfolio-scale problems against a regulated price. Khanyisa's are multi-tenancy and project-economics problems. Neither of them is the originator shape that published pharmaceutical reference material describes, and an architect who has internalised both has the two shapes they are most likely to meet.

The measures, and the honest warning about each

Every one of these will be asked for, and every one of them hides something.

Right-first-time batch rate. The proportion of batches completed without a deviation. It is the single most quoted manufacturing quality measure and the most sensitive to definition. Does a minor documentation deviation count? Does a deviation raised and closed during execution count? Does a batch reworked under an approved rework procedure count as right first time? Three sites answering those three questions differently will produce three incomparable numbers, and the gap between them will be larger than any real difference in performance.

Deviation ageing. Open deviations past target closure, by age band. **This is the single most reliable leading indicator of a site in trouble** [5], and it is visible from a dashboard in five minutes. It is also the measure most vulnerable to being managed rather than improved: a site under pressure closes the easy ones and the distribution tells you so. Watch the tail, not the mean.

Batch cycle time. Clean to clean, or order release to release for sale? The second is the one the business feels and the first is the one production controls. Both are legitimate; using one and labelling it the other is not. And the release portion of it is an architecture outcome rather than a production outcome — every manual reconciliation, nightly interface, hand-typed result and queue-based review adds days between manufacture and availability [5].

Enrolment rate and database-lock timeliness. Enrolment is behind, always, and the rate is a function of site selection decisions made a year earlier. Lock timeliness is the one clinical measure that is genuinely comparable across studies, because the lock is an evidenced event with a signature — provided nobody has soft-locked and then changed data, which is common enough to check before quoting the number [5].

Pharmacovigilance compliance rate. The proportion of expedited reports submitted within the statutory clock. Honest only if day zero is honest, and day zero in South Africa is **first receipt by any personnel of the holder, including sales representatives and contractors** [16]. A compliance rate computed from the date the case reached the safety department is a measure of the safety department, not of compliance.

On-time in-full delivery. Definitionally ambiguous in every industry and more so here, because "in full" interacts with batch status, expiry and customer-specific remaining-shelf-life terms. A line delivered from stock the customer contractually cannot accept is not in full.

Measure	Comparable across sites?	What makes it incomparable
Right first time	**No**	deviation-counting rules differ by site
Deviation ageing	**Yes**, with care	target closure periods differ; age bands must match
Batch cycle time	**No**	clean-to-clean against order-to-release-for-sale
Enrolment rate	**No**	indication, geography and competing studies dominate
Database-lock timeliness	**Yes**	only if soft locks are excluded
PV compliance rate	**Yes**, within a jurisdiction	day-zero definition and clock differ by country
On-time in-full	**No**	"in full" interacts with batch status and shelf life

> **A MEASURE DEFINITION REGISTER COMES BEFORE ANY DASHBOARD**
>
> This is the same discipline the manufacturing volume insisted on, and it is here for the same reason. For every measure: its name, its precise formula, the system that is authoritative for each input, the period convention, the inclusions and exclusions, the owner, and the date the definition last changed.
>
> It sounds like documentation. It is not. **It is the thing that makes a number mean the same thing twice**, and without it a multi-site quality dashboard is a machine for generating arguments in management review. Build the register first, agree it with quality assurance and regulatory, and only then build the dashboard. If the client will not fund the register, build the dashboard for one site and say plainly why you have not extended it.

What an architect does with this chapter

Model the site as a licence, a scope, an inspection history and a named pharmacist, with the registrations that point at it. The building is an attribute.

Model the registration with a validity period and a renewal state, because it expires.

Model a variation workflow with a timer, not only with an approval event, because deemed implementation on silence after 47 working days is a real pathway and the record of the silence is the authority to have acted.

Derive the commitment calendar from the registration register, never from a parallel list.

Treat registration lead time as a queue position, and say so in writing when a launch plan assumes a date.

Treat reliance as reduced assessment effort, not as recognition and not as a shorter calendar.

And build the measure definition register before the dashboard. The portfolio cadence is the demand signal for every system in the estate; a cadence you cannot measure consistently is a cadence you cannot improve, and the first thing a new architect can do that nobody will argue with is to make two sites' numbers mean the same thing.

THE CHAPTER IN TEN LINES

- A registered manufacturing site is a licence with a scope, an inspection history and a named Responsible Pharmacist; the building is an attribute.
- The Responsible Pharmacist must be a pharmacist registered with the Pharmacy Council, is dual-accountable to the regulator and the Council, is named on the licence with not more than one per licence, and owns the duty that processes remain in a validated state.
- Over-labelling an imported pack is a packaging activity and is therefore licensable, and a valid site licence at bid closure is a first-stage gate in a state tender.
- A sterile site is a utilisation business and an oral solid dose site is a changeover business, so their costing and capacity models are not variants of one model.
- The quality agreement governs contract manufacture in both directions, and client-specific change-control rights have to be enforced in workflow rather than in a procedure document.
- The choice between centralised and site-autonomous quality is really a master-data decision and is usually made as though it were a process decision.
- A stage gate is an authorisation to spend, conditional go is the commonest outcome, and the decision usually lives in a minute as prose.
- A clinical programme's dates move for reasons no system records, so the honest deliverables are a reason code on every date change and one harmonised milestone definition.
- The regulatory commitment calendar is the obligation set most companies cannot produce on demand, and it must be derived from an authoritative registration register.
- The current South African GMP guideline addresses neither the Annual Product Quality Review nor data integrity, but sites run reviews and PIC/S data-integrity expectations apply; the annual frequency comes from PIC/S, not from SAHPRA.
- Medicine registration expires after five years and is renewed six months early, phased across 2023 to 2035.
- Type IAIN variations are deemed implementable after 47 working days of regulator silence, which inverts how an approval workflow is normally built.
- Review targets are in working days: 360 for new chemical entities, biologicals and biosimilars, 250 for generics without clinical data, 200 for reliance.
- Clearing the backlog did not fix throughput: 2,691 generic applications were received against 568 finalised in 2024/25, so registration lead time is a queue position.
- Reliance is not recognition: SAHPRA does not recognise foreign approvals, it operates abridged and verified review and remains responsible for its own decisions.
- Section 21 ran 20,245 applications in 2024/25 on a three-working-day clock, about 80 a working day, and is the regulator's busiest workflow.
- A measure definition register comes before any dashboard, because without it a number does not mean the same thing twice.
- Several key roles are statutory and one person deep, including the Responsible Pharmacist and a resident pharmacovigilance officer, and none of them appears on an application inventory.
- A design whose maintenance requires scarce skills will not be maintained, which argues for fewer, more standard, better-supported systems.

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- **3.** SAHPRA, SAHPGL-INSP-02 v9, "Guidelines to Good Manufacturing Practice for Medicines", published 17 November 2025 — adoption of the PIC/S Guide to GMP and its prospective adaptations; South African Annex 16; Responsible Pharmacist and release checklist; verified negatives: no Annex 11 or computerised-systems content, no data-integrity section, no retention periods, and no detailed Annual Product Quality Review requirements (the review is named only as a shared responsibility of the heads of production and quality control, with no frequency). Research found this version superseded by v10 (13 April 2026) inside its stated validity window; re-check version and date before citing https://www.sahpra.org.za/wp-content/uploads/2025/11/SAHPGL-INSP-02_v9-Guidelines-to-Good-Manufacturing-Practice-for-Medicines.pdf
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CHAPTER 8

The process model

Fourteen processes run a pharmaceutical company, and the value of modelling them is almost entirely in the seams rather than the boxes. This chapter states the notation exception for the whole volume, teaches the four levels, draws eight level-2 flows with the systems named and the hand-off failures called out, and ends with the trap that makes GxP process modelling different from process modelling anywhere else.

Everything in this book that is architecture is modelled in ArchiMate 4. This chapter is the one exception, and the exception is worth stating once, here, for the whole volume.

The notation exception

Level-2 process models in this book are BPMN 2.0, not ArchiMate.

The reason is specific rather than stylistic. ArchiMate has no swimlanes. It can express that a role is assigned to a process, and that one process triggers another, but it cannot lay a flow out so that the **hand-off** is the visible thing on the page. In a pharmaceutical process model the hand-off is the whole point. Almost every failure in this chapter happens at a boundary: between the manufacturing execution system and the laboratory system, between the laboratory and quality assurance, between the quality system and regulatory affairs, between a sales representative's conversation and the safety database. A notation that cannot draw the boundary cannot draw the failure.

So: ArchiMate for the architecture — capabilities, roles, application components, services, business and data objects, technology. BPMN for the modelled flow of work, at level 2 only. And one rule to keep them from drifting apart: **every BPMN lane must correspond to an ArchiMate Role or an application component that exists in the model.** If a lane has no counterpart, either the lane is wrong or the application model is incomplete. That single constraint is what stops a process library and an architecture repository becoming two unrelated collections of pictures.

> Chapter 17 covers modelling pharmaceuticals in ArchiMate properly, including how to attach these process models to the architecture without duplicating them. Do not model the same flow twice in two notations. Model it in BPMN and reference it.

The four levels, and who reads each

Decide which level you are drawing before you draw. Most wasted modelling effort comes from drawing at the wrong one — usually level 3 when level 1 was asked for.

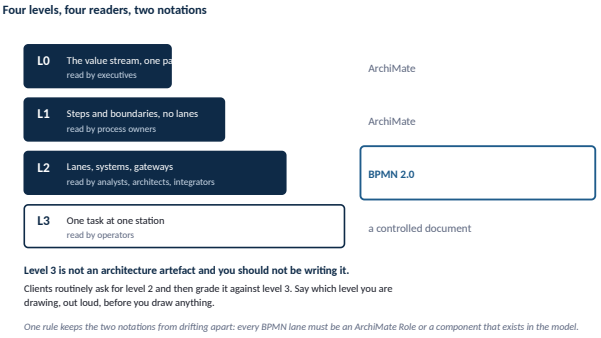


Figure 8.1 The four process levels, their readers and their notation. **Level 2 is BPMN; everything else that is architecture is ArchiMate**, and level 3 is a controlled document you should not be writing.

Level 0, the value stream. One page. Discover, develop, register, make, release, distribute, monitor — with the major material, document and information flows between them. No roles, no systems, no decisions. Its reader is an executive and its purpose is to agree scope and ownership.

Level 1, the process. One page per process, five to twelve steps, no swimlanes. Its reader is a process owner. Its purpose is to agree the steps, the sequence and — critically — the **boundaries**: where this process starts, where it ends, and which other process picks up. Most arguments that look like level-2 arguments are unresolved level-1 boundaries.

Level 2, the modelled flow. BPMN with lanes, with the systems named, with the gateways drawn. Its readers are the business analyst, the solution architect and the integration designer, and its purpose is to be *implementable*: this is the artefact from which interfaces, data requirements and test cases are derived. Everything drawn in the rest of this chapter is level 2.

Level 3, the work instruction. How one person does one task at one station, with the screen, the button, the sample size, the gowning sequence. Its reader is an operator, its owner is manufacturing or quality, and **it is a controlled document under the quality system with a revision and an approval.** It is not an architecture artefact and you should not be writing it. Say so clearly when asked, because clients routinely ask for level 2 and then grade it against level 3.

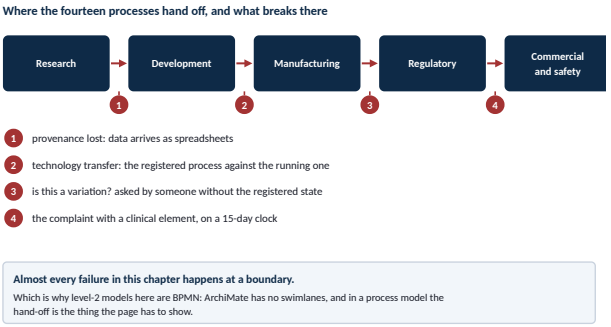


Figure 8.2 The four boundaries that cause the most trouble, in order. **Almost every failure in this chapter happens at a seam**, which is why the notation has to be able to draw one.

Fourteen end-to-end processes run a pharmaceutical company. Eight are drawn fully below. The other six are described in prose, because their value to an architect is in a handful of facts rather than in the shape of the flow. In each drawn process the systems are named and the hand-off failures are called out explicitly. **The value of this chapter is in the seams.**

A note on the names. The company and the systems in these models — Veldtkruin Pharmaceuticals and Khanyisa Bio-Development, with Kelpfield the enterprise system, Protea the manufacturing execution system, Aloe the laboratory system, Rooibos the quality system, Fynbos the regulatory information system, Sentinel the safety database, Marula the trial master file, Sable the clinical systems, Driftwood serialisation and Silverleaf the validated document store — are **invented**. They are there so the lanes have concrete labels instead of generic ones. Nothing about them is real.

Batch manufacture to release

This is the centrepiece, the longest, and the one that must be drawn. Quality, manufacturing, laboratory and regulatory systems all have to agree, and the releasing authority is the **Responsible Pharmacist** — not an EU Qualified Person [1][2].

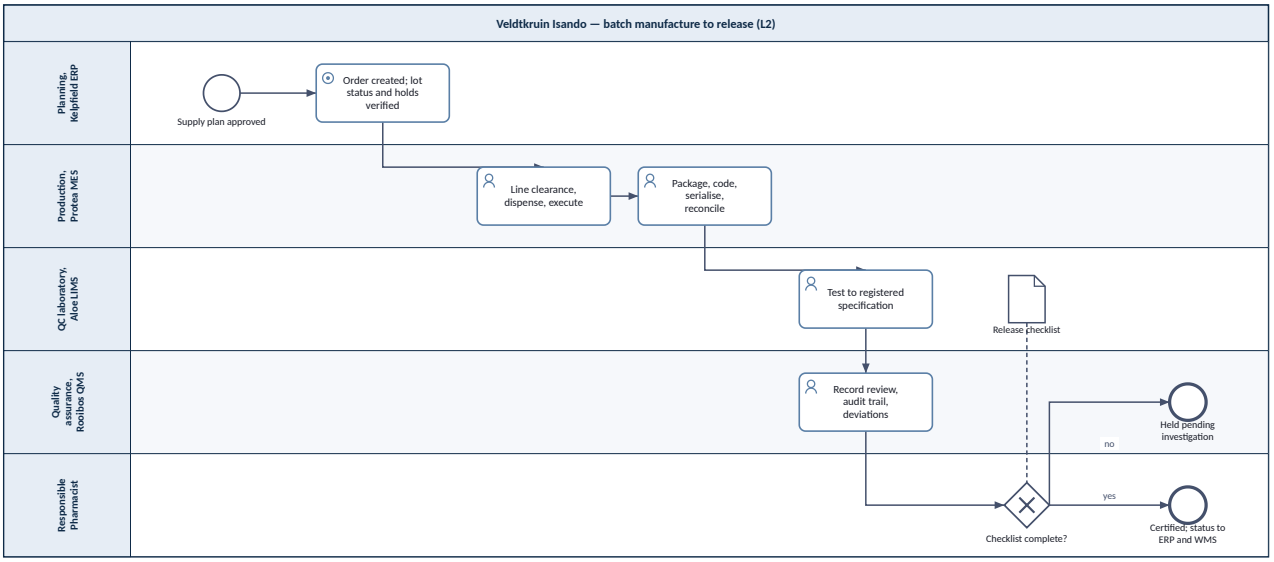


Figure 8.3 Batch manufacture to release, compressed from twenty-eight steps across five phases. The releasing authority is the **Responsible Pharmacist** [1][2]. Note the computed gateway and the release checklist as a named artefact, rather than a folder of documents to reconcile by hand.

The drawn flow compresses twenty-eight steps across five phases. Take them as phases, with the systems and the breaks.

Plan and prepare. The supply plan becomes a production order in the enterprise system against a released master recipe version, and the order is downloaded to the execution system where the recipe is instantiated as a batch and equipment and rooms are reserved. Material availability is checked against lot status, and — separately — against quality holds.

> *Hand-off failures here: the enterprise system's recipe version and the execution system's master recipe version differ, so the batch is made to a recipe nobody approved. A lot is released in the enterprise system and on quality hold in the quality system, so material moves while held. Operator qualification is checked from a training system queried nightly, so a newly trained operator cannot work and an expired qualification still can. And equipment calibration status is checked at periodic review rather than at execution, which is the wrong moment entirely.*

Execute. Dispensing with weighing equipment integrated to the execution system and material identity scanned. Processing with the execution system prompting, the control system running the process, process values to the historian and critical values into the batch record. In-process checks sampled to the laboratory. Deviations raised at the point of occurrence. Then packaging: component verification against the component master and the approved artwork version, coding, serialisation, aggregation, and reconciliation of components issued, used and destroyed.

> *Hand-off failures: the weight is hand-typed, which breaks attributability and invites a transposition error. The batch record captures a historian-interpolated or compressed value rather than the measured one, so a reviewer cannot reconcile the two. The in-process gate is advisory, so the step proceeds on a pending result. And the deviation is raised on paper later, because the system's workflow stops the line — so the record is not contemporaneous at precisely the point where contemporaneity matters.*

Test. Finished-product samples to the laboratory, tested per the **registered** specification version. Results from instrument to chromatography system to laboratory system, raw data to the scientific data archive. Out-of-specification results into a quality investigation — not a retest loop. Microbiological and sterility testing, whose longer cycle times set the release lead time for sterile product. Multi-level approval, then a certificate of analysis.

> *Hand-off failures: the laboratory specification and the registered specification differ, and only the registered one has legal force. The result is calculated in a spreadsheet, so the calculation is the unvalidated part of the chain. And where the system makes retesting easy and investigation hard, the laboratory tests until it passes, which is the most serious laboratory data-integrity finding there is.*

Review. Batch record review — by exception where the exception definitions are complete, and in full where they are not. Audit-trail review of the contributing systems. Deviation disposition, with every open deviation on the batch either closed or justified as not affecting it. Environmental monitoring results for the manufacturing period. Reconciliation of yield, components, serial numbers and label accountability.

> *Hand-off failures: review by exception not actually implemented, so reviewers read the whole record and the execution system has added cost without removing effort — the commonest reason these programmes disappoint. Audit-trail review not scoped at all, or scoped and not resourced, so release either waits on it or proceeds without it. And an open deviation with no batch impact assessment, which blocks release or, worse, does not.*

Release. The release checklist is assembled — **a named artefact in the South African guideline**, covering marketing authorisation conformity, GMP conformity, legal requirements, storage and transport conditions, sampling, quality control results and transmissible-agent status where relevant [2]. The

Responsible Pharmacist reviews and certifies. The electronic signature is applied with the meaning of signing explicit. Status changes to released in the authoritative system and propagates. The batch record is archived. And market-specific release follows where a destination requires importation testing or local release.

> *Hand-off failures: status propagated by nightly batch, so the warehouse cannot ship for hours after certification. And no decommissioning plan for the system holding the archived record, which matters because batch records are kept* ***one year after batch expiry or five years after certification, whichever is longer*** *— a computation, not a number.*

THREE ARCHITECTURAL FINDINGS CARRIED FROM THIS FLOW

Release is a join across at least six systems, and the join key is the batch number — which is not formatted the same way in any two of them. The enterprise system's batch, the execution system's batch, the laboratory system's sample lot reference, the serialisation system's lot, the historian's batch context tag and the quality system's batch field. **Normalising that key is the single highest-value integration task in a pharmaceutical estate**, and it is cheap compared with everything else on the roadmap.

The release decision has a precondition set, and the architecture should compute it rather than present it. The Responsible Pharmacist should see a state — all preconditions met, or these three are not — rather than a folder of documents to reconcile by hand. That is why the drawn flow has a gateway and a data object rather than a long queue of review tasks.

Release lead time is an architecture outcome. Every manual reconciliation, nightly interface, hand-typed result and queue-based review adds days between manufacture and availability. For a cold-chain or short-shelf-life product, those days are the business case.

Adverse event intake to expedited report

The most time-critical process in the estate, and the one whose first step is usually outside any system.

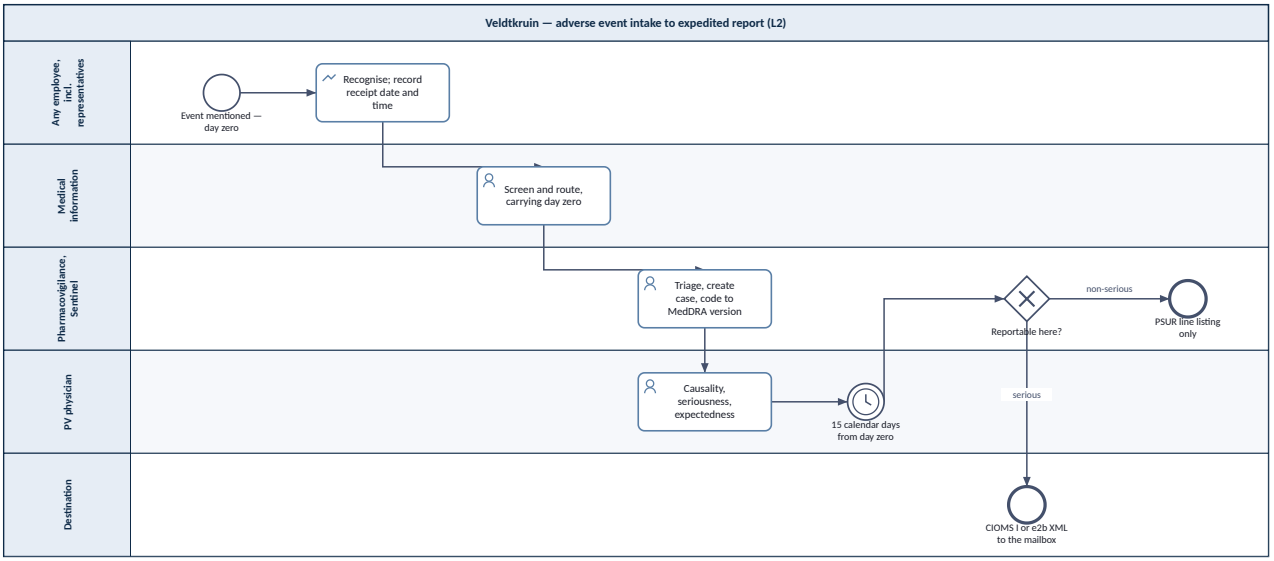


Figure 8.4 Adverse event intake to expedited report. **Day zero is first receipt by any employee, including sales representatives and contractors** [3], and the channel is a mailbox rather than a validated gateway.

The South African clocks are the regulator's own and are not the European ones. Serious adverse drug reactions and adverse events following immunisation — related or unrelated, expected or unexpected — within **15 calendar days**. Significant safety issues arising from a listed foreign regulator, and regulatory action on such issues, within **5 calendar days**. Changes in the nature, severity or frequency of risks within **5 calendar days**. Medication errors within **15 calendar days**, serious or not. Non-serious reactions are not routinely reported; they go into the periodic report's line listing and are submitted on request within 30 calendar days [3].

And the clock starts here:

> **DAY ZERO IS ANY EMPLOYEE'S RECEIPT**
>
> The clock starts when **any personnel of the holder of the certificate of registration, including sales representatives and contractors**, first receives the four minimum data elements [3].
>
> That one clause makes every employee an intake channel. It is why a field representative's call note is in the safety path, why a medical information ticket is a regulated intake record, and why "day zero" is a data field that has to be captured and defensible rather than inferred. **A compliance rate computed from the date the case reached the safety department measures the safety department, not compliance.**

Three corrections, because this is where authors overreach.

"SAHPRA requires E2B(R3)" is wrong. The operative guideline says reports **can also be submitted** in e2b format and **names no revision** — it does not specify R2 against R3. The accepted formats are **CIOMS I, the holder's own in-house form if it carries all minimum data elements, or e2b XML, with only one method per individual case** [3].

Trials are different. A clinical trial requires **CIOMS plus the regulator's own form, not E2B.** Do not design one intake-to-submission path and assume it covers both post-marketing and trial safety.

And foreign individual case reports are not routinely forwarded. They go only for significant safety issues or on request [3]. **What runs on the five-day clock is foreign regulatory *action*, not foreign cases** — which means the regulatory-intelligence feed is an input to the safety system rather than a separate reading exercise by somebody in regulatory affairs.

One more thing the drawn flow cannot show. **The channel is a mailbox, not a gateway.** The guideline sets out separate mailboxes per submission type, including a dedicated one for e2b XML files, and mentions no portal and no validated gateway [3]. So acknowledgement handling, retry and reconciliation sit entirely with the sender. **A submission register with its own per-destination state machine and manual acknowledgement reconciliation is a required component here, not a workaround** — and the commonest failure is that an unprocessed acknowledgement makes a rejection look like a submission.

Deviation to CAPA closure

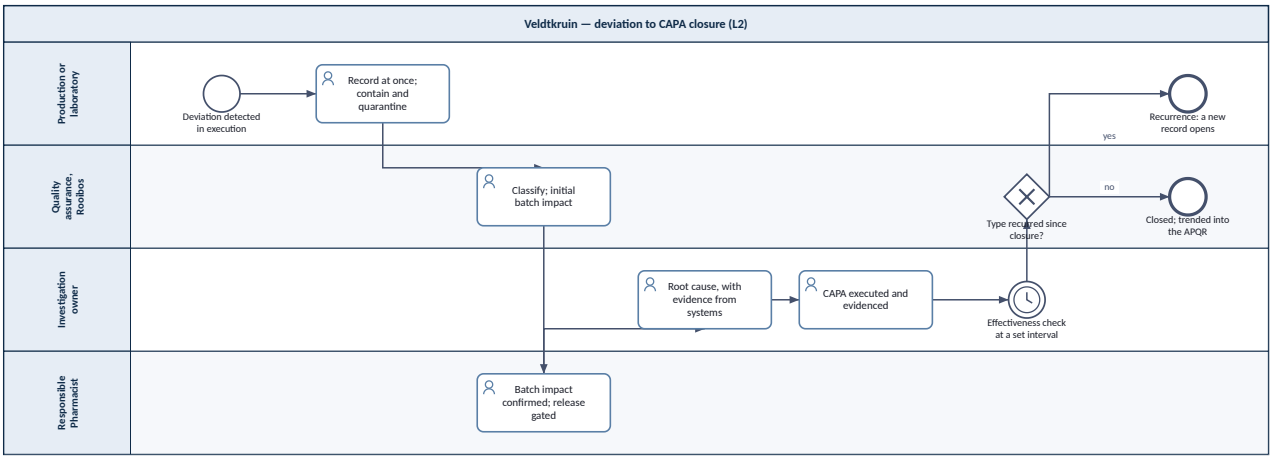


Figure 8.5 Deviation to corrective and preventive action. The effectiveness check is the step everyone drops, and **the only real measure is whether the deviation type recurred after closure**. Note the second end event instead of a loop back.

A deviation is detected — in execution, in the laboratory, in the warehouse, on inspection or from a complaint — recorded and classified with an initial batch impact, contained, investigated to root cause with evidence drawn from the contributing systems, and then a corrective and preventive action is determined. Where the action changes a validated system, a process or a registered detail, a change control is raised.

Two steps carry all the architectural weight.

The batch impact assessment is the step that gates release. It is the same step as step 20 in the flow above, seen from the other side, and the two processes must share one record rather than two.

> **CAPA CLOSURE IS NOT THE SAME AS FIXED**
>
> The effectiveness check is the step everyone drops. Where it exists at all it is frequently trivial — "confirmed with the department that the procedure was updated" — which checks nothing.
>
> **The only real effectiveness measure is whether the deviation type recurred after the action closed.** That is a query, not a form field, and it needs the deviation taxonomy to be stable enough to group by. A quality system with thousands of actions raised and closed on time, and no reduction in recurrence, is producing paper rather than control. Ask for the recurrence query in the first workshop. If nobody can run it, you have found the finding.

Note how the drawn flow ends. There is no loop back from the effectiveness check to the investigation, because a backward flow between two nodes in the same lane would overlay the forward flow and the page becomes unreadable. The honest modelling choice is **a second end event** — "recurrence" — which is also closer to what actually happens, since a recurrence is a new record with a reference to the old one rather than a rewind.

Change control

The process that makes every other GxP change possible, and the one that stalls programmes.

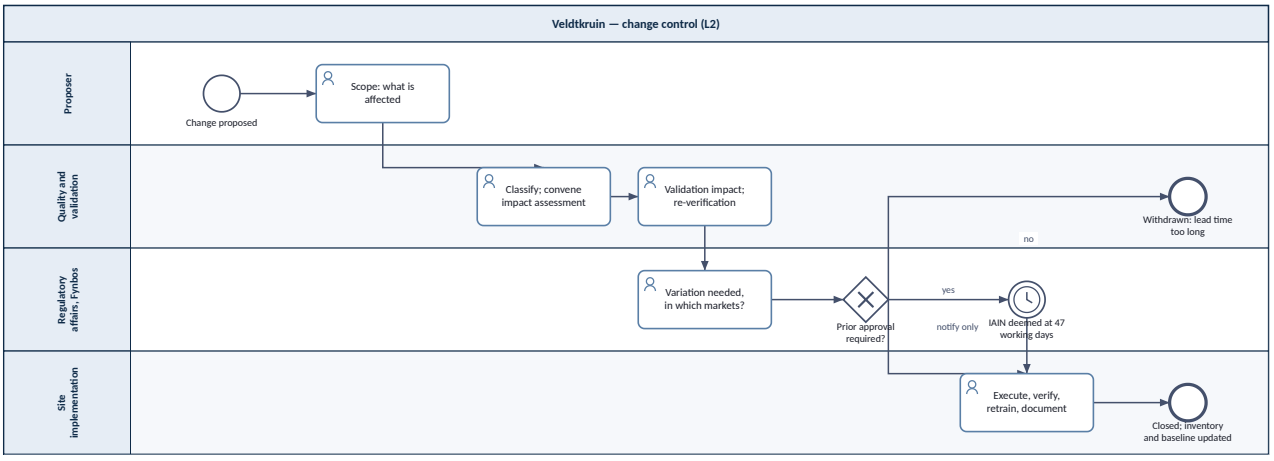


Figure 8.6 Change control: the process that makes every other GxP change possible and the one that stalls programmes. The timer is there because **a Type IAIN variation is deemed implementable after 47 working days of regulator silence**.

A change is proposed from anywhere — engineering, quality, IT, supply chain, a corrective action, a supplier notification, a software vendor's release. It is classified and scoped: what is affected, across equipment, process, facility, computerised system, specification, supplier, label and registered detail. Then the impact assessments run in parallel: GMP impact on product quality and the validated state; validation impact and what re-verification is proportionate; **regulatory impact — does this require a variation, and in which markets**; data-integrity impact on audit trail, access, retention and migration; and safety and labelling impact.

> *The hand-off failure that matters most:* ***the regulatory impact assessment is the one most often skipped, or done from memory****, by somebody who does not have the registered state in front of them. Years later a pre-approval inspection or a market audit finds that the running process is not the registered process. This is the master failure of the regulatory domain, and it is silent while it accumulates.*

Then the coupling that stalls programmes. **In many markets the change cannot be implemented until the variation is approved**, so the technical change and the regulatory change are one dependency rather than two parallel ones. The drawn flow therefore has a timer, not only an approval event — because a **Type IAIN** variation is deemed implementable after 47 working days of regulator silence, and a workflow built as "await approval" will hold that change forever while somebody implements it off-system.

The flow closes with execution, verification, documentation and training, and then the step nobody budgets: **the validated-system inventory and the periodic-review baseline are updated.** Skip it and the inventory is wrong, which means the answer to the first question an inspector asks — which of your systems hold GxP records — is wrong too.

Study start-up to first patient in

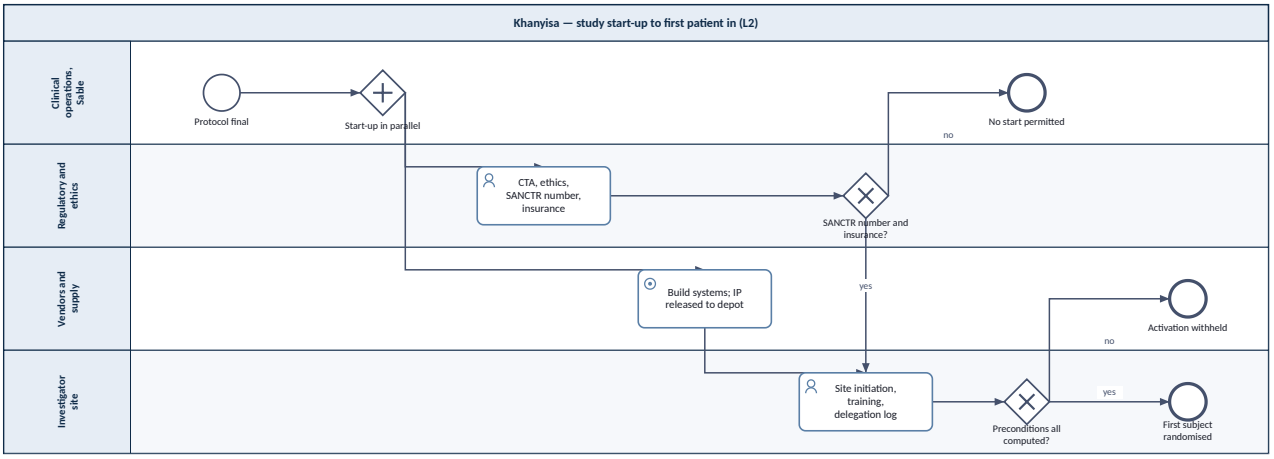


Figure 8.7 Study start-up to first patient in. **No trial may start without the SANCTR Department of Health number**, and clinical trial insurance is a rejection gate, so both sit in the activation precondition set.

Protocol final, then three things in parallel: regulatory and ethics submissions per country; the study built in systems — electronic data capture database and edit checks, randomisation scheme and supply strategy, outcome instruments licensed and configured, the trial management record and site plan, the trial master file structure and completeness model; and investigational product manufactured, labelled, released and shipped to depot. The drawn flow folds the vendor build and the supply release into one lane so that the page stays at four lanes and seven columns.

> *Hand-off failures: **four vendors building from four versions of the protocol**, which is the single commonest start-up defect. Label expiry and randomisation-system expiry disagreeing, so sites either destroy usable product or dispense expired product. And a site user active in the data capture system before their training record exists, which is an attributability and delegation finding.*

Two South African gates must be on the page, because missing either stops the study.

> **NO TRIAL MAY START WITHOUT THE SANCTR NUMBER, AND INSURANCE IS A REJECTION GATE**
>
> **No trial may start without the South African National Clinical Trials Register number issued by the Department of Health.** It is separate from the regulator's authorisation and separate from ethics approval, and it is a precondition rather than a formality.
>
> **And clinical trial insurance is a rejection gate** — an application without it is rejected rather than queried. Both belong in the activation precondition set as hard, computed conditions.

The last gateway in the drawn flow is the architectural question of the whole process. **Is site activation a computed state derived from verified preconditions, or a person ticking a box?** The preconditions are all knowable: authorisation received and filed with its date, ethics approval per site with its date, contract executed, training records complete, delegation log signed, product released and received, users provisioned. If activation is computed, a site cannot be activated early and the trial master file holds the evidence by construction. If it is a checkbox, then the specific test an inspector applies — **was a subject enrolled under a protocol version the site had not yet had approved** — becomes a question the company cannot answer quickly.

Subject screening to database lock

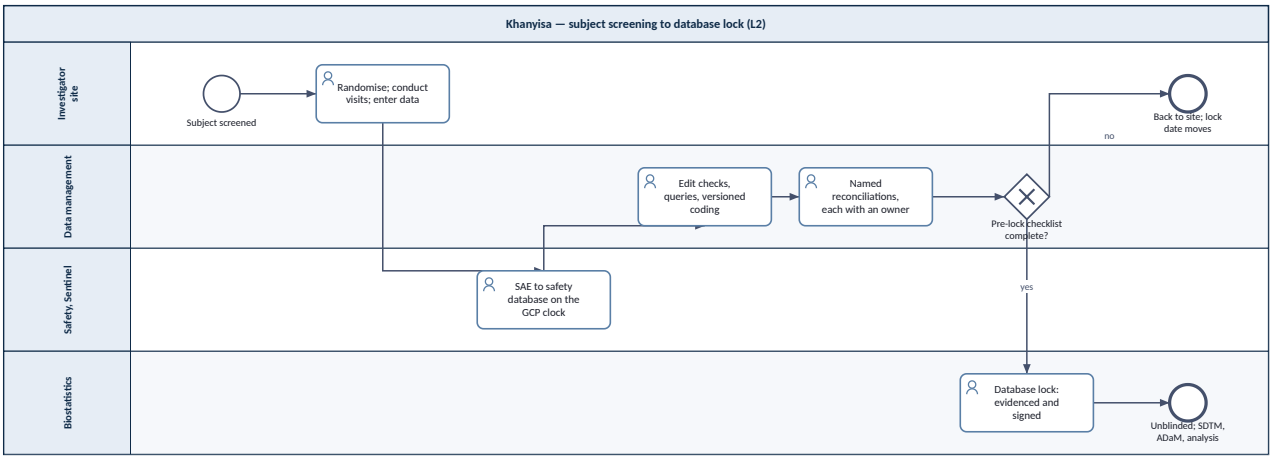


Figure 8.8 Subject screening to database lock. The reconciliations are each a named activity with an owner, and **an unlocked database is an architectural event rather than an administrative one**.

Screening with eligibility checks and recorded screen failures. Randomisation, where the randomisation system returns an assignment and a kit and the data capture system records the number rather than the arm. Visits conducted and data entered, with central laboratory and imaging data loaded rather than keyed. Edit checks firing, queries raised and answered. Serious adverse events reported to the safety database on the Good Clinical Practice timeline and into the data capture system on the site's own cadence — **two clocks on the same facts, which is why the two never reconcile without work.** Medical coding of events to MedDRA and medications to a drug dictionary, **with the version recorded**, because without the version the coding is unreproducible and re-coding historical data becomes a project rather than a refresh.

Then the reconciliations, and this is the part to insist on: **safety database to data capture; outcome instruments to data capture; randomisation system to data capture; laboratory to data capture; and drug accountability.** Each is a named reconciliation with an owner, and each is commonly done once, late, under time pressure, with discrepancies negotiated rather than resolved.

The pre-lock checklist is the computed gate: all queries closed, all reconciliations signed, all coding final, all external data loaded, all protocol deviations documented.

> **AN UNLOCKED DATABASE IS AN ARCHITECTURAL EVENT, NOT AN ADMINISTRATIVE ONE**
>
> A lock is a formal, evidenced event with a signature. The failure mode is the **soft lock**: the dataset is declared ready, the statistician starts, and then more data arrives. Every downstream output is now versioned against a moving base, and nobody can say which version of which table is in the submitted document.
>
> If a locked database is later unlocked, that is not a scheduling inconvenience. It invalidates the derived datasets, the tables, listings and figures, the analysis outputs quoted in the study report, and any submission content built from them. **Model the unlock as an event with downstream consequences and a required impact assessment.** The architecture question is whether the systems can tell you what was derived from the pre-unlock state. Usually they cannot.

Regulatory submission assembly to approval

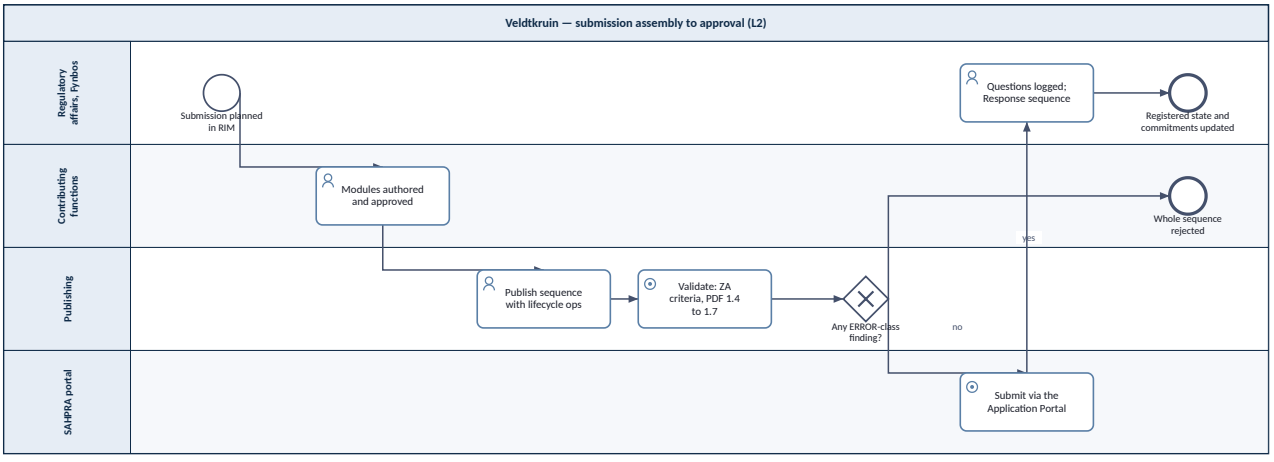


Figure 8.9 Submission assembly to approval. **South Africa is portal-only on format version 3.2.2, and an ERROR-class validation finding rejects the whole sequence** [4][5], which is why validation is drawn as a gate rather than a check.

Submission planned in the regulatory information system with type, markets, target dates and a content plan. Content authored or sourced — quality and manufacturing data, non-clinical, clinical, the summaries, and the regional module including the label. Content approved under electronic signature.

Then published: the sequence assembled with lifecycle operations — new, replace, append, delete — so that the cumulative view is correct.

South Africa's specifics are sharp and they belong on the page.

It is portal-only. The Application Portal is **the only valid method for submitting sequences** [4]. **It is on version 3.2.2 of the format**, per the national specification v3.1 of July 2024, which is based on the ICH eCTD 3.2.2 specification; **version 4.0 is not adopted**, with migration planned once other authorities have gained experience [4]. PDFs must be **version 1.4 to 1.7 with Fast Web View**, and earlier versions are an error [4]. Validation findings are classed ERROR, WARNING or INFO — and **an ERROR rejects the whole sequence** [4]. **Baselines start at sequence 0000, not 0001**, and an application not yet in the electronic format must submit Modules 1 to 5 as a baseline containing only previously submitted content, with an electronic declaration confirming that [5].

So pre-submission validation against the national criteria is a gate, not a courtesy — which is why the drawn flow has an explicit gateway with a rejection end event.

> *Hand-off failures: **lifecycle operations wrong** — a leaf marked new that should have been replace, so the cumulative view shows two versions of the same document and the reviewer reads superseded content. This is the characteristic error and it is **cumulative**: once the lifecycle is wrong, every subsequent sequence inherits the problem. **Validation run at the end**, on a sequence built over weeks, failing the day before the filing date on a PDF version or a file-naming breach. **Granularity decided late**, so a coarse-grained quality module means a small specification change republishes a large document. And the last one, which is the worst: **approval received and the registered state in the regulatory system not updated**, so the next change is assessed against a stale baseline.*

There is a practical consequence for any company filing in more than one region. The United States accepts 3.2.2 or 4.0, with 4.0 supported for new applications since 16 September 2024 [6], and Japan requires 4.0 for filings from April 2026. South Africa is 3.2.2 only. **A company filing in all three maintains two generations of the format at once, and the publishing tool must support both.** There is no cutover date to plan around.

Signal detection to label change

The chain that connects safety to the dossier, and the longest lag in the estate.

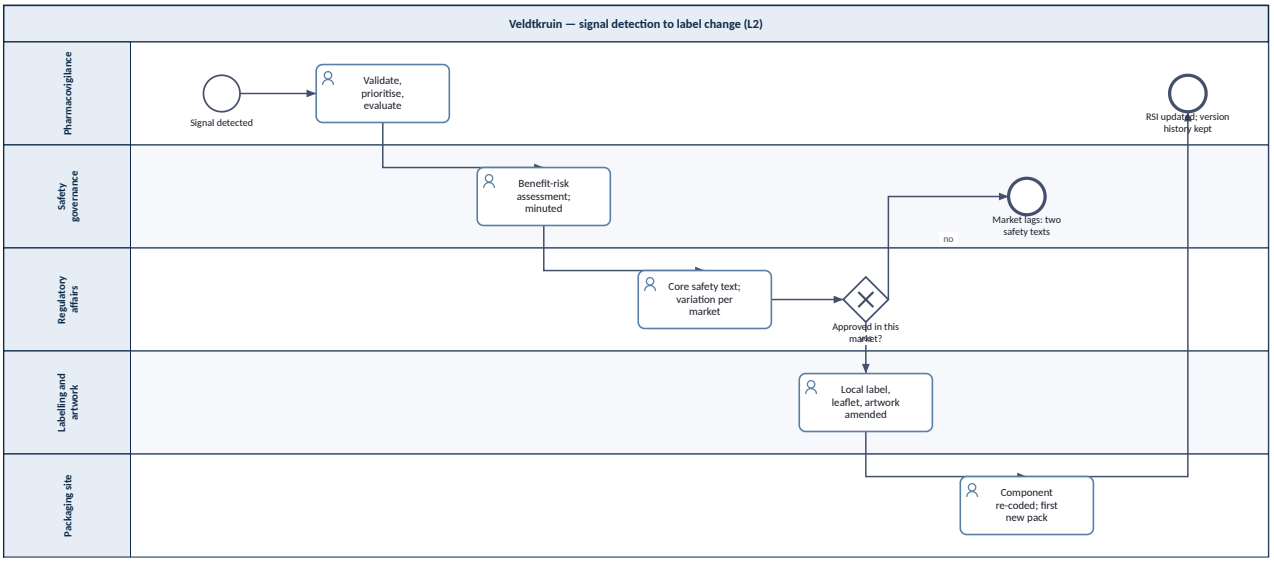


Figure 8.10 Signal detection to label change — the chain that connects safety to the dossier, and the longest lag in the estate. The second end event is the real-world one: a market lags and the same product carries two safety texts.

A routine detection run — disproportionality and time-to-onset over the safety database and external sources. The signal validated: is there a plausible new association, documented either way. Prioritised and assigned, opening the signal management record. Evaluated through case-series review, literature, an exposure estimate and biological plausibility. A benefit-risk assessment with a recommendation. A decision at a safety governance committee, **minuted, and the minute is inspectable.**

Then the long tail: core safety text drafted, variations prepared and submitted per market, approvals received on different dates, local labels and leaflets updated, then artwork, then packaging components, then the line.

> *Hand-off failures: **markets prioritised by commercial importance**, so small markets lag by years and the same product carries different safety text in different countries. **Text approved, artwork not updated**, so packs continue printing with the old leaflet — a classic recall cause. **Artwork updated, component not re-coded in the enterprise system**, so the production order consumes the old component number and the line runs old artwork with a new approval on file. And the step most often missed entirely: **the reference safety information is not updated**, which means expectedness assessments for new cases use the wrong version, and historical assessments stop being reproducible because the version history was not retained.*

The six described in prose

Artwork change to pack release. Triggered by a safety text change, a regulatory change, a pack change, a serialisation requirement or a brand change. Text approved per market, artwork brief raised with the approved text as the source of truth, artwork created, reviewed and proofed — **automated text**

comparison against the approved text, not against the previous artwork, plus braille, barcode and data-carrier verification — then approved under electronic signature and frozen, released to the printer with a version acknowledgement, and the packaging component created or revised in the enterprise system with an effective date. **That last step is the most common break in the whole chain**, and the printer's own prepress version quietly becoming the operative one is the second. The architecturally interesting part is the collision at the end: a regulatory text change with a short implementation deadline meeting an artwork and print lead time measured in weeks forces either a stock write-off or an unapproved pack in the market, and both decisions get made under pressure without any system that models the choice.

Serialisation commissioning to market report. Market requirement determined, numbers requested and allocated per site and per order, downloaded to the line and **buffered locally so the line keeps running if the link drops**, printed and verified with rejects decommissioned rather than reused, commissioned with the product identifier, batch and expiry associated, aggregated pack to case to pallet, reported upward to the enterprise system and outward to the destination market. Aggregation accuracy is the hardest operational problem and the one that generates partner disputes. **South Africa has no national verification scheme comparable to the European or United States regimes as far as the research could establish** [7], which means serialisation at a South African site is an **export capability scoped per destination market** — frequently present on some lines and not others. That is a very different design problem from a national mandate applied uniformly, and anyone scoping a serialisation programme for a South-Africa-only manufacturer should verify the current position before committing a budget.

Recall. The process that tests whether the estate's traceability is real. Trigger, assessment and classification, then the question that decides everything: **which batches, which markets, which customers.** That is a traceability query — batch genealogy upward to starting materials and downward to deliveries, warehouse records, serialisation event history where it exists, distributor records. **Where distribution is recorded at customer level and not at batch level, the scope cannot be narrowed and must be widened.** Then notification, where the contact list is master data nobody maintains; retrieval and reconciliation of quantity shipped against quantity returned per batch per customer, because the reconciliation percentage is what the authority asks for; disposition of returned stock; root cause and action; and a closure report. The periodic **mock recall** is the only honest test, and its architectural value is that it measures the time to produce a defensible distribution list — a direct read-out of integration quality. Two related facts: **where-used tells you which products could contain a lot; only genealogy tells you which do.** And destruction of recalled or expired product in South Africa is governed by **regulation 44, not regulation 27** [7], and requires prior permission from the regulator, a facility licensed under the waste legislation, a named witness with their designation recorded, and no disposal to municipal sewerage.

Annual product quality review. Scope and schedule set per product or product family; data extracted from the enterprise system, execution system, laboratory system, quality system, stability, calibration, supplier quality and the regulatory system; trends computed with control limits; reviewed and interpreted with manufacturing and regulatory input; conclusions on process consistency, specification appropriateness and validated state; actions raised as corrective actions or changes; approved and distributed, with the previous year's actions closed out. The characteristic failure is manual extraction on inconsistent bases, and the characteristic silent failure is actions that exist in the document only. Chapter 7 has the cadence and the cost; what matters here is that this is **a reporting and judgement activity over nine or ten other systems**, which makes it the best architectural stress test in the estate.

Inspection readiness. Notification received — or not, because an unannounced inspection is a real scenario. Scope anticipated. Document and data requests answered, and **the measure of readiness is the time to produce a named record**: a validated-system inventory, a batch record, a training record, a trial master file artefact, an audit-trail extract. Then system demonstrations, where an inspector sits at a terminal and asks to see an audit trail, a user list, a configuration setting and a deleted record. **A read-only inspector account with cross-system visibility is a design requirement, and it is almost never built in advance.** Observations received and responded to; remediation through action, change control and re-validation; post-inspection trending into the next cycle. One South African correction: **the regulator does not classify findings critical, major and minor with differentiated deadlines.** Its response guideline sets **one deadline, 30 calendar days, and no classification scheme** [7] — so do not design a response workflow with severity-driven due dates and then discover it has no basis. The architectural point is that readiness is not a project. It is a property of the estate, and every manual reconciliation in it is a place where the answer will be slow. **Slowness reads as loss of control.**

Compound registration to candidate selection. Structures proposed in silico, synthesis planned and executed in the laboratory notebook, the new entity submitted to compound registration where structure normalisation runs and a corporate number is assigned or matched, physical material recorded as a batch in a container, assays requested and plated, results captured from instrument to assay store to the structure-activity database, series progression reviewed, developability and early safety assessed, and then candidate selection as a governance decision. Three failures are worth carrying even though most South African companies will never run this process. **Normalisation rules disagreeing with the chemist's intent**, producing a duplicate or a wrong match. **Results attributed to the compound rather than to the batch**, so purity artefacts pollute an entire series. And the real one: **the series progression decision recorded only in a slide deck**, so the candidate data package with provenance to every cited result — the artefact that should exist — does not. The handover to development is then spreadsheets detached from provenance, and development regenerates what research already knew. **The compound registration system is the first authoritative identity for the thing that will eventually be a product, and it is almost never linked to the material master that will carry the same substance twelve years later.** That gap is a standing architectural finding.

Running a process-modelling workshop in a GxP environment

The mechanics are ordinary: the right people in the room, a wall, a clear statement of which level you are drawing, and a decision log. Four things are not ordinary.

Insist on the system at every step, by name and instance. Not "the LIMS" but which laboratory system, at which site, in which version, and whether it is the same instance as the one at the other site. In a pharmaceutical estate the answer is routinely that there are three, one per laboratory, and that is the finding.

Bring the validation lead into the first workshop, not the last. They know which systems are actually in a validated state and what a proposed change would cost in re-verification. Consulting them at the end converts a design into a re-work. They are also usually the only person who can tell you whether the interface you have just drawn is in anybody's validation scope, and the answer is frequently no.

Model the decision, not the approval. A step labelled "QA approves" tells you nothing. What is being decided, on what evidence, against what criterion, and what are the possible outcomes? This is the difference between a flow you can implement and a flow you can only admire.

And ask for the hand-off failures explicitly. The question that works is not "what goes wrong" but "when was the last time this step had to be redone, and why". People remember the specific occasion and will describe it accurately. They will generalise badly about causes and precisely about events.

> THE DOCUMENTED PROCESS IS A CONTROLLED DOCUMENT, SO THE GAP IS ITSELF A FINDING
>
> This is the trap that makes GxP process modelling different from process modelling anywhere else.
>
> In an ordinary organisation, the gap between the documented process and the actual process is interesting, slightly embarrassing, and the raw material of an improvement programme. In a GxP environment **the documented process is a controlled document, approved, with an effective date, and operating differently from it is a deviation.** The gap is not a finding you report; it is a finding an inspector would write.
>
> That changes how you ask the question, and you have to change it deliberately:
>
> - **Do not ask** "do you actually follow the procedure?" Nobody can answer that truthfully in a room with their quality manager in it, and asking it is close to asking somebody to self-report a deviation.
> - **Do ask** "where does the procedure not describe the situation you are in?", "which step takes longest and why", "what do you have to do that the procedure does not cover", and "what would you change if you could".
> - **Model to-be, not as-is, where the gap is likely.** An as-is model that documents an undocumented practice has created a record of a deviation, in your deliverable, with your name on it. A to-be model that incorporates the practice properly, routed through change control, fixes the same problem without creating the record.
> - **And agree the status of your own artefacts up front, in writing.** Is your process model an uncontrolled working document, or is it going into the quality system? If it is going in, it acquires an approval route, an effective date and a training obligation, and the project plan has to carry that. If it is not, say so on every page.
>
> Tell the client this in the kick-off, before the first workshop. An architect who discovers it in week six has already written the record.

Two closing points of craft. **Keep the models small.** Five lanes and seven columns is the practical limit of a page a human will read, and it is also, not coincidentally, the limit the house figure format imposes. A process that does not fit is usually two processes with an unresolved level-1 boundary, and splitting it is the right answer rather than shrinking the font.

And never draw a loop back into the same lane. Rework is real, and the honest way to model it is a second end event — "recurrence", "back to site", "held pending investigation" — because that is what the record actually shows. A rewind arrow suggests the process went backwards. It did not. A new record was created with a reference to the old one, and the audit trail will show exactly that.

THE CHAPTER IN TEN LINES

- Everything that is architecture is ArchiMate 4; level-2 process models are BPMN 2.0, because ArchiMate has no swimlanes and in a process model the hand-off is the point.
- Every BPMN lane must correspond to an ArchiMate Role or an application component that exists in the model, or the process library and the repository drift apart.
- Decide the level before you draw: level 0 for executives, level 1 for process owners, level 2 for implementers, and level 3 is a controlled document you should not be writing.
- Batch release is a join across at least six systems whose batch-number formats differ, so normalising that key is the highest-value integration task in the estate.
- The release decision has a precondition set the architecture should compute rather than present, and release lead time is an architecture outcome.
- In South Africa the releasing authority is the Responsible Pharmacist against a named release checklist, not an EU Qualified Person.
- Day zero for an adverse event is first receipt by any employee including sales representatives and contractors, so every employee is an intake channel.
- SAHPRA does not require E2B(R3): its guideline says reports can also be submitted in e2b format and names no revision, trials need CIOMS plus its own form, and the channel is a mailbox.
- Foreign individual case reports are not routinely forwarded; what runs on the five-day clock is foreign regulatory action.
- CAPA closure is not the same as fixed, and the only real effectiveness measure is whether the deviation type recurred after closure.
- Change control couples the technical change to the regulatory one, and a Type IAIN variation deemed implementable on silence after 47 working days needs a timer rather than an approval event.
- No trial may start without the SANCTR Department of Health number, and clinical trial insurance is a rejection gate.
- An unlocked database invalidates every derived dataset and output built from the locked state, so model it as an event with consequences.
- South Africa is portal-only on eCTD 3.2.2 with whole-sequence rejection on an ERROR finding, and baselines start at sequence 0000.
- The documented process is a controlled document, so the gap between documented and actual is itself a finding, which changes how you ask the question in a workshop.
- Never draw a loop back into the same lane: use a second end event, because that is what the record actually shows.

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- **7.** Internal research files for this volume, compiled 24 October 2026 — the fourteen end-to-end processes with their systems and hand-off failures, the system-class failure registers, and the South African corrections register (destruction under regulation 44 rather than regulation 27; a single 30-calendar-day inspection-response deadline with no critical, major or minor classification; no national serialisation or verification scheme found). Each item there carries its own primary citation or an explicit unverified marker

CHAPTER 9

The conceptual data model

Twenty-odd business objects, no attributes and no system names — and most of them are already decided for you, in regulation, with dates attached. This chapter works them one at a time and says what breaks when each is wrong.

What a conceptual model is here, and what it is not

A conceptual data model is business entities and the relationships between them. No attributes. No keys. No system names. The moment a box carries a field list, the workshop that was going to settle what a thing *is* has become a design review.

That restraint is harder here than anywhere else in this series, because here the argument is about the concepts rather than the implementation. The business holds four meanings for "product", three identifiers for one trial participant and two retention clocks for one record — and most of those are settled in regulation, with dates on them. Get this layer wrong and you produce a model a regulatory affairs manager will not recognise as true.

Every object below is worked the same way: its definition, its **grain** — one row is one *what* — its identity, its lifecycle, which system may assert it, and **what goes wrong when the model is wrong**. That last one is where the value lives. Anyone can list the entities; knowing that collapsing salt form into parent structure makes dose calculations wrong by the salt factor, and that it looks like a rounding difference, is what an architect is paid for.

Four rules run through the chapter and will not be repeated.

- **A pharmaceutical company's data model is four models wearing one coat**: science, regulatory, supply and quality. They share vocabulary and almost no keys, and nearly every large failure here is an attempt to make one the master of another — most often letting the enterprise resource planning system's material master define "product".
- **Almost nothing is a value; almost everything is a dated, scoped, authored assertion.** "This batch is released" is a person, on a date, on evidence listed, under an authority they held then, for a named market.
- **Coded data carries the version of its code system, or it means nothing.**
- **Irreversibility is a first-class attribute.** Database lock, batch certification, submission of a sequence and commissioning of a serial can be reversed in the world and never erased from the record. A model that implements them as status updates has failed.

> The regulated object in this industry is not the product. It is **the record about the product**. A batch that was made correctly and whose record is incomplete is, in regulatory terms, a batch that was not made correctly, and there is no evidential route back. Everything below follows from that one sentence.

The object that breaks most models: "a product" is four things

This is the error that costs most, and the one an architect arriving from banking or retail makes in the first week. Ask a commercial director how many products the company has and you get one number. Ask regulatory affairs and you get a number ten or twenty times larger. Ask the plant and you get a third. None is wrong: they are counting four different things by one word.

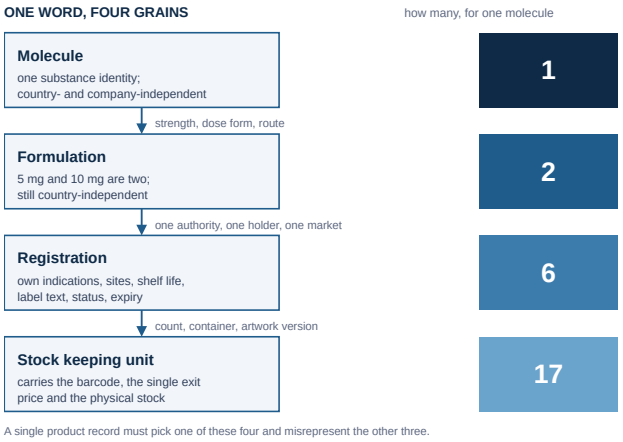


Figure 9.1 The four identities of a product. One molecule, two formulations, six registrations, 17 stock keeping units — and the business calls all four of them "a product". Counts are illustrative, for the invented Veldtkruin Pharmaceuticals.

Work the cardinality out loud, because the shape is the lesson. **The molecule** is one substance identity, country- and company-independent. **The formulation** — a strength, a dose form, a route — is country-independent too, but there are several per molecule: 5 mg and 10 mg are two, immediate release and modified release two more. **The registration** is one permission, one authority, one holder, one market, so one formulation in nine countries is nine registrations differing in approved indications, sites, shelf life and label text — and one can be suspended while the other eight are not. **The stock keeping unit** is a pack, and it sits at the fourth level and nowhere else: put it at the third and the warehouse cannot pick, put it at the second and the regulator cannot be answered.

> A single product record as the unit of everything will be rebuilt. Not refactored — rebuilt, because the grain cannot be split afterwards without re-keying every transaction that referenced it. If you are early enough in an engagement to insist on one thing, insist on this.

Veldtkruin Pharmaceuticals, the invented generics manufacturer this volume uses throughout and about which nothing is real, has 184 registrations and 498 stock keeping units across roughly 70 molecules, while its commercial report says "about 90 products". The dimension nearly always missing, even where the four levels are right, is **the dated link from pack to registration** — which answers "was this pack legally saleable in this country on that date", and that question arrives with a lawyer attached.

The science objects

Compound, substance, molecule — and the registration problem

A **compound** is a chemical or biological entity identified in the first instance by *structure*. It is not a product, not a material, not a batch, and not one thing with one name for its whole life.

Grain. One row is one distinct chemical entity as defined by the registration rules you chose, and choosing without knowing you chose is the common error. Mature registries hold three levels — a parent for the structure as drawn with no counter-ion and no solvate, a form for the salt, solvate or polymorph, and a batch — because biological activity belongs to the parent, solubility and dose belong to the form, and analytical results belong only to the batch.

Identity is enforced by canonical normalisation and a uniqueness check at registration: normalise the drawing for tautomer, charge, stereochemistry and counter-ion, then hash-match against existing parents. Two consequences. **Identity depends on the normalisation rules, so changing the rules changes identity retrospectively** — tighten stereo-perception three years in and structures that were one parent become two, and a registry that cannot say which rule version produced an identity cannot defend it. And **structure is not always known**: biologics, extracts and cell therapies are registered by definition rather than by graph, and the identity rule for a biologic is "same process, same source, same sequence", which makes **the process part of the identity** — the root of why a biosimilar is not a generic.

The registration problem is that the research identifier and the regulatory substance identifier solve different problems. The research number is company-internal, structure-derived and uniqueness-enforced; it stops two chemists registering one molecule twice. The regulatory substance identifier is global, authority-curated and definition-derived; it lets two regulators agree that two dossiers concern the same substance. Neither can do the other's job: no regulator accepts a key only you can resolve, and the regulatory identifier does not exist yet for a novel entity. So **the map between them is a first-class entity**, many-to-many and dated — and the dated part is what nobody builds, because external identifiers are reassigned, merged and deprecated by their issuing authorities. A safety database that resolved a substance code in 2019 and re-resolves it in 2026 can get a different answer with no change on its own side.

Batch against substance is the other half: the substance is the kind, the batch is the instance, and a model that lets a result point at a substance rather than a sample cannot say *which sample* when the result will not reproduce. The costly failure, though, is **the enterprise resource planning system minting substance identity**, because the first material master row for an active ingredient was created by a procurement clerk.

The clinical objects

Study and trial — the protocol as a controlled document

A **study** is a planned investigation with a protocol, conducted at sites, enrolling subjects, generating observations against a defined schedule. The study is not the protocol, not the database and not the project; those three describe it, hold its data and fund it.

Grain. One row is one study as identified by its protocol identifier. Not one per protocol version, not one per country submission, not one per database. Everything else hangs off it: protocol versions, study countries, sites, subjects, arms, epochs, planned visits, the schedule of assessments.

Identity. The protocol identifier must be immutable from first regulatory contact, because every external party keys on it — ethics committees, the regulator, the national register, the safety database, the laboratory, the randomisation system, every contract research organisation. A study also carries a registry identifier per public register, and in South Africa the Department of Health number, **without which the trial may not start** [7]. Those are dated external references, not alternate keys.

Four sub-objects are modelled wrongly in predictable ways.

- **A site is not a hospital.** It is one principal investigator at one location under one set of approvals, and one hospital can hold two sites in one study. The grain that gets lost is the delegation log — site, role, person, from-date, to-date — which is what a good clinical practice inspector asks for.
- **An arm is a planned treatment sequence**, and a subject's actual treatment is a different fact. Collapse the two and the analysis population is undefinable.
- **A visit has a planned identity and an actual occurrence.** Visit 3, week 12, window plus or minus three days belongs to the protocol version; the date it happened belongs to the subject. One visit table carrying both is the commonest clinical-data modelling error, and after it the schedule cannot be re-derived for an earlier version.
- **The schedule of assessments** is the matrix of which procedure happens at which visit for which arm under which protocol version. It generates the expected-data inventory, which makes "missing data" computable rather than an opinion.

Protocol amendment, which invalidates downstream structures

The usual instinct — "a document version, and we have document versioning" — is exactly wrong. The protocol does not merely describe the study. It **defines the structures the data conforms to**. Under South African good clinical practice an investigator may not implement a change without sponsor agreement and both ethics-committee and regulator approval, the one exception being a change to eliminate an immediate hazard, which must then be promptly documented [7].



Figure 9.2 Protocol amendment as a data event. One approved document version, and six downstream structures that have to be re-generated, re-computed or re-determined. Document versioning records the event and loses its meaning.

> **Protocol amendment is a data event, not a document revision.** Amend the schedule of assessments and every expected-visit row generated for visits not yet reached is void, along with the missing-data computation and the site payment schedule derived from it. Amend visit windows and out-of-window detection changes backwards as well as forwards. Amend an endpoint and the analysis plan, the analysis datasets, the derivations and the sample size all move. Amend the consent form and every enrolled subject acquires a determination about whether they must re-consent.

Four rules handle it. Everything the protocol defines is versioned **by protocol version**, not globally — there is no single current schedule of assessments, there is one per version with an effective date. Every subject carries the version they are treated under **as a dated series**, because subjects transition at different dates depending on when their site got approval and when they next attended. Collected data points carry the version in force when they were collected. And the amendment is an entity with a per-country, per-site approval fan-out, because "was this subject's visit conducted under an approved protocol version at that site on that date" is answered only from that table — and it is the first thing an inspector reads.

Subject — the most sensitive record the company holds

A **subject** is an individual enrolled, or screened for enrolment, in a study. The subject record in the sponsor's systems is deliberately **not the person**: a pseudonymous shell carrying identifiers, arm, dates and observations, with the link back to the human held separately.

There are at least three identifiers and they are routinely confused. The **screening number** is minted at consent, before eligibility is known, one per screening episode. The **randomisation number** is minted at randomisation, only for eligible subjects, and links to a treatment assignment and to kits. The **subject identifier** is the key used in the clinical database and in the submission.

Three traps follow. **Screen failures have screening numbers and no randomisation numbers**, so a model keyed on the randomisation number loses the population the enrolment funnel needs. **A re-screened person has two screening numbers and may have one randomisation**, so the screening number is not a subject key. And **the subject identifier must be unique across the whole submission, not the study** — the standard construction concatenates study, site and subject, because per-study uniqueness fails once datasets are pooled, and that is discovered at submission.

Then the identifier the sponsor should not hold: the site's own link from subject number to the patient's name and contact details. That is **the re-identification key**, and where it lives is the most consequential privacy decision in a clinical architecture. The site holds that log in the investigator site file and the sponsor never receives it.

> **Pseudonymised data is not outside the Protection of Personal Information Act.** Only permanent de-identification is, and South Africa's draft code of conduct for research advises researchers to assume that re-identification is possible [12]. Pseudonymised trial data in a sponsor's hands remains personal information, because the sponsor plus the site can re-identify. An architect who presents pseudonymisation as a compliance exit is wrong, and the error is load-bearing: it changes the cross-border position, the breach impact and the lawful basis.

Why the most sensitive record in the company? Identity, health condition, genetic data, biological samples, sometimes HIV status — held for ten years or more, shared with a contract research organisation, a laboratory, a biostatistics vendor, a safety vendor and at least one foreign regulator, across borders, under a consent given once, in a system the sponsor will replace twice before the retention clock runs out. It is the one entity where sensitivity, retention duration, cross-border exposure and the number of onward recipients all peak together.

Clinical data point — the grain and the audit trail

A **clinical data point** is one recorded observation about one subject at one time; the form, the dataset and the domain are containers for it. Clinical data is tabulated into **domains** — two-letter codes grouped into observation classes: interventions, events, findings, plus special-purpose and trial-design domains. **The grain is one row per observation**, and in the findings class one row per subject per test per timepoint, with the test named in one column and the result carried both as collected and as standardised.

Source data verification — the monitor comparing data in the capture system against the source document at the site — is an assertion about a data point, with its own grain. Since risk-based monitoring verifies only some points, *which* points were verified is itself evidence.

The audit trail is the system's record of every create, change and delete: who, when, old value, new value, reason. South African good clinical practice requires computerised systems to be access-restricted, requires changes to be documented automatically, and states that **no entered data may be deleted** [7]. So the clinical data store is **append-only at the value level**: a changed value is a new row with a coded reason and the previous row stays addressable. The trail must be readable independent of the application, because the ten-year retention clock outlives it. And **audit-trail review is an activity with its own records**, which inspectors ask for and which is routinely absent.

Database lock and unlock

Database lock is the point at which the clinical database for a study is declared complete and closed to change, so the analysis can run on a fixed dataset. It follows last subject last visit, query resolution, coding completion, reconciliation of serious adverse events against the safety database, external data reconciliation and a data review meeting.

> **A locked database that is later unlocked is an architectural event, not an administrative one.** The lock is the moment the database stops being a transaction system and becomes evidence, so **the locked state is an object** — study, lock identifier, lock timestamp, scope, authoriser, record counts, a manifest, and the extract produced from it. Without it there is no way to show the analysis dataset came from the locked state rather than from a later one.

Databases do get unlocked, legitimately. Unlock is a controlled change with a named requester, reason, scope, approver and planned re-lock, and **the pre-unlock snapshot is never overwritten** — a study with three unlocks has four locked states and four lineages, and "the database" stops being a meaningful phrase. If unlock happens after unblinding the integrity argument is materially different, because the people making the changes now know the treatment assignments.

The failure is specific and fatal: the sponsor is asked at inspection to reproduce a table from the clinical study report and cannot, because the database has moved and no immutable locked state was kept. Most data-integrity findings are recoverable; this one is not.

The safety objects

Adverse event and the individual case

An **adverse event** is any untoward medical occurrence in a patient or trial participant administered a medicinal product, **whether or not it is causally related** — the South African clinical trial safety guideline uses exactly that construction [9]. An **adverse drug reaction** adds causality: at least a reasonable possibility [9]. A **serious** event results in death, is life-threatening, requires or prolongs hospitalisation, causes persistent disability or a

congenital anomaly, or is otherwise medically important [9]. An **individual case safety report** is the exchangeable record of one patient's adverse experience at one point in time.

The grain is deep and nearly always collapsed: **case**, then **case version** — initial, follow-up one, follow-up two — then **event** within that version, then **coded term** under one dictionary version. One row per case is wrong, because cases are reported repeatedly as information arrives. One row per event is wrong, because one case has many events, each with its own seriousness and causality per suspect product. And the point of follow-up is that attributes change between versions.

> **The safety database is a bi-temporal store** — what we believed, as at when we believed it. Anything less cannot answer the only question that matters at a pharmacovigilance inspection: what did you know, and when did you know it?

And **expectedness is the trap**, because it is not a property of the event. South Africa's post-marketing guideline states that a fatal reaction is unexpected unless the South African labelling specifically says the reaction may be fatal [8]; the clinical trial guideline makes the same move against the product information or investigator's brochure [9]. So expectedness is computed from the coded term, the reference document, **the version of that document**, and the country. The same event can therefore be expected in one market and unexpected in another, which is why one case produces different obligations per country. Store an assessment without the document version it was made against and every historical reporting decision becomes indefensible.

The dictionary-version problem

This is the most under-modelled thing in pharmaceutical data, and it is worth the space.

The terminology used to code adverse events, medical history and indications for regulatory purposes is hierarchical: a lowest level term the verbatim maps to, a preferred term that is the analysis unit, then high level terms, high level group terms and system organ classes. Standardised queries are curated groupings used to find cases of a defined medical concept. **It releases twice a year, in March and September.** Version 29.0 came out in March 2026 [10] and version 29.1 later that year, its training material dated August 2026 [11]. Each release adds and modifies terms, and — the part that matters — demotes terms, merges terms and changes the primary system organ class of existing ones.

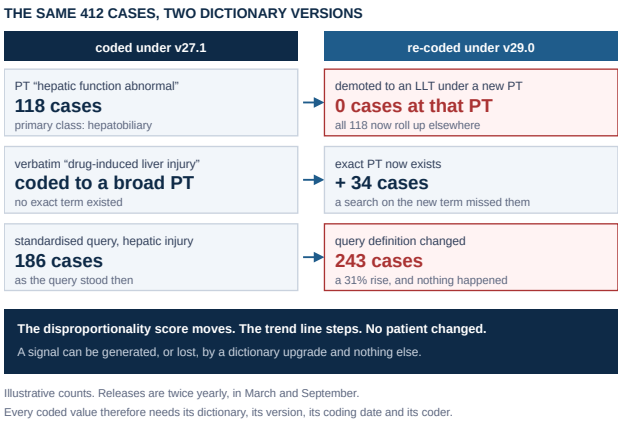


Figure 9.3 The dictionary-version problem, worked. The same cases under two dictionary versions. A demoted term, a newly available term and a changed standardised query, and three different numbers about one unchanged population.

Five mechanisms. A verbatim with no good match two years ago may have an exact preferred term now, so the old case sits on a less specific term and a search on the new one misses it. A preferred term can be **demoted** beneath a different one, so every case coded to it rolls up elsewhere and a trend line across that boundary steps for no clinical reason. A **primary class change** moves counts between classes in every class-level table ever produced, including submitted ones. **Non-current terms** are flagged rather than deleted, so filtering to current terms silently makes historical data un-matchable. And **query changes** are the mechanism by which **a signal can appear or disappear because of a dictionary upgrade and nothing else.**

> **Re-coding historical safety data is a project, not a refresh.** Full up-versioning at each release is the most expensive option, requires medical review of every changed mapping, and means previously submitted tables no longer reproduce from the live database unless the old coding is kept. Forward-only coding is cheapest and leaves several versions in the database at once, so every aggregate query must be version-aware — the usual state, and it must be *designed for*, not discovered. Keeping every coding generation and presenting a chosen version is the correct answer, and almost nobody does it from the start.

What the model needs is short to state. Every coded value carries its dictionary, its version, its coding date, its coder and its method; every aggregate report declares the version it was run under; and **the dictionary itself is a stored, versioned asset in the estate, not a library call to a vendor service**, because a report produced in 2023 must be reproducible in 2029. The drug dictionary used for concomitant and suspect medications has the same shape and the same problem.

Signal, the aggregate view, and the chain to a label change

A **signal** is information suggesting a new potentially causal association, or a new aspect of a known one, judged likely enough to justify verification. It is not a case, not a confirmed risk and not a label change. South Africa's guideline requires the holder to establish signal management as part of its

pharmacovigilance system, and significant safety issues go on a named notification form [8]. **Grain:** one product or substance, one medical concept, one signal identifier — the concept expressed as a term or a standardised query **at a stated dictionary version**.

Detection runs on an aggregate view structurally different from the case store, and **it needs an exposure denominator that is not in the safety system** — it comes from sales or shipment data converted to exposure estimates, so the safety datamart depends on commercial data and on the pack-to-product map. That is the most awkward integration in pharmacovigilance. The view must also be dictionary-version-coherent and reproducible as at a date, so it is snapshotted per reporting cycle rather than rebuilt in place.

The chain an architect should be able to draw from memory: cases received, coded at a dictionary version, aggregated with exposure, detected by a named and dated method, raised in the signal register, prioritised, assessed, then refuted or confirmed. If confirmed, the risk is characterised, a safety variation goes in **per market**, approvals land on different dates with different wording, the label is updated per market and version-dated, expectedness is recomputed for future cases, and risk-minimisation measures are implemented **and their effectiveness measured.**

The fan-out is where it breaks. One global signal becomes several market variations approved on several dates, so for a period there are two truths about what is expected, and the safety system must hold label versions per country with effective dates. And the signal register is often a spreadsheet, so the company cannot show an inspector the **population** of signals — including the refuted ones, which are what the inspector most wants, because a refuted signal with no documented basis is the finding.

The regulatory objects

Submission against dossier state

A **submission** is an assembled, transmitted package of documents and data sent to a regulator for a purpose. The electronic common technical document is its structure: five modules plus an XML backbone declaring every file, its location and its lifecycle operation.

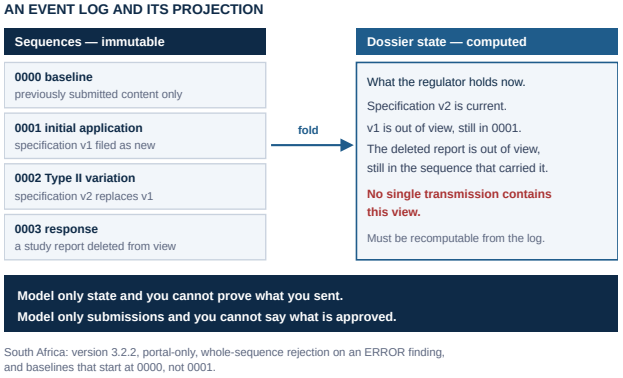


Figure 9.4 A submission is an assembled artefact; a dossier is a maintained state. The sequences are immutable events; the current view is a projection that must be recomputable from them.

Three grains are routinely collapsed. A **sequence** is one transmission, numbered, to one authority, for one application, **immutable once sent**. An **application or dossier** is a maintained, cumulative state: the current view. A **dossier content item** is one document in one module location, whose history is a chain of lifecycle operations — new, replace, append, delete. **Nothing is ever actually deleted**: delete removes a document from the current view, and the sequence that carried it stays intact forever.

> **The current view of a dossier is a computed projection over the sequence history** — a fold over an append-only log. The dossier is an event-sourced aggregate, the sequences are the events, the current view is a materialised projection, and the projection must be recomputable from the log.

The South African specifics are concrete and an architect will be asked for them. The specification is version 3.1 of July 2024 and is **based on version 3.2.2** of the international standard; it notes that version 4.0 exists and that South Africa intends to migrate later, and gives no date [4]. Submission is **portal-only**, and a finding of type ERROR **rejects the whole sequence** before anyone reads the science, so validation is a gate in the publishing pipeline and its results are records [4]. New applications start at sequence 0001 — but **a baseline from non-electronic content starts at 0000**, must be a separate sequence, and may contain only previously submitted content; and on transfer of an application the acquiring applicant's sequence must follow on and **must not restart at 0001** [4].

The submission is an event; the dossier is a state. A system that models submissions and not dossier state cannot answer "what is currently approved". One that models state and not submissions cannot answer "what did we tell them, and when". Both are required, and the second gets built first because it is the one the business asks for.

Commitments

A **commitment** is an obligation accepted in or alongside an approval: a post-approval study, a stability commitment on production-scale batches, a risk-management activity, a periodic report, a condition of a conditional registration. **Grain:** one commitment, to one authority, for one application, with its own due date, owner, discharge evidence and status. It is **not** an attribute of the authorisation, because one authorisation can carry a dozen with different owners on different dates. The **basis of the due date** matters: a date computed from an event such as first commercial batch needs that event to be observable in some other system.

Marketing authorisation — why one product is many authorisations

A **marketing authorisation** — in South Africa a certificate of registration — is a permission, granted by one authority, to one holder, to place one product on one market, subject to conditions. **Grain:** one product, one authority, one holder. What marketing calls "one brand in 40 countries" is 40 authorisations, diverging in registration number and date, approved indications, strengths and pack sizes, **approved manufacturing and testing sites**, approved specification and shelf life, product information per language, conditions, and status.

Three facts an architect should not guess.

Registration does expire. It is valid for five years, renewal must be submitted **six months before expiry**, products registered from October 2022 carry an expiry date on the certificate, the renewal programme is phased **2023 to 2035**, and failure to submit a complete, compliant renewal may result in expiration, after which a new registration application is required [5]. Renewal also requires an up-to-date variation history [5], which makes the variation register the evidence for a five-yearly legal event rather than a convenience.

Variations are not all prior-approval. Type IA is notify within 12 months of implementation. Type IB runs 87 working days, after which changes may be implemented though the regulator may still require reversal. Type II requires prior approval and is reviewed within 220 working days. And **Type IAIN requires at least 47 working days' notice before implementation, and if the regulator gives no opinion the variation is deemed implementable** [3].

> **"Deemed implementable" inverts how an approval workflow is normally built.** The absence of a response is itself the event the system must detect and act on. A workflow that waits for an inbound approval will wait forever, and the business will implement the change anyway, out of band, with no record. Model the clock, not the message.

The same change fans out into many variations. Changing an active ingredient supplier is one manufacturing decision and a prior-approval variation in every market that names that manufacturer, approved on different dates — so for a while the product is legally different in different markets while being physically identical, and the batch-release decision has to know that. **Lapsing is the state nobody models**, because it happens by inaction.

Identification of medicinal products as the identity layer

This standards family gives regulated products a structured, exchangeable identity: an authorised-product identifier, a country-independent pharmaceutical-product identifier, a substance identity layer, controlled vocabularies for dose forms, routes, units of presentation and packaging, and units of measurement.

The reason to care is structural rather than compliance-driven. It forces a company to state, in one place and in a controlled vocabulary, what it has always held implicitly across five systems: which substance, in which strength and dose form, in which pack, authorised to whom, in which country, manufactured where. **Every one of those facts already exists in the estate, and in most companies no two systems agree on all of them.** So: *this is a master data programme with a regulatory deadline attached.*

> Write the status honestly. **This has no go-live date.** What exists is a phased sequence of deadlines in one region, **two of which have already been extended**. And the edition years of the individual standards could not be sourced for this book — **only the substance standard, ISO 11238, was confirmed by name from an agency source.** Do not print edition years for the others, and do not repeat deadline dates from a vendor blog; cite the agency.

The South African connection arrives from an unexpected direction: the traceability guideline requires that product master data under this family be provided during new registration, or during amendments affecting its attributes, through the regulator's portal [1] — so it is in scope through the traceability route rather than the submission route.

The supply and quality objects

Batch and lot — genealogy and the electronic batch record

A **batch** is a defined quantity of material produced in one process or series of processes such that it is expected to be homogeneous. It is the fundamental unit of pharmaceutical traceability, quality and liability, and everything physical is batched: the active ingredient, each excipient, the carton, the bulk tablet, the blister, the finished pack.

Grain. One batch of one material at one site. A batch repacked is a new batch linked to its parent; a batch split across markets is one batch with several dispositions; and a bulk batch filled into several pack configurations yields several finished batches from one bulk, which breaks models assuming batch numbers are preserved up the chain. The guide South Africa has adopted requires each batch to be considered separate for sampling, testing and release [6], which forecloses testing one and releasing many. The batch number is on the label and in the barcode, so it is public and immutable from first printing.

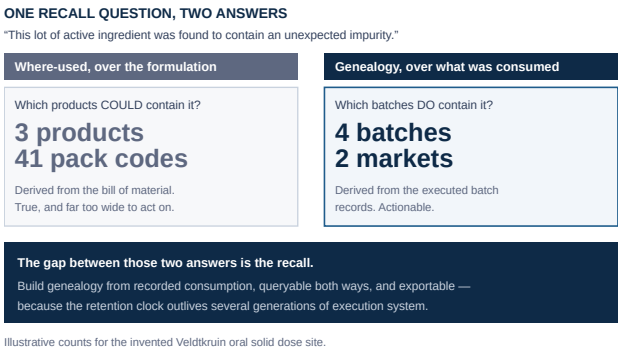


Figure 9.5 Where-used against genealogy. Where-used tells you which products could contain a lot. Only genealogy over recorded consumption tells you which ones do.

> **Where-used tells you which products could contain a lot; only genealogy tells you which do.** Where-used runs over the formulation and answers at the level of a product line. Genealogy runs over recorded consumption in the executed batch records and answers at the level of four batches. The difference is the difference between recalling a product and recalling four batches, measured in tens of millions of rand and in patient access.

Genealogy is a directed acyclic graph with quantities — parent batch, component batch, quantity, unit, operation, timestamp, equipment, operator — built **from the executed batch record, not from the bill of material**. It must be queryable in both directions, because the forward query serves recall and the backward query serves investigation, and resolvable for as long as the batch record must be retained — which means **the genealogy outlives several generations of manufacturing execution system**.

The **electronic batch record** is the executed instance of a **master batch record**: the master is one approved, versioned, effective-dated instruction for one product at one site, the executed record is one batch, append-only, every entry attributable, timed and signed. **When a batch starts, the master is copied into the executed record**, so a later change to the master does not alter what was done. A model that points the executed record at the current master has destroyed the evidence — the most serious data-integrity finding available.

Disposition, retest date and expiry date

Disposition is the recorded decision about what a batch may be used for. It is not a quality result: it is a decision taken on evidence, by a named person, with authority, at a time, for a scope — and the state set runs from quarantine through under investigation, approved for further processing, certified for sale, released with restriction, rejected, recalled and destroyed.

Three design rules. **Market-specific release is the normal case, not an exception** — specifications and approved shelf lives differ per authorisation, so the grain of the release decision is batch and market. **The release record must name the person, the checklist, the inputs considered and the date**; in South Africa the Responsible Pharmacist must certify, in a register or other appropriate record before release for sale, that the batch meets its registration conditions [13], and the good manufacturing practice guideline sets out those duties [14]. And **states are append-only**: a batch that goes quarantine, released, recalled has three dated rows.

	Expiry date	Retest date
Applies to	Finished products and materials with a shelf life	Active ingredients and some excipients
Meaning	After this date the material may not be used	After this date it may be re-tested and, if conforming, used for a further period
Extendable?	Only by an approved shelf-life variation	Yes, by retesting, within the rules
Modelled as	A date on the batch	A date **plus a retest history**, each retest producing a new date

Retest is a repeating event with its own results and its own date arithmetic, and a single expiry-date column cannot hold it. Treat retest as expiry and the company scraps active ingredient it could have used; treat expiry as retest and it uses material it should not have.

Specification, test, method, result and certificate of analysis

Five objects, each with its own grain, and the commonest quality-systems modelling error is to collapse them into a results table with a limit column. A **specification** is one version of the tests and acceptance criteria for one material, one purpose and one market. A **test** is one measurable property. A **method** is one version of the analytical procedure, with its own validation and transfer records. A **result** is one sample, one test, one method version, one determination. A **certificate of analysis** is immutable once issued, and a re-issue is a new version with a reason.

Specification versioning is where the authorisation meets the plant. Any change to accepted standards or specifications of a registered product should be submitted for prior approval [3], so the entity is a specification version with its approving variation, and the comparison of result to specification must use the version in force **for that market at the time of that batch's release**. Almost no system does this, and the symptom is a batch re-assessed years later against today's specification, reaching a conclusion that is not the one that was reached.

The adopted guide requires a procedure for investigating out-of-specification and out-of-trend results [6]. **Out-of-specification** investigation runs in two phases, a laboratory investigation conducted *before* the original data is invalidated and then a full investigation extending to manufacturing. The original result is never deleted, any retest is a new result with its justification recorded **in advance**, and invalidating the original must rest on a documented laboratory cause. **Out-of-trend** is statistical, so its trend-model parameters are themselves versioned.

Stability

A **stability study** stores samples of a batch under defined conditions and tests them at defined timepoints, to establish shelf life, retest period and storage statement.

The grain is a five-way cross product — study, batch, storage condition, timepoint, test — so three batches by four conditions by eight timepoints by 15 tests is 1,440 scheduled determinations for **one** product, over a long-term arm of 36 to 60 months. **It is a schedule before it is data**, so a missed timepoint is an event. **It spans method versions**, so the result must carry the method version. And **it justifies a dated claim in an authorisation**, so the chain from stability result to proposed shelf life to variation to approved shelf life to the expiry on every future batch must be traceable.

> Storage conditions are regulated reference data with a climatic-zone dimension, and this is exactly the detail an author asserts from memory and gets wrong. **South Africa's climatic-zone assignment and the long-term storage condition the regulator requires for registration stability data could not be sourced for this book**; both are marked unverified in the research behind it. South Africa spans more than one climate and the assignment is not derivable from geography. What is safe to say: **European stability data may not cover a South African filing**, so check the condition against the current registration stability guideline first.

Deviation, corrective action, change control and complaint

Four objects, one family. They cross-reference each other, and conflating any two produces a quality system that cannot be reported on.

Object	Grain	Closes with
Deviation	One departure from an approved procedure, specification or expectation	An investigation, a root cause, a product-impact decision, a disposition
Corrective and preventive action	One action plan, with tasks	Implementation **plus an effectiveness check**
Change control	One planned change, assessed before implementation	Implementation, verification and closure
Complaint	One report from outside about a product unit or batch	An investigation, a reply, a decision on reportability and recall

Two attributes are almost always missing. On a deviation, **affected batches as a proper many-to-many** — a list that can grow during investigation. On a change record, **the regulatory impact assessment**: does this change require a variation, and of which type? That is the most valuable cross-reference in a pharmaceutical quality system and the one most often absent, because change control lives in the quality system, variations live in the regulatory system, and nobody owns the join.

The effectiveness check is the step everyone drops. A corrective action is complete not when the action is done but when it is shown to have worked, after a defined interval, against a defined measure. The reason is mechanical: the record reaches "implemented", the implementer closes it, and the check falling due 90 or 180 days later is owned by someone else and has no transaction to attach to. So make it a **separate dated obligation**, make closure impossible without it, and report overdue checks rather than closure rate. Tellingly, **the regulator's own guideline on responding to an inspection report does not mention an effectiveness check at all** [16].

One complaint attribute carries disproportionate weight: whether it is a suspected adverse reaction. If it is, **it is also a pharmacovigilance case and both clocks run**. The intake process must fork, and a complaint system that cannot fork to safety is the commonest cause of a missed expedited report.

Material — and the approved-supplier question

A **material** is a kind of thing bought, held, consumed or produced. The active substance is named in the authorisation and so is its manufacturer; **primary packaging is part of the product's quality**, so a change to it is usually a prior-approval variation; and for secondary packaging **the artwork is the regulated content**, version-controlled per market and per language, which makes printed components the most change-prone materials in the estate.

Three distinct relationships are then routinely conflated into one "approved vendor" flag. **The approved manufacturer** is the site that makes the material; this is the one named in the authorisation, and changing it is a variation in every market that names it. **The approved supplier** is the entity you buy from, which may be a distributor or broker; qualification here is about supply-chain integrity and documentation, not chemistry. And **qualification is a property of the triple** — material, manufacturer site, receiving site — with a date range, an audit basis and a review date. The join nobody builds is **the list of authorisations that name this manufacturer for this material**, which turns "we are changing supplier" into a costed regulatory programme instead of a surprise.

Serialised unit

A **serialised unit** is one physical saleable item bearing a unique identifier. Serialisation turns the batch — a set — into a population of individually addressable members, with a hierarchy: item or retail pack, bundle or inner pack, case or trade item, and pallet or logistics unit.

Aggregation is the recorded parent-child relationship between a container and its contents: a **time-varying tree**. Disaggregation — opening a case to pick individual packs — breaks it, and the common error is to store the current parent as a field on the child, which cannot represent a unit that was in case A, then loose, then in case B. **The event log is the model; the current tree is a projection.**

Commissioning brings a serial into existence as a valid identifier; **decommissioning** retires it — dispensed, destroyed, exported, stolen, recalled. Commissioning is irreversible as a record: a serial once commissioned is never un-issued, even if the pack is scrapped. The regulator's traceability guideline states that the combination of trade item number and serial must be unique forever and can never be reused [1], and **"unique forever" is a data-architecture requirement with teeth**: it forbids any scheme whose space can be exhausted and recycled, and makes the serial generator a system of record with its own reconciliation obligations.

> **"South Africa requires unit serialisation with national repository reporting" is wrong three ways.** It is a **phased programme running 2025 to 2031**, **batch-level first** and unit-level later [1, 2]. It **specifies no transactional or verification repository** — what is required is the submission of *master data*, including trade item and location numbers, for the regulator's health product registration database [1]. And **imported products are staged two years ahead of domestic ones**: batch level at primary and secondary packaging from 1 October 2027 for imported and 1 October 2029 for domestic, unit level from 1 October 2029 for imported and 1 October 2031 for domestic [2]. That reverses the usual assumption. The guideline requires the relationship between identifiers at different packaging levels to be captured in the manufacturer's own electronic systems [1] — it does **not** impose a parent-child scan requirement.

Site and facility — the licence as a dated object

A **site** is a physical location where a regulated activity happens. In the regulatory model a site is not a building. It is **a licensed capability at an address**, and the licence enumerates what may happen there.

The facts that matter hang off the site at finer grains, each an entity rather than an attribute: the **licence** with its number, scope, dates and conditions; the **licence scope line**, because a product may only be made at a site whose scope covers its form; the **named person**, with **only one Responsible Pharmacist permitted per licence** [13]; the **inspection** with its findings; and the **authorisation reference**, one pair of site and authorisation where that site is named.

The licence is a dated object and the derived field is the whole point. A licence to manufacture is valid five years from issue, the holder must apply for renewal **at least 180 days before expiry**, and changes to the application particulars must be reported within 30 days [13]. Renewal therefore falls due six months before the date anyone has in mind, and a lapse stops production.

Import is narrower than architects expect. Medicines may enter South Africa only through **four named ports** — the Cape Town, Port Elizabeth and Durban airports or harbours, and O.R. Tambo International Airport — and **each consignment needs a signed MBR 20** lodged with a customs officer at the port, with the invoice, licence and permit attached, and also with the regulator [15]. Schedule 5 to 8 import permits are valid **six months** against a five-year licence [15]: two clocks, two renewal processes.

Inspection history is almost always a folder of PDFs; the structured minimum is an inspection entity with findings linked to their corrective actions. One correction: **the regulator's response guideline does not define a critical, major and minor classification scheme with differentiated deadlines.** It refers to deficiencies, non-conformity and serious non-compliance, and sets **one deadline, 30 calendar days from receipt of the report** [16].

The object most architects leave out: the validated system itself

Here is the claim, stated hard. **Most enterprise architects do not model the validated system as an object, and then cannot answer "which of our systems hold records subject to good practice regulation?" — the first question an inspector asks.** It is also the first question from a privacy assessment, a disaster-recovery review, an IT audit and any cloud-migration business case, so the absence is expensive four times over.

A **validated system** is a computerised system whose fitness for its regulated purpose has been documented, tested against specified requirements, approved, and maintained under change control for its operational life. **Validation is not a property of software.** It is a property of a configured instance of software, used for a defined purpose, in a defined process, by a defined organisation.

Grain. One system instance, in one environment, for one defined use — not one per application, not one per server, not one per vendor. Three traps: a productive instance and a test instance are different rows; a multi-tenant hosted instance is one row from the customer's side and carries a dependency on the vendor's quality system, which is a supplier-qualification object rather than a validation object; and **an interface between two validated systems is itself in scope and is frequently nobody's row.**

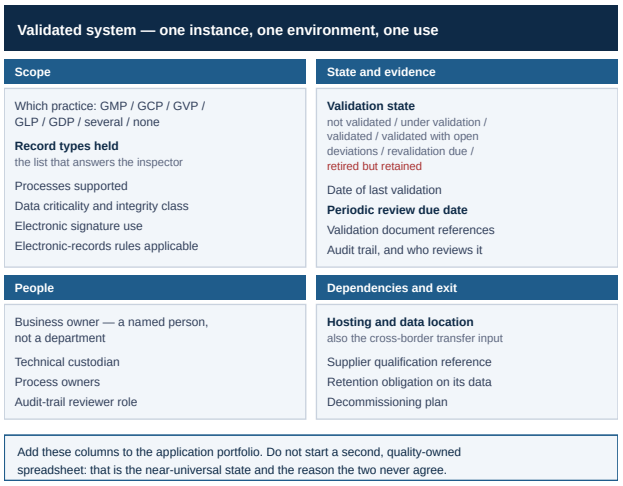


Figure 9.6 The validated-system object. The attribute set most application portfolios do not carry, and the reason "which of our systems hold regulated records?" takes three weeks to answer.

> The register for this object should be **the same register as the application portfolio**, with those columns added — not a separate quality-owned spreadsheet. The separate spreadsheet is the near-universal state and the reason the two disagree. If an architect owns the application portfolio and the quality function owns a validated-system list, the organisation has two inventories and no answer.

Controlled vocabularies as architectural dependencies

The vocabularies this estate depends on are **licensed products with versioned release cycles**, which makes them architectural dependencies rather than reference data. The adverse-event terminology releases twice yearly by subscription and is not redistributable [10, 11]. The drug dictionary releases on 1 March and 1 September by subscription, and **the United States regulator mandates it for studies starting after 15 March 2019** [17] — a paid product made mandatory by a regulator. Pharmacopoeial monographs are paid subscriptions too, and **the monograph edition is part of a specification's identity**.

Two boundaries the model must carry

Everything above is a construction. These two are prohibitions, and they belong in the conceptual model because they are statements about what must not be joined.

Trial data and commercial data must not be joined. Trial data is pseudonymised subject data governed by the protocol, the informed consent, good clinical practice and the ethics committee's approval. Commercial data is customer, script and dispensing data governed by privacy law and competition law. A "single patient view" spanning both is a consent and competition-law breach, not a feature. An architect who proposes it has produced a finding rather than a design, and the right move is to say so in the first workshop — the idea is attractive, it will be raised, and it is cheaper to kill early.

One enterprise platform spanning clinical, safety and commercial is wrong here, for four separate reasons with four different remedies.

Reason	What it is	Remedy
Blinding	Commercial access to kit-level or arm-level data breaks the blind	Access control plus report design
Personal health information	Subject and patient data is special personal information under South African law	Minimisation, pseudonymisation, access control
Promotional bias	A commercial function with unpublished trial outcomes can act on them	Information barriers and timing controls
Pharmacovigilance independence	Safety assessments must not be influenced by commercial interest	Organisational separation reflected in system access

The model, drawn

Here it is as one picture: the objects above as business objects, with the relationships that matter. The version that earns its keep is the one that fits on a page and that a regulatory affairs manager can correct in front of you.

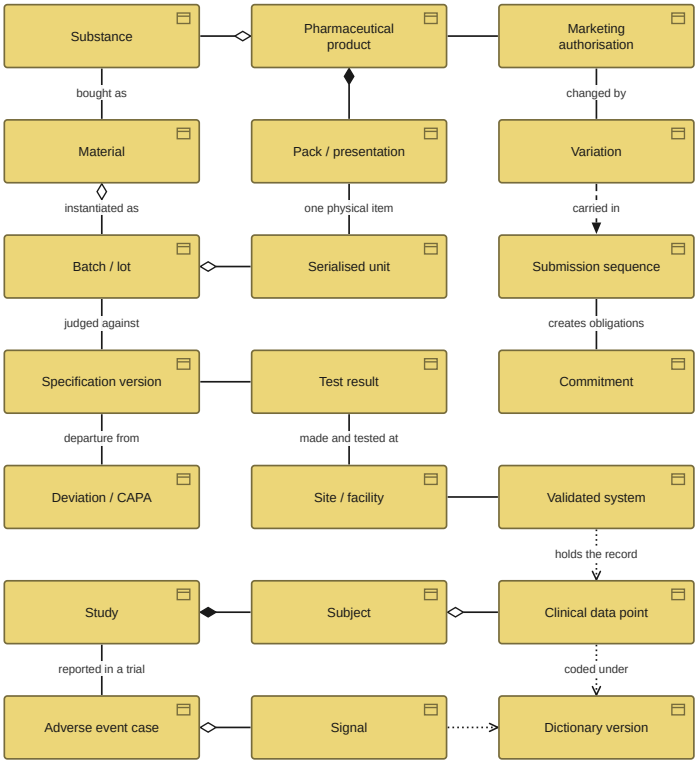


Figure 9.7 The conceptual model on one page. Twenty-one business objects and the relationships that matter: science at the top left, regulatory down the right, supply and quality through the middle, clinical and safety at the foot — and one dictionary-version object that both safety and clinical data depend on.

Draw it with your client's own object names. If the regulatory affairs manager says "we would never call it that", change the label and keep the box. If they say "that is two different things", you have found the next section of the model.

THE CHAPTER IN TEN LINES

- A conceptual model here is entities and relationships only; the moment a field list appears, the argument about what a thing is has been abandoned.
- A product is four things with different cardinality: molecule, formulation, registration and stock keeping unit, and a single product record will be rebuilt.
- Protocol amendment is a data event that invalidates downstream structures, and the per-site approval fan-out is the table an inspector reads first.
- Pseudonymised trial data is still personal information; only permanent de-identification is outside the Act, and the research code says to assume re-identification is possible.
- A locked clinical database that is later unlocked is an architectural event: keep every locked state, never overwrite one, and make every downstream artefact declare which it used.
- Every coded value carries its dictionary and its dictionary version, or the aggregate safety view cannot be defended, and a signal can appear from an upgrade alone.
- A submission is an event and a dossier is a state; model both, because each answers a question the other cannot.
- Where-used tells you which products could contain a lot; only genealogy over recorded consumption tells you which do.
- Model the validated system itself, in the application portfolio and not in a second spreadsheet, because its absence is expensive four times over.
- Trial data and commercial data must not be joined, and one enterprise platform spanning clinical, safety and commercial is wrong for four separate reasons.

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CHAPTER 10

The information model

Chapter 9 asked what the things are. This one asks how they are held, moved, mastered and reported on — in an industry where master data is a control parameter, data integrity is a design constraint, and a regulator has already decided part of your model.

What changes when you go logical

Chapter 9 asked what the things are. This chapter asks how they are held, moved, mastered and reported on — and that is where pharma stops resembling the industries an architect has already worked in.

Three differences drive everything below. **Master data is not housekeeping here; it is a set of control parameters**, so changing one is a regulated event. **Data integrity is not a policy; it is a design constraint** with named attributes, each of which obliges something specific of a system. And **the data standards at the regulatory interface are mandatory, dated and machine-enforced**, which means parts of the logical model are decided for you, by a regulator, before you start.

Nothing here is legal advice. Every regulatory passage below frames the obligation as *what the record has to show*, never *whether you comply*. The second question belongs to an admitted attorney or the regulator.

Before any of that, though, there is a short answer to "what does the logical layer add here". It adds six storage patterns, and almost every entity in chapter 9 lands on one of them. Knowing which is most of the design.

Pattern	What it holds	Which chapter-9 objects use it
Append-only value store	A new row per change, with a coded reason, the previous row still addressable	Clinical data points, test results, disposition states
Bi-temporal store	What we believed, as at when we believed it	Safety cases and their versions
Event log with a projection	The events are the truth; the current state is computed and recomputable	Submission sequences against dossier state; serial aggregation against the current tree
Versioned definition with an effective date	One row per version, never an update in place	Protocol versions, specification versions, method versions, label versions
Dated assertion	Who said what, about what, on what date, under what authority, for what scope	Batch release, source-data verification, causality, adequacy determinations
Immutable snapshot with a manifest	A frozen state preserved beside the live one	Database locks; the extract behind every submitted table

Two consequences fall out immediately. **Update-in-place is the exception in this estate, not the rule**, so a physical design built on mutable rows with a last-changed timestamp is wrong for most of the model and cannot be retrofitted. And **every one of those patterns needs its own query path**, because "the current value" and "the value as at a date" are different questions and a schema that answers only the first cannot be inspected.

Master data governance, and why it is a compliance matter

In most industries the master data argument is economic: duplicates cost money, broken hierarchies make reporting hard, somebody should own the supplier list. That argument is true here too and it is not the one to open with.

Open with "which of these fields are control parameters". That question gets quality and regulatory affairs into the room. "How many duplicate suppliers do we have" gets a data-quality dashboard nobody acts on.

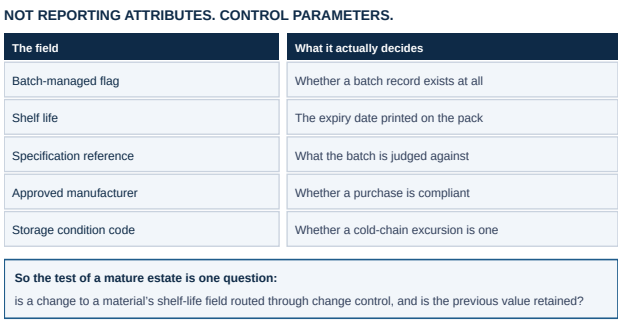


Figure 10.1 Master data as control parameters. Five fields that decide regulated behaviour rather than describe it — which is why a change to one belongs in change control rather than in a maintenance queue.

The ownership answers, object by object, are mostly uncontroversial once stated — and the wrong answer is always "whoever created the record first".

Object	Who owns creation	Who must approve	What the resource planning system may do
Compound or substance	Compound registration for internal identity; regulatory affairs for the external substance identity	Research informatics on structure rules	Cache only, never mint
Product hierarchy	Regulatory affairs	Commercial for brand grouping; supply for pack	Hold material and pack extensions keyed to the regulatory product
Site master	Quality and regulatory compliance	Supply chain; IT for location numbering	Hold plant and storage-location extensions
Supplier and manufacturer	Procurement for the commercial entity; **quality for the approved-manufacturer qualification**	Quality, and procurement cannot override it	Hold purchasing data only
Specification	Quality, derived from the approved dossier	Regulatory affairs for the approved limits	Reference it, never author it

Then the argument that makes this a compliance matter rather than a tidiness matter. It runs in four steps.

1. **Master data determines regulated behaviour.** The batch-managed flag determines whether a batch record exists at all. The shelf-life field determines the expiry printed on the pack. The specification reference determines what the batch is judged against. The approved-manufacturer record determines whether a purchase is compliant. Those are not reporting attributes.
2. **So a change to one is a change requiring assessment** — quality change control, not a maintenance ticket. The test of a mature estate: *is a change to a material's shelf-life field routed through change control, and is the previous value retained?* Usually not, and that is the finding.
3. **Master data must carry the same audit trail as transactional data**: who, when, old value, new value, reason. Most tooling records who and when. The old value and the reason are frequently absent, which makes the change unreviewable.
4. **Master data is the common cause of multi-batch, multi-market events.** One wrong master value affects every batch made after it, in every market, which makes a master-data error a *systemic* deviation rather than a batch deviation. The investigation then has to establish the value's effective-dated history — and a master record with no such history cannot support it, so the company cannot bound the impact.

> Under data-integrity expectations a master-data change is **an audited event**, which makes the governance workflow itself a validated process. If the approval workflow for a shelf-life change lives in an unvalidated ticketing tool, the control that protects the expiry date on the pack is outside the validated estate.

Data integrity as a design constraint, not a policy

The mnemonic is ALCOA, extended to ALCOA+. The mnemonic is not the point; **the design obligations are.**

Attribute	What it obliges of the design
Attributable	Unique individual accounts, no shared or generic logins, identity verified before issue — and the originating identity carried through every interface, so a record arriving from another system still names the person, not the service account
Legible	Durable formats, no proprietary binary that only a decommissioned version can open, readable renderings for the whole retention period, and the audit trail readable too
Contemporaneous	**The system clock is a regulated control**: synchronised, auditable, time zones declared, no local clock-setting rights; offline capture records the observation time, not the sync time, and makes the difference visible
Original	Identify the original explicitly — the instrument's raw file, not the printout — retain it, and document the verification of any copy used in its place
Accurate	Input validation, calculation verification, no silent rounding or unit conversion, reconciliation between systems, and error handling that fails loudly

Then the four that make it ALCOA+.

Attribute	What it obliges of the design
Complete	**No deletion.** Every injection including the failed ones, every trial run, every reprocessing, retained and visible. Deleting a bad result is the archetypal integrity fraud
Consistent	Chronological integrity across systems, one authoritative clock, event ordering preserved through queues — and no interface that reorders
Enduring	Archive strategy, media refresh, format migration, decommissioning plan. Not a laboratory computer's local drive
Available	Retrievable for review throughout the retention period, searchable, restorable — and the restore actually tested

> Two citation traps. **The PIC/S guide's Part I never uses the term "ALCOA"** [1]; its nearest statement is the documentation principle on accuracy, integrity, availability and legibility, plus contemporaneity and signed alterations. The detailed expectations live in the data-integrity guidance, PI 041-1 of 1 July 2021 [2], and in the equivalent national guidance [3]. **The verified South African ALCOA statement is ALCOA-CCEA, in the good clinical practice guidelines, for clinical source documents** [4] — attributable, legible, contemporaneous, original, accurate, plus complete, consistent, enduring and available. **That is a good clinical practice requirement, not a good manufacturing practice one.** A chapter that cites "PIC/S Chapter 4 ALCOA" is citing something that is not there in those words.

Audit-trail review is the operational burden nobody scopes. The obligation is not to *have* a trail. It is to review it — to look at recorded changes and satisfy a reviewer that nothing improper happened. A mid-size quality control laboratory running a chromatography system across a dozen instruments generates thousands of entries a week: every log-in, every method change, every integration adjustment, every reprocessing, every sequence edit.

Organisations cope in four ways, in increasing maturity. They do not, and the procedure says monthly while no review records exist. They review a documented sample, which is defensible if the sampling basis is risk-justified and indefensible if the percentage was chosen to fit the available hours. They review by risk-targeting: manual changes to results, changes to methods and specifications, deletions, changes to date and time settings, permission changes, overrides of out-of-specification flags. Or they extract the trail to a reviewable store with rules that surface the reviewable subset and a workflow that records each disposition — which works, and introduces a new problem, because **the review tool is itself a system handling regulated records and so needs validating.**

Three findings follow. Review is a **workflow** requirement, not a reporting one: the deliverable is a record that a competent, independent person looked at the right entries and concluded something, so a design that produces a PDF and no disposition record has built the wrong thing. Reviewer independence constrains the access model, because the reviewer must not be the person who performed the activity — which needs a role that can read across the laboratory without being able to perform laboratory work, and many systems do not have one. And **if review must complete before batch release, review capacity becomes part of the release lead time**, so a backlog is a supply problem and not only a compliance one.

> That last point is dated and unresolved, and must be written that way. **The version of the computerised-systems annex in force is still the January 2011 revision** [5]. A draft revision was consulted jointly from **7 July to 7 October 2025**, alongside a revised documentation chapter and a new annex on artificial intelligence, and **no final text or date had been published as at 24 October 2026** [5]. Commentary on the draft reports that audit-trail review becomes risk-based, targeted at manual changes, performed by someone not directly involved, and **normally completed before batch release** — which is where the release-path dependency comes from. Treat that as a draft reading, not a requirement, and re-check before you design a release gate around it.

The standards that actually constrain the data model

This is where pharma differs from every sibling industry in this series. Regulators mandate data standards as a condition of accepting a submission or a safety report, which means parts of the logical model arrive pre-decided, with dates attached and an automated gate at the far end.

WHAT IS MANDATED, AND WHAT IS ONLY PUBLISHED

Layer	Verified version, dated	Status
Collection what the form asks	not confirmed in this pass	published, not mandated
Tabulation submitted representation, one row per observation	**model v2.1, 10 Jun 2024** guide 3.4 pairs with v2.0 (29 Nov 2021)	**required at the gate** named datasets and tags; refuse-to-file consequence
Analysis traceable derivations	model stable; guide moves	**required at the gate**
Nonclinical animal toxicology	**guide v3.1.1**	required by one authority, not by South Africa
Dataset metadata the interpretability layer	**v2.1.11, 6 Apr 2026** x-releases, not a base 2.1	**required, every study**
Operational exchange metadata and audit trail	**v2.0, 23 Aug 2023** page contradicts itself	published; most estates still on 1.3

The authority's data standards catalogue — not the standards body's latest release — governs what you may submit. No version support matrix is printed here:
the catalogue could not be read for this book. Check it at the time of use.

Figure 10.2 The clinical data standards, by layer, with what is actually mandated. Verified versions and dates only. **No version support matrix is printed**: the authority's catalogue governs, and it could not be read for this book.

The clinical standards family splits by lifecycle stage. **Collection** standards govern what the case report form asks and what the field is called at capture. **Tabulation** governs the submitted representation, organised into domains. **Analysis** datasets are derived from the tabulation with the derivation traceable back to it. There is a nonclinical equivalent for animal toxicology, an operational exchange model, and a machine-readable metadata document that describes the datasets, variables, codelists and derivations — which is the thing that makes submitted data interpretable and is treated as an afterthought almost everywhere.

Four version facts an architect should carry, and none of them should be stated from memory.

- **The tabulation model and its implementation guide are different artefacts at different version numbers.** The model is at v2.1, published 10 June 2024; the implementation guide numbered 3.4 pairs with model v2.0 of 29 November 2021 [6]. "SDTM 3.4" names an implementation guide, not a model.
- **The nonclinical implementation guide is v3.1.1** [7], with separate extensions for reproductive, animal-rule and genetic-toxicology studies.
- **The metadata standard runs as x-releases**, currently v2.1.11 of 6 April 2026 [8]. Writing "v2.1" names a family, not a version.
- **The operational exchange model has a v2.0, published 23 August 2023, and the standards body's own page contradicts itself about it** — the prose calls an older 1.3.2 release the most current while the version table lists v2.0 as newest [9]. Cite the table, note the inconsistency, and expect most implementations to still be on 1.3.

> **The authority's data standards catalogue, not the standards body's latest release, governs what you may submit.** Support windows and requirement dates live in the catalogue, per standard, per version, and they close as well as open. The research behind this book could not read the catalogue — the file returned as binary and the site refused direct download — so **no version support matrix is printed here.** Read the catalogue and the standards pages together, at the time of use, and hold the per-study decision as metadata. Up-versioning mid-study is a project, not a configuration change.

What *is* enforceable is worth knowing precisely, because it is the clearest example in the series of a conformance contract on a data interchange. The United States authority's technical rejection criteria name specific datasets, a named metadata document and named file tags in named submission sections, enforced by refuse-to-file for new drug and biologic applications and refuse-to-receive for generics, for studies started after 17 December 2016 and, for commercial investigational applications, after 17 December 2017 [10]. **But the document the research could read lists some of the criteria's own effective dates as "to be determined"**, so the honest statement is that the criteria exist and are enforced — not that every one of them is live today.

> **Dataset-JSON has not replaced the long-standing transport format.** The authority requested comments, published 9 April 2025, closed 9 June 2025 with 42 comments received, and **made no change**; the notice describes the format as having long-term potential to replace the incumbent [11]. The open-source submission pilots are instructive here: one of them had to validate with the old format because the dominant conformance tool did not yet support the new one. Design the transport layer as **replaceable**, not as any one format's shape.

Safety exchange is the other mandated structure. The current generation is built on a healthcare-interoperability case-report standard, with explicit null flavours so that "not provided", "unknown" and "not applicable" are distinguishable, date precision as a property of the date rather than a coerced full date, structured causality per drug-and-event pair, and the dictionary version travelling in the message. Three design consequences run upstream: a source system storing a single null cannot populate it faithfully; an onset date is not a date column but a value with a precision; and a sender that cannot state its dictionary version per case cannot produce a conformant message. Acknowledgements are part of the structure, so **a case is reported when it is acknowledged, not when it is sent**, and an unacknowledged transmission is an open item with a clock still running.

Three dated facts about it. It has been **mandatory in the European Union since 30 June 2022, and that region's system does not accept the previous revision at all** [12]. **Its own implementation guide was at step 3 — public consultation — dated 18 July 2025** [13], so the standard is itself under revision. And **South Africa requires neither**: the post-marketing guideline says reports "can also be submitted" in that format using an XML file and **names no revision**, with only one submission method permitted per report [14], while for trials the guideline requires a council-format narrative form and the regulator's own annexure instead [15].

> "South Africa requires the current safety exchange revision" is the single most likely error in a South African pharmacovigilance design. It is the right thing to build towards. It is not, on this evidence, a local mandate — and the local channel is a mailbox rather than a gateway, which puts the whole confirmation, retry and reconciliation burden on the sender.

Two further version dimensions belong on the diagram and almost never appear. **Controlled terminology is versioned independently of the standards that reference it** — the dictionaries below, plus the code lists maintained alongside the safety message, each on its own calendar. And **conformance tooling is versioned independently of both**: the de facto validation tool's rule sets move on their own release cycle, which is how a sponsor ends up conformant to a standard and failing a check, or passing a check against a rule set two releases behind the one the reviewer runs. Three version axes, one submission.

One dictionary is mandated outright. The drug dictionary used for concomitant and suspect medications is a paid subscription product, released on 1 March and 1 September, and **the United States authority mandates it for studies starting after 15 March 2019** [16]. That is the clearest case in this book of a commercial licensed product made compulsory by a regulator, and it has consequences for contracts, for non-production environments and for any outsourced coding arrangement.

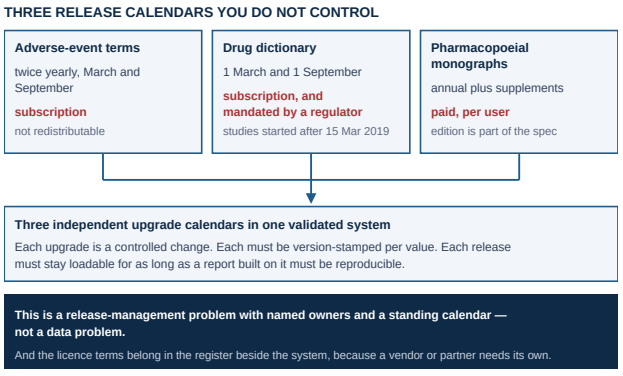


Figure 10.3 A dictionary is an architectural dependency, not reference data. Three independent release calendars inside one validated system, each a controlled change, each licensed by somebody else.

Healthcare interoperability enters at three places: real-world data ingress from health systems and registries, electronic product information work, and — newly — the regulatory product-data interface itself. The version position is counter-intuitive and gets misquoted constantly. **Release 4, version 4.0.1 of 27 December 2018, carries the normative content. Release 5, version 5.0.0 of 26 March 2023, is described by the standards body as "the current published version" and is trial-use**, with no release-5 content normative except what was already normative in release 4 [17]. The publication-history page **does not mention a release 6 at all**, and is **internally inconsistent about the 4B date** [17]. So integrate against release 4 for anything that must be stable, and say which release you mean every time.

Serialisation brings its own keys, and the keys are the integration contract: a trade item number identifies the item, trade item plus serial identifies the pack, a shipping container code identifies the logistics unit, and a location number identifies the party and the place. A company whose resource planning system does not hold trade item and location numbers as first-class master data cannot serialise, and discovering that is usually the first month

of a serialisation programme. The event standard for supply-chain events is at **release 2.0, which added JSON, JSON-LD and REST bindings alongside XML, and is published as ISO/IEC 19987:2024** [18] — so "it is XML" is out of date. Two parties both "using 2.0" may still disagree on event types, business steps, extensions, transport and master-data pre-exchange, so onboarding is per partner.

> **Never write that the standards body requires a two-dimensional carrier by 2027.** The 2027 milestone is a **voluntary industry capability ambition** of one national member organisation, with no day or month, described in its own material as a milestone and as a staged "crawl, walk, run" progression, and explicitly not requiring all capabilities by that date [19]. Pharmaceutical trade press has run headlines of the form "won't be optional for brands". That is the distortion. Carrier obligations for medicines come from national traceability regimes, not from the keys body.

The reporting layer, and the reconciliations that are always wrong

Everything above is held so that somebody can be answered. This is what they ask for.

The flagship is the **annual product quality review**: one review per product per year, per site, taking previous reviews into account, which the adopted guide says should normally be conducted and documented annually for all authorised products [1]. Two things about it are routinely misunderstood. It is **a document with a dated obligation, not a dashboard.** And its inputs span batch records, deviations, out-of-specification results, complaints, returns, recalls, stability, changes, validation status and contracts — which is to say, the whole of chapter 9. The common failure is that the review is written from whatever data was easy to get, and the gaps are not stated.

Then the operational measures. Each has a grain and a trap, and the trap is usually the definition rather than the arithmetic.

Measure	Grain	The trap
Right first time	One batch, binary: completed with no deviation, no out-of-specification result, no rework	The definition is local, so the number is not comparable between sites unless the deviation-classification rule is common. Publishing it without the definition is how a plant looks better than one that counts honestly
Deviation ageing	One open deviation, days since opened, banded	Needs a clock-stop convention for deviations waiting on a third party, and that convention is where the gaming happens. Report raw and clock-stopped
Batch cycle time	One batch, in segments: process, test, release	Three functions own three segments and each reports its own, so the total is nobody's number. The useful cut is the *waiting* time between segments

Measure	Grain	The trap
Enrolment	Screened, screen-failed, randomised, discontinued, per site per month	Needs the screening-against-randomisation distinction to be right, or the funnel is wrong — and needs the planned curve, or actuals mean nothing
Database-lock cycle time	Days from last subject last visit to lock, per study	Report it with the number of unlocks, or a fast lock is indistinguishable from a premature one
Pharmacovigilance compliance	**Obligations due in the period submitted on time, divided by obligations due in the period**	The denominator is obligations, not cases. Counting cases understates exposure, and the externally triggered obligations drop out of a case-driven denominator entirely
Stability compliance	Scheduled timepoints due in the period, tested within window	Needs the stability schedule to exist as data, not as a spreadsheet of results
Corrective-action effectiveness	Effectiveness checks due in the period that passed	Everyone reports closure rate instead, which measures administration
Obligation horizon	Dated external obligations falling due in the next 6, 12 and 24 months	Nobody owns the consolidated list, which is why it cannot be produced on request

> No benchmark value for any of these was sourced for this book. Industry targets circulate freely and none could be verified, so none is printed. Build the measure, baseline your own estate, and treat any number quoted at you as a claim with no provenance.

The reconciliation that matters most is three-way, and it is always wrong somewhere.

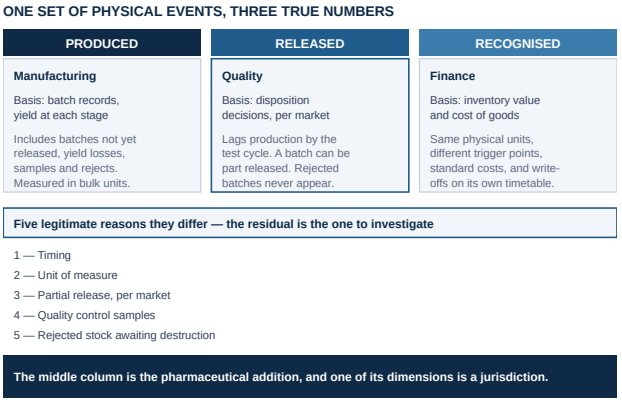


Figure 10.4 The three-way reconciliation. What the plant made, what quality released and what finance recognised, with the five legitimate reasons they differ. The residual is the number worth investigating.

What the plant made, what quality released and what finance recognised are three different numbers about the same physical events, maintained by three functions. They differ for five reasons, all legitimate, and the point of writing them down is that the residual difference is then the one worth investigating.

1. **Timing.** Release lags production by the testing cycle, and recognition moves at its own trigger points.
2. **Unit.** Production measures bulk units; release and sale are in sellable packs.
3. **Partial release.** A batch can be released to one market and not another, so one system counts it in full and another in part.
4. **Quality control samples.** Consumed physically, often not relieved from stock correctly, producing a persistent small variance everyone learns to ignore.
5. **Rejected stock awaiting destruction.** It sits in inventory carrying a value until the destruction certificate arrives — which makes **the destruction record a financial document as well as a quality one**.

The pharmaceutical addition to the general point is the middle column. Quantity and value are different facts about the same event, maintained by different functions and reconciled at period end, in any manufacturer. Here **a third fact — the regulatory disposition — sits between them, and it is market-specific**, so the reconciliation is three-way and one of its dimensions is a jurisdiction.

The deliverable that precedes any dashboard is the measure definition register. One row per measure: its name, its business question, its grain, its numerator and denominator in words, its source system per element, its clock-stop and exclusion rules, its owner, its publication cadence, and the date the definition last changed. Without it, two sites report right-first-time on different definitions and the comparison is meaningless; with it, the dashboard is a rendering rather than an argument. It is also the cheapest artefact in this chapter to produce and the one most often skipped.

Data migration in a validated estate

Migration is a validation activity, not an IT activity, and treating it as the latter is the most expensive mistake in pharmaceutical system replacement.

The reason is in the computerised-systems annex: where data is transferred to another format or system, validation should include checks that **data are not altered in value and/or meaning** during migration [5]. That is a verification obligation on *meaning*, not only on bytes. And migrated records keep their original retention obligations, their original audit trails and, where signed, their original signature bindings.

The deliverable set parallels any validation: a migration plan with acceptance criteria, rollback and the decision on what is migrated against archived against left in a read-only legacy system; a source data assessment that finds what is actually there, including orphans, test data, duplicates and records in states the model does not allow; a field-by-field mapping specification with transformation rules and the handling of values with no target; migration scripts treated as custom code; a verification protocol with reconciliation counts, full comparison on critical fields and **a demonstration that meaning is preserved**, which usually means a user confirming that a representative set of records reads correctly in the target; an audit-trail migration decision with its justification; a cutover plan; and a legacy disposition.

Six problems are specific to this estate and worth naming before anyone budgets.

1. **Audit trails rarely migrate faithfully.** The target's structure differs, and the source trail references users, configurations and states that do not exist in the target. Exporting the trail as a readable artefact and retaining it alongside is acceptable if declared, and loses the relational meaning.
2. **Signatures.** A compliant electronic signature is bound to the specific record such that it cannot be excised, copied or transferred to falsify another, and it carries the signer's printed name, the date and time, and **the meaning of the signing** — review, approval, authorship, responsibility. A signature without a declared meaning is a timestamp. That binding cannot be recreated in a new system, so the only honest options are to retain the signed rendering as a verified true copy with evidence of the original binding, and declare it. This is why document and quality-system migrations are the hardest in the estate.
3. **Open workflow items.** A half-investigated deviation, a change control awaiting approval, a corrective action with two tasks done and one open. In-flight state is harder than history, and the usual compromise — close in the old system, re-open in the new — creates records that misrepresent their own history.
4. **Specification and method versions.** A result referencing specification v4 migrated into a target holding v7 has lost its interpretability.
5. **Retention clocks do not reset.** A record created in 2009 and migrated in 2026 keeps its 2009-based clock. A target that computes retention from its own creation date gets this wrong by seventeen years.

6. **The migration itself must be auditable**: which records moved, when, under what version of which script, with what verification outcome, and what happened to the failures. A migration with no record of its exceptions is one whose completeness cannot be asserted.

> The economically correct default is **migrate the minimum, archive the rest, and decide at design time.** Most organisations migrate everything because deciding is hard, then find the migration is well over half the programme cost and that nobody has looked at the 2011 records since 2012.

Integration at the information-model level

Chapter 13 does the validation mechanics. What belongs here is the shape of the boundaries, because they are properties of the information model rather than of the network.

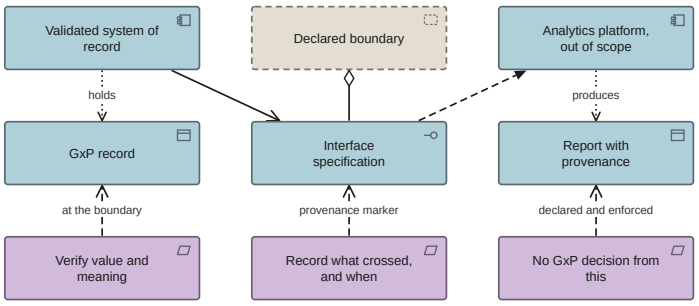


Figure 10.5 The validated-to-unvalidated boundary. An interface specification that is a validation deliverable, verification at the crossing, a provenance marker on every record — and an honest declaration that no regulated decision is made on the far side.

The validated-to-unvalidated boundary is a boundary of evidential quality. Data crossing outward loses its guarantees; data crossing inward imports risk. The trap is specific: a reporting platform is out of scope while it produces management reports, and comes into scope the day a trend chart from it is used to justify a specification. Two questions settle it — does anyone make a regulated decision from this output, and is the output cited in a regulated record? If either is yes, the pipeline needs verification of correctness, change control and traceability to source.

Three things sit at a boundary that works: **an interface specification that is a validation deliverable**, naming every field, its type, its unit, its mandatory status and its transformation; **verification at the boundary** — counts, checksums, business-rule checks — with the result recorded; and **a provenance marker on every record that crossed**, naming the source system, the extraction time and the originating user. The third is cheap at design time and impossible to retrofit, and its absence is how attribution gets laundered: a record created by a named person in one system arrives in the next as the integration service account, and attributability is destroyed by an interface.

And **declaring a system out of scope is a valid architectural act**, provided the declaration is honest and enforced. Read-only, one-way extraction to an analytics environment that explicitly does not support regulated decisions, with that limitation documented and the outputs labelled, obliges nothing.

Two further integration facts belong at this level. **Error handling must prove completeness, not hope for it.** In a validated estate a lost message is a lost record, so the design obligation is a reconciliation that demonstrates every expected record arrived — counts, sequence checks, a dead-letter queue somebody actually monitors — rather than a retry that assumes it. And **point-to-point sprawl has a specific cost here**: eleven systems and forty interfaces means forty separate error-handling designs and forty separate validation artefacts. The remedy is a declared canonical set and a mediation layer, which is fine as long as everyone understands that **the mediation layer is itself a validated system and must be budgeted as one.**

The clinical and commercial firewall was drawn in chapter 9 as a prohibition. At the information-model level it becomes a set of flows that must not exist and a small set that must be governed explicitly — the exposure estimates that give signal detection its denominator being the obvious one, built to a documented specification with a named approver.

Pseudonymisation sits at the boundary, and three design findings matter. **The key must be held where the re-identification right is**; a service that holds the key in the same platform as the data has achieved nothing. **Re-identification is sometimes an obligation, not a risk** — a subject must be contactable if a safety finding affects them, so a design that makes re-identification impossible has broken a duty. And **dates are the leak**: visit dates, birth dates and event dates in combination are identifying, so coarsening or shifting is the control and it changes the analysis. Cross-border transfer is the live local constraint, and the hosting question is therefore not only a validation question.

Then the question nobody contracts for. A mid-size company outsources most clinical conduct and much manufacture, so the data lives in someone else's validated systems while the regulatory accountability stays at home.

The question	Why it decides the architecture
Who owns the data?	Ownership determines the right to receive it, to use it for another purpose, and to require deletion
In what format, and when?	"All data in a usable electronic format" is not a specification. Conformant tabulation datasets with their metadata document are
Including the audit trail?	Almost always omitted, almost always needed. An exported dataset without its trail is not a complete record
Including metadata and configuration?	Edit checks, derivations, specification versions, user lists — without these the data cannot be interpreted
Who retains, how long, who pays?	A vendor's obligation ends with its contract; the sponsor's runs for decades
What access during conduct?	Read access to the live system, or a periodic extract? Oversight needs the former
What on insolvency or acquisition?	An escrow or a return trigger, or the records become an asset in someone else's estate

Integration patterns rank cleanly. Best: the sponsor's platform, the vendor's users — cleanest ownership, highest sponsor cost. Then the vendor's systems with a contracted, specified extract. Then periodic file transfer, which is the common state, where the specification is a template, the files drift and reconciliation is manual. Worst, and commoner than anyone admits: **the sponsor has PDFs — pictures of data rather than data.**

> **Interfaces are routinely in nobody's validation scope.** Each endpoint is validated; the interface between them belongs to an integration team outside the quality organisation. That is the most consequential gap in a pharmaceutical estate, and it is invisible on an application-portfolio diagram, because the thing that is missing is a row rather than a box.

Retention as a data-model constraint

In most industries retention is a policy setting. Here it is a computation, and **a single corporate retention period is always wrong.**

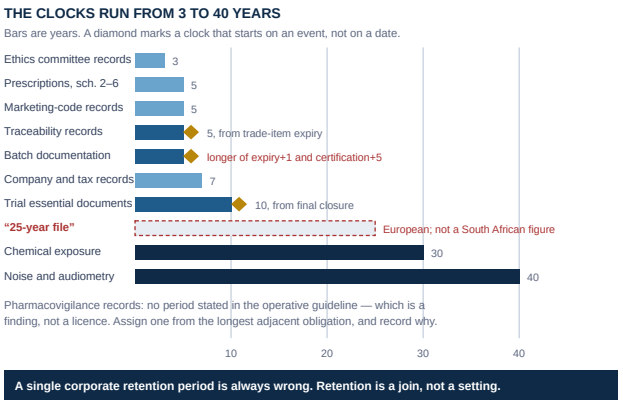


Figure 10.6 The retention clocks. Three to 40 years, four kinds of trigger, and three of the longest starting on an event rather than a date. Only periods verified for this book are shown.

The clocks run from four kinds of trigger — a fixed number of years, a product event, a study event and an employment event — and from three years to forty. Any single number is either unlawfully short for the longest clock or needlessly expensive for everything else. And because three of the longest clocks are event-triggered, *you cannot evaluate "one year after batch expiry" without the batch expiry date*. Only the periods verified for this book appear below; anything absent is absent because it was not verified.

Record class	Period	Trigger	Basis
Batch documentation	**1 year after batch expiry, or 5 years after certification, whichever is longer**	Batch expiry or certification	Adopted manufacturing guide [1]; mirrored in the local licensing guide [20]
Clinical trial essential documents, investigator and sponsor	**At least 10 years after final closure**, or at least 2 years after formal discontinuation of development	Trial closure or programme discontinuation	South African good clinical practice [4]
Research ethics committee trial records	**At least 3 years after trial completion**	Trial completion	South African good clinical practice [4]
Traceability and unique-identifier records	**Up to 5 years after the expiry of a trade item**	Trade item expiry	Traceability guideline [21]
Prescriptions, schedules 2 to 6, held by the last seller	**At least 5 years**	Date of sale	Medicines Act s 22A(6)(o) [22]

Record class	Period	Trigger	Basis
Pharmacovigilance records	**No period stated** in the operative guideline	—	Post-marketing guideline [14]
Marketing-code records and financial-transaction agreements	**At least 5 years**	—	Marketing code v19 [23]
Chemical exposure assessments and air monitoring	**Minimum 30 years**	—	Hazardous Chemical Agents Regulations 2021 [24]
Noise exposure, audiometry and medical surveillance	**40 years**	—	Noise-Induced Hearing Loss Regulations [25]
Company and accounting records	**7 years**	—	Companies Act 71 of 2008 s 24 [26]

Five readings of that table.

1. **Retention is a join, not a setting.** Batch documentation runs from batch *expiry*, traceability records from trade-item *expiry*, trial documents from *final closure or programme discontinuation*. The join is to data held in other systems, and a records platform that cannot consume those dates cannot implement the policy. **"Batch records: five years" is wrong** — it is a computation.
2. **The longest clocks sit on the least glamorous data.** Forty years for audiometry and thirty for chemical exposure, against ten for clinical trial documents. The privacy-heavy, inspection-heavy clinical data has a *shorter* clock than the occupational health file nobody thinks about.
3. **The obligation outlives the application by a factor of four or five.** The requirement is therefore an export format, with its reference data, readable without the source application — cheap to build at the start and impossible to reconstruct later. In pharma it is worse than elsewhere, because the export must carry **the audit trail** [4] and **the dictionary version**, and neither survives a naive extract.
4. **Two obligations can pull in opposite directions** and must be reconciled in the design. The draft research code contemplates indefinite retention where data is held only for research, while also requiring identifiable information to be destroyed or de-identified as soon as it is no longer needed [27] — against a ten-year floor for trial documents [4]. The only design that satisfies all three separates the identifiable layer from the research layer.
5. **Where an instrument states no period, that is a finding, not a licence.** Pharmacovigilance records, destruction records and import documents all lack a stated period in the documents retrieved. The architecture must assign one anyway, from the longest adjacent obligation, and **record the basis for the assignment** — because "we chose seven years" with no reasoning is the answer that fails an audit.

> **"Trial master file records are kept for 25 years" is a European figure**, routinely mis-attributed as a global requirement. The verified South African floor is **at least 10 years after final closure**, or at least two years after formal discontinuation of clinical development, for both investigator and sponsor, with **research ethics committee records at three years** after completion [4].

The modelling consequence is one sentence. **Retention is per record class, not per system and not per company.** So the entity is a record class carrying its period, its trigger type, its basis citation, the system that holds the trigger date, the disposal method, and the date the classification was last reviewed — and every record in the estate is mapped to exactly one class. Two attributes earn their keep immediately: the **basis citation**, because an assigned period with no reasoning fails an audit; and the **review date**, because guideline versions move faster than their stated validity. The research behind this book found one local guideline superseded five months inside its own window, which means **re-checking the citation is a standing job, not a publication-time task.**

Chapter 16 works through the regulation behind these clocks. What belongs in the information model is the classification, the join to the trigger date, and the honest admission that the obligation will outlive the system holding it.

THE CHAPTER IN TEN LINES

- In pharma master data is a set of control parameters, so open the governance conversation with which fields are controls, not with how many duplicates exist.
- ALCOA and ALCOA+ are design obligations rather than a slogan, and the system clock is one of the controls.
- The PIC/S guide's Part I never uses the term ALCOA; the verified South African statement is ALCOA-CCEA in the good clinical practice guidelines, so it is a GCP requirement and not a GMP one.
- Audit-trail review is an unscoped recurring cost, and under the draft annex it is normally completed before batch release, which turns review capacity into release lead time.
- The computerised-systems annex in force is still the January 2011 revision; the draft was consulted from 7 July to 7 October 2025 and has no published final text or date.
- The authority's data standards catalogue, not the standards body's latest release, governs what you may submit, so hold the per-study version decision as metadata.
- Dataset-JSON has not replaced the incumbent transport format, and the safety exchange standard's own implementation guide was itself at step 3 in July 2025.
- South Africa mandates neither the current safety exchange revision nor a gateway: the channel is a mailbox, so confirmation, retry and reconciliation are the sender's problem.
- Never write that the keys body requires a two-dimensional carrier by 2027; it is a voluntary capability ambition with no day or month.
- A single corporate retention period is always wrong: the clocks run from three to 40 years, three of them start on events, and the 25-year trial file figure is European.

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CHAPTER 11

The conceptual application model

The applications a pharmaceutical company needs, named by what they do rather than by product. It is the artefact you draw in your first week, before you know anyone's product names — and its real value is showing you what the company has no application for at all.

Draw this before you learn anyone's product names

In your first week at a pharmaceutical company somebody will hand you a list of systems. It will have sixty rows. Eleven of them will be spreadsheets with names like *RA Tracker v14 FINAL*. Four will be products the vendor stopped supporting in 2019. Nine will belong to a contract research organisation and will not be yours at all. And the list will tell you almost nothing, because a list of products is a list of answers to questions nobody wrote down.

The conceptual application model is the question list. It names the applications a pharmaceutical company **needs**, by what they do, with no product names in it. It is deliberately drawn before you know what the company owns, for two reasons that are specific to this industry.

The first is that a large part of the estate is somebody else's. The clinical stack is frequently the contract research organisation's; the manufacturing records for an outsourced product sit in a contract development and manufacturing organisation's systems. If you start from an inventory of what the company has licensed, those capabilities appear to be absent. They are not absent. They are present and not owned, which is a different and worse problem.

The second is that the validated boundary has to be drawn in capability terms, not product terms. Whether an application holds a GxP record determines its validation obligation, its change cadence, its access model and who is allowed to touch it. That determination attaches to what the application *does with which records*, and you can make it on a conceptual model. You cannot make it on a product list, because the same product is in scope in one laboratory and out of scope in the next.

> A conceptual application model names what the business needs an application to do, what data that application should master, and who uses it. It contains no products, no interfaces, no technologies and no instances. In pharma it also carries one more thing: whether the capability holds a regulated record.

This is the shortest chapter in Part III and the one with the highest ratio of value to length. Chapter 12 is the long one — what is actually installed, class by class. Read this first, because chapter 12 makes far more sense once you can see what each of those thirty-five system classes is *for*.

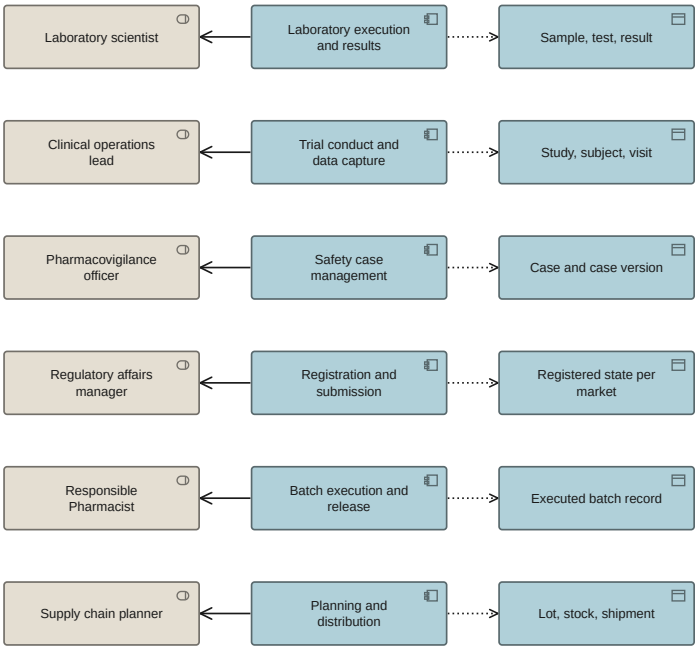


Figure 11.1 The conceptual application model: six domain capabilities, the role each serves and the object each masters. **No product names, and that is the point.** The seventh capability, the register of validated systems, cuts across all of them.

Seven domains, and one that no other industry has

Pharmaceutical application capability sorts into six business domains and one cross-cutting capability that exists only because of the regulatory regime.

Domain	What it is about	Dominant clock
Research	Molecules, assays, samples, predictions	The scientist's week
Clinical	Protocols, subjects, sites, blinded data	The study milestone
Safety	Cases, signals, labels	Calendar days, forever
Regulatory	Dossiers, registrations, commitments	The authority's queue
Quality and manufacturing	Batches, specifications, deviations	The batch and the shift
Supply and commercial	Lots, stock, prices, prescribers	The order and the month
The validated-system register	Which systems hold regulated records	The periodic review date

Those domains have different data, different regulators, different vendors and different tempos. Chapter 9 made the data argument; this is its application consequence. **Most pharmaceutical integration failures happen at the seams between these domains rather than inside them,** and the conceptual model's job is to make the seams visible early enough to design them.

Through the chapter, the table columns are the same four questions. *What it must do. What it should master* — the data for which it is the system of record. *What it must not master* — the data it will try to own and should not, which is where most estates go wrong. *Who uses it* — the job title, because an architect who cannot name the jobs cannot run a workshop.

The third column is the one to argue about. Every application in this industry arrives with the ambition to master more than it should, usually because the vendor's demonstration shows it doing so and because the function that bought it would rather not depend on another function's system. That is how a laboratory system ends up holding a specification that is registered somewhere else, and how an enterprise resource planner ends up holding a shelf life that was determined by a stability study it has never heard of. The conceptual model is where you say no, once, in writing, before the configuration decision gets taken by whoever is in the room.

Research

Research is the least regulated part of the estate and the most heterogeneous. Discovery is often entirely outside GxP. Good Laboratory Practice arrives with non-clinical safety studies, and the boundary between the two is a technology transfer in miniature.

Capability	What it must do	Masters	Must not master	Who uses it
Laboratory execution	Record experimental intent, procedure, conditions, observation, interpretation; witness and date it	The experiment	The sample result, the compound identity	Research scientists, project leaders
Sample and result management	Register samples, schedule and assign tests, capture results, compare to specification, approve	The sample, the test, the result	The experiment, the raw instrument file	Analysts, laboratory managers
Compound registration and inventory	Assign a permanent corporate identifier to a distinct entity; enforce uniqueness by structure; track physical material	The compound identifier and the structure it denotes; vial location and quantity	Assay results, batch quality	Chemists, biologists, compound managers
Scientific data management	Capture and archive raw instrument output with its context and a checksum, independent of the instrument's own software	The raw data file as generated	The interpreted result	Analysts, laboratory IT, archivists

Two separations in that table do real work. **The notebook owns the experiment; the sample manager owns the sample.** Where both claim the result, you have an ownership conflict that has to be settled explicitly — the usual settlement makes the sample system authoritative for anything with a specification and the notebook authoritative for exploratory work. And **the raw instrument file is a different object from the result derived from it.** A scientific data management capability exists because "where is the original data" has to have an answer that is not "on the instrument PC under the bench". Chapter 13 explains why that answer is a regulatory requirement and not merely good housekeeping.

There is a third thing in the research table that a conceptual model has to carry, and it is a regulatory point rather than a functional one. **The same capability can be inside the regime in one laboratory and outside it in the next.** A notebook used for discovery chemistry is usually not a GxP system at all. The same product, in the laboratory next door, supporting a non-clinical safety study that will go into a filing, is a Good Laboratory Practice system with a controlled time source, no post-hoc editing and an archive with a named archivist. The failure mode is quiet and arrives years late: a discovery instance starts holding data for work that ends up in a dossier, and nobody reclassified it. So the model records the GxP classification against the capability *and its use*, not against the product. That distinction is the seed of the validated-system register later in this chapter.

Two more capabilities belong in research and master nothing at all. Computational chemistry and in-silico prediction produce *predictions*, which are evidence, not records; treating a predicted value as a system of record is the category error that later shows up as a prediction cited as a measurement. Sequence and construct management owns the construct and its lineage, and becomes regulated at the cell bank, where a master cell bank is a GMP starting material with a documented identity chain back to the research construct.

Clinical

Capability	What it must do	Masters	Must not master	Who uses it
Trial planning and oversight	Select and activate countries and sites; plan and record monitoring; track enrolment, milestones, site payments	The site, the investigator, the visit, the monitoring activity	Subject data; the evidence file	Clinical operations, monitors, study managers
Clinical data capture	Collect case report form data from sites with edit checks, queries and an item-level audit trail	The trial data as reported by the investigator	Randomisation; the safety case	Site coordinators, data managers, monitors
Randomisation and trial supply	Assign treatment arms while holding the blind; get the right kit to the right site in time; manage expiry and unblinding	The randomisation assignment, the blind, the kit and its status	Subject clinical data	Clinical supply, pharmacists, unblinded statisticians
The trial master file	Hold the essential documents that let the conduct of the trial be evaluated, with expectations, completeness and timeliness	The evidence of compliant conduct	Any primary data	Clinical operations, quality assurance, inspectors
Statistical computing	Specify, program, execute, review and archive the analysis so it is reproducible years later	The analysis, its code, its outputs and the links between them	The study data, which comes from upstream	Statisticians, statistical programmers

The trial master file is the odd one. **It is a system of record for a claim about other systems' records.** It is second-order, and that is why it behaves unlike everything else in the estate and why chapter 12 gives it extended treatment: it is the artefact an inspector actually inspects.

Note what is missing from the randomisation row. The blind is not a feature of one application. It is an information-flow property of the whole estate, and it can be broken by a kit inventory report, a shipment manifest, a drug accountability log or a support ticket. A conceptual model that places "maintain the blind" inside one box has already lost the argument.

Safety

Capability	What it must do	Masters	Must not master	Who uses it
Case intake	Capture a report from any channel with a defensible day zero	The receipt event and its date	The assessed case	Call centre, representatives, medical information, safety associates
Case processing	Build and version the case: patient, product, event, narrative, seriousness, causality, expectedness	The case and the case version	The product label; the registration	Safety associates, safety physicians
Coding	Apply the medical and drug dictionaries at a stated dictionary version	The coded term and the version under which it was coded	The dictionary itself, which is licensed and external	Coders, safety data managers
Signal detection	Look across cases for evidence of an unestablished effect; manage a signal to a documented disposition	The signal, its evaluation, its decision and its actions	Causality in an individual case	Safety physicians, epidemiologists
Regulatory reporting	Compute who must be told what, by when, and evidence that it happened	The submission obligation and its state per destination	The case content	Safety operations, the local pharmacovigilance officer

Two South African facts reshape this domain and belong in the model, not in a footnote. First, **the fifteen-day clock starts when any personnel of the holder of the certificate of registration first receives the four minimum data elements — including sales representatives and contractors** [4]. That makes every commercial interaction an intake channel, which makes case intake an estate-wide capability rather than a departmental one. Second, **the submission channel is a mailbox, not a gateway** [4]. The guideline names separate mailboxes per submission type including a dedicated one for e2b XML, and does not name a portal or a transport standard revision [4]. Email provides no acknowledgement, no retry and no reconciliation, so the "regulatory reporting" capability has to supply all three itself. A submission register with a per-destination state machine is a required component here, not a workaround.

Regulatory

Capability	What it must do	Masters	Must not master	Who uses it
Registration management	Know what is registered, where, in what form, with what conditions, and what is pending	The registered state of the product in each market	Documents; the physical specification in the plant	Regulatory affairs managers, regulatory operations
Submission management	Assemble the dossier, validate it against the authority's criteria, manage the sequence lifecycle	The submitted dossier as the authority holds it	The source documents	Regulatory operations, publishers
Labelling and artwork	Manage approved text per market and the print-ready component through to the printer	The approved text version; the approved artwork version	The packaging component master in the plant	Labelling managers, artwork studio, packaging technologists
Commitment tracking	Hold every dated obligation accepted at approval or in response to a question, with owner and evidence	The commitment, its due date, its evidence and its closure	The work itself, which happens elsewhere	Regulatory affairs, quality, project managers
Product identity	Hold structured product data — substances, strengths, forms, routes, packaging, manufacturers — to a controlled vocabulary	Arguably nothing new; it is a view over data owned elsewhere	Everything, which is the trap	Regulatory data managers, master-data owners

Registration management is the most under-appreciated system of record in a pharmaceutical estate. The registered state is what the manufacturing process, the label, the specification and the pack must match. When those drift apart the result is product that is physically fine and legally non-compliant, and nobody notices until a pre-approval inspection. The commonest version of this failure is not a bad system; it is that nobody can state, from a system, the currently approved specification or approved site list for a given market without asking someone who remembers.

Product identity deserves its warning. Structured product data is a reporting obligation over data that lives in registration management, the enterprise resource planner, manufacturing and labelling. It needs that data at a granularity and consistency no single existing system holds, which means the implementation is a master-data programme wearing a reporting programme's name [6]. Model it as a view with an owner, not as a new master.

Quality and manufacturing

Capability	What it must do	Masters	Must not master	Who uses it
Quality management	Run deviations, out-of-specification investigations, corrective and preventive actions, change control, complaints, audits, controlled documents, training	The quality event; the controlled document; the training record	Results; batch execution; registered details	Quality assurance, every GMP function
Batch execution	Prompt and enforce manufacture step by step; verify operator, material and equipment; capture deviations at the point of occurrence	The executed batch record; equipment and room state	Material and financial inventory; the specification	Operators, supervisors, quality assurance reviewers
Laboratory quality control	Test raw materials, in-process, finished product and stability samples against the registered specification; issue the certificate of analysis	The sample, the result, the specification version applied, the disposition	The registered specification, which belongs to registration management	Analysts, QC managers, qualified reviewers
Stability management	Schedule and execute timepoints, trend results, determine and confirm shelf life	The stability study, its schedule and its conclusion	The test result, which belongs to quality control	Stability coordinators, QC managers
Calibration and maintenance	Hold the instrument register, intervals, tolerances, as-found and as-left values, due dates	The instrument and its calibration state	The measurement itself	Metrology, engineering, QC
Environmental monitoring	Monitor classified areas by location and time against grade limits, with trending and excursion management	The monitoring result by location and time	The product sample	Microbiologists, QA, facilities

Three notes on that set.

The quality management capability is the spine of a GMP estate — more than batch execution, more than the enterprise resource planner. Every other system's problems end up as records in it, and it holds the evidence of the state of control that a manufacturing licence depends on.

Environmental monitoring has a different grain from everything around it. Its data is location-and-time-series shaped, not sample shaped. Forcing it into a sample model, which is the natural thing to do when the laboratory system is already there, makes trending close to impossible. Model it separately even if it ends up configured inside the same product.

Calibration state is a precondition, not a report. An instrument out of calibration must not be usable for a GMP measurement, which means the calibration capability has to be queryable by the systems that consume measurements, at the moment they consume them. Most estates check it at periodic review instead, which is months too late.

Supply and commercial

Capability	What it must do	Masters	Must not master	Who uses it
Planning	Plan supply with shelf-life awareness, registration-constrained sourcing and expiry-aware lot selection	The plan	Lot status, which is a quality decision	Demand and supply planners, site planners
Inventory and distribution	Hold stock by location, lot and status with quarantine segregation; execute receipt, pick, pack, dispatch	Stock by bin, lot and status; the lot's commercial availability	The quality decision about the lot	Warehouse, logistics, distribution quality
Serialisation	Apply a unique identifier per saleable pack, record aggregation and movement events, report to the required destination	The serial number, its status, its parent and its event history	The lot's quality status	Packaging, serialisation operations, supply quality
Cold chain	Keep product within its labelled conditions, record that it stayed there, and support the excursion decision	The temperature history of a shipment and the excursion decision	The release decision	Distribution, quality assurance
Medical information	Answer unsolicited medical questions with approved responses, under a documented process	The enquiry, the response given, the approved response library	The product label	Medical information officers, medical affairs
Field medical and sales	Plan and record interaction with healthcare professionals; present only approved content; account for samples	The interaction; sample accountability	The adverse event case, which belongs to safety	Representatives, medical science liaisons, sales management

The two commercial rows are the ones architects under-scope. Both are intake channels into safety [4], and in a South African company both are inside the fifteen-day clock. The honest architectural answer is not to validate the whole customer relationship system; it is to validate the specific intake and hand-off functionality and declare the rest out of scope. Chapter 13 explains why declaring something out of scope is a legitimate architectural act, provided the declaration is honest and enforced.

South Africa deletes most of a capability here, and it is worth saying plainly because importing a North American reference architecture is a real and expensive mistake. **The private-sector price is a nationally regulated single exit price, one price per product, adjusted by notice** — so customer-level net-price management, rebate accrual and chargeback reconciliation, which are the bulk of a United States revenue-management estate, describe transactions that largely cannot lawfully occur here above the lowest schedule. What South Africa needs instead is a price object with a legally mandated effective date, and a tender capability with award validity and litigation state. Chapters 4 and 6 have the economics; the application consequence is that you build much less than the reference model suggests, and you build it differently.

The capability no other industry has: the register of validated systems

Every industry has an application portfolio. Pharma needs the same portfolio with a set of columns no other industry's portfolio carries, because **the system itself is regulated**.

The question an inspection opens with is *which of your systems hold GxP records*. Annex 11 expects an up-to-date listing of all relevant systems and their GMP functionality [1], and in South Africa that obligation arrives through SAHPRA's standing adoption of the PIC/S guide rather than through any SAHPRA document of its own [2][3]. Most architects do not model this, and then cannot answer the question. The absence is expensive four times over, because the same list is the first input to a privacy assessment, a disaster-recovery review, an IT audit and any cloud business case.

The capability has to hold, per system instance and per defined use: the GxP classification — manufacturing, clinical, pharmacovigilance, laboratory, distribution, several, or none; the record types held; the validation state; the date of last validation and the periodic review due date; a named business owner and a named technical custodian; whether electronic signatures are applied; whether an audit trail exists, is reviewable, and who reviews it at what frequency; the retention obligation on the data it holds; hosting and data location; and the plan for how its records stay readable after it is switched off.

> The grain is **one system instance, in one environment, for one defined use** — not one row per application. A productive instance and a validation instance are different rows. The same product validated for two different purposes has two validation scopes. And an interface between two validated systems is in scope and is almost always nobody's row at all.

Make this the application portfolio with extra columns, not a separate quality-owned spreadsheet. The near-universal state is two registers that disagree, and the disagreement is discovered by an inspector rather than by you. Chapter 13 takes the register apart attribute by attribute; here it is enough to say that it is a *capability* with an owner, and that putting it on the conceptual application model is how it acquires one.

Three questions that decide every pharmaceutical application architecture

Everything above is a vocabulary. These three questions are the design decisions, and they are taken whether or not anybody writes them down.

One: what is the system of record for the compound, the study, the batch and the submission?

Four objects, four different answers, and getting any of them wrong is expensive in a different way.

THE OBJECT	SYSTEM OF RECORD	THE PLAUSIBLE WRONG OWNER
The compound a distinct chemical or biological entity	Compound registration Permanent corporate number; structure-based uniqueness	The ERP material master holds a saleable item, not a structure; cannot answer "have we made this before?"
The study a protocol run on human subjects	Trial planning and oversight Protocol version, sites, milestones, oversight	Clinical data capture owns the subject data, not the study; and it is often the CRO's, not yours
The batch what was actually done to make it	Batch execution The executed batch record: steps, operators, materials, equipment, deviations	The ERP production order knows quantity, cost and lot number, and nothing about execution
The submission the dossier as the authority holds it	Submission management Sequence and leaf with its lifecycle operation: new, replace, append, delete	Document management holds the documents and knows nothing about what is currently approved

Figure 11.2 Four objects, four different systems of record — and the plausible wrong owner sitting next to each. **Name an owner for all four in writing.**

The reason this question is hard is that each object has a plausible wrong owner nearby. The enterprise resource planner's material master looks like the product. The clinical data capture system looks like the study. The publishing tool looks like the submission, because it holds the documents. In each case the plausible owner holds a *projection* of the object, and the projection is missing the attribute that matters: the structure, the protocol version, the execution record, the lifecycle operation.

Chapter 9 made the companion data argument — that "a product" is four things with different cardinality — and this is the same fault line seen from the application side. If you take one decision from this chapter, take this one, and write it down with a named owner against each of the four.

Two: where is the validated boundary?

Which applications hold GxP records, and therefore which are validated? The answer sets cost, change cadence and who may touch the system, so it is an architecture decision with a budget attached.

Two framings make the boundary drawable. **The regulated record, not the system, is the unit of scope.** You determine scope by asking which records in the system are GxP records and what decisions depend on them. A customer relationship system that records a sales call is out of scope; the same system, the moment a representative types a patient's symptom into a call note, has created a potential adverse event report and is in scope for intake and hand-off [4].

And **the boundary is a boundary of evidential quality, not a network boundary and not a security zone.** Data crossing it outward loses its guarantees. Data crossing inward imports risk. The trap is directional: a reporting platform is out of scope while it produces management information, and comes

into scope the day a trend chart from it is used to justify a specification. The settling questions are whether anyone makes a GxP decision from the output, and whether the output is cited in a regulated record.

Draw the boundary on the conceptual model, before any product is chosen. Chapter 13 gives the mechanics of what crossing it obliges.

One encouraging consequence, because this section otherwise reads as a list of costs: **declaring a capability out of scope is a valid architectural act**, provided the declaration is honest and enforced. A read-only, one-way extract into an analytics environment that explicitly does not support GxP decisions, with that limitation documented and its outputs labelled, obliges nothing. What is not available is the fudge — an unlabelled platform that everyone quietly uses for decisions nobody has written down.

Three: where is the clinical/commercial firewall?

This is the one an enterprise architect is most likely to get wrong, because the instinct trained by every other industry points the wrong way. **Trial data and commercial data must not be joined.** A single patient view spanning both is not an unbuilt feature; it is a consent and competition-law breach.

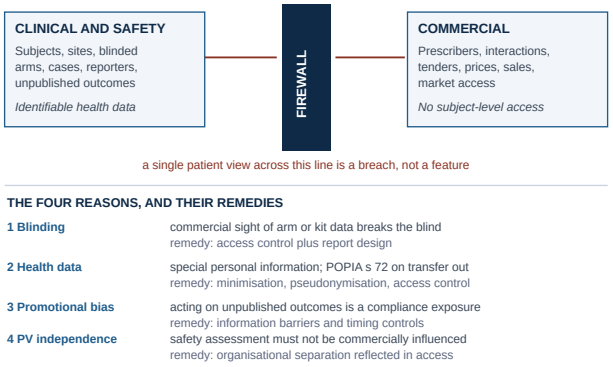


Figure 11.3 The clinical/commercial firewall. Four separate reasons, four different remedies — which is why treating it as **only** a privacy question fixes a quarter of the problem.

There is no single enterprise data platform that holds clinical, safety and commercial data under one access model, and any design that proposes one is wrong in this industry. The four reasons in the figure are genuinely separate and have different remedies, which is why treating this as "a privacy thing" produces a design that fixes a quarter of it. Blinding is answered by access control and report design. Personal health information is answered by minimisation, pseudonymisation and access control — and in South Africa by section 72 of POPIA, under which health information may not leave the country unless a lawful basis plus either adequate protection in the destination or POPIA-aligned contractual obligations is established [5]. Promotional bias is answered by information barriers and timing controls. Pharmacovigilance independence is answered by organisational separation reflected in system access, with an audit trail showing who could see what.

What works instead is domain platforms with controlled, purpose-specific, minimised flows between them, plus a small number of explicitly governed aggregate views — an exposure estimate to serve as a signal denominator, for instance — each built to a documented specification with a named approver. Saying this in week two saves a programme; saying it in month nine is a rebuild.

> Pseudonymisation is not an exit from the regime. Pseudonymised personal data is still personal data, and re-identification is sometimes an obligation rather than a risk: a subject must be contactable if a safety finding affects them. A design that makes re-identification impossible has broken a Good Clinical Practice duty, not satisfied a privacy one [5].

What the model is really for

Here is the honest note to close on. You draw the conceptual application model before you know what the company owns, and the point of drawing it first is not to classify what is there. It is to notice what is **not** there at all.

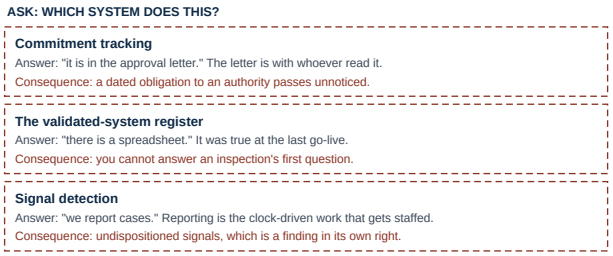


Figure 11.4 The three boxes that most often come back as a shrug. **Finding these is the whole point of drawing the model before you see the inventory.**

Run the model past the people who do the work — the regulatory affairs manager, the head of quality assurance, the validation lead who knows which systems are actually in a validated state — and ask one question per box: *which system does this?* Three answers come back as a shrug with remarkable regularity.

Commitment tracking. The commitments live in the approval letters, and the letters live with whoever read them. There is no application, and the first time anyone finds out is when a dated obligation has passed.

The validated-system register. There is a spreadsheet, it was accurate at the end of the last implementation project, and the periodic review dates in it are historical.

Signal detection. Cases are processed and reported, which is the clock-driven work that gets resourced. Looking across the accumulated cases for an unestablished effect is the work that does not, and a backlog of undispositioned signals is a finding in its own right.

None of those three is a product you buy in your first ninety days. All three are capabilities you can name in your first week, on one page, without knowing a single product name. That is what the conceptual application model is for.

> Chapter 18 walks the whole of this through **Veldtkruin Group**, the invented South African pharmaceutical group this book uses for its worked material — a generics manufacturer with two GMP sites alongside a small contract development business. **Nothing about Veldtkruin Group is real**, and it is deliberately a composite, because the difference between a generics estate and a contract developer's estate is the thing an architect most needs to see. Running the model in this chapter against Veldtkruin produces exactly the three shrugs above, and one more: its two sites answer the *validated boundary* question differently, because only one of them runs batch execution electronically.

THE CHAPTER IN TEN LINES

- A conceptual application model names what the business needs an application to do, with no product names in it.
- Draw it first because a large part of a pharmaceutical estate belongs to a CRO or a CDMO and will look absent on an inventory.
- The trial master file is a system of record for a claim about other systems' records, which is why it behaves unlike everything else.
- Registration management owns the registered state of the product in each market, and it is the most under-appreciated system of record in the estate.
- In South Africa every employee is a safety intake channel and the reporting channel is a mailbox, so the reporting capability must supply its own acknowledgement and retry.
- Pharma needs one capability no other industry needs: a register of validated systems, at the grain of one instance for one defined use.
- Name the system of record for the compound, the study, the batch and the submission; each has a plausible wrong owner next door.
- The validated boundary is a boundary of evidential quality, and the regulated record rather than the system is the unit of scope.
- One enterprise platform across clinical, safety and commercial is wrong here for four separate reasons with four different remedies.
- The model earns its keep by showing what the company has no application for: usually commitment tracking, the validated-system register and signal detection.

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CHAPTER 12

The application landscape

Thirty-five system classes, what each owns, whether it is validated, and above all what breaks. Plus the four things that are true across all of them — including the one that produces most audit findings: nobody has validated the interfaces.

What is actually installed

Chapter 11 drew what a pharmaceutical company needs. This chapter is what you will find. The gap between the two is most of the work.

Expect between thirty and sixty things that are plausibly systems. Expect one quality management system and one enterprise resource planner, and fight to keep it that way. Expect **several** instances of the laboratory, chromatography, batch execution, environmental monitoring and plant control systems — one set per site, often per laboratory. Expect a good part of the clinical set to be somebody else's. And expect the safety database to be the oldest system in the estate and the hardest to replace, because every change to it is a change to a reporting obligation.

> The most useful early question is not "what systems do you have". It is **"show me your validated system inventory, and show me the list of systems that are not on it."** The second list is where the risk is: the database in the stability laboratory, the workbook that calculates the assay result, the shared-drive list tracking deviations, the instrument with a local hard drive and no backup. Every regulator who has inspected a pharmaceutical site knows to ask for the second list.

Each class below gets the same six questions, because asking them in the same order is how you stop a workshop becoming a product demonstration. *What it is for. What it owns as system of record. Whether it is validated, and under which GxP. What it integrates with. The grain of its data. What breaks.* The last one is the longest, because it is the part you will walk into. Product names appear because the book should name real products, but **a name here is an example of who plays in a category, never a recommendation and never an endorsement.** Ownership and naming go stale faster than anything else in this industry, so every vendor claim carries a date; chapter 15 is the providers chapter where the selection questions live.

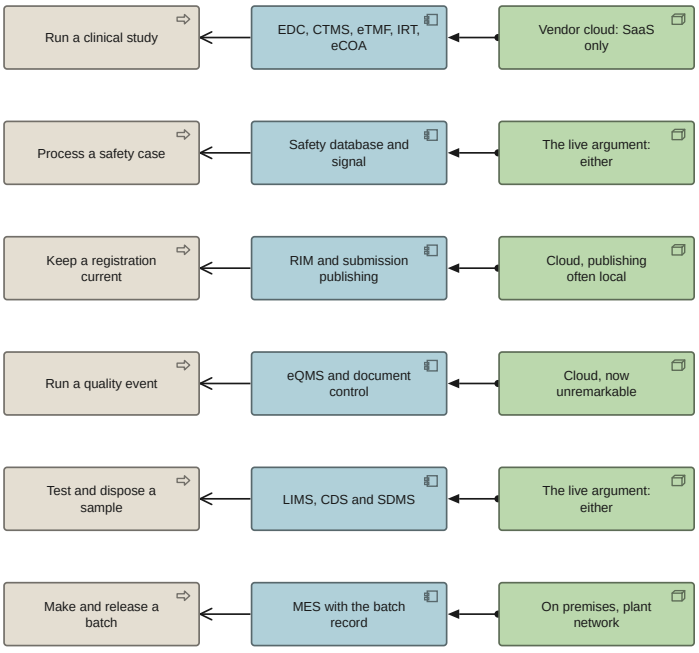


Figure 12.1 The estate as a layered view: the process, the applications that serve it, and where those applications actually run. **The hosting column is not an IT preference** — the execution path cannot cross a wide-area network, and the middle row is where the validated-cloud argument actually is.

Research

The research estate is the least regulated part of the company and the most varied. Discovery is frequently outside GxP altogether; Good Laboratory Practice arrives with non-clinical safety studies.

The electronic laboratory notebook (ELN) replaces the bound paper notebook: experimental intent, procedure, conditions, observation, interpretation, with a date and a witness. It owns the experiment as an intellectual and legal artefact — its original purpose was inventorship evidence for patents, which is why witnessing and date integrity are built in and why the signature model is author-plus-witness rather than author-plus-approver. Grain: one experiment, one scientist, one date. Not GxP in discovery; GLP where the work supports a safety study. Examples: Benchling, Dotmatics, Revvity Signals Notebook, LabArchives, SciNote.

What breaks. It gets adopted as a filing cabinet — a scanned page pasted in and signed — so the company has bought a document store at notebook prices. Witnessing lapses, and the patent-evidence case quietly evaporates until a priority dispute. A drawn structure is stored as a picture rather than a

connection table, which destroys the ability to answer *has anyone here made this compound before*. And instrument data is referenced by file path; the share is decommissioned four years later and the experiment is unverifiable.

The LIMS in research manages samples and the tests on them. Note the boundary: the notebook owns the experiment, the LIMS owns the sample. Grain: sample, test, analyte, result, with specification and version, analyst, instrument, method and version, and an approval state. Research-specific failures: methods under development change weekly, so a LIMS configured for controlled methods either blocks the science or gets bypassed; and sample naming is scientist-chosen, so the same material arrives under four names.

The scientific data management system (SDMS) captures, catalogues and archives **raw instrument output** with its context, independent of the instrument's own software. It is the answer to "where is the original data", and it owns the raw file as generated plus its metadata — instrument, method, operator, timestamp, sample identity, checksum. Yes, GxP, and it is the system that makes two of the data-integrity attributes achievable at all. Examples: Waters NuGenesis, Agilent OpenLab ECM, Thermo Chronicle, and a great deal of bespoke scripting.

What breaks. Most often it is not bought, so raw data lives on instrument PCs backed up inconsistently. Where it is bought it covers the instruments with a supported connector, and the gap is the older balance, the specialist analyser and the one-off rig — frequently the ones producing the critical number. The checksum is generated at ingestion rather than at creation, so anything that happened to the file on the way in is inside the trusted boundary. And it is treated as the backup rather than the archive of record, so original data ages out at ninety days.

Compound registration and inventory are two jobs sold together: a unique permanent corporate identifier for a distinct entity, with uniqueness enforced by structure normalisation; and physical material tracking. Grain: compound, then salt or form, then batch, then container, and confusing those four levels is the characteristic modelling error. Usually not GxP, and legally critical regardless, because a registration error propagates into a patent application. Examples: Dotmatics Register, CDD Vault, ChemAxon Compound Registration, Titian Mosaic.

What breaks. Duplicate registration, because the normalisation rules did not treat a tautomer, a stereocentre, a salt or a solvate the way the chemist did — two sets of assay data on "different" compounds that are the same molecule, unmergeable without breaking historical references. An assay result attributed to the compound when it came from a batch at 85% purity, so the structure-activity relationship is polluted by a potency artefact. And biologics forced into a small-molecule model: an antibody is a sequence, a set of modifications, a cell line, a process and a formulation.

In-silico and cheminformatics tools predict properties and activity before synthesis. **They are system of record for nothing**, and that is the architectural statement: computations over data owned elsewhere, producing predictions that are evidence rather than records. Grain: one prediction, one structure, one model version — and the model version is rarely captured, which makes predictions irreproducible within months. Not GxP; but where a model's output informs a filing, the model, its version, its training data and its qualification become part of the filing's evidence.

Bioinformatics and sequence management own the construct and the sequence with provenance and lineage. Grain: sequence, construct, clone, bank, batch — and lineage is a graph, not a tree. Not GxP in research; **it becomes regulated at the cell bank**, where a master cell bank is a GMP starting material whose identity chain back to the research construct must be documented. It fails when lineage breaks at the clone.

Preclinical study management — long called DIVA systems, for in-vivo data acquisition — runs non-clinical safety studies to GLP. **The animal is the subject.** Grain: study, group, animal, observation at a timepoint. GLP differs from GMP organisationally: a Study Director as a single point of study control, a study plan with controlled amendments, an archive with a named archivist, and an independent quality assurance unit that inspects study phases. Examples: Instem Provantis, Xybion Pristima, Ascentos, Climb. *What breaks:* observation windows missed, when a timepoint is a protocol requirement rather than a to-do item; paper in-life sheets transcribed later, still common because a clipboard works in an animal room and a tablet does not; and non-clinical submission datasets generated from the finished report rather than from the data, which fails conformance.

How the research estate fails as a whole. There is a cliff between research and development. Research runs on flexible, non-validated, scientist-owned tools; development needs controlled, validated, auditable ones. The transition — a compound becoming a candidate, an assay becoming a GLP method, a construct becoming a cell bank — **has no system.** It is a set of meetings and a document pack, and the data arrives in development detached from its provenance. Alongside it sits identifier proliferation: each system adopts a different one of the compound's five or six identifiers as primary, and the mapping lives in one person's spreadsheet.

Clinical

The clinical estate is the one most likely to be substantially somebody else's. Ownership, access and retention are therefore architectural questions before integration is.

One framing for the whole set, with the dates right. **ICH E6(R3) Good Clinical Practice is split and partly not yet in force.** The principles and Annex 1 have been in effect in the European Union since 23 July 2025; Annex 2 was adopted by ICH on 3 June 2026 and takes legal effect on 15 January 2027, as does the consolidated guideline [1]. Three errors to avoid: calling R3 a draft, treating Annex 2 as live, and generalising European dates to other regions.

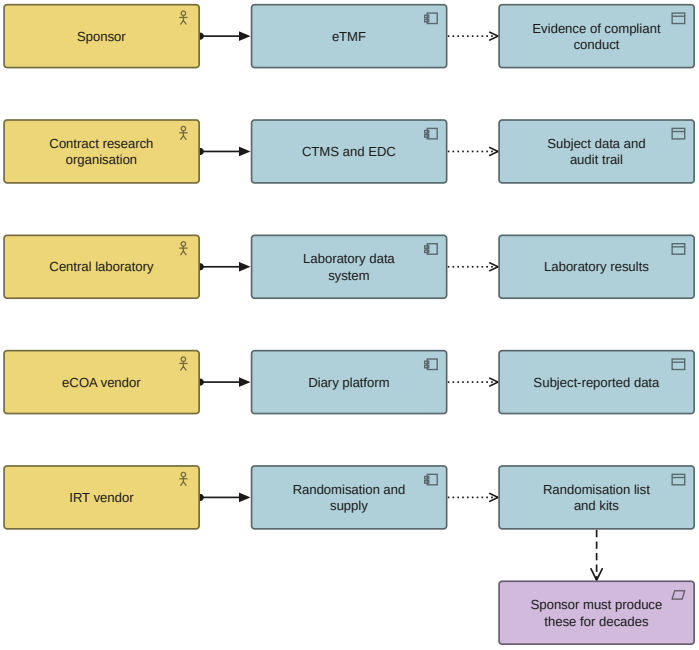


Figure 12.2 Who operates which clinical system, and what the sponsor must nevertheless be able to produce decades later. **The sponsor's accountability does not transfer with the data.**

The clinical trial management system (CTMS) runs the operational trial: site selection and activation, investigator records, contracts and budgets, monitoring visit planning and reports, enrolment tracking, site payments, milestones. It owns the site, the investigator, the visit, the monitoring activity and the milestone — **not the subject data.** GCP. Grain: study, country, site, activity, action item. Examples: Veeva CTMS, Medidata Rave CTMS, Oracle Siebel CTMS, IQVIA, Clario. *What breaks:* two instances, the sponsor's and the provider's, with no agreed master, so the governance meeting resolves activation dates by weekly manual reconciliation; monitoring reports written weeks after the visit, which is a contemporaneity failure in GCP terms; and the system used as a document store, so documents technically exist while the evidence file is incomplete at inspection.

Electronic data capture (EDC) collects case report form data from sites with edit checks, queries and an audit trail on every data point. It owns **the clinical trial data as reported by the investigator** — the dataset the filing rests on. GCP, with a United States electronic-records overlay interpreted by the October 2024 final guidance on electronic systems, records and signatures in clinical investigations [2]. Grain: study, site, subject, visit, form, item. **The item-level audit trail is the regulated object**, and the reason these databases are larger than their payload suggests. Examples: Medidata Rave EDC, Veeva EDC, Oracle Clinical One, Castor, OpenClinica, REDCap.

What breaks. The form designed from the protocol by someone who will not analyse the data, so fields are collected and never used. Mid-study database amendments splitting the dataset into pre- and post-amendment shapes — legitimate, sometimes unavoidable, routinely under-costed. Edit checks too aggressive, so a site forced to reject a value it knows is real enters a wrong one. **Laboratory data loading with unit mismatches** — conventional units sent, SI expected, conversion applied twice or not at all, which is the specific integration error that most often produces a clinically impossible value in a listing. And reconciliation with the safety database done once on a spreadsheet before lock, where discrepancies are negotiated rather than resolved.

eCOA and ePRO capture outcomes reported by the subject, the clinician or an observer. The system owns the subject's own report at the time they made it. **Where an instrument is validated and licensed, its wording and layout are fixed and cannot be altered** — a configuration constraint with legal force through both copyright and psychometric validity. GCP, and attribution is subtle: the subject is not a company user and has no company account, so attributability runs through device or credential binding plus an audit trail of the administration. Examples: Medidata Patient Cloud, Signant Health, Clario, uMotif, ClinOne. *What breaks:* device clock against server clock, where a diary entry's time is the whole point of a diary; offline entries synced later, where the recorded time may be the entry time or the sync time and the system does not show which; bring-your-own-device completion bias, **commonly reported as biting harder in South African multi-site trials than in European ones, though I could not source a quantified effect**; and a licensed scale reworded to fit a screen, which invalidates the instrument.

Randomisation and trial supply management (IRT or RTSM) assigns the subject to an arm while holding the blind, and gets the right kit to the right site in time with expiry management, resupply triggers, excursion handling and emergency unblinding. **The randomisation list is the single most access-controlled object in the clinical estate.** Both GCP and GMP. Grain: subject, randomisation, arm, kit, shipment, dispensation; and separately kit, lot, expiry. Examples: Suvoda, 4G Clinical Prancer, Signant, Endpoint Clinical, Calyx, Medidata RTSM.

What breaks. **The blind broken by a report** — a kit-level inventory listing, a shipment manifest with differing kit weights, an accountability log, a dispensing label. Blind integrity is an information-flow property of the whole estate, not a feature of one application, and this is the highest-consequence integration failure in clinical architecture. Emergency unblinding either too hard, so a workaround appears, or easy and unlogged. An expiry extension that reaches the system but not the site's physical labels. And a temperature excursion that quarantines nothing, so the kit is dispensed before the quality assessment completes.

The eTMF, at length, because it is the inspection artefact

The trial master file is the collection of **essential documents that permit the conduct of the trial to be evaluated and the quality of the data produced to be verified.** An electronic trial master file is that file with a structure, a completeness model and a quality workflow. Examples: Veeva eTMF, Montrium eTMF Connect, Phlexglobal PhlexEview, SureClinical, Florence.

Here is the property that makes it different from everything else in this chapter. **It is a system of record for a claim about other systems' records.** It is second-order, and that is why its own completeness is a regulatory fact rather than an administrative one.

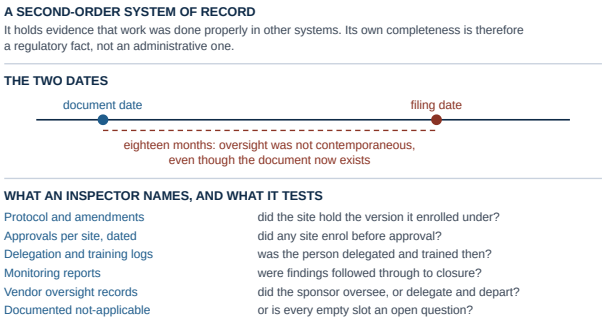


Figure 12.3 The eTMF as the inspection artefact. **Two dates per document, and the gap between them is the finding** — completeness and filing timeliness are deficiencies in their own right, whatever the trial's actual quality.

What an inspector actually asks for. Not "show me the TMF". Specific things, by name, on screen, in minutes:

1. The index or inventory, and then named artefacts from it.
2. The protocol and every amendment with approval dates — then evidence that each site had the current version before enrolling under it. The test is whether a subject was enrolled under a protocol version the site had not yet had approved.
3. Ethics committee and regulatory approvals per site, with dates, cross-checked against each site's first-subject-in date. A site that enrolled before approval is a critical finding.
4. Delegation-of-authority logs and training records — then evidence that the person who performed an activity was delegated to do it and trained on the procedure then in force.
5. Monitoring reports, and whether findings were followed to closure.
6. Safety reporting evidence: that expedited reports met their timelines and that investigators and ethics committees were notified.
7. Investigational product accountability and the chain for the blind.
8. Vendor oversight records, showing the sponsor oversaw rather than delegating and departing.
9. The trial master file plan, and the completeness and quality-review records against it.

> **Completeness and filing timeliness are inspection findings in their own right**, independent of whether the trial was actually conducted well, because a sponsor cannot demonstrate oversight of what it cannot show. A document filed eighteen months after it was created shows that contemporaneous oversight did not happen, even though the document now exists — which is why inspectorates look at the gap between document date and filing date, and why "we can produce it on request" is not the standard. The file is to be *maintained*, not assembled.

The industry structure is the **TMF Reference Model**: a hierarchy of zones, then sections, then artefacts, with metadata and expectations per artefact. It began under a pharmaceutical information association and is now affiliated with CDISC, which describes the model as becoming part of the CDISC organization [3]. **Its current version number, zone count and artefact count could not be sourced in the research behind this book, so none is printed here** — the structure is stable and is what to design against. A great deal of live collateral still attributes the model to its former home, which is four years out of date [3].

Grain: trial, zone, section, artefact, document, with version, status, expected or received, and **two dates — document date and filing date.** An empty expected slot is a question, so the discipline is a *documented not-applicable*.

What breaks. Filed at the end, in the three months before an inspection, so every filing date postdates the activity by a year and the inspector reads the filing dates. The completeness model left at its out-of-the-box expectation set, so the system reports 98% complete against the wrong list. Metadata entered inconsistently, which makes completeness reporting unusable. Quality review as a volume queue, where a reviewer approving sixty documents an hour is not reviewing them and the review record then asserts something untrue. Sponsor oversight of the outsourced file by assertion, discovered at study close when it cannot be reconciled. And signatures that do not survive migration, which makes this among the hardest migrations in the estate.

The rest of the clinical set

The clinical data management system and clinical data warehouse bring together data from capture, central laboratories, diaries, randomisation, imaging and sensors into an integrated, cleaned, analysable study database and produce the locked dataset. They own the integrated database and the cleaning decisions applied to it — **not the source data**, which stays authoritative for its own values. GCP, and the transformations are the regulated part. Examples: eClinical Solutions elluminate, Saama, Veeva CDB, SAS. *What breaks:* mapping logic undocumented, so nobody can say why a value differs from its source; cleaned data overwriting reported data, destroying the chain back to the investigator; and a dataset soft-locked, then changed, so every downstream output is versioned against a moving base.

The safety database captures, codes, assesses, narrates, reports and follows up individual case safety reports for the life of the product and beyond. It owns **the case** — a living record, created on initial receipt with minimum information and then accumulating follow-up versions for years. **The grain is case, then case version, and reporting obligations attach to versions, not to cases.** Good Pharmacovigilance Practice, with the longest retention horizon in the estate. Examples: Oracle Argus Safety, ArisGlobal LifeSphere Safety, Veeva Safety, Ennov Safety.

What breaks. **Day zero wrong or indefensible** — the case was mentioned to a sales representative three weeks before it reached the safety department, so every timeline downstream is wrong and the late report is itself a compliance failure. Intake channels not inventoried: a website form, a partner's email, a patient-support programme's call log, each lawful, each unconnected. **Dictionary version drift** — the medical dictionary releases twice yearly, so cases coded under different versions are not directly comparable and aggregate analyses that ignore version are quietly wrong. Expectedness assessed against the wrong label version. And, most commonly of all, **a safety database nobody can replace**: twenty years of cases, bespoke configuration, a version three releases behind, and a migration nobody will fund.

Signal detection looks across accumulated cases for evidence of an unestablished effect and manages a signal to a documented disposition. It owns the signal, its evaluation, its decision and its actions — an inspectable artefact under Good Pharmacovigilance Practice. Examples: Oracle Empirica Signal, ArisGlobal LifeSphere Signal, Qinecsa. *What breaks:* disproportionality treated as causality, when a high reporting ratio is a hypothesis generator confounded by reporting bias, indication and co-medication; signals detected and never dispositioned, because detection is automatable and evaluation is not; and the denominator unknown, since reporting rates need exposure and exposure is estimated from sales with assumed daily doses.

The statistical computing environment is where the analysis is specified, programmed, executed, reviewed and archived. It owns **the analysis and its provenance** — code, derivation, output and the links between them. GCP, and the specific obligation is reproducibility: the same inputs and programs must produce the same outputs years later, on demand. Examples: SAS Life Science Analytics Framework, Domino, Posit Workbench and Package Manager. *What breaks:* package versions unpinned, so an analysis cannot be reproduced — the commonest failure in organisations moving to open-source statistical tooling, and the control is a frozen, versioned repository with a documented snapshot per study; and derivations living in a table program rather than in the submitted analysis datasets, so a reviewer cannot reproduce the number from what was filed.

Medical writing and submission authoring produce the protocol, investigator's brochure, analysis plan, study report and summary documents under version control. *What breaks:* numbers typed into text that no longer match the tables after a re-run; and no component reuse across a programme, so a safety-text change must be made twelve times and is made eleven.

Regulatory

Regulatory information management (**RIM**) knows what is registered, where, in what form, with what conditions, and what must be done by when to keep it that way. It owns **the registered state of the product in each market.** Grain: product, presentation, market registration, submission, sequence, commitment — and the product-presentation-registration triple is the hard part, because one formulation is registered as different presentations in different markets with different names, pack sizes and approved texts. Examples: Veeva Registrations, ArisGlobal LifeSphere Regulatory, Extedo, Samarind, Ennov — plus large amounts of spreadsheet in mid-size companies.

What breaks. **Registered state unknown**, the commonest and most serious failure in the domain: nobody can state, from a system, the currently approved specification, site list or shelf life for a market without asking a person who remembers. A tracker maintained by one regulatory affairs manager in which the real answers live, parallel to an under-populated RIM. And correspondence answered from a manager's own mailbox, so a commitment made in an email is undiscoverable at the next inspection.

Submission publishing assembles the dossier into the common technical document structure, produces a valid sequence with its backbone, validates it against the authority's criteria, and manages the lifecycle so each sequence correctly replaces, appends to or deletes content in the cumulative view. It owns **the submitted dossier as the authority holds it** — a different thing from the documents in document management. Examples: Lorenz docuBridge, Extedo eCTDmanager, Veeva Submissions Publishing, Certara GlobalSubmit, Freyr.

South Africa's facts here are specific and unforgiving. **The ZA-SAHPRA eCTD specification is version 3.1 of July 2024, based on the ICH 3.2.2 specification; version 4.0 is not yet adopted and SAHPRA plans to migrate once other authorities have gained experience** [4]. **The application portal is the only valid method for submitting sequences**, with the current portal live from 1 July 2024 [4][5]. **PDFs must be version 1.4 to 1.7 with fast web view; an earlier version is an ERROR class finding and rejects the whole sequence** [4]. eCTD has been the only accepted format for orthodox and biological applications since 1 April 2022, and **baselines start at sequence 0000, not 0001** [4][5].

What breaks. **Lifecycle operations wrong** — a leaf marked new that should have been replace, so the cumulative view shows two versions and the reviewer sees superseded content. This is the characteristic error and it is cumulative: once the lifecycle is wrong, every later sequence inherits the problem. Validation run the day before the filing date, which with whole-sequence rejection is not a cosmetic risk [4]. And granularity decided late, so a coarse Module 3 means a small specification change republishes a large document.

Identification of medicinal products data management holds and submits **structured** product data to authorities: substances, strengths, forms, routes, packaging, manufacturers, authorisation details, with controlled terminology for each. It is system of record for arguably nothing new — it is a reporting obligation over data owned in RIM, the enterprise resource planner, manufacturing and labelling — **which is exactly why it is hard: the implementation is a master-data programme wearing a reporting programme's name.** Do not quote a single go-live date. The European obligation arrives as a sequence of data-enrichment deadlines: structured pack sizes for one product class extended to June 2026, manufacturer data for all non-centrally-authorised products at December 2026, pack sizes extended to June 2027 — **two of those already extended** — plus an interface upgrade to a healthcare-interoperability standard in late 2026 [6]. South Africa has no such obligation [4][5]. *What breaks:* treated as a submission project, so the data goes stale the day after submission; and no owner for the structured product record.

Labelling and artwork management covers two linked things. Labelling manages the regulated *text* per market under version control, with core safety text cascading to local versions. Artwork manages the *printed component* — layout, graphics, barcodes, braille, print-ready files — through design, review, approval, proofing and release to the printer. Grain: product, market, label text version, artwork version, packaging component, print revision. **The chain from approved text to the component on the line is the critical path, and the most common break is between the artwork system and the enterprise resource planner's component master.** Examples: Esko WebCenter, Loftware, Kallik Veraciti, GLAMS.

What breaks. Text approved and artwork not updated, so packs continue printing with the old leaflet — a classic recall cause. Artwork updated and the component not re-coded, so the line runs old artwork with a new approval on file. Proofing by eye instead of automated comparison against the approved source. Braille height, barcode grade and data-matrix readability not measured at proof and at press. And a short regulatory implementation deadline meeting a multi-week print lead time, which forces either a stock write-off or an unapproved pack.

Commitment tracking holds obligations accepted as a condition of approval or in response to a question: post-approval stability, additional validation batches, a paediatric plan, a renewal date. It owns the commitment, its owner, its due date, its evidence and its closure. GxP in effect, because failure to meet a commitment is a compliance failure with authorisation consequences. *What breaks:* commitments living in the approval letter and nowhere else; and a due date recorded without the evidence requirement, so it is closed with nothing to show. In most mid-size companies there is no application here at all.

How the regulatory estate fails as a whole. The registered state and the actual state diverge, silently. Manufacturing improves a process, changes a supplier, revises a specification — each legitimately, each under change control — and *does this require a variation* is answered by someone without the registered state in front of them. Years later a pre-approval inspection finds that the running process is not the registered process. Underneath it: the dossier is documents, the registered state is data, and nobody owns the data.

Quality and manufacturing

The quality management system (QMS) runs deviations, out-of-specification investigations, corrective and preventive actions, change control, complaints, recalls, audits, document control, training, risk management, supplier quality and management review. It owns the quality event in all its forms plus the controlled document and the training record. **It is the spine of a GMP estate** — because every other system's problems become records in it, and because it holds the evidence of the state of control that a licence depends on. Grain: event, investigation, root cause, action, effectiveness check, closure. Examples: Veeva quality applications, MasterControl, Honeywell TrackWise, ETQ Reliance, Qualio, Dot Compliance, AmpleLogic — and in the South African mid-market a material amount of collaboration-platform lists and desktop databases. Two ownership notes with dates: **TrackWise is Honeywell's, and which Honeywell depends on the Automation separation, unconfirmed as complete**; and **ETQ is Octave's, Octave having become an independent public company in May 2026** [7].

What breaks. **CAPA as a ticket queue** — thousands raised, closed on time, with no reduction in recurrence, because root-cause analysis is a dropdown and the real cause is systemic. The diagnostic question is whether the same deviation type recurs after its action closed; if it does, the system is producing paper rather than control. **Deviation backlog**, the most reliable leading indicator of a site in trouble and visible from a dashboard in five minutes. Effectiveness checks skipped, where "confirmed the procedure was updated" has checked nothing. Document control as a folder, so the operator at the line works to an uncontrolled printout. Training recorded as read-and-understood, which is a record of reading and not of competence. And a complaint with a clinical element triaged by a quality analyst, reaching pharmacovigilance days later and consuming most of the fifteen-day clock [8].

The manufacturing execution system (MES) and the electronic batch record (EBR). The system executes manufacture — dispensing, charging, processing, in-process checks, equipment and room status, line clearance, operator and material verification, deviation capture at the point of occurrence — and produces the **electronic batch record**, the executed record of what was actually done to make this batch. That record is the central GMP artefact of the whole company. Grain: order, batch, phase, step, parameter, with operator identity, equipment identity, material lot, timestamp and a signature where review is required. Examples: Körber Werum PAS-X, Rockwell PharmaSuite, Siemens Opcenter Execution Pharma, Emerson Syncade, and paper-on-glass in smaller plants.

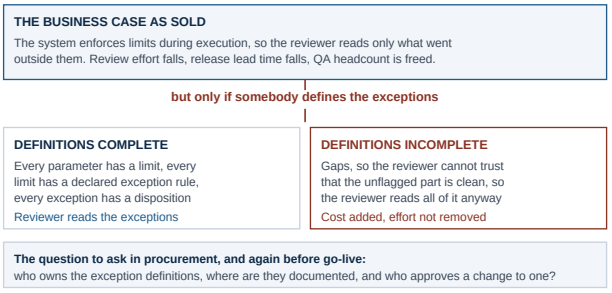


Figure 12.4 The trap that disappoints most electronic batch record programmes. **Review by exception comes with the exception definitions, not with the system** — and defining them is governance work, not configuration work.

Carry one finding about these programmes above all others. **Review by exception comes with exception definitions, not with the electronic batch record.** The commercial case for an EBR is that the system enforces limits during execution so a reviewer examines only the exceptions. Where the definitions are incomplete, reviewers read the whole record anyway and the system has added cost without removing effort. **This is the single most common reason EBR programmes disappoint, and it is a configuration and governance failure rather than a product failure.**

What else breaks. The recipe maintained in two places — master recipe in the execution system, process parameters in a specification document or in the control system — drifting apart, when the authoritative source must be declared and enforced. Deviation capture bypassed, because the workflow stops the line. The paper fall-back either unvalidated and improvised or validated and never practised. Review queues as the release bottleneck, so release lead time becomes a function of quality assurance headcount. And operator experience ignored: gloved hands, a mask, a gown, a small panel and a workflow designed in an office — operators find the fastest compliant path, and if that path involves pre-signing, the record is no longer contemporaneous.

The LIMS in quality control tests raw materials, in-process samples, finished product, stability samples and environmental samples against registered specifications and produces the certificate of analysis. It owns the sample, the test, the result, the specification version applied and the disposition. Grain: sample, test, analyte, replicate, result, with method and version, analyst, instrument, specification and version, and a multi-level approval. Examples: LabWare, LabVantage, STARLIMS — **which is Francisco Partners', not Abbott's** [7] — Thermo SampleManager, Sapio.

What breaks. Specifications maintained in the laboratory system *and* in the enterprise resource planner's inspection plans, drifting, so a lot passes in one and fails in the other. Specification and method versions not recorded against results, so a 2021 result cannot be reproduced. Partial instrument integration, so critical results are hand-typed from a printout. **Out-of-specification handling as a retest loop**: where the system makes retesting easy and investigation hard, the laboratory tests until it passes, which is the most serious laboratory data-integrity finding there is. And rounding applied at report time, so stored value, reported value and specification comparison can disagree — the rounding rule is a GMP decision and belongs in the configuration.

Stability management runs the studies that establish and confirm shelf life: protocol, chambers, scheduled timepoints, pull and test, trending, and the statistical determination. The schedule is the operationally critical object, because **a missed timepoint cannot be recovered** — the sample can only be pulled at that age once. One South African point to verify rather than assume: storage conditions depend on climatic zone, and **South Africa is not in the same zone as Western Europe.** The research behind this book could not source SAHPRA's zone classification, so treat it as a must-check; it determines chamber configuration, study design, and whether European stability data suffices for a South African filing. *What breaks:* the schedule in a spreadsheet; and a chamber excursion not linked to the studies inside it.

The annual product quality review is the periodic review of all batches of a product to verify process consistency and the appropriateness of current specifications. It is system of record for nothing underneath it — **it is a reporting and judgement activity over nine or ten other systems, which makes it the best architectural stress test in the estate.** Its characteristic failure is three months of manual extraction per product, by hand, for every product, every year. **For a South African generics manufacturer with a couple of hundred registered presentations that consumes a team**, and the reviews are late — the 2025 review completed in late 2026, so its conclusions cannot inform anything.

Three supporting classes that are preconditions rather than reports. Supplier quality holds the approved supplier for a material, the quality agreement and the audit history — and fails when procurement's approved-vendor list and quality's approved-supplier list differ, so an order can be raised against a vendor quality has not approved, and when a material is qualified but the shipping site is not. **Calibration and maintenance** holds the instrument register, intervals, tolerances and results — and its signature failure is that **only "pass" is recorded and the as-found value is not**, which destroys the only input a retrospective impact assessment has when the instrument is later found out of tolerance. **Environmental monitoring** covers viable and non-viable counts, personnel monitoring, pressure, temperature and humidity in classified areas against grade limits; the governing European text is Annex 1, fully applicable from 25 August 2024, whose contamination-control-strategy requirement reshaped monitoring programmes materially [9]. **Its grain is location and time, not sample.** It fails when the sampling plan lives in a document and execution in a spreadsheet, and when continuous and discrete monitoring sit in separate systems with no common time base.

Serialisation and track-and-trace applies a unique identifier to each saleable pack, records aggregation into cases and pallets, records movement events, and reports to the required destination. Usually split: a site-level system manages numbers on the line, an enterprise-level system manages allocation, partner exchange and regulatory reporting. The identifier and event standards are GS1's — a trade item number, a serialised pack identity in a data matrix, a logistics unit code and a location code — with EPCIS as the event standard. **EPCIS 2.0 was released in June 2022 and is not only XML: it added JSON and JSON-LD bindings and REST bindings for capture and query** [10]. Examples: TraceLink, Antares Vision and rfxcel, Systech, SAP ATTP, Movilitas.

South Africa has no national pharmaceutical serialisation and verification scheme of the European or United States kind — none was found in the research behind this book, and the submission channel remains document-based, which is consistent with there being none. If that holds, the consequence is distinctive: **a South African manufacturer serialises because it exports to regimes that require it, not because its home market does.** Serialisation is an export capability, scoped per destination market, frequently present on some packaging lines and not others. The two overseas models differ too: the European scheme is publish-to-hub, the United States model peer-to-peer event exchange between trading partners [11][12]. A manufacturer exporting to both needs both.

What breaks. **Aggregation**, the hardest operational problem in serialisation and the one that generates partner disputes: a case opened for a check and resealed without re-aggregating, a pallet reworked, a short case shipped against a full-case record. Number allocation collisions between two sites or a site and a contract packer, which produce duplicate serials in a national hub. Partner onboarding as the long pole, because each partner's event implementation differs in profile, vocabulary and transport, so "we support EPCIS 2.0" does not mean two parties can exchange [10]. And line speed: print, verify, aggregate and report at hundreds of packs a minute. **This is the one place where a compliance system sits directly in the critical path of a physical process at speed**, which is why serialisation projects are plant projects, not IT projects.

Commercial and adjacent

Customer relationship management for field medical and sales manages interaction with healthcare professionals: call planning and recording, sample distribution and accountability, approved-content presentation, medical science liaison insights, consent. It owns the interaction and **sample accountability**, a regulated record. Partly GxP: **the adverse-event intake path is the pharmacovigilance-relevant part**, and the South African rule makes it unavoidable, because the fifteen-day clock starts when any personnel of the holder — including sales representatives and contractors — receives the four minimum data elements [8].

What breaks. **Adverse-event intake by good intention**: the representative is trained to report within twenty-four hours by email, some do, and the system offers no structured capture, no mandatory triage, no timer and no evidence of completeness — so the company cannot demonstrate that every mentioned event was captured. The architectural answer is a structured, mandatory, in-system capture path with a hard hand-off to safety and a timestamp at first mention, and it is resisted because it slows a call. Also: consent not modelled per channel and per purpose, which under POPIA is not a consent record at all [13].

Medical information answers unsolicited medical questions with approved standard responses where they exist. It owns the enquiry, the response given and the approved response library, and its intake paths into safety and complaints are the regulated parts. *What breaks:* triage for adverse events applied by a non-clinical agent under call-time pressure; a response library out of date against the current label, so an approved answer is wrong; and out-of-hours enquiries handled by an outsourced service whose records are not integrated, which is both a completeness gap and a clock risk [8].

Market access, pricing and tender manages price across markets and channels. In most countries that means list price, reference price, negotiated and tender prices, rebates, discounts and reimbursement status. **In South Africa the private-sector price is a single exit price — one price per product, nationally — which deletes most of what a pricing system does elsewhere.** There is no customer-level net price to manage, and the transactions that produce a United States gross-to-net estate are largely unlawful here above the lowest schedule. What remains is a price object with a legally mandated effective date, and a tender capability where **an award is a contract with a validity and a litigation state** rather than secure revenue. *What breaks:* pricing conditions maintained in the enterprise resource planner by finance and in a pricing system by market access, diverging; and tender prices with expiry dates nobody tracks.

Transfer-of-value reporting records and discloses payments and transfers to healthcare professionals and organisations. Its characteristic failure is **recipient identity resolution** — the same physician as three records, so the disclosed aggregate is wrong. The South African disclosure position runs through industry self-regulation and the statutory provisions on bonusing and sampling, and the research behind this book could not source it primarily — treat it as a must-check.

The enterprise resource planner and supply planning hold the commercial and material backbone: material and lot master, bill of material, production order, purchasing, inventory, costing, ledger, sales order, delivery, invoicing, and alongside them demand, supply and distribution planning. **In pharma**

it additionally owns the lot's commercial availability status, which is where it meets the quality systems. GxP for GMP-relevant master data, lot status, and the genealogy and distribution records a recall depends on. Examples: SAP S/4HANA and ECC, Oracle, Microsoft Dynamics 365, and **SYSPRO in the South African mid-market** — South African-founded in 1978, majority-acquired by Advent International following an August 2024 announcement, with completion unconfirmed in the research behind this book.

Four pharmaceutical-specific points. **Shelf-life-driven planning**: finished product with a twenty-four-month shelf life, a quarantine-to-market lead time of weeks, and a customer requirement for minimum remaining shelf life at delivery — standard material requirements planning cannot do it. **Batch determination**: first-expiry-first-out rather than first-in-first-out, which is configuration with regulatory consequences. **Registration-constrained sourcing**: a market may only receive product made at a registered site, and the system usually does not know the site-market matrix because that lives in RIM, so the constraint is enforced by a planner's memory. And **mixed-mode manufacture in one instance**: an active-ingredient plant is a process plant with recipes and a many-to-many lot genealogy graph, while formulation and packaging are closer to discrete manufacture.

Cold chain and distribution move product within its labelled conditions, monitor that it stayed there, and support the decision when it did not. It owns the temperature history of a shipment and the excursion decision. Good Distribution Practice. Examples: Controlant, Berlinger, Sensitech, Tive, Elpro. *What breaks:* excursion data arriving after the product, so a logger read on receipt tells you two days late that the shipment failed — real-time telemetry solves the information problem and creates a decision problem, because somebody must make a quality judgement in transit; mean kinetic temperature not calculated, so systems that alarm on threshold alone over-quarantine; and **the last mile out of scope**, with delivery to a rural clinic unmonitored. In a South African context that is the materially weakest link and the one least likely to be in the architecture.

The warehouse management system runs receipt, putaway, storage by condition, picking, packing, dispatch and cycle counting, with quarantine, released, rejected and retained-sample areas segregated and lot and expiry control throughout. *What breaks:* status segregation physical but not systemic, or systemic but not physical; expiry-based picking overridden manually; and retained and reference samples held under warehouse control rather than under the reference-and-retention-samples regime, revised in PE 009-18 with effect from 24 September 2026 [14].

How the commercial estate fails as a whole. The safety path is the thinnest thread in the widest funnel: thousands of commercial interactions a week, any one of which may contain an adverse event, feeding a safety function sized for a fraction of that volume through intake paths that are mostly email. The South African clock starting at any employee's receipt makes that an architectural problem rather than a training problem [8].

Four cross-cutting realities

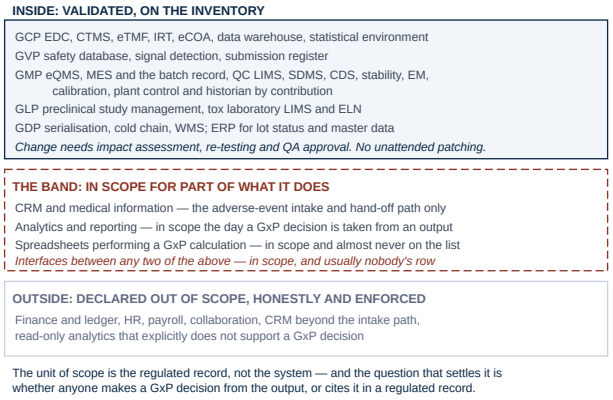


Figure 12.5 The validated boundary across the estate. **The hard part is the dotted band**: systems that are in scope for one function and out of scope for the rest, and the interfaces that cross the line and belong to nobody.

Interfaces are routinely in nobody's validation scope

This is the single most common audit gap in a pharmaceutical estate, and it is structural rather than careless. Each endpoint system is validated by the function that owns it. The interface between them belongs to an integration team that is not in the GxP organisation at all. So effort goes into the systems' functions and almost none into the place where the data actually gets corrupted.

What gets corrupted, specifically. **Units, precision and rounding** — a result transmitted at full precision and stored at two decimal places, a unit converted by a hard-coded factor, a concentration transmitted without its unit because both systems "know" it. **Timestamps and time zones** — three systems contributing to one batch record, three clocks, two zones, and an audit trail that cannot establish the order of events; in pharma the sequence of events is itself a regulated fact. **Identity** — the same batch, subject, product or supplier under different keys in every system. **Attribution** — the record is created by a named person, the interface posts it onward as the integration service account, and the receiving audit trail says the service account created it. And **error handling**, which is observed rather than designed: the message is lost, or retried forever, or dead-lettered into a queue nobody monitors. **In a validated estate a lost GxP message is a lost record.**

The pattern that works has three parts at a declared boundary. An **interface specification that is a validation deliverable**, naming every field, its type, its unit and its transformation. **Verification at the boundary**, recorded. And **a provenance marker on every record that crossed**, naming source system, extraction time and originating user. The third is cheap at design time and impossible to retrofit.

The sponsor usually has PDFs, not data

When a contract research organisation or a contract development and manufacturing organisation hands work back, **the data lives in someone else's validated systems and the regulatory accountability does not transfer with it.** These are the questions that must be answered in the contract and almost never are.

Question	Why it matters
Who owns the data?	Ownership determines the right to receive it, to reuse it, and to require deletion
In what format, and when?	"A usable electronic format" is not a specification. A named standard with its metadata document is
Including the audit trail?	Almost always omitted, almost always needed. An export without it is not a complete record
Including metadata and configuration?	Edit checks, derivations, specification versions, user lists — without these the data cannot be interpreted
Who retains, how long, who pays?	The provider's obligation ends with its contract; the sponsor's runs for decades

Question	Why it matters
What access during conduct?	Read access to the live system, or a periodic extract? Oversight requires the former
What on insolvency or acquisition?	An escrow or a data-return trigger — or the records are an asset in an estate
Who may be audited, and by whom?	The sponsor's right to audit the provider, and its sub-processors
How are quality events shared?	A contract manufacturer's deviation on your batch is your deviation for release

The patterns, in descending order of quality: **the sponsor's systems with the provider's users** — cleanest ownership, highest sponsor cost; **the provider's systems with a contracted, specified extract** — workable where the specification is real; **the provider's systems with periodic file transfer**, the common state, where the files drift and reconciliation is manual; and **PDFs**, the state in which the sponsor has no data at all. Decide which of the four you are in, in writing, at study or campaign start. The transfer specification written at study close is the characteristic failure, and its consequences are unrecoverable.

The platform consolidation story, and why the framing every deck uses has broken

Most pharmaceutical technology decks describe a three-way platform contest: one vendor consolidating from the content and quality end, one defending a long-established clinical and safety portfolio, one holding the data-capture centre of gravity. **That framing needs two amendments and one correction.**

THE FRAMING MOST DECKS STILL USE

One content-and-quality platform vendor, one incumbent clinical and safety portfolio, one data-capture leader. Three independent multi-domain platforms.

WHAT THE DATED EVIDENCE SHOWS	
Dassault Systèmes	owns Medidata; agreed 23 July 2026 to buy ArisGlobal expected to close H2 2026 — completion unconfirmed
Siemens	completed Dotmatics 1 July 2025, $5.1bn; platform Luma
Oracle	Argus 2026.1 GA Mar 2026, Clinical One 26.3 GA Aug 2026, eTMF 26.1 GA Sep 2026, with a published EOL calendar
Veeva	the one large pure-play; FY2026 revenue $3.195bn, year ended 31 Jan 2026. Current product names are not "Vault X"

Figure 12.6 Consolidation as at October 2026, dated. **Treat the ArisGlobal completion as unconfirmed** — if it closes, the three-way framing every pharmaceutical deck uses stops describing the market.

Amendment one. Dassault Systèmes owns Medidata, and on **23 July 2026** Nordic Capital announced an agreement to sell **ArisGlobal** — whose platform the seller describes as an integrated, cloud-native suite spanning regulatory, safety and quality — to Dassault Systèmes, **expected to close in the second half of 2026** [15]. **Treat completion as unconfirmed; it is the single highest-priority fact to re-check in this chapter.** If it closes, Dassault holds both a clinical platform and a safety, regulatory and quality platform, and the three-way framing stops describing the market.

Amendment two. Siemens completed its acquisition of Dotmatics on **1 July 2025 at $5.1bn enterprise value** — that figure is in dollars because the source is — naming **Luma** as the platform [16]. That is research informatics rather than clinical operations, but it puts a fourth industrial-software major into life sciences alongside its own plant-floor presence.

The correction. "Oracle is legacy in clinical" is unsupported. Oracle's own published general-availability and end-of-life schedule shows Argus safety at 2026.1 generally available in March 2026, Clinical One at 26.3 in August 2026 and eTMF at 26.1 in September 2026, with dated end-of-life entries across the portfolio [17]. That is an actively maintained portfolio with a published forced-upgrade calendar — and the calendar is the thing to ask about in procurement, because in a validated estate **a vendor's upgrade cadence is your revalidation cadence.**

One caution on names. **Veeva's own August 2026 Development Cloud product-sheet contents do not use the names "Vault CDMS", "Vault RIM" or "Vault QualityDocs"**, and list no quality or medical products at all [18]. Those are the names most architects use. They could not be verified as current, so this chapter hedges them and so should your slides.

Do not rank the vendors and do not predict the outcome. The useful consequence is that **the number of independent multi-domain platforms is falling, and the buyer's bargaining power with it** — so contractual protections on exit, data portability and upgrade deferral matter more in a consolidating market than in a fragmenting one.

Two estates, and almost everything published describes the second

	SA MID-SIZE GENERICS	ORIGINATOR OR AFFILIATE
Products	100 to 300 presentations	few, high margin
Research	none: formulation and BE only	full stack, usually at parent
Clinical	BE at a local CRO; no eTMF	full clinical estate
Regulatory	RIM is often a spreadsheet; publishing outsourced	platform RIM, in-house publishing
Quality	eQMS if modern; shared drives and desktop databases if not	enterprise eQMS, often one global instance
Manufacturing	one or two sites; ERP plus LIMS, often no MES; paper batch records	MES with the batch record; serialisation everywhere
Laboratory	the CDS is the critical system; SDMS often absent	full LIMS, SDMS and CDS
Safety	a local PV officer; CIOMS forms by email	global safety database; the affiliate is a local node
Validation	done by a consultancy once and not maintained; no periodic review	a validation function, a VMP, periodic review on a schedule
Commercial	tender and single exit price	full commercial stack
Binding constraint	people: one of each role	the parent's global template

Figure 12.7 Two estates, not two sizes of one. The claims here are **practitioner consensus rather than measurement** — read them as what you are likely to find, not as a survey.

A mid-size South African generics manufacturer and a research-based originator have different estates, not differently sized versions of one estate. The comparison in the figure comes from the research behind this book and **its estate-shape claims are flagged as practitioner consensus rather than measurement** — read them as judgement about what you are likely to find, not as a survey.

For the generics manufacturer the highest-value interventions are unglamorous and cheap: a real validated-system inventory; a registration tracker that is a system rather than a spreadsheet; specification master data held once; a release checklist as a system object; and a document-control system that actually controls documents. None is a platform purchase.

For the affiliate of an originator the highest-value intervention is the opposite — explaining to the parent why a global template does not fit a market with a Responsible Pharmacist rather than a Qualified Person [19], a nationally regulated single price, a portal-and-mailbox regulator [4][8] and a POPIA constraint on moving health information out of the country [13]. The affiliate cannot choose its systems. It can choose how well the parent understands what it is deploying.

THE CHAPTER IN TEN LINES

- Ask for the validated system inventory and then for the list of systems that are not on it; the second list is where the risk is.
- The notebook owns the experiment, the LIMS owns the sample, and the SDMS owns the raw instrument file; each is a separate system of record.
- The eTMF is a system of record for a claim about other systems' records, and its completeness and filing timeliness are findings in their own right.
- Review by exception comes with the exception definitions, not with the electronic batch record; that is why most EBR programmes disappoint.
- The registered state and the actual state diverge silently, and the regulatory estate's master failure is that nobody owns the registered-state data.
- Interfaces are routinely in nobody's validation scope, and that is the commonest audit gap in a pharmaceutical estate.
- When a CRO or CDMO hands work back the sponsor often has PDFs rather than data, because the transfer specification was written at study close.
- Dassault agreed on 23 July 2026 to buy ArisGlobal, closing in H2 2026 and unconfirmed; if it closes the three-way platform framing breaks.
- Oracle is not legacy in clinical on its own published GA and EOL calendar, and Veeva's 2026 collateral does not use the Vault product names most architects say.
- A South African generics estate and an originator's estate are different estates, not two sizes of the same one.

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CHAPTER 13

GxP, validation and data integrity

No other industry in this series has a regime where the system itself is regulated. A system that holds a GxP record is validated — so its lifecycle, change control, access model, audit trail and decommissioning are all regulated, and "just patch it" is not available. This is the chapter that makes a pharmaceutical estate different from every other estate you have modelled.

The system itself is regulated

Everything else in this book has a counterpart in another industry. A bank has retention rules, a retailer has an audit obligation, a manufacturer has traceability. This chapter does not have a counterpart, and it is the reason a pharmaceutical estate is not a normal estate.

Here is the whole of it in one sentence, and the rest of the chapter unpacks it.

> **A system that holds a GxP record is validated — which means its lifecycle, its change control, its access model, its audit trail and its decommissioning are all regulated.** "Just patch it" is not available. Neither is an unattended dependency bump, nor an auto-updating tenant the vendor upgrades on its own cadence, nor a configuration change made by an administrator without a change record.

Walk through what that costs you, part by part, because every one of these is something an architect normally treats as free.

Change. A change to a validated system requires an impact assessment, a documented risk evaluation, re-testing proportionate to the risk, approval by Quality Assurance, and an update to the validation documentation set. A GxP system upgrade is a regulatory project with an IT component, not the other way round.

Access. The access model is a regulated control rather than an IT hygiene matter. Unique individual accounts, no shared logins, documented authority levels, periodic access review, and segregation of duties between the person who performs an action and the person who approves it are requirements, not good practice.

The audit trail. Every creation, modification and deletion of a GxP record must be captured with who, what, when, the previous value, the new value, and — for a change to a previously recorded value — a reason. The trail must be computer-generated, independent of the operator, not editable, and retained as long as the record. And it must be *reviewed*, by people, which is the operational burden nobody scopes.

The lifecycle. Validation starts at the user requirement and ends at decommissioning — and decommissioning is itself a validated activity, because the records must remain retrievable and readable for their full retention period after the system that created them is switched off.

The supplier. Validation obligations extend to the software supplier and to any hosting provider, through supplier assessment and audit. That is the real reason validated cloud is still an argument rather than a settled position.

One framing before any detail, because it is architecture material and not legal advice. **Every regulatory passage in this chapter describes what the record has to show, never whether you comply.** Compliance is a judgement a regulator makes about a company, with its own procedures and its own training, and nobody can make it from a book. What an architect can do is make the evidence producible. That is the job, and it is enough work for a career.

Which GxP, and for which record

Two scoping errors cost more than everything else in this chapter, and both are made in the first week.

The first error is treating "GxP" as one requirement set. It is a family. Good Manufacturing Practice governs manufacture and quality control. Good Clinical Practice governs trials in humans. Good Laboratory Practice governs non-clinical safety studies. Good Distribution Practice governs the supply chain. Good Pharmacovigilance Practice governs safety surveillance after launch. Good Documentation Practice is the record-keeping discipline cutting across all of them. The same software product carries different obligations in a GLP toxicology laboratory and a GMP quality-control laboratory, with a different inspectorate and a different organisational model behind each. So the first question about any system is not *is it validated* but **which GxP, and for which record**.

The second error is making the system the unit of scope. It is not. **The regulated record is.** You determine scope by asking which records in the system are GxP records and what decisions depend on them. A customer relationship system that records a sales call is out of scope. The same system, the moment a representative types a patient's symptom into a call note, has created a potential adverse event report and is in scope for intake and hand-off. Nothing about the software changed; the record changed.

That is why **the scoping exercise, not the testing exercise, is where validation effort is won or lost.** Organisations that get scoping wrong spend heavily on low-risk functions because the template says so, and leave the high-risk interfaces uncovered. And scoping is an architecture activity, done on a model, before procurement.

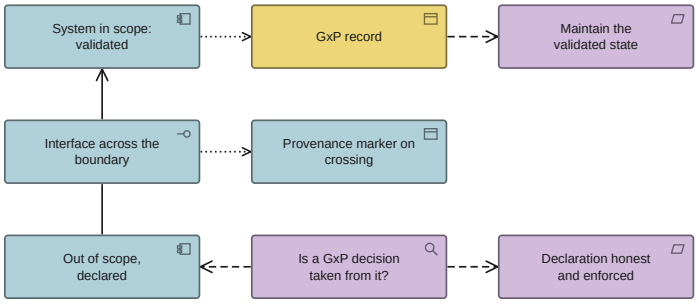


Figure 13.1 The validation boundary, drawn in ArchiMate. The requirement is the load-bearing element: **it is the record that pulls the system into scope**, which is why a system can be in scope for one of the things it does and out of scope for the rest.

Computerised system validation, as it is actually practised

Validation is documented evidence that a system does what it is specified to do, consistently, and will continue to. It is not testing, though testing is part of it. It is a *claim* the company makes, backed by a document set, that a named system in a named configuration on a named infrastructure is fit for a named intended use — plus a process for keeping that claim true.

Three consequences follow immediately. **The claim is specific to a configuration**, so changing the configuration means the claim no longer covers it. **The claim is specific to an intended use**, so "we already validated that laboratory system" does not cover using it for a new product with a new specification type. And **the claim has to be maintained**, because a system validated in 2019 on an operating system that went out of support in 2023 is not in a validated state however good the original binder was.

The V-model, and how it actually runs

The V-model is the organising picture: specification descending the left limb, verification ascending the right, build or configuration at the point. The pairing practitioners state is user requirements against performance qualification, functional specification against operational qualification, design specification against installation qualification [1].

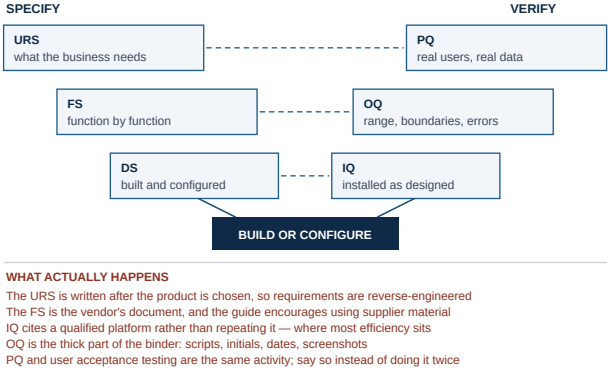


Figure 13.2 The V-model as practised. The pairings on the right are the test an inspector applies; the notes underneath are what actually happens. **Traceability, not the shape, is what makes it work.**

Traceability is the mechanism that makes the V work. A requirements traceability matrix maps every requirement to its specification, its design element, its test case and its test result, and an inspector will pull a requirement at random and walk the chain in both directions.

Now the honest version, because the drawing and the practice differ in five places.

The user requirements specification is written late and badly. In practice it is often written after a product has been selected, which inverts the model: the requirements are reverse-engineered from the product's capabilities so the product can pass. This is extremely common, is not in itself a finding, and is the single biggest reason validation document sets are thick and useless. *That prevalence claim is practitioner consensus — it is not something a company publishes about itself, and I could not source it.*

The functional specification is usually the vendor's document, and for a configured commercial product that is correct: GAMP 5 Second Edition explicitly encourages making use of supplier knowledge, experience and documentation [2].

Installation qualification is largely infrastructure, and is increasingly reused. A qualified platform — qualified operating-system build, qualified database, qualified virtualisation layer — lets each application cite the platform qualification rather than repeat it. This is where most of the available efficiency in a traditional approach sits.

Operational qualification is where the volume is: pre-written step-by-step scripts, executed by a named person, initialled and dated line by line, with screenshots as objective evidence and deviations raised where actual results differ from expected.

Performance qualification blurs into user acceptance testing. Where they are the same activity with two names, say so in the validation plan rather than running it twice.

The validation master plan and the deliverable set

The validation master plan is the site-level or company-level document that says how validation is done here: scope, roles, the system inventory and its risk classification, the lifecycle model, the deliverable set required at each risk level, supplier assessment, the change-control linkage, the periodic-review cycle, and the approach to legacy systems. **It is the first document an inspector asks for in a computerised-systems inspection**, because it tells them what to hold the company to.

Below it, per system, a validation plan scopes that system's effort, and the deliverable set typically runs: validation plan; system and architecture description; risk assessment at system and function level; supplier assessment, with an audit report where risk warrants; user requirements, functional and design or configuration specifications; the traceability matrix; installation, operational and performance qualification protocols, executed with evidence and deviation records; a data migration plan and report where data is migrated; standard operating procedures for operation, administration, backup, restore, disaster recovery, security administration, audit-trail review and periodic review; training records for all users and administrators; and a validation summary report carrying the release-for-use statement.

That list is where unnecessary cost lives. A Category 1 infrastructure component does not need a user requirements specification. A Category 3 instrument driver does not need a design specification. Organisations that produce the full set for every system are not being careful; they are converting assurance budget into paper.

Periodic review

Periodic review asks, at a defined interval, whether the system is still in a validated state. It is not a re-validation. It is a documented check, and in the draft Annex 11 revision it is promoted from a single paragraph to its own chapter with twelve review items and a risk-justified frequency [3] (*secondary source; see the status warning below*).

What it actually examines: changes since the last review and whether each went through change control; deviations and incidents involving the system; **the current configuration against the documented configuration**, which is the drift check and the one that most often finds something; infrastructure and platform currency; backup and restore evidence, and specifically evidence that a *restore* was tested rather than that backups completed; user access review for leavers, privilege creep and generic accounts; audit-trail review records — performed, at the stated frequency, by someone appropriately independent; patch status; supplier status; training currency; and whether the original intended use still matches the actual use.

The outcome is a report with actions, and the actions are the point. A periodic review that finds nothing on a ten-year-old system is itself a finding.

Decommissioning, which nobody budgets

Switching off a validated system is a regulated activity, because the records it holds must remain **available, readable and intelligible** for their retention period. For a batch record under the PIC/S guide that is one year after expiry of the batch or at least five years after certification, whichever is longer [4] — a computation, not a number. For clinical trial records it runs to decades.

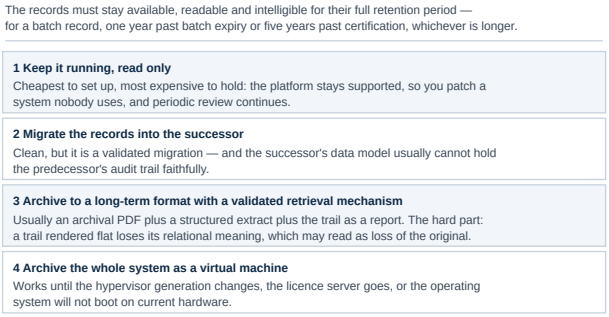


Figure 13.3 Four ways to switch off a validated system, and what each costs. **The decision is really made at implementation**, by whether anyone specified how the records would be got out.

The practical finding, and it is the most valuable thing in this section: decommissioning cost is a function of decisions made at implementation, and almost nobody makes those decisions at implementation. An architect's single highest-value contribution to a new GxP system is to specify, at design time, how its records will be got out of it in fifteen years — the export format, the reference data that makes it interpretable, and the mechanism that will render it.

Seven pieces of folklore, punctured

Each of these is said confidently in pharmaceutical IT meetings, and each is wrong or backwards.

"Validation is a project phase." It is a **maintained state**. Periodic review and change control are operational activities, not project closedown tasks, and the draft Annex 11 frames this explicitly as a shift from project deliverable to operational task [3].

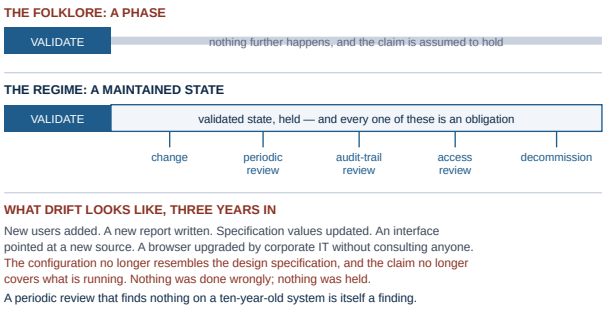


Figure 13.4 Validation is a state held over time, not an event. **The gap between the configuration as designed and the configuration as running is what periodic review exists to catch** — and what a checkbox review routinely misses.

"More documentation is more assurance." The explicit position of GAMP 5 Second Edition's critical-thinking emphasis [2] and of the computer software assurance movement [5] is the opposite. A six-hundred-page package whose traceability matrix was maintained by hand and no longer matches the test scripts passes an audit of its existence and fails an audit of its content.

"You cannot do Agile in a validated environment." You can. GAMP 5 Second Edition recognises that specification and verification are not strictly linear and supports iterative and incremental methods [2]. **What you cannot do is ship without the evidence.** The thing Agile changes is when the evidence is produced, not whether.

"Infrastructure is validated." Infrastructure is qualified; applications are validated [1]. The words are not interchangeable, and using them interchangeably is the fastest way to tell a room of quality people that you have not done this before.

"Hybrid paper-and-electronic systems are non-compliant." Hybrids are legitimate when declared. What a compliant hybrid needs: an explicit statement of which artefact is the record; a documented, verifiable link between the paper and electronic parts — page count, checksum, sequence number, and the draft Annex 11 names checksums as a safeguard [3]; control of printing so a printout cannot be produced, discarded and re-produced without trace; retention of the electronic part even where the paper is the record; and the Annex 11 printout clause satisfied [6]. They fail when the declaration is absent — then neither version is clearly the record, and both are incomplete. The commonest case is the laboratory worksheet: result calculated in a spreadsheet, printed, signed, filed, and the spreadsheet overwritten next time.

"You cannot put a GxP system in the cloud." You can, on five conditions, all of which have to be contracted and evidenced:

1. **A documented shared-responsibility split** — who qualifies what, in writing.
2. **Change notification with a deferral or verification window**, so the vendor's release cadence does not silently end your validated state.
3. **Access to the provider's audit evidence**, standing in for a company audit you will not get.
4. **A data-export and exit mechanism with a tested format**, because records outlive subscriptions.
5. **For personal health data, a lawful cross-border basis** — in South Africa, section 72 of POPIA, under which health information may not leave the country unless there is a lawful basis plus either adequate protection in the destination or POPIA-aligned contractual obligations [7].

Where any of the five is missing, the honest answer is that the system is not in a maintainable validated state in that hosting model. That sentence is worth writing plainly, because it is the one most often fudged. And the alternative is not automatically safer: an unpatched on-premises laboratory system on an out-of-support operating system is not a validated state either, and periodic review will say so.

"Data migration is an IT activity." It is a **validation activity**, and treating it otherwise is the most expensive mistake in pharmaceutical system replacement. Annex 11's project-phase clause is explicit: where data is transferred to another format or system, validation should include checks that data are **not altered in value and/or meaning** during migration [6]. Meaning, not just bytes.

Six hard problems to name at the start of any migration. **Audit trails rarely migrate faithfully**, because the target's structure differs and the source's trail references users, configurations and states that do not exist in the target; exporting it as a report and retaining it alongside is acceptable if declared, and loses the relational meaning. **Signatures** cannot have their binding recreated in a new system, so you retain the signed rendering as a verified true copy with evidence of the original binding — which is why quality-system and trial-master-file migrations are the hardest in the estate. **Open workflow items** — a half-investigated deviation, a change control awaiting approval — are harder than history, and the usual compromise of closing in the old system and re-opening in the new creates records that misrepresent their own history. **Specification and method versions**, because migrating a result without its specification version destroys its interpretability. **Retention clocks do not reset**: a record created in 2009 and migrated in 2026 keeps its 2009-based obligation, and systems that compute retention from the creation date in *that* system get it wrong by seventeen years. And **the migration must itself be auditable** — which records moved, when, under what script version, with what verification outcome, and what happened to the ones that failed.

The economically correct default: **migrate the minimum, archive the rest, and decide at design time.** Most organisations migrate everything because deciding is hard, then find the migration is most of the programme cost and that nobody has looked at the 2011 records since 2012.

GAMP 5, with the version facts right

ISPE GAMP 5: A Risk-Based Approach to Compliant GxP Computerized Systems, Second Edition, was published in July 2022 and runs to 404 pages, produced by the ISPE GAMP Community of Practice [2]. The first edition dates from 2008.

> **GAMP 5 is a guide and has no legal force anywhere.** It is published by a trade body. Regulators reference it, inspectors recognise it, and "we followed GAMP 5" is a credible position — but the obligation is to the regulation: Part 11, Annex 11, or the PIC/S guide's Annex 11. An architect who says "GAMP 5 requires..." is wrong in a way an inspector will notice.

What the Second Edition changed, in its publisher's own description [2]: greater emphasis on service providers and on making use of supplier documentation; recognition that **specification and verification are not strictly linear**, with support for iterative and incremental methods — which is the opening for Agile and the reason the V-model is now one option rather than *the* model; increased use of software tools and automation, meaning automated testing as assurance evidence; explicit emphasis on **critical thinking** by experienced subject-matter experts, in a dedicated appendix; discussion of the United States computer software assurance activity; and expanded appendices on blockchain, artificial intelligence and machine learning, cloud computing and open-source software.

A second correction, because it is one of the commonest: there is no GAMP Category 2. Category 2 covered firmware in earlier versions and was withdrawn; firmware now falls into the other categories depending on whether it is configured [1]. An author who writes "Categories 1 to 5" is reproducing a pre-2008 scheme.

Category	What it is	Typical approach
1	Infrastructure software: operating systems, database engines, middleware, network monitoring	Record version and configuration; verify correct installation. Infrastructure is **qualified**
3	Non-configured products used as supplied; instrument firmware at default settings	Risk-based testing against intended use, plus supplier assessment
4	Configured products: quality systems, enterprise resource planners, laboratory systems, execution systems, data capture	Test the configuration and the business process it supports against a controlled configuration specification
5	Custom applications: bespoke code, interfaces, macros, scripts, reports	Full lifecycle with requirements, design, code review, unit and integration testing, full traceability

Two practical points. **The category is assigned to a component, not to a system** — a real laboratory-system implementation is Category 4 for the product, Category 5 for its bespoke interfaces and custom reports, and Category 1 for the platform beneath, and the effort is the sum. And **the category is a starting point for effort estimation, not a substitute for risk assessment**: a Category 3 instrument producing a release-critical result deserves more scrutiny than a Category 5 report nobody decides anything from.

Computer software assurance, and the correction that matters

Computer software assurance is a risk-based reframing of validation that puts assurance effort where patient risk is, and explicitly permits unscripted and exploratory testing where risk is low. The complaint it answers is real: validation as templated rewards document volume, so a low-risk function gets a fully scripted test with screenshots because that is the template, and a high-risk function gets the same, which means the high-risk function is under-examined relative to its risk.

Now the correction, and it is one of the most commonly mis-stated facts in the field.

> **The FDA's "Computer Software Assurance for Production and Quality System Software" is final, issued September 2025, docket FDA-2022-D-0795, from CDRH and CBER — and it relates to the Quality System regulation at 21 CFR Part 820. That is medical devices, not drug GMP.** It supersedes Section 6 of the FDA's earlier "General Principles of Software Validation" [5]. Pharmaceutical manufacturers adopt the *thinking* widely and defensibly, but a pharmaceutical company citing this guidance as its authority for a Part 211-scope system is citing a device guidance by analogy and should say so. I could find no equivalent instrument for drug GMP under Parts 210 and 211.

Two further points. **It is not a replacement for validation and does not remove the obligation** — the deliverable is still documented evidence of fitness for intended use; what changes is how much of that evidence is a scripted test and how much is a risk rationale, an unscripted test record, a supplier's test result or an automated one. And **it is aligned with GAMP 5 Second Edition rather than competing with it** [2][5].

There is no European equivalent instrument. The nearest movement is the quality-risk-management emphasis running through the draft Annex 11 revision — and a consultancy reading of that draft argues it requires executed test scripts and traceability as a pass/fail criterion while allowing alternatives that provide the same or higher level of control [3], which, if the final text retains it, would be a materially more conservative posture than the American one.

21 CFR Part 11 and EU GMP Annex 11

Part 11 is a United States regulation in force since 1997, applying where a company chooses to keep records required by a predicate rule in electronic form, or to use electronic signatures in place of handwritten ones. **It has no independent record-keeping requirements of its own.** The requirement to keep a batch record comes from the drug GMP regulations; Part 11 governs *how*, if you keep it electronically.

Its substance, in the order an architect needs it: validation of systems to ensure accuracy, reliability and the ability to discern invalid or altered records; **accurate and complete copies in both human-readable and electronic form, suitable for inspection, review and copying** — the clause that means "can you give the inspector the data in a usable electronic format", and the one many systems fail; record protection for ready retrieval through the retention period; limiting access to authorised individuals; secure, computer-generated, time-stamped audit trails that do not obscure previously recorded information; operational system checks enforcing permitted sequencing; authority and device checks; training requirements and written policies holding individuals accountable for actions under their electronic signatures; and controls over system documentation.

Two things about Part 11 are widely forgotten.

Its scope was narrowed in 2003. An FDA guidance announced a narrow interpretation and enforcement discretion for several provisions, in response to industry criticism that a literal reading imposed cost without patient benefit. The practical result is that Part 11 is enforced against the predicate-rule requirement, with the audit-trail, record-copy and validation provisions carrying most weight, rather than as a uniform checklist. **An architect working from the 1997 text alone will overstate the obligation.**

For clinical investigations the operative document is newer. The FDA's final guidance "Electronic Systems, Electronic Records, and Electronic Signatures in Clinical Investigations: Questions and Answers" was issued as final in October 2024, with the notice of availability in the Federal Register of 2 October 2024, docket FDA-2017-D-1105 [8]. It addresses sponsor and investigator use of electronic systems, third-party service providers, and records held by third parties. Design a clinical system for a United States filing from that document, not from Part 11 in the abstract.

> **No product can be "Part 11 compliant."** Part 11 applies to records and signatures, not to software. Compliance is a property of the system *as implemented, configured, administered and operated* by the company, together with its procedures and training. A product can be *capable* of supporting compliance — and the characteristic failure is that the capability ships switched off: audit trail not enabled because it costs performance and disk, unique accounts not enforced because the shop floor shares a login, electronic signature configured without the meaning of signing.

Annex 11 of the EU GMP Guide covers computerised systems, and the version in force is still the revision of January 2011 [6]. It applies to all forms of computerised system used in GMP-regulated activities — there is no election. Its substance runs across a principle requiring the application to be validated and the infrastructure qualified; a general section on risk management, personnel roles and suppliers and service providers; a project-phase section requiring validation documentation, **an up-to-date listing of all relevant systems and their GMP functionality**, traceable user requirements, evidence of appropriate test methods including data limits, boundaries and error handling, and verification of value and meaning on migration; and an operational section covering accuracy checks, built-in checks, audit trails, change and configuration management, periodic evaluation, security, incident management, electronic signatures, batch release, business continuity, archiving and printouts [6].

Where they differ

Dimension	21 CFR Part 11	EU GMP Annex 11
Legal nature	A regulation, enforceable, cited in its own right	An annex to a guideline, given force through member-state transposition; cited as GMP non-compliance
Trigger	Applies when you *choose* electronic records or signatures for predicate-rule records	Applies whenever a computerised system is used in a GMP activity — no election
Audit trail	**Required** for records subject to the rule: secure, computer-generated, time-stamped	**Risk-based**: consideration to be given on a risk assessment. The draft makes it effectively mandatory where users can change data, settings or access [3]
Supplier and service provider	Largely silent; handled through validation and documentation control	**Explicit section**: formal agreements, defined responsibilities, competence assessment, review of off-the-shelf documentation
Signature mechanics	**Prescriptive**: two identification components, session rules, a certification letter to the FDA, printed meaning	**Brief**: same impact as handwritten, permanently linked, with time and date

Dimension	21 CFR Part 11	EU GMP Annex 11
Printouts	Addressed as accurate and complete copies in human-readable and electronic form	**Explicit clause**: a batch-release printout must indicate whether any data has changed since original entry
Business continuity	Not addressed as such	**Explicit clause**
Periodic evaluation	Not explicit; emerges through validation maintenance	**Explicit clause**
Batch release	Not addressed	**Explicit clause** on certification by a validated system — and the point at which South Africa diverges
Scope breadth	Records and signatures	The whole lifecycle, infrastructure, people and suppliers

The honest summary: Annex 11 is shorter, broader and more lifecycle-oriented; Part 11 is narrower, more prescriptive about signatures, and legally sharper. A system designed for Annex 11 usually needs signature mechanics added for Part 11. A system designed for Part 11 usually needs supplier agreements, periodic evaluation, business continuity and the printout clause added for Annex 11. Design for both from the start; retrofitting either is expensive.

The Annex 11 revision: dated and unresolved

This is the live item, and the discipline is to date the statement and refuse to predict.

- **The version in force remains the January 2011 revision.** The European Commission's own EudraLex Volume 4 index lists Annex 11 as "Computerised Systems, revision January 2011" and carries no draft, consultation, adoption or entry-into-operation entry for a revision [6].
- **A draft was issued for public consultation jointly by the European Commission and PIC/S, running from 7 July 2025 to 7 October 2025.** The consultation covered **three documents together**: a revised Chapter 4 on documentation, a revised Annex 11 on computerised systems, and a **new Annex 22 on artificial intelligence**. The drafts were prepared by a joint inspectors' working group [9][10].
- **No final text and no date have been published.** PIC/S's own news page, whose newest items at retrieval are from 2026, does not report the outcome, adoption or entry into force of any of the three [9]. A consultancy reading states the working group's workplan targets handing final text to the Commission toward the end of 2026, about half a year later than originally planned, with a transition period after publication during which the 2011 version continues to apply [3] — *a single secondary source presenting its own interpretation, and not a date to plan against.*

What the draft appears to change, with that caveat throughout: growth from five pages to nineteen; codification of expectations previously found only in questions-and-answers documents and data-integrity guidance; quality risk management across the whole lifecycle; **security expanded from four paragraphs on access control to around twenty requirements** covering patching, firewalls, penetration testing and disaster recovery, which makes IT security explicitly a GMP topic; identity and access requirements including multi-factor authentication for critical remote access; **alarm handling as its own chapter**; **audit trails mandatory rather than risk-considered** where users can change data, settings or access, with review that is risk-based, targeted at manual changes, performed by someone not directly involved and **normally completed before batch release**; electronic signatures requiring re-authentication at signing; periodic review as a programme with twelve items; and data governance including the ALCOA++ formulation sitting in the **Chapter 4** draft rather than in Annex 11, which matters for where you look [3][10].

Two cautions for an architect. **The audit-trail clause as drafted makes audit-trail review a gating step in the release process with a defined latency**, which it very often is not today. And if the final text retains prescriptive controls such as authentication factors and password rules, those will age badly inside a GMP annex revised roughly once a decade — a criticism the consultancy source itself makes [3]. Note also that the same package carries a new Annex 22 on artificial intelligence in manufacture, covering model selection, training, validation, defined intended use, performance metrics, training-data quality, test-data management and continuous oversight including change control [9][10]. **It is unresolved on the same timetable, and anything you build with a model in a GMP decision path should be designed on the assumption that it will be inspected against something.**

Data integrity

ALCOA began as a mnemonic for the attributes of reliable data and was extended to ALCOA+ by adding four more. **The design obligations are the point, not the mnemonic.**

Attribute	What it obliges by design
Attributable	Unique individual accounts; **no shared or generic logins**; identity verified before an account is issued; and the identity carried through every interface, so a record arriving from another system names the originator and not the service account
Legible	Durable format; no proprietary binary only a decommissioned version can open; readable renderings available for the retention period; the audit trail readable as well as present
Contemporaneous	**The system clock is a GxP control** — synchronised, auditable, with time-zone handling declared; no backdating; offline capture recording the *actual* observation time, with the system making it visible when that differs from the sync time
Original	Identify the original record explicitly — the instrument's raw data file, not the printed result; retain it; where a copy is used, verify it and document the verification. **This is the attribute laboratory instruments routinely fail**
Accurate	Input validation; calculation verification; no silent rounding or unit conversion; second-person review where risk warrants; reconciliation between systems; error handling that fails loudly

Attribute	What it obliges by design
Complete	**No deletion.** All injections including failed ones, all trial runs, all reprocessing, all aborted sequences — retained and visible. Deleting a "bad" result is the archetypal data-integrity fraud
Consistent	Chronological integrity across systems; one authoritative clock; event ordering preserved through queues and batch interfaces; no interface that reorders
Enduring	Archive strategy, media refresh, format migration, readable rendering, a decommissioning plan — not a laboratory PC's local drive
Available	Retrievable within a reasonable time for an inspector; searchable; restorable; and the restore actually tested

Two corrections that matter, and the second is the one to carry into a South African engagement.

> **PIC/S PE 009 Part I never uses the term "ALCOA"** in the edition read for this book [4]. The formulation lives in data-integrity guidance, not in the GMP guide's chapters; the guide's nearest statements are its documentation principle on accuracy, integrity, availability and legibility, its clause requiring records to be made at the time of each action, and its clause requiring alterations to be signed and dated [4]. A book or a slide citing "PIC/S Chapter 4 ALCOA" is citing something that is not there in those words.
>
> **The verified South African ALCOA statement is ALCOA-CCEA, and it is in the South African Good Clinical Practice guidelines for clinical source documents** — attributable, legible, contemporaneous, original, accurate, plus complete, consistent, enduring and available, with source documents to be signed and dated visit by visit [11]. **That is a GCP requirement, not a GMP one.** Get the jurisdiction of your citation right; auditors notice.

The guidance stack behind data integrity, in order of usefulness to an architect. **PIC/S PI 041-1, "Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments", 1 July 2021, 63 pages** [12] — the most operationally detailed, and the most relevant to South Africa because SAHPRA is a PIC/S participating authority. It went through drafts in 2016 and 2018 before the 2021 final, so **a citation to "PI 041 (2016)" or "(2018)" is citing a draft.** Then the MHRA's "'GXP' Data Integrity Guidance and Definitions", Revision 1, March 2018 [13] — short, readable, and the source of the clearest published definitions of raw data, true copy, data lifecycle and audit trail, applying across GxP rather than GMP alone. The World Health Organization and the FDA each publish data-integrity guidance as well; *the research behind this book could not verify their current reference numbers and dates, so none is printed here.* ALCOA++ appears in the draft Chapter 4, and the additional element is usually described as traceability — *but the draft's own definition could not be sourced, so treat the "++" as in flight* [3].

Audit-trail review, the most under-resourced obligation in the regime

This deserves its own treatment because it is the single most under-estimated recurring cost in a validated estate.

The obligation is not to *have* an audit trail. It is to review it — to look at the recorded changes and satisfy a reviewer that nothing improper happened. The volume is the problem. A mid-size quality-control laboratory running a chromatography data system across a dozen instruments generates thousands of trail entries a week: every log-in, every method change, every integration adjustment, every reprocessing, every sequence edit, every result approval.

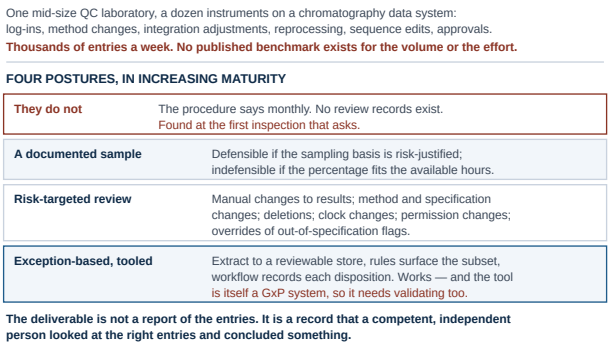


Figure 13.5 The audit-trail review burden. **Having a trail is not reviewing it**, and under the Annex 11 draft the review is normally completed before batch release — which puts it on the release critical path with a latency nobody has costed.

How organisations actually cope, in increasing order of maturity. **They do not** — the procedure says monthly, no review records exist, and it is found at the first inspection that asks. **A documented sample** — a defined percentage reviewed by procedure, defensible if the sampling basis is risk-justified and documented, indefensible if the percentage was chosen to fit the available hours. **Risk-targeted review** — only the entry types that matter: manual changes to results, changes to methods and specifications, deletions, changes to time and date settings, access-permission changes, overrides of out-of-specification flags. This is the posture the draft Annex 11 describes, and it adds that the review should be performed by someone not directly involved and **normally completed before batch release** [3]. **Exception-based review with tooling** — trail data extracted to a reviewable store, with rules that surface the reviewable subset and a workflow recording the reviewer's disposition of each item. That works, and it introduces a new problem: **the review tool is itself a GxP system handling GxP records, so it needs validating** — and if it is a cloud analytics platform, the whole five-condition argument above applies to it.

Three findings for an architect, and they are design findings rather than process ones.

1. **Audit-trail review is a workflow requirement, not a reporting requirement.** The deliverable is not a report of the entries; it is a record that a competent, independent person looked at the right entries and concluded something. Any design producing a PDF and no disposition record has built the wrong thing.
2. **It is a release-path dependency with a latency.** If review must complete before batch release, review capacity becomes part of release lead time, and a backlog is a supply problem as well as a compliance one.
3. **Reviewer independence constrains the access model.** The reviewer must not be the person who performed the activity, which means a role that can read the trail across the laboratory without being able to perform laboratory work — a role many products do not have out of the box.

Say the quiet part plainly: nobody scopes the capacity for this. I looked for a published benchmark of typical trail volumes or the full-time-equivalent cost of review at a mid-size site and found none, which is itself the finding. It makes an audit-trail volume extract from a client's chromatography system and quality-control laboratory system over a representative month one of the most worthwhile early discovery exercises on any pharmaceutical engagement — and it is cheap.

Electronic signatures

What makes one compliant, assembled from both regimes: **unique to one individual**, never reused, with identity verified before allocation; **bound to the specific record** so it cannot be excised, copied or transferred to falsify another — which rules out a signature as an image in a document; **carrying the printed name, the date and time, and the meaning of the signing**, because a signature without a declared meaning is a timestamp; **at least two distinct identification components** for non-biometric signatures, both required for the first signing in a continuous session and at least one thereafter; **backed by a written policy** holding individuals accountable, so repudiation is not available; and, for the United States, **certified to the FDA in writing before first use** — a one-time letter per organisation, not per system, and very frequently forgotten by companies entering that market.

What does not qualify, and is routinely attempted: a scanned wet signature pasted into a document; an "approve" button recording a user identity and no meaning; a shared departmental account approving on behalf of a person; an envelope-signing service applied to a PDF produced from a system whose own record remains unsigned and editable; an approval in a ticketing system with no binding to the record it approves.

The architectural consequence that is easy to miss: because a signature must be bound to the record, a signed record cannot be migrated without breaking the binding — unless the migration carries the signature's full manifestation and the evidence of the binding, or the signed rendering is retained as a verified true copy.

PIC/S, SAHPRA, and where the rules actually live

For a South African reader this is the most useful section in the chapter, and it contains the chapter's single most quotable finding.

The Pharmaceutical Inspection Co-operation Scheme lists 53 jurisdictions as participating authorities, shown as 57 authority entries because four countries have two each. SAHPRA is listed with accession to the scheme in July 2007 [14]. Accession predates SAHPRA itself: the authority that acceded was its predecessor, and the members page carries the current name against the 2007 date [14].

Three facts that place South Africa correctly.

South Africa acceded in July 2007 — earlier than the United States FDA (January 2011), Japan (July 2014), Brazil (January 2021) and Saudi Arabia (July 2023) [14]. So "SAHPRA only adopts PIC/S because it is not a member" is wrong twice over: it is a member, and it has been one for longer than

several markets an architect might assume lead it. The newest accession in the list is Jordan, January 2026 [14], which both dates the retrieval and shows the scheme is still growing.

PIC/S membership means mutual confidence in inspection outcomes, not mutual recognition. A SAHPRA GMP certificate carries weight with other participating authorities; it is not a licence to market, and it does not mean one member accepts another's batch certification. An import into South Africa has its own requirements regardless of where it was released.

The PIC/S GMP guide is PE 009, and the current version is PE 009-18, adopted 22 September 2026 by written procedure of the PIC/S Committee and in force from 24 September 2026, superseding PE 009-17 [15]. The change is the incorporation of the revised Annex 19 on reference and retention samples. The European index independently lists that revised annex as applicable from the same date [6], which corroborates it from the other side.

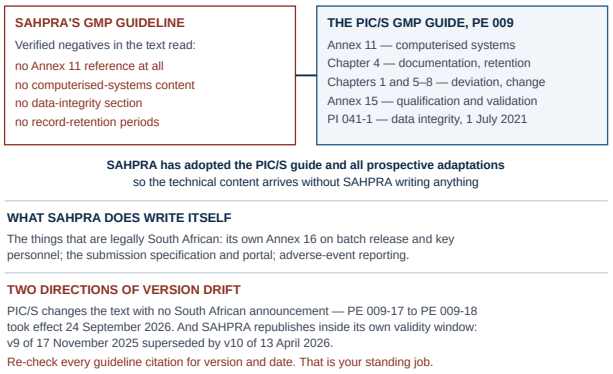


Figure 13.6 Where a South African manufacturer's GMP requirements actually live. **The three verified negatives on the left are the proof** — and the reason an architect who designs to the SAHPRA website specifies no validation at all.

Do not cite SAHPRA for GMP detail

SAHPRA's GMP guideline **adopts the PIC/S Guide to GMP**, and the version the research behind this book read said so plainly: it had adopted the guide "and all prospective adaptations as prescribed by the PIC/S" [16]. **The adaptation is specific and narrow: PIC/S Annex 16 is replaced by a South African Annex 16** set out in the guideline's own section 4, which carries the batch-release and key-personnel content [16].

Now the proof, and it is the chapter's most quotable finding. Three verified negatives, checked against the full text:

- **No reference to Annex 11 and no computerised-systems content at all.** The nearest statements are general — correct and effective record keeping for pharmacy records, and training records that should be kept [16].
- **No data-integrity section, and the guideline never uses the term "data integrity"** [16][17].
- **No record-retention periods.** Its only record-keeping obligations are specific and unrelated: registers of certain scheduled substances balanced quarterly, and records of consultants with name, address, qualifications and period of service [16].

> **Therefore the PIC/S guide, not SAHPRA's guideline, is where batch-record retention, deviation handling, change control and the computerised-systems annex actually live for a South African manufacturer.** This is verified from SAHPRA's own current text rather than inferred. The pattern is consistent: SAHPRA writes its own rules for the things that are legally South African — who releases a batch, how you submit a dossier, how you report an adverse event — and adopts PIC/S for the technical content of GMP. **An architect who looks only at the SAHPRA website will find the submission and pharmacovigilance rules, find no computerised-systems requirements, and conclude wrongly that South Africa has none.**

Two consequences, both of which belong in the model.

The annual product quality review. The current SAHPRA guideline mentions the review only as a shared responsibility of the heads of production and quality control, with no detailed requirements, and it does not carry a frequency [17]. South African sites nevertheless run them, and PIC/S expectations apply because SAHPRA adopted the guide. **The annual frequency comes from the PIC/S guide, which says such reviews should normally be conducted and documented annually for all authorised products** [4] — not from SAHPRA. Do not infer from SAHPRA's silence that either the review or data integrity is optional.

Version drift is the standing risk, in two directions. Because SAHPRA adopts prospective adaptations, **the applicable text changes when PIC/S changes it — PE 009-17 to PE 009-18 on 24 September 2026, for instance [15] — without a South African announcement.** And SAHPRA's own guideline versions move faster than their stated validity: the research behind this book found **version 9 dated 17 November 2025 superseded by version 10 dated 13 April 2026** inside its own five-year window, and its two research passes disagree about which text is operative [16][17].

> **Every SAHPRA guideline citation in this book, and in your own work, needs a version-and-date re-check before you rely on it.** Re-checking is the reader's standing job, not the author's. Three generations of a South African GMP guideline are discoverable on the public web at once, which is a documented hazard for anyone researching this area.

And here is the architecture gap to name out loud: PIC/S publications, not SAHPRA's website, are where the requirements change — and in most mid-size South African companies nobody owns watching them. That is not a process gap to be fixed with an email reminder. It is a named capability with an owner, a watch list, an assessment path into change control, and a place to record what was assessed and concluded. Put it on the model. It costs almost nothing and its absence means a company does not know when its own requirements have moved.

The Responsible Pharmacist

South Africa's releasing authority is the Responsible Pharmacist, and this is not a naming difference from the European Qualified Person. The substance differs in four ways that reach the systems.

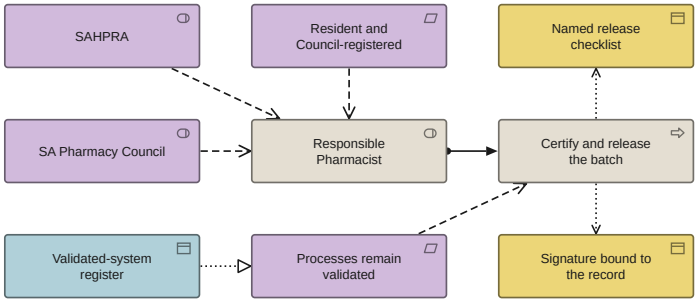


Figure 13.7 The Responsible Pharmacist's release duty. Answerable to **both** SAHPRA and the Pharmacy Council, resident in the Republic — and carrying the one clause that reaches straight into the estate: **processes must remain in a validated state.**

Dual accountability. The guideline lists the Responsible Pharmacist as key personnel, **responsible both to SAHPRA and to the Pharmacy Council** [16]. A Qualified Person answers to the medicines regulator. Two professional bodies, two disciplinary routes.

A validated-state duty. The guideline assigns the Responsible Pharmacist the duties of ensuring that all batches released for sale, supply or export meet the marketing authorisation, GMP and legal requirements; that batch release considers storage and transport conditions, sampling and quality-control results; **that manufacturing and testing processes remain in a validated state**; and that a self-inspection programme is active and current [16]. Read that third item again. **In South Africa, keeping systems in a validated state is explicitly a Responsible Pharmacist duty, not an IT matter.** It makes validation status a release-relevant fact, and it gives the Responsible Pharmacist a legitimate interest in the validated-system inventory, periodic-review status and open change records. A release dashboard that does not surface validation status is incomplete against the duty as written.

A named release checklist. The guideline assigns a pharmacist or other legally authorised person responsibility for release for sale of the finished product, **supported by a release checklist covering all important release criteria** [16]. It is a named artefact in the guideline. **It should be a system object with a state, not a word-processor document** — and its items are the integration requirements for the release step: authorisation conformity, GMP conformity, legal requirements, storage and transport conditions, sampling, quality-control results.

Residency and registration. The Responsible Pharmacist must reside in the Republic and be registered with the Pharmacy Council [16], with not more than one per licence. **So the release signature cannot be delegated offshore to a shared-services centre** — which rules out a design pattern that works in other industries and in other pharmaceutical functions. And because the signature is a pharmacist's professional act with professional-conduct consequences, the signature control is load-bearing in a way a commercial approval is not: the system must make the *content* of what is being signed unambiguous at the point of signing.

What the architect is actually expected to produce

Not a validation methodology — that belongs to the validation lead, and in most South African mid-size companies that person is the scarcest in the building. Four artefacts, and they are architecture artefacts.

One: a register of validated systems, with GxP classification and validation state. This is the deliverable that answers an inspection's first question. **Make it the application portfolio with extra columns, not a separate quality-owned spreadsheet** — the near-universal state is two registers that disagree, and the disagreement is discovered by an inspector rather than by you.

VALIDATED SYSTEM — one row per instance, per environment, per defined use	
GxP classification	GMP / GCP / GVP / GLP / GDP / several / none
GxP record types held	the list that answers the inspector's first question
Validation state	not validated / under validation / validated / validated with open deviations / revalidation due / retired but retained
Documentation references	URS, FS, DS, risk assessment, IQ/OQ/PQ, matrix, summary report, release-for-use approval
Last validation date	and the periodic review due date
Business owner	a named person, not a department
Technical custodian	a named person
Electronic signatures	applied or not, and under what control
Audit trail	present / reviewable / review frequency / reviewer
Part 11, Annex 11 scope	which applies, and to which records
Retention obligation	and whether it can be met without the system
Hosting and data location	also the POPIA section 72 input
Supplier qualification	for anything vendor-hosted
Decommissioning plan	how its records stay readable afterwards
Make it the application portfolio with these columns added, not a separate quality spreadsheet.	

Figure 13.8 The validated-system register as an object. **The grain is one instance, one environment, one defined use** — and the row nobody owns is the interface between two systems that are each validated.

The grain is **one system instance, in one environment, for one defined use.** Not one row per application. A productive instance and a validation instance are different rows. The same product validated for two different purposes has two validation scopes. A multi-tenant hosted instance is one row from your side, carrying a dependency on the vendor's own quality system, which is a supplier qualification object rather than a validation one. **And an interface between two validated systems is in scope and is almost always nobody's row.**

Two: a validation boundary diagram. What is inside, what is outside, what is inside for part of what it does, and every crossing named. At each declared crossing: an interface specification that is a validation deliverable; verification at the boundary with the result recorded; and a provenance marker on every record that crossed. Declaring something out of scope is a valid architectural act provided the declaration is honest and enforced — and the inverse trap is directional, because a reporting platform comes into scope the day a trend chart from it justifies a specification.

Three: a change-control path that works. A high-volume change pipeline — a cloud quality system releasing quarterly, a customer system releasing three times a year — meeting a validation process that takes six weeks per change produces one of two outcomes: the organisation falls behind the vendor's supported version, or it quietly stops assessing changes. Both are findings. The answer is a pre-agreed, risk-tiered change framework with automated regression evidence, **agreed with Quality before the platform is bought** — which is almost never when it is discussed.

Four: an audit-trail review plan that is resourced. Which trails, which entry types, at what frequency, by which independent role, with what disposition record, through which tool, and whether that tool is itself validated. With a number of hours attached and a named person who has them.

Those four are the chapter's deliverable. If you produce nothing else in your first ninety days, the first and the fourth will change what the company knows about itself — because the first tells it which of its systems hold regulated records, and the fourth tells it what it has promised to do about them and whether anyone is doing it.

THE CHAPTER IN TEN LINES

- A system that holds a GxP record is validated, which puts its lifecycle, change control, access model, audit trail and decommissioning under regulation.
- The regulated record rather than the system is the unit of scope, so the first question is which GxP applies and to which record.
- Validation is a maintained state rather than a project phase, more documentation is not more assurance, and infrastructure is qualified while applications are validated.
- Agile and hybrids are both permitted, and cloud is permitted on five contracted conditions; what is never permitted is shipping without the evidence.
- Migration is a validation activity requiring verification of value and meaning, and decommissioning cost is set by decisions taken at implementation.
- GAMP 5 is a guide with no legal force, published July 2022 in 404 pages, there is no Category 2, and no product can be Part 11 compliant.
- The FDA computer software assurance final guidance of September 2025 is a device guidance under Part 820; pharma adopts it by analogy and should say so.
- The Annex 11 version in force is still January 2011, and the draft consulted from 7 July to 7 October 2025 has no published final text or date.
- Having an audit trail is not reviewing it, and PIC/S PE 009 Part I never uses the term ALCOA: the verified South African statement is ALCOA-CCEA in GCP.
- Do not cite SAHPRA for GMP detail, re-check every guideline citation for version and date, and produce the register, the boundary diagram, a workable change-control path and a resourced audit-trail review plan.

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- **4.** PIC/S, PE 009-17 (Part I), Guide to Good Manufacturing Practice for Medicinal Products, 25 August 2023 — the edition read for this book (batch documentation retention at §4.11 — one year after batch expiry or at least five years after certification, whichever is longer; product quality review normally annual at §1.10; Chapter 4 documentation principle; the term ALCOA is not used in Part I). PE 009-18 came into force 24 September 2026 and was not read; re-check each clause and the absence of the term against it before citing https://picscheme.org/docview/11332
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- **7.** goLegal, "New POPIA Regulations: Processing of health information", 10 April 2026 (legal-commentary secondary source; health information as special personal information, section 32(2) duty of confidentiality, section 72 restriction on cross-border transfer) https://www.golegal.co.za/popia-regulations-health/
- **8.** US Federal Register, Vol. 89 No. 191, 2 October 2024 — notice of availability of the final guidance "Electronic Systems, Electronic Records, and Electronic Signatures in Clinical Investigations: Questions and Answers", docket FDA-2017-D-1105 https://www.govinfo.gov/content/pkg/FR-2024-10-02/html/2024-22562.htm
- **9.** PIC/S, "Joint stakeholders consultation on the revision of Chapter 4 and Annex 11 of the PIC/S GMP Guide and on the new Annex 22" (consultation 7 July to 7 October 2025, run jointly by the European Commission and PIC/S; drafts prepared by the joint EMA and PIC/S Inspectors' Working Group; no outcome, adoption or entry into force reported on the page) https://picscheme.org/en/news/joint-stakeholders-consultation-on-the-revision-of-chapter-4
- **10.** Regulatory Affairs Professionals Society, "European Commission proposes updated GMP guidelines for AI", 9 July 2025 (trade-press secondary source: consultation opened 7 July with comments to 7 October; scope of the Annex 11 revision and the new Annex 22 on artificial intelligence; no finalisation timeline stated) https://www.raps.org/resource/european-commission-proposes-on-updated-gmp-guidel.html
- **11.** Department of Health, South African Good Clinical Practice: Clinical Trial Guidelines, third edition (ALCOA-CCEA for clinical source documents at §5.10.1.2, with source documents signed and dated visit by visit) https://www.sahpra.org.za/wp-content/uploads/2021/06/SA-GCP-2020_Final.pdf
- **12.** PIC/S, PI 041-1, "Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments", 1 July 2021, 63 pages (drafts in August 2016 and November 2018 preceded the final) https://picscheme.org/docview/4234
- **13.** MHRA, "'GXP' Data Integrity Guidance and Definitions", Revision 1, March 2018 https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/687246/MHRA_GxP_data_integrity_guide_March_edited_Final.pdf
- **14.** PIC/S, "Members" — participating authorities list (53 jurisdictions, 57 authority entries; South Africa, SAHPRA, accession to the PIC/S Scheme July 2007; US FDA January 2011; Japan July 2014; Brazil January 2021; Saudi Arabia July 2023; newest entry Jordan, January 2026) https://picscheme.org/en/members
- **15.** PIC/S, "Revision of PIC/S GMP Guide (PE 009-18)" — adopted 22 September 2026 by written procedure of the PIC/S Committee, in force 24 September 2026, superseding PE 009-17, incorporating the revised Annex 19 https://picscheme.org/en/news/revision-of-pics-gmp-guide-pe-009-18
- **16.** SAHPRA, SAHPGL-INSP-02 v9, Guidelines to Good Manufacturing Practice for Medicines, 17 November 2025 (adoption of the PIC/S Guide to GMP and all prospective adaptations at §2.1; South African Annex 16 at §4; Responsible Pharmacist at §§4.1.2, 4.1.2.5, 4.1.2.6 and 4.1.3.2; verified negatives: no Annex 11 reference, no computerised-systems content, no data-integrity section, no record-retention periods) https://www.sahpra.org.za/wp-content/uploads/2025/11/SAHPGL-INSP-02_v9-Guidelines-to-Good-Manufacturing-Practice-for-Medicines.pdf
- **17.** SAHPRA, SAHPGL-INSP-02 v10, Guidelines for Good Manufacturing Practice of Medicines, 13 April 2026 (replaces v9; states the PIC/S adoption at §2.1; mentions the product quality review only as a shared responsibility of the heads of production and quality control, with no detailed requirements; the term "data integrity" does not appear) https://www.spotlightnsp.co.za/wp-content/uploads/2026/05/final-gmp-guideline-for-publication_20260424.pdf

CHAPTER 14

The reference architecture

Pharma will hand you data standards more prescriptive than anything you have seen, and no enterprise model whatsoever. This chapter establishes exactly what is mandatory, exactly where every standards body stops, and what to do about the gap.

An architect arriving in pharma from another sector asks the question they have always asked: where is the industry model? Two things then happen. Somebody hands them a stack of standards far more prescriptive than anything they have seen before — dataset names, message schemas, controlled vocabularies, dated cliff edges, automated gateway checks that will refuse a filing outright. And nobody hands them an enterprise model at all, because there is not one.

Both of those facts are true at once, and the whole value of this chapter is in not blurring them.

Two different questions that sound like one

There are two separate questions hiding inside "does pharma have a reference architecture?"

The first is: **has anyone specified what the data has to look like?** In pharma the answer is an emphatic yes, and more tightly than in any other industry in this series. Regulators do not merely recommend data standards. They name datasets, name a metadata document, name file tags, name the sections of a submission those must appear in, give a date, and wire the check into the machine that receives the filing.

The second is: **has anyone specified how to organise the enterprise behind it?** Capabilities, value streams, business services, an application portfolio taxonomy, integration patterns, a technology reference model. The answer there is no. Not partially, not in draft. Nothing.

Manufacturing, the sibling volume in this series, found the opposite shape of problem: no industry reference model, plus a control-systems hierarchy standard and an interoperability binding that nobody had reconciled with each other. Pharma inherits that gap at the plant and adds a mandatory regulatory data layer on top.

> Pharma has mandatory, dated, machine-enforced data standards at the regulatory interface — and no enterprise reference architecture whatsoever. Those are different problems. An architect who conflates them will either over-build (treating a tabulation model as a target-state application architecture) or under-build (waiting for an industry model that is never going to arrive).

The single comparative sentence an architect from another sector needs is this. **Banking has BIAN and insurance has ACORD — industry bodies that publish a decomposition of the enterprise itself, services and all. Pharma has nothing of that kind, and instead has something those sectors do not: a set of data contracts at the regulatory boundary that a regulator will enforce by refusing your submission.** The constraint in pharma is tighter than a reference architecture. It is just much narrower.

What is genuinely mandatory

This is the part most pharma IT material gets wrong in both directions, so it is worth being precise and dated. Five obligations below are real, in force and enforced. Everything else in the chapter is guidance, convention or convergence.

FDA's Technical Rejection Criteria for Study Data. FDA does not publish a sentence saying "CDISC is required". It publishes a Data Standards Catalog that names supported and required standards and versions [1][2], and it enforces conformance at the electronic submission gateway through the Technical Rejection Criteria [3]. Those criteria state that a failing submission will be rejected and the sponsor notified, with **refuse-to-file** for new drug and biologics licence applications and **refuse-to-receive** for abbreviated applications [3]. The criteria name specific datasets — a Trial Summary (TS), a Demographics (DM), a subject-level analysis dataset (ADSL) — a named metadata document, `define.xml`, and named file tags, each in named sections of Modules 4 and 5 of the submission [3]. The dating is specific: standardised study data is required for studies that started after **17 December 2016** for new drug, biologics licence and abbreviated applications, and after **17 December 2017** for a commercial investigational new drug application; a Trial Summary dataset is required even for studies that started before those dates [3].

> Two cautions on the Criteria. First, the document this research could read lists the effective dates of the four criteria themselves as "TBD", with FDA undertaking to give thirty days' notice before they take effect [3] — so the honest statement is that the criteria exist and conformance is enforced, not that all four are live today. Second, validation does **not** apply to every section of Modules 4 and 5; the Criteria carry an explicit exemption list [3]. A blanket "everything must be SDTM" rule over-builds. And the Catalog's own contents could not be read — the PDF returned as binary and the site refused direct download [2] — which is why there is no version support matrix in this book. The Catalog, not CDISC's latest release, governs what you may submit.

E2B(R3) for safety messages into Europe. The EU has required the E2B(R3) format for all individual case safety reports and suspected unexpected serious adverse reactions submitted to EudraVigilance since **30 June 2022**, and EudraVigilance **does not accept E2B(R2) at all** [14]. This is as hard an edge as exists in the industry: no degraded mode, no legacy channel, no transition window left.

WHODrug, which is a paid subscription, made mandatory by a regulator. The Uppsala Monitoring Centre's own page states that "the US FDA mandates the dictionary for studies starting after 15 March 2019", and that Japan's PMDA recommends it; it is released twice a year, on 1 March and 1 September, under subscription [20]. It is the clearest example in pharma of a commercial licensed product that a regulator has made a condition of acceptance, which is why dictionary licensing turns up as an architectural dependency rather than a procurement footnote.

EU identification of medicinal products obligations. The European Medicines Agency's product management service carries dated submission obligations: structured manufacturer data and pack sizes for products on the Union list of critical medicines by **June 2026**; structured manufacturer data for all other nationally authorised products by **December 2026**; pack sizes for the remainder by **June 2027** [10]. Two of those were extended in September 2025 [10]. Note what is absent: a go-live date for the service itself. Phased obligations, and no single switch.

PMDA and electronic common technical document version 4.0. Japan's transition period for the former format ended **31 March 2026**; applications filed from April 2026 must use v4.0, while those already filed in v3.2.2 continue in v3.2.2 for their lifecycle [21].

Body / authority	Obligation	In force from	What it constrains
FDA	Technical Rejection Criteria: named datasets TS, DM, ADSL; define.xml; file tags; refuse-to-file	studies started after 17 Dec 2016 / 17 Dec 2017	**a structure**
EMA	E2B(R3) for all case reports; R2 not accepted at all	30 June 2022	**a message**
FDA / UMC	WHODrug, a paid subscription dictionary, mandated	studies starting after 15 Mar 2019	**a vocabulary**
EMA / ISO	Structured product and pack data over FHIR; two deadlines already extended	Jun 2026, Dec 2026, Jun 2027 (no go-live date)	**a master data structure**
PMDA	Submission format v4.0 for new filings; v3.2.2 lifecycles continue	filings from Apr 2026	**a transport format and taxonomy**

Figure 14.1 What is actually mandatory, and what each obligation constrains. Five obligations, dated, with the kind of thing each one binds. Read the right-hand column: every entry is a data artefact. Retrieval date 24 October 2026.

The finding: where every one of them stops

Read the five obligations again and ask what each one actually constrains.

The Technical Rejection Criteria constrain a **structure** — which datasets, with which names, in which section. E2B(R3) constrains a **message**. WHODrug constrains a **vocabulary**. Electronic submission format versions constrain a **transport format and a document taxonomy**. The identification-of-medicinal-products obligations constrain a **master data structure and its expression over an interface**.

Not one of them constrains a capability. Not one constrains a process, a business service, an application, an integration pattern or a technology component. Nothing in the mandatory layer tells a pharmaceutical company how many systems it should have, what a quality capability decomposes into, where the boundary between regulatory information management and document management should fall, or which business services a commercial affiliate needs.

That is not an accident of this particular set of five. It holds across every body that publishes anything here. CDISC specifies submission datasets and their metadata [5]. ICH specifies guidelines, message formats and submission structure [11][14]. ISO specifies product and substance identification [10]. EMA specifies master data domains and an interface [10]. FDA specifies a catalogue and a gateway check [1][2][3]. GS1 specifies identifiers and event structures [19].

Six independent bodies, with different constituencies, different funding models and different governance, all stop at exactly the same boundary. When that happens it is telling you something about what an industry has been able to agree on. Interfaces, yes. Organisation, no.

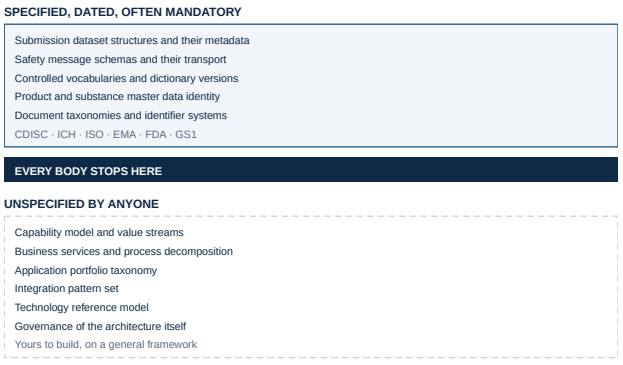


Figure 14.2 The boundary every body stops at. Six independent standards bodies specify everything above the line and nothing below it. The layers below the line are where an enterprise architecture lives, and they are the architect's to construct.

The practical consequence is a reordering of effort. Do not spend the first month looking for the industry model. Spend it enumerating the data contracts at the boundary — which ones bind you, for which submission or report type, from what date, enforced how — and then build the enterprise view deliberately on a general framework. The last section of this chapter sets out how.

The counter-evidence, reported fairly

There is real counter-evidence to the claim above, and a book that buried it would be teaching the reader the wrong habit. There are three pieces.

CDISC calls USDM a reference architecture

The strongest objection is that the industry's own standards body reaches for the term. CDISC's digital data flow page describes the **Unified Study Definitions Model** as "a reference architecture meant to serve as a standard model for building conformant study definition technologies" [7]. Digital Data Flow is a TransCelerate initiative; CDISC develops the model with TransCelerate and runs the governance group that manages changes to it; **USDM v4.0 was released on 3 June 2025** and comprises a logical model, API specifications, controlled terminology, conformance rule specifications and an implementation guide [7].

That is not a trivial artefact. It is probably the best-specified canonical model in the industry. So apply a test rather than dismissing the language.

An architect's reference architecture provides, at minimum, three things: a decomposition of the domain into building blocks, defined relationships between them, and guidance on instantiating them in a specific organisation. USDM provides a logical model, an interface contract, a terminology and a conformance regime — **for one entity class, the study definition** [7]. The phrase is being used in its software-engineering sense: a reference implementation architecture that conformant products should follow. It is not being used in its enterprise-architecture sense, and CDISC does not claim it is.

> The transferable skill here is worth more than the pharma fact. Every time somebody hands you a "reference architecture", in any industry, ask what its scope is. One entity class, one document class, one message — those are canonical models and interface contracts, and they are extremely useful. An enterprise decomposition is a different object. Both get the same name, and the confusion costs programmes real money.

CDISC 360i is a generation chain, not an architecture

The most architecturally ambitious programme in pharma standards work is **CDISC 360i** — and the first correction is the name, because "CDISC 360" is not what the current initiative is called. CDISC's own page describes it as digitalising study design through analysis, with four stated principles (traceability, reuse, consistency, readiness for artificial intelligence), a semantic backbone of three concept layers — biomedical, analysis and derivation — from which five downstream specifications are generated, and four parallel workstreams. **Phase 2 is the current stage, running through 2026** [6]. Its flagship demonstration has a single amendment to an assessment frequency propagating to sixteen connected artefacts [6].

The page **never uses the term "reference architecture"** [6]. That is the honest reading: 360i tells you how to *derive* downstream specifications from upstream concepts. It is closer to model-driven engineering than to architecture content. It gives you no capability model, no portfolio taxonomy, no service catalogue, no technology layer.

Use it as a proof point rather than a model. The sixteen-artefact demonstration is the best available public evidence that a pharmaceutical information architecture can be made traceable end to end [6]. And note the Data Transfer Agreement workstream specifically: a structured, machine-readable data transfer agreement is an integration contract expressed as a governed artefact, which is a pattern worth stealing in any industry.

The consortia produce standards, not enterprise models

The **Pistoia Alliance** is a pre-competitive collaboration on shared research and development data problems. Its current projects page lists, with descriptions, work on digital analytical methods built on the Allotrope Simple Model, an open-source instrument data converter repository, in-vivo efficacy data standards, chemical exchange formats, ontology projects, and a pharmacovigilance systems and processes standards initiative described as open and vendor-neutral [18]. Those are data standards, converters, ontologies and templates. None is an enterprise model. The one an architect should watch is the pharmacovigilance initiative, because it is explicitly aimed at **systems and processes** rather than at a data format [18] — which, if it delivers, would be the first vendor-neutral process-and-system standard in the safety domain.

TransCelerate is a member-funded collaboration of biopharmaceutical companies; its most substantial architectural output is USDM, scoped to the protocol [7]. The EU's **Innovative Medicines Initiative** and its successor the **Innovative Health Initiative** fund collaborative projects; this research did not examine their portfolios and makes no claim beyond one general caution that applies to all consortium output. Ask who maintains a deliverable after the funding stops. Most grant deliverables have no answer.

The TMF Reference Model, which is the closest thing and is genuinely useful

One artefact deserves its own section, because it is the nearest thing in pharma to something an architect recognises, and because it is useful on Monday morning rather than philosophically interesting.

The **Trial Master File** is the collection of essential documents that evidences the conduct of a clinical trial and compliance with good clinical practice. It is an obligation, not an optional artefact: a regulator inspecting a trial inspects the Trial Master File. The **TMF Reference Model** is a taxonomy reference model for it — not software, not a regulation, not a filing system, but an agreed, extensible classification of the documents a Trial Master File should contain and how they should be organised [8].

It is CDISC's, and it has been since April 2022. From the initiative's own charter: it began in 2009 under a Drug Information Association special interest area committee for document management, formally separated from that association in 2016, and **became part of CDISC in April 2022** [8]. Governance is a steering committee plus a **change control board** whose only job is to manage changes to the model, operating under CDISC bylaws [8]. The scope covers clinical research and **medical device investigations** [8], which matters for combination-product companies. The charter's own version history is internally inconsistent — the 2017 charter revision already described the move to CDISC while the body text dates it to 2022 [8] — so cite the body text and do not build an argument on the earlier date.

Writing "the DIA TMF Reference Model" is four years out of date, and vast quantities of live consultancy collateral, vendor marketing and training material still say it.

> The model's **current version number, its zone count and its artefact count are not printed in this book**, because the CDISC charter states none of them and no primary confirmation could be obtained [8]. Secondary and vendor material points at a version 3 line and an eleven-zone structure, and that is exactly the kind of figure that is wrong by the time a book prints. Get the number from the model workbook's own cover sheet on the day you need it.

The structure is conventionally a three-level hierarchy: **zones** as broad groupings of trial conduct, **sections** subdividing them, and **artefacts** as the individual document types, each with metadata and expectations about whether it is sponsor-held, site-held or both, and about when in the trial lifecycle it should exist.

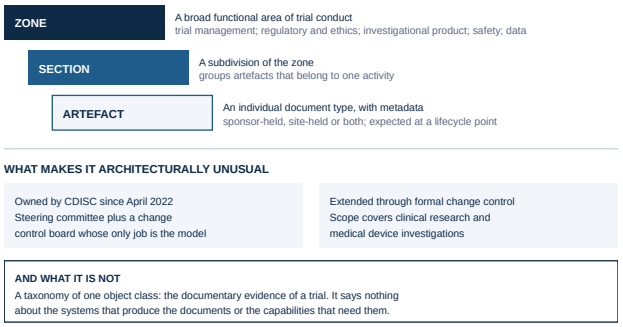


Figure 14.3 The TMF Reference Model's three-level shape, described generically. The shape and the governance are what an architect needs. The version number, the zone count and the artefact count are deliberately absent: they could not be sourced, and they are the part that goes stale first.

Four reasons it earns its space, and they are the reasons to read this section twice.

One: it is a maintained, versioned, vendor-neutral taxonomy of a business object class, under change control. That is architecturally rare. Anyone who has tried to govern a shared taxonomy inside one company will recognise how hard it is to sustain one across an industry for sixteen years.

Two: it is a procurement instrument. Every electronic trial master file product claims alignment with it, which converts a demo into a checkable question: show me your mapping to the current model version, artefact by artefact; show me where you have extended it; show me where you have not implemented it. That generalises to every category in chapter 15.

Three: it is a worked example of extending a reference taxonomy without forking it. The model is explicitly extended through formal change control [8], and real sponsors maintain organisation-specific extensions on top of the industry baseline. A governed industry baseline plus a governed local extension, with a defined route by which local extensions can be proposed upstream, is exactly the shape an enterprise architecture function needs for its own metamodel. It is much easier to teach from a concrete example than in the abstract, and this is the best concrete example pharma offers.

Four: it is the one place where a document taxonomy and a regulatory inspection meet on the same object. A trial master file finding is, in effect, a finding against the information architecture: the document existed but was filed in the wrong place, or did not exist, or existed in three versions with no indication which was current. If you ever need to argue that document taxonomy is an architecture concern rather than a records-management chore, this is the argument.

And what it is not: an enterprise reference architecture. It covers one object class — the documentary evidence of a trial — and says nothing about the systems that produce the documents, the capabilities that need them or the technology they run on. A book that presented it as pharma's reference architecture would be overselling it. A book that omitted it would be missing the most architecturally useful artefact the industry maintains.

What actually exists, body by body

What follows is the inventory, with current versions as at the research retrieval date of 24 October 2026. **Never state one of these from memory.** Every version line in pharma moves, several of them twice a year, and a confident version number in a slide deck is the commonest way an architect loses credibility in a regulatory affairs workshop.

CDISC

The foundational standards fall into layers that correspond to stages of a clinical data lifecycle. **CDASH** is collection-level: what a case report form should capture and how fields should be named so the downstream tabulation is mechanical rather than interpretive. **SDTM** is the submission tabulation model for clinical data. **SEND** is the SDTM-family model for nonclinical animal toxicology studies. **ADaM** is analysis-ready data derived from SDTM, carrying traceability from a tabulated observation to an analysis result. **ODM** is the operational data model, including audit trails. **Define-XML** is the machine-readable data definition document that accompanies a submission's datasets. **Dataset-JSON** is a JSON serialisation positioned as a possible successor to the SAS v5 transport format [4].

Three version facts matter and are routinely stated wrongly.

The **model and the implementation guide are different artefacts at different versions, and the guide trails the model.** CDISC's own page gives **SDTM v2.1, dated 10 June 2024** as the newest model version, and **SDTMIG v3.4** as the most recent implementation guide, related to **SDTM v2.0 of 29 November 2021** [5]. "SDTM 3.4" does not exist. SDTMIG 3.4 does.

Dataset-JSON has not replaced the SAS transport format. FDA requested comments — published 9 April 2025, closed **9 June 2025, forty-two comments received** — and made no change, describing the format as having long-term potential to replace the incumbent [4]. FDA's study data page notes that its drug and biologics centres are testing it [1]. It is not required and it is not settled.

For ADaM, CDASH and ODM this research obtained **no primary version confirmations**, only secondary material [5]. So read the authority's catalogue and the standards body's pages **together**, because they answer different questions. One tells you what has been published; the other tells you what you are allowed to submit.

> Budget for CDISC membership in any pharmaceutical architecture programme. The standards pages are publicly reachable and the mapping guide to healthcare interoperability formats is explicitly free [5][9], but CDISC's own 360i page refers to a members-only webinar [6], which establishes that at least some material is gated. The membership fee structure and exactly which artefacts are member-gated could not be confirmed.

ICH

The International Council for Harmonisation produces guidelines adopted by its regulatory members, in four families: **Q** for quality, **S** for safety (nonclinical toxicology), **E** for efficacy (clinical) and **M** for multidisciplinary topics that cut across. Guidelines progress through steps, of which two matter: **Step 4** is adoption by the ICH Assembly, meaning the harmonised text is final, and **Step 5** is implementation by each regional regulator, which is when it binds anyone. **The gap between Step 4 and Step 5 is where almost every date error in pharmaceutical compliance writing lives.**

E6(R3), good clinical practice, is the single most important date set here, and it is split. Its Principles and Annex 1 **came into effect in the EU on 23 July 2025**, replacing E6(R2), which applied until 22 July 2025 [11][12]. **Annex 2, covering non-traditional designs including decentralised trials and real-world data use, was adopted by ICH on 3 June 2026 and by the Committee for Medicinal Products for Human Use on 25 June 2026, and comes into effect on 15 January 2027** [11][13]. Three errors to avoid, all common: calling E6(R3) a draft; treating Annex 2 as live; and generalising the EU dates to other regions. The positions of FDA, the UK regulator, PMDA and SAHPRA on E6(R3) were not confirmed by this research, and for South Africa in particular it is a live question a sponsor needs answered rather than assumed.

What E6(R3) lands on systems: risk-based quality management and risk-proportionate monitoring, which means the trial management system, the data capture system and the data review tooling must support a *defined* risk framework rather than uniform source-data verification, and must evidence the risk decisions; computerised-systems expectations covering validation proportionate to risk, data integrity, audit trails, access control and a system inventory; and data governance across the lifecycle, including data originating outside the sponsor's systems [11].

E2B(R3) is the safety message, covered above. One further fact an architect should hold: the thing you actually implement against carries its own revision and date line — the EU implementation guide is Revision 2 of 7 April 2021, with its business rules spreadsheets last updated 29 August 2024 [14], and the ICH implementation guide was reported still at Step 3 as at 18 July 2025, which this chapter's own sources do not independently confirm. **"The standard is final" and "the thing you implement against is final" are different statements.**

M4 and M8 give the common technical document and its electronic form. The module structure — Module 1 regional administrative, Module 2 summaries, Module 3 quality, Module 4 nonclinical, Module 5 clinical — is what makes FDA's Technical Rejection Criteria expressible at all, because those criteria are stated in terms of section numbers inside Modules 4 and 5 [3]. **The module tree is the de facto filing taxonomy of the regulatory domain, and a document management system that does not model it natively will model it badly in metadata.**

On **version 4.0 of the electronic submission format**, the position is regional and the likeliest error is to write that it becomes mandatory in 2026. It is **mandatory in Japan and not in the US or the EU**. FDA's centres have accepted it since 16 September 2024 and FDA has published **no requirement date**; the EU made it optional for new centrally authorised applications from 22 December 2025, recommends it from Q1 2027 and makes it mandatory from **Q1 2028** [21]. All of that is from a secondary trade source and needs re-verification per region [21].

> The most consequential fact about submission format version 4.0 is not a date. It is the **lifecycle rule**: an application already filed in v3.2.2 stays in v3.2.2 for its life [21]. A regulatory information management estate must therefore run **two submission formats in parallel for years**, and a migration plan that assumes a cutover is wrong before it starts.

M11 is the structured protocol, and it is final and not binding. FDA's notice of availability was published 22 May 2026, finalising drafts of 22 December 2022 and 6 June 2025, after endorsement by regulatory agencies in November 2025; there are three documents — a guidance, a template and a technical specification — and FDA states the guidance is not binding [15]. Architecturally, M11's technical specification is the first harmonised, regulator-endorsed data model for a clinical protocol. Combined with USDM [7] it makes a protocol a queryable object rather than a document, which is the precondition for everything in the "digital protocol" story. It is not mandatory. Say so when somebody tells you otherwise.

Of the Q family, **Q9(R1) on quality risk management, Q10 on the pharmaceutical quality system, Q12 on lifecycle management, Q13 on continuous manufacturing, Q14 on analytical procedure development and Q2(R2) on analytical validation** are all relevant to architecture — and **their current statuses and dates are not stated in this book, because they could not be verified.** ICH's own guideline listing pages returned only page metadata and could not be read in this research pass. The practical lesson is itself useful: **for status facts about an ICH guideline, go to a regional regulator's page rather than ICH's own**, because the regional pages render and they state regional effective dates, which the harmonisation body's pages do not.

Quality risk management deserves one sentence on its own, because it does more work than it looks. It is the mechanism that licenses *proportionality* everywhere else in the regime — risk-based validation, risk-based monitoring, risk-based change control. **An architect who cannot point to the documented risk assessment that justifies the validation depth of a given system has not implemented a risk-based approach; they have implemented an undocumented one.**

ISO identification of medicinal products, and the EU implementation

The family is conventionally described as five standards: **11615** for the regulated authorised medicinal product, **11616** for the pharmaceutical product below it, **11238** for substances, **11239** for dose forms, units of presentation, routes of administration and packaging, and **11240** for units of measurement. **This book prints no edition years for them**, because only 11238 was confirmed by name, from the European agency's own page [10]; the other four appear only in secondary material.

The EU implementation is partial, phased and live. The agency's substance, product, organisation and referential master data covers four domains; referentials and organisations launched in June 2017, and the product and substance services support EU implementation of the international standards [10]. The first product-service release covers **a subset of the authorised medicinal product part** of the standard; the preparatory phase completed March 2024; the product user interface went live in May 2024 read-only; nationally authorised products were released read-only in September 2024; a public application programming interface beta arrived in June 2026 with a final version expected early 2027; and **the substance service user interface is not yet available** [10]. The deadlines are the three dated obligations above.

Two architecturally load-bearing facts sit inside that. First, **the data standard is HL7 FHIR**: the agency states plainly that FHIR is the data standard and interface basis for the product service, replacing the former submission format [10]. That is a regulatory master-data use of FHIR, not a clinical-care one, and it is the fact most architects miss. Second, the legacy format **coexists** — no retirement date is given, and pack sizes must still go through it where applicable [10].

The design question this forces on any company selling into Europe is the classic mastering question: **do you re-shape the internal product master to the regulatory model, or keep an internal model and maintain a mapped projection?** Both are defensible. The second is more common and generates a permanent reconciliation obligation that must be owned by a named function, not by a project.

HL7 FHIR

Two version facts, stated carefully because the body's own pages are inconsistent. **R4 carries the normative content, while R5 is the "current published version" and is trial-use.** The organisation's page does not mention R6 and is inconsistent about R4B's date. "We will build on the current FHIR version" is an ambiguous sentence: name the release, and say whether you need normative content.

FHIR enters pharma by two independent paths. One is the regulatory interface above [10]. The other is the research-and-care boundary: real-world data originates in health systems where FHIR is the exchange standard, and getting it into submission-grade form means mapping FHIR resources to clinical research structures. **The FHIR to CDISC Joint Mapping Implementation Guide was released jointly on 1 September 2021, is freely available, and maps FHIR to three CDISC standards — CDASH, SDTM and LAB** [9]. It passed both bodies' balloting processes [9]. Do not call it emerging; it is five years old.

Vulcan is the FHIR accelerator for the clinical-research and clinical-care boundary, and one of its projects connects CDISC, HL7 and ICH M11 to digitise protocol exchange [24]. Name it and point the reader at it; this research could not confirm its current project list.

The convergence is the instructive part. Three bodies approaching one object — the protocol — from three directions: regulatory structure, research data model and healthcare interoperability. Roughly four years from first draft to final guidance [15], and still not mandatory. That is what it takes to get an industry-wide canonical object.

GS1 in serialisation

The keys and what each does. **GTIN** identifies the product at a packaging level, and is what a national verification system looks up to know what a pack claims to be. **SSCC** identifies a logistics unit — a pallet, a case — and is what lets a warehouse receive a pallet without opening it. **GLN** identifies a party or location: a manufacturing site, a wholesaler, a dispensing pharmacy. **SGTIN** is a GTIN plus a serial number, the identifier of an individual saleable unit — the object serialisation exists to create, and the one that breaks naive data models, because it multiplies the item master by the production volume. **EPCIS** is the event-capture standard: what happened, to which object, when, where and why; **EPCIS 2.0 is JSON, JSON-LD and REST as well as XML, and has an ISO adoption as ISO/IEC 19987:2024** [19][25]. **GS1 Digital Link** makes an identifier resolvable as a web address, at **1.1.3** [19].

The two national verification models are genuinely different designs, and the contrast teaches a real lesson about where verification state lives. **The EU model is hub-and-spoke under the Falsified Medicines Directive**: a central hub receives pack data from authorisation holders and distributes it to national verification systems, where pharmacies and wholesalers verify and decommission packs at the point of dispense. **The US model is distributed under the Drug Supply Chain Security Act**: no central hub, with trading partners exchanging transaction information, history and statements peer-to-peer or through commercial networks, and verifying product identifiers on request.

The architectural contrast: the EU model centralises the **state of a pack** (active or decommissioned) in a national repository, so the authoritative answer to "is this pack valid?" lives outside the enterprise, reachable only over a network the enterprise does not control. The US model distributes the **history of a pack** across trading partners, so the authoritative answer to "where has this pack been?" must be assembled from counterparties. One is a synchronous lookup against a utility; the other is a federated query across a supply network. They imply different integration patterns, different availability requirements — a pharmacy cannot dispense if the national system is down, while a US wholesaler can ship and reconcile later — and different failure modes. A manufacturer serving both markets needs both designs.

> **Never write that GS1 requires a two-dimensional barcode by 2027.** The sunrise-2027 material is a **voluntary industry capability ambition with no day and no month**, framed as a milestone, with its own publisher stating that it is not necessary to implement all two-dimensional capabilities by that date. The page itself could not be retrieved in this research pass, and search instead returned pharmaceutical trade headlines of the form "Won't be Optional for Brands" [22][23]. That is the distortion to avoid. **Carrier obligations on pharmaceutical packs come from the US and EU statutory regimes, not from GS1.**

MedDRA and WHODrug, as licensed products with release cycles

MedDRA is the standardised medical terminology for regulatory communication about medicinal products, structured as a multi-level hierarchy from system organ class down through group and high-level terms to preferred and lowest-level terms. It is maintained under ICH, and it is a **licensed product requiring a subscription**. It releases **twice yearly**: training material is dated 3 March 2026 for **version 29.0** and 21 August 2026 for **version 29.1** [26]. The dictionary's own homepage returned only page metadata and could not be read in this pass [26], which is a practical trap worth knowing about.

WHODrug Global is the drug dictionary covering products from nearly 150 countries, maintained by the Uppsala Monitoring Centre, released on 1 March and 1 September, delivered as UTF-8 character-separated files under subscription [20]. It is the dictionary FDA mandates [20].

Two external vocabularies on independent cadences, plus — in the EU safety implementation — a third dependency on dose-form and route terminology sourced from the regional referential service [14]. **That is three external vocabulary version dependencies in a safety database, each with its own release calendar, each requiring a controlled upgrade in a validated system.** Dictionary versioning is therefore a release-management problem belonging in the architecture's change-management model with named owners and a standing calendar. Chapter 9 works through what version changes do to the data; what matters here is that they are architectural dependencies, not configuration.

The validation and quality layer

GAMP 5 is the ISPE guide on a risk-based approach to compliant computerised systems. Second edition, **published July 2022, 404 pages**, priced at $540 for members and $1,125 for non-members [17] — and it is a paid guide with no legal force. It emphasises critical thinking by experienced subject-matter experts, says specification and verification can be iterative and incremental rather than strictly linear, and carries expanded appendices on cloud computing, machine learning, blockchain and open-source software [17]. Regulators reference it; inspectors expect an approach consistent with it; a company is inspected against the applicable electronic-records rules, not against GAMP. **The software category numbers are not printed in this book** because the current scheme could not be confirmed from the publisher's page [17].

PIC/S harmonises inspection across participating authorities and maintains the PE 009 guide series. **PE 009-18 came into force on 23–24 September 2026** [16]. Note the asymmetry in its companion documents: qualification and validation recommendations were refreshed to **1 October 2026**, while the data management and integrity guidance — **PI 041-1, dated 1 July 2021** — is five years old [16]. Data-integrity expectations are therefore being reset through the revision of the computerised-systems annex rather than through the data-integrity guidance, and that tells an architect where to look. The publications listing does not mark superseded entries [16], which is a real trap for anyone citing a document number from a search result.

WHO prequalification is a market-access gate as consequential as a stringent authority's approval for a manufacturer serving donor-funded programmes. The current procedure is **Technical Report Series No. 1067, Annex 4, 2026**; the foundational good manufacturing practice annexes are much older — **No. 986, Annex 2, 2014** for main principles and **No. 957, Annex 2, 2010** for active ingredients [23].

Where a company also makes devices

ISO 13485 is the quality management system standard for medical devices, structured for regulatory purposes rather than business improvement, and deliberately not aligned to the high-level structure shared by the general management-system standards [22]. **Its current edition year is not printed here**, because the research surfaced both a 2016 catalogue listing and third-party speculation about a 2026 edition and could not resolve it [22].

FDA's quality management system regulation replaced its former quality system regulation with effect from **2 February 2026**, updating a part of its rules that had not been significantly revised since 1996 [22]. **The ISO 13485 edition incorporated by reference is not stated in the source and must not be written from memory** [22].

For a company making both medicines and devices, or combination products — a prefilled syringe, an autoinjector, an inhaler — one estate now carries a manufacturing regime inspected against the PIC/S or EU guide and a device regime assessed against a device quality standard, with overlapping but non-identical record obligations. Name the specific collisions rather than gesturing at complexity: unique device identification against pharmaceutical serialisation, usually built on the same identifier family but with different databases, attribute sets and submission routes; a design history record against a batch record, one per design and one per batch, which a single platform usually models one of well and the other badly; device risk management against pharmaceutical quality risk management, two frameworks that are not the same framework, where the honest answer is two risk registers; and

device vigilance against pharmacovigilance, with different triggers, timelines and message formats. The TMF Reference Model already covers device investigations [8], which is the one place the two worlds share a governed artefact.

Where a GMP plant meets the ERP

At the plant boundary, pharma borrows from general process manufacturing, and the facts here are reused from the manufacturing volume's research rather than re-established [25].

ANSI/ISA-95.00.01-2025 is the current Part 1, announced 10 April 2025, superseding the 2010 edition; its designation marks it as based on the international equivalent with modifications that the release does not enumerate, so a contract or conformance clause must name which document it means [25]. **There is no ISA-95 conformance scheme** — no tool, product or integration can be certified conformant, and anyone claiming otherwise is making a marketing claim [25]. **B2MML, the canonical XML binding, is at version 7 from 24 November 2020, built to the 2018/19 editions, and is MESA International's rather than ISA's** [25] — so the binding predates the current Part 1. **ISA-88 part designations and dates are not stated in this book**, because the committee's own page lists no parts [25].

What that means specifically at a pharmaceutical plant is the useful part. **ISA-88's batch model and the GMP batch record are the same object seen from two sides.** ISA-88 gives the recipe hierarchy and the procedural model mapped against an equipment hierarchy; the GMP regime demands a batch record, the complete reviewed manufacturing record of a specific batch. **The executed control recipe, with its parameter values, its deviations and its operator actions, is the raw material of the batch record.** That is the single most important integration statement in a pharmaceutical plant architecture, and it is why pharmaceutical execution systems are built around electronic batch records rather than around scheduling. An architect who selects a pharmaceutical execution system on scheduling capability has selected on the wrong axis.

And use ISA-95 to **structure the conversation** about what crosses the boundary between enterprise and plant — then say plainly that it does not tell you where the batch record lives, who holds master recipe authority, or how a deviation propagates to a batch disposition. Those three decisions actually matter in a GMP plant, and no standard makes them for you.

Artefact	Confirmed at 24 Oct 2026	Not printed here, and why
SDTM	model **v2.1, 10 Jun 2024**; **SDTMIG v3.4**, which relates to model v2.0 of 29 Nov 2021	"SDTM 3.4" does not exist
SEND, Define-XML	**SENDIG v3.1.1**; **Define-XML v2.1.11, 6 Apr 2026**	ADaM, CDASH and ODM versions: no primary confirmation obtained. ODM v2.0 exists and the publisher's own page contradicts itself
Dataset-JSON	comments closed 9 Jun 2025, 42 received, **no change made**	any claim that it has replaced the SAS transport format
ICH E6(R3)	Principles and Annex 1 effective in the EU **23 Jul 2025**; Annex 2 adopted **3 Jun 2026**, effective **15 Jan 2027**	other regions' adoption positions, SAHPRA's included
ICH Q family	the topics: Q9(R1), Q10, Q12, Q13, Q14, Q2(R2)	all statuses and dates: the publisher's listing pages returned only metadata
eCTD v4.0	mandatory in Japan from **Apr 2026**; EU mandatory **Q1 2028**; FDA accepting since 16 Sep 2024	an FDA requirement date: none published
ISO IDMP	**ISO 11238** confirmed by name; the five standards and their scope	edition years for 11615, 11616, 11239, 11240
HL7 FHIR	**R4 normative**; **R5 current published and trial-use**	R6: the publisher's page does not mention it, and is inconsistent on R4B's date
GS1	Digital Link **1.1.3**; EPCIS 2.0 as **ISO/IEC 19987:2024**	any 2027 carrier mandate: there is none
Dictionaries	MedDRA twice yearly, **v29.0 Mar 2026**, v29.1 later in 2026; WHODrug 1 Mar and 1 Sep	fee bands and language availability
Validation layer	**GAMP 5 2nd ed., Jul 2022**; **PE 009-18 in force 23–24 Sep 2026**; **PI 041-1 dated 1 Jul 2021**	GAMP software category numbers; the revised computerised-systems annex
Plant boundary	**ANSI/ISA-95.00.01-2025**; **B2MML v7, 24 Nov 2020**, MESA's not ISA's	ISA-88 part designations; and note there is no ISA-95 conformance scheme

Figure 14.4 Current versions at the research retrieval date of 24 October 2026, and the ones this book will not print. The right-hand column is the honest half of the table. Re-check every line before you quote it.

So build it

The enterprise architecture of a pharmaceutical company has to be **constructed**, and the TOGAF content metamodel plus a small industry extension is exactly the instrument for that. Here are the steps, in the order they work.

One: enumerate the mandatory data contracts first, as a dated table. Columns: standard, who mandates it, for which submission or report type, from what date, enforced how, owner. The five obligations in this chapter go in the mandatory rows. CDISC 360i, USDM, the TMF Reference Model, GAMP 5 and ICH M11 go in the recommended rows [6][7][8][17][15]. This single table converts a standards topic into a set of architectural constraints with owners and dates, and it is the most useful artefact an architect can produce in week one.

Two: model the external version dependencies explicitly. MedDRA, WHODrug, regional dose-form terminology, CDISC controlled terminology, the implementation guide version, the authority's data standards catalogue and the submission format all have independent version lines [5][14][20][26][2][21]. Put them in the architecture as **managed external dependencies with release calendars and named owners**, not as configuration.

Three: treat each regulatory interface as a bounded context with a published contract. The submission gateway, the safety gateway and the product master-data gateway each have a schema, a conformance rule set and a rejection behaviour. Design them as integration boundaries with their own validation and their own test harness, not as incidental outputs of internal systems.

Four: write the validation strategy before the target-state diagram. An architecture for a regulated estate without a validation strategy, an audit-trail design and a change-control model is not an architecture. It is a drawing. Chapter 13 is the working half of this step.

Five: take the TOGAF content metamodel as the base and add six industry object types. Compound, Study, Batch, Regulatory Submission, GxP Computerised System and Adverse Event. Not sixty. Six. Each one earns its place by being the subject of a question an inspector asks and a generic metamodel cannot answer.

Six: govern the extension the way the TMF Reference Model is governed. A named owner, a change board, a version number, a documented route for proposing changes, and a rule that a new type must be justified by a question it answers. The industry has one good example of that discipline; copy it.

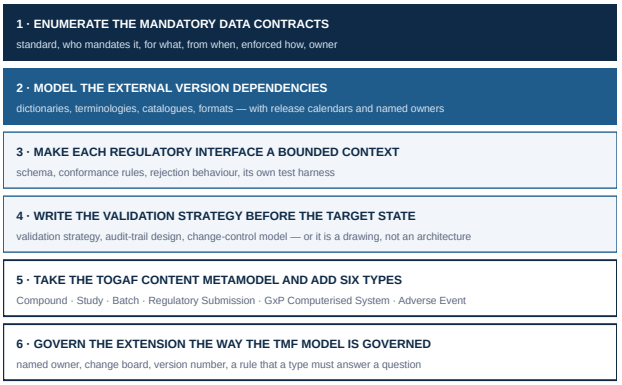


Figure 14.5 The construction sequence. Contracts first, then dependencies, then boundaries, then the validation strategy, and only then the content metamodel and its extension. Running this order backwards is the commonest way a pharmaceutical architecture programme produces documents nobody uses.

The six types are not arbitrary. Each attaches to something a generic metamodel leaves unanswerable.

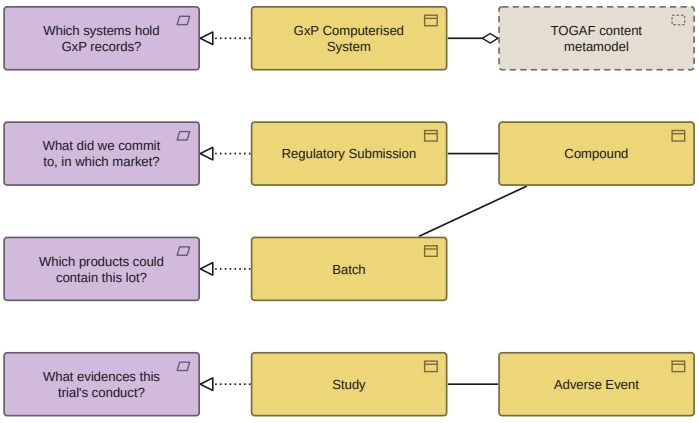


Figure 14.6 Why six types and not sixty. Each industry object type exists because it is the subject of a question an inspector asks and a generic content metamodel cannot answer. The Requirement elements on the left are the questions; the Business Objects are the types that answer them. A compound is registered in submissions and made into batches; an adverse event arises in a study.

The tooling gap, which is the opportunity

One last finding, and it is a negative one, which is why it is useful.

No enterprise architecture tool was found, at this retrieval date, to publish a pharmaceutical metamodel extension offering Compound, Study, Batch or Lot, Regulatory Submission, GxP Computerised System or Adverse Event as first-class object types [27][28]. The manufacturing volume's research found the same absence for manufacturing objects [25]. An exhaustive negative cannot be proven from public pages, so the honest form of the statement is "none found at this retrieval date", not "none exists".

The consequence is specific. Tools ship generic metamodels — the framework's content metamodel, the notation, or a vendor's own starter model — plus extension capability. The industry-specific object types a pharmaceutical architect needs most, the ones that make a repository recognisable to a regulatory affairs audience, are exactly the ones nobody ships. So **every pharmaceutical architecture repository is a local construction on a generic base.** No shared vocabulary between companies, no reusable reference content, no benchmarking, and a new architect's first six months spent learning one company's idiosyncratic extension.

That absence is the opportunity. It is also the honest answer to the question this chapter opened with. There is no model to find. There are contracts to honour at the boundary, and an enterprise view to build deliberately behind them — and the second of those is architecture work in the ordinary sense, which is the good news.

THE CHAPTER IN TEN LINES

- Pharma has mandatory, dated, machine-enforced data standards at the regulatory interface and no enterprise reference architecture at all; those are different problems and must not be blurred.
- Five obligations are genuinely mandatory and dated: FDA's Technical Rejection Criteria, EU E2B(R3) from 30 June 2022, WHODrug for FDA studies starting after 15 March 2019, the EU product-data deadlines of 2026 and 2027, and eCTD v4.0 in Japan from April 2026.
- Every one of those constrains a structure, a message, a vocabulary, a transport format or a document taxonomy; none constrains a capability, a process, a service, an application or a technology component.
- CDISC does call USDM a reference architecture, but it models one entity class, so the phrase is being used in its software-engineering sense; apply the scope test to every reference architecture you are handed.
- CDISC 360i never uses the term at all, and is a metadata generation chain with reference implementations rather than an architecture.
- The TMF Reference Model is CDISC's, not DIA's, since April 2022; it is a genuinely governed, versioned, change-controlled taxonomy of one document class, and the best available model for how to govern an architecture extension.
- Never state a pharmaceutical version number from memory; the authority's data standards catalogue, not the standards body's latest release, governs what you may submit.
- eCTD v4.0 is mandatory in Japan and not in the US or the EU, and existing v3.2.2 lifecycles stay v3.2.2, so parallel formats run for years and a cutover plan is wrong.
- GS1 does not require a two-dimensional carrier by 2027; carrier obligations come from the US and EU statutory serialisation regimes.
- The enterprise architecture has to be constructed: contracts first, then version dependencies, then interfaces as bounded contexts, then the validation strategy, then the content metamodel plus six industry object types, governed like the TMF model.
- No EA tool was found at this retrieval date to ship Compound, Study, Batch, Regulatory Submission, GxP Computerised System or Adverse Event as first-class objects, and that absence is the opportunity.

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CHAPTER 15

Technology providers

Who plays in each category, what distinguishes entrants, and which facts will be wrong first. Examples, not recommendations — and every ownership claim dated, because eight of them changed inside two years.

Every product named in this chapter is an example of who plays in a category. None is a recommendation and none is an endorsement. Inclusion is not a blessing and omission is not a judgement. The categories are the content; the names are illustrations chosen because a reader is likely to recognise them. The firm that produced this book is a preferred partner of one of the enterprise architecture vendors named later, which is stated here rather than buried, and that vendor is covered on exactly the same terms as its competitors.

There are also no scores, no quadrants, no "leader" positions and no market-share figures anywhere in this chapter, deliberately. Pharmaceutical software is unusually prone to those claims, because the buyer population is small, concentrated and heavily surveyed, and they go stale fastest and embarrass hardest. The only scale figures here are an acquirer's own stated deal value and a company's own reported revenue, both dated.

Every ownership and product-name claim below carries a date, and the retrieval date for all of it is **24 October 2026**.

What goes stale first, in order

This matters more in pharma than in the other volumes of this series, because the ownership of life-sciences software has churned violently — eight changes of control inside roughly two years, all dated in the second figure below [4][5][7][8][9][10][11][12][24].

Rank	What goes stale	Half-life	Evidence from this research pass
1	Any quadrant placement, "leader" position or market-share claim	months	excluded from this chapter entirely, on purpose
2	Regulatory in-force dates	a year or two	two European product-data deadlines extended in September 2025; a submission-format transition closed 31 March 2026
3	Ownership of life-sciences software assets	about two years	eight changes of control inside two years, listed in the figure below
4	Product names	a few years	one vendor's division renamed with both names live in its own documentation; another's quality product names unverifiable against its own August 2026 collateral
5	Dictionary versions	six months	two dictionaries, both twice-yearly; "we are on version X" is true for at most six months
6	South African local-capacity claims	one acquisition	capacity in the specialist categories is a handful of firms and a few dozen named consultants

Figure 15.1 What goes stale first in a pharmaceutical providers chapter, in order. The top three rows are why every claim in this chapter carries a date. All retrieved 24 October 2026.

Asset	Now belongs to	Date and status
ArisGlobal (LifeSphere)	Dassault Systemes, if it closes	**agreed 23 July 2026, expected to close in H2 2026 — completion UNCONFIRMED and the highest-priority re-check**
Dotmatics (platform: Luma)	Siemens	completed 1 July 2025, $5.1bn enterprise value
Alfabet	Bizzdesign	closed 9 January 2025, bought from Software GmbH, after the MEGA International acquisition
HOPEX / MEGA International	Bizzdesign	MEGA no longer exists independently; HOPEX sold as Bizzdesign Hopex
ETQ (Reliance)	Octave	spin-off named 17 June 2025; Octave an independent public company May 2026
STARLIMS	Francisco Partners	acquired from Abbott Laboratories; date and consideration unverified
TrackWise	Honeywell	via the Sparta Systems acquisition; which Honeywell depends on a separation unconfirmed as complete
LeanIX	SAP	completed November 2023
SYSPRO	Advent International, majority stake	definitive agreement announced 13 August 2024; completion unverified

Figure 15.2 Eight changes of control in roughly two years, dated. This is the figure to re-check before quoting any ownership sentence in this chapter. The top row is the one still in flight.

> **The search-engine layer is the main reason pharmaceutical IT decks contain stale vendor facts.** Many searches for this research returned pages that exist to rank for "best X 2026" and that confidently assert ownership, pricing and capability with no sources. **Ownership for roughly thirty-five vendors across laboratory informatics, regulatory publishing, serialisation and validation tooling could not be confirmed from a primary source at all.** This chapter therefore treats those categories at category level, naming examples and describing what distinguishes entrants, and does not print owners or product versions it could not verify. That honesty is worth more to you than a confident list, because the confident list is where the error comes from.

The categories

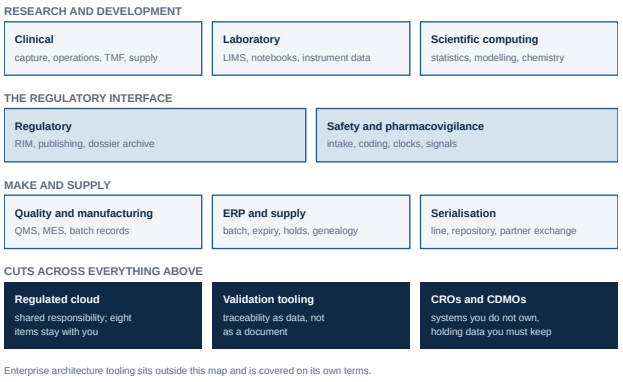


Figure 15.3 The eleven provider categories, by where in the business they sit. The bottom band cuts across all of them, which is why it is the band most often left out of a vendor shortlist.

Clinical — capture, operations, evidence, supply

The clinical category covers the systems that run a trial: capturing subject data (electronic data capture, and the broader clinical data management system that adds data review, coding and query management), managing the operational trial (sites, visits, monitoring, payments, enrolment), holding the evidence (the electronic trial master file), randomising and managing trial supply, collecting from patients directly, and activating sites.

What distinguishes entrants. First, **platform breadth against specialist depth** — and the architectural question is not which is better but **where you want your integration seams**. A single platform moves the seams to the platform's edge; buying the best tool per function puts a seam between every pair of systems. Both are defensible; only one is usually costed honestly. Second, **who configures a study and how fast**, which is a function of metadata-driven configuration: a platform that can ingest a structured study definition and generate its own configuration is on a different cost curve from one where a study is hand-built. Third, **validation posture** — how much validation the vendor performs, what evidence they hand over, and how often they force an upgrade. Fourth, and most under-represented in reference diagrams, **whether the contract research organisation brings its own**: in a sponsor–CRO relationship the CRO often supplies the capture and operations systems, which means the sponsor's architecture has to accommodate systems it does not own, for trials it does own, with data it must archive for decades.

Clinical — examples only, not recommendations

Medidata, Dassault Systèmes' since 2019, known for the capture product that defined the category's modern shape [21][22]. **Veeva**, whose distinguishing property is that clinical, regulatory and safety content sit on one document-and-data platform with one security and lifecycle model; its own August 2026 collateral names, among others, Veeva eTMF, Veeva CTMS, Veeva Study Startup, Veeva RTSM, Veeva EDC and Veeva eCOA [1]. **Oracle Life Sciences** (formerly Oracle Health Sciences — both names appear in live Oracle material [3]), known for InForm, Siebel CTMS, Clinical One and Central Coding. **IQVIA**, which combines contract research services, a very large health-data estate and software in one commercial relationship — one accountable party or one concentration risk, depending entirely on your view. And specialists in one function: **Suvoda** in randomisation, trial supply and electronic consent; **Signant Health** and **Clario** at the patient-facing and instrumented-endpoint end; **Castor** with a self-service, lower-cost positioning used by academic and investigator-initiated trials; **OpenClinica**, notable for an open-source origin in a validated-system category. *Current product names and ownership for most of the specialists could not be verified from vendor sources and are not printed.*

Safety and pharmacovigilance

A safety system intakes adverse event reports from many channels, triages and codes them against a medical dictionary and a drug dictionary, assesses seriousness, causality, expectedness and reportability, generates and transmits regulatory reports within the right clocks, tracks acknowledgements, and detects signals across the accumulated case base. It also holds product and licence reference data, because reportability cannot be computed without it. **This is the most time-critical and least forgiving system class in pharma**: clocks run in calendar days from day zero, and a missed clock is an inspection finding.

Four differentiators, and the third is the one demos never cover. **Intake automation**, where every vendor now claims machine assistance and the real question is what evidence they provide that the automation is validated and auditable. **The reportability rules engine** — who must be told what, by when, as a function of product, licence, market, event and regulation — and whether the buyer can maintain the rules without the vendor. **Dictionary version management**: ask how the dictionary version is stored per coded term, what happens on a version upgrade, and what happens to historical signals. And **where licence and product data comes from** — whether the safety system holds its own product dictionary or consumes one from the regulatory information system, which is a genuine architectural fork with long consequences.

Safety and pharmacovigilance — examples only

Oracle Argus, the safety database that became the de facto reference implementation of the case message for a generation; Oracle's own published schedule gives Argus Cloud Services 2026.1 generally available in March 2026, with 8.4.1 and 8.4.0 reaching end of life in September 2026 [2] — which is a forced-upgrade calendar in a validated system, and the model of the disclosure to demand from everyone. **ArisGlobal LifeSphere**, the integrated regulatory, safety and quality suite, known for pushing automated case processing harder and earlier than the category norm, and **agreed on 23 July 2026 to be sold by Nordic Capital to Dassault Systèmes** [4]. **Veeva Safety**, whose architecturally interesting feature is putting safety *documents* on the same platform as safety *cases* [1]. **Oracle Empirica** for signal detection [2]. **Ennov** and **Sarjen** at different price points, frequently seen in mid-size and emerging-market deployments; *ownership and current product names unverified.*

Regulatory — information management, publishing, archive

Regulatory information management holds the registration status of every product in every market: what is authorised, under what conditions, with what commitments, with what variations pending, with which documents supporting each. Submission publishing assembles and validates a dossier and transmits it. The dossier archive retains what was submitted, in the form it was submitted, for the life of the product plus the retention period.

For an architect this is the most under-modelled domain in pharma. It is where product master data, document management, workflow, external reference data and country-specific variation all meet, and it is routinely implemented as a document system with spreadsheets alongside. Three checkable questions. Can the tool **produce and validate both submission format versions, maintain v3.2.2 lifecycles indefinitely, and handle regional Module 1 variation** — a concrete requirement that belongs in every regulatory procurement from now until roughly 2030 [19]. Is **a registration a first-class data object or a document attribute**: the biggest quality differentiator in the category and the hardest to see in a demo, so ask to see the data model for a registration, a variation and a commitment. And can the tool produce the European structured product-data submissions over the regulator's own interface, against the June 2026, December 2026 and June 2027 dates [18] — a yes-or-no question with a date on it.

Regulatory — examples only

Veeva, whose August 2026 collateral names Veeva Registrations, Veeva Submissions, Veeva Submissions Publishing and Veeva Submissions Archive [1]. **ArisGlobal**, whose suite spans regulatory alongside safety and quality, and which is in flight to Dassault Systèmes [4]. **Lorenz** and **Extedo**, European publishing specialists rather than platforms. **Certara**, primarily a biosimulation and regulatory-science company with a regulatory writing and submissions arm. **Freyr**, a regulatory services provider with software, strong in the long tail of country-specific requirements — the part of the domain that platforms model worst, and which publishes South-Africa-specific readiness material [16]. *Product names, versions and ownership for the publishing specialists and the services firms could not be verified; search returned almost exclusively comparison content, so this book treats them at category level.*

Quality and manufacturing — quality systems, execution, batch records

A quality management system runs deviations, corrective actions, change control, audits, complaints, supplier quality and training, and holds the controlled documents with their lifecycle and signatures. A manufacturing execution system runs production: dispensing, weighing, line clearance, in-

process checks, equipment status, and above all **the electronic batch record**. The two meet at the deviation, raised in the plant and resolved in the quality system, and at the batch disposition, which is the decision the whole regulated estate exists to support.

The decisive question for execution systems: **is the electronic batch record the product's core object, or a report?** A pharmaceutical execution system is a batch-record engine with execution attached. A general one is a scheduling and execution engine with a batch report attached. The test is whether the system can produce a review-by-exception batch record that quality assurance will release on, with every deviation linked, without a parallel paper copy. Then: **who holds master recipe authority** — the execution system, the control system, the enterprise system or a product lifecycle tool — where the commonest failure is two systems both believing they hold it. And **installed-base gravity in automation** is a legitimate reason to buy, and should be named as such rather than dressed up as a functional comparison.

One deployment fact deserves stating plainly rather than apologetically. **Execution systems remain substantially on-premises or hybrid, and that is the clearest remaining on-premises case in the whole pharmaceutical estate.** A regulated line cannot stop because a wide-area link dropped, and the batch record cannot be unavailable at the moment of dispensing.

Quality and manufacturing — examples only

Quality: **MasterControl**, a quality-first suite that has extended into execution — a quality vendor moving into execution rather than the reverse, which is architecturally interesting. **TrackWise** and **TrackWise Digital**, long the default enterprise quality system in large pharma, **Honeywell's** since its acquisition of the maker, with which Honeywell owns it depending on a corporate separation this research could not confirm as complete [11][24]. **ETQ Reliance**, a highly configurable quality system used across regulated industries — **an Octave asset, not a Hexagon one**, since Octave became an independent public company in May 2026 [9][24]; ETQ announced Reliance AI in a release dated 27 January 2026 [9]. **Veeva**, whose quality products could not be verified as current names, because its August 2026 development-cloud collateral lists no quality products at all [1]. Execution and the historian: **Werum PAS-X**, Körber's, the category's long-standing pharmaceutical execution system built around the batch record; **Siemens Opcenter**; **Rockwell PharmaSuite**; **Emerson Syncade**; and **AVEVA PI**, Schneider Electric's, whose relevance here is the time-series record that a batch record and a continued-process-verification programme both depend on [24].

Laboratory — information management, notebooks, instrument data

A laboratory information management system manages samples, tests, specifications, results, out-of-specification investigations, stability programmes, instrument calibration and analyst qualification. In a quality control laboratory it is the system that decides whether a batch meets specification. An electronic laboratory notebook records experimental work, where the object is a scientist's reasoning rather than a test against a specification. Separately, instrument data systems capture raw output, and integration layers move data between instruments, notebooks, the management system and analytics.

The distinction between the two system classes matters and is routinely blurred by vendors. A management system is specification-driven and its output is a disposition; a notebook is narrative-driven and its output is a record of what was done. Products claiming both usually lean one way.

The real cost driver is **instrument integration**: whether the vendor ships connectors, whether a third party supplies them, and whether a connector change is a validated change. Then **how well the out-of-specification investigation is modelled** and how it links to the quality system; **stability programme support**, a distinctive long-running multi-condition workload and a reliable differentiator; and a vendor's position on **open instrument data standards**, the first differentiator here that is not a feature claim [15].

Laboratory — examples only, and the weakest-evidenced category

LabWare and **LabVantage**, enterprise management systems with long pharmaceutical installed bases and the consequence of their configurability: implementations are projects, not deployments. **Thermo Fisher Scientific**, whose informatics sits alongside a very large instrument business — natural advantages in integration, natural questions about neutrality. **STARLIMS**, **Francisco Partners', acquired from Abbott Laboratories** [10]: selection papers that still call it an Abbott product are wrong. **Benchling**, cloud-native research and development, designed cloud-first in a category whose incumbents were not. **Dotmatics**, **Siemens'** since **1 July 2025 at $5.1bn enterprise value**, with **Luma** named as the platform [5][6] — so "Siemens is a plant-floor vendor in pharma" is now an incomplete sentence. **Revvity Signals**, **IDBS** with its bioprocess-development strength, and **Sapio Sciences**. And **Scitara**, whose category is instrument-to-system connectivity rather than a system of record — a genuinely separable problem worth naming on its own. *Ownership and current product names for most of the above could not be verified from vendor sources; search returned almost entirely comparison content.*

ERP and supply

Enterprise resource planning in pharma does what it does everywhere, plus batch and lot management with genealogy, shelf-life and expiry-driven allocation, quality holds that block movement, country-specific regulatory status that gates sale, pack-level serialisation data, and cold-chain and controlled-substance handling. **The distinguishing requirement is that inventory is not fungible**: two units of the same stock-keeping unit with different batch numbers are different objects with different dispositions, different expiries and potentially different market authorisations.

Four differentiators. **Batch and lot modelling depth, and whether genealogy is bidirectional** — forward trace (where did this batch go) and backward trace (what went into it) must both be answerable inside a recall window, and mock recall timing is that rare and precious thing, a non-functional requirement with a number attached. **Quality integration**, where whether a hold in the quality system blocks movement natively or by interface determines whether the hold is reliable. **Serialisation strategy** — in the enterprise system, in a specialist platform, or both. And **scale fit**, the one category where too big is as real a failure mode as too small.

ERP and supply — examples only

SAP S/4HANA, the default at the large end, usually also the serialisation repository and often the master for product and batch data, which makes it the centre of gravity for traceability. **Oracle**, whose enterprise application family and life-sciences application portfolio are not the same product family and should not be conflated [2]. **Infor** and **QAD** in process-manufacturing-focused and mid-market positions. **SYSPRO**, founded **1978**, headquartered in the United Kingdom and South Africa, about **400 employees**, with **Advent International announcing a definitive agreement to acquire a majority stake on 13 August 2024** [12] — figures from the acquirer's release, with completion unverified. For a South African manufacturer it is the one product in the category with local engineering and local support depth.

Scientific computing, statistics and modelling

Three distinguishable things that get lumped together: statistical computing for submissions, where the requirements are reproducibility, traceability and the ability to hand a regulator something they can run; model-informed drug development, where model-based evidence increasingly supplements clinical data in a dossier; and computational chemistry and biology.

Is the statistical environment actually changing? Here is the honest assessment. What is established: the transport format of record remains SAS-defined, and that has not changed [17][20]. R has been used to produce submission packages accepted by FDA, repeatedly — a WebAssembly pilot submitted **20 September 2024** and described as the first publicly available submission package with a WebAssembly component received by the drug evaluation centre; containers submitted in summer 2025; a Dataset-JSON pilot submitted in autumn 2025, reworked and resubmitted in early January 2026 — and FDA has **expanded the R file formats it accepts through the submission portal** [13]. **The gating constraints the practitioners themselves report are validation tooling and reviewer-side infrastructure, not statistical capability**: the conformance checker did not support the newer dataset format, so the team validated with the old one in agreement with FDA [13]. What is not established and must not be asserted: **any quantitative share of submissions by tool.**

So: anyone who tells you R has replaced SAS is ahead of the evidence, and anyone who tells you R cannot be used is behind it. The architectural point is that **open source in a regulated context does not remove cost; it relocates it from licence to qualification** — which is precisely what the commercial R tooling vendors sell, and the validation guide's second edition explicitly added an open-source appendix [14].

Scientific computing — examples only

SAS, the historical and still-dominant statistical environment for regulatory work, and the reason the transport format of record is what it is [17] [20]. **R and the R Consortium's submissions working group**, the open-source route, now demonstrated into FDA with dates [13]. **Posit** and **Domino Data Lab**, commercial tooling for building a validated statistical computing environment — which is where the relocated cost lands. **Certara** in physiologically-based simulation and regulatory science; **Schrödinger** in physics-based molecular modelling; **Dassault BIOVIA** in scientific informatics, distinct from Medidata — so that with Medidata and, pending, ArisGlobal, **one industrial software company's life-sciences footprint would span discovery informatics, clinical, and safety and regulatory** [4][21]. *Product lists and versions unverified across this box.*

Serialisation and track-and-trace

Generate and manage serial numbers; commission and aggregate them at pack, case and pallet level; print and verify codes on the line; hold the serialised event history; and exchange data with trading partners and national verification systems. The distinguishing requirement is **volume and latency at the line**, plus exchange with systems the enterprise does not control.

Differentiators: **line-level against enterprise-level centre of gravity** — most programmes need both and buy them from different places, which makes the line-to-repository interface the critical seam. **National system connectivity coverage**, and how fast a vendor adds a new one, which is a moving target as more countries legislate. **Network against point-to-point**, which is the architecturally distinctive idea in the category. And **whether serialised data is retained for the record or merely transmitted**, because retention and retrieval obligations interact with retention periods measured in years after expiry.

Serialisation — examples only

TraceLink, known for a network model: rather than point-to-point integration between trading partners, a shared network with one connection per participant [23]. That is an N-to-one answer to an N-to-N integration problem, and the trade-off is dependence on the network operator. **Antares Vision Group**, which acquired **rfxcel**, spanning line-level inspection hardware through to cloud traceability software. **SAP Advanced Track and Trace for Pharmaceuticals**, the serialisation repository inside an SAP estate, chosen primarily where SAP is already the enterprise system. **Systech**, closer to the printer and the camera than to the enterprise repository. **Movilitas**, an implementation and extension player rather than a product owner. **OPTEL**. *Ownership and current product names could not be verified for any of these except SAP; treat the category at category level.*

Contract research and contract manufacturing, as a provider category

This belongs here rather than in a services footnote, for one reason. **A sponsor that outsources a trial or manufacturing does not thereby outsource the architecture.** It acquires systems it does not own, holding data it must retain, under validation it did not perform, in an estate it cannot inspect without a contractual right.

The specific consequences. **The trial master file is the sponsor's obligation even when the research organisation maintains it**, so the sponsor needs either its own repository with contributor access or a contractual right to the content in a usable form at defined intervals and at close-out — and the TMF Reference Model is the instrument for specifying that, because "deliver these artefacts, mapped to this model version, with this metadata" is a far better clause than "provide TMF documents" [27]. **Data transfer agreements are integration contracts**, and the fact that CDISC's current programme has

a workstream building a structured, machine-readable one tells you how often they are prose describing a file transfer [26]. **The research organisation's capture system becomes the sponsor's archive problem**, because retention runs decades and the relationship does not. **Contract manufacturing batch records are regulated records of the sponsor's product**, with the same shape of problem. And **supplier qualification extends to those systems** [14], so the sponsor's validation strategy has to say how.

Contract research and manufacturing — examples only

IQVIA and **Parexel** are the names a reader will recognise in contract research, and the major contract development and manufacturing organisations in the other half. *Current ownership, technology portfolios and scale for all of them are unverified in this research, so this book treats the category as a set of architectural consequences with named examples, not as a vendor comparison.* The useful move is not to compare them but to write the clauses: content delivered against a named model version, configuration artefacts owned by the sponsor, audit rights, and archive handover at close-out.

Regulated cloud and validation-lifecycle tooling

All three major cloud providers publish guidance for regulated systems, and the shape is the same: a shared-responsibility model. One provider's own material states that the regulated company "remains ultimately accountable for demonstrating that a GxP system is compliant, even if a supplier is responsible for parts of that system", supplies a quality management system, certifications and audit reports, and then **enumerates what remains with the customer**: supplier assessment, managing provider-made changes to services in use, incident response for outages, portability on changing provider, qualification of the landing zone and the building blocks and the application including installation qualification, physical access control for on-premises provider equipment, and oversight of any managed service provider [25].

> **"We moved to the cloud so validation is the vendor's problem" is false, and the falsity is quantifiable: the provider's own list of what remains with the customer runs to eight items** [25]. Use that list in procurement. If a vendor's answer is shorter than it, they have not read their own provider's material.

The consequence that actually bites is the second item. The customer must manage changes the provider makes to services in use [25]. In a validated system, a supplier-initiated change to a service you depend on is a change-control event you did not schedule. That is the structural tension between continuous cloud delivery and regulated change control, and no contract fully resolves it. The mitigations are architectural: prefer services with long-term-support or version-pinned options, keep the regulated critical path on the narrowest possible set of services, and monitor provider change feeds as a compliance activity rather than an operational one.

Note the dating problem even here: the page this research read is marked "for historical reference only" with no publication date, while a July 2026 edition of the same whitepaper exists [25].

Regulated cloud and validation tooling — examples only

The hyperscalers publish the shared-responsibility material [25]. Then a category of **qualification accelerators** — pre-built documentation, infrastructure-as-code landing zones and test evidence packages — whose architectural caution is that an accelerator supplies documentation for *a* configuration: **its value is inversely proportional to how much you customise it**, and that trade-off should be made at the start rather than discovered at the first audit. Then **validation-lifecycle management tools**: **ValGenesis** and **Kneat**, both digitising the validation documentation set — requirements, risk assessments, protocols, execution, traceability matrices, periodic review — rather than running it in documents [28]. The two market against each other directly, which is evidence of a two-horse perception and not of either's merit. *Product names, versions and ownership are unverified across this box.* Three questions that cut through: does the tool hold requirement-to-test traceability **as data or as a document** — a platform that produces PDFs has digitised the printing, not the validation; does it **schedule and evidence periodic review**, the obligation most often missed in practice; and **is the tool itself validated, and by whom**.

Enterprise architecture tooling, on identical terms

An architecture tool holds a repository of objects and relationships, renders views, supports impact and lifecycle analysis, and governs change to the repository. In a pharmaceutical context it must additionally cope with a validated-system inventory, regulated classification as an attribute of an application, and regulatory obligations as traceable objects.

The firm behind this book is a preferred partner of Orbus Software. OrbusInfinity is therefore covered accurately, without flattery, and with the same questions asked of it as of everyone else. That is the only treatment that is useful to a reader and the only one that is honest. No preference is expressed in this chapter, for Orbus or against anyone.

Three facts apply to every vendor equally, and all three are about the evidence rather than about any product.

One. The Open Group's public ArchiMate tool certification register could not be read. It returned unrendered template placeholders, with static text still referencing conformance to ArchiMate 3.0.1 or 3.1, and in this research pass the address failed with a client error [29][24]. **Therefore no tool — OrbusInfinity included — can be described in this book as a certified ArchiMate 4 tool on the strength of a public register.** That is a statement about the register, not about any vendor. Nor could any competitor be verified at ArchiMate 4.

Two. ArchiMate 4's exact publication date is not printed here. The specification is document C260; the announcement page is datelined 27 April with no year, the discussing blog is dated 20 May 2026, and the publications address was disallowed to automated retrieval [30][24].

Three. Shipping content for the previous major version shortly after a new one is normal, and is not a deficiency. OrbusInfinity's own capability page, at this retrieval date, claims **full ArchiMate 3.2 notation** with layered viewpoints and starter packs; **BPMN 2.0** templates and stencils; **TOGAF** with preconfigured architecture development method phases, a content metamodel and governance artefacts, **with no TOGAF version number on that page**; ITIL processes mappable in the same repository, no version given; a starter metamodel described as built on the vendor's practices and on unnamed leading frameworks; extension examples covering technology portfolio, technical debt and sustainability; and configurable object types, relationships and attributes, with **"Changes apply without disrupting live data"** [31]. All of that is a **dated vendor claim** and is reported as such. An earlier research pass recorded the same page claiming ArchiMate 3.2 and TOGAF 10; the TOGAF version is now absent from it [31][24].

Here are the selection questions. **Put them to every vendor in the same words, Orbus included.**

1. Which **version of the ArchiMate specification** does your current release implement, and can you show me the element and relationship set in the tool?
2. Which **version of TOGAF** does your preconfigured content implement, and is the content metamodel the framework's own or yours?
3. Can I **extend the metamodel** with new object types, relationships and attributes **without vendor services**, and what happens to live data when I do?
4. Can I hold **validation state** and **regulated classification** as governed attributes with their own lifecycle, and report on applications by validation state?
5. Can I model a **regulatory obligation** as an object and trace it to the controls and systems that satisfy it?
6. What is your **upgrade cadence**, and what is my exposure if the repository is itself in regulated scope?
7. What does **exit** look like: can I extract the full repository, with relationships and attribute history, in a documented format?

> Question three is the one that matters most for a pharmaceutical extension, and Orbus answers it on its own public page: configurable object types, relationships and attributes, with changes applying without disrupting live data [31]. **That is a good answer. Every competitor should be asked for it in the same words, and their answer compared with that one.** Nothing in that sentence is a recommendation; it is an observation about which published claim happens to be checkable.

Enterprise architecture tooling — examples only

OrbusInfinity (Orbus Software), a repository-based platform with a Microsoft ecosystem heritage [31]. **SAP LeanIX**, **SAP's since November 2023**, known for application-portfolio management with fast time-to-value and a software-as-a-service model, historically lighter on formal notation [24] — and an architecture-tool shortlist that lists it as an independent challenger to SAP is describing a market that ended in 2023. **Bizzdesign**, now the owner of **Horizzon, HOPEX and Alfabet**, following the MEGA International acquisition and the **Alfabet acquisition from Software GmbH, closed 9 January 2025**, giving a combined group of roughly **EUR 110 million revenue, about 2,000 customers and over 600 employees**; **MEGA International no longer exists as an independent vendor** and HOPEX is sold as Bizzdesign Hopex [7][8]. **Do not write "Software AG Alfabet", and do not treat MEGA as a separate vendor. ARIS**, still known for process-led architecture, now a separate business from Alfabet, with its current corporate owner unverified [7][24]. **Sparx Systems Enterprise Architect**, a low-cost, deep, desktop-rooted modelling tool with very broad notation support, whose trade-off is repository governance at enterprise scale. **Ardoq**, a data-driven, graph-oriented approach emphasising broad contribution over specialist modelling. **Avolution Abacus**, known for quantitative analysis and scoring over the repository — which in pharma could in principle be turned on validation-state and regulated-risk attributes. **Archi**, the free, open-source ArchiMate tool: the cheapest way to learn the notation properly, with no repository governance, which makes it an excellent teaching tool and a poor enterprise platform. *Ownership is unverified for Ardoq, Avolution and ARIS, and Archi's ArchiMate 4 support is unverified.*

And the category gap, which is the finding. **No architecture tool was found at this retrieval date to ship a pharmaceutical metamodel extension offering Compound, Study, Batch or Lot, Regulatory Submission, GxP Computerised System or Adverse Event as first-class object types** [31][32]. An exhaustive negative cannot be proven from public pages, so the honest form is "none found", not "none exists". Tools ship generic metamodels plus extension capability; the industry-specific types that make a repository recognisable to a regulatory affairs audience are exactly the ones nobody ships. That is why chapter 14's six-type extension exists, and it is said here as an explanation rather than as a sales point.

The South African provider landscape

Four structural facts come before any vendor name.

One: the export map scopes the architecture, not the domicile. A South African manufacturer selling only domestically faces the local regulator. One exporting to Europe faces EU manufacturing practice, European serialisation and potentially structured product data [18]. One exporting to the United States faces that regulator, its supply-chain security regime and the submission data regime [33]. One serving donor-funded African markets faces WHO prequalification. **The export map is a commercial decision made outside IT, and it determines the regulatory surface the estate has to meet.**

Two: the local regulator's adopted manufacturing-practice baseline is a South African adaptation of the PIC/S guide, not the EU guide directly [34], and the local guideline is silent on computerised systems, the computerised-systems annex and data integrity [34]. Those expectations have to be taken from the international texts [33]. A compliance architecture built on "we follow EU practice" is not wrong, but it is not what the local regulator has adopted, and the *justification* for a validation approach has to be constructed rather than cited.

Three: local capacity correlates with how general-purpose a product is, not with how good it is. That is the pattern across every category above, and it is the single most useful thing to know before planning a programme.

Four: time zone is an asset for European-supported systems and a liability for American ones. South Africa is two hours ahead of coordinated universal time, so European support overlaps the local working day almost entirely and American support overlaps the local afternoon and evening. A genuine and under-used selection criterion.

Category	Local capacity (judgement)	Basis
ERP	**Real and deep**	a third-party directory claims over a hundred South African partners for one vendor; another is part-headquartered here
Cloud and infrastructure	**Real**	three in-country regions; a large local managed-services industry; at least one publicly referenceable pharmaceutical cloud engagement
MES, automation, historian	**Real, through the automation channel**	arrives with the control vendor's established local presence
Laboratory	**Partial**	no primary source obtained; vendor-specific
EA tooling	**Real but thin, consultancy-mediated**	no primary source obtained
Clinical	**Thin to absent locally; delivered from Europe, India or the US**	one platform vendor's own partner page names no Africa-presence top-tier partner and its regional structure does not include Africa — which is not proof that no capacity exists
Safety, regulatory	**Thin locally; services-led**	regulatory services firms publish South-Africa-specific readiness material, which evidences a services rather than a product presence
Serialisation, validation tooling	**Partial to thin**	no primary source obtained; packaging-equipment and direct-vendor channels

Figure 15.4 Local implementation capacity, by category. The assessments in the middle column are this book's **judgement**, not findings, and the basis column says what each one rests on. Where the basis is thin the assessment should be treated as a hypothesis to test by direct enquiry.

Local-partner dependence as an architectural risk

For most of the specialist categories, local capacity is a handful of firms and a few dozen named consultants. **Treat that as a nameable architectural risk with mitigations, not as a complaint.**

The risk has four shapes. **Single-supplier concentration at the implementation layer**: the vendor may be global, and the people who know your configuration may be four individuals at one firm. **Key-person risk is the dominant risk, not vendor risk** — emigration, a competitor's offer or a single resignation can remove the only person who understands why a validated configuration is the way it is. **Validation knowledge is harder to replace than technical knowledge**: anyone can learn a product, but knowing why a particular risk assessment justified a particular test depth, and being able to defend it to an inspector, is institutional memory. And **documentation debt compounds the dependence** — if the validation documentation only makes sense to the person who wrote it, the dependence is structural rather than contractual.

Six mitigations, all concrete. Require the validation documentation to be **self-standing and readable by a competent stranger**, and make that an acceptance criterion tested by having someone outside the project read it. **Own the configuration artefacts, not just the system** — configuration export, in a documented format, in your repository, under your version control, as a contractual deliverable at each release. **Name two routes to support, contractually, before go-live**, typically the local partner plus either the vendor directly or a second-region partner, and test the second route once a year. **Keep a current system inventory with regulated classification and validation state in the architecture repository**, so that "what do we depend on and who supports it" is answerable without the partner — the single most valuable architecture deliverable in an estate with thin local capacity. **Budget for knowledge transfer as a line item with a deliverable**, not as an assumption. And **prefer configuration over customisation harder than you would elsewhere**, because customisation deepens the dependence and widens the validation surface at the same time.

Rand exposure in the licence line

Almost every product in this chapter is priced in dollars or euros; the exceptions are the locally headquartered enterprise system and local services [12]. **For a South African buyer the licence line is a foreign-exchange exposure and the services line largely is not**, which changes the shape of a business case: the cost of ownership carries a currency-risk component that the internal rate of return usually ignores.

Four consequences worth stating plainly. **A multi-year subscription commitment in hard currency is an unhedged liability** — a three- or five-year commitment denominated in dollars is a forward foreign-exchange position taken by an IT department, which argues for shorter terms or for contractual rand pricing even at a nominal premium. **Consumption-based cloud pricing is worse for budgeting than subscription under a volatile currency**, not better, because the quantum moves as well as the rate. **Per-user pricing interacts badly with local salary levels**, which systematically pushes South African buyers towards fewer named users — and in a quality system fewer named users means more shared accounts, and **shared accounts are a direct data-integrity violation under the attributability requirement**. That is a real, specific, local architectural consequence of a commercial model. And the one genuine structural advantage: **validation cost is largely local and largely rand-denominated**, so the expensive part of a regulated implementation — the documentation, the testing, the qualification — can be done at local rates, provided the capability exists.

> This book will not print prices. No vendor's South African pricing, rand denomination availability or local-entity contracting arrangements could be verified, and a printed price is the fastest-ageing fact in a procurement conversation.

Cloud regions, and the check this book teaches instead of a table

The region identifiers are checkable and are confirmed: **an opt-in region in Cape Town that must be enabled per account** under one provider, a Johannesburg region under a second, and Johannesburg and Cape Town regions under a third [35]. The source gives no launch dates and no availability-zone counts [35].

This book will not print which services are available in which South African region, because per-service availability could not be confirmed and it changes monthly. What it prints instead is the check. For each regulated workload, before committing to in-country residency:

1. **Name the services the design depends on**, including the ones nobody thinks about — identity, key management, logging, backup, the managed database engine *and version*, the container service, the monitoring stack.
2. **Check each one against the provider's own region-services table, on the day**, and record the date you checked. That table is the only authority and it changes.
3. **Check the availability-zone count** in the target region, because a two-zone region constrains the high-availability design differently from a three-zone one.
4. **Confirm an opt-in region is actually enabled on the account** [35], and that every automation, policy and landing-zone guardrail accounts for it. An opt-in region is a real operational trap: organisation-wide service control policies and infrastructure-as-code written for default regions can silently exclude it.
5. **Check whether the provider's regulated-systems evidence package covers the specific region and the specific services** [25]. A certification that covers a service globally is not the same as one that covers it in a region, and a qualification scope statement names regions.
6. **Record the result as a dated architecture decision**, because when it changes you need to know what you decided on what evidence.

Health-data residency under POPIA

Chapter 16 handles the law. The provider question here is narrower and has four answers that hold regardless of how the current regulatory position resolves.

Section 72 of the Protection of Personal Information Act is the cross-border gate for personal information generally, and health-specific regulations that came into force on 6 March 2026 restate it specifically for health information [36]. Whatever those regulations' covered categories turn out to mean for a manufacturer, authorisation holder or sponsor — and that is genuinely unresolved — section 72 applies.

The architectural answer is usually not "keep it in South Africa"; it is "transfer it lawfully and prove it". Pharmaceutical data flows are inherently cross-border, because the regulator receiving the safety report or the submission is frequently in another jurisdiction. An architecture that tries to prevent cross-border flow of safety data has misunderstood the obligation.

Group policy or operational convenience alone is explicitly flagged as unlikely to comply [36], which rules out the commonest de facto justification in a multinational affiliate — "the group database is in Europe" — as a standalone basis.

And **safeguards must cover physical records too** [36]. In a regulated environment paper is still real: signed consent forms, executed batch records, archived source documents. An information-security architecture scoped to electronic records only does not meet the obligation as reported.

Support hours against a twenty-four-hour plant and a global trial

Two different problems with different answers. **For the plant**: the systems in the critical path have an availability requirement set by the production schedule, not by office hours, and a 22:00 line stoppage is nobody's working day. So the real question is not "what are the vendor's support hours" but **"what is the documented local workaround when the system is unavailable, and is it a validated workaround?"** A regulated plant cannot improvise. **The paper-based contingency batch record is an architectural artefact that must be designed, approved, validated and periodically tested** — and it is the thing most target-state architectures omit. Escalation to a vendor in another hemisphere during a batch deviation is not a recovery plan.

For the trial: no single system sits in a production-line critical path, but safety reporting clocks run in calendar days and do not pause for a weekend or a public holiday — and South African public holidays do not coincide with European or American ones. A safety operating model with a single-country team has a coverage gap on every local holiday, and the architecture should make the clock state visible rather than relying on a person remembering. Vendor support hours matter less than **configuration-change turnaround**, because the operationally painful dependency in a trial is a mid-study change,

which is a validated change, which is a queue. Where systems come from a research organisation, the support relationship sits outside the sponsor's control entirely, which means specifying the service levels in the contract, which means IT has to be in that negotiation.

> Both answers are architecture, not operations, and both are usually written after the first incident rather than before it.

Restating the framing

Every vendor named in this chapter is an example of who plays in a category. None is a recommendation and none is an endorsement. Inclusion is not a blessing; omission is not a judgement.

There are no scores, no quadrants, no "leaders" and no market-share figures here, deliberately — those claims go stale fastest and embarrass hardest, and in a market this concentrated they are also the least informative.

Ownership and product names in life-sciences software change faster than books print. Every attribution in this chapter carries a date, and the retrieval date is 24 October 2026. Re-verify before repeating. **The single highest-priority re-check before you quote anything here is whether the ArisGlobal transaction has completed** [4], because if it has, the "one clinical vendor against one enterprise vendor against one capture vendor" framing that every pharmaceutical deck still uses is structurally out of date: one company would hold both a clinical platform and a safety, regulatory and quality platform, which is the same multi-domain position another vendor built organically [1][4][21].

And the thing to take from this chapter rather than from any box in it: **the number of independent multi-domain platforms in pharma is falling, and in a consolidating market a buyer's bargaining position comes from contractual protections rather than from choice.** Exit, data portability, record reproducibility after the relationship ends, and upgrade deferral. Those four clauses are worth more than a feature comparison, and they are the ones nobody negotiates until year two.

THE CHAPTER IN TEN LINES

- Every vendor in this chapter is an example of who plays in a category, never a recommendation and never an endorsement; there are no quadrants and no market-share claims anywhere in it, deliberately.
- Market-share claims, leader positions and analyst placements go stale fastest, then regulatory in-force dates, then ownership, then product names, then dictionary versions.
- Dassault Systemes agreed on 23 July 2026 to buy ArisGlobal, closing in the second half of 2026; completion is unconfirmed and it is the highest-priority re-check, because if it closes one company holds both a clinical platform and a safety, regulatory and quality platform.
- Siemens completed the Dotmatics acquisition on 1 July 2025 at 5.1 billion dollars enterprise value and the platform is Luma, so Siemens is now a scientific-informatics vendor and not only a plant-floor one.
- Alfabet is Bizzdesign's since 9 January 2025 and MEGA International no longer exists independently; ETQ is Octave's, TrackWise is Honeywell's, STARLIMS is Francisco Partners' and LeanIX is SAP's.
- Ownership for roughly thirty-five vendors in laboratory, regulatory publishing, serialisation and validation tooling could not be confirmed, so this book treats those at category level and says so.
- R is already in FDA submissions with dates attached; the gating constraints practitioners report are validation tooling and reviewer infrastructure, not statistical capability.
- No tool, OrbusInfinity included, can be called a certified ArchiMate 4 tool on the strength of a public register, because the register could not be read; and shipping 3.2 content shortly after a new major version is normal, not a deficiency.
- Put the same seven selection questions to every architecture vendor in the same words, Orbus included; no preference is expressed in this chapter.
- No architecture tool was found at this retrieval date to ship a pharmaceutical metamodel extension as first-class object types.
- In South Africa, local capacity correlates with how general-purpose a product is rather than with how good it is, and local-partner dependence is a nameable architectural risk with six concrete mitigations.
- Do not print per-service cloud availability by region; teach the check, record the date you checked it, and confirm an opt-in region is actually enabled on the account.
- Moving to the cloud does not move validation to the vendor: one provider's own list of what remains with the customer runs to eight items.

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CHAPTER 16

Regulation, risk and what the inspector asks for

Eleven bodies inspect a South African pharmaceutical company, and each one asks to see a record. This chapter names the instrument, dates the obligation, and says where the evidence has to live — as architecture, not as legal advice.

READ THIS BEFORE ANYTHING ELSE IN THE CHAPTER

This is architecture material, not legal advice. It tells an architect what entities, columns, retention clocks and evidence trails a repository needs, because an inspector will ask for them. It cannot tell anyone whether a company complies or what a provision means in a contested case. **Every obligation below is framed as what the record has to show, never as whether you comply.**

One standing caution. Regulator guidelines change faster than their own stated validity suggests: this research found one marked valid for five years that had been superseded five months after publication. **Re-checking the version and date of every guideline citation is the reader's standing job, not a one-off.**

The regulated object in pharma is not the product. **It is the record about the product.** A batch made correctly whose record is incomplete is, in regulatory terms, a batch that was not made correctly. Every obligation here is a record obligation, and the question is always the same: *where does the evidence live, at what grain, with what dates, and who can produce it years later?*

The spine: Act 101 of 1965 and SAHPRA

The **Medicines and Related Substances Act 101 of 1965** is the spine, with its **General Regulations** carrying most of the operative detail [1]. Be careful which generation of those regulations you read: several are on the open web, and the set relevant to current practice was published **25 August 2017** [11].

The provisions an architect meets most often: **s 15(6)** makes a registration valid for five years [2]; **s 18** requires each batch to be checked [3]; **s 21** authorises import and sale of a specified quantity of an *unregistered* medicine for a specified period [4]; **s 22A** carries scheduling and the sale controls [5]; **s 22C(1)(b)** is the licence to manufacture, import or export [4]; **s 22G** is understood to carry the pricing machinery; **regulation 44** governs destruction [6]; and **regulation 53** requires prior approval for changes to accepted standards or specifications [7].

SAHPRA, the South African Health Products Regulatory Authority, is the regulator for health products, replacing the former Medicines Control Council.

SAHPRA's establishment date and establishing Act could not be sourced by this research. They are commonly given as 1 February 2018 and the Medicines and Related Substances Amendment Act 14 of 2015, with Act 72 of 2008 in the chain, and that may well be right — but the commencement proclamation was not retrieved, so this book does not state it. A competent author will assert it confidently. Check the gazette.

What SAHPRA verifiably does: **registration** with five-yearly renewal [2]; **variations** [7]; **licensing** of manufacturers, importers, exporters, wholesalers and distributors [3][4]; **scheduling** recommendations to the Minister [5]; **clinical trial authorisation** [11]; **pharmacovigilance** [8]; **inspection** across manufacturing, wholesaling, clinical and pharmacovigilance practice [9]; **traceability** [10]; and **destruction approvals** [6]. **SAHPRA acceded to the PIC Scheme in July 2007** [12] — before the United States, Japan and Brazil — so adopting the PIC/S manufacturing guide is a harmonisation choice by a member, not a workaround by a non-member.

Body	Over what	Primary instrument
SAHPRA — four inspection functions plus registration, licensing and destruction	manufacturing, wholesaling, clinical, pharmacovigilance; registration and variation; scheduling recommendations	Medicines and Related Substances Act 101 of 1965 and its General Regulations
South African Pharmacy Council	the profession; pharmacy premises; the Responsible Pharmacist's registration	Pharmacy Act 53 of 1974 and rules under s 35A
Information Regulator	lawful processing of personal and special personal information	Protection of Personal Information Act 4 of 2013 and the 2026 health regulations
Department of Employment and Labour	exposure, noise, machinery, major hazard installations	Occupational Health and Safety Act 85 of 1993 and the 2021 chemical agents regulations
SARS	declarations, duty relief, licensed premises	Customs and Excise Act 91 of 1964
Ethics committee and the national research ethics council	trials at the institution; committee registration	National Health Act 61 of 2003, Chapter 9, and GN R.719 of 19 September 2014
Marketing Code Authority	promotional conduct of member companies	a voluntary code, v19 of June 2026, hooked to s 18C and General Regulation 42
Competition Commission	tender conduct, pricing, agreements	Competition Act 89 of 1998
Environmental authorities	effluent, emissions, waste, water use	NEMA 107 of 1998; Air Quality Act 39 of 2004; Waste Act 59 of 2008; National Water Act 36 of 1998

Figure 16.1 Who regulates a South African pharmaceutical company, and under which instrument. Eleven inspecting bodies, four of them inside one regulator. The right-hand column is the instrument to cite — and to re-check before citing.

What the record has to show. A **regulatory obligations register** keyed on `(authority, instrument, provision, obligation, owner, evidence location)`, plus the three dated registers hanging off the Act: authorisations, licences and commitments. **The provision column is the one always missing.**

Licensing, and the Responsible Pharmacist

A manufacturing licence issues under s 22C with general regulations 23 and 24 [3]. The application needs the fee, proof from the **South African Pharmacy Council** covering the owner's particulars, the Responsible Pharmacist's registration and the pharmacy recording, **and a national health department premises licence**; staff qualifications; and a **Site Master File** under regulation 23 [3]. The dated obligations: renewal **every five years, at least 180 days before expiry**; particulars changes **within 30 days**; and batch documentation kept **at least one year after batch expiry, or five years after certification, whichever is longer** [3].

The Responsible Pharmacist is the accountable individual, and South Africa has no EU-style Qualified Person. The holder appoints one **named on the licence**, with **not more than one per licence**; candidates are expected to be pharmacists **registered with the Council**; and the duties include ensuring each batch is checked under s 18 and certifying, **in a register or other appropriate record before release for sale**, that the batch meets its registration conditions [3]. Under the manufacturing-practice guideline the role must ensure every released batch meets the authorisation's terms, practice and legal requirements, taking account of storage and transport conditions, sampling and quality-control results [13]. And the duty most architects miss: **ensuring that processes remain in a validated state** is part of the role, which makes validation state a fact the signer needs to see.

Model the role as **Responsible Pharmacist**, with the Council registration number and status as attributes, because a lapse there invalidates the condition the licence rests on.

What the record has to show. A **licence register** with number, class, scope lines, issue and expiry dates, the derived renewal-due date at expiry minus 180 days, the **named Responsible Pharmacist with a from-date and the Council reference**, the conditions, particulars-change notifications with dates, the Site Master File version and the inspection history. Plus a **batch-release decision record at `(batch, market)` grain** naming the person, the date and the inputs considered [3][13].

Manufacturing practice — and why not to cite SAHPRA for the detail

SAHPRA has adopted the PIC/S manufacturing-practice guide, and its own guideline carries only the local additions, including a South African replacement for the guide's certification annex [13]. That has a consequence for citation practice, because this research checked the local guideline for each of the following and found them absent: **batch record retention, batch numbering, dedicated deviation and change-control requirements, detailed product quality review requirements, any data-integrity section, and any computerised-systems annex** [13]. The substantive rules live in the adopted guide [14]: **batch documentation kept one year after the expiry of the batch it relates to, or at least five years after certification, whichever is the longer**; **product quality reviews "should normally be conducted and documented annually"**; root cause analysis on deviations; change control with approval before implementation; and records enabling "the complete history of a batch to be traced".

Three cautions. **The annual frequency of the product quality review comes from the adopted guide, not from SAHPRA**, whose guideline mentions the review only as a shared responsibility with no detail [13][14] — do not infer that it is optional. **The term "ALCOA" is not used in the adopted guide's Part I at all** [14]; the verified South African ALCOA statement is **ALCOA-CCEA in the good clinical practice guidelines for clinical source documents** [15], a clinical requirement rather than a manufacturing one. And **the guide's own revision moves**: the current revision came into force **23–24 September 2026** [16], while the local guideline's document history records adoption under a revision from 1 January 2017.

Registration, variation and renewal

The dossier is a **maintained state**, not a submitted document, and the clocks are what make that true.

Review timelines are in working days, not months: **360** for new chemical entities, biologicals and biosimilars, **250** for generics without clinical data, **200** for reliance pathways [17][18]. **360 working days is roughly 17 to 18 calendar months of regulator time**, before any sponsor response. Quote working days and convert explicitly.

Clearing a backlog did not fix throughput. In 2024/25 SAHPRA received **2,691 generic applications and finalised 568** [17]: a capacity-constrained queue, not a backlog that was cleared. And **section 21 is not a rare exception — 20,245 applications on a three-working-day clock** in the same year, SAHPRA's busiest workflow by volume [17] — while **section 21 supply is not subject to the Single Exit Price**, which makes it a separate pricing path in the model.

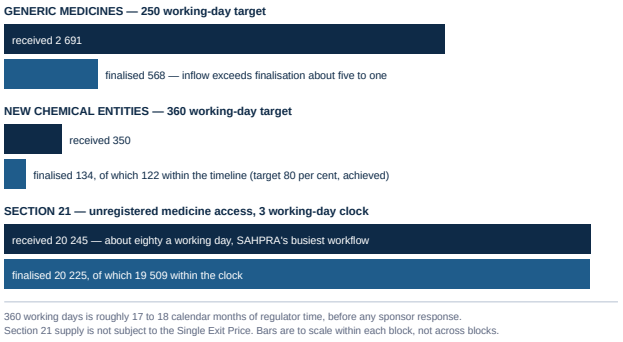


Figure 16.2 The queue arithmetic, from SAHPRA's own 2024/25 report to Parliament. Targets were met and the generic queue still grew: 2,691 received against 568 finalised. Registration lead time is a capacity-constrained queue, not a backlog that was cleared.

Reliance is not recognition. SAHPRA does not recognise another regulator's approval. It operates **reliance** — abridged and verified review using a recognised authority's assessment reports — and **"remains responsible and accountable for decisions taken"** [18]. Recognition is a separate pathway still being negotiated, with an unpublished framework. One rule is worth modelling: **if no full assessment reports arrive from the reference authority within three months of the request, the application is reviewed as a full review** [18] — another regulator's responsiveness inside a launch plan.

Registration expires. Validity is **five years**; renewal goes in **six months before expiry**; the programme is phased **2023 to 2035**; failure to submit a complete renewal may result in expiration, after which **a new registration application is required**; and no variations may be in flight during renewal review [2].

Variations invert how approval workflows are normally built. Type IA is notify-within-twelve-months; **Type IAIN requires 47 working days' notice and is deemed implementable if SAHPRA gives no opinion**; Type IB runs 87 working days, after which changes may be implemented though reversal may still be required; only Type II requires prior approval, at 220 working days [7]. Parallel submissions are not acceptable, and an application without an amended quality information summary **will be rejected** [7].

"Deemed implementable on silence" means the absence of a response is an event the system must detect. A workflow waiting for an approval message will wait for ever. Model the 47-working-day expiry as a scheduled, computed event with its own evidence record, because the implementation date is what matters downstream.

On the channel: **South Africa is on electronic common technical document v3.2.2, portal-only**, with PDF 1.4 to 1.7 and **whole-sequence rejection on error**. **A baseline built from non-electronic content starts at sequence 0000**, as a separate sequence containing only previously submitted content; new applications start at 0001; and on transfer the acquirer must not restart at 0001 [19]. There is **no adopted date for version 4.0** [19].

Single Exit Price

South Africa regulates the price at which medicines enter the private market through a **Single Exit Price**: one price per pack, published, adjusted periodically by the Minister. Two things are routinely stated wrongly.

One: it does not apply to the State. The government source says so directly [20]. State procurement runs through tender, at tendered prices, entirely outside the regime. **This is the most commonly misstated fact in South African pharma.**

Two: it is not a maximum and not a recommended price — it is the only price. For a manufacturer or importer it is the single price at which a medicine may be sold to **any** private buyer: the same to a large retail group, an independent pharmacy, a dispensing doctor and a private hospital, with no volume

discount, no rebate, no bonus. **So the customer-specific pricing machinery every enterprise system ships with must be switched off for this class, and the control that prevents a discount being applied is a compliance control, not a commercial one.**

There were two adjustments in 2026: the ordinary annual **1.47% in January**, and an extraordinary **2.88% effective 1 October 2026**, applying to all medicines registered and listed in the national database as at 30 September 2026, including all related pack sizes [20][21]. **Any 2026 pricing statement mentioning only one is wrong**; one outlet rendered the January figure as 1.4%, so use 1.47% [21].

Then the two mechanics most often inverted. **Wholesalers take no mark-up.** Instead **the manufacturer pays the wholesaler a logistics fee out of the exit price**; the amount varies considerably, appears as a column in the national pricing database, and where a seller owns its own wholesaler it earns that income undisclosed to buyers [21]. So **exit price times volume is gross revenue, not net** — the one genuine gross-to-net item in the local private market, and the percentage could not be sourced. And **the dispensing fee is charged in addition to the exit price, not out of it**, calculated on the price **excluding** value-added tax, with different bands for pharmacists and dispensing doctors [21][22].



Figure 16.3 Where the money actually goes under the Single Exit Price. The logistics fee comes out of the exit price; the dispensing fee is added on top of it. Reversing either is the commonest error in a South African pharmaceutical pricing model.

> **Do not quote the February 2025 gazette's dispensing-fee bands as current**: those were published for comment and differ from the bands later published [22][23]. And **the gazette notice for the 2.88% adjustment, and the section under which adjustments are made — commonly cited as s 22G — could not be verified**.

What the record has to show. A **price record at pack grain, dated**, with the logistics fee separate; a **control that blocks any customer-specific price variance**, with exceptions logged; a **State-versus-private channel flag on the sale**; and an audit trail of price changes, because an adjustment applied to the wrong pack set or date is a non-compliance affecting every invoice after it.

Scheduling, Schedules 0 to 8

Schedule is a **per-substance, per-strength, per-indication, dated classification that drives transaction legality** [5]. It is not a product label. It can differ by strength and pack size for the same substance, and **rescheduling is a 250-working-day variation class**, so it is a dated change with an effective date [7].

Schedule	Who may sell or supply	Record obligation	Public advertising
0	any open shop, including general dealers, and pharmacies	none specific	permitted
1	pharmacists and supervised staff; manufacturers and wholesalers; prescribers	non-manufacturer sellers record the prescribed particulars of **each sale**	permitted
2	pharmacists, with counselling; no prescription	particulars of **every sale** in a permanent record; prescription kept 5 years at the last seller	**not to the public**
3, 4	authorised prescribers, dispensed on prescription	as above; repeats not longer than six months	not to the public
5	prescribers; low to moderate dependence potential	**specified Schedule 5: register balanced quarterly, within 14 days**; enhanced records by all sellers	not to the public
6	prescribers, close medical supervision; no repeats without a new prescription	**register balanced quarterly, within 14 days**	not to the public
7	no recognised medical use; scientific purposes only	destruction witnessed by an inspector or two pharmacists	prohibited
8	**the Director-General only**, to practitioners with special permission	as Schedule 7; import and export permits valid six months	prohibited

Figure 16.4 Schedules 0 to 8, and what each obliges in records and control. Schedule is a dated classification per substance, strength and presentation that drives transaction legality — not a label attribute. Rescheduling is a 250-working-day variation.

Two obligations out of the figure are worth stating in prose, because both are computable controls. **Specified Schedule 5 and Schedule 6 require manufacturers and wholesalers to keep a register balanced at 31 March, 30 June, 30 September and 31 December, with balancing completed within 14 days after each date** [5]. And the trade control: a **Director-General permit for Schedule 5 to 8 substances, valid six months from issue**, against a **licence valid five years** [4] — **two clocks, two renewal processes**, and an architecture tracking only the licence will miss the permit.

What the record has to show. The dated classification per `(substance, strength, presentation)` with the amending gazette reference, and the quarterly-balanced register at `(substance, schedule, site, period)` grain with opening balance, movements, closing balance, a reconciliation to the physical count and the 14-day evidence [5].

Clinical trials

The instruments: Act 101; the **National Health Act 61 of 2003**, Chapter 9, ss 71 to 73; the **Regulations Relating to Research with Human Participants**, GN R.719 in GG 38000 of **19 September 2014**; and **South African good clinical practice, third edition, 2020** [11][15][24].

SAHPRA's clinical trial committee reviews applications and also authorises importation of unregistered medicines for trials; scope covers phases I to IV plus bioequivalence studies, and SAHPRA and a registered ethics committee may review in parallel [11]. Applications go in on predetermined committee dates with the prescribed fee **and proof of clinical trial insurance — applications without insurance are rejected** [11]. **An ethics committee reviewing a trial must be registered with the national council**, which audits committees on a five-year cycle; approval must be written, dated and obtained before the trial begins, and **lasts a maximum of 12 months, renewable** [15][11].

No trial may start without the register. Sponsors must register all South-Africa-based trials on the national clinical trials register, and **"no trial may commence without this Department of Health number"**; the trial must be **fully registered before participants are enrolled** [15][11].

The retention floor is ten years, not twenty-five. Essential documents must be retained **at least 10 years after final closure, or at least two years after formal discontinuation of clinical development of the investigational product**, for investigator and sponsor alike, and **ethics committees retain their own records for at least three years after trial completion** [15]. The twenty-five-year figure is European and is routinely mis-attributed as a global requirement.

> **The day-count timelines for reporting serious adverse events to South African ethics committees could not be sourced, and that is a real gap.** The trial safety guideline does not address committee timelines [25], and the good clinical practice guidelines require investigators to report to committees without stating a day count in the text retrieved [15]. **Do not design a safety workflow with an assumed number in it.** Design it with the number as a sourced, configurable parameter per committee, and capture where it came from.

What the record has to show. The study spine plus per-site approval states with dates; the **register number before first enrolment**; the amendment fan-out with per-site approval and implementation dates; **per-subject consent at version grain**; the delegation log; the essential-document set with a **10-year clock from final closure**; and audit trails with no deletion of entered data [15].

Pharmacovigilance

The operative instrument is SAHPRA's post-marketing reporting guideline [8]. The clocks, verified: **serious adverse reactions occurring in South Africa — related or unrelated, expected or unexpected — within 15 calendar days of receipt**, with follow-up on the same clock; **significant safety issues within 5 calendar days**; **regulatory action taken abroad within 5 calendar days of first local knowledge**; a change in the nature, severity or frequency of reactions within 5 calendar days; and medication errors within 15 calendar days [8].

Day zero is the hard architectural fact. The clock starts when the company first became aware — awareness by **any employee or contractual partner**, including the call centre, a sales representative and a market-research vendor, not arrival in the safety database. So the architecture needs a **receipt date captured at the point of first contact and transported forward**; resetting the clock to the data-entry date is the most serious data defect in this domain. Submission is **by mailbox rather than gateway**, with **only one submission method permitted per case** [8].

Five corrections follow, each of which a competent author is likely to get wrong.

"SAHPRA requires E2B(R3)" is wrong. The guideline says reports "can also be submitted ... in the e2b format using an 'xml' file" and **names no revision** [8]. For trials the requirement is **CIOMS format plus SAHPRA's own form**, not E2B [25].

"SAHPRA requires a pharmacovigilance system master file" is wrong. The term does not appear [8]. What is required is a system with signal management, risk management and evaluation of risk-minimisation effectiveness, plus "full documentation covering all procedures and activities" [8]. A master file satisfies that; it is not the stated requirement.

"SAHPRA mandates MedDRA" is not stated in the operative guideline, where the dictionary appears only inside a line-listing definition [8]. It is unavoidable in practice; the mandate is elsewhere.

Foreign individual case reports are not routinely forwarded to SAHPRA [8]: what runs on the five-day clock is foreign **regulatory action**. That is the opposite of the European and American default, and it changes the volume of the submission pipeline by an order of magnitude.

Periodic safety reports have two different triggers. They are prepared per the European reference-date list where applicable, or **annually** if the product is not on it — but **submitted only on request**, as a registration condition, or with a conditions amendment, and then within 30 calendar days [8]. **Preparation and submission are separate obligations, and most safety systems model only one.**

Two further requirements with architectural weight. The qualified person for pharmacovigilance **and the back-up "should reside in South Africa"** [8]. And surveillance continues **until six months after the expiry date of the last marketed batch** [8] — computed from batch data in the supply systems, and dropped the moment a product is deactivated. **No fixed record retention period is given** [8]: a finding, not a licence.

Inspection response

SAHPRA does not classify findings as critical, major and minor with differentiated deadlines. Its response guideline refers to "deficiencies", "non-conformity" and "serious non-compliance", and sets **one deadline — 30 calendar days from receipt of the report** — with extensions only by prior written grant [9]. The classification scheme exists in the international inspection framework, not here. So **build one 30-calendar-day clock from receipt, per finding, with a documented extension object**, and let severity be a field for internal prioritisation rather than a driver of the external deadline.

Pharmacy law

The **Pharmacy Act 53 of 1974** establishes the **South African Pharmacy Council** and the regulation of the profession [26]. Its subordinate instruments — practice regulations, good pharmacy practice rules under s 35A, and the board notices on Responsible Pharmacist registration criteria and on scope of practice — were identified by notice number and date but not retrieved, so their content is not stated here [26][27].

Three points the retrieved evidence does support. **The Council's registration is an input to the SAHPRA licence** [3], so the architecture needs the registration number and status as attributes of the named person, and must detect lapse. **"Who may do what" is a scope-of-practice question with data consequences** — a system that records "pharmacist" as a role without the registration category cannot enforce the scheduling rules [5]. And **good pharmacy practice reaches into waste**: SAHPRA's destruction guideline states that medicine waste must also comply with the Council's rules [6] — exactly the cross-instrument link an obligations register has to carry. What the record has to show is a **person register** at `(person, Council number, category, status, dates)` plus the dated `(person, role, site, from, to)` history, because "who was the Responsible Pharmacist on the date this batch was released" will be asked years later.

POPIA and health data

This is the longest section in the chapter because it is where an architectural decision — where a database sits, who the operator is, what the lawful basis is, whether the data is pseudonymised — determines the legal position, and where the legal position is genuinely unsettled.

Health information about an identified person is **special personal information**, and the default under the **Protection of Personal Information Act** is that processing it is **prohibited**. Everything else is an exception. So the architect's job is not to ask whether you are allowed to do this. **It is to make the system able to show which exception it is relying on, for which processing operation, with which safeguards, and since when.** Five provisions do the work: **s 26** prohibits [28], **s 27** is the general authorisation [29], **s 32** is the health-specific authorisation [30], **ss 57 and 58** cover prior authorisation [31], and **s 72** governs transfers out of the country [32].

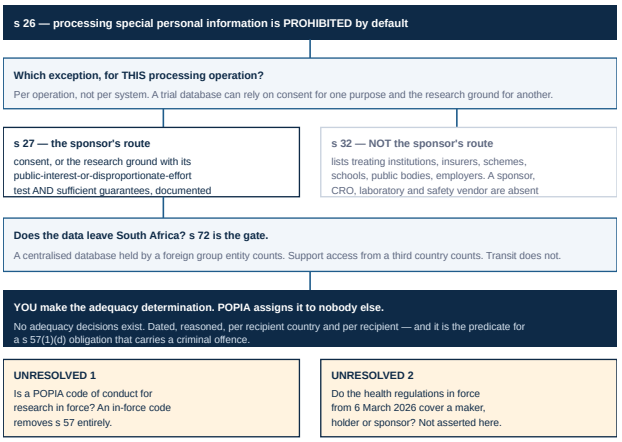


Figure 16.5 The decision path for health data under POPIA, as the architecture has to hold it. Each diamond is a dated, reasoned record, not a judgement made once. The two shaded boxes are the unresolved questions: hold them as unresolved, with an owner and a review date.

Section 27, and the research ground

Section 27(1) lifts the prohibition where the processing has the data subject's **consent**; is necessary to establish, exercise or defend a legal right; is necessary under international public law; is for **historical, statistical or research purposes** subject to conditions; concerns information deliberately made public; or complies with ss 28 to 33 [29]. The research ground's conditions are what the architecture has to evidence: the purpose must **serve a public interest and the processing be necessary for it**, *or* it must be **impossible or involve a disproportionate effort to ask for consent — and sufficient guarantees** must be provided so that privacy is not adversely affected to a disproportionate extent [29].

Three design requirements fall out. **The lawful basis is per processing operation, not per system**: a trial database may rely on consent for the primary purpose and on the research ground for a secondary analysis, separately recorded. **"Sufficient guarantees" is an evidence requirement** — the pseudonymisation design, the access controls, the minimisation decisions and the retention limits *are* the guarantees, documented as such. And **"impossible or disproportionate effort" is a determination with a date and a reasoning**, which makes it the record a company relying on secondary use of trial data will be asked for.

Section 32 does not cover a sponsor

Section 32(1) lifts the prohibition for medical professionals, healthcare institutions and social services; insurers, medical schemes and their administrators; schools; bodies managing a child's care; public bodies in connection with detention; and administrative bodies, pension funds, employers or their agents [30].

Notice who is not in that list. A pharmaceutical company, a contract research organisation, a central laboratory and a safety-database vendor are not among the enumerated parties. The sponsor's route is therefore **s 27** — consent, or the research ground — not s 32, which is the route for the treating institution and the insurer. So **the sponsor's lawful basis rests on the consent it obtained or on a research determination it made**, which puts the consent-version model and the sufficient-guarantees record at the centre of its position. Section 32(5) additionally restricts processing information about inherited characteristics unless a serious medical interest prevails or the processing is for research [30], which matters to any study collecting genetic data.

The 2026 health-data regulations

New regulations on the processing of health information by certain responsible parties came into force on 6 March 2026, signed 27 February 2026 and published in **Gazette No. 54268, Notice No. 7198**, with **no transitional period** [33]. They are made under s 32(6), and the analysis of the final text describes them as "more detailed rules" and little else [34]. The operative content as reported: special personal information may be processed only where s 27 is satisfied, with each activity mapped to a documented justification; safeguards must cover **both physical and electronic records**; health data may go abroad **only if a condition in s 72(1) is met**, with group policy or operational convenience unlikely to suffice; and penalties run to **fines up to R10 million** [33].

> **Whether a pharmaceutical manufacturer, a marketing authorisation holder or a clinical trial sponsor falls within the eight covered categories could not be verified, and must not be asserted either way.** The categories as reported are insurance- and employment-facing: insurers, medical schemes, scheme administrators, managed healthcare organisations, administrative bodies, pension funds, employers, and institutions working for employers, administrative bodies or pension funds [33]. Settled only by the regulations' own text in the gazette. **Do not write that these regulations bind pharmaceutical companies. Do write that section 72 remains the cross-border gate regardless.**

And the correction that will catch anyone writing from 2025 commentary. Between the draft of **26 September 2025** and the final text, the following were **removed**: the **Legitimate Interest Assessment**; the dual-authorisation requirement; the **retention, destruction and de-identification provisions**; the **cross-border transfer notification provisions**; and the **safeguard provisions** [34][35]. References to sex life information went, and the employer definition was broadened [34]. **Anyone writing that the health regulations impose retention and destruction rules and require a legitimate-interest assessment is describing the draft, not the law** — the single most likely factual error in any 2026 write-up of South African health-data law. Where they do touch a pharmaceutical architecture: **as an employer**, which puts occupational health records in scope, and **as a counterparty to schemes and insurers**.

Section 57, and the question nobody has answered

The Regulator's guidance note lists the prior-authorisation categories as **unique identifiers** used beyond their collection purpose to link information held by other responsible parties; criminal-behaviour checks for third parties; credit reporting; **transfer of special personal information, or children's personal information, to a third party in a foreign country that does not provide an adequate level of protection**; and any other processing the Regulator identifies as carrying particular risk [31]. **Health is not a standalone category** — it appears only inside the definition of special personal information and becomes relevant through the cross-border limb — and **research data is not mentioned at all** [31].

The clocks sit on a project schedule, not just a compliance register. Applications go in **before processing**, and notified processing may not begin until the Regulator completes its investigation or confirms there will be none; it must state **in writing within four weeks** whether it will investigate in detail, and if so finalise **within a maximum of thirteen weeks**; failing to notify is an offence, as is processing before the investigation concludes; and the fine reaches **R10 million** [31]. So the real question is whether a global trial, a cloud-hosted data capture system or a safety database hosted abroad engages prior authorisation — and on the Regulator's own description it does **if** the transfer is to a third party in a country without an adequate level of protection [31]. Which turns everything on adequacy.

Section 72, and the adequacy burden POPIA puts on you

Section 72(1) prohibits transferring personal information to a third party in a foreign country unless one of five grounds applies: the third party is subject to a law, binding corporate rules or binding agreement providing **an adequate level of protection** that effectively upholds principles for reasonable processing and includes provisions substantially similar to section 72 on **onward transfers**; the data subject consents; the transfer is necessary to perform a contract with the data subject; it is necessary for a contract concluded in the data subject's interest; or it is for the data subject's benefit where consent is not reasonably practicable and would likely be given [32].

> **POPIA has no counterpart to the GDPR's adequacy decisions. It assigns no body the task of assessing adequacy, so the responsible party bears that burden itself** [32] — which makes the adequacy determination **the predicate for a s 57(1)(d) obligation that carries a criminal offence**. A South African architecture must therefore *produce* an adequacy determination per recipient country and per recipient, dated and reasoned; a European one does not. **The most consequential structural difference in the chapter.**

Two scoping points cut in opposite directions. **Data merely passing through servers in another country is not a transfer.** But **a centralised database held by a foreign group entity can be treated as a transfer to an operator** [32]. The second catches global sponsors: no export event, no file moved — the South African site typed into a system whose database is abroad. **The transfer is architectural, not operational, and it happens continuously.** Note too that section 72 **does not cover transfers by operators** [32], and that consent is characterised as a backdoor rather than a sustainable basis [32]. For a cloud-hosted system add two facts: **data location differs per environment**, and **vendor support access from a third country is a transfer** even if the data never leaves its region.

What the record has to show. A **transfer register** at `(processing operation, data categories, recipient entity, recipient country, s 72 ground, adequacy determination with date and reasoning, instrument and its onward-transfer clause, sub-processors, environments and data locations, s 57 determination, review date)` — **the same register as the validated-system register with four columns added**, which is the argument for building one, not two.

The research code of conduct, which is the highest-value open question

A Regulator-issued, in-force code of conduct for a sector removes the need to apply for prior authorisation in that sector [31]. For a research-active organisation that is the difference between a four-to-seventeen-week regulatory process on the critical path of every study start-up and no process at all.

Whether such a code is in force for research could not be established, and it is the single highest-value unresolved question in this chapter. The code this research located is a **draft**: its commencement clause reads "This Code will come into effect on [insert date once agreed with the Information Regulator]", and it still carries placeholders [36]. Settled by the Regulator's register of codes of conduct — **check it before you design a study start-up process, because the answer changes the whole prior-authorisation analysis.**

The draft's positions are useful even unapproved, and two correct common errors. **"Broad consent is fine for research" is wrong**: consent must be voluntary, specific, informed and explicit, and **obtaining consent to "conduct research" "will not be sufficient"** [36] — so consent is per specifically defined study, which reinforces a per-version consent model and makes secondary use a separate determination. And **pseudonymised data is not**

outside the Act: only **permanent de-identification** takes data outside it, and the draft advises researchers to **"assume that re-identification is possible"**, requiring pseudonymisation of high-risk research unless there is a compelling reason not to [36].

The draft also carries a retention tension worth designing for now. Research records **may be retained indefinitely if retained only for research purposes**, while identifiable information must be **destroyed or de-identified as soon as possible** once no longer needed [36] — against good clinical practice's **ten-year floor** [15]. **Those three rules are satisfiable together only by a design that separates the identifiable layer from the research layer and retires them on different clocks** — a pseudonymisation architecture justified by retention rather than by privacy, which survives a change in the privacy position.

Occupational health and the plant

The operative instrument for a pharmaceutical plant is the **Regulations for Hazardous Chemical Agents, GN R.280 in GG 44348 of 29 March 2021**, made under s 43 of the **Occupational Health and Safety Act 85 of 1993**, which **replaced the earlier hazardous chemical substances regime** [37]. The dated obligations: **risk assessment every two years**; **exposure assessment and air-monitoring records for a minimum of 30 years**; **engineering-control test records for at least three years**; and **safety data sheets reviewed at least every five years** [37]. Separately, under the noise-induced hearing loss regulations: **noise monitoring at 40 years** and **medical surveillance records including baseline audiograms at 40 years** [38]. Medical surveillance under the chemical regulations is covered but **no retention period is stated there** [37].

Two pharmaceutical-specific additions. **Potent compound handling**: these compounds have exposure limits in the nanogram-per-cubic-metre range, so the limit is **a company-derived value with an author, a date, a toxicological basis and a review cycle** — and it is simultaneously an occupational-health control parameter **and a cleaning-validation acceptance criterion**. One number governs "is the operator safe" and "is the next product free of carryover", so holding it in a spreadsheet is two regulatory exposures in one cell. And plant employees' surveillance records are **health data under the privacy Act, held for decades, in a system nobody classifies as a privacy system**.

Environment, and destruction

The framework is the **National Environmental Management Act 107 of 1998**, with the **Air Quality Act 39 of 2004** and its emission standards [39]; the **Waste Act 59 of 2008** with its classification regulations and landfill norms [40]; and the **National Water Act 36 of 1998**, with authorisable water uses in s 21 [41]. An active-ingredient plant is a fine-chemicals plant, with the effluent, solvent-vapour and hazardous-waste profile of one. **Whether South Africa imposes any active-ingredient-specific or antibiotic-specific effluent limit could not be established** and is not asserted here.

The destruction regime is verified and specific — and **it is governed by general regulation 44, not regulation 27**; SAHPRA amended its guideline in October 2025 specifically to reflect that renumbering [6]. The rules: destruction only at a **waste treatment facility authorised under the Waste Act**; **no disposal into municipal sewerage**; the item denatured and unsalvageable; **Schedule 5 or 6 destruction witnessed by an inspector, a pharmacist or a chief-executive-authorised person, and Schedule 7 or 8 by an inspector, two pharmacists or an authorised person**; and a facility certificate naming the medicine or its schedule, the quantity, the date and **the name and designation of the witness** [6]. **SAHPRA's prior permission is required**, on its destruction form with **exact quantities, batch numbers and expiry dates**, the waste company, the reason and the named pharmacists; complete applications are processed within **10 working days** [6].

> **The destruction certificate is simultaneously a quality record, a controlled-substance record, a waste record, an inventory document and a financial document.** It is the best worked example in this book of why a pharmaceutical architecture cannot be partitioned by function. Model it **at batch grain**, because regulation 44 and the application form both demand batch numbers and expiry dates, and link it to the disposition that condemned the stock, the serialised units decommissioned, the inventory relief and the write-off.

Trade and customs

Imports and exports of medicines need a licence under **s 22C(1)(b)**, none being permitted without one under **s 22C(6)**; it is **valid five years** and every application requires a responsible pharmacist [4]. **Imports must enter through one of four named ports**: Cape Town, Port Elizabeth and Durban airports or harbours, or O.R. Tambo International Airport [4]. **Per consignment**, form **MBR 20** must be completed, **personally signed**, and submitted to a customs officer at the port with the invoice, the licence and the permit attached, and also to the chief executive's office [4]. Permit holders provide annual returns and must notify **within 10 days in writing** if the quantity imported falls short of the quantity permitted [4]. Note what the guideline does **not** cover: active ingredients specifically, the revenue service, or the trade administration commission [4].

On customs, the reusable insight is a sequencing trap. Specific drawbacks refund duties paid on goods used in exported products, but **applications may only be submitted after export, with proof of it; the export declaration must already show the correct customs procedure code and drawback item before the goods leave South Africa**; and claims must be lodged **within six months of export** [42]. **The entitlement is created by a field on a declaration filed before departure, and no amount of later documentation recovers it.** And the sentence worth carrying into every engagement: **the duty-relief claim and the recall trace are the same query with different filters** [42] — both are substantiated only by production genealogy.

Serialisation and traceability

"South Africa requires unit serialisation with national repository reporting" is wrong three ways. The instrument is a SAHPRA guideline rather than a regulation, effective October 2025, with its roadmap in a stakeholder communication of 29 September 2025 [10][43]. On that evidence the programme is **phased 2025 to 2031**; it is **batch-level first and unit-level later**; and **no national verification or transactional reporting repository is specified** — the stated submission is of **master data** at registration or amendment [10][43]. Record-keeping runs to **five years after the expiry of a trade item** [10].

And the deadlines are earlier for imported products than for domestic ones — batch level 1 October 2027 against 1 October 2029, unit level 1 October 2029 against 1 October 2031 [43] — the reverse of the usual assumption.

Competition and marketing

The **South African Code of Marketing Practice for Health Products** is maintained by the Marketing Code Authority, at **version 19, approved in June 2026**, and it is **voluntary self-regulation** [44]. Its statutory hook is **s 18C** of Act 101, with **general regulation 42** carrying the advertising requirements, and **where the code and a law conflict, the law prevails** [44]. The content with architectural weight: no **inducement** that is not legally due, and no personal incentive to a healthcare professional in an advertisement; **honoraria and fee-for-service** paid at **fair market value**; all financial transactions recorded in an agreement signed by both parties; and **records retained at least five years** [44].

> **The code cannot set fair market value, and the reason is interesting.** It states that **competition law prevents the Authority prescribing an amount**, so **each company must set and substantiate its own** [44]. The methodology is therefore a dated, documented, per-service, per-seniority reference dataset owned by the company — and because it derives from the company's own data it is itself a competition-sensitive document.
>
> **Whether the code imposes a public transfer-of-value disclosure obligation of the European or American kind could not be verified** — the relevant chapter was headings only in the text retrieved [44]. Given that the code is voluntary self-regulation, "mandatory public disclosure" is the wrong framing in any case. **Do not assert it.**

Rebates are prohibited above Schedule 0, but the regulations defining what is prohibited were never finalised — comment rounds ran in 2014 and 2017 — so the extent of co-marketing fees, shelf fees and off-invoice arrangements is unknown [21]. The control exists and the definition does not, so the safe design logs the transaction types and lets compliance characterise them rather than hard-coding a prohibition list.

Competition exposure attaches to **tender conduct in the State channel**, where the Single Exit Price does not apply and price is set by bid. The live example: on 11 and 12 February 2026 it was reported that the Competition Commission is investigating three named companies after a health department complaint alleging collusive practices in state medicine tenders, with no specific conduct attributed to any company [45]. Three controls follow: tender pricing decisions need a documented internal basis **independent of any competitor-sourced information**; contact with competitors, including through trade associations, must be logged; and the fair-market-value substantiation must come from the company's own data.

Who inspects, and what they ask for

Eleven bodies, what each asks to see, and where the evidence lives, split into two tables with the same columns.

Body	What it inspects	What it asks to see	Where the evidence lives
SAHPRA — manufacturing	Licensed manufacturing, packaging and testing sites	Licence and scope; Site Master File; master and executed batch records; batch genealogy; deviations with affected batches and root cause; change control with regulatory and validation impact; corrective actions and their **effectiveness checks**; out-of-specification investigations; the annual product quality review; **the Responsible Pharmacist's release decision with its inputs**	Licence register; master and executed batch record pair; genealogy both ways; deviation with its affected-batch link; change record linked to the variation register; `(batch, market)` disposition
SAHPRA — clinical	Trials, sponsors, sites, research organisations	Protocol and amendments with approvals **per site, dated before the data collected under them**; the register number before first enrolment; **consent per subject at version grain**; the delegation log; the audit trail with no deleted data	Study spine with protocol version and amendment fan-out; `(subject, consent version, signed date, witness)`; `(site, role, person, from, to)`; append-only data point with audit trail
SAHPRA — pharmacovigilance	Authorisation holders' safety systems	**The qualified person and back-up with South African residency evidence**; receipt dates traceable to first contact; compliance against the 15- and 5-day clocks; **the signal register including refuted signals**; risk-minimisation **effectiveness evaluation**; the six-months-after-last-batch-expiry determination	Bi-temporal case store; obligation table; signal register; label version per country; **dictionary version on every coded value**; batch expiry feed
SAHPRA — wholesaling	Wholesalers, importers, distributors	Not established: the guidelines were identified but not retrieved. Expect licence and scope; temperature excursions; consignment traceability; recall capability	Licence register; consignment records; the returns-to-complaint-to-destruction chain
SAHPRA — destruction	Destruction of medicines and scheduled substances	The application with **exact quantities, batch numbers and expiry dates**; SAHPRA's permission; the facility's Waste Act authorisation; **the witness's name and designation**	Destruction record at batch grain, linked to disposition, decommissioning, inventory relief and write-off
South African Pharmacy Council	Pharmacies, premises, pharmacists	Registration of the pharmacy and **the recording of the premises**; the Responsible Pharmacist's registration and category	Person register at `(person, Council number, category, status, dates)` plus `(person, role, site, from, to)`

The second half, same columns:

Body	What it inspects	What it asks to see	Where the evidence lives
Information Regulator	Lawful processing of personal and special personal information	The processing register with **lawful basis per operation citing the provision**; consent at version grain; the sufficient-guarantees documentation; operator agreements with their onward-transfer terms; **the transfer register with a dated, reasoned adequacy determination per recipient country**; the breach register	Processing register; transfer register, which is the validated-system register with four columns added; consent per version; key custody record
Department of Employment and Labour	Hazardous chemical agents, noise, machinery	Risk assessments **every two years**; exposure monitoring with a **30-year** floor; audiometry with a **40-year** floor; **the dated employee-to-exposure-zone assignment**; safety data sheets with their five-yearly review	Exposure repository at `(exposure group, date)`; surveillance at `(employee, test date)`; employee-to-zone history; compound safety dataset
SARS (customs)	Declarations, duty relief, licensed premises	Tariff classification with its basis; declarations **linked to the order, receipt, batch and despatch**; duty-relief registers reconciling imported inputs to exported outputs; **drawback claims within six months of export, with the procedure code already on the declaration**	Declaration repository; **duty-relief reconciliation built on production genealogy**
Ethics committee and the national council	Trials at the institution; committee registration	Written, dated approval **before** commencement, renewed at most every twelve months; every amendment; consent documents in the approved versions and languages; **ministerial consent for non-therapeutic research with minors**, with its 60-day outcome notice	Site approval states with dates; amendment fan-out; consent-version register; obligation register entries for the renewal and the annual report
Marketing Code Authority	Promotional conduct of member companies	Promotional material with its approvals and the product-information version checked against; **financial-transaction agreements signed by both parties, retained five years**; fair-market-value substantiation	Promotional approval register; transfer-of-value register; fair-market-value methodology with versions
Competition Commission	Tender conduct, pricing, agreements	Tender bids with their pricing basis and internal approvals; contact with competitors; any competitor-information inputs and their provenance	Tender register with pricing basis; competitor-contact log

Now read down the right-hand column of both halves. **The same six structures answer nearly every inspector.**

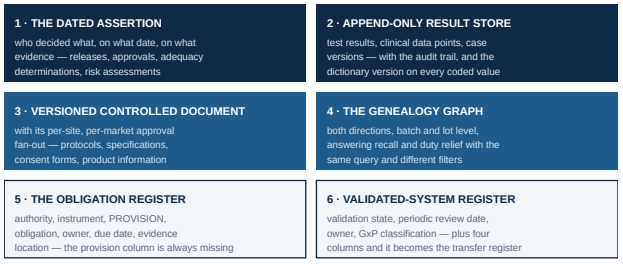


Figure 16.6 The six structures that answer nearly every inspector in the table. Read them against the right-hand column of both halves above: almost every entry there is one of these six, seen from a different angle.

Build those six well and the inspection readiness of the whole enterprise follows. Build eleven inspector-specific repositories and none of them will be current.

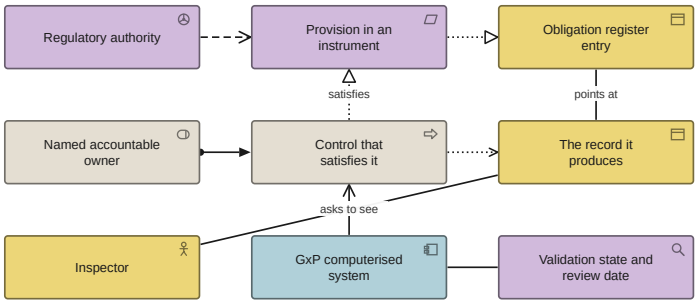


Figure 16.7 The obligation-to-evidence chain, as one traceable path. A provision in an instrument is realised by a control, which a system supports, which produces the record an inspector asks for. If any link is missing from the repository, the answer to "show me" is a search of someone's mailbox.

The retention clocks

A single corporate retention period is always wrong in a pharmaceutical company. The clocks below run from four kinds of trigger — a fixed number of years, a product event, a study event and an employment event — and range from **three years to forty**. Any single number is either unlawfully short for the longest clock or needlessly expensive for everything else, and either way it cannot be applied: **you cannot evaluate "one year from batch expiry" without the batch expiry date, so retention is a computation over operational data, not a policy setting.** Only verified periods appear.

Record	Period	Trigger	Instrument
Batch documentation	**1 year after batch expiry, or 5 years after certification, whichever is longer**	Batch expiry or certification	Adopted manufacturing guide, Part I §4.11; mirrored in the licensing guide [3][14]
Clinical trial essential documents	**At least 10 years after final closure, or 2 years after formal discontinuation of clinical development**	Trial closure or programme discontinuation	Good clinical practice §5.10.5 and §6.4.6 [15]
Ethics committee trial records	**At least 3 years after trial completion**	Trial completion	Same, §4.3.6 [15]
Traceability records	**Up to 5 years after the expiry of a trade item**	Trade item expiry	Traceability guideline §3.1.2 [10]
Prescriptions, Schedules 2 to 6, at the last seller	**At least 5 years**	Date of sale	s 22A(6)(o) [5]
Specified Schedule 5 and Schedule 6 register	No period stated; **balanced quarterly, completed within 14 days**	Quarter end	s 22A(6)(n) [5]

The second half, same columns:

Record	Period	Trigger	Instrument
Pharmacovigilance records	**No period stated** — a finding, not a licence	—	Post-marketing guideline [8]
Safety surveillance, not a retention period	Until **6 months after the expiry date of the last marketed batch**	Last batch expiry	Post-marketing guideline [8]
Marketing code records and financial-transaction agreements	**At least 5 years**	—	Code v19 §§3.7.1, 3.3.13.1.2 [44]
Chemical exposure assessments and air monitoring	**Minimum 30 years**	—	Hazardous Chemical Agents Regulations 2021, reg 9(e) [37]
Noise monitoring, and audiometric and medical surveillance records including baseline audiograms	**40 years**	—	Noise-induced hearing loss regulations, reg 11(e)–(f) [38]
Employee name, occupation, time worked, remuneration	**3 years from the last entry**	—	Basic Conditions of Employment Act 75 of 1997, s 31 [46]



Figure 16.8 The retention clocks to scale, with their triggers. Three of them are event-triggered, which is why retention is a computation over operational data rather than a policy setting. A single corporate period is always wrong.

Four readings of that table.

One: three of the longest clocks are event-triggered, running from batch *expiry* [14], trade-item *expiry* [10] and *final closure or programme discontinuation* [15]. **Retention is therefore a join, not a setting**, and the join is to data held in other systems — so a records platform that cannot consume those dates cannot implement the policy.

Two: the longest clocks sit on the least glamorous data, and the obligation outlives the application by four or five times — so what is needed is an export format carrying the **audit trail** and the **dictionary version** [15].

Three: "batch records: five years" is wrong. The rule is one year after the batch's expiry *or* at least five years after certification, whichever is longer [14] [3]. For a three-year shelf life the governing clock is certification plus five; for a five-year shelf life, expiry plus one. **A computation, not a number.**

Four: where the instrument states no period, that is a finding, not a licence. Pharmacovigilance records, destruction records and import documents all lack a stated period in the documents retrieved [8][6][4]. Assign one from the longest adjacent obligation and **record the basis** — because "we chose seven years" with no reasoning is the answer that fails an audit.

And again, at the end

This chapter is architecture material, not legal advice. Every obligation in it is framed as what the record has to show, and nothing in it tells anyone whether a company complies or what a provision means in a contested case.

Three standing jobs follow, and they are the architect's rather than the lawyer's. **Re-check the version and date of every guideline citation. Hold the unresolved questions as unresolved in the model** — whether the 2026 health regulations cover your entity, whether a research code of conduct is in

force, what the ethics-committee reporting clock is — with an owner and a review date against each. And **build the six structures rather than the eleven repositories**, because the six are what every inspector in the table is asking to see.

THE CHAPTER IN TEN LINES

- This is architecture material and not legal advice; frame every obligation as what the record has to show, never as whether you comply.
- Re-checking the version and date of every regulator guideline citation is a standing job: this research found one superseded five months into a five-year validity window.
- South Africa's accountable batch-release individual is the Responsible Pharmacist, Council-registered and named on the licence with not more than one per licence, who certifies each batch and must ensure processes remain in a validated state; there is no EU-style Qualified Person.
- Do not cite SAHPRA for manufacturing-practice detail: its guideline adopts the PIC/S guide and contains no batch-record retention, no computerised-systems content, no data-integrity section and no retention periods.
- SAHPRA timelines are in working days, 360 for new chemical entities and biologicals, 250 for generics without clinical data and 200 for reliance, and reliance is not recognition.
- Registration expires at five years and is renewed six months early; Type IAIN variations are deemed implementable on silence after 47 working days, so the absence of a response is an event the system must detect.
- The Single Exit Price does not apply to the State, there were two adjustments in 2026 at 1.47 per cent and 2.88 per cent, wholesalers take no mark-up but are paid a logistics fee out of the exit price, and the dispensing fee is charged in addition to it.
- Pharmacovigilance day zero is any employee's receipt including sales representatives and contractors; SAHPRA does not require E2B revision 3, does not require a system master file, does not mandate MedDRA in the operative guideline, and does not routinely take foreign individual case reports.
- SAHPRA sets one inspection-response deadline of 30 calendar days from receipt and no critical, major or minor classification scheme.
- Health information is special personal information under section 26, a sponsor's route is section 27 and not section 32, and POPIA has no adequacy decisions so the responsible party makes the adequacy determination itself.
- The 2026 health regulations removed the legitimate interest assessment, the retention and destruction rules, the cross-border notification and the safeguards that the 2025 draft contained; anyone writing from 2025 commentary states obligations that do not exist.
- Whether a POPIA code of conduct for research is in force is the highest-value unresolved question here, because an in-force sector code removes the section 57 prior-authorisation requirement entirely.
- Destruction is governed by regulation 44, not regulation 27, and needs prior permission, a Waste Act facility, a named witness with a designation, and no municipal sewerage.
- A single corporate retention period is always wrong: the clocks run from three to forty years, three are event-triggered, and batch records are one year after batch expiry or five years after certification, whichever is longer, which is a computation.
- The same six structures answer nearly every inspector, so build those rather than eleven inspector-specific repositories.

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- **26.** Pharmacy Act 53 of 1974 (assented 9 October 1974), establishing the South African Pharmacy Council; and the Good Pharmacy Practice rules published under s 35A of that Act — identified by title and notice, not retrieved for this research https://pharmcouncil.co.za/Media/Default/Pharmacy%20Act/Pharmacy%20Act%2053%20of%201974.pdf
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- **28.** Protection of Personal Information Act 4 of 2013, s 26 — Prohibition on processing of special personal information, retrieved 24 October 2026 https://popia.co.za/section-26-prohibition-on-processing-of-special-personal-information/
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- **32.** J Coetzee — Cross-Border Data Flows and the Protection of Personal Information Act 4 of 2013, Part II: The Data Transfer Provision, [2024] PER 48 (SAFLII) (the five s 72(1) grounds; adequacy read as essentially equivalent with no body assigned the task and no counterpart to the GDPR's adequacy decisions; binding corporate rules at s 72(2); consent as a backdoor and not sustainable; the necessity grounds applying to occasional transfers; transit not a transfer; a centralised database held by a foreign group entity treatable as a transfer to an operator; s 72 not covering transfers by operators), retrieved 24 October 2026 https://www.saflii.org/za/journals/PER/2024/48.html
- **33.** ITLawCo — POPIA health data regulations 2026: What you need to know (Regulations relating to the Processing of Data Subjects' Health Information by Certain Responsible Parties, 2026; signed 27 February 2026; Gazette No. 54268, Notice No. 7198; in force 6 March 2026 with no transitional period; eight categories of responsible party and their applicable operators; Regulation 4 s 27 justification per activity; Regulation 5 safeguards over physical and electronic records and disposal; Regulation 6 gating cross-border transfer on s 72(1), with group policy or operational convenience unlikely to comply; fines up to R10 million) — secondary legal commentary, not legal advice, retrieved 24 October 2026 https://itlawco.com/popia-health-data-regulations-2026/
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- **36.** Academy of Science of South Africa — POPIA Code of Conduct for Research. A DRAFT with no effective date: clause B.3.3 reads "This Code will come into effect on [insert date once agreed with the Information Regulator]" and the text still carries placeholders. Cited for its substantive positions only: s 27(1)(d) and s 35(1)(d) as the most common authorisations; further processing via s 15(3)(e); permanent de-identification taking data outside the Act while researchers should "assume that re-identification is possible"; pseudonymisation required for high-risk research; "simply obtaining consent to 'conduct research' will not be sufficient"; research records retainable indefinitely if retained only for research while identifiable information is destroyed or de-identified as soon as possible, retrieved 24 October 2026 https://www.assaf.org.za/wp-content/uploads/2023/04/ASSAf-POPIA-Code-of-Conduct-for-Research.pdf
- **37.** Regulations for Hazardous Chemical Agents, GN R.280 in GG 44348, 29 March 2021, made under s 43 of the Occupational Health and Safety Act 85 of 1993 (risk assessment every two years; reg 9(e) exposure and air-monitoring records minimum 30 years; reg 9(g) engineering-control records at least three years; reg 9(a) medical surveillance with no stated period; reg 14A(1)(a)(iii) safety data sheet review at least every five years; the 2020/2021 short-title oddity) — reused from the Manufacturing volume research pass, 16 October 2026 https://www.gov.za/sites/default/files/gcis_document/202103/44348rg11263gon280.pdf
- **38.** Noise-Induced Hearing Loss Regulations under the Occupational Health and Safety Act 85 of 1993, reg 11(e)–(g) (noise exposure assessments and monitoring 40 years; medical surveillance records including baseline audiograms 40 years; training records while the employee remains at that workplace) — reused from the Manufacturing volume research pass, 16 October 2026
- **39.** National Environmental Management: Air Quality Act 39 of 2004, with the listed activities and minimum emission standards (GN 893 in GG 37054 in operation from 22 November 2013, amended by GN 551 in GG 38863 effective 12 June 2015) and the National Atmospheric Emission Reporting Regulations 2015 — reused from the Manufacturing volume research pass, 16 October 2026 https://iaiasa.co.za/wp-content/uploads/2024/02/NEMAQA.pdf
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- **43.** SAHPRA — Communication to Stakeholders, Issue No. HPA16-2025/26, 29 September 2025 (traceability guideline publication and the implementation roadmap: batch level 1 October 2027 for imported and 1 October 2029 for domestic products; unit level 1 October 2029 for imported and 1 October 2031 for domestic; SSCC by 1 October 2030; real-time capture language; the declaration letter on a SAHPRA template), retrieved 24 October 2026 https://www.sahpra.org.za/wp-content/uploads/2025/10/HPA16-202526-Industry-Comms-for-Traceability-Guideline-publication.pdf
- **44.** Marketing Code Authority — The South African Code of Marketing Practice for Health Products, Code and Guideline version 19, approved at the annual general meeting in June 2026 (§3.31 and §3.6.1 voluntary self-regulation; §3.1.4 the s 18C hook; §3.3.2 the law prevails on conflict; §3.3.7 and §5.2.12 inducements; §3.20 honoraria at fair market value; §3.4.1 competition law preventing the Authority prescribing a fair market value; §3.3.13 and §3.3.13.1.2 disclosure and signed financial-transaction agreements; §3.7.1 five-year retention; the

transparency chapter was headings only in the text retrieved), retrieved 24 October 2026 https://www.marketingcode.co.za/wp-content/uploads/2026/07/MCA-Code-of-Marketing-Practice-v19-2026.pdf

- **45.** Syenza News — Pharmaceutical Tender Investigation Sparks Regulatory Scrutiny in South Africa, 12 February 2026, citing Business Day of 11 February 2026 (the Competition Commission investigating three named companies following a health department complaint alleging collusive practices in state medicine tenders; the reporting does not attribute specific conduct to specific companies and the investigation is early-stage) — secondary news source, retrieved 24 October 2026 https://news.syenza.com/pharmaceutical-tender-investigation-south-africa-regulatory-scrutiny/
- **46.** SAICA retention-of-records guide (Companies Act 71 of 2008 s 24 seven years; Tax Administration Act 28 of 2011 s 29(3)(a) five years with the s 32 extension; VAT Act 89 of 1991 five years; Basic Conditions of Employment Act 75 of 1997 s 31 three years from the last entry and s 29(4) three years after termination; Employment Equity Act s 26 five years after expiry of the plan) — reused from the Manufacturing volume research pass, 16 October 2026

CHAPTER 17

Modelling pharmaceuticals in ArchiMate

The notation is general-purpose and this industry's objects are not. This chapter teaches ArchiMate 4 through the places where a pharmaceutical model and the notation do not fit — and says plainly which three of them cannot be drawn at all.

The ground rules, stated once

ArchiMate 4 gives you 42 elements across seven domains, 11 relationship types and two kinds of junction. That is the whole vocabulary. It is enough for most of what a pharmaceutical company does, and it is not enough for about ten things, which is what makes this chapter worth your time.

Three rules before any of that.

The element is `Role`. There is no layer-prefixed variant of it in ArchiMate 4 — no business-prefixed Role, no application-prefixed Service. The layer-specific names that older versions carried were collapsed, so the element that holds *Responsible Pharmacist*, *Qualified Person*, *principal investigator* and *marketing authorisation holder* is called `Role` and nothing else. Give it a layer prefix in a review and you will be corrected, and the correction matters more than it sounds: half the arguments about pharmaceutical models are arguments about whether a thing is a person, a job or a legal accountability, and a notation that has one word for the job makes you answer the question.

There is no Constraint element. Express a constraint as a `Requirement`. "Only the named Responsible Pharmacist may certify a batch" is a Requirement [9]. "No reporting view may join trial data to sales data" is a Requirement. "A GxP record must remain readable for one year after batch expiry or five years after certification, whichever is longer" is a Requirement [5]. If you find yourself reaching for a Constraint element because the thing is a prohibition rather than an obligation, write the prohibition as the obligation it implies and move on.

Never reproduce a specification. This applies more widely in pharma than anywhere else in this series. The ArchiMate specification is licensed. So are the CDISC foundational standards, the pharmacopoeial monographs, the GAMP 5 guide — a 404-page paid ISPE publication, not a regulation [10] — and the two dictionaries the industry cannot work without: MedDRA, a subscription product released twice a year [3], and WHODrug Global, also by subscription, which the US regulator nonetheless mandates for studies starting after 15 March 2019 [2]. Describe them, point at them, draw your own picture. Do not transcribe a domain list, an artefact taxonomy, a term hierarchy or a category table into a client deliverable.

> This chapter teaches the notation through this industry's problems rather than in the abstract. It assumes you can already model. If you want the element set, read the specification [1].

Where the notation holds, so we can stop talking about it

Most of a pharmaceutical company models cleanly, and it is worth saying so before the complaints start.

The organisation maps. `Business Actor` for the legal entity and the department, `Role` for the job, `Collaboration` for the cross-functional body that signs things off — a safety management team, a change control board, a bid committee. `Facility` for a site and `Location` for an address, which is a distinction pharma needs because a site in the regulatory model is a licensed capability at an address and not a building [9].

Capability, value stream, process, function and service map. So does the application layer: `Application Component` for a system, `Application Interface` for what it exposes, `Data Object` for what it holds. So does technology: `Node`, `Device`, `System Software`, `Communication Network`, `Path`, `Artifact`. Motivation maps well enough that the motivation view is the single most useful thing you will draw in a regulated estate, because in pharma the drivers are external, dated and citable.

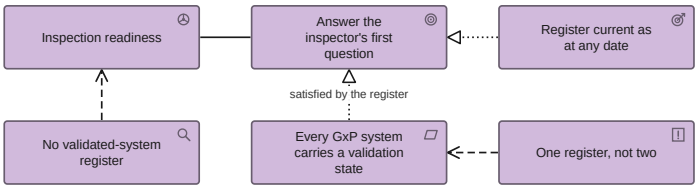


Figure 17.1 The motivation view is the one that pays in a regulated estate, because the drivers are external, dated and citable. There is no Constraint element — the constraint is the Requirement.

What follows is the rest. Ten or so objects where the fit is poor, bad, or absent — and for three of them the honest answer is that they do not belong in an ArchiMate model at all. Saying so is not a failure of the notation. It is the difference between a model a regulatory affairs manager trusts and a drawing they politely ignore.

The four identities of a product

A product is four things with different cardinality, and the data chapter works this through in full. For modelling purposes the four are:

- the **molecule** — a substance, identified by structure internally and by an International Nonproprietary Name and a substance identifier externally;
- the **formulation** — substance plus strength plus dose form plus route, which is country-independent;

- the **registration** — that formulation as authorised to one holder in one country under one name, with its own number, its own approved text, its own approved sites and its own conditions;
- the **SKU** — a pack of a registered product in a specific container and count, which is what carries a barcode and a price.

One molecule becomes several formulations, each of which becomes many registrations, each of which becomes several SKUs. The counts are not small: a mid-size South African generics company with 71 molecules can hold 184 registrations and sell 498 SKUs.

ArchiMate has one `Product` element and no way to express that cardinality chain. You have three bad options and one tolerable one.

The tolerable one is to stop trying to make `Product` mean all four. Use `Product` for the two levels that behave like a marketable offering — the formulation and the SKU — and `Business Object` for the two that are facts about a regulated entity: the substance and the registration. Then draw the chain with Aggregation, and put the dated map between internal and external identifiers on the diagram as a `Data Object`, because that map is the asset.

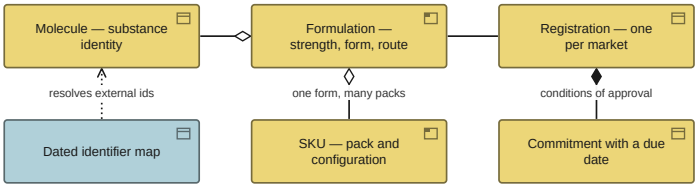


Figure 17.2 The best available approximation of the four identities of a product. `Product` carries the two levels that behave like an offering; `Business Object` carries the two that are facts about a regulated entity. The cardinality is not expressible here — it lives in the conceptual data model.

> The approximation above loses the cardinality. The diagram says a SKU belongs to a formulation; it does not say there are 498 of them, or that three SKUs on this formulation exist only because a tender wanted a three-month patient-ready pack. **The real model lives in the conceptual data model, drawn in crow's-foot notation where cardinality is expressible, and the ArchiMate view points at it.** An architect who tries to carry cardinality in an ArchiMate view ends up with labels like "1..n" on Aggregation relationships, which is not what Aggregation means and will not survive contact with anyone who knows the notation.

Compound and substance

A compound in the research sense is a chemical entity under investigation, identified by structure. A substance in the regulatory sense is the same chemistry as a regulated entity, identified by definition and curated by an authority. Neither identifier can do the other's job: no regulator will accept a key only you can resolve, and no registry can use a regulatory identifier that does not yet exist for a novel entity.

So which is it: a Business Object, or a Data Object?

`Business Object` is right when you are modelling what the business means — "substance" as a thing regulatory affairs, research and supply all talk about and disagree about. `Data Object` is right when you are modelling the realisation — the row in the compound registry, with its normalisation rule version and its uniqueness check.

What is lost either way is the same thing, and it is the interesting part. **Structure-based identity depends on the normalisation rules, so changing the rules changes identity retrospectively.** Tighten stereo-perception three years in and structures that were one parent become two. A registry that cannot say which rule version decided an identity cannot defend the identity. Neither `Business Object` nor `Data Object` has anywhere to put "identity under rule set v3, asserted 2023-08-14". It is a property, and properties are where the pharmaceutical content of an ArchiMate model actually lives.

Study and protocol — and the version the notation cannot carry

A study is an investigational undertaking. A protocol is the controlled document that defines it. They are not the same object and a model that uses one box for both will mislead everyone who reads it.

Model the study as a `Business Object` and the protocol as a `Business Object` too, or as a `Deliverable` if you are modelling the work that produces it. Then try to express the thing that matters, and watch it fail.

The protocol does not describe the study. It defines the structures the data conforms to. An amendment can change the schedule of assessments, the visit windows, the eligibility criteria, the endpoints, the dosing rule, the arms, a laboratory method, the number of countries, and the consent form. Each of those invalidates something already built: expected-visit rows for visits not yet reached, out-of-window deviation detection, the statistical analysis plan and its derivations, the randomisation schedule, the kit design, every enrolled subject's consent state [7].

ArchiMate has no native way to express a version, and no way at all to express an amendment that invalidates downstream structures. There is no versioning relationship. `Specialization` is not it. Nesting a "v3.0" box inside a "Protocol" box says something about composition, not about succession. The best you can do is honest and ugly:

- one `Business Object` for the protocol, with its versions as properties and a documented naming convention;
- `Association` to the structures it defines — schedule of assessments, visit definition, arm definition — each as its own `Business Object`, so that at least the invalidation surface is visible;
- a `Requirement` saying that every collected data point carries the protocol version in force when it was collected;
- and a line in the view's notes saying that the per-country, per-site amendment approval fan-out is held in the clinical data model and not here.

> The question a Good Clinical Practice inspector asks is: *was this subject's visit conducted under a protocol version that was approved at that site on that date?* That question is answered from a four-way dated table, not from a diagram. Know which of your artefacts answers it, and say so on the view.

Subject — a type, never an instance

A trial subject is the most sensitive record in the company. Identity, health condition, often genetic data, sometimes HIV status, held for at least 10 years after final closure under South African Good Clinical Practice [7], shared with a contract research organisation, a central laboratory, a statistics vendor and at least one foreign regulator.

Modelling it is easy and the rule around it is absolute.

Model `Business Object` "Trial subject" — the type. Never an instance. No "Subject 1042". No specimen diagram populated with a plausible-looking patient. No screenshot of a case report form with a date of birth on it, even an invented one, in a deliverable that will be emailed.

This is not a modelling preference. A view that could identify a subject is a privacy problem. Health data is special personal information under section 26 of the Protection of Personal Information Act [8], and pseudonymised trial data in a sponsor's hands is still personal information, because the sponsor plus the site can re-identify. South Africa's draft research code of conduct tells researchers to assume re-identification is possible [8]. A view with a site identifier, a visit date and a rare condition on it is identifying even with no name anywhere near it.

The practical rules, which belong in your modelling standard and not in a conversation:

- entity types only, in every clinical view;
- no sample data in any view that leaves the repository;
- the re-identification key appears in the model as a `Business Object` held by the **site**, with a `Requirement` that the sponsor does not receive it — modelling the custody decision is useful, modelling the key's contents is not;
- and if a client asks for an example view with real subject data in it because it will be easier to review, the answer is no, with the reason given once.

Batch and genealogy — and the one that is not drawable

A batch is a defined quantity of material expected to be homogeneous. It is the unit of traceability, quality and liability in this industry, and everything physical is batched: the active ingredient, each excipient, the printed carton, the bulk tablet, the finished pack.

The type models fine. `Business Object` "Batch", with `Association` to material, site, specification version and disposition. Good.

Genealogy does not model at all.

A genealogy is the recorded graph of what was actually consumed into what: *this* lot of active ingredient, in *this* quantity, at *this* time, on *this* equipment, into *this* batch, which was filled into *these* finished batches. It is a directed graph of **instances** with quantities on the edges, built from the executed batch records rather than from the bill of material. ArchiMate models types. There is no instance layer. You cannot draw a genealogy in ArchiMate, and an attempt to do it produces a picture of four boxes that is true of one batch and false of the next.

> The distinction worth more than any diagram here: **where-used tells you which products *could* contain a lot; only genealogy tells you which ones *do*.** The first answer is a product line. The second answer is four batches. The difference between those two answers is the difference between a recall that costs tens of millions of rand and one that does not.

So name the type in the ArchiMate model, state the relationship as an `Association` labelled "genealogy", and point at the data model and the export format that carry it. The export format matters because batch documentation must be retained for one year after batch expiry or five years after certification, whichever is longer [5] — a computation, not a number — and that obligation outlives several generations of manufacturing execution system.

Adverse event and the coded term

An adverse event is any untoward medical occurrence in a patient given a medicinal product, whether or not it is causally related. A case is one patient's safety story. A case version is one report of that case at a point in time. An event is one reaction within that version. A coded term is one MedDRA term for that event, under one MedDRA version.

The first three levels model acceptably as nested Business Objects. The fourth defeats the notation completely.

A term coded under one MedDRA version does not necessarily mean the same thing under another. MedDRA releases twice yearly, in March and September — version 29.0 in March 2026, 29.1 later the same year [3]. Each release adds terms, and also demotes them, merges them and changes which system organ class is primary for a term. So a preferred term can stop existing as a preferred term, and every case coded to it now rolls up somewhere else; a trend line drawn across that boundary has a step in it that is pure artefact. Standardised groupings of terms change by release too, which is the specific mechanism by which **a safety signal can appear or disappear because of a dictionary upgrade and nothing else**.

The notation cannot carry any of that. There is no element for a version, no relationship for supersession, and no way to say that the identity of a coded value includes the version of the code system. What you can do:

- `Business Object` "Coded term", with a `Requirement`: every coded value carries dictionary, dictionary version, coded date, coder and method;
- `Data Object` "MedDRA release", modelled as a held, versioned, archived asset rather than as a call to a vendor service — because a report produced in 2023 must be reproducible in 2029, and that needs the 2023 dictionary to still be loadable;

- an `Assessment` on the view recording which version the estate is actually on, dated, because that statement is true for at most six months.

Re-coding history is a project with medical review in it, not a refresh. Say that on the view, in words, because the diagram will not say it for you.

Regulatory submission against dossier state

This one is worth working through slowly, because it is the clearest case in the industry of two objects that look like one.

A **submission** is an event. It is an assembled package transmitted to one authority on one date, numbered as a sequence, and **immutable once sent** — nothing in a sent sequence is ever altered. South Africa runs on the electronic Common Technical Document at version 3.2.2, portal only, with whole-sequence rejection on a structural error; new applications start at sequence 0001 and a baseline built from non-eCTD content starts at 0000 [4].

A **dossier** is a state. It is what the regulator currently holds for one application, which no single transmission contains, and which changes as each sequence's lifecycle operations are applied: new, replace, append, delete. Nothing is ever actually deleted — "delete" removes a document from the current view while the sequence that carried it stays intact forever. **The current view of a dossier is a computed projection over an append-only log of sequences.**

In ArchiMate the submission is comfortable: `Deliverable` for the thing a work package produces, or `Artifact` if you are modelling the physical transmitted package with its backbone file. Either is defensible. `Deliverable` reads better to a business audience; `Artifact` is more honest about what is on the wire.

The dossier is not comfortable. The closest element is `Plateau` — a relatively stable state of the architecture at a point in time — and the fit is poor in three specific ways:

1. `Plateau` is an implementation-and-migration element meant for *your* architecture's states, not for an external party's record of your product. Using it means borrowing the element for a different job and saying so.
2. `Plateau` has no notion of being computed from a log. It is asserted, not derived.
3. There is one dossier per application per authority, and a mid-size generics company has 184 of them. A notation element that invites one box per state does not scale to that.

What I do, and I will defend it only as the least bad option: **model the dossier as a `Business Object` with a documented state property set, and use `Plateau` only when you are drawing a specific product's regulatory position across a transition** — before a variation, after the variation is implementable, after it is implemented in every market. Then add the `Requirement` that carries the actual design: the current view must be recomputable from the sequence history. That one requirement is worth more than the diagram, because it is the sentence that stops someone building a dossier-state table with an update statement in it.

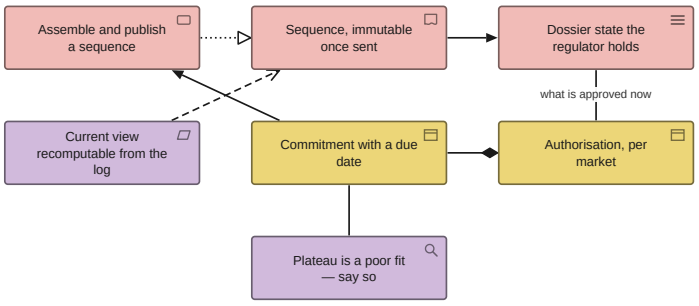


Figure 17.3 The submission-and-commitment view, one of the three an architect has to invent. The submission is an event and models well; the dossier is a maintained state and does not. The Requirement carries the design the picture cannot: nothing is ever deleted, only superseded, and the current view is a projection over the sequence log.

Marketing authorisation — a contract with no element

A marketing authorisation is a legal instrument: a permission granted by one authority to one holder to place one product on one market, subject to conditions, with a validity period and a variation history. In South Africa it is a certificate of registration, valid five years, renewable with the application in six months before expiry [4].

ArchiMate has no Contract element. It had one in earlier versions, in the business layer, as a specialisation of a business object. It does not have one now. So use `Business Object`.

What is lost is the contractual semantics, and in this industry that loss is not cosmetic:

- **Parties.** A contract has at least two. A `Business Object` has none, so the holder, the authority and the named sites become `Association` relationships with no indication that they are parties rather than references.
- **Validity.** A contract has a term. Registration is valid five years, renewal is due at six months before expiry, and lapsing is the state nobody models because it happens by inaction. Validity becomes a property.
- **Conditions.** An authorisation carries commitments — a post-approval study, a stability commitment, a specification to be tightened. Each is a dated obligation with an owner and an evidence requirement, and each is its own `Business Object` associated with the authorisation, because one authorisation can carry a dozen with different owners in different functions.
- **The fan-out.** Changing an active-ingredient supplier is one manufacturing decision and n variations across n markets, approved on n dates. Between the first and last approval the product is legally different in different markets while being physically identical, and the batch release decision has to

know that. The notation shows none of this; a `Requirement` can at least state that release is decided per market.

One South African specific worth a sentence on the view: a Type IAIN variation — a minor one notified with a waiting period — is **deemed implementable** if the authority gives no opinion within 47 working days [6]. The absence of a response is an event the system must detect and act on, which inverts how an approval workflow is normally built. If you model variations as a process waiting for an inbound approval, you have built something that waits forever.

Specification, test, result — and the fact that `Value` is not a number

Five objects, each with its own grain: specification, test, method, result, certificate of analysis. They model adequately as Business Objects, with the method's own version as a property, and the result at determination grain rather than reportable-value grain, because an out-of-specification investigation is about the determinations.

Then you hit the wall that the manufacturing volume in this series hit, and it bites harder here.

ArchiMate has no metric element, and `Value` is not a number. `Value` is the worth of something to a stakeholder — "faster release", "assured supply". It is not a place to put 98.5%, or 45 mg, or "not less than 90.0 per cent and not more than 110.0 per cent of the labelled amount". So:

- a **specification limit** has no home in the notation;
- a **yield** has no home, and in pharma the point is sharper than elsewhere, because a yield outside the validated range is a deviation *even when it is higher* — continuous-improvement framing imported from general manufacturing is wrong here;
- a **pharmacovigilance compliance rate** has no home, and its definition is the whole argument: obligations due in the period that were submitted on time, divided by obligations due in the period — not cases, obligations, because one case generates many obligations in many markets and a case-driven denominator silently drops the five-day ones triggered by another regulator's action [11].

The answer is the same one the manufacturing volume reached, and it is a separate artefact: a **measure definition register**. One row per measure, with its name, its formula, its grain, its numerator and denominator definitions, its owner, its source systems, and the version of the definition with an effective date. The ArchiMate model holds a `Business Object` "Measure definition" and Association relationships to the processes that produce the inputs. The numbers live where numbers live.

> If a client asks you to put target values on an ArchiMate view — "release in two days", "99% compliance" — put them in the caption or in the register, not in a `Value` element. The moment a number is inside a notation element, someone will build a report that reads it, and the definition behind it will not travel.

The validated system itself

This is the pharmaceutical object with no sibling precedent anywhere else in this series, and it is the one most architects do not model at all. The consequence is specific: they cannot answer *which of our systems hold GxP records?* — which is the first question an inspector asks, and also the first question asked by a privacy assessment, a disaster-recovery review and any cloud business case. The absence is expensive four times over.

A validated system is not a product. **Validation is a property of a configured instance of software used for a defined purpose in a defined process by a defined organisation**, held in a validated state for the system's operational life, under change control. One row is one instance in one environment for one defined use. The productive instance and the test instance are different rows. An interface between two validated systems is itself in scope and is frequently nobody's row.

Now try to model it. What you need to express is:

Fact about the system	What it is, in notation terms
GxP classification — GMP, GCP, GVP, GLP, GDP, or none	a property
GxP record types held	a property, or an Association set
Validation state	a property with a lifecycle
Date of last validation, periodic review due date	properties, and dated obligations
Part 11 and Annex 11 applicability	a property, and a judgement
System owner, custodian, process owners	Role associations
Audit trail present, reviewable, reviewed by whom	properties
Hosting and data location	properties, and a cross-border input
Retention obligation on the data it holds	a computation

Nine rows, and only one of them is an element. **Validation state is not an element, a relationship or a view. It is a property, and ArchiMate's property mechanism is a key-value bag with no lifecycle of its own.**

So the honest answer, which you should give plainly:

- model the system as `Node` or `Application Component` as you would anywhere else;
- carry a **documented property set** on it, with the nine facts above, governed as attributes in the repository rather than as text in a description field;
- draw a `Grouping` around the validated boundary, so that scope is at least visible;

- add Requirement elements for the obligations that attach — change assessed before implementation, audit trail reviewed before release, records readable after decommissioning;
- and say, in writing, that **the notation was not designed for this and the register is the real artefact.**

The register should be the same register as the application portfolio with these columns added. The near-universal state is two registers — an architecture portfolio and a quality-owned spreadsheet — and they disagree. That disagreement is a finding waiting to be written.

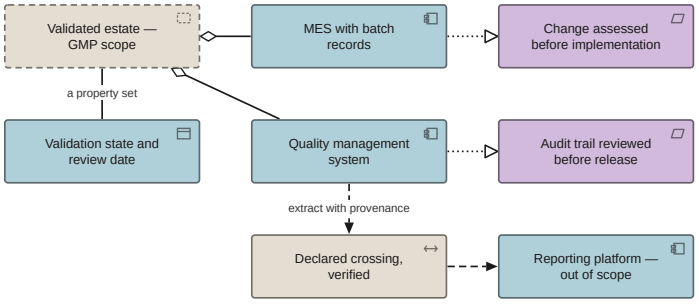


Figure 17.4 The validated-boundary view. The Grouping makes scope visible and enforces nothing. Validation state is a property, not an element, which is why the caption has to say that the register is the real artefact.

> No enterprise architecture tool was found, on public pages at the research retrieval date of 24 October 2026, to ship a pharmaceutical metamodel extension with Compound, Study, Batch, Regulatory Submission, GxP Computerised System or Adverse Event as first-class object types [12]. An exhaustive negative cannot be proved from public pages, so read that as "none found", not as "none exists". The practical consequence either way is that every pharmaceutical repository is a local construction on a generic base, and the same question goes to every vendor: can I hold validation state and GxP classification as governed attributes with their own lifecycle, and can I report on applications by validation state? Ask it in those words.

The validated boundary and the clinical/commercial firewall

Two boundaries, both drawn with `Grouping`, both carrying a warning.

The validated boundary. Draw a line around the validated estate. Data crossing it outward loses its guarantees; data crossing inward imports risk. The boundary is not a network boundary and not a security zone — it is a boundary of evidential quality. The trap is directional: a data warehouse that was out of scope while it produced management reports comes **into** scope the day a trend chart from it is used to justify a specification. The test is two questions. Does anyone make a GxP decision from this output? Is the output cited in a regulated record? Either yes, and the pipeline needs verification of correctness, change control and traceability to source.

The pattern that works, modelled: a `Path` for each declared crossing, an `Artifact` or `Application Interface` for the interface specification — which is itself a validation deliverable naming every field, type, unit and transformation — and a `Requirement` that every record which crossed carries a provenance marker naming source system, extraction time and originating user. The third is cheap at design time and impossible to retrofit.

The clinical/commercial firewall. Trial data about identifiable subjects must not reach commercial functions. Safety case data must not either. There are four separate reasons and they have four different remedies: blinding, personal health information, promotional bias, and the independence of the pharmacovigilance function. Which means **there is no single enterprise data platform holding clinical, safety and commercial data with uniform access, and any design that proposes one is wrong in this industry.** What works is domain platforms with narrow, purpose-built, governed flows — the exposure denominator for signal detection being the canonical one, since the safety datamart genuinely needs sales volumes to compute a rate.

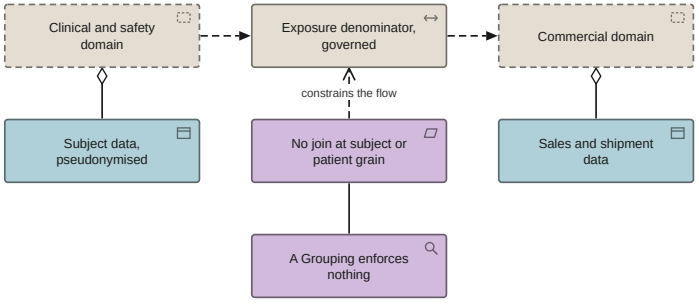


Figure 17.5 The clinical/commercial firewall view. One governed flow is permitted — the exposure denominator a safety datamart needs to compute a rate. The Assessment on the view is there to stop anyone mistaking the Grouping for a control.

> **A `Grouping` has no semantics and enforces nothing.** It is a box with a dashed border. It does not restrict a relationship, propagate a property, or stop anyone building the join. Four things enforce the firewall: access control, report design, organisational separation, and an audit trail showing who could see what. The Grouping's job is to make the boundary visible so that a reviewer notices when a proposed interface crosses it. Treating it as a control is the commonest way an architect produces a diagram that is both correct and useless.

The GMP plant where it meets the enterprise resource planning system

The plant end reuses the manufacturing conventions and should be kept lighter here than that volume kept it, because in a pharmaceutical engagement the plant is rarely the problem you were hired for.

`Node` for the server and for the logical platform. `Device` for physical equipment with a processor in it — the dispensing terminal, the serialisation unit on the packaging line, the environmental monitoring sensor set. `System Software` for the batch execution engine, the historian software, the control system runtime. `Communication Network` for the plant control network, and `Path` for the logical exchange between the plant and the enterprise systems above it.

Two pharmaceutical specifics worth the ink.

First, **the execution path cannot cross a wide-area link.** The batch engine, the control layer, the historian primary and the line-level serialisation system are on the plant floor because the plant must keep running when everything above it is gone. Model that as a `Requirement` on the view, not as a preference in a document.

Second, **the batch record must capture the measured value, not a historian-interpolated or compressed one.** A reviewer who cannot reconcile the batch record against the historian has a data-integrity question, not a reporting inconvenience. That is a `Requirement` too, and it belongs on the interface, which is where it will otherwise be forgotten.

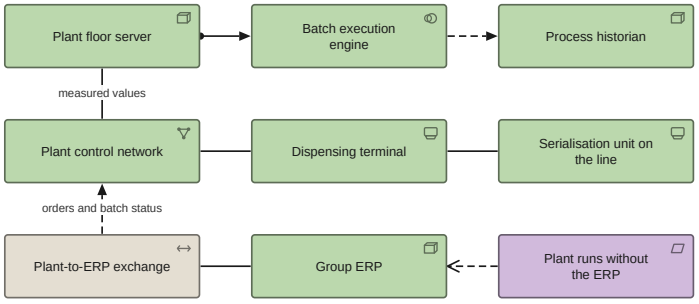


Figure 17.6 The GMP plant where it meets the enterprise systems, kept deliberately light. The Requirement is the load-bearing element: the plant runs when everything above it is gone.

What has no ArchiMate home at all

Collecting the honest answers in one place, because a client will ask and the answer should be immediate rather than discovered.

The object	What carries it instead
Batch genealogy — a graph of instances	The batch-record data model, plus an export format that outlives the system
A dictionary version on a coded value	The safety database, per value, plus a held dictionary archive
A specification limit, a yield, a compliance rate	The measure definition register — *Value* is not a number
Validation state, GxP class, review date, Part 11 applicability	Properties on the portfolio — the validated-system register
A protocol version, and the amendment fan-out	The clinical data model: protocol version, country, site, date
A subject instance	Nowhere, deliberately — types only in every view
A retention clock	A computation over batch expiry and study closure dates
A tender award's litigation state	The contract register — an award is not master data

Figure 17.7 The objects with no ArchiMate home, and what carries them instead. Say this list out loud in the first fortnight of an engagement. It is the fastest way to be believed.

Three of those are worth repeating as a rule. **A genealogy is a graph of instances, so it is not drawable in ArchiMate. A dictionary version is part of a value's identity, and the notation cannot carry it. Validation state, periodic review date, GxP classification and Part 11 applicability are properties,**

not elements. Say all three early in an engagement. An architect who says them on day three is credible; one who is still drawing round them in month three is not.

The viewpoints that actually get used

Fourteen standard viewpoints earn their place in a pharmaceutical engagement, plus three you have to invent because no standard viewpoint covers them. For each: what it is for, what may be in it, what feeds it, and — the column everyone skips — what it cannot show.

Viewpoint	What it is for	What it cannot show
Organisation	Who is accountable, and for which licence	That the Responsible Pharmacist is named on the licence and there is only one per licence [9]
Motivation	The external, dated, citable drivers	Which guideline version is operative today
Goal realisation	From an executive sentence to a requirement	Whether the requirement is testable
Capability map	What the group can do, heat-mapped	That one capability is run twice, differently, in two divisions
Value stream	Where value accumulates, stage by stage	The decade between spend and revenue
Layered	One slice from role to infrastructure	Anything at instance level
Process co-operation	Which process hands to which	The hand-off itself — see the BPMN exception
Information structure	The entity set and its relationships	Cardinality, keys, versions, dictionary versions
Application co-operation	Which system talks to which, and about what	Whether the interface is in anybody's validation scope
Application usage	Which role uses which system for what	Whether that role's access is segregated from approval
Technology	Nodes, devices, networks, what runs where	Latency, and what survives a link failure
Implementation and deployment	Which work package touches which component	Revalidation effort
Migration	Plateaus and the moves between them	Exit criteria, unless you write them as Requirements
Requirements realisation	Which element satisfies which obligation	Evidence — the obligation register holds that

The three you invent:

The validated-boundary view. A `Grouping` for the validated estate with its members, their GxP classification and validation state as displayed properties, every declared crossing as a `Path`, and the obligations as Requirement elements. Fed by the validated-system register and the interface inventory. It cannot show whether the periodic reviews are actually current — that is a report, run from the register, and the view should say where to find it.

The clinical/commercial firewall view. Two Grouping elements, the governed flows between them as Path elements, and a `Requirement` per flow stating what may and may not be joined. Fed by the application inventory, the access model and the reporting-layer inventory. It cannot show that the firewall is holding. Only an access review can.

The submission-and-commitment view. `Work Package` to `Deliverable` for the sequence, the dossier state alongside it, the authorisation and its commitments as Business Objects, and Requirement elements for immutability and recomputability. Fed by the regulatory information management system, the approval letters and — be honest about this — the spreadsheet that is the real commitment register in most companies. It cannot show the due dates. Those belong in one obligation register with the licence renewals, the periodic review dates, the corrective-action effectiveness checks and the pharmacovigilance clocks, because they are one family and five trackers is how a commitment gets missed.

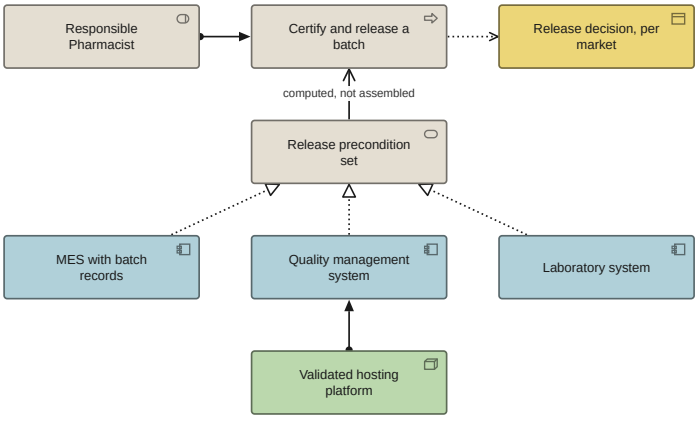


Figure 17.8 A layered slice through batch release. Eight elements, one role, one decision. A view that needed 40 elements would be two views or a report.

The BPMN exception, stated once

Everything architectural is ArchiMate. **Level-2 process models are BPMN 2.0**, for one reason: ArchiMate has no swimlanes, and in a pharmaceutical process the hand-off *is* the point. Batch release is a join across at least six systems and four functions, and the four days it takes are spent in the gaps between them, not inside any one step. A notation that cannot show who is holding the work cannot show where the days go.

Keep BPMN to a maximum of five lanes and seven columns. Beyond that it does not fit a page, and a process model nobody can read on paper is a process model nobody validates. Where a loop would run backwards into the same lane and overlay the forward path, replace it with a second end event and name it — "held pending deviation closure" is more useful than an arrow curving back anyway.

The discipline that keeps a view useful

Four habits, in descending order of how often they are broken.

Element count. A view stops communicating somewhere between 15 and 25 elements, and in a regulated estate the lower end is right, because every element on the view invites a question with a regulatory consequence. If a view needs 40 elements it is two views, or it is a report. The test: can the person whose job the view describes find themselves on it in under 10 seconds?

Naming conventions. Agree them in week one, write them down, and apply them to the repository rather than to the view. In pharma the conventions that pay for themselves are: roles named as the industry names the job, never as a job title the company invented; systems named by what they do with the product name in brackets, because the product name will change and the function will not; processes named verb-first; and objects named with their qualifier, because "product" unqualified means four different things and "authorised product (per market)" means one.

Keeping the view and the repository in step. A view is a query over the repository, not a drawing. The failure is always the same: a view is exported to a slide, the slide is edited, the edited slide becomes the version people quote, and six months later the repository and the circulating picture disagree. The fix is unglamorous — regenerate rather than edit, and date every exported view in its own footer.

And the pharmaceutical one, which has no analogue in the other volumes. If a view is itself a controlled document, changing it is a change-control event. An architecture view attached to a validation package, a Site Master File, a submission module or a standard operating procedure is inside the quality system. Changing it needs a change record, an assessment, approval and a document version. That is correct and it is also slow, and it is the reason **most pharmaceutical architecture views should deliberately not be controlled documents.**

Draw the line explicitly and early. A small number of views are controlled because a regulated document needs them — a data-flow diagram in a validation summary report, a system diagram in a Site Master File. Everything else is working architecture: dated, versioned in the repository, circulated, and explicitly marked "not a controlled document". Get that marking wrong in the other direction and you have done something worse than slowing yourself down: you have put an uncontrolled drawing inside a regulated record, where an inspector will find it and ask which version was operative.

> Decide, for every view, whether it is inside the quality system or outside it. Mark it. The default should be outside, and the exceptions should be few and named.

THE CHAPTER IN TEN LINES

- ArchiMate 4 has 42 elements, 11 relationships and junctions; the element is Role, with no layer-prefixed variant of it.
- There is no Constraint element: write the constraint as a Requirement, including the prohibitions.
- A product is four things with different cardinality, and the notation has one Product element, so the real model is the conceptual data model and the view points at it.
- A protocol amendment invalidates downstream structures, and the notation has no way to express a version or a supersession.
- A subject is drawn as a type, never an instance; a view that could identify a subject is a privacy problem, not a style choice.
- A batch genealogy is a graph of instances, so it is not drawable in ArchiMate at all - name the type and point at the data model.
- A coded term's identity includes the dictionary version, and the notation cannot carry it.
- A submission is an event and models as a Deliverable; a dossier is a maintained state and Plateau fits it poorly.
- Validation state, GxP classification, periodic review date and Part 11 applicability are properties, so the validated-system register is the real artefact.
- A Grouping has no semantics and enforces nothing; it makes a boundary visible so a reviewer notices a crossing.
- Level-2 process models are BPMN 2.0, because ArchiMate has no swimlanes and in this industry the hand-off is the point.
- If a view is itself a controlled document, changing it is a change-control event, which is why most pharmaceutical architecture views should deliberately not be controlled documents.

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CHAPTER 18

Veldtkruin Group, end to end

One invented South African group, two archetypes under one roof, and an engagement run from the brief to the hand-over. Everything the book has taught, applied once.

> **Nothing about Veldtkruin Group is real.** It is an invented South African pharmaceutical group, built as a composite so that two archetypes can be shown under one roof. Every person, site, system and number in this chapter is invented and internally consistent, and none of it describes any company. The regulatory facts around it are real and sourced; the company is not.

1. What the architect walks into

The brief arrives from Riaan Opperman, the group chief information officer, as a page and a half. It says that Veldtkruin Group needs an application strategy; that the estate has grown by acquisition and nobody has a current picture of it; that the board has asked for a quality dashboard; and that the group would like to consolidate onto fewer platforms over three years. There is a budget number and a start date.

Thandeka Mabuza, the group chief executive, names three symptoms in the first meeting.

One. The last Good Manufacturing Practice inspection at the Isando site took three weeks to prepare for and produced findings she found embarrassing rather than serious. She wants to know why the company cannot just be inspection-ready.

Two. Batch release takes too long. She has been told four days and she does not believe that the four days are doing anything.

Three. The group cannot answer questions. When the Brackenfell site's second shift was being considered, nobody could say how many products were approved to be made there, and the answer took eleven days to assemble.

Three things turn out to be true instead, and all three are visible inside the first fortnight.

The first is that there is no register of validated systems, so the company cannot say which of its systems hold records that a regulator cares about. That is not an IT housekeeping gap. It is the first question an inspector asks, and it is also the first question asked by a privacy assessment, a disaster-recovery review and any business case for moving a system.

The second is that the group is two different companies and the estate has been designed as though it were one. Veldtkruin Pharmaceuticals is a generics manufacturer whose economics are portfolio-scale against a gazetted price. Khanyisa Bio-Development is a contract developer and manufacturer whose economics are fee-for-service against a campaign, and which holds client data it does not own. A design that fits one fails at the other, in both directions.

The third is that the group's dated obligations live in five places, none of which is a register. Post-approval commitments are in a spreadsheet on Dr Nolwazi Dube's drive. Licence renewal dates are in two site managers' calendars. Periodic review dates for validated systems are nowhere, because there is no register to hold them. Corrective-action effectiveness checks are a field on a closed record. Pharmacovigilance clocks are in Sentinel and are the only ones computed properly.

None of those three is in the brief. All three outrank it.

How the fortnight was spent

Not on the org chart. Nine interviews, in this order, each of about 90 minutes.

Anita Govender, validation lead, first — because she knows which systems are actually in a validated state as opposed to which were validated once, and that distinction is the whole estate. She produced the quality spreadsheet, apologised for it, and named four systems she believed were out of validation. **Dr Nolwazi Dube, group head of regulatory affairs**, second, for the registration portfolio: which products are registered where, on what conditions, with what expiry, and which planned business change is blocked on a variation. She also produced the commitment spreadsheet, and the request that mattered was: *show me the spreadsheet*, not *show me the system*.

Marieke Visser, quality assurance, for the deviation and change-control backlog, which is the best single proxy for whether a site is in control. **Faizel Adams**, Responsible Pharmacist at Isando, for what cannot be changed — the most useful answer of the fortnight, because he said the four days were made of two things, a reconciliation and a queue, and that only one of them was his.

Then **Dr Sipho Ndlovu** on the safety database and its dictionary, **Lerato Phiri** on the trials and the client agreements, the Isando supply planner on batch sizes fixed by registration and active-ingredient lead times measured in months, the market access manager on how the Single Exit Price is actually administered and what the logistics-fee arrangement is with each wholesaler, and **Riaan Opperman** last rather than first, because by then the questions were worth his time.

Johan Kriel, the group's lead enterprise architect, sat in all nine. That was deliberate. He is the person who will still be there in eighteen months, and the register is his to own.

2. The group as it is

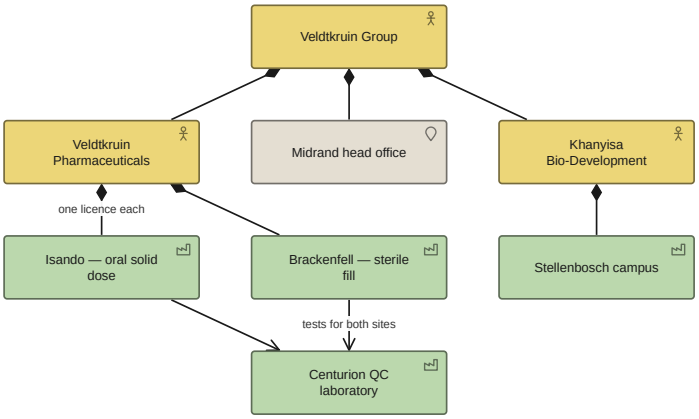


Figure 18.1 Veldtkruin Group as it is constituted — two divisions, two archetypes, one shared quality control laboratory. Invented. The shared laboratory is the structure that makes the two divisions' problems collide.

Veldtkruin Pharmaceuticals. Revenue about R2.1 billion, of which roughly 62% is private-market sales at the Single Exit Price, 34% is state and institutional tender, and 4% is export into the region. 71 molecules, **184 registrations** held with the regulator, and **498 active SKUs** once pack sizes and configurations are counted — 27 of those exist only because a tender wanted a three-month patient-ready pack with its own artwork. Eleven registrations are held and not marketed, because reinstatement is slower than retention; each one still consumes a stability slot, an annual retention fee and an annual product quality review.

Two licensed sites. **Isando**, oral solid dose, tablets and capsules plus secondary packaging, running about 1,510 batches a year. **Brackenfell**, sterile fill and finish, small-volume injectables for hospital and tender supply, running about 300. **About 1,810 batches a year across the group**, roughly seven a working day. A separate quality control laboratory at **Centurion** tests for both sites. Head office is in **Midrand**.

All 64 active ingredients are imported, from 31 suppliers, and four of them are single-sourced with no qualified alternate. Three molecules carry the public tender exposure.

Faizel Adams is the Responsible Pharmacist at Isando, named on the licence — and there may be only one named per licence [3]. Marieke Visser runs group quality assurance. Their two names between them cover most of what cannot be changed quickly.

Khanyisa Bio-Development. Revenue about R265 million on the **Stellenbosch** campus: contract manufacturing and packaging, development and analytical services, and clinical trial supply. It holds **no marketing authorisations of its own**. It is a named manufacturing site on **38 clients' registrations**, and it runs a small clinical-stage portfolio of two compounds of its own. About **310 batches a year** across about 95 distinct products, so the average product runs three batches a year and the site is changeover-dominated: changeover and cleaning consume roughly 38% of available suite time, which is the most important economic fact about the business.

Lerato Phiri runs clinical operations. Twenty-two studies are active at any time across about 85 South African investigator sites.

	Veldtkruin Pharma	Khanyisa Bio-Dev
Archetype	Generics manufacturer	Contract developer and manufacturer
Revenue	About R2.1bn	About R265m
Earns	Margin on product, at a gazetted price	Fee for service, against a campaign
Cost object	The product, the SKU, the batch	The campaign and the project
Registrations	184 of its own; 498 SKUs	None of its own; named on 38 clients'
Batches a year	About 1,810, few products	About 310, about 95 products
Binding constraint	Price ceiling and QA capacity	Changeover — about 38% of suite time
Whose data	Its own	Clients', held but not owned
A GMP finding is	Its own problem	38 registration holders' problem at once

Figure 18.2 Two archetypes under one roof. A design that fits the left column fails in the right one. The row that matters most is the last: who the finding belongs to.

The point of the pairing. One GMP finding at Khanyisa is simultaneously a problem for 38 registration holders, which is why its quality agreements are its most important commercial documents and why clients audit it about nine times a year. One GMP finding at Isando is a problem for Veldtkruin alone. The same event has a completely different blast radius, and therefore a completely different architecture around it: Khanyisa needs multi-tenant segregation inside one quality system and one laboratory system, with 38 different change-approval rights enforced in workflow rather than described in a procedure. Veldtkruin needs portfolio-scale reporting against a price it cannot change.

The estate. Kelpfield is the group enterprise resource planning system, one instance, both divisions. Protea is the manufacturing execution system with electronic batch records at Isando and nowhere else; Brackenfell runs on paper batch records with a validated document store, Silverleaf, holding the scanned copies. Aloe is the laboratory system at Centurion and Stellenbosch. Rooibos is the quality management system, one instance, both divisions, which is where the multi-tenancy problem bites. Fynbos is the regulatory information management system, under-populated. Sentinel is the safety database. Marula is the electronic trial master file. Sable is the clinical trial management system and the electronic data capture system at Khanyisa, hosted offshore by its vendor. Driftwood is the serialisation platform, commissioned for the imported-product dates and not yet carrying serials.

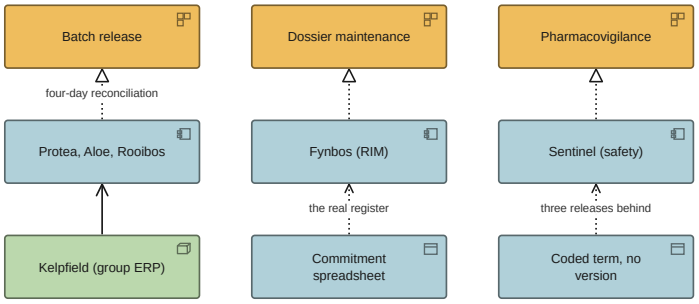


Figure 18.3 The as-is, at the level that changes the conversation. Two of the nine boxes are the ones people actually rely on: a spreadsheet standing in for a registration and commitment register, and a coded term with no dictionary version on it.

Two observations about that picture that are worth more than the picture.

The enterprise resource planning system is minting product identity. The first material master row for every imported active ingredient was created by a procurement clerk, which means Kelpfield and Fynbos have different answers to "what is this substance" and there is no rule about which wins. Regulatory affairs has quietly built its own register in a spreadsheet, and that spreadsheet is the one people actually use.

The interfaces are in nobody's validation scope. There are, on Riaan's own count, 31 of them between GxP systems. Not one has an interface specification that is a validation deliverable naming every field, its unit and its transformation. Four of them are a nightly file drop onto a share.

Where it runs

The hosting split is not where a cloud-first architect would draw it, and most of it is correct.

Already cloud, uncontroversially: Sable, Marula and Fynbos, because their vendors are software-as-a-service only and there is no on-premises option to argue about. Rooibos too.

On premises, and correctly so: Protea, the control layer at both sites, the historian primary, the serialisation line units and the building and environmental monitoring systems. The reason is physical rather than cultural. **The execution path cannot cross a wide-area link**, and the plant has to keep running when everything above it is gone. An in-process gate that depends on a link is a line stoppage waiting for a fibre cut.

Where the argument actually is: the middle. Aloe, Sentinel, Silverleaf, the stability system and the chromatography data system at Centurion. These are not execution-path systems, so the physical argument does not apply; they are validated GxP systems, so the validation argument does. The position worth defending is that cloud is appropriate for a validated system where five things are contracted and evidenced — a documented shared-responsibility split; change notification with a deferral or verification window; access to the provider's audit evidence; a tested data-export and exit mechanism; and, for personal health data, a lawful cross-border basis under section 72 [6]. **Where any of the five is missing, the system is not in a maintainable validated state in that hosting model**, and that sentence is the one most often fudged.

Veldtkruin has none of the five in writing for Sable, which is Finding 8 arriving from a second direction. It has four of the five for Rooibos, the missing one being the tested export.

At the data level

Three sentences that between them explain most of the estate's behaviour.

The word "product" means four different things inside the group and nobody qualifies it. Commercial means a brand family. Kelpfield means a material and a pack. Fynbos means a registration in a country. Sentinel means a substance. When the board asks how many products Veldtkruin has, four functions give four answers and all four are correct — and the gap between the largest and the smallest is a factor of more than ten.

Almost nothing in the estate is stored as a dated, scoped, authored assertion, and almost everything should be. "This batch is released" is not a flag. It is a person, on a date, on evidence listed, under an authority they held on that date, in respect of a named market. Rooibos holds all five for a release. Kelpfield holds a status field, and Kelpfield is the system the rest of the business reads.

Coded data does not carry the version of its code system. That is Finding 3 for Sentinel, and the same defect in a quieter form sits in Aloe, where a specification limit is held without the pharmacopoeial edition it came from, and in the stability schedule, where a result is held without the version of the analytical method that produced it. A 60-month stability study crosses a method revision, and a shelf-life extrapolation drawn through an unrecorded methodological step change is an extrapolation through a discontinuity.

3. The findings

Eight, each stated so that somebody inside the company could check it in a day.

Finding	The check
1. No validated-system register, so no answer to which systems hold GxP records	Reconcile the two lists: 212 and 61 entries, 44 overlap, 17 on neither
2. Trial master file incomplete and filed late	Re-scope the expectation set: 94% becomes 71%; median filing gap 94 days
3. Dictionary three releases behind; historical coding inconsistent	Ask the loaded version, then when data was last re-coded
4. Commitments in a spreadsheet; obligations not producible on demand	41 rows against 184 registrations; 12 letters yielded 7 missing
5. Reporting layer joins trial and sales data	Read the dashboard's query and its grain
6. Release waits on a four-day manual reconciliation	Measure the lead time; count the batch-key formats
7. Tender revenue modelled as secure	Ask where the award's validity and litigation state are held
8. Cross-border health data with no adequacy determination	Ask for the determination per recipient country, dated and reasoned

Figure 18.4 The eight findings on one page, each with the check that confirms it. A finding that cannot be verified by somebody who already works there is an opinion.

Finding 1 — there is no register of validated systems, so nobody can answer which systems hold GxP records. Two lists exist. Riaan's application inventory has 212 entries and no validation or GxP column. Marieke's quality spreadsheet has 61 entries, a validation date for 44 of them, and no system owner. The overlap is 44 entries. Seventeen systems named in site procedures appear on neither list, including the spreadsheet that calculates one assay result at Centurion and an Access database that holds the stability schedule. **Evidence to check: ask for both lists and reconcile them. It takes an afternoon.**

Finding 2 — the trial master file at Khanyisa would not survive a Good Clinical Practice inspection. Marula reports 94% complete, against an expectation set that was never configured for the designs actually running. Re-scoped against the trials as designed, completeness is 71%. The median gap between a document's own date and the date it was filed is 94 days; for 19 documents it is over a year. There is no documented not-applicable, so every absent artefact is an open question. **Completeness and filing timeliness are findings in their own right**, independent of how well the trials were actually run, because a sponsor cannot demonstrate oversight of what it cannot show. Essential documents must be kept at least 10 years after final closure [4].

Finding 3 — Sentinel's dictionary version has not been upgraded in three releases, so historical coding is inconsistent. MedDRA releases twice a year; version 29.1 followed 29.0 of March 2026 [5]. Sentinel codes against a version three releases back. Cases coded under different versions sit in one database with no version on the value, and the company has been coding forward-only without having designed for it. The consequence is precise: a count of events under a preferred term is not comparable across the version boundary, because terms get demoted, merged and reassigned to a different system organ class at each release — so **a signal can appear or disappear because of a dictionary upgrade and nothing else**. Dr Sipho Ndlovu knows this and has said so twice in writing. **Evidence to check: ask which version is loaded, then ask when historical data was last re-coded and what the medical review of the changed mappings concluded.**

Finding 4 — post-approval commitments exist only in a spreadsheet, so the obligations cannot be produced on demand. The spreadsheet has 41 rows for 184 registrations. Twelve approval letters sampled at random produced seven commitments that are not in it, two of them with due dates expressed relative to an event — first commercial batch — that happens in a system regulatory affairs does not watch. A missed commitment is a breach of an approval condition, and the remedies available run from a question to suspension. **Evidence to check: ask for a single list of every dated external obligation with its owner and its evidence. Almost nobody can produce one, and the inability is the mechanism by which commitments are missed.**

Finding 5 — the clinical/commercial firewall is breached by a reporting layer. A dashboard built eight months ago for Khanyisa's own clinical-stage portfolio joins subject-level data extracted from Sable with Kelpfield sales data, to show "patient reach by province". It was built by two people with good intentions and a reporting tool. It joins trial data to commercial data at a grain that identifies, and it does so for a division that holds client data it does not own. That is a consent problem and a competition-law problem before it is an architecture problem, and it is a **finding, not a feature**. The remedy is not one remedy: blinding, personal health information, promotional bias and the independence of the safety function are four separate reasons with four different controls.

Finding 6 — batch release depends on a manual reconciliation that takes four days. Protea, Aloe and Rooibos are reconciled by hand before Faizel Adams sees a release pack. The batch key is formatted three different ways across the three systems plus Kelpfield. Audit-trail review across the contributing systems is in a procedure, is not resourced, and is not in the release path at all — which matters, because the direction of travel in the computerised-systems guidance is towards that review being completed before release [7]. Thandeka was right that the four days are not doing anything; she was wrong that they are therefore removable without doing the work.

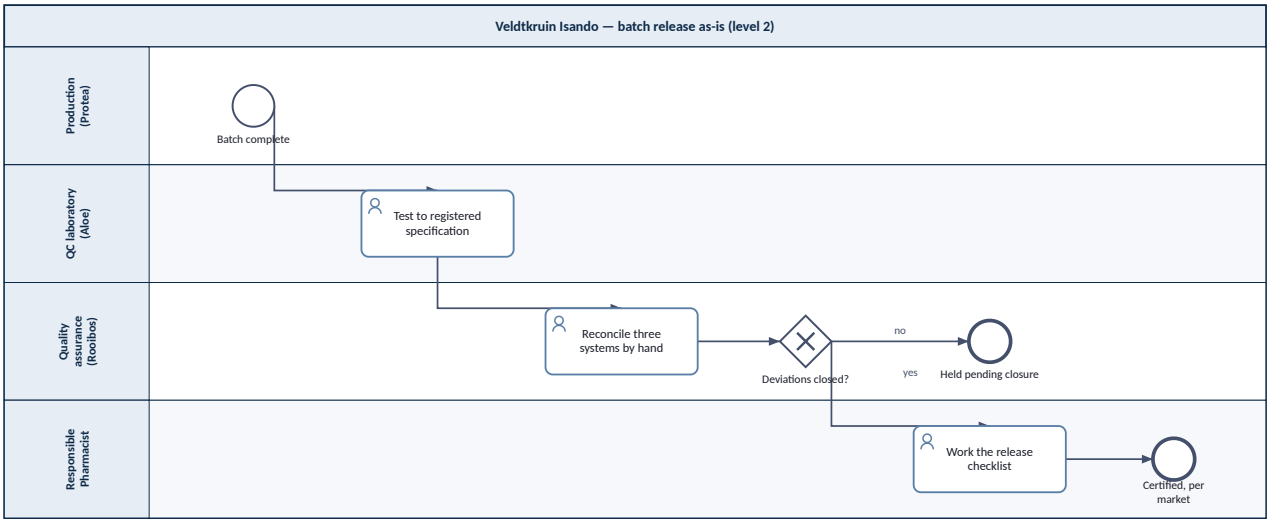


Figure 18.5 Batch release as it runs today at Isando, drawn in BPMN because the hand-off is the point. The four days are spent in the reconciliation step and in the queue in front of it, not in any of the tests.

Finding 7 — tender revenue is modelled as secure, and a tender award is a contract with a litigation state. Three molecules carry the exposure and the plan treats the award as master data. On **22 October 2026** the High Court set aside the one-month and three-month pack awards of a R15.5 billion antiretroviral tender as unlawful and unconstitutional, ordered the award redone within 120 days, and made the set-aside effective 90 days after the new awards, with current suppliers continuing to deliver until then [1]. Separately, on **17 September 2026** the Constitutional Court struck down the entire Public Procurement Act 28 of 2024, without suspending the invalidity [2] — and **which framework now governs health-product tenders is genuinely unresolved.** The plant is buying active ingredient on a six-to-twelve-month lead time against an award that has a validity, a challenge window and a litigation status. Model it that way or the plant holds inventory against an award that no longer exists.

Finding 8 — cross-border trial data moves with no adequacy determination on file. Sable is hosted outside South Africa by its vendor, and client sponsors receive data under their own agreements. Section 72 of the Protection of Personal Information Act gates that transfer, and the structural difference from European practice is the one that matters: **POPIA has no adequacy decisions.** It assigns no body the task of assessing adequacy, so the determination falls to the responsible party itself — which makes that determination the predicate for a section 57(1)(d) obligation carrying a criminal offence [6]. There is no determination on file for any recipient country. Pseudonymised trial data in a sponsor's hands is still personal information, and the draft research code tells researchers to assume re-identification is possible [6].

Two things that were not findings

Worth stating, because an engagement's credibility rests as much on what the architect declines to call a defect.

Brackenfell runs paper batch records with scanned copies in Silverleaf, and that is legitimate. Hybrid paper-and-electronic systems are allowed where the record is declared, the link between the paper and the electronic copy is verifiable, and a readable copy can be produced. The site has all three. A recommendation to put electronic batch records at Brackenfell because Isando has them would be a preference dressed as a finding, and it would cost a revalidation.

The eleven dormant registrations are a business decision, not a data defect. They are maintained because reinstating a lapsed registration is slower than retaining one, and each one's annual cost — a retention fee, a stability slot, an annual review — is known. The architecture question is not "why are these here" but "can the company produce the fully loaded cost to serve per SKU, including regulatory and quality overhead, so that the pruning decision is made on evidence". It cannot. That is a gap in the costing model, and it belongs in the measure definition register rather than in the findings list.

> Every one of those eight is checkable from inside the company in under a day, by someone who already works there. That is the test of a finding. A finding that depends on the architect's judgement alone is an opinion, and an opinion will not survive the first executive who dislikes it.

4. What it costs to be wrong

This section is where engagements lose their credibility, because the temptation is to produce a number. There is no published benchmark for most of what matters here — not for audit-trail review effort, not for release cycle time, not for quality-system metrics — and a consultant's industry figure in a pharmaceutical board pack is usually an invention with a footnote.

So: four costs, in the terms the executive committee already uses, and the person inside the business who can calculate each one.

The cost that is already in the company's own numbers. Veldtkruin produces **184 annual product quality reviews a year**, scheduled by product anniversary at about 15 a month. Each takes a quality officer four to six working days to assemble, from the laboratory system, the enterprise system, the quality system and three spreadsheets. That is roughly **1,000 quality person-days a year**, or about five full-time people doing data assembly. Khanyisa's equivalent is worse in kind rather than in size: it produces reviews **to each client's template under each quality agreement**, so 38 formats across 95 products, at about two and a half days each. **Marieke Visser can confirm both numbers from her own resourcing model this week.** The review is an

inspection document — an inspector asks for the current ones early and reads them for whether the company notices its own trends — and its annual frequency comes from the adopted international guide, not from the local guideline [8].

The cost of release lead time. Every manual reconciliation, nightly interface, hand-typed result and queue-based review sits between manufacture and availability. Four days on 1,810 batches is a working-capital and a service-level number, and for the sterile products at Brackenfell with shorter shelf lives it is also a shelf-life number, because two of Veldtkruin's largest private customers contractually require 12 months' remaining shelf life at delivery. **The chief financial officer can price the working capital; Marieke can measure the actual lead time rather than the asserted one. Nobody should price it from a benchmark.**

The cost of an obligation missed. This one has no rand figure and should not be given one. The remedies available to a regulator for a breached approval condition run from a question to suspension of a registration, and a suspension removes a product's revenue entirely until it is lifted. **Dr Nolwazi Dube can quantify the exposure the honest way: list the registrations by revenue, and state which of them carry commitments whose evidence the company cannot currently produce.** That list is the business case and it does not need a multiplier.

The cost of the firewall breach. Also not a rand figure. The exposure is an administrative fine of up to R10 million under the health-information regulations that came into force on 6 March 2026 [9], the possibility of criminal liability on the section 57 route [6], and — more likely to arrive first — a client of Khanyisa discovering that its data was joined to a commercial dataset, which is a quality-agreement breach and a terminated contract. **Lerato Phiri can name the affected clients and read their agreements. That is a one-day task and it is more useful than a risk score.**

And the cost that hides inside a habitual variance. What the plant made, what quality released and what finance recognised are three different numbers about the same physical events, maintained by three functions, and they differ for legitimate reasons: production counts batches not yet released and measures them in bulk units; quality counts disposition decisions **per market**, so a batch can be released for one market and not another; finance values the same units, moves at different trigger points and writes off rejects on its own timetable. The three-way reconciliation at Veldtkruin is not defined anywhere. It is performed, informally, by one financial manager who knows where the differences usually are — quality control samples consumed and not relieved from inventory, partly released batches counted in full by one system and in part by another, and rejected batches sitting in inventory at value until a destruction certificate arrives. **A real loss — diversion, unrecorded destruction, a mis-stated yield — would hide comfortably inside that habitual variance, and nobody would see it.** That is a cost with no figure attached and a named owner who can put one on it.

> Where the research has no number, say there is none and name the person who can calculate it. An architect who produces a sourced "there is no public benchmark, and here is who inside your business can work it out" is more credible at the second meeting than one who produced a figure at the first.

5. The target and the transition

The target, in six sentences and six paragraphs

Product identity is owned where it is asserted. Regulatory affairs owns the product hierarchy from substance to authorised product to pack; Kelpfield holds the material and pack extensions keyed to it and **mints nothing**; the dated map between internal and external identifiers is a maintained asset rather than a reference-data chore. The test of whether this has really happened is whether a change to a material's shelf-life field is routed through change control with the previous value retained. Today it is not.

Dated obligations are one model with typed subclasses. Commitments, renewals, periodic reviews, effectiveness checks and reporting clocks are one family, held once, reported as a horizon.

The release decision is computed. The preconditions are evaluated by the architecture and presented as a state to a named person who holds a legal accountability; the batch key is one key; audit-trail review is a resourced step with a latency inside the path rather than a procedure outside it.

The two divisions get two shapes on shared platforms. One quality system, with client segregation and 38 change-approval rights enforced in workflow; one laboratory system with the same property; reporting that is per-division by construction, so a Khanyisa client's data cannot appear in a Veldtkruin view by accident.

The regulatory interface is a bounded context with a published contract. The submission gateway, the safety mailbox and the traceability master-data submission each have a structure, a conformance rule set and a rejection behaviour — South African sequences reject whole on a structural error — so they are designed as integration boundaries with their own validation and their own test harness, not as incidental outputs of internal systems.

Nothing joins clinical to commercial except one governed flow. The exposure denominator the safety datamart needs, built to a documented specification, with a named approver and an access model that an audit can test.

Three transition states, each with written exit criteria, because a transition state without exit criteria is a wish with a date on it.

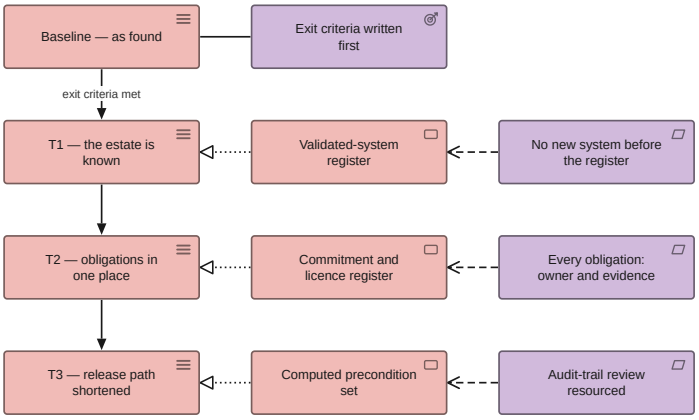


Figure 18.6 Three transition states, each gated by written exit criteria. The Requirements on the right are the gates. The year allows two of the three, and saying so in month one is the point.

Transition 1 — the estate is known. The validated-system register exists, as columns on the application portfolio rather than as a separate quality-owned spreadsheet. Exit criteria, written before the work starts:

1. Every system in the estate has a GxP classification, the record types it holds, a validation state, a named business owner and a named technical custodian, a periodic review due date, its hosting location, and the retention obligation on its data.
2. The two existing lists are reconciled to zero unexplained entries, and the 17 systems on neither list are either on the register or formally declared out of scope, with the declaration signed.
3. Interfaces between GxP systems are rows on the register, not gaps between rows.
4. Quality assurance and information technology both agree, in writing, that this is the single register.

Transition 2 — every dated obligation is in one register. Commitments, licence renewals, registration renewals, periodic review dates, corrective-action effectiveness checks, stability timepoints and the pharmacovigilance clocks are one family of objects — dated promises to external parties with evidence requirements — and the correct move is to notice the family rather than build five trackers. Exit criteria:

1. The company can produce, inside one working day, every dated external obligation falling due in the next 90 days, with its owner, its due-date basis and the evidence that will discharge it.
2. No obligation is owned by a vacant role.
3. Obligations whose due date is relative to an event — first commercial batch, last marketed batch expiry — have that event's source system named, and the event is actually observed there.
4. Licence renewals appear **180 days before expiry**, because that is when the obligation really falls due [3].

Transition 3 — the release path is shortened and the firewall is enforced. Exit criteria:

1. The release precondition set is **computed and presented as a state**, not assembled as a folder: all preconditions met, or these three are not. Faizel Adams sees a state.
2. The batch key is normalised across Protea, Aloe, Rooibos and Kelpfield. This is the single highest-value integration task in the estate and it is unglamorous enough that nobody has done it.
3. Audit-trail review is risk-targeted, resourced, performed by someone not directly involved, and inside the release path with a measured latency.
4. An access review demonstrates that no commercial role can read data at subject or patient grain, and the exposure denominator the safety datamart needs is rebuilt as one governed flow with a documented specification and a named approver.

The decisions that must be taken before any of it. Five, and they are decisions rather than analysis:

- **Who owns the register.** One register with two owners is the state the company is already in. The answer that works is one register, one accountable owner, with quality assurance holding the approval right over the GxP columns.
- **Whether Marula stays.** Not yet. **Do not propose replacing a validated system before the register exists**, because the business case cannot be written without knowing what records it holds and what the retention obligation on them is.
- **Whether Sable's hosting changes.** The adequacy determination is required either way, so do that first; the hosting decision is downstream of it, not a substitute for it.
- **Whether Rooibos stays one instance for both divisions.** Probably yes, with 38 client change-approval rights enforced in workflow — but that is a real design decision with a real cost and it should be made explicitly rather than inherited.
- **Whether Protea is upgraded this year.** It is two major versions behind. **A GxP system upgrade is a regulatory project with an information technology component**, not the reverse, and Anita Govender should size the revalidation before anyone commits to a date.

6. The roadmap and what the year allows

Three calendars bound the delivery year and none of them is negotiable.

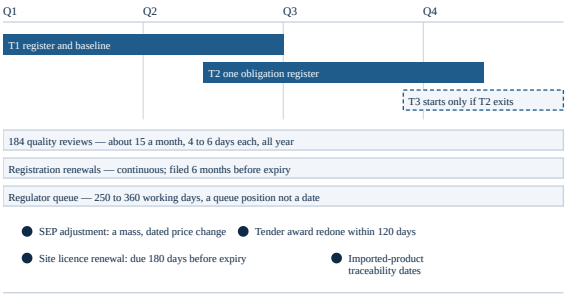


Figure 18.7 The delivery year against the calendars that bound it. The quality review cycle runs all year at about 15 reviews a month, which is why quality capacity rather than budget decides what the year holds.

The regulatory commitment calendar. Renewals, periodic safety reports, variation commitments, post-authorisation study milestones and licence renewals, each with a date and a legal consequence. Registration is valid five years and the renewal application is due six months before expiry; the renewal programme is phased to 2035 and failure to submit a complete renewal may result in expiration, after which a new registration application is required [10]. For 184 registrations that is a continuous flow, not an annual event.

The regulator's queue, as a planning input rather than a complaint. Review timelines are expressed in **working days to finalisation**: 360 for new chemical entities, biologicals and biosimilars, 250 for generics without clinical data, and 200 for the reliance pathways [11]. 360 working days is roughly 17 to 18 calendar months of regulator time before any sponsor response time. And clearing a backlog did not fix throughput: in 2024/25 the authority **received 2,691 generic applications and finalised 568** [11] — inflow exceeding finalisation by roughly five to one on that stream. **A launch plan that assumes a timeline is assuming a queue position.** Veldtkruin's three pending generic registrations are not a schedule input the architecture can influence; they are an external dependency to be modelled with a distribution rather than a date.

The annual product quality review cycle. Fifteen reviews a month, four to six quality person-days each, running all year. That is the binding constraint on the whole programme, and the arithmetic is worth doing out loud: the people who would validate a new system, write the user requirements, review the audit trails and approve the change records are the same people who are assembling those reviews. **Quality assurance capacity, not budget, is what the year actually allows.**

Two scheduled events sit on top. The Single Exit Price adjustment is a **mass data-change event across every pack with a hard effective date**, and 2026 had two of them — the ordinary 1.47% in January and an extraordinary second adjustment of **2.88% effective 1 October 2026** [12]. Treat it as this sector's equivalent of a tax-rate change. And the traceability programme has dated milestones in which **imported products are staged two years ahead of domestic ones** — batch-level identification on primary and secondary packaging from **1 October 2027** for imported product and **1 October 2029** for domestic, with unit-level serialisation later still [13]. That ordering is counter-intuitive and it changes which of Veldtkruin's products Driftwood has to cover first.

One sequencing consequence is worth drawing out, because it is the kind of thing that only appears when the calendars are laid over each other. Driftwood's scope has been planned around Veldtkruin's own manufactured product, which is the later date. The earlier date applies to imported product — and Veldtkruin imports finished injectables for over-labelling, which makes those the first SKUs in scope. **The traceability programme's first deliverable is therefore about the products the project plan currently treats as out of scope.** Nobody had noticed, because the two facts live in two different functions' documents.

The honest arithmetic. A delivery year of about 44 usable weeks, bounded by a quality function already committed to 1,000 person-days of review assembly, inside a regulatory calendar with non-movable dates, with an upgrade that is a regulatory project and a tender whose award may be redone within 120 days. **That year holds two of the three transition states, not three.** Saying so in month one is the single most valuable thing in the engagement, because the alternative is saying it in month nine.

7. What the architect hands over

At the end of the first quarter, seven artefacts and one argument.

1. **The validated-system register**, populated, with the two old lists retired and the declarations of out-of-scope signed. This is the deliverable that made all the others possible, because building it is how the estate was actually learned.
2. **The obligation register**, with every dated external obligation, its owner, its due-date basis and its evidence — and the 90-day view that proves it works.
3. **The baseline architecture**, counted rather than asserted: 212 applications resolved to the real number, 31 interfaces documented, GxP scope stated, hosting and data location per system.
4. **The eight findings**, each with the evidence a named person inside the company verified, and each with a remedy owner.
5. **Three transition states with written exit criteria**, and the five decisions, with a recommendation on each and the consequence of deferring it.
6. **The measure definition register**, started: release lead time, review assembly days, obligation ageing, interface inventory, and audit-trail review latency — each with its formula, its grain and its owner, so that when the dashboard is eventually built it reports something defined.
7. **A short modelling standard**: entity types only in clinical views, no subject-level data in any view, the naming conventions, and the rule about which views are controlled documents and which deliberately are not.

The argument is the one that reframed the brief: **the group is two companies, the first deliverable is a register rather than a dashboard, and the year allows two transition states.**

What was deliberately not done, and the reasons given in writing:

- **No dashboard.** The board asked for one in month one. What it got in month one was a list of twelve measures with no agreed definitions, and a recommendation to define them first. A dashboard over undefined measures makes a quality system look healthy while it repeats the same deviations.
- **No platform consolidation proposal.** Three vendors had already been approached before the engagement started. Consolidation cannot be specified before the register exists, and a consolidation that moves GxP records is a validation and migration programme in which the value **and the meaning** of every migrated field has to be verified.
- **No target-state diagram beyond the transition states.** A five-year target architecture for an estate whose GxP scope was unknown in month one would have been a drawing.
- **No change to Sentinel.** The dictionary upgrade and the historical re-coding is a project with medical review in it, owned by pharmacovigilance, sized by Dr Ndlovu. The architect's contribution was to establish that the version is a first-class attribute of every coded value, and to stop the reporting team building trend charts across the version boundary in the meantime.
- **No opinion on National Health Insurance.** It is judicially suspended rather than being implemented, and the architect said in writing that no investment in this plan is sequenced on an assumed commencement date. It is a scenario, not a roadmap item.

THE CHAPTER IN TEN LINES

- Nothing about Veldtkruin Group is real; the regulatory facts around it are sourced and the company is not.
- The brief asked for a dashboard and a consolidation; the three things actually true were a missing validated-system register, two archetypes under one roof, and dated obligations scattered across five places.
- A generics manufacturer and a contract developer are different businesses, and a GMP finding at the contract site is 38 registration holders' problem at once.
- Eight findings, each checkable in a day by somebody who already works there - that is the test of a finding.
- Where there is no published benchmark, say so and name the person inside the business who can calculate the number.
- The cost of a missed commitment is stated as a list of registrations by revenue whose evidence cannot be produced, not as a rand figure with a multiplier.
- Three transition states, each with exit criteria written before the work starts, and five decisions that come first.
- Three calendars bound the year: the commitment calendar, the regulator's queue, and the annual product quality review cycle.
- Quality assurance capacity rather than budget decides what a delivery year holds in a regulated estate.
- What was deliberately not done matters as much as what was: no dashboard, no consolidation before the register, no target state beyond the transition states.

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CHAPTER 19

Your first ninety days

The chapter to come back to. What to find out in week one, what to count before you model anything, what to say when you do not know, and the six things not to do.

Before week one: what to read, and what to put down

Four documents, in this order, and they are usually all obtainable before you start.

The Site Master File, for each licensed site. It is the document a site gives an inspector to describe itself: the licence and its scope, the activities performed, the premises, the equipment, the quality system, the outsourced activities. It is the most accurate single description of a pharmaceutical site that exists, it is maintained because it has to be, and most architects have never heard of it.

The last inspection report and the company's response. Findings tell you what the regulator thought; the response tells you what the company committed to, and the response is due within 30 calendar days of receipt with extensions only by prior written grant. Then ask the question nobody asks: **was the effectiveness of those corrections ever checked?** The regulator's own response guideline does not require an effectiveness check, so if the company does one it is because it wants to — and if it does not, the same finding will recur.

The quality manual and the validation master plan. The second names the systems the company believes are in scope, which gives you a first version of the register before you have spoken to anyone.

The registration register — and then ask for the spreadsheet, because the spreadsheet is usually the authoritative one. What you are looking for in it is not the product list. It is three columns that are usually missing: the expiry date of each registration, the conditions attached to it, and the markets each pack is actually approved for. Those three turn a product list into a constraint set, and the constraint set is what the rest of the engagement runs against.

What to put down: the previous consultant's target-state diagram, the vendor platform decks, and any document whose title contains the word "transformation". None of them will tell you which systems hold GxP records, and all of them will give you an opinion before you have a fact.

Week one: find out what is validated

Not the org chart. Not the application inventory. **The validated-system register.**

A validated system is a computerised system whose fitness for its regulated purpose has been documented, tested against specified requirements, approved, and is **maintained** under change control for its operational life. The register is the list of them, with — at minimum — the GxP classification of each, the record types it holds, its validation state, a named business owner and technical custodian, the date of last validation, the periodic review due date, its hosting location, and the retention obligation on the data inside it.

If the register exists, read it and then test it: pick three systems, ask for the validation summary report and the last periodic review, and see what comes back. If it does not exist, **that is your first deliverable**, and building it is how you learn the estate. You will have to talk to every function, open every site's procedure set, find the spreadsheet that calculates an assay result and the database that holds the stability schedule, and decide — with quality assurance, not alone — which of them hold records a regulator cares about. There is no faster way to understand a pharmaceutical company, and there is no deliverable with a better ratio of effort to standing.

The validation lead is the single most useful first interview. Not the chief information officer, not the head of quality, not the architect you are replacing. The validation lead knows which systems are *actually in a validated state* as opposed to which were *validated once*, and that distinction is the estate. They will also tell you, in the first twenty minutes, what a proposed change would cost in revalidation — which is the number every later conversation depends on.

#	Who	What only they can give you
1	Validation lead	Which systems are in a validated state, and what a change would cost to revalidate
2	Regulatory affairs manager	What is registered where, on what conditions, with what expiry — and the spreadsheet
3	Quality assurance head	The deviation and change-control backlog — the best proxy for control
4	Responsible Pharmacist	What cannot be changed, and what the release days are actually made of
5	Head of pharmacovigilance	The clocks, the intake channels, and the dictionary version
6	Clinical data manager or ops head	Where every data defect in the trial estate is, and what the agreements say
7	Supply planner	Batch sizes fixed by registration, ingredient lead times, what a tender does
8	Market access and pricing	How the exit price is administered, and the logistics-fee arrangements
9	Chief information officer	The money, the contracts, and what they have already promised the board

Figure 19.1 The interview order for a first pharmaceutical engagement. The validation lead first, and the chief information officer last rather than first — by then the questions are worth their time.

> **Validation is a maintained state, not a phase.** A system validated in 2021 whose periodic review lapsed in 2023 and which has had eleven configuration changes since is not a validated system. It is a system with a validation history. Learning to hear the difference in how someone answers is the most transferable skill in this chapter.

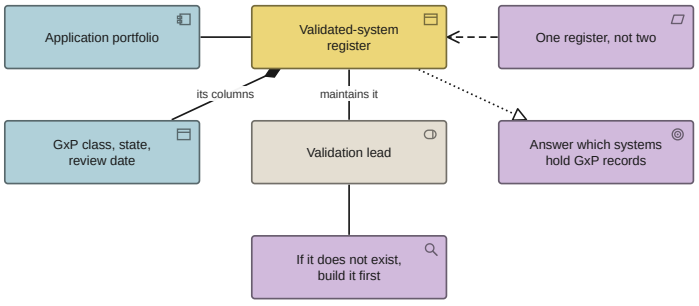


Figure 19.2 The validated-system register, as an object. The columns are properties rather than elements, which is why the register and not the view is the deliverable. Note the Requirement: one register, with quality holding the approval right over the GxP columns.

Two things the register is not. It is not a quality-owned spreadsheet parallel to the architecture portfolio — that is the near-universal state, and the two always disagree. And it is not a list of applications. **A productive instance and a test instance are different rows**, because the validated state applies to the productive one; a multi-tenant hosted instance is one row from your side with a dependency on the vendor's own quality system; and **an interface between two validated systems is itself in scope and is frequently nobody's row.**

The checks that take an afternoon

Thirteen questions. Each can be answered in a few hours by someone who already works there. Between them they change what the engagement is about, and most of them have been answered wrongly in the brief you were given.

The check	What 'no' means
1 Register of validated systems?	The inspector's first question has no answer
2 Periodic reviews current?	Validated once, not maintained
3 Dictionary version current and re-coded?	Aggregate safety counts are not comparable
4 Commitment list on demand, with dates?	An approval condition will be breached by inaction
5 Trial file complete and filed on time?	A finding in its own right, either way
6 Audit-trail review planned and resourced?	Having a trail is not reviewing it
7 Interfaces in anybody's validation scope?	Units, timestamps and attribution are lost unseen
8 Any reporting join of trial and sales data?	A consent and competition-law breach
9 Adequacy determination per country, on file?	The predicate for a criminal-offence-bearing duty
10 Retention by clock, not by single period?	Clocks from 3 to 40 years, three event-triggered
11 Measure definition register?	The dashboard will report undefined numbers
12 Guideline versions re-checked, with dates?	One was superseded inside its own validity window
13 Responsible Pharmacist and licence dates?	Renewal falls due 180 days before expiry

Figure 19.3 The thirteen afternoon checks. Each is answerable in a few hours by someone who already works there. The right-hand column is what the answer 'no' means.

1. Is there a register of validated systems, with GxP classification and validation state? If not, **you cannot answer the first question an inspector asks** — which of your systems hold GxP records. That absence is also why a privacy assessment, a disaster-recovery review and any cloud business case will each start from scratch.

2. Which systems hold GxP records, and which of those have had a periodic review in the last cycle? Periodic review is a dated obligation, usually at a risk-based frequency. Reviews lapse silently, because nothing stops working when one does.

3. Is the dictionary version in the safety database current, and when was historical data last re-coded? MedDRA releases twice a year, in March and September [1]. Terms are added, demoted, merged and reassigned to a different system organ class at each release, so a count under a preferred term is not comparable across a version boundary. Re-coding history is a project with medical review in it, not a refresh — and **a signal can appear or disappear because of a dictionary upgrade and nothing else.**

4. Can the company produce its post-approval commitment list on demand, with due dates? Ask for a single list of every dated external obligation, with its owner and the evidence that will discharge it. Almost nobody can produce one, and the inability is the mechanism by which commitments are missed. Watch for a due date expressed relative to an event — first commercial batch, last marketed batch expiry — because that date is computed from data held in a system the owner does not watch.

5. Is the trial master file complete, and filed contemporaneously? Completeness and timeliness are inspection findings in their own right, independent of how well the trial was run, because a sponsor cannot demonstrate oversight of what it cannot show. Two specific tests: re-scope the completeness model against the trials as actually designed, and measure the gap between each document's own date and the date it was filed. A document filed eighteen months late shows that contemporaneous oversight did not happen, even though the document now exists.

6. Is there an audit-trail review plan, and is it resourced? Having an audit trail is not reviewing it. A mid-size quality control laboratory running a chromatography data system across a dozen instruments generates thousands of trail entries a week. The obligation is to review them — risk-targeted, by someone not directly involved — and the direction of travel in the computerised systems guidance is that the review is normally completed before batch release [2]. **Nobody scopes the capacity.** Ask who does it, how many hours a month they have for it, and what they did with the last exception they found.

7. Are interfaces in anybody's validation scope? Count them, then ask for the interface specification of any three. The interface is where units, precision, timestamps, attribution and completeness are lost. "Both systems are validated" is not an answer.

8. Does any reporting layer join trial and commercial data? If it does, that is **a consent and competition-law breach, not a feature**, and there are four distinct reasons with four different remedies: blinding, personal health information, promotional bias, and the independence of the pharmacovigilance function. Read the query, not the dashboard.

9. Is there an adequacy determination on file for every cross-border transfer of health data? This is the one most architects miss, and it is specific to this jurisdiction. **The Protection of Personal Information Act has no counterpart to a European adequacy decision.** It assigns no body the task of assessing adequacy, so the determination falls to the responsible party itself — which makes that determination **the predicate for a section 57(1)(d) obligation carrying a criminal offence** [3]. Ask for it per recipient country, dated and reasoned. Note also that pseudonymised trial data in a sponsor's hands is still personal information; only permanent de-identification takes data outside the Act, and the draft research code tells researchers to assume re-identification is possible [3].

10. What is the retention policy, and which of the clocks does it breach? A single corporate retention period is always wrong in a pharmaceutical company. The verified clocks run from three years to forty, and three of the longest are event-triggered rather than year-triggered:

Record	Period	Runs from
Batch documentation	1 year after batch expiry **or** 5 years after certification, whichever is longer [4]	A product event
Clinical trial essential documents	At least 10 years after final closure, or 2 years after formal discontinuation [5]	A study event
Ethics committee trial records	At least 3 years after trial completion [5]	A study event
Traceability and unique-identifier records	Up to 5 years after the expiry of a trade item [6]	A product event
Noise exposure and audiometry	40 years	A fixed period
Chemical exposure and air monitoring	30 years	A fixed period

So **retention is a computation over operational data, not a policy setting** — you cannot evaluate "one year after batch expiry" without the batch expiry date. "Batch records: five years" is wrong, and a corporate seven-year policy is simultaneously unlawfully short for exposure records and wrong in kind for trial documents. Where an instrument states no period at all — and several do not — the architecture must assign one from the longest adjacent obligation **and record the basis for the assignment**, because "we chose seven years" with no reasoning is the answer that fails an audit.

11. Is there a measure definition register, or did the company go straight to a dashboard? Ask for the definition of three measures: right-first-time, deviation ageing, and the pharmacovigilance compliance rate. If right-first-time has no published definition the number is not comparable between the company's own two sites. If deviation ageing has no clock-stop convention for items waiting on a third party, that convention is where the gaming happens. And if the compliance rate's denominator is cases rather than **obligations due in the period**, it understates the exposure and silently drops the five-day obligations that are triggered by another regulator's action rather than by a case arriving [7].

12. For each guideline the company relies on: which version, and when was it last checked? This is not pedantry. The research behind this book found a South African guideline — the Good Manufacturing Practice guideline, version 9 of 17 November 2025 — **superseded by version 10 of 13 April 2026, inside its own stated five-year validity window** [8]. A guideline with a five-year validity statement can be republished in five months. **Re-checking the version and date of every guideline the architecture depends on is your standing job, not a one-off.**

13. Who is named as the Responsible Pharmacist on each licence, and when does each licence expire? There may be **not more than one Responsible Pharmacist named per licence**, that person is dual-accountable to the regulator and to the Pharmacy Council, and they must sign or certify for each batch — including a duty to ensure processes **remain in a validated state** [9]. The licence runs five years and renewal must be applied for at least 180 days before expiry [9], so the obligation falls due six months before the date anyone has in mind. Ask who covers the role on leave, and what the delegation arrangement is. It is an access-control design problem with a legal consequence.

The ninety-day shape

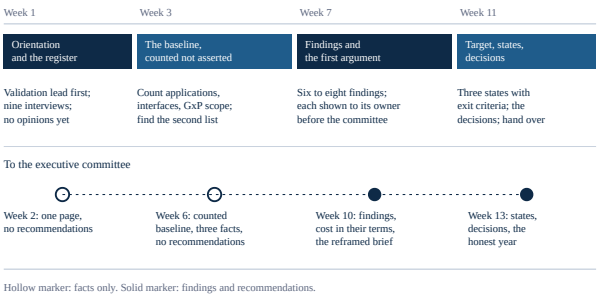


Figure 19.4 The ninety-day shape. Two of the four executive sessions deliberately carry no recommendations. Counting comes before advising, because here every change has a regulatory consequence.

Weeks 1 to 2 — orientation and the register. Nine interviews, validation lead first. The register started, as columns on the application portfolio rather than as a new artefact. No opinions yet. What goes to the executive committee at the end of week two is **one page**: what you are doing, who you have spoken to, and the one thing you already know — most usefully, whether a register exists. Nothing else. An architect who presents conclusions in week two is guessing, and everyone in the room will know.

Weeks 3 to 6 — the baseline, counted rather than asserted. Resolve the application list to a real number. Count the interfaces. State the GxP scope. Record hosting and data location per system. Reconcile the register against the quality function's own list until every difference is explained. Find the systems that are on neither list, because **that is where the risk is**: the spreadsheet that calculates a result, the departmental database holding a schedule, the instrument with a local hard drive and no backup. Every regulator who has inspected a pharmaceutical site knows to ask for that second list.

At the end of week six the executive committee gets **the counted baseline and three facts**, with no recommendations attached. Facts, with numbers, that nobody in the room disputes. This is the meeting that buys the next one.

Weeks 7 to 10 — the findings and the first argument. Six to eight findings, each traceable to evidence someone inside the business verified, each with a named remedy owner. **Show every finding to the person whose area it concerns before it goes anywhere near the executive committee.** Not for approval — for correction. Half of them will be refined and one will usually be wrong, and finding that out privately is worth more than being right in public.

The first argument goes with them: what this engagement is actually about, which will not be what the brief said. At the end of week ten the executive committee gets the findings, the cost framed in the terms it already uses, and the reframed brief.

Weeks 11 to 13 — the target, the transition states and the decisions. Three transition states with written exit criteria. Five or so decisions that have to be taken before any of it, each with a recommendation and the consequence of deferring it. And the arithmetic of what the year actually allows, said out loud, because in a regulated estate the binding constraint is usually quality assurance capacity rather than budget — the people who would validate a new system are the same people assembling the annual product quality reviews.

At the end of week thirteen the executive committee gets the transition states, the decisions and the honest year. Then you hand over, with a register that somebody internal owns.

> Two of the four executive sessions carry no recommendations at all. That is deliberate. In an industry where every architectural change has a regulatory consequence, an architect who recommends before counting has produced a preference, and preferences do not survive a quality review.

How to run the workshops

Three habits, specific to a regulated estate.

Name the jobs as the industry names them. Responsible Pharmacist, not "release manager". Qualified Person only if you are talking about Europe, because South Africa does not use the title and using it signals that your reference material is foreign. Principal investigator, not "lead doctor". Marketing authorisation holder, not "product owner". An architect who cannot name the jobs cannot run the workshop, and the room works this out in the first ten minutes.

Put the validation lead in every workshop that touches a system, from the first one. Not consulted at the end. They are the person who can say, in the room, that the change being sketched on the board is a six-month revalidation — which saves the four weeks the group would otherwise spend designing it.

Do not ask a quality officer for requirements in week two. Ask them what they do all day, what they have to produce, who signs it and what they do when the data is not there. Requirements in this industry come out of obligations, and obligations come out of documents — the licence, the approval letter, the quality agreement, the protocol, the guideline. Start from the document and the requirement writes itself; start from a blank requirements template and you will get a wish list that quality assurance cannot approve.

What ninety days will not achieve

Say this at the start, because it is the difference between a second engagement and a disappointed client.

Ninety days gets you a known estate, an obligation register, verified findings, three transition states and a set of decisions. It does not get you a migrated system, a shortened release path, a re-coded safety database or a dashboard. **None of those is a ninety-day object**, because each one is a validated change with a quality approval, a testing burden and a people constraint that the architect does not control.

What ninety days *can* achieve beyond the artefacts is one operational improvement that needed no validated change at all. There is usually one available, and it is usually either a normalised key, a report definition agreed between two functions that had two versions of it, or a resourced audit-trail review step that takes a day out of release. Pick one. Deliver it. It is the proof that the register was worth building.

What to say when you do not know

Pharmaceutical regulation is genuinely uncertain in places, and the uncertainty is not evenly distributed: the data standards at the regulatory interface are precise and dated, while the things around them move. The research behind this book carries **175 items it could not verify**, and the three with the most architectural consequence are worth naming:

- **Whether a research code of conduct under the privacy Act is in force.** An in-force sector code would remove the prior-authorisation requirement for the sector entirely, which changes the whole cross-border analysis for a trial [3]. The draft has no effective date.
- **Which procurement framework now governs health-product tenders**, after the entire Public Procurement Act 28 of 2024 was struck down on 17 September 2026 with invalidity not suspended [10]. Existing legislation continues to apply; exactly which, in practice, is unresolved.
- **The final text of the revised computerised-systems annex.** A draft was consulted jointly from 7 July to 7 October 2025 and **no final text or date had been published** as at 24 October 2026 [2]. The in-force version is still the one from January 2011.

The discipline is simple to state and hard to hold. **Say "I will find out, and I will record the date I found it."** Then do exactly that, and treat **a dated, sourced answer as the deliverable** — because in this industry it is one. A line in a register reading *"Annex 11 revision: no final text published; checked 24 October 2026 against the regulator's own consultation page; re-check quarterly; owner: validation lead"* is worth more to a client than a confident sentence, and it is the only form of the answer that stays true.

Three habits make that work. Put the retrieval date next to every regulatory claim in every deliverable. Keep a single open-items register with the item, what would settle it, who owns the re-check and when it is next due — the same structure as the obligation register, because it is the same kind of object. And never launder an unverified thing into a confident one by removing the hedge during a tidy-up pass; that happens at the end of engagements, in the formatting, by someone being helpful.

> **Do not go looking for the pharmaceutical reference architecture, because there is not one.** What exists is a set of **mandatory, dated, machine-enforced data standards at the regulatory interface** — named datasets and file tags in named submission sections, enforced by refusal to file [11]; a safety message format mandatory in one region since 30 June 2022, where the previous revision is not accepted at all [12]; a commercial subscription dictionary made mandatory by a regulator [13] — and **nothing above them.** Six independent bodies specify at the data and document level and none at the enterprise level. There is no industry capability model, no application portfolio taxonomy, no service catalogue and no technology reference model. The nearest things are a governed taxonomy for one document class and a canonical data model for one entity class whose owner calls it a reference architecture [14]. The constraint at the boundary is *tighter* than a reference architecture would be; it is just much narrower. Go and find the mandatory data contracts, and build the rest yourself on a general framework.

What not to do

Six, each of which has cost somebody a quarter.

Do not promise a dashboard in month one. The board will ask. What it can have in month one is a list of measures with no agreed definitions, and a recommendation to define them first. A dashboard over undefined measures lets a quality system look healthy while it repeats the same deviations — which is exactly what a closure-rate metric with no effectiveness check does.

Do not propose to replace a validated system before the register exists. The business case cannot be written without knowing what records the system holds, what the retention obligation on them is, and what the migration would have to verify. And migration in this estate is a validation activity requiring verification that **value and meaning** are both unaltered — audit trails, signatures, open workflow state, specification versions and retention clocks are each a specific hard problem.

Do not assume a GxP upgrade is an information technology project. "We can just upgrade it" does not exist here. A change to a validated system needs an impact assessment, a documented risk evaluation, re-testing proportionate to the risk, quality approval, and an update to the validation documentation set — and it may need a regulatory variation. **It is a regulatory project with an information technology component**, and the sizing comes from the validation lead, not from the vendor.

Do not model the documented process without asking how it differs from the actual one. This is the one with a twist. In most industries that gap is an inconvenience. Here, **the documented process is a controlled document, so the gap is itself a finding** — if people are not working to the procedure, either the procedure is wrong or the practice is. Ask the question in that form, to the person doing the work, and write down both answers.

Do not import a United States revenue-management architecture into a South African generics manufacturer. The gross-to-net machinery that architecture exists to serve — rebates, chargebacks, copay programmes, formulary payments — has no local analogue, and the transactions that produce it are largely prohibited here above the lowest schedule. In the private market there is **one price per pack**, set by the manufacturer's own declaration, adjusted by a gazetted maximum percentage, the same for a large hospital group and a single pharmacy — so customer-specific pricing for the regulated class is not a missing feature, it is a compliance control. The one genuine deduction is the logistics fee the manufacturer pays its wholesalers out of the exit price, and **the typical percentage could not be sourced** [15]. Build for that, not for a rebate engine.

And do not treat National Health Insurance as a roadmap item. It is judicially suspended rather than being implemented: an order recorded on 24 February 2026 prohibits proclaiming or implementing any provision until the public-participation challenges resolve; the Constitutional Court heard those challenges on 5 to 7 May 2026 and reserved judgment; it struck down the certificate-of-need provisions only, not the whole Act [16]. **Treat it as a scenario.** The defensible preparatory work is capability that is worth having regardless of its fate — patient and provider master data quality, coding standards, dispensing data integrity, formulary management, and the ability to supply at different price points to different channels. Any proposal that sequences investment on an assumed commencement date should be marked unsound, in writing.

The ninety-day checklist

Print this. The left column is what you do; the right column is the test that it is done.

Weeks 1 to 2	Done when
Interview the validation lead, first	You can name the systems that are in a validated state, and those that were validated once
Obtain or start the validated-system register	Every system has a GxP class, record types, state, owner, review date, hosting
Obtain the regulatory affairs registration register — including the spreadsheet	You have seen the spreadsheet, not only the system
Obtain the quality function's own system list	You can state the difference between it and the portfolio
Send the executive committee one page	It carries no recommendations

Weeks 3 to 6	Done when
Count the applications, the interfaces and the GxP scope	The numbers are counted, not asserted, and someone internal verified them
Find the systems on neither list	The spreadsheets, departmental databases and standalone instruments are named
Record hosting and data location per system	The cross-border transfers are a list
Run the thirteen afternoon checks	Each has a written answer with a date and a source
Present the counted baseline and three facts	Nobody in the room disputes the numbers

Weeks 7 to 10	Done when
Write six to eight findings	Each is checkable in a day by someone who already works there
Show each finding to the person whose area it concerns	At least one has been corrected
Frame the cost in the executive committee's own terms	Where there is no benchmark, you have named who can calculate it
Open the open-items register	Every unverified item has what would settle it, an owner and a re-check date
Present the findings and the reframed brief	The brief you were given has visibly changed

Weeks 11 to 13	Done when
Write three transition states	Each has exit criteria written before the work starts
List the decisions that come first	Each has a recommendation and the cost of deferring it
Do the arithmetic of the year	You have said out loud how many transition states the year holds
Start the measure definition register	Three measures have formulas, grains and owners
Write the modelling standard	Types only in clinical views; which views are controlled documents and which are not
Hand over the register to a named internal owner	It is theirs, and they have changed something in it

THE CHAPTER IN TEN LINES

- Week one is the validated-system register, not the org chart; if there is no register, building it is the first deliverable and is how you learn the estate.
- The validation lead is the most useful first interview, because they know which systems are in a validated state as opposed to which were validated once.
- Thirteen checks take an afternoon each and change what the engagement is about.
- Completeness and filing timeliness of the trial master file are inspection findings in their own right.
- Having an audit trail is not reviewing it, and nobody scopes the review capacity.
- A single corporate retention period is always wrong: the clocks run from 3 to 40 years, three are event-triggered, and batch-record retention is a computation.
- POPIA has no adequacy decisions, so the determination falls to the responsible party and is the predicate for an obligation carrying a criminal offence.
- A guideline with a five-year validity statement can be republished in five months, so re-checking versions and dates is a standing job.
- Say I will find out and record the date I found it, and treat a dated, sourced answer as the deliverable.
- Do not look for the pharmaceutical reference architecture; there is none. There are mandatory data standards at the regulatory interface and nothing above them.
- No dashboard in month one, no replacement before the register, no GxP upgrade treated as an IT project, and no NHI on the roadmap.
- The documented process is a controlled document, so the gap between it and the actual process is itself a finding.

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APPENDIX A

The cross-maps

Ten tables that let you translate between the vocabularies you will meet in your first month — and, where there is no translation, say so.

This appendix is a set of translation tables. Each sits between two vocabularies you will meet in the same week — the regulatory affairs manager's and the ArchiMate metamodel's, the quality manager's and the ERP's, a European reference architecture's and a South African licence's. None is a model. They are the crib sheets that stop a workshop stalling on a word.

Each carries a note on where it breaks down, because a translation table that pretends to be complete does more harm than no table. Where a mapping is approximate it says so; where there is none it says that too, which is the more useful answer.

Pharmaceutical object to ArchiMate 4 element

An architect's first instinct on arriving in pharma is to find the ArchiMate element for each new noun. For about two thirds of the vocabulary that works, and the honest treatment of the rest is what separates a model a regulatory affairs manager recognises from one they politely ignore. The element is `Role`; there is no separately named business-layer variant of it [1]. There is no Constraint element; express a constraint as a `Requirement`. And `Value` is not a number, which matters here more than anywhere else in this series, because pharma is full of numbers that look as though they belong in a model and do not.

Pharmaceutical object	ArchiMate 4 element	What the element cannot carry
Compound, substance, formulation	`Business Object`	The normalisation rule version that decided identity; the salt factor
Certify and release a batch; receive a report	`Business Process`, `Business Event`	That release is per market; that the receipt timestamp starts a legal clock
Marketing authorisation, certificate of registration	`Contract`	Its conditions as separately dated obligations
Site, laboratory, depot	`Facility`	The licence scope lines
Responsible Pharmacist, investigator	`Role`	Council registration, residency, one per licence
Safety database, LIMS, eTMF	`Application Component`	Validation state, GxP classification
Batch record, protocol, specification	`Representation`	Approval signatures, effective dates per site
A GxP obligation	`Requirement`	The instrument and provision it comes from

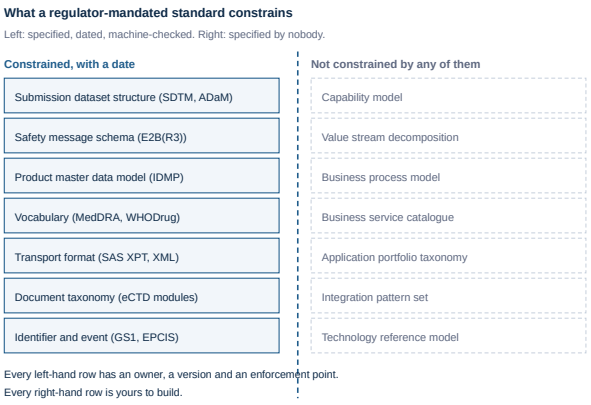


Figure 20.1 What a regulator-mandated standard constrains, and what it leaves to you. **Every left-hand row has an owner, a version and an enforcement point. Every right-hand row is yours to build.**

Now the harder half. These are the objects where the element is approximate, or where there is no element at all and an architect who forces one has quietly lost the thing that mattered.

Pharmaceutical object	Nearest element	Why it does not work
Study, clinical trial	`Business Object` plus `Work Package`	The definition and the undertaking are different things and neither element holds both; the schedule of assessments belongs to a protocol version, not to the study
Batch, lot	`Business Object` or `Material`	A batch is an instance and ArchiMate models types. Modelling 1,810 batches a year is not modelling
Individual case safety report	`Business Object`	A case is bi-temporal — what we believed, as at when. The element is one state
Signal	`Business Object`	A signal is a case set plus a hypothesis plus a history, and the history is the valuable part
eCTD sequence and dossier state	`Representation` plus `Business Object`	The dossier is a projection over an append-only sequence log, and ArchiMate has no fold
Dictionary version dependency	`Business Object` with an `Association`	The dependency is on an external licensed asset with its own release calendar
Batch genealogy	**none**	**A genealogy is a directed graph of instances with quantities. It is not drawable in a type model at all; a "genealogy" box is the register, not the genealogy**
Specification limit, acceptance criterion	**none**	**There is no metric element and `Value` is not a number. A limit of not less than 80% dissolved in 30 minutes has nowhere to live**
Yield, right-first-time, compliance rate	**none**	**Same reason. Repository attributes of a measured object, never elements**
Validation state, periodic review date, GxP classification, Part 11 applicability	**none**	**Properties of an `Application Component`. As elements they give you one box per attribute and no information**
Retention period	**none**	**Three of the longest clocks are computations over operational data, not durations**

> **Model the types, carry the instances in the repository, keep the numbers as attributes.** An architect who models the batch has the wrong grain; one who models the batch *record*, with a genealogy relationship to itself, has it right and should then stop drawing.

The four identities of a product

The most useful table in the volume. When a business user says "we have 180 products", ask which of these four they mean. The answer is usually "180 registrations, about 450 SKUs, about 60 molecules" — three numbers from three systems that have never been reconciled.

Identity	What it is	Who owns it	Cardinality	Mastered in
Molecule	The active substance, by structure and by a substance identifier under ISO 11238 [2]	Research informatics internally; regulatory affairs externally	One to many formulations	Compound registration, then the regulatory system
Formulation	Molecule plus excipients, at a strength, dose form and route. Country-independent; carries a PhPID under ISO 11616 [2]	Development, then regulatory affairs	One to many registrations	Regulatory information management
Registration	One formulation authorised to one holder in one country, under one name, at named sites, to a named specification, with its own label. Carries an MPID under ISO 11615 [2]	Regulatory affairs, unambiguously	One to many packs	Regulatory information management
SKU, pack	The registration in a container and count, with a barcode, a Single Exit Price and a stock-ledger identity	Supply chain and commercial	One to many serialised units	ERP, keyed to the regulatory product

Where it breaks down: the ratio is many-to-many in every direction, not the clean one-to-many implied here. And the dimension almost always missing is the **dated** link from pack to authorisation — which answers "was this pack legally saleable here on this date", the question a recall asks.

GxP regime to the record it compels

A regime becomes an architecture constraint at exactly the point where it says what the company must be able to produce. Until then it is a policy. The right-hand column is therefore the one to read.

Regime	Governs	Who inspects (South Africa)	Records it compels
GMP — manufacturing	Making, packing and testing to the authorisation	SAHPRA, against the adopted PIC/S guide [3]	Master and executed batch records; genealogy both ways; deviations with affected batches; change control with regulatory impact; out-of-specification investigations; stability data; the annual product quality review; the Responsible Pharmacist's release decision; distribution records per batch
GCP — clinical	How a trial is conducted, documented and overseen	SAHPRA and the ethics committee	Protocol and amendments with per-site approvals dated before the data collected under them; the SANCTR number before first enrolment; consent at version grain; the delegation log; source documents meeting ALCOA-CCEA [4]; an audit trail with no deletions; the lock record
GLP — non-clinical	Safety studies a filing relies on	The receiving authority	The study plan and amendments; the in-life record; raw data with its archive; the study director's statement; the quality assurance inspection record
GVP — pharmacovigilance	The holder's safety system	SAHPRA pharmacovigilance inspection	The case database with receipt dates traceable to first contact by any employee; the 15 and 5-day clocks; the signal register **including refuted signals**; risk-management plans with effectiveness evaluation; dictionary version control
GDP, GWP — distribution	Storage, transport, wholesaling	SAHPRA good wholesaling inspection	The wholesale licence and scope; temperature excursions; consignment traceability; recall capability

Where it breaks down: the regimes overlap on one system more often than they divide. A complaint that turns out to be an adverse reaction is a quality record and a safety case at once, and both clocks run.

System class to what it masters

Thirty-five system classes appear in a large estate; twelve carry the mastering arguments that matter. Read the third column first — what a system is *not* allowed to assert settles a design review.

System class	System of record for	Not the master of	Contested with
ELN	The experiment: intent, procedure, observation, witness	Compound identity	Compound registration
LIMS	Sample, test, result at determination grain, certificate of analysis	The specification's approved limits, which descend from the dossier	QMS, regulatory system
CTMS	Site, investigator, monitoring visit, payment, milestone	Collected subject data; the study identifier	EDC, study master
EDC	Case report form data and its audit trail	Randomisation; the safety case; the study spine	IRT, safety database
eTMF	The documentary evidence that the trial was run properly	Anything else — though its completeness is itself a finding	Document management
Safety database	The individual case safety report, versioned	Trial data as collected; authorisation status	EDC, regulatory system
RIM	Authorisation, dossier state, submission plan, commitment, variation	The sequences; the source documents	Publishing archive, document management
QMS	Deviation, CAPA, change control, complaint, audit, training	The variation a change requires; the batch disposition	RIM, ERP
MES with electronic batch records	The executed batch record	The master recipe's approval; the release decision	ERP, QMS
ERP	Material, pack, batch stock by location and status, cost, ledger	Substance and product identity, specification limits, approved-manufacturer qualification	RIM, QMS, MDM
Serialisation platform	The serial number, its aggregation and movement events	The batch; the pack master data it reports	MES, ERP
Statistical computing environment	Nothing. It is a derived environment	Any value that could be corrected in it rather than at source	—

Three objects are habitually mastered twice. **Product and material identity** is claimed by the ERP, because a clerk created the first material master row, and by the regulatory system, where authorisation lives; the ERP may cache and must never mint. **The serious adverse event** sits in the EDC as collected and in the safety database as a case, under two keys. **Batch status** is asserted by the ERP, the LIMS and the QMS; the disposition decision is the quality system's, at `(batch, market)` grain.

Regulator-mandated standard to what it constrains

This table converts a standards topic into architectural constraints with owners and dates. Dates move, so each row says where it came from; the research behind this book carries a retrieval date of 24 October 2026. A version support matrix is deliberately absent: the authority's own data standards catalogue governs what you may submit, not a standards body's latest release, and the research could not read the current catalogue [5].

Standard	Mandated by	For what	From when	Constrains
CDISC SDTM and its implementation guide	FDA, through its data standards catalogue, enforced at the gateway by the Technical Rejection Criteria; also PMDA [5][6]	Study data in named eCTD sections of an NDA, BLA, ANDA or commercial IND	Studies started after 17 December 2016, and 17 December 2017 for a commercial IND [6]	A **data structure** — named datasets in named locations
CDISC ADaM	Same mechanism; criterion 1736 names `ADSL` [6]	Analysis datasets, same submission types	Same dates [6]	A **data structure** plus traceability
CDISC SENDIG	Same mechanism	Non-clinical study data	Per the catalogue [5]	A **data structure**
Define-XML	Same mechanism; criterion 1736 names `define.xml` [6]	Every study in an electronic submission	Same dates [6]	A **metadata document**
E2B(R3)	EMA; EudraVigilance does not accept E2B(R2) [7]	All case reports and serious unexpected reactions to EudraVigilance	30 June 2022 [7]	A **message**

Standard	Mandated by	For what	From when	Constrains
eCTD, under ICH M4 and M8	Each authority regionally: PMDA from April 2026 for v4.0, FDA no requirement date, EMA mandatory Q1 2028 [8]; South Africa on v3.2.2, portal-only [9]	Marketing applications and their lifecycles	Per region. Existing v3.2.2 lifecycles stay v3.2.2, so formats run in parallel for years [8]	A **document taxonomy** and a **transport format**
ISO IDMP	EU, phased: critical medicines June 2026, other non-centrally-authorised products December 2026, remaining pack sizes June 2027, two already extended [10]. In South Africa through the traceability guideline [11]	Product master data submitted to the authority	**No go-live date; only phased deadlines** [10]	A **master data model**
WHODrug Global	FDA, explicitly, for a **paid subscription dictionary** [12]	Medication coding in study data	Studies starting after 15 March 2019 [12]	A **vocabulary**
MedDRA	ICH-aligned regulators in practice, carried inside the E2B(R3) message; **not stated** in SAHPRA's operative guideline [13]	Event, condition and indication coding	Twice-yearly releases [14]	A **vocabulary**
GS1 identifiers and EPCIS	DSCSA and the Falsified Medicines Directive — **not GS1**, which publishes voluntary specifications. South Africa phases identifiers 2025–2031 [11]	Pack identification, aggregation, event capture	Per jurisdiction [11]	An **identifier scheme** and an **event vocabulary**

Now the column that makes the point. Every one of these constrains a structure, a message, a vocabulary, a transport format, a taxonomy or an identifier scheme. **Not one constrains a capability, a value stream, a business process, a business service, an application component or a technology component.** Six independent bodies, with different constituencies, funding and governance, all stop at the same boundary.

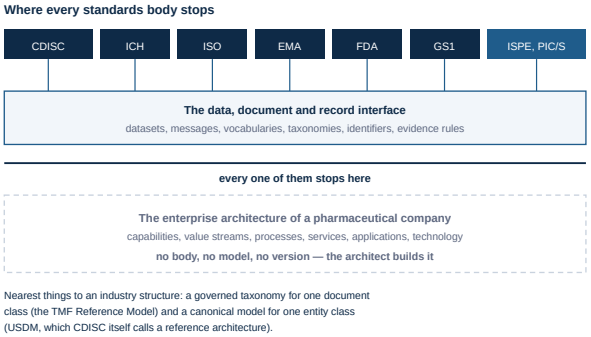


Figure 20.2 Six independent bodies, one boundary. **When bodies with different constituencies, funding and governance all stop at the same line, that line is telling you what the industry has been able to agree on — interfaces, not organisation.**

> The two nearest things to an industry reference structure both fail the same scope test. The CDISC TMF Reference Model is a genuinely governed, versioned, change-controlled industry taxonomy — of one document class [15]. CDISC describes its Unified Study Definitions Model as "a reference architecture meant to serve as a standard model for building conformant study definition technologies" [16] — a reference architecture for a study definition technology, not for a pharmaceutical enterprise. Apply that same test to every reference architecture you are handed, in any industry.

Dictionary and vocabulary to its versioning and licensing

Dictionary	Domain	Cadence	Licensing	Architectural consequence
MedDRA	Events, conditions, indications	**Twice yearly**; v29.0 March 2026 [14]	Subscription through the MSSO, fee-banded; not redistributable	Every coded value carries its version or means nothing. A term split between versions divides a signal silently
WHODrug Global	Products, ingredients, ATC classes	Twice yearly, 1 March and 1 September [12]	Subscription through the Uppsala Monitoring Centre; not redistributable — and FDA-mandated [12]	The clearest case of a regulator making a paid product compulsory. Your CRO needs its own licence
SNOMED CT	Clinical terms	Twice-yearly international release plus national extensions	Member-country licensing. **Whether South Africa is a member could not be sourced**	Real-world data arrives coded in it; mapping to CDISC structures is a project
LOINC	Laboratory tests and observations	Twice yearly	Free, with attribution and no modification of codes	Central-laboratory transfers are where unit and test-code mismatches enter
UCUM	Units of measure	Infrequent	Free, permissive	Storing the code beside the display unit cannot be retrofitted; historical strings are ambiguous
Pharmacopoeial monographs, EDQM terms	Specifications, dose forms, routes	Annual plus supplements	Paid, per user	The **monograph edition** is part of a specification's identity; the EU message sources dose form and route from EDQM via SPOR [7]

The consequence applies to all six rows: **these are dependencies, not reference data.** Reference data you own and maintain. A dependency has an external owner, a release calendar you do not set, a licence constraining who may hold it, and a version that is part of the meaning of every value coded against it. Put them in the architecture as managed external dependencies with named owners and a standing upgrade calendar, archive each release alongside the data coded against it, and budget the licence before the integration design. The terms surprise architects latest and hardest: a non-production environment may not be allowed to carry the dictionary, and an offshore coding vendor needs its own subscription.

Retention clock by record class

A single corporate retention period is always wrong in a pharmaceutical company. The clocks below run from three years to forty, from four kinds of trigger, and **three of the longest are event-triggered** — which makes retention a join to operational data rather than a policy setting. You cannot evaluate "one year after batch expiry" without the batch's expiry date.

Record class	Period	Trigger	Instrument
Batch documentation	**One year after batch expiry, or five years after certification by the authorised person, whichever is longer**	Batch expiry, or certification	PIC/S PE 009 Part I §4.11 [17], mirrored in SAHPRA's licensing guide [18]
Trial essential documents, investigator and sponsor	At least 10 years after final closure, or two years after formal discontinuation of development	Closure, or discontinuation	SA GCP 2020 §§5.10.5, 6.4.6 [4]
Ethics committee trial records	At least 3 years after completion	Trial completion	SA GCP 2020 §4.3.6 [4]
Traceability and unique-identifier records	Up to 5 years after a trade item's expiry	Trade-item expiry	SAHPRA traceability guideline §3.1.2 [11]
Prescriptions, Schedules 2 to 6, at the last seller	At least 5 years	Date of sale	Medicines Act s 22A(6)(o) [19]
Pharmacovigilance records	**No period stated**; surveillance itself runs until 6 months after the last marketed batch expires	—	SAHPRA pharmacovigilance guideline [13]

Record class	Period	Trigger	Instrument
Noise exposure, audiometry including baseline audiograms	**40 years**	—	Noise-Induced Hearing Loss Regulations reg 11
Chemical exposure assessments and air monitoring	**30 years**	—	Hazardous Chemical Agents Regulations 2021 reg 9(e)
Corporate, tax and VAT records	7 years; 5 years from submission for tax and VAT, extended under audit	—	Companies Act 71 of 2008 s 24; Tax Administration Act 28 of 2011

Three readings. **The longest clocks sit on the least glamorous data** — forty years for audiometry against ten for trial documents. **The obligation outlives the application by a factor of four or five**, so the deliverable is an export format with its reference data, readable without the source system. And **where the instrument states no period, that is a finding, not a licence**: assign one from the longest adjacent obligation and record the basis.

Regulator to evidence

The same six structures answer nearly every inspector.

Body	Inspects	Asks for first	Evidence lives in
SAHPRA — GMP	Manufacturing, packing, testing sites	The licence and its scope; the Site Master File; the master and executed batch record pair	Licence register; batch record pair; genealogy both ways; deviation with its affected-batch list; release at `(batch, market)`
SAHPRA — GCP	Trials, sponsors, sites, CROs	Per-site approvals dated before the data collected under them; the SANCTR number; consent at version grain	Study spine with protocol versions and the amendment fan-out; consent register; append-only data points; the locked-state object
SAHPRA — pharmacovigilance	Holders' safety systems	The system documentation; the resident responsible person; the signal register **including refuted signals**	Bi-temporal case store; obligation table at `(case version, recipient, type)`; signal register; dictionary version per value
SAHPRA — good wholesaling	Wholesalers, importers, distributors	The licence and scope; temperature excursions; consignment traceability	Licence register; consignment and batch records; the returns-to-destruction chain
SAHPRA — regulatory compliance	Destruction of medicines and scheduled substances	Quantities, batch numbers, expiry dates; the authorised facility; the **named witness and designation**	Destruction record at batch grain, linked to disposition, decommissioning and write-off [21]
Pharmacy Council	Pharmacies, premises, pharmacists	The pharmacy's registration and recording of the premises; the Responsible Pharmacist's category	Person register at `(person, Council number, category, status, dates)`; the licence register's named-pharmacist field

Body	Inspects	Asks for first	Evidence lives in
Information Regulator	Lawful processing of personal and special personal information	The processing register with the lawful basis **citing the provision**; a dated, reasoned adequacy determination per recipient country	Processing register; transfer register, which is the validated-system register plus four columns; consent per version
Ethics committee, NHREC	Trials at the institution; committee registration	Dated approval **before** commencement, renewed at most every 12 months; approved consent versions and languages	Site approval states; amendment fan-out; consent-version register with language and readability
Employment and Labour	Occupational health at the plant	Risk assessments every two years; exposure and noise monitoring against the 30 and 40-year floors; **the dated employee-to-zone assignment**	Exposure repository at `(exposure group, date)`; surveillance repository; zone assignment history
SARS, customs	Declarations, duty relief, licensed premises	Declarations linked to order, receipt, batch and despatch; drawback within six months of export; the signed MBR 20	Declaration repository; duty-relief reconciliation **built on production genealogy**
Marketing Code Authority; Competition Commission	Promotional conduct; tender conduct and pricing	Material with its approvals and the product-information version checked against; bids with their pricing basis	Promotional approval register; transfer-of-value register [20]; tender register with pricing basis

The six are: the dated, authored, scoped assertion; the append-only result with its audit trail; the versioned controlled document with its per-site approval fan-out; the genealogy graph; the obligation register with due dates and evidence; and the validated-system register. **Build those six well and inspection readiness follows across the enterprise. Build eleven inspector-specific repositories and none of them will be current.**

Archetype to design consequence

Archetype	System of record shifts to	Cost object	Regulatory burden	Design consequence
Originator	Research informatics and the regulatory system, across two decades	The programme, not the product	Heaviest: GLP, GCP, GMP, GVP, GDP	Traceability from a research observation to a label to a safety signal, across six systems of record
Generics	The regulatory system and the quality system	The SKU, against a regulated price ceiling	GMP, bioequivalence, full pharmacovigilance, variations	Portfolio administration at scale: artwork, change-control throughput, and whether a SKU is still worth having
Biosimilar	Analytical data management	The programme, then the campaign	Comparability plus biologics GMP, closer to an originator than a generic	The regulatory argument *is* the analytical dataset
CDMO	The client's registration, which the CDMO does not hold	**The campaign**	Full GMP, no authorisation of its own; exposure spread across every client's registration	Multi-tenancy under GMP: one quality system, one MES, many clients, absolute segregation

Archetype	System of record shifts to	Cost object	Regulatory burden	Design consequence
CRO	The sponsor's systems as much as its own	The study, and the billable hour	**GCP, not GMP** — a different inspection regime entirely	Sponsor-system multi-tenancy and data hand-back: a complete, validated, audit-ready dataset and trial master file at study end
Consumer health	The commercial estate	The brand and the trade term	Lowest and jurisdictionally variable; advertising to the public is permitted and separately regulated	Trade promotion beside a regulated-price estate: the control that stops a rebate reaching a prescription line *is* the architecture
API manufacture	The master file and the customer-qualification register	The kilogram, and the campaign	API GMP plus support of every customer's registration	Change propagation: which registrations in which countries name you as the source, and what each must file
Compounding	Pharmacy practice records	The prescription and the preparation	Pharmacy Act, not SAHPRA licensing; **not a licensed category here** [22]	Traceability with no authorisation to anchor it

Most real companies are two or three of these at once, and the estate then contains the architectural problems of each. The conflicts between them are where the defects are.

South African instrument to its EU or US analogue

This table stops an imported reference architecture doing damage. Read the last column: where the analogue does not exist, a model built elsewhere contains an estate for transactions that cannot occur here, and is missing an obligation that can stop a product being sold.

South African instrument	What it does	EU analogue	US analogue	Same thing?
Responsible Pharmacist	Named on the licence, one per licence, Council-registered, accountable to both the Council and SAHPRA; certifies each batch before release and must ensure processes **remain in a validated state** [18][23]	The Qualified Person	**None.** Release is a quality-unit responsibility with no named statutory individual	**No.** Nearest to the QP, but a different qualification, a different regulator pair and an explicit validation-state duty. The most important vocabulary divergence in the book
Single Exit Price	One gazetted price per pack: the **only** price at which a manufacturer or importer may sell to any private buyer. Does not apply to the State, which buys by tender [24]	National reference pricing and reimbursement, state by state	Gross-to-net: list less rebates, chargebacks and fees	**No analogue at all.** The transactions producing gross-to-net are largely prohibited here above Schedule 0, so a US revenue-management architecture builds an estate for transactions that cannot occur
Reliance	Abridged and verified review on a recognised authority's assessment report, with SAHPRA **remaining responsible for its own decision** [25]	Mutual recognition; the decentralised procedure	None of this shape	**No.** Reliance is not recognition. Recognition is a separate pathway still being negotiated, its framework unpublished [25]
SAHPRA traceability phasing	GS1 identifiers, phased 2025–2031, **batch level first**, **imports two years ahead of domestic**, **no transactional repository specified** [11]	The Falsified Medicines Directive: a central hub, national systems, decommissioning at dispense	DSCSA: no hub, interoperable tracing between trading partners	**No.** Neither a hub nor a peer-to-peer obligation. What is required now is master data and internal linkage, which is a different design

South African instrument	What it does	EU analogue	US analogue	Same thing?
POPIA	Health information is special personal information; s 27 is the sponsor's route, not s 32; s 72 gates cross-border transfer; **the responsible party itself determines adequacy**	GDPR, with **Commission adequacy decisions**, standard clauses and binding corporate rules	No general federal statute; sectoral and state rules	**Partly.** The structure rhymes; the machinery does not. In the EU adequacy is decided for you; here you decide it, date it and reason it, and carry a criminal-offence-bearing obligation on the determination
Section 57 prior authorisation	Notify the Regulator before certain processing, including transfer of special personal information to an inadequate jurisdiction; it replies in 4 weeks, finalises in 13	The impact assessment and prior consultation — not the same instrument	**None**	**No.** A South Africa-specific critical-path item for study start-up. An in-force sector code would remove it entirely
Section 21	Authorised supply of a **specified quantity of an unregistered medicine** for a specified period — 20,245 applications in 2024/25 on a three-working-day clock [26]	Named-patient and compassionate use	Expanded access	**Approximately.** The shape is familiar, the volume is not, and **Section 21 supply is not subject to Single Exit Price** [24]

> The rows marked "none" are where imported reference architectures fail silently. They do not throw an error. They produce a target state that is complete by its own lights, with customer-specific pricing machinery that must be switched off, a recognition workflow for a pathway that does not exist, a serialisation repository nobody requires, and no adequacy-determination document — which is the one artefact the Information Regulator asks for.

How to use this appendix

Three habits make these tables worth their page count. **Carry the four identities table into the first workshop**, and ask which of the four "product" means until the room qualifies it unprompted. **Date every row you reuse.** And **when a table has no row for your question, say so out loud** — the absence of a mapping is the finding, and the architect who names the gap is trusted where one who approximates is corrected later, in public.

THE CHAPTER IN TEN LINES

- About two thirds of pharmaceutical vocabulary has a good ArchiMate 4 home; the honest treatment of the rest is what makes the model credible.
- A genealogy is a graph of instances with quantities, so it is not drawable in a type model at all.
- There is no metric element and Value is not a number, so a specification limit and a yield belong in attributes, not elements.
- Validation state, periodic review date and GxP classification are properties of an application component, not elements of their own.
- Product means four things with different cardinality: molecule, formulation, registration and SKU. Ask which one every time the word is used.
- A regime becomes an architecture constraint at the point where it says what record the company must produce.
- Mandated standards constrain structures, messages, vocabularies, transport formats, taxonomies and identifiers, and not one of them constrains a capability, process, service, application or technology component.
- Dictionaries are external licensed dependencies with their own release calendars, not reference data you own.
- A single corporate retention period is always wrong: the clocks run from three to forty years and three of the longest are event-triggered computations.
- Where a South African instrument has no EU or US analogue is exactly where an imported reference architecture fails without throwing an error.

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APPENDIX B

Glossary

Pharmaceutical vocabulary has four overlapping dialects that use the same words differently. This is the lookup list, with both meanings where there are two.

Pharmaceutical vocabulary is the barrier to entry. It is not that the words are hard; it is that there are four overlapping dialects — discovery, clinical, regulatory, and quality and manufacturing — and they use the same words differently. A scientist, a clinical operations lead, a regulatory affairs manager and a Responsible Pharmacist can all say "release" in one meeting and mean four things, and nobody notices, because each hears their own meaning.

This appendix is for lookup, so it runs alphabetically; the vocabulary chapter earlier organises the same material by dialect, which is the better way to learn it. Where a word means two things, both are given and the entry says which conversation you are in. Where the industry genuinely disagrees, it says so rather than picking a side. And a few terms carry a warning instead of a definition, because the obvious reading is wrong.

A

ADaM — Analysis Data Model: the CDISC layer of analysis-ready datasets derived from tabulation data. Its rule is traceability — every analysis value must be explainable from the row it came from.

Adverse drug reaction (ADR) — an adverse event for which a causal relationship with the medicine is at least a reasonable possibility. The difference from an adverse event is causality, and that one word changes which clock runs.

Adverse event (AE) — any untoward medical occurrence in a patient or participant given a medicine, **whether or not it is causally related**. The "whether or not" is the point: this is an observation, not a conclusion.

Aggregation — the recorded parent-and-child relationship between a container and its contents. Model it as an event log, not a current-parent field, or the first case anyone opens breaks it.

ALCOA, ALCOA+ — attributable, legible, contemporaneous, original, accurate, extended with complete, consistent, enduring and available. **The industry disagrees on the attribute count**, a European draft has introduced "ALCOA++", and the PIC/S guide's Part I never uses the term [1].

ALCOA-CCEA — the South African clinical variant, and the only verified statement of the mnemonic in a local instrument: it appears in the national good clinical practice guidelines for source documents [2]. A clinical requirement, not a manufacturing one.

Annex 11 — the computerised-systems annex of the European GMP guide, adopted in substance through PIC/S. **The version in force is still the January 2011 revision**; a draft was consulted from 7 July to 7 October 2025 with no published final text or date [3].

Annual Product Quality Review (APQR) — a yearly documented review of a product's quality record: batches, deviations, out-of-specification results, complaints, returns, recalls, stability, changes. The annual frequency comes from the PIC/S guide, not from SAHPRA's own [1].

Active pharmaceutical ingredient (API) — the substance that does the therapeutic work. Named in the authorisation together with its manufacturer, so changing where it comes from is a regulatory event in every market naming the old source.

Active substance master file (ASMF, DMF) — a dossier an ingredient maker files so customers can reference it without seeing its restricted part. It is why a change at the ingredient maker propagates into every customer's registration.

Arm — a *planned* treatment sequence in a trial. A subject's actual treatment is a different fact, because deviations and dose reductions make them differ, and collapsing the two makes the analysis population undefinable.

Artwork — the regulated graphic content of a carton, label or leaflet, version-controlled against the approved product information per market and language. In a wide generics portfolio it is the most change-prone material in the estate.

Audit trail — the system's own computer-generated record of every create, change and delete: who, when, old value, new value, reason. It must be independent of the user, which rules out logs written by the code path the user controls.

Audit trail review — actually reading audit trails, risk-targeted, by someone not directly involved, with records of what was reviewed and found. **Having an audit trail is not reviewing it**, and almost nobody scopes the capacity.

Authorised Person — whoever certifies a batch for release. The title varies by jurisdiction, and the PIC/S retention clock is expressed in terms of certification by this person [1].

B

Batch, lot — a defined quantity made in one process, expected to be homogeneous. "Lot" is commoner in North American usage and in clinical supply, "batch" in European and South African manufacturing. **Two traps**: a batch is the unit of release and liability, a clinical-supply lot the unit of kit labelling and expiry; and a batch repacked is a *new* batch linked to its parent.

Batch record — two objects, one name. The **master batch record** is the approved, versioned instruction for a product at a site; the **executed** record is the append-only record of one batch, copied from the master at the start. Pointing the executed record at the current master destroys the evidence.

Batch release — the decision that a batch may be sold into a named market, taken by a named person on listed evidence at a stated time. Its grain is `(batch, market)`, because specifications and shelf lives differ by authorisation.

Bioequivalence — the demonstration that a generic delivers the same active substance to the bloodstream at the same rate and extent as the reference product. It replaces efficacy trials, and is not the claim "the same product".

Biologic — a medicine made in or from a living system. Its identity includes its process, which is why a change of cell line or purification is a change of substance rather than a manufacturing tweak.

Biosimilar — a biologic shown **similar**, not identical, to a reference biologic, on an analytical comparability package plus comparative clinical work. **Not a generic biologic**: erosion is slower and driven by jurisdiction-specific substitution rules.

Blinding, masking — withholding treatment assignment from subjects, investigators or both. Architecturally an access-control requirement across the capture system, the randomisation system and the statistical environment at once.

C

Campaign — a run of one product at a contract manufacturer between changeovers. **It is the cost object of a CDMO**, where a product company's is the product and an originator's is the programme.

CAPA — corrective and preventive action: a plan arising from a deviation, complaint, audit or trend. It closes on implementation **plus a passed effectiveness check**, which is the step everyone drops.

CDASH — the CDISC collection standard: what a case report form asks and what each field is called at capture, so the downstream tabulation is mechanical rather than interpretive.

CDISC — the Clinical Data Interchange Standards Consortium, which publishes the clinical data standards regulators require: CDASH, SDTM, ADaM, SEND, ODM and Define-XML [4]. It also hosts the TMF Reference Model.

Certificate of analysis (CoA) — an attested summary of a batch's results, issued to a recipient. A published artefact: immutable once issued, and a re-issue is a new version with a stated reason.

Certificate of registration — the South African marketing authorisation. **It expires**: five years, renewal due six months before expiry, phased across the portfolio from 2023 to 2035 [5].

Change control — prospective assessment and approval of a change before it is made, then verification that it worked. Its load-bearing attributes are the regulatory impact assessment (does this need a variation?) and the validation impact assessment.

Clinical data point — one recorded observation about one subject at one time. Forms, datasets and domains are containers for it, and the submitted shape is tall and thin, not wide.

Commissioning — two dialects. In serialisation, bringing a serial number into existence as a valid identifier, irreversible as a record. In engineering, setting equipment to work before formal qualification begins.

Commitment — an obligation accepted in or alongside an approval: a post-approval study, a stability commitment, a periodic report. One authorisation can carry a dozen, with different owners and dates.

Comparability — showing that a biologic made one way is the same as one made another way, across a process change or against a reference product. In biosimilars the analytical dataset *is* the regulatory argument.

Compounding — a pharmacist preparing a medicine for an individual patient or in anticipation of need. **Not a licensed category in South Africa**: such operations are licensed as community pharmacies, which SAHPRA does not license, and draft guidelines from June 2023 were never finalised [6].

Consent, informed consent — the subject's documented, voluntary agreement, given on a specific approved version of a form. **Consent is a state with versions, not an event**, and a form change creates a re-consent determination per enrolled subject.

Contract development and manufacturing organisation (CDMO) — paid a fee for service to develop and make product for clients who hold the registrations. **It earns no margin on product** and appears as a named site on every client's registration.

Contract research organisation (CRO) — paid to run trials, or parts of them, for a sponsor. **Not a kind of CDMO**: good clinical rather than good manufacturing practice, labour rather than capacity, the study rather than the batch.

Common Technical Document (CTD, eCTD) — the five-module structure of a marketing application: regional administration, summaries, quality, non-clinical, clinical. The electronic form adds an XML backbone declaring every file and its lifecycle operation.

CTMS — clinical trial management system: the operational record for sites, investigators, monitoring visits, payments and milestones. It holds no collected subject data and should not own the study identifier.

D

Database lock — the point at which a trial's database is closed to change so the analysis runs on a fixed dataset. **The locked state is an object** with its own identity, manifest and preserved snapshot, not a status field.

Dataset-JSON — a JSON serialisation for exchanging study datasets, positioned as a possible successor to the long-standing transport format. **It has not replaced SAS XPT**: comments closed on 9 June 2025 with 42 received and no change made [7].

Decommissioning — two dialects. In serialisation, retiring a serial number as dispensed, destroyed, exported or recalled. In validation, retiring a system — which **is not switching it off**, because its records must stay readable for their retention period.

Define-XML — the machine-readable metadata document accompanying submitted datasets: domains, variables, codelists, derivations. It is what makes submitted data interpretable, and it is treated as an afterthought almost everywhere.

Deviation — two dialects. In manufacturing, a departure from an approved procedure or specification, investigated to root cause with a list of affected batches. In clinical research, a **protocol deviation** at a site: a different object, a different workflow.

Discovery — the research stage that finds and optimises candidate molecules before anything reaches a person. Mostly outside the regulated regimes, which is why its data discipline is usually the weakest in the company.

Disposition — the recorded decision about what a batch may be used for: quarantine, under investigation, released for further processing, certified for sale, restricted, rejected, recalled, destroyed. Append-only, and per market.

Dispensing fee — what the final seller may charge for dispensing in South Africa. **Charged in addition to the Single Exit Price, not out of it**, on the price excluding value-added tax, with different bands for pharmacists and dispensing doctors [8]. Do not quote proposed bands as current.

Dossier — the body of evidence a regulator holds for a product. **The submission is an event; the dossier is a state** — so the current view is a computed projection over the whole sequence history.

DSUR — development safety update report: the annual safety report for a product in clinical development, due within a year of study approval and annually after [9].

E

E2B(R3) — the current standard message for exchanging individual case safety reports. **Mandatory in the European Union since 30 June 2022**, where the safety database refuses the previous revision outright [10]; its implementation guide was at an intermediate step as at 18 July 2025 [11].

E6(R3) — the current international good clinical practice guideline, **split into Principles plus Annexes**. Principles and Annex 1 took effect in the European Union on 23 July 2025; **Annex 2, adopted 3 June 2026, takes effect 15 January 2027** [12].

eCOA, ePRO — electronic clinical outcome assessment and patient-reported outcome: the subject's own report, on a device. Licensed instruments, which makes the licence a dependency of the study build.

EDC — electronic data capture: the record for case report form data and its audit trail. Built per study, often outsourced per study and decommissioned per study, which is why it makes a poor study master.

Effectiveness check — the dated evaluation, after a defined interval against a defined measure, of whether a corrective action worked. Make it a separate obligation, not a field on the action, or it never happens.

Electronic batch record (EBR) — the executed batch record held in a manufacturing execution system. **Review by exception comes with a governed set of exception definitions, not with the system** — the commonest reason these programmes disappoint.

ELN — electronic laboratory notebook: the record of an experiment's intent, procedure, observation and witness. Not the record for compound identity, and a result should point at a sample rather than a substance.

EPCIS — the GS1 standard for capturing supply-chain events: what happened, to which object, when, where and why. **EPCIS 2.0 is JSON, JSON-LD and REST as well as XML**, with an international adoption as ISO/IEC 19987:2024 [13].

eTMF — the electronic trial master file. **It is the inspection artefact**, and its completeness and filing timeliness are findings in their own right, independent of how well the trial was run.

Excipient — an ingredient other than the active substance. Some carry their own regulatory sensitivities — animal-derived, allergenic, novel — and a change to a functional one is usually a major variation.

Expectedness — whether a reaction is already described in a named reference document at a named version: the local label, or the investigator's brochure in a trial. **The same event can be expected in one market and unexpected in another.**

Expiry date — the date after which a finished product may not be used. Extendable only by an approved shelf-life variation, and **never for batches already released** without regulatory agreement.

F

Formulation — the active substance plus excipients, at a strength, in a dose form, by a route. Country-independent, and the second of the four identities a "product" can mean.

Fair market value (FMV) — the defensible rate paid to a healthcare professional for services. The South African marketing code states that **competition law prevents it prescribing an amount**, so each company sets and substantiates its own [14].

G

GAMP 5 — the ISPE guide to a risk-based approach to compliant computerised systems, second edition July 2022, 404 pages. **A paid industry guide with no legal force**; there is **no Category 2**, and the second edition does not mandate the V-model [15].

Genealogy — the record, built from executed batch records, of which component batches actually went into which product batches, both directions. The difference from where-used is the difference between recalling a product line and four batches.

Generic — a medicine with the same active substance as an off-patent reference product, authorised on an abbreviated dossier plus bioequivalence. **A different registration, at different named sites, to its own specifications**, with its own stability and safety obligations.

GCP — good clinical practice: the regime governing how trials are conducted, documented and overseen. The regime a contract research organisation lives under, and completely different from manufacturing's.

GDP, GWP — good distribution practice and, in South African usage, good wholesaling practice: the regimes over storage, transport and wholesaling, where temperature excursions and consignment traceability live.

GLN — GS1 Global Location Number: the identifier of a party or location. South Africa's traceability guideline requires one per relevant location or legal entity [16].

GLP — good laboratory practice: the regime for non-clinical safety studies a filing relies on. It does **not** govern quality-control laboratories, which sit under good manufacturing practice — a trap for anyone reading the words literally.

GMP — good manufacturing practice: the regime for making, packing and testing medicines to their authorisation. In South Africa the substantive text is the adopted PIC/S guide, not SAHPRA's own guideline, which carries only local additions [17].

Good Pharmacy Practice — the standards the South African Pharmacy Council sets for pharmacy premises and practice under the Pharmacy Act 53 of 1974. A separate inspection regime from SAHPRA's, over separate premises.

Gross-to-net — the gap between list price and what the manufacturer realises after rebates, chargebacks, discounts and fees. Enormous in the United States. **It has no South African analogue**, because the transactions producing it are largely prohibited above Schedule 0.

GTIN — GS1 Global Trade Item Number: the identifier of a product at a packaging level. What a verification system looks up to know what a pack claims to be.

GVP — good pharmacovigilance practice: the regime over an authorisation holder's safety system, its reporting clocks, its signal management and its risk-management plans.

GxP — the collective term for the regulated practices. A system's **GxP classification** decides which inspectorate cares about it, and is the attribute most application portfolios lack.

H

Hybrid system — one where part of the record is paper and part electronic. **Hybrids are legitimate when declared**: the record identified, the link verifiable, the printout requirement met. Undeclared hybrids are the finding.

I

ICH — the International Council for Harmonisation, producing guidelines in four families: Quality, Safety, Efficacy, Multidisciplinary. A guideline binds nobody until a regional regulator implements it, and **that gap is where almost every date error lives**.

ICSR — individual case safety report: the exchangeable record of one patient's adverse experience at one point in time. Reported repeatedly as information arrives, so the grain is the case *version*.

IDMP — the ISO family giving products a structured identity: 11615 for the authorised product, 11616 for the pharmaceutical product, 11238 for substances, 11239 for dose forms, 11240 for units. **No go-live date, only phased deadlines** [18].

Investigational medicinal product (**IMP**) — the product under test in a trial, including comparators and placebo. Made and released under GMP, labelled for the blind, with an expiry the randomisation system must agree with.

Indication — the condition a medicine is approved to treat, per authorisation. Indications diverge across markets for the same molecule, which is why claims and expectedness assessments are country-specific.

INN — International Non-proprietary Name: the generic name of a substance, assigned by the World Health Organization. A global name, which is why an internal compound number cannot do its job.

Interchangeability — whether a biosimilar may be substituted for its reference product, and by whom. **Jurisdiction-specific**, decided by substitution rules rather than by the approval, and the main reason biosimilar erosion is slower.

IQ, OQ, PQ — installation, operational and performance qualification: staged evidence that equipment or infrastructure was installed, operates and performs as specified. **Infrastructure is qualified; applications are validated.**

IRT, RTSM — interactive response technology, or randomisation and trial supply management: the record for the randomisation list, the kit assignment and the blind.

K

Kit — a packaged, labelled unit of investigational product assigned to a subject at a visit. Kit design, label expiry and dispensing logic all change when an amendment changes the dose, which makes an amendment a supply event.

L

Label — three dialects. The **regulatory label** is the approved product information text per market and language; the **physical label** is the printed component; the **investigational label** is the blinded clinical-supply label, which is not the material master's description.

LIMS — laboratory information management system: the record for samples, tests, results at determination grain and certificates of analysis. Not the author of the specification's approved limits, which descend from the dossier.

Loss of exclusivity (**LOE**) — the date a product loses protection and generic entry becomes possible. **Not the same as patent expiry**: exclusivity is a stack of patents, data and market protections, per country, and litigated.

Logistics fee — what a South African manufacturer pays a wholesaler **out of** the Single Exit Price, in place of a mark-up. **The typical percentage could not be sourced**, and it is the only genuine gross-to-net item in the local private market.

LOINC — a free coding system for laboratory tests and observations, with attribution and no-modification conditions. Central-laboratory transfers usually fail on a test-code or unit mismatch, and this fixes half of it.

M

Marketing authorisation — permission from one authority to one holder to place one product on one market, subject to conditions. **One brand in forty countries is forty authorisations**, differing in indications, sites, specifications, shelf life and label text.

Marketing authorisation holder (MAH) — the legal entity holding an authorisation and carrying its obligations, including pharmacovigilance. Often not the manufacturer, and often not the entity the commercial organisation thinks of as "us".

MBR 20 — the South African form signed per consignment of imported medicines, submitted to a customs officer at the port and to the authority. Imports may enter only through four named air and sea ports [19].

MCA — the Marketing Code Authority, which administers the voluntary South African code of marketing practice for health products, version 19 approved June 2026 [14]. Self-regulation hooked to the Medicines Act, not a statute.

MCC — the Medicines Control Council, SAHPRA's predecessor. The name still appears in old certificates, legacy dossiers and system reference data, and is a reliable marker of how old a record is.

MedDRA — the Medical Dictionary for Regulatory Activities: the hierarchy used to code events, conditions and indications. **Twice yearly; version 29.0 in March 2026** [20]. Licensed, not free.

MES — manufacturing execution system: in pharma, built around the electronic batch record rather than around scheduling. An architect who selects one on scheduling capability has selected on the wrong axis.

Molecule — the informal word for the active substance, and the coarsest of the four identities of a product. One molecule can carry twenty registrations across strengths and dose forms.

Monograph — a pharmacopoeial specification for a substance or preparation, revised on its own cycle. **The monograph edition is part of a specification's identity**, so a specification without its edition cannot be judged against.

N

NHI — National Health Insurance, South Africa's single-purchaser financing reform. **Judicially suspended, not being implemented**: an order of 24 February 2026 prohibits implementing any provision until the public-participation challenges resolve, and the Constitutional Court reserved judgment in May 2026 [21]. A scenario, never a roadmap item.

NHREC — the National Health Research Ethics Council, which registers and audits South African research ethics committees under the National Health Act 61 of 2003 on a five-year cycle.

Non-serious — an adverse event meeting none of the seriousness criteria. In South Africa non-serious reactions are **not routinely reported** individually; they appear in periodic report line listings, or on request within 30 calendar days [22].

O

ODM — the CDISC Operational Data Model, for operational clinical data and metadata including audit trails. **Version 2.0 exists and CDISC's own page contradicts itself about which release is current**; most implementations remain on the 1.3 line [23].

OOS — out of specification: a result outside its acceptance criterion. The investigation has two phases, the original result is **never deleted**, and any retest needs its justification recorded in advance.

OOT — out of trend: a result inside specification but outside the expected behaviour of the process or stability profile. A statistical concept, so the trend model's parameters are themselves versioned.

Originator — a research-based company that discovers and develops new medicines and sells them under exclusivity. The heaviest regulatory load of the eight archetypes, and usually present in South Africa only as a commercial and regulatory affiliate.

P

Pack, presentation — an authorised product in a specific container and count. Carries the barcode, the Single Exit Price and the stock-ledger identity, and is the level at which serialisation applies.

> **PART 11 COMPLIANT**
>
> **No product can be "Part 11 compliant", and a vendor claiming it is telling you about their marketing rather than their software.** The United States rule applies to **records and signatures**, not to software: compliance is a property of a system as implemented, configured, administered and operated, so it is something a company achieves rather than something it buys. Two facts to carry: the scope was narrowed by guidance in 2003, and the final guidance on electronic systems in clinical investigations issued in October 2024 [24].

> **PATIENT, SUBJECT**
>
> Not synonyms, and the dialect matters. In a **trial** the person is a **subject**, pseudonymous in the sponsor's systems. In **safety** and **commercial** data the person is a **patient**, and may be identifiable. Joining the two to build a "single patient view" is a consent and competition-law breach rather than a feature — one of the few hard architectural boundaries in the industry.

Periodic review — the scheduled re-examination of a validated system to confirm it is still fit, still controlled and still documented. A dated obligation that lapses silently when nobody models the validated system as an object.

PhPID, MPID — the pharmaceutical product identifier and the medicinal product identifier under IDMP. The first is country-independent, the second country-specific: the four-identities problem in standard form.

PIC/S — the Pharmaceutical Inspection Co-operation Scheme, which harmonises inspection and maintains the PE 009 GMP guide. **PE 009-18 came into force on 23 to 24 September 2026** [25]. Membership means **mutual confidence in inspection outcomes, not mutual recognition**.

Pharmacovigilance (PV) — detecting, assessing, understanding and preventing adverse effects. Architecturally a set of clocks, each with a start event, a duration, a recipient, a format and a compliance measure.

Pharmacovigilance System Master File (PSMF) — the document describing a holder's safety system, required in the European Union. **The term does not appear in South Africa's operative pharmacovigilance guideline** [22], which requires a documented system without naming it.

POPIA — the Protection of Personal Information Act 4 of 2013. Health information is special personal information, a sponsor's route is section 27 rather than section 32, section 72 gates cross-border transfer, and **there are no adequacy decisions** — the responsible party determines that itself [26]. Health-data regulations came into force on 6 March 2026 [27].

Preclinical, non-clinical — the studies done before and alongside human testing: toxicology, pharmacology, pharmacokinetics. Where the regime is good laboratory practice and the data standard is SEND.

Primary packaging — the packaging in direct contact with the product. **Part of the product's quality**, with extractables and leachables data in the dossier, so changing it is normally a major variation.

Product — the word that causes the most damage here, because it means four things with different cardinality: **molecule**, **formulation**, **registration**, **SKU**. Regulatory affairs means the third, the ERP the fourth, the safety database the second, commercial reporting a brand family. **Always ask which.**

Protocol — two dialects. In clinical research, the controlled document defining a study — and an amendment to it is a **data event** that invalidates downstream structures, not a document revision. In quality and validation, a pre-approved test plan.

> **PSEUDONYMISATION**
>
> **Pseudonymised data is not outside POPIA, and treating it as outside is the commonest privacy error in clinical architecture.** Only *permanent* de-identification takes data outside the Act. Trial data keyed on a subject identifier stays personal information in the sponsor's hands, because the sponsor plus the site can re-identify, and the draft research code tells researchers to **assume re-identification is possible** [28]. So the re-identification key stays with the site, and a sponsor that takes it "for convenience" has changed its legal position without recording a decision.

PSUR, PBRER — the periodic safety report for a marketed product. In South Africa it is **prepared** per the European reference-date list, or annually otherwise, but **submitted only on request** or as a registration condition, then within 30 calendar days [22]. Preparation and submission have different triggers.

Q

QMS, eQMS — the quality management system: the record for deviations, corrective actions, change control, complaints, audits, training and controlled documents. Expect one instance and fight to keep it that way.

QPPV — the qualified person responsible for pharmacovigilance. In South Africa the holder needs a responsible person **and a back-up resident in the Republic** [22], which is a staffing constraint before it is a system one.

Qualification — three dialects. **Equipment and infrastructure qualification** is the IQ, OQ, PQ chain. **Supplier qualification** is a property of the triple `(material, manufacturer site, receiving site)`. **Personnel qualification** is training and competence. They share a word and nothing else.

Qualified Person (QP) — the European Union's statutorily qualified individual who certifies batches and carries personal liability. **South Africa does not use this role**, which is the most important vocabulary divergence in this book.

Quality risk management (QRM) — the documented method that licenses proportionality everywhere else: risk-based validation, monitoring and change control. An architect who cannot point to the risk assessment justifying a system's validation depth has not implemented a risk-based approach.

R

Randomisation — allocating an eligible subject to an arm from a pre-generated schedule. The randomisation number is not the subject key: screen failures have screening numbers and no randomisation number, and a re-screened person can have two of the first.

Recall — withdrawal of product already distributed. Its scope is set by genealogy and distribution records, which is why those two structures separate a four-batch recall from a product-line recall.

> **RECOGNITION**
>
> **South Africa does not recognise another regulator's approval, and this word is misused more than any other in local regulatory conversation.** SAHPRA operates **reliance** — abridged and verified review on a recognised authority's assessment report — and **remains responsible for its own decision** [29]. **Recognition** is a separate pathway still being negotiated, its framework unpublished, and **mutual recognition** is a third thing that PIC/S membership does not provide [25]. A plan assuming a European approval can be presented and accepted is planning for a pathway that does not exist.

Registration — in South African usage, the marketing authorisation itself, and the third of the four identities. "We have 180 registrations" and "we have 180 products" are different statements.

Reliance — a pathway using another authority's assessment report to shorten the local review without transferring the decision. SAHPRA's target is 200 working days, against 360 for a new chemical entity and 250 for a generic without clinical data [30].

Responsible Pharmacist (RP) — South Africa's accountable batch-release individual: a Council-registered pharmacist **named on the licence, one per licence**, accountable to both the Council and SAHPRA, certifying each batch and ensuring processes **remain in a validated state** [17][31].

Retest date — the date after which a material, typically an active substance, **may be retested and, if conforming, used for a further defined period**. Not an expiry date, so a repeating event with its own results and arithmetic.

Right-first-time (RFT) — the share of batches completed with no deviation, out-of-specification result or rework. The definition is local, so publish the definition or do not publish the number.

Risk management plan (RMP) — the document setting out a product's identified and potential risks and the measures to minimise them. The measures' **effectiveness** must be evaluated, which is the step most companies drop.

S

SAE — serious adverse event: one resulting in death, life-threatening, requiring or prolonging hospitalisation, causing persistent disability or a congenital anomaly, or otherwise medically important. **The criterion matters, not just the flag**, because it is the audit evidence.

SAHPRA — the South African Health Products Regulatory Authority: registration, variations, licensing, scheduling recommendations, trial authorisation, pharmacovigilance, inspection, traceability, destruction approvals and section 21 access. **A full PIC/S participating authority since July 2007** [25].

SANCTR — the South African National Clinical Trials Register. It issues the Department of Health number **without which a trial may not start**, and full registration is required before participants are enrolled [2].

SAPC — the South African Pharmacy Council, which registers pharmacists and pharmacies and records premises under the Pharmacy Act 53 of 1974. The Responsible Pharmacist is accountable to it as well as to SAHPRA.

SAS XPT — the long-standing transport format for submitted study datasets, still the format despite a credible successor. The clearest available lesson in how slowly a mandated transport layer changes [7].

Schedules 0 to 8 — the South African classification of substances by safety and abuse potential, driving who may sell and prescribe, whether it may be advertised publicly, and what records must be kept. **Specified Schedule 5 and 6 registers are balanced quarterly, within 14 days of quarter end** [32].

SDTM — the CDISC Study Data Tabulation Model: submitted clinical data in two-letter domains. **The model is v2.1 of 10 June 2024; 3.4 is the implementation guide, not the model** [4]. Its grain is one row per observation.

Section 21 — the South African provision authorising sale of a **specified quantity of an unregistered medicine for a specified period**. Not rare: 20,245 applications in 2024/25 on a three-working-day clock [30], and **not subject to the Single Exit Price** [33].

SEND — the Standard for Exchange of Non-clinical Data: the SDTM-family model for animal toxicology studies, with the implementation guide at v3.1.1 [34].

Single Exit Price (SEP) — the gazetted price that is **the only price** at which a South African manufacturer may sell to any private buyer: no volume discount, no rebate, no bonus. **It does not apply to the State**, which buys by tender. **There were two adjustments in 2026**: 1.47% in January and an extraordinary 2.88% from 1 October 2026 [33].

Serialisation — giving each saleable pack a unique identifier so individual items are distinguishable. It turns the batch from a set into a population of addressable members, and multiplies the item master by the production volume.

Serialised unit — one physical pack bearing a unique identifier. South Africa requires product code, expiry, batch and serial at primary, secondary and trade-item level, and the code-and-serial combination **must be unique forever and never reused** [16].

Shelf life — the period from manufacture or packing for which a product meets its specification, approved per authorisation. Market-specific dated reference data, and the input to every future batch's expiry computation.

Signal — two dialects. A **safety signal** is information suggesting a new potentially causal association, or a new aspect of a known one, likely enough to justify verification; it is **not a case, not a confirmed risk and not a label change**. In engineering it is a measured value on a wire.

Site — two dialects. In manufacturing, a **licensed capability at an address**, the licence enumerating what may happen there. In clinical research, **one principal investigator at one location under one set of approvals** — so one hospital can hold two sites.

SKU — stock-keeping unit: the pack as the supply chain and the ledger know it, and the fourth identity. A mid-size generics company can hold hundreds of registrations and over a thousand of these.

SMQ — Standardised MedDRA Query: a curated grouping of terms for finding cases of a defined medical concept. Its definition changes by release, so counts are **not comparable across dictionary versions**.

SNOMED CT — a comprehensive clinical terminology, licensed by member country with national extensions. How real-world data arrives, and mapping it to clinical research structures is a project rather than a configuration.

Source data verification (**SDV**) — a monitor comparing captured data against the source document at the site. Under risk-based monitoring not everything is verified, so **which points were verified is itself evidence**.

> SPECIFICATION
>
> In quality, a **specification** is an approved, effective-dated, change-controlled set of tests and acceptance criteria for one material, purpose and market, traceable to an approved variation. In information technology, it describes what software should do. The dialects collide in a validation project, where both are "the spec" and only one has legal force. The conformance question is always *which version was in force for that market when that batch was released*, which almost no system answers correctly.

SSCC — Serial Shipping Container Code: the identifier of a logistics unit such as a pallet or case. What lets a warehouse receive a pallet without opening it.

Stability study — storing samples under defined conditions and testing at defined timepoints, to set shelf life, retest period and storage statement. A **schedule before it is data**, and a long-term arm crosses method revisions.

Subject — an individual enrolled, or screened for enrolment, in a study. The sponsor's record is a pseudonymous shell, and the sponsor usually cannot tell whether two subjects in two studies are the same person.

Substance — the regulated entity behind a molecule, independent of company: the non-proprietary name, the monograph name, the ISO 11238 identifier. Authority-curated, which is why an internal compound number cannot be submitted.

Supplier qualification — the documented basis on which a supplier or manufacturer may be used. **Three relationships are routinely conflated into one "approved vendor" flag**: the approved manufacturer, the supplier you buy from, and the qualified triple.

SUSAR — suspected unexpected serious adverse reaction: the trial-stage category driving the fastest clocks. In South Africa, 7 calendar days for a local fatal or life-threatening case and 15 for other serious unexpected reactions [9].

T

Tender — how the South African State buys medicines, entirely outside the Single Exit Price regime. **An award is not secure revenue**: on 22 October 2026 the High Court set aside the one-month and three-month pack awards of a R15.5bn antiretroviral tender, ordering a redo within 120 days [35].

TMF Reference Model — a governed, versioned, change-controlled industry taxonomy of trial master file content. **CDISC's, not DIA's, since April 2022** [36]; its version, zone count and artefact count could not be sourced and are not printed here.

Traceability — the ability to follow a product and its components through the chain in both directions. In South Africa the subject of its own guideline, which also brings IDMP master data into local scope through the registration route [16].

Transfer of value — a payment or benefit to a healthcare professional or organisation. Agreements must be signed by both parties and kept five years under the voluntary code; **whether public disclosure is required could not be established** [14].

U

UCUM — the Unified Code for Units of Measure: a free, computable encoding of units. Storing the code beside the display unit cannot be retrofitted, because historical unit strings are ambiguous.

Unblinding — releasing treatment assignment, normally after database lock and normally irreversible for that study. An unlock *after* unblinding needs a different integrity argument, because those making changes now know the assignments.

Up-versioning — re-coding a database to a newer dictionary version. **A project, not a refresh**: it needs medical review of every changed mapping, and previously submitted tables no longer reproduce unless the old coding is kept.

USUBJID — the unique subject identifier in submitted data. **It must be unique across the whole submission, not the study**, which is why the standard construction concatenates study, site and subject.

V

> VALIDATION
>
> **Validation is a maintained state, not a project phase, and more documentation is not more assurance.** A system is validated for a defined purpose, configuration and process, and stays validated only through change control, periodic review and ongoing evidence. **"We can just upgrade it" does not exist in a GxP environment**: an upgrade is a regulatory project with an IT component. **Agile is permitted; shipping without evidence is not.** And the word has three dialects — **process**, **computerised system** and **analytical method** validation — sharing a principle and almost no artefacts, so a project that does not say which it means delivers the wrong documents.

Validated state — the condition of a process or system shown fit for purpose and kept that way. In South Africa, keeping manufacturing and testing processes in a validated state is explicitly a Responsible Pharmacist duty [17].

Validated system register — the inventory of computerised systems with their GxP classification, record types held, validation state, periodic review date, named owners and retention obligations. **It should be the application portfolio with those columns added**, not a quality-owned spreadsheet.

Variation — a classified, dated change to an authorisation. In South Africa: Type IA notified within 12 months, Type IAIN notified 47 working days ahead and **deemed implementable on silence**, Type IB reviewable within 87 working days, Type II requiring prior approval within 220 working days [37]. In statistics the word means spread; only the first can stop a sale.

Visit — two objects, one name. The **planned** visit belongs to a protocol version, with a number, a nominal day and a window; the **actual** visit belongs to the subject. One table carrying both is the commonest clinical data modelling error.

W

Where-used — the query over the bill of material telling you which products *could* contain a component. **Only genealogy tells you which ones do**, and the gap between the two answers is measured in tens of millions of rand.

Wholesaler — a licensed distributor of medicines. In South Africa wholesalers **take no mark-up on the Single Exit Price**; the manufacturer pays them a logistics fee out of the exit price instead.

WHODrug Global — the licensed product dictionary maintained by the Uppsala Monitoring Centre, released 1 March and 1 September. **The United States regulator mandates it for studies starting after 15 March 2019** — the clearest case of a regulator making a paid product compulsory [38].

Working days — the unit South African regulatory timelines are expressed in, and the reason they are routinely misquoted. **360 working days is roughly 17 to 18 calendar months** of regulator time, before any sponsor response [30].

X

XEVMPD — the European Union's earlier extended product-data submission format, being succeeded by the product management service on an HL7 FHIR basis. **No retirement date has been published**, and pack sizes must still go through it where applicable [18].

Y

> YIELD
>
> **Higher yield is not better, and this is the most reliable way to tell whether a dashboard was designed by someone who understands GMP.** A registered process has a **validated range**, and a yield outside it is a deviation requiring investigation **even when it is higher**, because a sustained change in performance means the process has drifted from the state that was validated and registered. Continuous-improvement framing imported from general manufacturing is wrong here: if the company wants a better process, it changes the process under change control and files a variation.

Z

ZA eCTD — the South African electronic submission specification, **based on international version 3.2.2, portal-only**. Baselines from non-electronic content start at sequence 0000, new applications at 0001, and a structural error rejects the whole sequence [39].

Zone — two dialects. A **climatic zone** sets the long-term storage condition for registration stability data, and **South Africa's assignment could not be sourced**. A **TMF zone** is a top-level grouping in the trial master file taxonomy.

THE CHAPTER IN TEN LINES

- Pharmaceutical vocabulary has four dialects — discovery, clinical, regulatory, and quality and manufacturing — and the same word often means different things in two of them.
- Product means four things with different cardinality: molecule, formulation, registration and SKU. Always ask which.
- Validation is a maintained state, not a project phase, and it has three dialects of its own.
- Higher yield is not better under GMP: a yield outside the validated range is a deviation even when it is higher.
- SAHPRA operates reliance, not recognition, and PIC/S membership gives mutual confidence rather than mutual recognition.
- No product can be Part 11 compliant, because the rule applies to records and signatures rather than to software.
- Pseudonymised trial data remains personal information under POPIA; only permanent de-identification leaves the Act.
- A subject is not a patient, and joining trial data to commercial patient data is a breach rather than a feature.
- Where the industry genuinely disagrees, this glossary says so instead of picking a side.
- Every term with a version or a date in it needs re-checking before it goes in front of a client.

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- **23.** CDISC — ODM standard page (ODM v2.0 of 23 August 2023; the page's prose and its version table disagree about which release is current) https://www.cdisc.org/standards/data-exchange/odm
- **24.** US Federal Register, 2 October 2024 — notice of availability of the final guidance Electronic Systems, Electronic Records, and Electronic Signatures in Clinical Investigations, docket FDA-2017-D-1105 https://www.govinfo.gov/content/pkg/FR-2024-10-02/html/2024-22562.htm
- **25.** PIC/S — Revision of the PIC/S GMP Guide (PE 009-18, adopted 22 September 2026, in force 24 September 2026) and Members page (SAHPRA accession July 2007) https://picscheme.org/en/members
- **26.** J Coetzee — Cross-Border Data Flows and the Protection of Personal Information Act 4 of 2013, Part II: The Data Transfer Provision [2024] PER 48 https://www.saflii.org/za/journals/PER/2024/48.html
- **27.** Bowmans — South Africa: POPIA Health Information Regulations cross the finish line, 12 March 2026; goLegal — New POPIA Regulations: Processing of health information, 10 April 2026 https://bowmanslaw.com/insights/south-africa-popia-health-information-regulations-cross-the-finish-line/
- **28.** Academy of Science of South Africa — POPIA Code of Conduct for Research (draft, no effective date) https://www.assaf.org.za/wp-content/uploads/2023/04/ASSAf-POPIA-Code-of-Conduct-for-Research.pdf
- **29.** SAHPRA — SAHPGL-BAU-01 v5, Reliance Guideline, effective 6 August 2025 https://www.sahpra.org.za/wp-content/uploads/2025/08/SAHPGL-BAU-01_v5-Reliance-Guideline.pdf
- **30.** SAHPRA — 2024/25 Annual Report Presentation to Parliament, 8 October 2025 https://static.pmg.org.za/251008SAHPRAs_PRESENTATION_-_8_OCTOBER_2025.pdf
- **31.** SAHPRA — SAHPGL-LIC-01 v3, Guide on How to Apply for a Licence to Manufacture, June 2022; South African Pharmacy Council, Responsible Pharmacist overview https://www.sahpra.org.za/wp-content/uploads/2022/06/SAHPGL-LIC-01-Guide-On-How-to-Apply-for-A-Licence-To-Manufacture_DN.pdf
- **32.** SAHPRA — SAHPGL-CEM-NS-02 v3, Guideline to the Scheduling of Substances and Medicines, effective 30 May 2022 https://www.sahpra.org.za/wp-content/uploads/2022/05/SAHPGL-CEM-NS-02_v3-Guideline_Scheduling-of-Substances-and-Medicines.pdf
- **33.** SAnews — Health department sets increase for registered medicines (2.88% effective 1 October 2026; the SEP does not apply to the State); SA Health News on the second 2026 adjustment https://www.sanews.gov.za/south-africa/health-department-sets-increase-registered-medicines
- **34.** CDISC — SEND standard page (SENDIG v3.1.1, 30 March 2021) https://www.cdisc.org/standards/foundational/send
- **35.** Daily Maverick and GroundUp — High court sets aside multibillion-rand ARV contracts over procurement irregularities, 22 October 2026 https://www.dailymaverick.co.za/article/2026-10-22-high-court-sets-aside-multibillion-rand-arv-contracts-over-procurement-irregularities/
- **36.** CDISC — TMF Reference Model Initiative Charter V4.0 (part of CDISC since April 2022; no version, zone or artefact count stated) https://www.cdisc.org/sites/default/files/2023-09/CDISC_TMFRM_Initiative_Charter_V4.0.pdf
- **37.** SAHPRA — SAHPGL-HPA-06 v7, Guideline for Variations Addendum for Human and Veterinary Medicines, 30 September 2025 https://sahpra.org.za/wp-content/uploads/2025/12/SAHPGL-HPA-06_v7-Guideline-for-Variations-Addendum-for-Human-and-Veterinary-Medicines.pdf

- **38.** Uppsala Monitoring Centre — WHODrug Global (released 1 March and 1 September; subscription; FDA mandate for studies starting after 15 March 2019) https://who-umc.org/whodrug/whodrug-global/what-is-whodrug-global/
- **39.** SAHPRA — ZA-SAHPRA eCTD Specification and Guidance for Module 1 and Regional Information, v3.1, July 2024 https://ectd.sahpra.org.za/assets/files/Specifications/2.21-ZA-SAHPRA-eCTD-Specifications-v3.1.pdf

APPENDIX C

Standards, bodies and where to find them

What each standard is, who publishes it, what version is current with a date, how you get it — and the one procedural skill that keeps this appendix from going stale.

This appendix answers four questions about each standard and body an architect will meet: what it is, who publishes it, what version is current with a date, and how you get it and on what terms. It adds a fifth, which matters more than the other four: **whether you will genuinely meet it**, because a surprising number of standards named in pharmaceutical IT decks never appear in a real programme.

Two rules govern every number below. **No version here is stated from memory.** Each came from a publishing body's own page or a regulator's own page, with a retrieval date, and the research behind this book carries a cut-off of 24 October 2026. And where a version, a date or a status **could not be sourced, this appendix says so and prints nothing** — which happens far more often than a confident-looking reference list would suggest.

The bodies

Body	Produces	Access	Will you meet it?
CDISC	SDTM, SEND, ADaM, CDASH, ODM, Define-XML, Dataset-JSON; 360i; USDM with TransCelerate; hosts the TMF Reference Model	Standards pages public; at least some material members-only; **fee structure not sourced** [1]	Yes, if the company runs or submits trials
ICH	The Q, S, E and M guideline families; owns MedDRA through the MSSO	Guidelines free. **ich.org's own guideline listing pages returned only page metadata** [2]	Yes, constantly
ISO	The IDMP family; ISO 13485; the management-system standards; adoptions of EPCIS	Paid catalogue	Through a regulator's implementation, not directly
HL7	FHIR; the Vulcan accelerator; the FHIR-to-CDISC mapping with CDISC	Free [3]	Yes, through the European product master data service and real-world data
GS1	GTIN, SSCC, GLN, serialised GTIN, EPCIS, Digital Link, the General Specifications	Standards free at the reference site; member organisations charge for numbers	Yes, wherever packs are identified
ISPE	GAMP 5; the Baseline Guides; Good Practice Guides	**Paid**: GAMP 5 second edition at USD 540 for members, USD 1,125 otherwise [4]	Yes, by convention rather than by law
PIC/S	The PE 009 GMP guide series; the PI-series guidance and aide-memoires	Free; the publications listing **does not mark superseded entries** [5]	Yes, and in South Africa it is the operative GMP text
WHO	Technical Report Series annexes; the prequalification procedure	Free [6]	If the company sells into donor-funded or African public markets

Body	Produces	Access	Will you meet it?
SAHPRA	Registration, variations, licensing, scheduling, trial authorisation, pharmacovigilance, inspection, traceability, destruction, section 21	Free, on its own site	Yes, weekly
South African Pharmacy Council	Pharmacist and pharmacy registration; premises recording; Good Pharmacy Practice rules under the Pharmacy Act 53 of 1974	Free	Yes, through the Responsible Pharmacist
Information Regulator	POPIA guidance notes; prior-authorisation process; the 2026 health-data regulations	Free	Yes, for any clinical or patient-facing data
NHREC	Registration and audit of research ethics committees under the National Health Act 61 of 2003	Free	Only if the company runs trials in South Africa
MedDRA MSSO	MedDRA, twice yearly	**Licensed subscription**; **meddra.org's homepage returned only metadata** at the research retrieval date [7]	Yes, if there is a safety database
Uppsala Monitoring Centre	WHODrug Global, 1 March and 1 September	**Licensed subscription** [8]	Yes, and the United States regulator mandates it
TransCelerate	Member initiatives, of which Digital Data Flow and USDM are the substantial ones	Member-funded; **portfolio not sourced**	Rarely, and then through CDISC
Pistoia Alliance	Research data standards, ontologies, templates; a pharmacovigilance systems and processes initiative	Member collaboration; project pages public [9]	Rarely. Worth watching, not planning on
Marketing Code Authority	The voluntary South African code of marketing practice, version 19 approved June 2026 [10]	Free	Yes, if the company promotes to healthcare professionals
The Open Group	TOGAF; ArchiMate, whose fourth version is document C260	Specification licensed. **The publications page was robots-disallowed and the ArchiMate tool certification register failed to load** [11]	Yes — it is your own notation

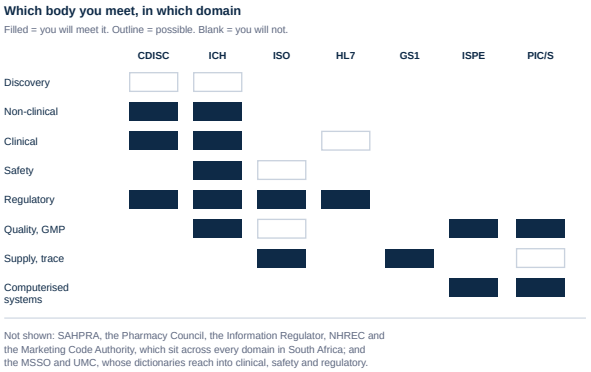


Figure 22.1 Which body an architect meets, by domain. **The gaps matter as much as the marks: no body appears in a column for capability, process or application architecture, because none of them covers it.**

CDISC

The clinical data standards, and the only place in this book where a standards body's output is enforced by an automated gate.

Artefact	Version, with its date	Note
SDTM model	**v2.1, 10 June 2024**	The *model*. Not 3.4 [1]
SDTM implementation guide	**v3.4**, which relates to SDTM **v2.0 of 29 November 2021**	The guide trails the model, and they are decoupled [1]
ADaM model, ADaM implementation guide	Model **v2.1, 17 December 2009**; guide **v1.3, 29 November 2021**	From CDISC's own ADaM page [12]. A separate research pass could not confirm these and marked them unverified — **re-check before printing them in a client document**
SENDIG	**v3.1.1, 30 March 2021**, plus SENDIG-DART v1.2 and SENDIG-Genetox v1.0, both 28 June 2023	[12]
CDASH	**Not sourced.** Nothing is printed here	Settled by the CDISC foundational standards page
ODM	**v2.0, 23 August 2023** — and **CDISC's own page contradicts itself** about which release is current; most implementations remain on the 1.3 line	[13]
Define-XML	**v2.1.11, 6 April 2026.** It runs as x-releases; there is no base v2.1 release date	[14]
Dataset-JSON	Published for review, piloted, and **it has not replaced SAS XPT**	Comments closed 9 June 2025 with 42 received and no change made [15]

The sentence that matters more than the table. The authority's own data standards catalogue governs what you may submit, not CDISC's latest release. The United States catalogue is the operative document for a filing there, **the research could not read it** — the linked PDF returned as binary and direct download was refused — so **no version support matrix appears in this book** [16]. Go to the catalogue, read it yourself, and record the date you read it.

What is genuinely mandatory is narrower and sharper than "the FDA requires CDISC". It is the **Technical Rejection Criteria for Study Data**, which name a trial summary dataset, a demographics dataset, an analysis subject-level dataset and a `define.xml`, in named sections of Modules 4 and 5, with refuse-to-file for new drug and biologics applications and refuse-to-receive for abbreviated ones, for studies started after 17 December 2016 and after 17 December 2017 for a commercial investigational application [17]. **The document the research could read lists some criteria effective dates as "TBD"**, so say the criteria exist and are enforced and do not print a complete in-force list.

ICH

The guideline families — Quality, Safety, Efficacy and Multidisciplinary — adopted by regional regulators. A guideline reaches Step 4 when the Assembly adopts the harmonised text and Step 5 when a regional regulator implements it, and **the gap between the two is where almost every date error in pharmaceutical compliance writing lives**.

E6(R3), good clinical practice, is split and partly not yet in force. Principles and Annex 1 took effect in the European Union on **23 July 2025**. **Annex 2 was adopted by ICH on 3 June 2026**, by the European committee on 25 June 2026, published at Step 5 on 15 July 2026, and **takes effect on 15 January 2027**, as does the consolidated guideline [18]. Three errors to avoid: calling R3 a draft, treating Annex 2 as live, and generalising the European dates to other regions. **The positions of the United States, United Kingdom, Japanese and South African authorities were not sourced.**

E2B(R3), the safety message, has been mandatory in the European Union since 30 June 2022, and the European safety database does not accept the previous revision at all [19]. Note the asymmetry: the mandate is firm while **the implementation guide was itself at Step 3 as at 18 July 2025** [20]. A mandated message whose guide is still moving is a normal state of affairs in this industry and the architecture should expect it.

M4 and M8 give the Common Technical Document and its electronic form. Version 4.0 is **mandatory in Japan for applications filed from April 2026**, the transition for the former format having ended 31 March 2026. **The United States regulator has published no requirement date**, though it has accepted the new version since 16 September 2024. The European agency has it optional from 22 December 2025, recommended in the first quarter of 2027 and **mandatory in the first quarter of 2028** [21]. The consequential fact is not a date but a lifecycle rule: **an application already filed in the older**

version stays in it for its life, so a regulatory estate runs two submission formats in parallel for years and a migration plan that assumes a cutover is wrong.

M11, the structured protocol, is final and not binding. The United States notice of availability published 22 May 2026, finalising drafts of 22 December 2022 and 6 June 2025, with three documents — guidance, template and technical specification — and the agency states it is not binding [22]. It is the first harmonised, regulator-endorsed data model for a protocol, which is the precondition for everything in the digital-protocol argument.

The Q family: name the topics, print no statuses. Q9(R1) on quality risk management, Q10 on the pharmaceutical quality system, Q12 on lifecycle management, Q13 on continuous manufacturing, Q14 on analytical procedure development and Q2(R2) on analytical validation are all architecturally relevant. **The step and effective dates for every one of them are unverified** in the research behind this book, because ICH's own quality and efficacy listing pages returned only page metadata [2]. Go to the European agency's or the United States agency's guideline pages instead: they render, and they state regional effective dates, which ICH's own pages do not.

ISO IDMP

Five standards, conventionally described together: **ISO 11615** for the authorised medicinal product, **11616** for the pharmaceutical product, **11238** for substances, **11239** for dose forms, units of presentation, routes and packaging, and **11240** for units of measurement. **Only ISO 11238 was confirmed by name, from the European agency's own page; the edition years of all five are unverified and none is printed here** [23].

IDMP has no go-live date. What it has is a phased European implementation with dated obligations: structured manufacturer data and pack sizes for products on the critical-medicines list by **June 2026**, manufacturer data for other nationally authorised products by **December 2026**, and remaining pack sizes by **June 2027** — **two of which have already been extended once**, in September 2025 [23]. No other region was found to be implementing at comparable depth.

HL7 FHIR

Two independent paths put FHIR inside a pharmaceutical architecture: the European product management service uses it as its data standard and application programming interface basis, replacing the older extended product-data format, and real-world data arrives from health systems in it.

R4 (4.0.1, 27 December 2018) carries the normative content; R5 (5.0.0, 26 March 2023) is the "current published version" and is trial-use. HL7's own publication history **does not mention R6 and is internally inconsistent about R4B's date**, giving 27 December 2018 in one table and 28 May 2022 in another [24]. Build against R4 unless something specific requires otherwise, and say which release every interface targets.

GS1

Artefact	Version, with its date	Note
General Specifications	**Release 26.0 (2026)**	Always quote the release with its year
EPCIS	**2.0, released June 2022** with its vocabulary companion; **adopted as ISO/IEC 19987:2024**	**2.0 added JSON, JSON-LD and REST bindings as well as XML** [25]. Textual identity between the GS1 release and the ISO edition is unverified
Digital Link	**1.1.3**	Makes a GS1 identifier resolvable as a web address [26]
Healthcare GTIN allocation rules	**Not sourced.** Nothing is printed	They govern when a product change requires a new identifier — a master-data rule with regulatory consequences

> **SUNRISE 2027**
>
> **Never write that GS1 requires a two-dimensional carrier by 2027.** The correct sentence is this: it is a **voluntary industry capability ambition published by GS1 US, with no day and no month**, which the organisation's own page calls a milestone, states that it is not necessary to implement all two-dimensional capabilities by that date, and organises as "crawl, walk, run" [27]. The page could not be retrieved in the second research pass, and search instead returned trade headlines of the form "Won't be Optional for Brands" [28]. **Pharmaceutical carrier obligations come from the United States supply chain security statute and the European falsified medicines regime, not from GS1.**

GAMP 5, and the ISPE guides

GAMP 5, second edition, published July 2022, 404 pages, by ISPE through its GAMP community of practice. USD 540 for members and USD 1,125 otherwise [4]. Three corrections, all of which an author working from general knowledge will get wrong.

- **It is a guide with no legal force anywhere.** A company is inspected against Annex 11 or the United States electronic records rule, not against GAMP. Regulators reference it; inspectors expect an approach consistent with it; nothing more.
- **There is no GAMP Category 2.** It covered firmware and was withdrawn, leaving categories 1, 3, 4 and 5. **The current category scheme in the second edition is unverified** — ISPE's own product page does not describe it — so **do not print category numbers**.
- **The second edition does not mandate the V-model.** It states that specification and verification can be iterative and incremental, which is what makes an agile approach arguable.

The **Baseline Guides** are paid facility and system design guidance. **The current volume list and the current edition of each could not be sourced**, so none is printed. The one that intersects an IT architecture most directly is commissioning and qualification, because it governs how equipment and its control systems are qualified — which decides whether a change to a programmable controller is a qualification event.

PIC/S

Document	Date	Note
PE 009-18, the GMP guide	Adopted 22 September 2026; **in force 23 to 24 September 2026**	The change is the revised Annex 19 on reference and retention samples. The two research passes give 23 and 24 September; check which [5]
PI 006-4, qualification and validation recommendations	**1 October 2026**	Current
PI 041-1, data management and integrity	**1 July 2021**	Not its 2016 or 2018 drafts [5]
PI 054-1, risk-based change management	15 July 2021	
PI 040-1, classification of GMP deficiencies	1 January 2019	Where the critical, major and minor scheme actually lives

Three facts worth carrying. **PE 009 Part I never uses the term "ALCOA"** — its nearest statement is a principle on accuracy, integrity, availability and legibility, plus contemporaneity and signed alterations [29]; the verified South African statement of the mnemonic is **ALCOA-CCEA in the national good clinical practice guidelines**, which makes it a clinical requirement rather than a manufacturing one. **SAHPRA acceded to PIC/S in July 2007** — before the United States authority in January 2011, Japan in July 2014 and Brazil in January 2021 [30] — so "SAHPRA only adopts the PIC/S guide because it is not a member" is wrong. And **membership means mutual confidence in inspection outcomes, not mutual recognition** of authorisations or batch certification.

Note also that **the publications listing does not mark superseded entries** [5], which is a practical trap for anyone citing a PI number from a search result.

Computerised systems: Part 11, Annex 11, and computer software assurance

21 CFR Part 11 is the United States rule on electronic records and signatures: when an electronic record may substitute for paper and what a signature must satisfy. Its scope was narrowed by guidance in 2003, and the final guidance on electronic systems, records and signatures in clinical investigations issued in **October 2024** [31]. **The current text of the rule was not re-confirmed by the research behind this book**, so no clause numbers or scope language are quoted here. And **no product can be "Part 11 compliant"**: the rule applies to records and signatures, not to software.

EU GMP Annex 11, computerised systems, is the biggest live watch item in this appendix. The version in force is still the revision of January 2011. A draft was consulted jointly from **7 July to 7 October 2025**, alongside a revised Chapter 4 and a new Annex 22 on artificial intelligence, and a PIC/S concept paper on the revision is public [32]. **There is no published final text and no published date.** Write it as dated and unresolved, do not predict, and design for the stricter reading — because if it becomes more prescriptive on audit-trail review, cloud and artificial intelligence, a large class of decisions currently made on risk-based judgement becomes constrained.

The United States computer software assurance guidance went final in September 2025, docket FDA-2022-D-0795. It is a device guidance from the device and biologics centres under **21 CFR Part 820**, not drug good manufacturing practice [33]. Pharmaceutical companies adopt its thinking by analogy, and should say that they are doing so. **Saying that it applies to drug manufacturing is wrong.**

The licensed dictionaries

MedDRA is maintained by the maintenance and support services organisation under ICH, released **twice yearly**, with **version 29.0 released in March 2026** and training material for 29.1 dated 21 August 2026 [7]. It is a **licensed subscription product**; the fee bands could not be sourced, and **meddra.org's homepage returned only page metadata** at the research retrieval date, which is itself a practical trap.

WHODrug Global is maintained by the Uppsala Monitoring Centre, released **1 March and 1 September**, covering products from nearly 150 countries, delivered as UTF-8 comma-separated files under subscription. In the organisation's own words, **the United States authority mandates the dictionary for studies starting after 15 March 2019** [8]. It is the clearest example available of a regulator making a paid subscription product compulsory, and the architectural consequences — your contract research organisation needs its own licence, your non-production environments may not be allowed to hold it — arrive during contract negotiation and are always late.

The TMF Reference Model

A maintained, versioned, change-controlled, vendor-neutral taxonomy of trial master file content, with a steering committee and a change control board. **It is CDISC's, not DIA's, and has been since April 2022** [34]; large volumes of live consultancy and vendor collateral still say otherwise. Note that the charter's own history is internally inconsistent, since a 2017 charter version already described the move.

Its version number, its zone count and its artefact count could not be sourced and are not printed here. The charter states none of them. Secondary material points at a 3.x line and eleven zones; no primary confirmation was obtained. Describe the three-level shape — zones, sections, artefacts — and the governance, because the enumeration is the part that will be out of date by the time this book is read.

The device boundary, and the management-system standards

ISO 13485 is the quality management system standard for medical devices, deliberately not aligned to the common high-level structure the way the general management-system standards are. **Its current edition year is unverified and is not printed**: the research surfaced both a 2016 catalogue listing and third-party speculation about a 2026 edition. The United States device quality regulation was replaced by a quality management system regulation **effective 2 February 2026**, updating Part 820, and **the ISO 13485 edition it incorporates by reference could not be sourced** [35].

Three management-system dates, reused from this series' manufacturing research and verified there: **ISO 9001:2026 published 16 September 2026**, with the 2015 edition withdrawn; **ISO 14001:2026 published 15 April 2026**; and **ISO 45001:2018 is still the current edition**, with a revision in progress.

At the plant boundary: ISA-95 and ISA-88

ANSI/ISA-95.00.01 is the 2025 edition, announced 10 April 2025, superseding the 2010 one. Two things follow that a plant-integration programme should know. **There is no ISA-95 conformance scheme** — no tool, product or integration can be certified against it, and anyone claiming otherwise is making a marketing claim. And **the canonical XML binding predates the current part**: B2MML version 7, released 24 November 2020, was built to the 2018 and 2019 editions, and **it is MESA International's, not ISA's**, royalty-free with attribution. **ISA-88 part designations and dates are unverified** and are not printed.

South African instruments

Instrument	Status, with its date	Note
Medicines and Related Substances Act 101 of 1965	Consolidated to Government Gazette 40869 of 26 May 2017	Its **General Regulations** carry most of the operative detail; the relevant generation is the set published 25 August 2017, and **an earlier 2003 set still circulates online** — citing the wrong one is easy [36]
The SAHPGL guideline series	Many documents, each with its own version and five-year validity	See the warning below
SA GCP 2020, 3rd edition	Published 2020; **no exact publication date in the retrieved text**	The source of the 10-year trial retention floor and of ALCOA-CCEA [37]
ZA eCTD specification	**v3.1, July 2024, based on the international v3.2.2**	**Portal-only**; PDF 1.4 to 1.7 required; an error rejects the whole sequence; baselines start at sequence 0000. **No v4.0 date** [38]
Transparent pricing regulations	Single Exit Price machinery, commonly cited to section 22G	**The gazette notice numbers for the 2026 adjustments, and the methodology, could not be sourced** [39]
Pharmacy Act 53 of 1974	With regulations on the practice of pharmacy and Good Pharmacy Practice rules under section 35A	The Responsible Pharmacist's other regulator [40]
POPIA, Act 4 of 2013	Health-data regulations **in force 6 March 2026**, signed 27 February 2026, Gazette 54268, Notice 7198, **no transitional period**	**Whether a pharmaceutical manufacturer, authorisation holder or sponsor falls within the eight covered categories is unverified and must not be asserted either way** [41]
Occupational Health and Safety Act, with the Hazardous Chemical Agents Regulations 2021	GN R.280, Gazette 44348, 29 March 2021	Where the 30-year chemical exposure clock comes from

> THE STANDING SAHPGL WARNING
>
> **SAHPRA's guideline versions move faster than their own stated validity, and every citation in this book needs a version-and-date re-check before it goes in front of a client.** The research found **SAHPGL-INSP-02 version 9, dated 17 November 2025, superseded by version 10, dated 13 April 2026** — inside its own five-year validity window — and **the two research passes disagree about which text is operative**, one treating v9 as current and the other v10. There are also three generations of a South African GMP guideline discoverable on the open web at once. Treat re-checking the guideline index as a standing job, not a one-off task.

How to check a standard's version

This is the one procedural skill that keeps an appendix like this from going stale, and it is worth more than the tables above.

1. **Go to the publishing body's own page for that standard, not to a search result.** Search returns resellers, consultancy summaries and "best X in 2026" content, all of which carry versions that were current at some unstated point.
2. **For *status and effective dates*, go to the regional regulator instead.** ICH's own listing pages did not render at all in the research behind this book [2], and even when they do they state no regional effective dates. The European and United States agency pages render and do state them.
3. **Read the document's own cover sheet and its document-history table.** That is where a superseded version admits it has been superseded, and where a version dated inside another's validity window shows up.
4. **Check whether the listing you are reading marks superseded entries.** The PIC/S publications listing does not [5]. A listing that does not mark them will happily show you a document that was replaced five years ago.
5. **Record the URL and the retrieval date next to the fact, in the same sentence or the same cell.** A version printed without a retrieval date is a liability; a version printed with one is evidence of when you looked.
6. **Never state a version from a vendor blog, a trade headline or a conference deck.** Two of the worst errors this appendix corrects — the two-dimensional carrier "requirement" and the European product data deadlines — entered circulation exactly that way.
7. **For a licensed standard, check the catalogue entry rather than a reseller's listing**, and budget the licence while you are there.

A closing statement about what is not known

There are 175 unverified items in the research behind this book. Some are small. Several are not: the current contents of the United States data standards catalogue, the edition years of five identification standards, the current text of an electronic records rule, a reference model's version number, and whether a South African code of conduct for research is in force.

The honest practice is three steps and it has no shortcut. **Find out** — go to the primary source and read it. **Record the date** you read it, beside the fact. **Cite it**, so that the next person can see what you saw and when. An architect who does that produces material that ages visibly and can be repaired. An architect who launders an unverified item into a confident sentence produces material that ages invisibly, which is worse, because nobody knows which parts to distrust.

THE CHAPTER IN TEN LINES

- Never state a standard's version from memory; every version in this appendix came from a publishing or regulatory body's own page, with a retrieval date.
- The authority's own data standards catalogue governs what you may submit, not a standards body's latest release.
- ICH E6(R3) is split: Principles and Annex 1 took effect in the EU on 23 July 2025, and Annex 2 takes effect on 15 January 2027.
- eCTD v4.0 is mandatory in Japan and not in the United States or the European Union, and existing lifecycles stay on the older version for years.
- GAMP 5 is a paid industry guide with no legal force, it has no Category 2, and its second edition does not mandate the V-model.
- EU GMP Annex 11 in force is still the January 2011 version; a 2025 draft has no published final text or date.
- The computer software assurance guidance is a device guidance under Part 820, not drug good manufacturing practice.
- GS1 does not require a two-dimensional carrier by 2027; that is a voluntary industry capability ambition with no day or month.
- SAHPRA guideline versions move faster than their stated validity, so every citation needs a version-and-date re-check.
- There are 175 unverified items in the research behind this book: find out, record the date, cite it.

SOURCES FOR THIS CHAPTER

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APPENDIX D

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The consolidated source list, grouped by domain, with what each one could and could not tell us — and the nine sources most worth re-checking before any of this reaches a client.

The consolidated source list, grouped by domain and deduplicated: about 256 citations across four research files reduce to the list below, because many sources carry several facts and several were cited twice. Where a file flagged a source as secondary, the primary is preferred here.

Four conventions. **Every entry was retrieved on 24 October 2026** unless it says otherwise; a few come from this series' manufacturing research of 16 October 2026 and are marked (*reused*). **A † marks a source seen only as a headline or a secondary report** — treat such a fact as weaker, and never build an argument on one alone. **Figures carry their own as-at dates**, which govern where they differ from a retrieval date. And **several official pages returned only their shells, refused direct download or failed with a client error**; that is recorded rather than hidden, because it tells you what you will have to obtain another way.

A standing caution: a source consulted in October 2026 may have moved. If a link fails, search the publishing body for the document title, not the URL.

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Research files behind this book

- Internal — the four pharmaceutical research files of 24 October 2026: the business and South African market; systems and processes; standards and providers; data and South African regulation. About 172,000 words, 256 citations.
- Internal — the manufacturing research files of 16 October 2026, reused for ISA-95 and ISA-88, B2MML, the ISO management-system editions, EPCIS as an ISO adoption, the GS1 specifications and Sunrise reading, cloud regions, and several South African retention clocks.

The sources most worth re-checking before this goes in front of a client

The research flags these, in this order. Each is a fact on which a recommendation could rest.

1. **Whether the Dassault and ArisGlobal transaction has closed.** If it has, Dassault owns both Medidata and ArisGlobal — a clinical platform and a safety, regulatory and quality platform — and **the three-way clinical-platform framing every pharmaceutical IT deck uses is wrong.**
2. **Every per-standard date in the United States data standards catalogue.** The PDF returned as binary and the site refused direct download, so no version support matrix appears here.
3. **Whether the eCTD technical rejection criteria are all in force.** The document read lists some effective dates as "TBD".
4. **The TMF Reference Model's version, zone count and artefact count.** The charter states none of them.
5. **Whether a POPIA code of conduct for research is in force.** An in-force sector code removes the section 57 prior-authorisation requirement entirely.
6. **Which procurement framework now governs health-product tenders**, after the Public Procurement Act 28 of 2024 was struck down on 17 September 2026, invalidity not suspended.
7. **Whether the Constitutional Court has ruled on the NHI public-participation challenges**, heard 5 to 7 May 2026 with judgment reserved.
8. **The operative version of each SAHPGL guideline cited.** One was superseded inside its own validity window, and the two research passes disagree about which text is current.
9. **The Annex 11 revision's final text and date.** Consulted to 7 October 2025; nothing final published.

The categories most likely to have moved

Reading this more than a few months after October 2026, assume five groups have changed.

- **Standards versions.** Dictionaries release twice a year, metadata standards move quarterly, and guideline steps and regional effective dates move without notice.
- **Vendor ownership.** Six changes landed in this sector inside two years, one still open at the research cut-off.
- **SAHPRA guideline versions.** The fastest-moving category, and where a stale citation does the most damage, because a client will recognise it.
- **Single Exit Price adjustments.** Two in 2026 alone, one extraordinary.
- **The litigation states of National Health Insurance and the procurement framework.** Both live, both with judgments reserved.

THE CHAPTER IN TEN LINES

- Every source here was retrieved on 24 October 2026 unless the entry says otherwise.
- A dagger marks a source seen only as a headline or a secondary report; never build an argument on one alone.
- Figures in this book carry their own as-at dates, which govern where they differ from a retrieval date.
- Several official pages returned only their shells or refused direct download, and that is recorded rather than hidden.
- The highest-priority re-check is whether the Dassault and ArisGlobal transaction has closed, because it breaks the usual clinical-platform framing.
- No version support matrix appears in this book, because the authority catalogue that governs it could not be read.
- An in-force POPIA code of conduct for research would remove the prior-authorisation requirement entirely, so its status matters more than its length.
- Which framework governs health-product tenders is unresolved after the Public Procurement Act was struck down.
- SAHPRA guideline versions move faster than their stated validity, so every citation needs a date re-check.
- A source consulted in October 2026 may have moved: search the publishing body for the document title, not for the URL.